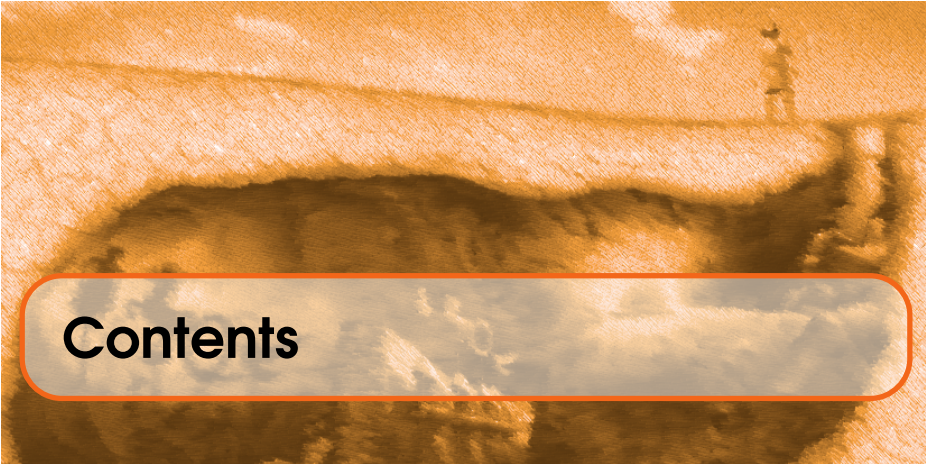


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v1.0

To my wife Justi, my son Daniel,
and my two daughters Teresa and Lucía.



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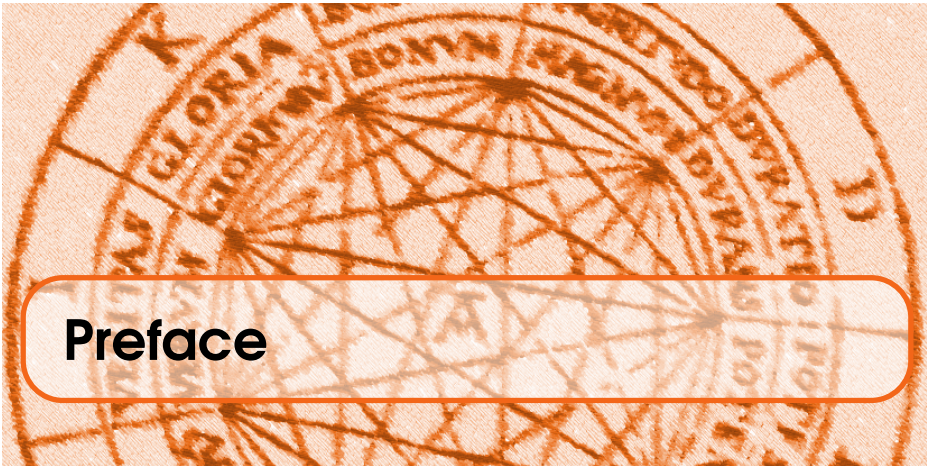
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Preface

*Perfection is achieved not when there is nothing more to add,
but when there is nothing left to take away.*
Antoine de Saint-Exupéry

We live in an era that places a high value on knowledge and celebrates scientific advancement. The development of vaccines in record time, the detection of gravitational waves, and the use of artificial intelligence to model protein structures are just a few examples of the extraordinary progress science has made in recent years. However, this celebration of progress should not obscure the structural limitations within the scientific enterprise itself. Despite its accomplishments, science is constrained by the methodological and conceptual frameworks it employs. For example, the "publish or perish" culture in academia often results in a flood of low-quality publications, which can dilute the impact of meaningful scientific work. Another major constraint lies in the allocation of research funding, which often fails to prioritize investigations with the highest potential societal return.

This book is driven by a central aspiration: to overcome the methodological constraints of science in order to challenge established knowledge, explore the unknown, and progress towards the limits of scientific understanding. Our aim is to investigate not only well-known unsolved problems but also those that remain

outside our current awareness, the known unknowns and the unknown unknowns. But this pursuit is not merely an academic curiosity; it is a practical necessity. Rather than developing a purely theoretical framework, our goal is to create an approach that can be applied in real-world scientific contexts.

Our assumption is that to explore and ultimately reduce what we do not yet know, we need a structured framework for understanding ignorance itself. To this end, we propose a rigorous mathematical theory, named the Theory of Nescience, grounded in the principles of computability, randomness, and artificial intelligence. The theory builds upon a striking insight: perfect knowledge implies randomness. Although this may initially appear counterintuitive, given that science is traditionally associated with structure, order, and explanation, it becomes clearer when we consider the process of theory refinement. As scientific understanding improves, our descriptions of phenomena become more concise. Eventually, a theory becomes so concise that it cannot be simplified any further. Its description becomes incompressible, which aligns with the mathematical definition of a random string.

Randomness sets the ultimate limit to our knowledge, since a random description cannot be refined any further and, assuming the theory is accurate, our understanding must be perfect. Far from being a handicap, the constraints that randomness imposes on knowledge pave the way for new possibilities in science and technology. By comprehending these limitations, we can address the most challenging open problems and discover completely new research topics.

From this theoretical foundation emerges a new mathematical framework for understanding nescience, or human ignorance, and the processes by which knowledge is acquired, organized, and refined. Some of the contributions of this book are:

- A mathematical theory to quantify our lack of scientific knowledge, based on computability and randomness, and on the assumption that measuring how much we do not know is easier than measuring how much we do know.
- A new collection of metrics to measure how much we do not know, whether about individual research topics or broader research areas, and how new research contributions either reduce or fail to reduce that ignorance.
- A practical framework to address some of the most relevant challenges in science, such as defining what constitutes perfect knowledge, identifying the limits of science, and discovering new, previously unknown, research topics.
- A software library that combines the Theory of Nescience with principles of artificial intelligence to automatically derive new knowledge and make predictions based on data.
- A web-based application for exploring, organizing, and discovering

promising unknown research topics.

The Theory of Nescience challenges us to rethink the aims of scientific inquiry. Instead of striving to maximize what we know, it encourages us to minimize what we do not know. By shifting our focus from accumulation to reduction, we open the door to more meaningful measures of scientific progress. The chapters that follow lay the theoretical groundwork for this new approach, one that begins with ignorance and ends in knowledge.

Research Agenda

In this section, we present a focused list of research questions aimed at addressing important gaps in our understanding of science and the scientific method. Our goal is to stimulate thoughtful investigation by drawing from a variety of academic perspectives. By pursuing these questions, we hope to push the boundaries of human knowledge and inspire innovative solutions that can benefit both scientific inquiry and society more broadly.

Can we provide a quantitative characterization of our ignorance regarding a research topic? Developing such a metric would give us a way to measure not just what we know, but how much we are still missing. This would be especially useful for evaluating scientific contributions, as we could measure how much each new idea or publication actually improves our understanding. By combining this with relevance metrics, we would gain a powerful tool for assessing the true value of scientific work.

Can we compare the extent of our ignorance across disparate scientific fields? If this is possible, we could evaluate and prioritize research based on how much there is left to understand, regardless of the field. This approach could help distribute scientific attention and funding more effectively, highlighting areas of science that are in greater need of exploration while still valuing fundamental research.

Can we establish a systematic procedure to enhance our knowledge? While the scientific method has long served as the cornerstone of research, it varies significantly across disciplines and often lacks clear, measurable criteria for success. Developing a unified and precise approach could improve the consistency and efficiency of scientific discovery. If successful, it would allow us to evaluate and refine how knowledge is generated, leading to more reliable and accelerated progress across all areas of inquiry.

Can we devise a method for discovering new, previously unknown, and intriguing research entities and problems? Many major discoveries arise from questions we had not thought to ask. If we could build a systematic way to identify these unknown unknowns, we could unlock entirely new areas of research and significantly accelerate scientific progress.

What constitutes perfect knowledge? Understanding what it means to know something completely is essential to defining the limits of science. If we could precisely define and recognize perfect knowledge, we would know when a field is complete and when it is time to shift our focus elsewhere. Clarifying whether such completeness is always possible would also help set realistic goals for scientific inquiry.

What effort is required to fully comprehend an unfamiliar subject? Measuring the cost of understanding could fundamentally change how we plan and fund research. For example, knowing the minimal effort needed to improve our grasp of cancer treatment by a small percentage would help us allocate resources more wisely. Even an estimate would be useful for setting priorities and managing expectations.

Do some research topics inherently possess a higher degree of complexity than others? This question addresses a long-standing debate about intellectual differences across disciplines. If we could objectively measure topic complexity, we might dispel myths of superiority and develop more targeted educational strategies. It would also help explain why some areas advance faster than others.

How can we differentiate between science and pseudoscience? Many fields present themselves as scientific but lack rigorous foundations. A mathematically grounded method for identifying what qualifies as science would have major implications for public policy, education, and research funding. It would also help protect scientific integrity by distinguishing valid inquiry from unsupported speculation.

Each chapter in this book engages with these questions from both theoretical and practical standpoints. Some offer detailed responses, while others lay the groundwork for future research. This openness is by design. The Theory of Nescience is not a closed doctrine but a platform for further exploration, an invitation to investigate the limits of science and to imagine new pathways of understanding. In this light, the book is not merely a treatise but a research agenda, a provocation to conventional thinking, and a call to those who are driven to explore the boundaries of knowledge itself.

Origins of the Theory of Nescience

It was in 1991, when I was eighteen years old, that I first encountered the statement, "*Computers are useless, they can only give you answers.*" This quote, attributed to Pablo Picasso, struck me as profoundly true. Of course, Picasso was referring to the early calculators of his time, not modern computers. But the underlying idea remains relevant: machines are designed to process predefined tasks, not to generate new and meaningful questions on their own. That realization stayed with me for years. However, it wasn't until 2014, more than two decades later, that I began to explore the implications of Picasso's observation from a

practical and computational perspective.

The core ideas behind the Theory of Nescience came together during one particularly restless night. Concepts such as miscoding, nescience, relevance, or the unknown unknown suddenly aligned. In hindsight, my long-standing interests in information theory and Kolmogorov complexity had likely prepared me for this moment. These disciplines proved essential in articulating the mathematical foundation for a theory that had first emerged as a series of intuitive insights.

Over the following weeks, I conducted a series of computer experiments to test these concepts. The results were promising, but the theory needed time, several years, to develop into a rigorous mathematical framework. Initially, my goal was quite focused: to devise a method for identifying interesting and underexplored scientific questions. But as I examined how ignorance, or nescience, changes over time within scientific disciplines, my scope expanded. I began to wonder whether it was possible to define perfect knowledge and to determine whether such a state could be formally described.

This line of inquiry led to an unexpected and striking insight: perfect knowledge could be expressed in terms of randomness. Specifically, when a theory's description cannot be compressed further, when it becomes algorithmically random, it may indicate that the theory is as complete as possible. This realization broadened my original focus into a general framework for studying the structure of ignorance, the development of understanding, and the mechanisms that drive scientific progress.

Encouraging early feedback from colleagues and researchers helped me refine the theory and explore new applications, particularly in data science and machine learning. Its potential to shed light on long-standing questions in the philosophy of science was especially motivating. For instance, it offered new ways to compare our understanding across fields like mathematics and sociology, and to address the problem of how to distinguish between science and pseudoscience.

Continued mathematical development eventually led to a highly surprising discovery: in certain types of topics additional research can actually increase our ignorance. In these cases, there may exist a critical threshold beyond which further investigation does not improve understanding, but instead introduces greater uncertainty.

This deepened my belief that grappling with ignorance is as fundamental as pursuing knowledge. The Theory of Nescience represents my response to Picasso's challenge: not merely to construct systems that answer questions but to build frameworks capable of identifying which questions are most worth asking.

These developments and insights have shaped what has become a comprehensive and mature theory, one that I hope will not only reframe how we understand knowledge, but also inspire readers to question more deeply, think more broadly, and search more boldly. The chapters that follow present this

journey in both its theoretical depth and practical implications.

About the Book

The Theory of Nescience draws on concepts from multiple academic disciplines, including computability, complexity theory, artificial intelligence, and the philosophy of science. Despite the breadth of its foundations, this book is designed to be self-contained. Readers are expected to have only a basic understanding of first-year calculus and some experience with programming. The content is crafted to serve a wide technical audience, including mathematicians, computer scientists, engineers, and other scientifically inclined readers. The mathematical level is suitable for graduate students and advanced undergraduates.

The book is organized into three main parts: Foundations, Applications, and a collection of appendices on Prerequisites. Readers who already have a background in the mathematics covered in the Prerequisites may wish to begin directly with the Foundations. However, we recommend at least a brief review of the notation and key concepts introduced in those appendices. Once readers are familiar with the core ideas presented in the Foundations, they can proceed to the Applications. A detailed understanding of the underlying formalism is not essential; a general grasp of the main concepts and results is sufficient to engage with the practical examples and insights explored in that part of the book.

Chapter 1 Introduction provides a gentle entry point to the theory of nescience, presenting a brief overview of its main concepts and results. While it avoids the use of advanced mathematics, the ideas are introduced in a semi-formal manner. Although it is not recommended as a substitute for the full theoretical exposition, readers who find the mathematics challenging may choose to read only this chapter before proceeding directly to the Applications.

PART I Foundations presents a comprehensive account of the theory of nescience, including formal definitions of its core concepts and proofs of key theoretical results. This section forms the core of the book. Readers with prior knowledge in computability, complexity, information theory, probability, and artificial intelligence may choose to begin here directly.

Chapter 2 Entities introduces the objects about which scientific knowledge is sought. It defines the notion of a research entity, distinguishes between different kinds of entities, and explains why the set of entities under investigation must be fixed relative to a particular domain. The chapter also studies relations between entities and introduces research areas as structured collections of entities that can be analyzed jointly.

Chapter 3 Representations examines how research entities can be encoded as finite strings of symbols. It introduces the representation oracle, the distinction between decodable, minimal, admissible, extended, and effective representations, and the notion of knowable and unknowable entities. The chapter also studies

how representations may be refined, reduced, or combined, thereby providing the formal basis for measuring errors in the way entities are encoded.

Chapter 4 Descriptions studies how representations can be modeled by computable descriptions. It defines valid, minimal, and effective descriptions, explains their role as scientific models, and introduces operations for combining, simplifying, and conditioning descriptions. The chapter provides the formal bridge between representations and models, preparing the ground for the later measurement of mismodel, inaccuracy, surfeit, and nescience.

Chapter 5 Miscoding explores the challenge of representing both abstract and concrete research entities as strings of symbols for research purposes. It formally introduces the concept of miscoding and examines its properties. Miscoding serves as a measure of the error introduced by inaccurate or inappropriate encodings of the entities being studied. The chapter also discusses strategies for minimizing the errors introduced by poor representations, thereby improving the accuracy and utility of symbolic encodings in research.

Chapter 6 Mismodel introduces a general measure of how poorly a description models a given representation. It decomposes this defect into the deficiency and surplus of a description, studies the theoretical properties of mismodel, and introduces relative and area-level variants. The chapter also examines how mismodel can be reduced, providing the formal foundation for later practical approximations such as inaccuracy and surfeit.

Chapter 7 Nescience serves as the core of the book, presenting the fundamental elements of the theory. It defines science as a nonlinear multiobjective optimization problem in which miscoding and mismodel are minimized simultaneously. A new metric, nescience, is introduced as a function of these two components. The chapter also explores key properties of this metric, including its evolution over time, the concept of perfect knowledge (zero nescience), and methods for identifying our current best model. In addition, it introduces a definition for the frontier of human knowledge and offers a characterization of what lies beyond that boundary.

Chapter 8 Supervised Miscoding introduces a computable approximation of miscoding for settings in which admissible representations are not directly available, but labeled effective representations are. It develops a supervised proxy based on datasets whose samples are associated with entity labels, studies its main properties, and examines feature-level, joint, partial, and dataset-level variants. The chapter provides a practical bridge between the theoretical notion of miscoding and its application to machine learning and empirical data analysis.

Chapter 9 Inaccuracy introduces inaccuracy as a practical approximation of one component of mismodel: the extent to which a description fails to reconstruct the intended representation. It generalizes the classical notion of error so that it can be applied not only to empirical datasets, but also to abstract

research topics. The chapter studies the theoretical properties of inaccuracy, introduces a conditional variant, and investigates how inaccuracy can be reduced. It also examines how miscoding, understood as error in the representation itself, influences the apparent inaccuracy of a description.

Chapter 10 Surfeit introduces surfeit as a practical approximation of the other component of mismodel: the surplus contained in a description beyond what is needed to reconstruct the intended representation. It studies how unnecessary descriptive complexity can indicate an incomplete understanding of a research topic or area, since poor understanding is often reflected in unnecessarily long or redundant descriptions. The chapter also introduces a conditional variant of surfeit, develops practical approximations applicable in real-world scenarios, and examines the relationship between surfeit and inaccuracy.

Chapter 11 Interesting Questions presents a methodology for identifying new research ideas, based on combinatorics and computational creativity, and focusing on how to address challenging open problems. It introduces two new metrics to measure a topic's relevance and its applicability to existing problems, and examines the properties of these metrics. The chapter also outlines a systematic approach to uncovering what lies hidden in the unknown unknown, illustrating how previously unrecognized research directions may be revealed.

Chapter 12 Effort introduces effort as an orthogonal concept to nescience, measuring the computational cost required to reduce our lack of knowledge. It defines effort in terms of research transitions, studies component-wise, marginal, expected, and conditional forms of effort, and examines the structure of effort landscapes. The chapter also explores thermodynamic lower bounds, monetary costs, applications to machine learning, and the use of effort to compare the difficulty of different research topics and areas.

PART II Applications presents a collection of practical uses of the concept of nescience in areas such as machine learning, the philosophy of science, and the discovery of new research topics. The included examples have been selected to illustrate the broad applicability of the theory, spanning from abstract research questions to more tangible problems grounded in datasets.

Chapter 13 Machine Learning explores how the theory of nescience can be applied to entities represented by collections of measured samples. We introduce `mnpilib`, a software library that can be used to analyze datasets, select relevant features, identify optimal model hyperparameters, and compute the errors of trained models. In addition, the chapter presents novel machine learning algorithms, including an innovative method for deriving optimal decision trees and automating the construction of machine learning models.

Chapter 14 Analysis of Science investigates how well we understand current research topics by applying the proposed metrics. Our aim is to assess the degree of understanding across different areas of science. To this end, we compare

research topics within the same academic discipline, as well as across multiple disciplines. The chapter also addresses major open questions in the philosophy of science, including the demarcation problem (how to distinguish science from pseudoscience) and the nature of scientific progress.

Chapter 15 Discovering the Unknown demonstrates the practical application of the theory of nescience for identifying promising research questions. Specifically, we show how nescience can be leveraged to generate new research ideas aimed at solving the most difficult open problems. We propose a methodology for identifying new research topics, that is, for uncovering what lies hidden in the unknown unknown. Multiple examples of such research questions and novel research areas are provided.

Part III Prerequisites introduces the mathematical foundations necessary to quantify our lack of knowledge about a research topic and to assess the randomness of a string. Its primary aim is to establish consistent notation, formally define key concepts, and present important theoretical results. While no prior background is required to follow the material, readers are encouraged to consult the standard references provided at the end of each chapter for deeper understanding.

Chapter 16 Discrete Mathematics offers a summary of the fundamentals of discrete mathematics needed to understand the more advanced topics discussed in the book. This chapter serves as a quick review of these concepts without providing formal definitions or proofs. Topics covered include sets, relations, strings, graphs, and counting methods. A section on linear algebra (matrices and vectors) is also included.

Chapter 17 Discrete Probability introduces the foundational concepts of probability related to discrete events. Topics covered include conditional probability, random variables, distribution characterization, common distributions, and large random samples. This chapter aims to equip readers with the necessary background in probability to understand the more advanced statistical learning discussed later in the book.

Chapter 18 Computability presents a formal definition of the concept of algorithm. It introduces the idea of a universal Turing machine and shows that certain well-defined mathematical problems cannot be solved by computers. The chapter also examines the essential tool of an oracle Turing machine.

Chapter 19 Coding explores how codes function and how they enable us to compress texts by eliminating redundant patterns without losing essential information. It shows that there is a limit to how much a text can be compressed using this technique, and that this limit is determined by the entropy of the source. The important relation between optimal codes and discrete probabilities is also studied.

Chapter 20 Complexity introduces an absolute metric known as Kolmogorov

complexity, which measures the amount of information contained in a string by calculating the length of the shortest computer program that can produce it. The chapter studies the properties of this metric in detail and explores the relationship between string complexity and randomness.

Chapter 21 Learning offers a concise overview of the field of statistical learning, presenting key results from statistical inference and machine learning. It also examines the relationship between codes and probabilities, focusing on practical approaches that apply the concept of minimum string length. Additionally, the chapter introduces the concepts and notation associated with nonlinear multiobjective optimization problems.

Chapter 22 Philosophy of Science provides a brief introduction to the study of science from a philosophical perspective. It reviews key concepts such as scientific representations, models, and theories, and identifies the essential components that any formal theory of science should include. The chapter also offers an overview of our current understanding of how the scientific method works.

Acknowledgements

I would like to express my gratitude to everyone who has contributed their comments and ideas to the development of the theory of nescience. In particular, I am grateful to Antonio Fernández, Vincenzo Mancuso, and Paolo Casari, who believed in and supported this project from its very beginning when it was merely a far-fetched idea. Others who have provided contributions and valuable feedback include Héctor Cordobés, Luis F. Chiroque, Agustín Santos, Marco Ajmone, Pablo Rojo, Manuel Cebrián, Andrés Ortega, Emilio Amaya, Mattis Choumanivong, Alexander Lynch, Andrés Carrillo, Simon Bihoreau and Ricardo Hernando. Early versions of the `mnp1ib` library described in Chapter 13 have been partially funded by the IMDEA Networks Institute, the European Union's Horizon 2020 research and innovation programme under grant agreement No 732667 RECAP, and Nokia Spain through the project NetPredict.

The ideas presented in this book build on a long history of thought. I draw inspiration from philosophers such as Plato, Occam, Popper, Feyerabend, and Wittgenstein; from polymaths like Llull, Leibniz, and Newton; and from the mathematical foundations established by Turing, Church, Post, Shannon, Solomonoff, Chaitin, Kolmogorov, and many others. My contribution lies in connecting and reinterpreting some of these insights within a unified framework. The References section at the end of each chapter highlights the sources that have most directly influenced my thinking.

I would like to express my sincere gratitude to the open source community. This book would not have been possible without the extensive use of open

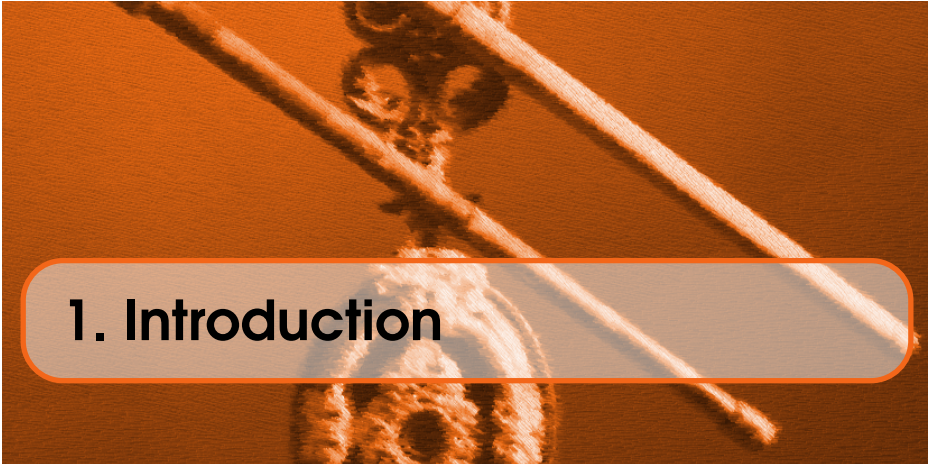
source tools, which provided both the technical foundation and the flexibility required throughout its development. This support extended from the writing process, carried out using $\text{T}_{\text{E}}\text{X}$, $\text{L}_{\text{A}}\text{T}_{\text{E}}\text{X}$, and TikZ , to the data analysis and machine learning experiments, powered by Python and libraries such as `scikit-learn`, `statsmodels`, `pandas`, `NumPy`, and `Matplotlib`. The contributions of countless developers who share their work openly have been indispensable. Their dedication to building and maintaining high-quality, freely available software is a testament to the collaborative spirit of scientific progress.

I would also like to acknowledge the role of ChatGPT in the preparation of this book. As an AI assistant, ChatGPT provided valuable support throughout the review process, helping to refine ideas, clarify language, structure arguments, and polish technical content¹. While all decisions and final content were ultimately my own, the ability to engage in thoughtful dialogue, receive constructive feedback, and explore alternative phrasings and formulations greatly enhanced the clarity and coherence of the work. I am grateful for this tool, which has proven to be both efficient and intellectually stimulating.

Finally, I am deeply grateful to my parents for providing me with opportunities they themselves never had, and to my wife and three children, who give my life its deepest meaning.

¹The following prompts have been used in the review process:

- Review the English and clarity of the following text.
- Clarify the following paragraph.
- Propose an example to clarify the following concept.
- Provide a draft proof for the following proposition.
- Review the following section and find logical inconsistencies and mathematical errors.
- Draw a figure in TikZ .



1. Introduction

*If presented with a choice between indifferent alternatives,
then one ought to select the simplest one.*

Occam's razor principle

We find ourselves in an age where knowledge has become one of our most valued resources, driving both economic progress and societal well-being. Science continually provides us with deeper insights and practical solutions, shaping our daily lives through innovations ranging from advanced medical therapies to transformative digital technologies. However, the path of scientific exploration is neither straightforward nor free from significant barriers. Despite remarkable progress, our current scientific methodologies remain bounded by inherent limitations that restrict our ability to fully understand and address complex issues.

These limitations manifest in various ways, including fragmented research efforts, rigid disciplinary boundaries, and an often narrow approach to funding allocation, which tends to overlook bold, high-risk projects with transformative potential. Moreover, the conventional metrics of scientific success frequently incentivize incremental progress rather than genuinely groundbreaking discoveries, limiting the pace and scope of innovation.

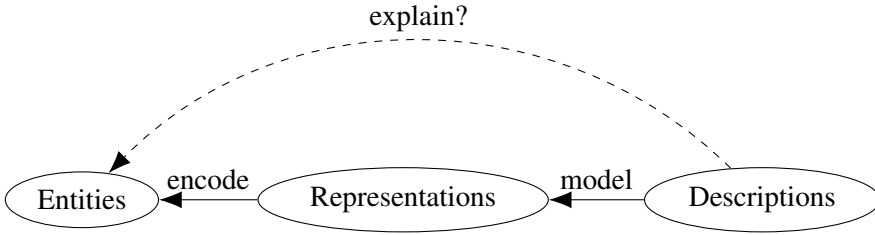


Figure 1.1: The Problem of Understanding

The *theory of nescience* emerges from the critical recognition of these challenges and the need to overcome them. It advocates a renewed approach to exploring the unknown, driven by curiosity and the willingness to question accepted knowledge rigorously. By systematically addressing the limitations inherent in contemporary scientific practice, our objective is to provide robust frameworks that significantly enhance our capability to solve real-world problems and catalyze meaningful advancements across diverse fields.

1.1 Scientific Knowledge

The pursuit of knowledge begins with identifying the *entities* we seek to understand, entities that are often extraordinarily diverse. Mathematicians focus on abstract concepts, biologists on living organisms, and engineers on machines. Our quest for understanding is fundamentally driven by our need to predict outcomes, solve problems, and navigate the complexities of the world around us. For instance, we know that applying sufficient heat to wood will ignite a fire, or that a fever may indicate a viral infection. Practical problem-solving, therefore, hinges crucially on our ability to recognize patterns and create simplified models of these entities. These models empower us to interact beneficially with the world.

Yet, while ideally we strive to build precise *descriptions* capable of fully reconstructing the entities we study, in reality, such perfection often eludes us, particularly with abstract entities. Consequently, we rely on *representations*, texts or datasets capturing essential details of these entities. Physicists might represent an entity through experimental results; computer scientists through measured data; sociologists through observed facts. Figure 1.1 illustrates this critical interplay: although we aim to directly describe entities, practically, our descriptions are based on constructed representations.

Understanding the effectiveness of our scientific knowledge requires examining two fundamental sources of error. First, the representation of an entity may fail to encode the entity faithfully, either because relevant information

is missing or because irrelevant information has been included. We call this error *miscoding*. Second, even when a suitable representation has been chosen, the description or model used to explain it may fail to capture its structure correctly or economically. We call this error *mismodel*.

Taken together, miscoding and mismodel hinder our search for clear understanding and reliable prediction. By combining these two fundamental sources of error into a single measure, known as *nescience*, we quantify our lack of knowledge about an entity. This provides a systematic framework for identifying whether our ignorance arises primarily from inadequate representations, inadequate descriptions, or both, and for guiding the progressive reduction of that ignorance.

1.2 Entities

The theory of nescience begins with the identification of the objects about which knowledge is sought. Scientific inquiry is directed toward a wide variety of such objects: planets, particles, languages, ecosystems, social institutions, mathematical structures, technological systems, and many others. In this book, we refer to these objects of investigation as *entities*. An entity may be concrete, such as a chemical compound, a biological cell, or a physical process, or abstract, such as a mathematical function, an algorithm, or a cultural practice.

The range of entities that may become objects of scientific investigation is broad and historically variable. Advances in instrumentation, changes in practical concerns, and developments in conceptual frameworks can all modify the entities that are regarded as scientifically relevant. What unifies these cases is not a common ontological nature, but their role in scientific practice: they are objects that researchers attempt to represent, describe, model, explain, or predict. In this sense, the theory assumes that at least part of the unknown is, in principle, knowable by scientific means (see Section 22.2). The central problem is then to characterize the conditions under which such understanding is possible, and to identify the ways in which it may fail.

A first conceptual difficulty concerns the status of the set of entities under consideration, denoted by $\mathcal{E}$. This set should not be understood as a universal set containing all possible objects, but as a domain-relative collection fixed by the scope of a particular investigation. The theory of nescience therefore avoids the assumption that there exists a single unrestricted set of all entities. Such an assumption would conflict with standard set-theoretic considerations, including Cantor's theorem and related paradoxes such as Russell's paradox, both of which are discussed in Section 2.1.

Accordingly, each application of the theory begins by fixing a non-empty set $\mathcal{E}$ of research entities. The nature of this set depends on the domain under

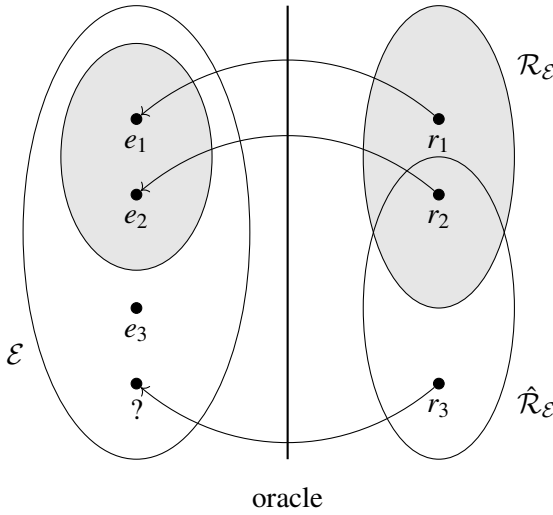


Figure 1.2: Entities and Representations.

study. In mathematics, it may consist of classes of abstract structures such as groups, functions, or graphs. In biology, it may contain organisms, cells, genes, or ecosystems. In the social sciences, it may include social practices, institutions, or economic systems. In computer science, it may contain algorithms, data structures, programs, or computational processes.

Once such a domain-relative set of entities has been fixed, the theory studies how its elements may be represented, how those representations may be described, and how failures in representation and description give rise to nescience. The purpose is not to provide a universal ontology of all possible objects, but to establish a general framework that can be applied to well-delimited domains of inquiry.

1.3 Representations

In many scientific contexts, entities cannot be investigated directly. This is especially clear for abstract entities, but it also occurs in empirical domains, where the object of study is accessed only through observations, measurements, diagrams, datasets, or other symbolic records. For this reason, scientific inquiry typically proceeds by means of *representations*: finite strings or symbolic structures that stand in for the entities under investigation and make them available for analysis, comparison, and modeling.

In the theory of nescience, representations are treated as finite binary strings. This restriction is not intended to exclude diagrams, texts, equations, datasets, or experimental records, but to emphasize that such objects must ultimately be

encoded in a form that can be stored, communicated, and processed. Different scientific contexts may therefore use different representation styles. A biological organism may be represented by a genome, a set of anatomical measurements, or a collection of images; a physical system may be represented by experimental data; and an abstract object may be represented by a formal expression, an algorithm, or a proof. A single entity may thus admit several representations, each emphasizing different aspects of the entity and serving different investigative purposes.

A natural first idea would be to introduce an encoding function from entities to representations. Such a function would assign to each entity $e \in \mathcal{E}$ a string r representing it. However, this approach is too strong for the purposes of the theory. In general, the set of entities $\mathcal{E}$ is not effectively enumerable, and we do not usually possess a direct procedure for transforming an entity into a faithful representation. Instead, the theory adopts the opposite perspective: it begins with finite strings and asks whether a given string contains enough information to reconstruct an entity.

This motivates the introduction of a representation oracle. Relative to a fixed set of entities $\mathcal{E}$, the oracle determines whether a string $r \in \mathcal{B}^*$ is a decodable representation of some entity in $\mathcal{E}$. When this is the case, the oracle is assumed to have an internal reconstruction function that associates the string with the entity it represents. The oracle is not assumed to be computable or available in practice. Its role is formal: it provides an ideal reference point for distinguishing strings that genuinely represent entities from strings that do not.

The set of representations associated with $\mathcal{E}$ is therefore not directly accessible in scientific practice. In real research, we work instead with effective representations, denoted by $\hat{\mathcal{R}}_{\mathcal{E}}$, composed by strings that are currently taken, within a given scientific context, to represent entities of interest. These effective representations may be correct, incomplete, misleading, or assigned to the wrong entity. They constitute the practical material from which scientific models are built, but they do not guarantee faithful reconstruction of the entities they are intended to represent.

It is important to distinguish this practical notion from stronger theoretical notions. Some representations may be decodable by the oracle; some may be minimal, in the sense that no shorter representation reconstructs the same entity; and some may be admissible, in the sense that they contain no semantically irrelevant independent information. Scientific practice rarely begins with such ideal representations. Instead, it proceeds by refining effective representations: adding missing information, removing irrelevant information, correcting misleading encodings, and sometimes discovering that a representation does not refer to the intended entity at all.

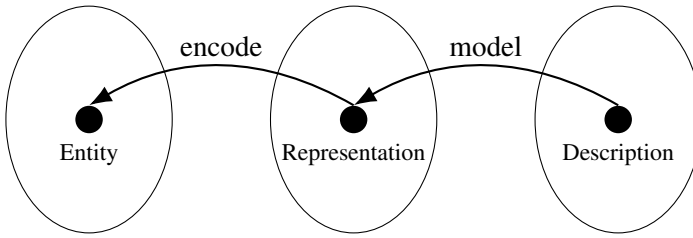


Figure 1.3: Entities, representations, and descriptions.

■ **Example 1.1** If the entities under study are animals, one possible representation style is to encode them by means of their DNA. Such a representation may be highly informative for many biological purposes, since it captures inherited structure and supports the study of species, traits, and evolutionary relations. However, DNA does not encode all aspects of an individual animal. It does not include its life history, acquired injuries, learned behavior, or environmental interactions. Thus, DNA may be an adequate representation for some research questions, such as the study of species-level traits, but insufficient for others, such as the study of a particular individual. ■

Because representations are finite strings, not every entity can necessarily be represented. This limitation is especially clear when the set of entities is uncountable. For instance, if $\mathcal{E}$ is the set of real numbers, then only countably many of them can be represented by finite binary strings. Most real numbers therefore cannot be reconstructed from any finite representation. This illustrates a basic limitation of representation-based knowledge: the space of possible entities may exceed the space of available finite encodings.

A second difficulty arises because the effective representations used in practice may fail to correspond to the entities they are intended to represent. A string may appear to be a representation of one entity while in fact tracking another, or it may contain only fragmentary information about several entities at once. Such cases are central to the theory of nescience, because they show that ignorance may arise not only from poor descriptions or models, but also from failures at the level of representation itself. These failures will be studied in Chapter 5.

1.4 Descriptions

Once entities have been represented by finite strings, the next problem is to describe those representations in a form suitable for explanation, prediction, and further reasoning. In the theory of nescience, descriptions are not applied directly to entities, but to their representations. A description is therefore a formal

object that attempts to reconstruct, summarize, or model a representation. This distinction is important: scientific models do not usually give direct access to the entities themselves, but operate through the symbolic, empirical, or computational representations available at a given stage of inquiry.

Descriptions change over time as scientific understanding develops. A representation that initially requires a long and irregular description may later be explained by a shorter and more structured one. Scientific progress can therefore be understood, in part, as the search for descriptions that capture the regularities present in representations while avoiding unnecessary assumptions and irrelevant detail.

■ **Example 1.2** Consider the historical development of descriptions of macroscopic physical motion. Aristotelian physics, Cartesian mechanics, Newtonian mechanics, and relativistic mechanics may be understood as successive attempts to describe representations of physical phenomena with increasing generality and precision. Each framework provides a different way of reconstructing and organizing the empirical representations available at a given historical stage. The point is not that later descriptions simply replace earlier ones in all contexts, but that they may reduce mismodel by accounting for a wider range of representations with greater accuracy or economy. ■

The notion of description must be restricted in order to avoid the ambiguities associated with unrestricted definability. Classical paradoxes such as the Berry paradox illustrate the difficulties that arise when descriptions are treated as arbitrary linguistic expressions. To avoid such difficulties, the theory of nescience requires descriptions to be computationally meaningful. A valid description is a finite string that encodes an effective procedure capable of producing, or reconstructing, a representation. Here, “effective” means that the reconstruction can be carried out by a computer without requiring human interpretation.

This computational restriction connects scientific description with the theory of computation. In the present framework, a description is represented by a pair $\langle TM, a \rangle$, where TM is a Turing machine and a is an input string. The Turing machine captures the regularities, rules, or model structure used to generate the representation, while the input string contains the remaining information required for reconstruction. This decomposition reflects familiar distinctions in scientific practice, such as a model together with its initial conditions, a theory together with auxiliary assumptions, or a general algorithm together with a particular instance.

The interpretation of the pair $\langle TM, a \rangle$ depends on the domain of application. In physics, the Turing machine may encode a system of equations or a simulation procedure, while the input may encode initial conditions or parameter values. In machine learning, the Turing machine may encode a trained model or prediction

rule, while the input may encode data required to reconstruct the observed outputs. In mathematics, it may encode a formal procedure together with the assumptions or objects to which the procedure is applied. In all cases, the description is evaluated by its capacity to reconstruct the relevant representation.

Figure 1.3 summarizes the relation between entities, representations, and descriptions. Entities are accessed through representations, and representations are modeled by descriptions. The set of all possible descriptions is denoted by $\mathcal{D}$. However, not every finite string is a valid description: to qualify as a description, a string must encode a computational procedure of the appropriate kind. Moreover, not every valid description is an adequate description of a given representation. A representation may admit many descriptions, differing in length, accuracy, generality, and explanatory value.

A central theoretical ideal is the search for minimal descriptions: descriptions that reconstruct a given representation with no unnecessary information. This ideal is closely related to Kolmogorov complexity, since the shortest program that outputs a string gives a formal measure of the information required to describe it. However, as discussed in Appendix 20, Kolmogorov complexity is not computable in general. Consequently, there is no general algorithm that, given a representation, returns its shortest description.

This incomputability has direct consequences for scientific modeling. In practice, researchers cannot usually know whether a proposed model is minimal, nor whether it is the best possible description of the representation under study. Scientific inquiry must therefore proceed by means of effective descriptions: descriptions actually available within a given scientific context, without any guarantee that they are minimal or fully adequate. These effective descriptions are refined through comparison with representations, correction of failures, simplification of unnecessary structure, and replacement by more successful alternatives.

The theory of nescience uses this framework to identify the sources of failure that arise in scientific knowledge. Errors may occur at the level of representation, when a string fails to encode the intended entity adequately; this is the problem of miscoding. Errors may also occur at the level of description, when a model fails to reconstruct a representation correctly or economically; this is the problem of mismodel. The following sections introduce these components and explain how they are combined into the measure of nescience.

1.5 Miscoding

The preceding sections distinguished between entities and their representations. In practice, however, the representations available to researchers are rarely guaranteed to be correct, complete, or free from irrelevant information. Scientific

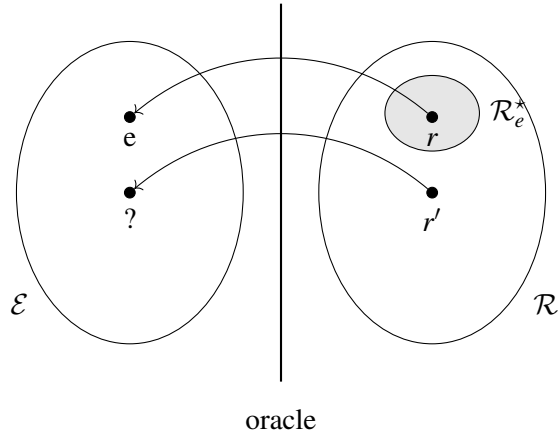


Figure 1.4: Miscoding of representations.

inquiry typically begins with effective representations: finite strings that are currently taken to represent an entity, even though they may fail to coincide with any admissible representation of that entity. The purpose of miscoding is to measure this failure at the representational level.

Let $e \in \mathcal{E}$ be an entity and let r' be an effective representation that is intended to represent e . Ideally, r' would belong to the set of admissible representations of e , denoted $\bar{\mathcal{R}}_e$. In that case, the representation would contain sufficient information about the entity and would not contain independent information irrelevant to it. In general, however, r' may deviate from $\bar{\mathcal{R}}_e$ in two different ways: it may omit information that is required to represent the entity adequately, or it may include information that does not belong to the entity at all. Miscoding is introduced to quantify these two forms of representational failure.

The first component of miscoding is representational deficiency. A representation is deficient when it lacks information required to obtain an admissible representation of the entity. Formally, this deficiency can be understood in terms of the shortest program needed to transform the effective representation r' into an admissible representation $r \in \bar{\mathcal{R}}_e$. If this transformation requires a long program, then substantial information is missing from r' . In that case, the representation is poor not because it is too long, but because it fails to contain the information needed to reconstruct the entity adequately.

The second component is representational surplus. A representation may contain information that is not required to represent the entity and is not recoverable from the rest of the representation. Such information is not merely redundant; it may be semantically extraneous. For example, an empirical dataset may contain variables that are unrelated to the phenomenon under study, or a

historical description may include assumptions that later turn out to belong to a different explanatory framework. This surplus can mislead the construction of descriptions, because any model built from the representation may be forced to account for structure that is not genuinely part of the entity.

Miscoding combines these two components. Deficiency measures how much information must be added or corrected in order to obtain an admissible representation. Surplus measures how much independent information must be removed because it does not contribute to representing the entity. A representation is therefore well coded only when it is sufficiently informative and sufficiently free from semantically irrelevant content. In this sense, miscoding is not simply a measure of error in the ordinary empirical sense, but a measure of the distance between an effective representation and the class of admissible representations of the intended entity.

A complication arises because admissible representations are generally not unique. The same entity may admit several different representations, each emphasizing different aspects of the entity or serving different scientific purposes. Therefore, miscoding cannot be defined by comparing an effective representation with one arbitrarily chosen correct representation. Instead, it must be defined relative to the closest admissible representation of the entity.

■ **Example 1.3** Consider the entity e corresponding to the mathematical constant π , the ratio between the circumference and diameter of a circle. One representation of this entity is the Wallis product,

$$2 \left(\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdots \right),$$

while another is the Leibniz series,

$$4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right).$$

These two strings have very different syntactic forms, but both represent the same mathematical entity. It would therefore be misleading to regard one as highly miscoded merely because it differs from the other. What matters is not similarity to a fixed representation, but distance from the set of admissible representations of the entity. ■

This point is important because the set $\bar{\mathcal{R}}_e$ of admissible representations of an entity is generally inaccessible in practice. Researchers usually work with effective representations, while the admissible representations serve as an ideal theoretical reference. The representation oracle introduced earlier provides the formal device needed to make this reference precise. It determines, at the theoretical level, which strings genuinely reconstruct entities and which strings

fail to do so. Miscoding is therefore an ideal quantity: it is defined by comparison with admissible representations, but it cannot usually be computed exactly.

There is a further difficulty. Since effective representations are assigned within scientific practice, they may be associated with the wrong entity. A string may be treated as a representation of one entity while in fact tracking another, or it may fail to correspond to any genuine entity at all. Such cases are not peripheral; they occur whenever scientific practice introduces a candidate entity that is later rejected, reinterpreted, or replaced.

■ **Example 1.4** In 1961, the Soviet physicist Nikolai Fedyakin informed about experimental results that appeared to indicate the existence of a new form of water, later called polywater. This alleged substance was described as having unusual physical properties, including a higher boiling point, a lower freezing point, and increased viscosity. Subsequent investigations showed that these effects were caused by contamination. The effective representations associated with polywater therefore did not correspond to a distinct form of water, but to experimental artifacts produced by impurities. ■

Miscoding captures this kind of representational failure. It concerns the relation between an entity and the strings used to represent it, prior to the construction of a model or description. A low-miscoding representation provides a suitable basis for description; a high-miscoding representation may lead even a well-constructed model toward a misleading conclusion. For this reason, reducing miscoding is a fundamental part of reducing nescience: before asking whether a description models a representation adequately, we must ask whether the representation itself adequately encodes the entity under investigation.

1.6 Mismodel

The preceding section considered errors that arise at the level of representation. Even if an entity e is represented by a sufficiently adequate string r , a second source of nescience remains: the representation may not be described correctly. In the theory of nescience, this source of error is called *mismodel*. Whereas miscoding concerns the relation between an entity and its representation, mismodel concerns the relation between a representation and its description.

A description is intended to reconstruct, explain, or model a representation. Ideally, for a representation r , we would have a description d capable of reconstructing r completely and without unnecessary information. In practice, however, the descriptions available to us are effective descriptions: finite strings or computational procedures that are currently used to model a representation, without any guarantee that they are minimal or fully adequate. Mismodel measures the extent to which such a description fails to coincide with an ideal

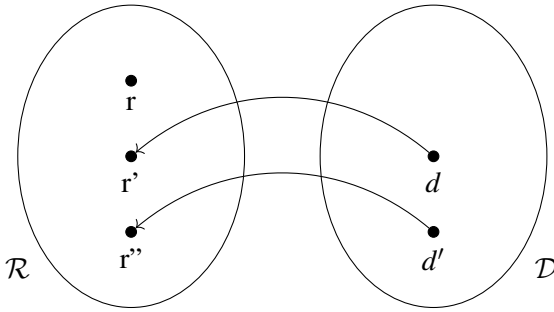


Figure 1.5: Inaccuracy of descriptions.

description of the representation.

This failure has two components. The first is *descriptive deficiency*: the description does not contain enough information to reconstruct the representation. In such a case, the output of the description differs from the intended representation, and additional information is required to correct it. The second is *descriptive surplus*: the description contains more information than is needed to reconstruct the representation. In that case, the description may be accurate in the sense that it produces the right representation, but it remains unnecessarily complex. Mismodel combines these two aspects: a good description should be both adequate and economical.

Figure ?? illustrates this situation. The ideal description d^* reconstructs the intended representation r , whereas the effective description d' may reconstruct a different string r' , or may reconstruct r only by using unnecessary descriptive structure. The mismatch between d' and d^* is not merely a matter of prediction error. It concerns the degree to which the available description fails to capture the structure of the representation in an adequate and minimal way.

The theoretical definition of mismodel is based on the distance between an effective description and the set of minimal descriptions of the representation. If the effective description lacks information required to reconstruct the representation, then one must add corrective information; this corresponds to descriptive deficiency. If the effective description contains independent information not required by any minimal description, then one must remove unnecessary structure; this corresponds to descriptive surplus. A description has low mismodel only when it is close to a minimal description in both directions: it omits little and adds little.

This formulation generalizes two more familiar notions. The first is *inaccuracy*. Inaccuracy approximates the deficiency of a description by measuring how far the output of the description is from the intended representation. In empirical contexts, this may resemble the ordinary notion

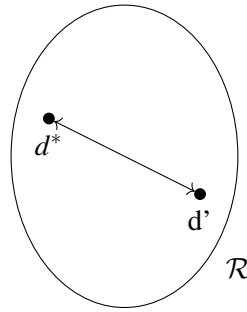


Figure 1.6: Surfeit of a model.

of model error: a model predicts values that differ from the observed data, and additional information is required to account for the discrepancy. The second is *surfeit*. Surfeit approximates the surplus of a description by measuring the extent to which the description is longer or more complex than necessary. A highly accurate model may still have high surfeit if it achieves its accuracy by introducing excessive parameters, ad hoc assumptions, or irrelevant structure.

■ **Example 1.5** Consider a dataset representing measurements of gravitational acceleration. A model may predict a constant acceleration of approximately 980.000 cm/s^2 , while the observed measurements vary around this value. The discrepancy between the model output and the measured representation provides a practical approximation of descriptive deficiency: additional information is required to reconstruct the observed data exactly. If the complete dataset requires 1,620 bits to encode, while the residual information needed after using the model requires 453 bits, then the corresponding inaccuracy can be estimated as

$$\frac{453}{1620} = 0.27.$$

This value does not compute mismodel exactly, but it approximates one of its components: the information still needed to reconstruct the representation given the model. ■

Surfeit captures the complementary case. Suppose two descriptions reconstruct the same representation, but one does so by using a compact law while the other uses a long list of special cases, auxiliary clauses, or unnecessary parameters. The second description may be accurate, but it is not minimal. Its defect lies not in failing to produce the representation, but in containing avoidable descriptive complexity. From the standpoint of nescience, such surplus matters because an unnecessarily long description indicates that the regularities of the representation have not been fully captured.

The distinction between deficiency and surplus is important because the two components may move in opposite directions. A more complex model may reduce inaccuracy by fitting the representation more closely, while increasing surfeit by introducing unnecessary structure. Conversely, a simpler model may reduce surfeit but increase inaccuracy by failing to reconstruct relevant details. The problem of modeling is therefore not merely the minimization of prediction error, but the search for descriptions that balance adequacy and economy.

In principle, the best description of a representation would be a minimal description: a shortest effective procedure capable of reconstructing it. This ideal connects *mismodel* with Kolmogorov complexity, since the shortest program that outputs a string provides a formal measure of its descriptive content. However, such minimal descriptions are generally unknown. Moreover, Kolmogorov complexity is not computable in general. Consequently, *mismodel* is an ideal theoretical quantity rather than a directly computable metric.

For this reason, practical applications rely on approximations. Inaccuracy estimates the information still needed to correct the output of a description, while surfeit estimates the unnecessary complexity present in the description itself. Together, they provide computable or semi-computable proxies for the two components of *mismodel*. These proxies are imperfect, but they allow the theory to be applied in domains such as machine learning, empirical science, and the analysis of scientific descriptions.

Mismodel therefore captures the second fundamental source of nescience. Even when the representation of an entity is adequate, our description of that representation may remain deficient, excessive, or both. Reducing *mismodel* means improving descriptions so that they reconstruct representations more faithfully and more economically. In this sense, scientific progress involves not only finding better representations of entities, but also finding better descriptions of the representations already available.

1.7 Nescience

Nescience is an old-fashioned English word meaning lack of knowledge or awareness. In this book, the term is used in a specific technical sense. Whereas ignorance often refers to the absence of knowledge that is already available and could in principle be acquired, nescience refers to the absence of knowledge that has not yet been obtained, formulated, or incorporated into scientific understanding. The purpose of the theory of nescience is to provide a formal framework for measuring this lack of knowledge.

The preceding sections introduced two fundamental sources of nescience. The first is *miscoding*, which concerns the relation between an entity and its representation. A representation may fail because it omits information

needed to encode the entity adequately, or because it contains information that is not genuinely associated with the entity. The second is *mismodel*, which concerns the relation between a representation and its description. A description may fail because it does not reconstruct the representation adequately, or because it contains unnecessary descriptive structure. Thus, miscoding measures representational failure, while mismodel measures descriptive failure.

Given an entity e , scientific inquiry may be understood as the search for a pair (r, d) , where r is a representation of the entity and d is a description of that representation. The quality of such a pair depends on both components. A good representation must encode the entity with low miscoding, and a good description must model the representation with low mismodel. Nescience is introduced as a measure that combines these two quantities into a single assessment of how poorly the pair (r, d) captures the entity under investigation.

The formal notion of mismodel is generally difficult to evaluate directly, because it refers to minimal descriptions, which are typically unknown and are related to the non-computability of Kolmogorov complexity. For this reason, practical applications rely on two complementary approximations. *Inaccuracy* approximates descriptive deficiency: the extent to which the output of a description fails to reconstruct the intended representation. *Surfeit* approximates descriptive surplus: the extent to which a description contains unnecessary structure beyond what is required to reconstruct the representation. Together, inaccuracy and surfeit provide practical ways to estimate mismodel when minimal descriptions are not accessible.

The two components of nescience are often in tension. Reducing miscoding may require changing the representation of the entity, for example by introducing new variables, collecting additional data, or adopting a different symbolic encoding. Such a change may improve the representation, but it may also make the existing description inadequate, thereby increasing mismodel. Conversely, improving a description may reduce mismodel relative to the current representation, while leaving the representation itself poorly aligned with the entity. In scientific practice, progress often requires adjusting both the representation and the description, rather than optimizing one while keeping the other fixed.

For this reason, nescience is not merely a measure of model error. Traditional approaches often assume that a representation has already been fixed and then focus on selecting a model that fits it as well as possible. This is a useful strategy in many contexts, but it neglects the possibility that the representation itself may be defective. The theory of nescience makes this additional source of error explicit. It treats scientific understanding as depending jointly on how entities are represented and how those representations are described.

The search for knowledge can therefore be formulated as an optimization

problem over pairs (r, d) . For each candidate pair, miscoding evaluates the adequacy of the representation, while mismodel evaluates the adequacy and economy of the description. A pair is preferable to another when it reduces one of these quantities without increasing the other. In general, however, there may be trade-offs between them. This leads naturally to a Pareto view of scientific progress: a pair (r, d) is Pareto optimal if no alternative pair improves either miscoding or mismodel without worsening the other.

The Pareto perspective is useful because there may be no single best pair independently of context. Different scientific objectives may justify different balances between representational fidelity and descriptive economy. For example, an exploratory study may tolerate a more complex representation if it captures aspects of the entity that were previously ignored. A predictive application may prefer a description that performs well on available data, even if the representation remains provisional. A theoretical investigation may prioritize a more economical description, even at the cost of leaving some empirical detail unexplained. The theory of nescience does not eliminate these choices, but it makes their structure explicit.

In practical applications, it may still be necessary to select a single pair (r, d) from a set of candidates. This requires a utility function or aggregation rule that ranks candidate pairs according to the aims of the investigation. The specific form of this function is not fixed by the theory itself. In machine learning, for instance, one may combine practical estimates of miscoding, inaccuracy, and surfeit to compare models and representations. In other scientific contexts, different weighting schemes may be appropriate, depending on whether the priority is prediction, explanation, simplicity, robustness, or experimental feasibility.

The central point is that nescience provides a unified framework for analyzing the incompleteness of scientific knowledge. It identifies whether our lack of understanding arises primarily because we have not represented the entity adequately, because we have not described the representation adequately, or because both failures occur simultaneously. By distinguishing miscoding from mismodel, the theory offers a structured way to evaluate scientific progress and to guide the refinement of both representations and descriptions.

1.8 Perfect Knowledge

In the theory of nescience, scientific progress is understood as the progressive reduction of the gap between an entity, its representation, and its description. This reduction can proceed in two complementary ways. First, researchers may improve the representation of the entity, for example by collecting better data, refining experimental procedures, introducing more adequate variables, or adopting a more suitable symbolic encoding. Second, they may improve the

description of that representation, for example by formulating a better model, eliminating unnecessary assumptions, or finding a more concise explanatory structure. In this sense, scientific inquiry proceeds through the iterative proposal and evaluation of pairs (r, d) , where r is a representation of the entity and d is a description of that representation.

We say that *perfect knowledge* of an entity has been achieved when there exists a pair (r, d) whose nescience is equal to zero. Since nescience is determined by miscoding and mismodel, this condition requires both quantities to vanish. Zero miscoding means that the representation r adequately encodes the intended entity: it neither omits information required for representing the entity nor contains independent information irrelevant to it. Zero mismodel means that the description d adequately and economically reconstructs the representation: it neither fails to account for the representation nor contains unnecessary descriptive structure.

Perfect knowledge is therefore not merely a state in which a model fits the available data. A model may fit a representation well while the representation itself remains defective. Conversely, a representation may be adequate while the available description remains deficient or unnecessarily complex. Perfect knowledge requires both levels to be resolved simultaneously: the entity must be properly represented, and the representation must be properly described.

This definition also shows why perfect knowledge is an ideal limiting notion rather than an ordinary empirical condition. In practice, miscoding and mismodel cannot usually be computed exactly. Miscoding depends on the relation between effective representations and admissible representations of the entity, which are generally inaccessible. Mismodel depends on the relation between effective descriptions and minimal descriptions of the representation, which are also generally unknown. Moreover, the identification of minimal descriptions is connected with Kolmogorov complexity, which is not computable in general. For this reason, claims of perfect knowledge cannot normally be established by direct calculation.

The connection between perfect knowledge and randomness can be explained by the link with minimal descriptions. If a description is minimal, then it cannot be compressed further without losing the ability to reconstruct the relevant representation. In the language of algorithmic information theory, such a description is incompressible relative to the representation it describes. Thus, at the ideal limit, perfect knowledge corresponds to a situation in which all removable regularity has already been captured by the description, and no further compression or simplification is possible.

This does not mean that every random or incompressible string constitutes perfect knowledge. A random string may be meaningless, may fail to represent the intended entity, or may describe the wrong representation. Randomness is

relevant only after the representational and descriptive relations have been fixed. The description must be minimal with respect to an adequate representation of the intended entity. Otherwise, incompressibility alone does not imply understanding.

It is also important to note that perfect knowledge need not be unique. A single entity may admit several admissible representations, and each such representation may admit one or more minimal descriptions. Different scientific contexts may therefore realize perfect knowledge through different pairs (r, d) . For example, one representation may be more convenient for prediction, another for explanation, and another for computation. If each representation adequately encodes the entity and each corresponding description has zero mismodel, then these pairs may constitute distinct but equally valid realizations of perfect knowledge.

In practical scientific work, the objective is not usually to prove that perfect knowledge has been reached, but to determine whether successive changes reduce nescience. Since exact values are generally unavailable, researchers must rely on approximations, proxies, and domain-specific criteria. A new representation or description may temporarily increase one component while decreasing another, and it may not be immediately clear whether the overall change constitutes progress. For this reason, the reduction of nescience should be understood as a long-term tendency rather than as a monotonic sequence of guaranteed improvements.

Perfect knowledge may also be unattainable in some domains. There may be structural limitations in the available representations, intrinsic complexity in the entity, or trade-offs between representational adequacy and descriptive economy. In such cases, inquiry may converge not to zero nescience, but to a frontier of best available pairs (r, d) , where any further improvement in one component requires deterioration in another. When this occurs, progress may require a change of perspective: a new representation, a new descriptive language, or a reformulation of the entity under investigation.

Thus, perfect knowledge functions as a theoretical reference point. It defines the state toward which the reduction of nescience is directed, even when that state cannot be reached or recognized in practice. By specifying the simultaneous disappearance of miscoding and mismodel, the concept clarifies what it would mean to understand an entity completely within the framework of the theory.

1.9 Unknown Unknowns

Scientific inquiry is not limited to solving problems that have already been formulated. It also involves the discovery of entities, questions, and relations that are not yet part of the current body of scientific knowledge. These cases belong to the domain of the *unknown unknowns*: not merely unsolved problems, but

possible objects of inquiry whose existence, relevance, or significance has not yet been recognized.

Within the theory of nescience, the unknown unknowns can be approached through the distinction between entities that are effectively known and entities that are not yet effectively known. An entity is effectively known when scientific practice has assigned to it at least one effective representation. Conversely, an entity may be knowable in principle while remaining effectively unknown because no representation has yet been constructed, adopted, or recognized for it. The problem of discovering unknown unknowns is therefore the problem of identifying candidate entities for which no effective representation is currently available, or whose available representations have not yet been interpreted as belonging to a distinct entity.

A naive approach would be to generate arbitrary binary strings and ask whether any of them constitutes, or approximates, a representation of a previously unknown entity. This strategy expresses the idea of discovery by blind search. However, the space of finite strings is far too large for such an approach to be useful in practice, and the majority of the generated strings would have no scientific meaning. The theory therefore seeks more structured procedures for exploring the boundary between what is already effectively known and what may still lie outside current scientific awareness.

To make this problem tractable, it is useful to distinguish between different regions of knowledge. Some entities are effectively known and relatively well understood: they have established representations and descriptions with comparatively low nescience. Other entities are effectively known but still poorly understood: they have available representations, but their associated nescience remains high. These constitute known unknowns. Beyond them lie possible entities for which no effective representation is yet available, or for which existing strings have not yet been recognized as representations of a distinct object of inquiry. These constitute the unknown unknowns.

The boundary between the effectively known and the effectively unknown may be called the *knowledge frontier*. This frontier is not a sharp metaphysical boundary, but a practical and historical one. It depends on the representations, descriptions, classifications, and research practices available at a given time. As scientific work progresses, the frontier may shift: new representations are constructed, old representations are reinterpreted, and entities that were previously invisible to scientific practice become objects of inquiry.

Identifying this frontier in practice is difficult. Scientific knowledge is distributed across publications, datasets, taxonomies, theories, and disciplinary traditions. Moreover, different fields may use different terminologies for related entities, or may classify the same entity under different conceptual schemes. For this reason, the discovery of unknown unknowns cannot be reduced to a

simple enumeration of already known research topics. It requires the analysis of existing representations, the relations among entities, and the degree of nescience associated with different regions of inquiry.

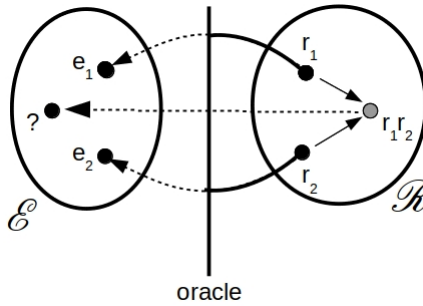


Figure 1.7: Discovering new research entities.

One strategy developed in this book is based on combining existing representations. If two effective representations correspond to entities already studied in different contexts, their joint representation may define a new candidate entity or reveal a previously unnoticed relation. This procedure does not guarantee discovery: many combinations will be trivial, meaningless, or already known. Nevertheless, it provides a systematic way to explore the space of possible research directions without relying on blind search over arbitrary strings.

The motivation for this strategy is that scientific novelty often arises from recombination. New entities and questions can emerge when representations from different domains are placed in relation to one another. In the language of the theory, joint representations may allow us to construct candidate entities located near or beyond the current knowledge frontier. The significance of such candidates must then be evaluated by further investigation: one must determine whether the joint representation corresponds to a coherent entity, whether it admits useful descriptions, and whether studying it is likely to reduce nescience.

■ **Example 1.6** One may combine a representation from theoretical computer science with a representation from phenomenology. For instance, joining the notion of a minimum-complexity computer program with the notion of self-awareness suggests the candidate research entity “minimum-complexity self-aware computer.” This expression does not by itself establish a well-defined scientific object, but it identifies a possible direction of inquiry: whether there exists a meaningful lower bound on the complexity required for a computational system to exhibit self-awareness. ■

Figure 1.8 illustrates this idea schematically. The inner region represents entities for which available representations and descriptions yield relatively

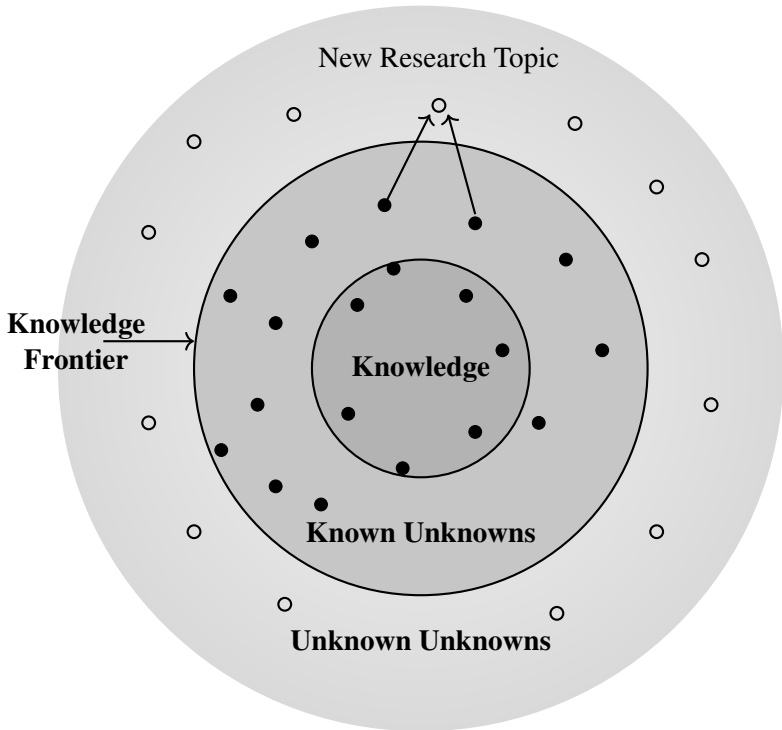


Figure 1.8: The structure of knowledge.

low nescience. The surrounding region corresponds to known unknowns: entities already recognized as objects of inquiry, but still poorly represented or poorly described. The outer region represents unknown unknowns: possible entities not yet effectively incorporated into scientific practice. The knowledge frontier separates the effectively known region from what remains outside current scientific recognition.

The theory suggests two practical criteria for guiding the search beyond this frontier. First, candidate entities are more likely to be fruitful when they are generated from entities that still have high nescience. Poorly understood entities often lie near the boundary of current knowledge, and their combination may reveal gaps or relations that have not yet been investigated. Second, candidates are more likely to be novel when they arise from the combination of distant research areas. Interdisciplinary combinations reduce the probability that the resulting entity has already been studied, especially when the relevant fields have developed independently.

These criteria do not provide a mechanical solution to discovery. A generated candidate may be ill-defined, scientifically uninteresting, or impossible to

represent adequately. The role of the theory is more modest: it provides a systematic way to propose and prioritize candidate entities for further investigation. Once a candidate has been generated, it must still be assessed by asking whether it can be represented, whether it admits meaningful descriptions, whether it connects to existing entities, and whether its study is likely to reduce nescience.

Unknown unknowns are therefore not treated as mysterious objects beyond all analysis. They are approached as potential research entities that have not yet entered the set of effectively known entities. By using nescience, representation, and joint construction as guiding concepts, the theory provides a disciplined method for exploring regions of inquiry that would otherwise remain dependent on chance, intuition, or accidental disciplinary contact.

References

The theory of nescience builds upon several well-established foundations in information theory, computability, algorithmic complexity, and the philosophy of science. The following references provide the theoretical background and conceptual tools necessary to understand and situate the ideas introduced in this chapter.

Sipser's book [Sip12] is a widely respected introduction to formal languages, automata, and computability theory. It lays the groundwork for understanding which descriptions are computationally feasible, an essential aspect of the theory of nescience, as it assumes that knowledge must be computable to be meaningful.

Li and Vitányi's book [LV13] is a comprehensive volume that presents the theory of Kolmogorov complexity, which formalizes the idea of description length using the shortest program capable of generating a given object. The concepts developed by Li and Vitányi are central to the theory of nescience, especially in defining metrics such as inaccuracy, miscoding, and surfeit based on algorithmic information theory.

Cover and Thomas's book [CT12] is a foundational text in information theory, providing the mathematical framework for understanding concepts such as entropy, mutual information, and data compression. It serves as a rigorous yet accessible introduction to how information can be quantified, transmitted, and encoded, ideas that relate to the notions of description length in the theory of nescience.

Chalmers's book [Cha13] offers a critical and historical introduction to the philosophy of science. It examines the assumptions, limits, and methodologies of scientific practice. This perspective is invaluable for situating the theory of nescience within the broader discourse about how scientific knowledge is constructed, evaluated, and refined over time.



2. Entities

*We are all agreed that your theory is crazy.
The question which divides us is whether it is crazy enough.*
Niels Bohr

The theory of nescience begins with a simple question: what is the object about which knowledge is being sought? Before one can measure ignorance, compare explanations, or evaluate scientific progress, one must first identify the entities under investigation. These entities constitute the domain to which the theory will be applied.

In scientific practice, the entities studied by researchers vary enormously. They may be physical objects, such as planets, cells, or chemical compounds; phenomena, such as combustion, infection, or economic inflation; formal structures, such as groups, algorithms, or probability distributions; or theoretical constructs introduced to organize and explain observations. What unifies these cases is not a single metaphysical nature, but their role in inquiry: they are things about which researchers attempt to construct representations, collect evidence, formulate models, derive explanations, or make predictions.

For this reason, the theory of nescience does not begin with a universal collection of all possible objects. Such a collection would be mathematically problematic and philosophically unclear. Instead, every application of the theory

starts by fixing a domain-specific set of entities, denoted by $\mathcal{E}$. The elements of $\mathcal{E}$ depend on the research context. In one application, $\mathcal{E}$ may be a set of mathematical structures; in another, a collection of biological organisms, diseases, physical systems, technologies, datasets, or research topics. The theory is therefore domain-relative: it does not attempt to quantify ignorance about everything at once, but about a specified collection of entities.

This chapter introduces the notion of a research entity and clarifies the formal role played by the set $\mathcal{E}$. We first explain why $\mathcal{E}$ must be fixed relative to a domain of inquiry and why universal sets are excluded. We then distinguish several types of entities that occur in scientific practice, including knowable and unknowable entities, concrete and abstract entities, observable and non-observable entities, and real and fictitious entities. This classification is not intended as a complete ontology, but as a practical taxonomy for understanding the kinds of objects that may become targets of representation and modeling.

Finally, we introduce the concept of a research area. A research area is treated formally as a subset of $\mathcal{E}$, although in practice meaningful areas are usually unified by some scientific criterion, such as a common kind, level of organization, mechanism, method, or explanatory purpose. This allows the theory of nescience to be applied not only to individual entities, but also to broader domains of inquiry. The concepts introduced in this chapter provide the foundation for the following chapters, where entities are accessed through representations and studied through descriptions.

2.1 Entities

The theory of nescience studies the limits of knowledge within a given domain of investigation. Any such analysis requires a precise specification of the objects about which knowledge may be obtained. These objects will be called *entities*.

Definition 2.1.1 A *research entity* is anything about which scientists attempt to construct representations, collect evidence, formulate models, or derive explanatory or predictive statements.

According to this definition, a research entity may be an object, phenomenon, structure, or concept that can become the subject of systematic investigation within a given research domain.

We assume that there exists a non-empty set of *entities* that we seek to understand.

Notation 2.1. The set of research entities under consideration is denoted by $\mathcal{E}$.

The elements of $\mathcal{E}$ are defined relative to the particular domain of knowledge in which the theory of nescience is applied. Depending on the context, this

domain may correspond to a broad discipline such as mathematics, physics, engineering, or the social sciences, or it may be restricted to a specific subarea within one of these fields.

From a formal perspective, $\mathcal{E}$ should be regarded as a fixed but generally inaccessible collection: although its elements are determined by the underlying domain of inquiry, there is in general no effective (computable) procedure for accessing, deciding membership in, or enumerating its elements.

The set $\mathcal{E}$ contains both *knowable* and *unknowable* entities (see Section 2.2 below for further discussion of the types of entities allowed in the theory of nescience).

The possibility of using universal sets is excluded; that is, the existence of a set ξ containing all possible objects cannot be assumed. The problem with universal sets is that their existence is incompatible with Cantor's theorem (see Example 2.1). Cantor's theorem proves that the power set $\mathcal{P}(\xi)$, consisting of all possible subsets of ξ , has more elements than the original set ξ . This contradicts the assumption that ξ includes everything. The set $\mathcal{E}$ must be explicitly fixed and domain-scoped; that is, it must denote the set of entities under consideration in a given research area, rather than an ambiguous or open-ended collection.

■ **Example 2.1 — Cantor's theorem.** Cantor's theorem proves that for any set A , $d(A) < d(\mathcal{P}(A))$ ¹. Consider $f : A \rightarrow \mathcal{P}(A)$, a function that maps each element $x \in A$ to the set $\{x\} \in \mathcal{P}(A)$; since f is injective, we have $d(A) \leq d(\mathcal{P}(A))$. To substantiate that the inequality is strict, let's assume there exists a surjective function $g : A \rightarrow \mathcal{P}(A)$ and consider the set $B = \{x \in A : x \notin g(x)\}$. As g is surjective, there must exist a $y \in A$ such that $g(y) = B$. This, however, raises a contradiction, $y \in B \Leftrightarrow y \notin g(y) = B$. Consequently, the function g cannot exist, therefore $d(A) < d(\mathcal{P}(A))$. ■

Not all conceivable sets are acceptable, as some may give rise to paradoxes. Take, for example, Russell's paradox, which proposes a set R consisting of all sets that are not members of themselves. The paradox arises when we try to discern whether R is a member of itself (see the Example 2.2). To avoid such problems, the theory of nescience is based on the Zermelo-Fraenkel set of axioms, along with the Axiom of Choice (or ZFC). The Axiom of Separation (if P is a property with parameter p , then for any set x and parameter p there exists a set $y = \{u \in x : P(u)\}$ that includes all those sets $u \in x$ that have property P) allows the use of this notation only to construct sets that are subsets of already existing sets. A more extensive Axiom of Comprehension (if P is a property, then there exists a set $y = \{u : P(u)\}$) would be required to allow sets like the one proposed by Russell's paradox. Russell's paradox arises from the use of an unrestricted comprehension principle. In the axioms of ZFC, and in the theory of nescience,

¹Here $d(\cdot)$ denotes cardinality, see Appendix 16.

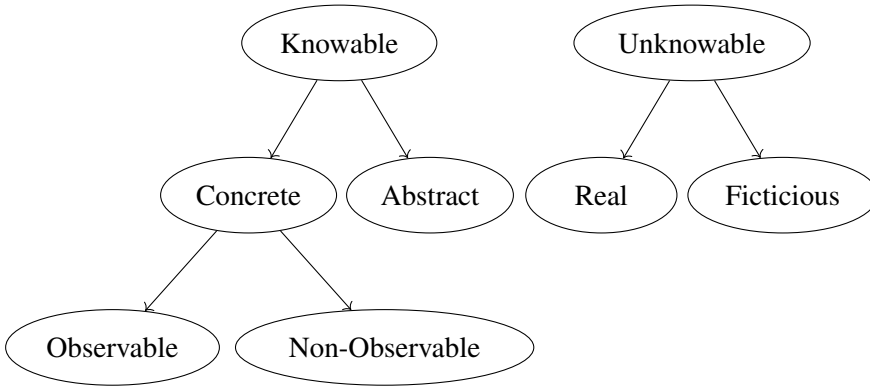


Figure 2.1: Classification of Research Entities

the axiom of comprehension is considered false.

■ **Example 2.2 — Russell’s paradox.** Suppose R is the set of all sets not members of themselves, such that $R = \{x : x \notin x\}$. The contradiction arises when querying if R is a member of itself. If R is not a member of itself, by its own definition, it should be; conversely, if R is declared to be a member of itself, its definition dictates it should not be. Symbolically, this can be written as $R \in R \Leftrightarrow R \notin R$. ■

2.2 Types of Entities

In scientific practice, researchers work with a wide variety of entities: physical systems, measurable phenomena, theoretical constructs, and formal objects. Some of these entities can be directly observed, others are inferred indirectly from experimental effects, and still others exist only within formal frameworks. Figure 2.1 provides a practical classification of research entities that will be useful throughout the book. This classification is not intended to be philosophically exhaustive or mathematically rigorous. Its purpose is illustrative rather than foundational. The boundaries between categories are not sharp, and in real scientific practice they often overlap or evolve over time. The diagram should therefore be read as a working taxonomy that helps clarify the kinds of objects that can become subjects of representation and modeling within the theory of nescience, rather than as a definitive ontological scheme.

Knowable entities

We distinguish between knowable and unknowable entities. A *knowable entity* is one for which, at least in principle, a sufficiently accurate scientific representation can be constructed. This does not mean that the entity is fully understood or

perfectly described, but that it is accessible to systematic investigation. Scientists can refine its representations by incorporating new empirical data, fixing incorrect assumptions, or removing irrelevant information. Over time, this iterative process reduces error in the encoding of the entity and improves the predictive and explanatory power of the associated models.

Knowable entities can be either concrete or abstract. *Concrete entities* are physical systems or phenomena. They exist in space and time and can interact with instruments or other physical systems. These interactions allow researchers to design experiments, collect data, and test hypotheses about their properties. Even when the entity itself is not directly accessible, its physical effects can be detected through controlled interventions and measurement protocols.

Concrete entities can be divided into observable and non-observable. *Observable entities* can be directly detected, either by the naked eye or with instruments that extend our senses, such as microscopes, telescopes, sensors, or imaging devices. Observability does not require unaided perception, but operational accessibility: the entity must produce stable signals that can be recorded, quantified, and compared across experiments. This operational criterion is what allows observable entities to serve as reliable anchors for empirical validation and model testing.

■ **Example 2.3** A bacterial colony growing on a Petri dish is a knowable, concrete, observable entity. It can be visually inspected, counted, measured, and experimentally manipulated. Researchers can quantify its growth rate, analyze its morphology, test its response to antibiotics, or sequence its genetic material. Independent laboratories can reproduce the same experimental conditions and compare results, allowing alternative descriptions of the colony's behavior to be evaluated and refined. Because it produces stable and measurable effects (such as changes in size, color, density, or metabolic activity) its properties can be encoded in increasingly precise representations. ■

Some concrete entities are *non-observable*, that is, they cannot be directly observed, yet their existence can be inferred from stable, reproducible experimental effects. In such cases, the entity does not appear itself in measurement data; rather, it is postulated as the best explanation for a consistent pattern of observations. Its properties are reconstructed indirectly through mathematical models, statistical analysis, or controlled interventions.

■ **Example 2.4** An electron is a knowable, concrete, non-observable entity. We do not observe electrons directly, but we detect their presence through cloud chamber tracks, spectral lines, scattering experiments, and electrical measurements. Their charge, mass, and spin are not accessed by direct perception, but are inferred from highly reproducible quantitative effects, such as discrete energy levels in atoms, interference patterns, and precisely measured deflections

in electromagnetic fields. Competing theoretical descriptions of electrons are evaluated by comparing predicted outcomes with experimental data. Although the electron itself is never seen as a localized object in the classical sense, the stability and precision of the phenomena associated with it allow researchers to construct progressively refined representations. ■

Abstract entities do not have a physical presence. They are defined within formal systems or theoretical frameworks. Although they do not interact physically with instruments, they mediate our understanding of concrete systems by offering compact and manipulable representations. Equations, probability distributions, algorithms, state spaces, and optimization functions allow researchers to formalize assumptions, derive consequences, and compute quantitative forecasts that can later be confronted with empirical data. In many cases, progress in a discipline results from introducing a new abstract entity that reorganizes existing observations under a more coherent structure, reduces descriptive complexity, or increases predictive accuracy.

■ **Example 2.5** The notion of a Turing machine (see Definition 18.1.1) is a knowable, abstract entity. It is not a physical object, but it is precisely defined, can be formally analyzed, and is used to prove theorems about computability and complexity. Its components are specified with exact rules, allowing rigorous reasoning about what can and cannot be computed. Although no physical device perfectly realizes the infinite tape assumed in the definition, the abstraction captures essential structural properties shared by all digital computers. Because its behavior is fully determined by formal rules, competing representations can be unambiguously compared, and proofs about decidability, universality, and incompleteness can be carried out with mathematical precision. ■

Unknowable entities

An *unknowable entity* is an entity for which no reliable and stable representation can be constructed within any scientific framework. In this sense, unknowability is structural rather than temporary: it does not arise from current technological limitations or lack of data, but from the absence of a possible operational connection between the entity and reproducible procedures. An entity is structurally unknowable when no experiment, measurement protocol, or formal derivation could, even in principle, discriminate between alternative representations or progressively reduce error in its encoding. Under this definition, unknowable entities fall outside the domain of scientific research.

Unknowable entities can be classified as real or fictitious. Some entities are *real* in the sense that they belong to physical reality, yet remain structurally beyond our capacity to investigate or represent with scientific precision. Even if the entity exists independently of our knowledge, no reproducible procedure could

in principle establish reliable constraints on its properties. As a consequence, scientific representations cannot converge, and competing encodings cannot be meaningfully evaluated. These entities may be logically conceivable and even physically instantiated, but they remain outside the domain of entities that can be stabilized through empirical or formal refinement.

■ **Example 2.6** Consider the exact microphysical state of the universe at a precise instant of time, including the exact position and momentum of every particle with infinite precision. If physical reality is continuous (as assumed in standard classical models), then such a state is a real entity: it corresponds to an actual configuration of the world. However, it is structurally unknowable. No finite physical system can measure, store, or encode an infinite amount of information with arbitrary precision. Moreover, no experimental protocol could discriminate between two candidate representations that differ only beyond any physically realizable resolution. Therefore, although the entity may exist, no reliable and stable scientific representation of it can in principle be constructed. ■

Fictitious entities are introduced hypothetically but do not correspond to any reproducible phenomenon. Such entities may arise as speculative explanatory devices, heuristic constructions, or artifacts of incomplete theories. Unlike real but unknowable entities, fictitious entities do not merely escape representation due to structural limits; rather, they fail to anchor representations to observable effects or formal necessities. They are usually abandoned when alternative frameworks explain the same phenomena without invoking them, or when continued experimental effort fails to produce reproducible consequences associated with their existence.

■ **Example 2.7** The luminiferous ether, once postulated as the medium for light propagation, is an example of an unknowable and fictitious entity. It was introduced in the nineteenth century to account for the apparent need of a mechanical medium through which electromagnetic waves could travel, in analogy with sound waves in air. However, despite increasingly precise experimental efforts to detect the Earth's motion relative to this supposed medium, no consistent or reproducible effect was ever observed. Eventually, the development of special relativity rendered the ether unnecessary by providing a coherent theoretical framework that explained electromagnetic phenomena without invoking any background medium. ■

2.3 Research Areas

Scientific inquiry is rarely confined to a single entity. Researchers usually study collections of entities that belong to a common domain, share relevant properties, appear at a similar level of organization, or participate in a common explanatory

framework. Such collections will be called *research areas*. The concept is useful because many scientific questions are not directed at isolated entities, but at families of entities considered together: biological species, chemical compounds, mathematical structures, diseases, algorithms, economic systems, or technological artifacts.

Definition 2.3.1 Let $\mathcal{E}$ be a set of entities. We define a *research area* as a subset of entities

$$\mathcal{A} \subseteq \mathcal{E}.$$

The definition is intentionally simple. A research area is formally treated as a subset of the set of entities under consideration. However, in scientific practice, not every subset of $\mathcal{E}$ constitutes a meaningful research area. Scientifically useful areas are usually unified by some intelligible criterion. The entities in an area may belong to the same kind (see Section XX), appear at the same level of organization, participate in the same mechanism, share relevant attributes, be studied by similar methods, or contribute to the same explanatory objective. The exact criterion depends on the domain in which the theory of nescience is applied.

■ **Example 2.8** If $\mathcal{E}$ is the set of entities studied in biology, then possible research areas include mammals, viruses, metabolic pathways, cancer types, or ecosystems. These areas are not arbitrary collections. Mammals are grouped by biological classification; viruses by their biological organization and replication mechanisms; metabolic pathways by their functional role in cellular processes; cancer types by pathological and molecular criteria; and ecosystems by the interaction of organisms with their environment. Each case reflects a different principle of scientific grouping. ■

In Chapter 4, we will study how a research areas can be characterized through descriptions that jointly account for the representations of its members. From this perspective, a research area is not merely a set of entities selected in advance, but may also be understood through the common descriptive structure that makes those entities scientifically intelligible as a group.

Research areas may also have subareas.

Definition 2.3.2 Let $\mathcal{E}$ be a set of entities, and $\mathcal{A} \subseteq \mathcal{E}$ a research area. We define a *research subarea* of the area $\mathcal{A}$ as a subset of entities

$$\mathcal{B} \subseteq \mathcal{A}.$$

This definition gives rise to hierarchies of areas, in which broad domains are progressively refined into more specialized subdomains.

■ **Example 2.9** Within the area of mammals, one may distinguish primates,

rodents, carnivores, or marine mammals. Within the area of machine learning, one may distinguish supervised learning, unsupervised learning, reinforcement learning, or representation learning. ■

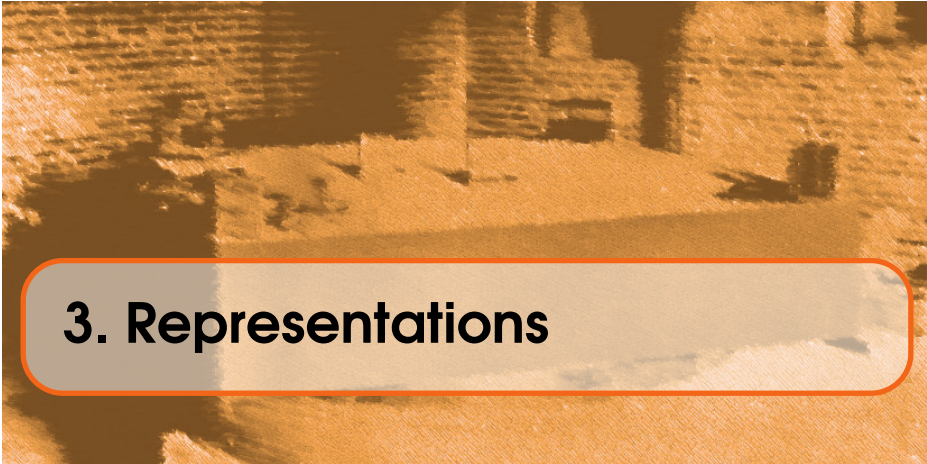
It is important, however, that such hierarchies need not be unique. The same set of entities may be organized in different ways depending on the scientific purpose. A particular biological domain may be organized taxonomically, grouping organisms into species, genera, families, and higher biological categories. The same organisms may also be grouped ecologically, according to habitat, trophic role, or environmental adaptation. Both organizations may be scientifically legitimate, because they answer different questions.

This flexibility is important for the theory of nescience. The value of a research area depends on the purpose of the investigation. A collection of entities may be appropriate for one question and inappropriate for another. If the goal is to study evolutionary ancestry, a taxonomic area may be useful. If the goal is to study disease transmission, an area based on host-pathogen interaction may be more appropriate. If the goal is to compare predictive models, an area may be defined by the availability of common datasets, common variables, or common modeling assumptions.

Research areas therefore play a practical role. They allow the theory of nescience to be applied not only to individual entities, but also to broader domains of inquiry. Once a research area $\mathcal{A}$ has been fixed, one may ask how well the entities in $\mathcal{A}$ are represented, how accurately they are described, how much unnecessary complexity is present in their descriptions, and how much nescience remains in the area as a whole. In later chapters, these questions will lead to area-level versions of miscoding, mismodel, and nescience.

2.4 References

For more information about Russell's paradox, Cantor's theorem, and universal sets, refer, for example, to [Jec13]. This book also covers the Zermelo-Fraenkel system of axioms, including the Axiom of Choice.



3. Representations

The map is not the territory.
Alfred Korzybski

Scientific inquiry is typically organized around research entities that serve as the objects of investigation. In the theory of nescience, these entities are collected into a set $\mathcal{E}$ whose elements may be physical, abstract, or even hypothetical. Within this framework, representations provide a means to reconstruct, compare, and combine such entities. However, this task is inherently challenging: entities may possess infinite structure, be empirically inaccessible, or remain only partially understood.

To address this difficulty, we decompose the representation problem into two complementary components. The first concerns the representation of entities. In particular, we study the conditions under which a finite string $r \in \mathcal{B}^*$ represents an entity $e \in \mathcal{E}$, the circumstances under which the entity can be reconstructed from such a string, and the relations that may exist between distinct representations of the same entity. These issues constitute the subject of the present chapter. The second component concerns the description of representations. In this case, the objective is to extract from a given representation a concise model capable of supporting human reasoning and prediction. This complementary problem will be examined in Chapter 4.

The analysis developed here relies on the notion of a *representation oracle*. The oracle establishes the semantic relation between strings and entities by specifying, through an internal reconstruction function, which strings encode elements of $\mathcal{E}$. This abstraction separates the syntactic domain $\mathcal{B}^*$ from the ontological domain $\mathcal{E}$ while making explicit the assumptions required for reconstruction.

Within this framework we study structural properties of representation oracles, including minimality and comparability, and examine the extent to which background knowledge may be internalized by the oracle. These properties determine how much information must be carried explicitly by the representation and how much may be supplied by the oracle.

Finally, we classify representations according to their formal properties. In particular, we distinguish decodable, minimal, admissible, extended and effective representations, and analyze how these categories arise in scientific practice. This classification reflects the progressive refinement of knowledge: early stages of inquiry rely on fragmentary and imperfect encodings, whereas deeper understanding tends to produce increasingly accurate and structured representations. In this way, the theory of nescience places representation at the center of scientific development while clarifying its formal constraints and epistemic limits.

3.1 Representation Oracle

The representation of the entities composing the set $\mathcal{E}$, which may not be effectively definable or enumerable, raises significant challenges (see Section 22.4 for an overview of proposed solutions in the scientific literature, and their limitations). This book presents a particular approach to this problem by dividing scientific representation into two closely related subproblems: the encoding of entities by means of representations, and the modeling of these representations by means of descriptions. From this perspective, scientific models do not explain entities directly, but rather explain them through their representations (see Figure 1.1 in the Introduction reproduced here for ease of reference). In this section we focus on encoding; the role of descriptions is addressed in Chapter 4.

In Kolmogorov complexity, the representation problem is usually approached by assuming that the set $\mathcal{E}$ is well defined and countable, and that a total encoding function $f : \mathcal{E} \rightarrow \mathbb{N}$ exists, analogous to a Gödel numbering. The theory of nescience similarly adopts the idea that entities can be encoded as numbers, or equivalently as finite strings. Our approach, however, departs from this tradition by dropping the requirement that $\mathcal{E}$ be explicitly definable or enumerable.

Rather than postulating an encoding function with domain $\mathcal{E}$, we take as primitive the well-defined set of finite strings and assume the existence of an

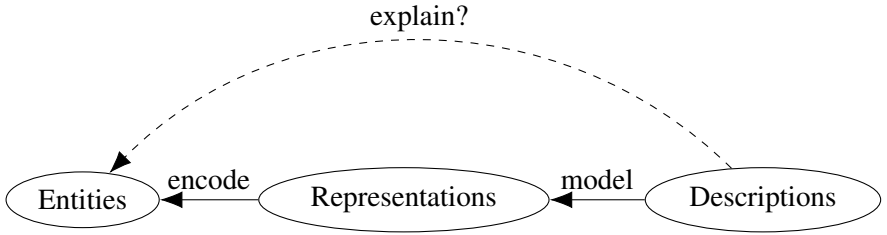


Figure 3.1: The Problem of Understanding

abstract oracle that internally associates certain strings with entities. More precisely, we introduce an oracle $\mathcal{O}_{\mathcal{E}}$ that acts as an abstract decision device, determining whether a string r contains sufficient information to reconstruct an entity.

For the purposes of this book, and without loss of generality, we restrict attention to binary strings as the medium of representation.

Definition 3.1.1 Given a set of entities $\mathcal{E}$, a *representation oracle* is a total decision oracle

$$\mathcal{O}_{\mathcal{E}} : \mathcal{B}^* \longrightarrow \{\text{true}, \text{false}\}$$

with the following semantics: there exists an oracle-internal partial *reconstruction function*

$$\delta_{\mathcal{O}} : \mathcal{B}^* \rightharpoonup \mathcal{E}$$

whose domain coincides with the set of strings accepted by the oracle, that is,

$$\text{dom}(\delta_{\mathcal{O}}) = \{r \in \mathcal{B}^* : \mathcal{O}_{\mathcal{E}}(r) = \text{true}\}.$$

Equivalently, for every $r \in \mathcal{B}^*$,

$$\mathcal{O}_{\mathcal{E}}(r) = \text{true} \iff \delta_{\mathcal{O}}(r) \text{ is defined.}$$

When defined, $\delta_{\mathcal{O}}(r)$ yields a unique entity of $\mathcal{E}$ reconstructed from r without any additional information. The function $\delta_{\mathcal{O}}$ is not assumed to be computable or explicitly characterizable, and serves only to fix the semantic interpretation of the oracle.

The oracle $\mathcal{O}_{\mathcal{E}}$ is an idealized construct. For most collections of entities, it cannot be realized in practice, and no explicit decoding procedure associated with it can be provided. The role of $\mathcal{O}_{\mathcal{E}}$ is purely formal: it serves as a reference device for delimiting what representations of entities are in principle obtainable within the framework. Accordingly, $\mathcal{O}_{\mathcal{E}}$ is treated as a black box and is defined

only through its input-output behavior. In this respect, $\mathcal{O}_{\mathcal{E}}$ is analogous to the oracle in an oracle Turing machine (see Definition 18.5.1).

Within this framework, only elements of $\mathcal{B}^*$, that is, finite binary strings, are eligible to serve as representations of entities (see the discussion of the ontology problem in Section 22.4). Other forms of representation (such as physical models, diagrams, or verbal descriptions) are admissible only insofar as they can be transcribed into binary strings. No distinction is drawn between scientific representations and other types of representations (see the discussion of representational demarcation in Section 22.4). Finally, it is essential to note that $\delta_{\mathcal{O}}$ is a partial function: not every binary string represents an entity.

In general, the internal oracle need not be unique for a given collection of entities, reflecting the plurality and ambiguity inherent in real-world scientific representation.

Proposition 3.1.1 Let $\mathcal{E}$ be a set of entities, let $\delta_{\mathcal{O}}$ be a reconstruction function for an oracle $\mathcal{O}_{\mathcal{E}}$, and let $\pi : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a bijection. Define the partial function $\delta_{\pi} : \mathcal{B}^* \rightarrow \mathcal{E}$ by $\delta_{\pi}(x) = \delta_{\mathcal{O}}(\pi(x))$ for every $x \in \mathcal{B}^*$ such that $\delta_{\mathcal{O}}(\pi(x))$ is defined. Then δ_{π} is also a reconstruction function for $\mathcal{E}$.

Proof. Since $\pi : \mathcal{B}^* \rightarrow \mathcal{B}^*$ is a total bijection and $\delta_{\mathcal{O}} : \mathcal{B}^* \rightarrow \mathcal{E}$ is a partial function, the composition $\delta_{\pi} = \delta_{\mathcal{O}} \circ \pi$ is again a partial function from $\mathcal{B}^*$ to $\mathcal{E}$.

More explicitly, for each $x \in \mathcal{B}^*$, the value $\delta_{\pi}(x)$ is defined if and only if $\pi(x)$ lies in the domain of $\delta_{\mathcal{O}}$. Hence

$$\text{dom}(\delta_{\pi}) = \{x \in \mathcal{B}^* : \pi(x) \in \text{dom}(\delta_{\mathcal{O}})\} = \pi^{-1}(\text{dom}(\delta_{\mathcal{O}})).$$

Whenever $\delta_{\pi}(x)$ is defined, we have

$$\delta_{\pi}(x) = \delta_{\mathcal{O}}(\pi(x)) \in \mathcal{E},$$

so δ_{π} associates binary strings with entities of $\mathcal{E}$. Therefore δ_{π} is also a valid reconstruction function for $\mathcal{E}$. ■

If π is not the identity, then in general δ_{π} differs from $\delta_{\mathcal{O}}$, because it decodes strings after systematically transforming them by π . Thus a given set of entities may admit multiple reconstruction functions.

Rather than deploying individual oracles for each set $\mathcal{E}$, we could have employed a universal decision device. This is a machine $\mathcal{U}_{\mathcal{O}}$ that, given the encoding of an oracle $\mathcal{O}_{\mathcal{E}}$ and a string r as input, computes $\mathcal{U}_{\mathcal{O}}(\langle \mathcal{O}_{\mathcal{E}}, r \rangle) = \mathcal{O}_{\mathcal{E}}(r)$. Universal machines are particularly applicable to the universal set ξ , encompassing all entities (see Section 2.1). However, the theory of nescience is explicitly domain-relative: the choice of $\mathcal{E}$ determines which entities are under investigation and the oracle $\mathcal{O}_{\mathcal{E}}$ fixes the admissible style in which the entities

are encoded. We work with entity sets $\mathcal{E}$ corresponding to different areas of knowledge, each associated with a single oracle $\mathcal{O}_{\mathcal{E}}$ that is most suitable to our current knowledge and practical needs.

■ **Example 3.1** Consider the case when the entities under study are animals. Initially, one might use detailed physical descriptions of animals as encodings. In this scenario, the internal reconstruction function would be a hypothetical machine capable of reconstructing the original animal from its description. As our understanding of biology advances, we might instead adopt an alternative encoding based on the animals' DNA. Both of these encodings serve as representations of the entities. ■

As illustrated in Proposition 3.1.1 and Example 3.1, the entities in $\mathcal{E}$ can be encoded in multiple ways (see the problem of style in Section 22.4). Different oracles accommodate different encoding schemes. In practical terms, some strings provide more adequate representations of an entity than others (see the standard of accuracy in Section 22.4). The optimal representation style depends on the type of questions we aim to answer.

Notation 3.1. We denote by $\mathfrak{D}_{\mathcal{E}}$ the class of all representation oracles over the set of entities $\mathcal{E}$.

Representations must not only correspond to entities but must also be relevant to them (see the requirement of directionality below in this section, in Chapter 5, and in Section 22.4).

We employ an oracle function rather than an oracle relation $\mathcal{R}_{\mathcal{O}} \subseteq \mathcal{B}^* \times \mathcal{E}$, not merely to associate strings with their corresponding entities, but to reconstruct an entity from its representation. The internal oracle's ability to recover original entities underpins our capacity to make hypotheses about entities based on their representations (see surrogative reasoning in Section 22.4). According to the theory of nescience, scientific inquiry involves not only learning how to encode entities properly but also understanding the mechanisms by which internal reconstruction functions decode them. In this sense, the purpose of encoding entities in the theory of nescience differs fundamentally from that in Shannon's information theory (see Appendix 19), as illustrated in Example 3.2.

■ **Example 3.2** Take, for instance, a set $\mathcal{E}$ consisting of two books: "The Ingenious Gentleman Don Quixote of La Mancha" and "The Tragedy of Romeo and Juliet". We might encode the first book with the string "0" and the second with the string "1". While these strings allow us to uniquely identify each book within the set, they do not qualify as valid encodings within the theory of nescience. In information theory, the goal is to uniquely identify an object based on a reference, assuming mutual agreement between the sender and the receiver about the mapping from references to objects. In contrast, the theory of nescience

seeks representations that preserve the richness and detail of the original entities. For example, it would be impossible to hypothesize about Cervantes' influence on Shakespeare using only the strings "0" and "1". ■

One possible response to the limitation discussed in Example 3.2, where entities are encoded using overly simplistic strings that fail to capture their internal structure, would be to require the set $\mathcal{E}$ to be infinite, as is done in Kolmogorov complexity. However, while requiring $\mathcal{E}$ to be infinite prevents all entities from being encoded using uniformly short strings, it does not prevent the use of representations that, although longer, remain arbitrary and devoid of semantic content. That is, they do not allow us to make meaningful inferences about the original entities based solely on their representations. To enable such reasoning, the encoding must preserve essential structural and semantic features of the entities, not just their identity.

3.2 Decodable Representations

Given a finite string r , the oracle determines whether r contains sufficient information to reconstruct a unique entity without additional input. If so, the oracle internally associates r with that entity; otherwise, the string is deemed non-representational.

Definition 3.2.1 Let $\mathcal{O}_{\mathcal{E}}$ be a representation oracle over the set of entities $\mathcal{E}$, with reconstruction function $\delta_{\mathcal{O}} : \mathcal{B}^* \rightarrow \mathcal{E}$. We define the set of *decodable representations* of the oracle as

$$\mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^{\top} = \{r \in \mathcal{B}^* \mid \mathcal{O}_{\mathcal{E}}(r) = \text{true}\}.$$

We say the finite binary string $r \in \mathcal{B}^*$ is a *decodable representation* if $r \in \mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^{\top}$.

Equivalently, we have that

$$\mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^{\top} = \text{dom}(\delta_{\mathcal{O}}).$$

Within the theory of nescience, decodable representations constitute the primary formal point of access to entities, since representations are the only mechanism at our disposal for accessing the entities under investigation.

Notation 3.2. When the set of entities $\mathcal{E}$ and the oracle $\mathcal{O}$ are clear from the context, we may write $\mathcal{R}_{\mathcal{O}}^{\top}$ or simply $\mathcal{R}^{\top}$.

The subset $\mathcal{R}^{\top}$ of representations an oracle can decode depends on the reconstruction function implemented by the oracle, and therefore on the representation style associated with that function.

■ **Example 3.3** Let $\mathcal{E}$ be the set of formal languages over a fixed alphabet (see Section 16.2). A representation of a language is a finite string describing a rule or mechanism that generates or recognizes that language. Different representation styles correspond to different classes of computational models, and therefore to different reconstruction capabilities for the oracle.

For example, a *finite-automaton style*, in which representations describe finite automata. The oracle reconstructs the language recognized by the automaton. In this case, the decodable representations correspond to the class of *regular languages*. A *pushdown-automaton style*, in which representations describe pushdown automata or context-free grammars. The oracle reconstructs the language generated by the grammar. This representation style can decode all context-free languages, which strictly include the regular languages. And a *Turing-machine style*, in which representations describe Turing machines. The oracle reconstructs the language recognized by the machine. This representation style can decode any recursively enumerable language.

Each representation style corresponds to a different oracle and therefore to a different subset of knowable entities. Since regular languages are strictly contained in the context-free languages, which in turn are contained in the recursively enumerable languages, the associated oracles have strictly increasing reconstructive power. ■

An entity may admit more than one decodable representation under a given oracle. It is therefore useful to isolate, for each entity $e \in \mathcal{E}$, the subset of decodable representations that reconstruct exactly that entity.

Definition 3.2.2 Let $\mathcal{O}_{\mathcal{E}}$ be a representation oracle with reconstruction function $\delta_{\mathcal{O}} : \mathcal{B}^* \rightarrow \mathcal{E}$. For any entity $e \in \mathcal{E}$, we define the set of *decodable representations* of e as

$$\mathcal{R}_{\mathcal{O},e}^{\top} = \{r \in \mathcal{R}_{\mathcal{O}}^{\top} \mid \delta_{\mathcal{O}}(r) = e\}.$$

The set $\mathcal{R}_{\mathcal{O},e}^{\top}$ collects all the oracle-accepted strings that have the same semantic target. In this way, it makes explicit that the relation between entities and representations is generally one-to-many: a single entity may be reconstructed from several distinct strings, reflecting redundancy, variation in encoding, or differences in descriptive detail within a fixed representation style.

Notation 3.3. When the oracle is clear from the context, we may write $\mathcal{R}_e^{\top}$ instead of $\mathcal{R}_{\mathcal{O},e}^{\top}$.

We distinguish between *knowable* and *unknowable* entities (see the gray areas in Figure 3.2; for example, the entity e_2 is not encoded by any string). An entity is said to be knowable if the oracle can reconstruct it from at least one decodable

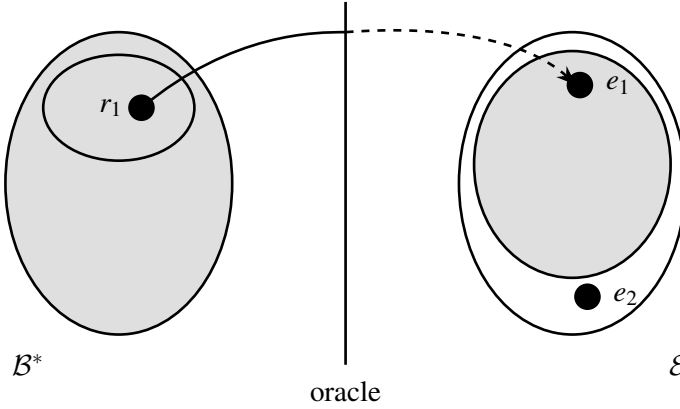


Figure 3.2: Encodings and Entities

representation string. Knowability is therefore always relative to the particular oracle under consideration.

Definition 3.2.3 We say that an entity $e \in \mathcal{E}$ is *oracle knowable* with respect to the oracle $\mathcal{O}_{\mathcal{E}}$ if there exists at least one $r \in \mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^{\top}$ such that $\delta_{\mathcal{O}}(r) = e$. An entity $e \in \mathcal{E}$ is *oracle unknowable* if it is not oracle knowable with respect to $\mathcal{O}_{\mathcal{E}}$.

An entity that is unknowable with respect to one oracle, for example because that oracle's internal reconstruction mechanism or representation style is too weak, may become knowable with respect to another oracle that adopts a different style or has greater reconstructive capacity.

■ **Example 3.4** Consider Example 3.3. The language

$$L_1 = \{a^n b^n \mid n \geq 0\}$$

cannot be recognized by any finite automaton, and therefore it is not decodable under the finite-automaton representation style. However, it can be generated by a context-free grammar, and thus becomes decodable under the pushdown-automaton style.

Similarly, the language

$$L_2 = \{a^n b^n c^n \mid n \geq 0\}$$

is not context-free and therefore cannot be decoded under the pushdown-automaton style, but it can be recognized by a Turing machine. Consequently, it becomes decodable under the Turing-machine representation style. ■

We are interested in the set of all those entities that are knowable, that is, the entities for which there exists at least one representation that decodes the entity.

Definition 3.2.4 Given a representation oracle $\mathcal{O}_\mathcal{E}$ over the set of entities $\mathcal{E}$, we define the set of *oracle-knowable entities* by

$$\mathcal{E}_\mathcal{O}^\top = \left\{ e \in \mathcal{E} \mid \exists r \in \mathcal{R}_\mathcal{O}^\top \text{ such that } \delta_\mathcal{O}(r) = e \right\}.$$

The fact that representations are finite binary strings imposes an immediate cardinality constraint on oracle-mediated knowledge. Even if the oracle is idealized and non-computable, it can only decode countably many representations, and therefore can make at most countably many entities knowable.

Proposition 3.2.1 For any oracle $\mathcal{O}_\mathcal{E}$, the set of oracle-knowable entities $\mathcal{E}_\mathcal{O}^\top$ is countable.

Proof. By definition,

$$\mathcal{E}_\mathcal{O}^\top = \{ \delta_\mathcal{O}(r) \mid r \in \mathcal{R}_\mathcal{O}^\top \}.$$

Since $\mathcal{R}_\mathcal{O}^\top \subseteq \mathcal{B}^*$ and $\mathcal{B}^*$ is countable, the set $\mathcal{R}_\mathcal{O}^\top$ is countable. The image of a countable set under a function is countable. Therefore $\mathcal{E}_\mathcal{O}^\top$ is countable. ■

This limitation has an immediate consequence for sufficiently large domains of inquiry. If the set of entities under consideration is uncountable, then no oracle based on finite representations can make all of them knowable.

Proposition 3.2.2 If the set of entities $\mathcal{E}$ is uncountable, then for any oracle $\mathcal{O}_\mathcal{E}$,

$$\mathcal{E} \setminus \mathcal{E}_\mathcal{O}^\top \neq \emptyset.$$

Proof. By Proposition 3.2.1, the set $\mathcal{E}_\mathcal{O}^\top$ is countable. Since $\mathcal{E}$ is uncountable, $\mathcal{E}_\mathcal{O}^\top$ cannot exhaust $\mathcal{E}$. Hence

$$\mathcal{E} \setminus \mathcal{E}_\mathcal{O}^\top \neq \emptyset.$$

■

■ **Example 3.5** Consider the case in which the entities under study are real numbers, so that $\mathcal{E} = \mathbb{R}$. Since the set of finite binary strings $\mathcal{B}^*$ is countable, any oracle $\mathcal{O}_\mathcal{E}$ can decode at most countably many representations. Consequently, the set of oracle-knowable real numbers

$$\mathcal{E}_\mathcal{O}^\top = \{ \delta_\mathcal{O}(r) \mid r \in \mathcal{R}_\mathcal{O}^\top \}$$

is countable. Because the set of real numbers $\mathbb{R}$ is uncountable, there must exist real numbers that cannot be reconstructed from any finite binary representation under the oracle. ■

Example 3.5 illustrates a fundamental limitation of representation-based knowledge. In domains where the set of possible entities is uncountable, the space of entities necessarily exceeds the space of finite representations available to us. As a consequence, even an idealized oracle cannot make all entities knowable through finite descriptions.

3.3 Minimal Representations

Once a representation oracle $\mathcal{O}$ and the notion of decodable representation have been defined, it becomes possible to compare different representations of a given entity. In general, an entity $e \in \mathcal{E}$ may admit multiple distinct decodable representations, since different binary strings may contain sufficient information for the oracle to reconstruct the same entity. These representations may differ significantly in length, reflecting the presence of symbols that are not strictly required for reconstruction.

From a theoretical perspective, it is therefore natural to identify those representations that encode an entity in the most economical manner. Intuitively, such representations are those for which no shorter decodable representation of the same entity exists under the oracle: if a shorter representation existed that allowed the oracle to reconstruct the same entity, the longer one would not be minimal. We refer to these encodings as *minimal representations*. They form one of the most stringent classes of representations in the theory of nescience, since they require shortest-length decodability for a given entity under a fixed oracle.

Definition 3.3.1 Let $\mathcal{E}$ be a set of entities, and $\mathcal{O}_{\mathcal{E}}$ be an oracle with reconstruction function δ . We say that the decodable representation $r \in \mathcal{R}^{\top}$ is a *minimal* representation if there is no decodable representation $s \in \mathcal{R}^{\top}$ such that

$$l(s) < l(r) \wedge \delta(s) = \delta(r)$$

We denote by $\mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^*$ the set of minimal representations for $\mathcal{E}$.

Because the internal reconstruction mechanism of the oracle is abstract and, in general, inaccessible, it is natural to regard the oracle as a black box and to characterize it solely through its observable input/output behaviour. Within this perspective, the set of minimal representations provides a particularly informative external description of the oracle. It consists of those decodable representations that achieve minimal length for each reconstructible entity, thereby indicating both which entities the oracle can decode and the shortest forms from which they can be recovered.

Notation 3.4. When the set of entities $\mathcal{E}$ is clear from the context we write $\mathcal{R}_{\mathcal{O}}^*$,

and when the oracle is also clear we simply write $\mathcal{R}^*$.

An entity may admit multiple minimal representations, each of equal length and decodable by the oracle. It is therefore convenient to consider the collection of all minimal representations that reconstruct the same entity.

Definition 3.3.2 Let $\mathcal{O}$ be a representation oracle with reconstruction function $\delta : \mathcal{B}^* \rightarrow \mathcal{E}$, and let $e \in \mathcal{E}$ be an entity. We define the *set of minimal representations of e* as

$$\mathcal{R}_e^* = \{r \in \mathcal{R}^* \mid \delta(r) = e\}.$$

In other words, $\mathcal{R}_e^*$ is the set of all minimal representations that decode the entity e under the oracle $\mathcal{O}$.

■ **Example 3.6** Let $\mathcal{E}$ be a set of entities consisting of chemical compounds, and let $\mathcal{O}_{\mathcal{E}}$ be a representation oracle whose reconstruction function δ maps valid molecular encodings to their corresponding molecular structure (e.g., as a labeled graph of atoms and bonds). Assume that the oracle accepts strings written in a non-canonical linear encoding scheme in which any traversal-compatible ordering of atoms is permitted, provided that the resulting string encodes the correct connectivity. In particular, different linearizations of the same molecular graph may yield distinct valid strings. Consider the molecule ethanol, denoted by $e \in \mathcal{E}$. Under such a non-canonical encoding, the same molecule may admit multiple valid representations of equal length. For instance,

$$r_1 = \text{CCO}, \quad r_2 = \text{OCC}.$$

Both strings are valid encodings under the chosen convention and satisfy

$$\delta(r_1) = \delta(r_2) = e.$$

Moreover, they have the same length, reflecting two different admissible linearizations of the same molecular structure. Assuming that no shorter valid encoding exists for this molecule within the given encoding scheme, both r_1 and r_2 are minimal representations of e , that is,

$$r_1, r_2 \in \mathcal{R}_e^*.$$

■

Before studying the properties of minimal representations, it is important to establish that such representations always exist for entities that the oracle can decode. Intuitively, if an entity admits at least one decodable representation, then among all such representations there must be one of minimal length. The following proposition formalizes this observation.

Proposition 3.3.1 Let $\mathcal{O}$ be a representation oracle and let $e \in \mathcal{E}$ be a oracle-knowable entity. Then there exists at least one minimal representation $r \in \mathcal{R}_e^*$.

Proof. Since e is decodable, there exists at least one representation $r_0 \in \mathcal{R}^\top$ such that $\delta(r_0) = e$. Hence the set

$$A_e = \{l(r) : r \in \mathcal{R}^\top, \delta(r) = e\} \subseteq \mathbb{N}$$

is non-empty. Because $\mathbb{N}$ is well ordered, the set A_e admits a minimum element, say m . Choose a representation $r \in \mathcal{R}^\top$ such that $\delta(r) = e$ and $l(r) = m$. By definition of m , there exists no representation $s \in \mathcal{R}^\top$ such that $\delta(s) = e$ and $l(s) < l(r)$. Therefore, r is a minimal representation of e , and thus $r \in \mathcal{R}_e^*$. ■

This result shows that every decodable entity admits at least one minimal representation. In general, however, the minimal representation of an entity need not be unique: several distinct strings may achieve the same minimal length while still allowing the oracle to reconstruct the same entity.

■ **Example 3.7** Let $\mathcal{E}$ be a set of entities consisting of finite undirected graphs with labeled vertices, and let $\mathcal{O}_{\mathcal{E}}$ be a representation oracle whose reconstruction function δ interprets binary strings as adjacency matrices encoded row by row. A string $r \in \mathcal{B}^*$ is decodable if it encodes an integer n (the number of vertices) followed by n^2 bits specifying the adjacency matrix. The oracle ignores any additional trailing bits beyond this structure.

Consider the graph $e \in \mathcal{E}$ consisting of two vertices connected by a single edge. The following strings are all decodable representations of e :

$$r_1 = [n = 2][0110][0000], r_2 = [n = 2][0110][\text{metadata}], r_3 = [n = 2][0110]$$

They satisfy $\delta(r_1) = \delta(r_2) = \delta(r_3) = e$. However, r_1 and r_2 contain additional symbols that are ignored by the reconstruction function, and therefore play no role in determining the entity. Among these representations, r_3 has minimal length. ■

The sets $\mathcal{R}_e^*$ introduced above collect the minimal representations associated with each individual decodable entity. The next proposition shows that these sets exhaust the whole set of minimal representations: every minimal representation belongs to exactly one such class, namely, the class determined by the entity reconstructed by the oracle.

Proposition 3.3.2 Let $\mathcal{E}$ be a set of entities and $\mathcal{O}_{\mathcal{E}}$ be a representation oracle with reconstruction function δ . We have that

$$\mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^* = \bigcup_{e \in \mathcal{E}^\top} \mathcal{R}_e^*.$$

Proof. First, let $r \in \mathcal{R}_{\mathcal{O}_\varepsilon}^*$. Since r is a minimal representation, it is in particular decodable, so $\delta(r)$ is defined. Let $e = \delta(r)$. Then $e \in \mathcal{E}_\mathcal{O}^\top$ by definition of knowable entity. Moreover, since $r \in \mathcal{R}^*$ and $\delta(r) = e$, the definition of $\mathcal{R}_e^*$ gives $r \in \mathcal{R}_e^*$. Hence

$$r \in \bigcup_{e \in \mathcal{E}_\mathcal{O}^\top} \mathcal{R}_e^*,$$

and therefore

$$\mathcal{R}_{\mathcal{O}_\varepsilon}^* \subseteq \bigcup_{e \in \mathcal{E}_\mathcal{O}^\top} \mathcal{R}_e^*.$$

Conversely, let

$$r \in \bigcup_{e \in \mathcal{E}_\mathcal{O}^\top} \mathcal{R}_e^*.$$

Then there exists some $e \in \mathcal{E}_\mathcal{O}^\top$ such that $r \in \mathcal{R}_e^*$. By definition of $\mathcal{R}_e^*$, this means that $r \in \mathcal{R}_{\mathcal{O}_\varepsilon}^*$ and $\delta(r) = e$. In particular, $r \in \mathcal{R}^*$. Therefore

$$\bigcup_{e \in \mathcal{E}_\mathcal{O}^\top} \mathcal{R}_e^* \subseteq \mathcal{R}_{\mathcal{O}_\varepsilon}^*.$$

■

Minimal representations correspond to entities through the reconstruction function of the oracle. It is therefore natural to ask whether a single minimal representation could correspond to more than one entity. The following proposition shows that this cannot occur: the sets of minimal representations associated with different entities are disjoint.

Proposition 3.3.3 Let $e_1, e_2 \in \mathcal{E}$ with $e_1 \neq e_2$. Then

$$\mathcal{R}_{e_1}^* \cap \mathcal{R}_{e_2}^* = \emptyset.$$

Proof. Suppose, for contradiction, that there exists a string $r \in \mathcal{R}_{e_1}^* \cap \mathcal{R}_{e_2}^*$. By definition of $\mathcal{R}_{e_1}^*$ we have that $\delta(r) = e_1$, and by definition of $\mathcal{R}_{e_2}^*$ we have $\delta(r) = e_2$. Since δ is a function on its domain, it assigns a unique value to each $r \in \mathcal{B}^*$ for which it is defined. Hence $e_1 = e_2$, contradicting the assumption that $e_1 \neq e_2$. ■

This proposition shows that the sets $\mathcal{R}_e^*$ form a collection of pairwise disjoint subsets of $\mathcal{R}^*$. Consequently, each minimal representation identifies a unique entity under the oracle, and the reconstruction function δ induces a natural

partition of the set of minimal representations according to the entities they encode.

A natural question is whether the notion of minimal representation depends on the particular coding scheme used to write representations as binary strings. The following proposition shows that minimality is invariant under any computable bijection that merely renames representations without altering their lengths. In this sense, minimal representations are stable under effective recodings of fixed size.

Proposition 3.3.4 Let $\mathcal{O}_\varepsilon$ be a representation oracle with reconstruction function $\delta_{\mathcal{O}}$, and let $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a total computable bijection such that

$$l(f(x)) = l(x) \quad \text{for all } x \in \mathcal{B}^*.$$

Define the transported reconstruction function

$$\delta'_{\mathcal{O}}(x) := \delta_{\mathcal{O}}(f^{-1}(x)).$$

Let $\mathcal{R}_{\mathcal{O}_\varepsilon}^*$ be the set of minimal representations induced by $\delta_{\mathcal{O}}$, and let $\mathcal{R}_{\mathcal{O}'_\varepsilon}^*$ be the corresponding set induced by $\delta'_{\mathcal{O}}$. Then, for every $r \in \mathcal{B}^*$,

$$r \in \mathcal{R}_{\mathcal{O}_\varepsilon}^* \iff f(r) \in \mathcal{R}_{\mathcal{O}'_\varepsilon}^*.$$

Proof. Fix $r \in \mathcal{B}^*$.

First observe that decodability is preserved by f . Indeed, by definition of $\delta'_{\mathcal{O}}$,

$$\delta'_{\mathcal{O}}(f(r)) = \delta_{\mathcal{O}}(f^{-1}(f(r))) = \delta_{\mathcal{O}}(r).$$

Hence $\delta_{\mathcal{O}}(r)$ is defined if and only if $\delta'_{\mathcal{O}}(f(r))$ is defined, and in that case both reconstruct the same entity.

Assume first that

$$r \in \mathcal{R}_{\mathcal{O}_\varepsilon}^*.$$

Then r is a minimal representation, so $\delta_{\mathcal{O}}(r)$ is defined and there exists no decodable representation $s \in \mathcal{B}^*$ such that

$$l(s) < l(r) \quad \text{and} \quad \delta_{\mathcal{O}}(s) = \delta_{\mathcal{O}}(r).$$

Suppose, for contradiction, that $f(r)$ is not minimal with respect to $\delta'_{\mathcal{O}}$. Then there exists some $t \in \mathcal{B}^*$ such that

$$l(t) < l(f(r)) \quad \text{and} \quad \delta'_{\mathcal{O}}(t) = \delta'_{\mathcal{O}}(f(r)).$$

Let $s = f^{-1}(t)$. Since f is a bijection, $s \in \mathcal{B}^*$, and by the length-preserving property,

$$l(s) = l(f(s)) = l(t) < l(f(r)) = l(r).$$

Moreover,

$$\delta_{\mathcal{O}}(s) = \delta_{\mathcal{O}}(f^{-1}(t)) = \delta'_{\mathcal{E}}(t) = \delta'_{\mathcal{E}}(f(r)) = \delta_{\mathcal{O}}(r).$$

Thus s is a shorter decodable representation of the same entity as r , contradicting the minimality of r . Therefore $f(r)$ is minimal with respect to $\delta'_{\mathcal{E}}$.

Conversely, assume that

$$f(r) \in \mathcal{R}_{\mathcal{O}'_{\mathcal{E}}}^*.$$

By the same argument applied to the bijection f^{-1} , which is also total, computable, bijective, and length-preserving, it follows that r is minimal with respect to $\delta_{\mathcal{O}}$.

Hence

$$r \in \mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^* \iff f(r) \in \mathcal{R}_{\mathcal{O}'_{\mathcal{E}}}^*.$$

■

This proposition shows that minimality is not tied to a particular binary coding, but only to the reconstruction structure induced by the oracle and to the relative lengths of equivalent encodings. In particular, any effective relabeling that preserves length leaves the class of minimal representations unchanged up to the induced bijection.

Finally, it is useful to isolate the smallest amount of algorithmic information that any decodable representation of e must contain. This quantity provides a lower bound on the irreducible descriptive complexity of the entity relative to the decodability criterion.

Definition 3.3.3 Let $e \in \mathcal{E}_{\mathcal{O}}^{\top}$ be an entity and $\mathcal{O}_{\mathcal{E}}$ be a representational oracle. We define the *representational complexity floor* of e given the oracle $\mathcal{O}_{\mathcal{E}}$ as

$$\kappa_e = \min_{r \in \mathcal{R}_e^{\top}} K(r).$$

The representational complexity floor measures the least Kolmogorov complexity among all decodable representations of e . In this sense, it captures the minimum amount of algorithmic information required to represent e .

3.4 Minimal Oracles

Ideally, representation oracles should be minimal in size, meaning that all information needed to reconstruct an entity is carried by the representation string rather than being hard-wired into the oracle. A non-minimal oracle can simplify the representation problem by offloading complexity to its internal representation function, but it also forces researchers to understand not only the representation itself but the oracle's internal reconstruction process, something that, given the abstract nature of oracles, is generally not accessible in practice.

Definition 3.4.1 Let $\mathcal{O}_A$ and $\mathcal{O}_B$ be two oracles defined over the set of entities $\mathcal{E}$. We say are said that the oracles $\mathcal{O}_A$ and $\mathcal{O}_B$ are *comparable*, denoted by $\mathcal{O}_A \sim \mathcal{O}_B$, if

$$\mathcal{E}_{\mathcal{O}_A}^\top = \mathcal{E}_{\mathcal{O}_B}^\top.$$

The comparability relation defined above does more than merely identify pairs of oracles with the same knowable-entity set: it partitions the entire class $\mathfrak{D}_{\mathcal{E}}$ into structurally homogeneous blocks. Within each block, all oracles agree on which entities are decodable, and differ only in how those entities are represented or reconstructed.

Proposition 3.4.1 The relation $\sim$ is an equivalence relation on $\mathfrak{D}_{\mathcal{E}}$.

Proof. Reflexivity is immediate since $\mathcal{E}_{\mathcal{O}}^\top = \mathcal{E}_{\mathcal{O}}^\top$. Symmetry holds because equality is symmetric. Transitivity holds because if $\mathcal{E}_{\mathcal{O}_A}^\top = \mathcal{E}_{\mathcal{O}_B}^\top$ and $\mathcal{E}_{\mathcal{O}_B}^\top = \mathcal{E}_{\mathcal{O}_C}^\top$, then $\mathcal{E}_{\mathcal{O}_A}^\top = \mathcal{E}_{\mathcal{O}_C}^\top$. ■

The equivalence relation $\sim$ partitions the space of oracles into disjoint classes, each class collecting all oracles that make exactly the same entities decodable. The comparability class of an oracle thus isolates differences in representation mechanisms and internal structure while holding fixed the underlying set of decodable entities.

Definition 3.4.2 Let $\mathcal{O} \in \mathfrak{D}_{\mathcal{E}}$ be an oracle. We define the comparability class of $\mathcal{O}$, denoted by $[\mathcal{O}]$, as follows:

$$[\mathcal{O}] = \{\mathcal{O}' \in \mathfrak{D}_{\mathcal{E}} \mid \mathcal{O}' \sim \mathcal{O}\}$$

When an oracle contains hard-wired information, it can reconstruct an entity from a representation that leaves part of that information implicit, representations that an oracle without that hard-wired knowledge would be unable to decode. This property provides a natural basis for ordering oracles: we compare the sets of distinct minimal representations they accept, since an oracle with more hard-wired information admits a larger set of minimal decodable representations.

Definition 3.4.3 Let $\mathcal{O}_A$ and $\mathcal{O}_B$ be two comparable oracles. We say that $\mathcal{O}_A$ *reduces* to $\mathcal{O}_B$, denoted by $\mathcal{O}_A \triangleright \mathcal{O}_B$, if

$$\left(\forall e \in \mathcal{E}_{\mathcal{O}_A}^\top, \mathcal{R}_{\mathcal{O}_B, e}^* \subseteq \mathcal{R}_{\mathcal{O}_A, e}^* \right) \wedge \left(\exists e \in \mathcal{E}_{\mathcal{O}_A}^\top, \mathcal{R}_{\mathcal{O}_B, e}^* \subsetneq \mathcal{R}_{\mathcal{O}_A, e}^* \right).$$

An oracle is less minimal only when its internal knowledge makes it possible to reconstruct an entity from a strictly shorter encoding that omits information which another comparable oracle would require explicitly.

■ **Example 3.8** Consider representations of chemical compounds. A non-minimal oracle may incorporate extensive background chemical knowledge (e.g., periodic table, standard valences, and common structures). Under such an oracle, a compound could be represented by a short string such as its common name (e.g., *water*, *ethanol*), from which the oracle reconstructs the full molecular structure. In this case, most of the information required for reconstruction is implicit in the oracle rather than explicit in the representation. By contrast, a more minimal oracle requires the representation to encode the relevant structure directly, for instance by specifying atoms and bonds. ■

The binary relation of reduction defines a non-strict partial order over the sets of all comparable oracles as next proposition shows.

Proposition 3.4.2 Let $\mathfrak{O}_{\mathcal{E}}$ be the set of comparable oracles over the set of entities $\mathcal{E}$. We have that the reduction relation $\triangleright$ is:

- 1 *irreflexive*: $\neg(\mathcal{O} \triangleright \mathcal{O})$ for all $\mathcal{O} \in \mathfrak{O}_{\mathcal{E}}$,
- 2 *transitive*: if $\mathcal{O}_A \triangleright \mathcal{O}_B$ and $\mathcal{O}_B \triangleright \mathcal{O}_C$, then $\mathcal{O}_A \triangleright \mathcal{O}_C$, for all $\mathcal{O}_A, \mathcal{O}_B, \mathcal{O}_C \in \mathfrak{O}_{\mathcal{E}}$

Proof. Irreflexivity follows from the fact that no set is a strict subset of itself. Transitivity follows from transitivity of strict set inclusion. ■

While the relation $\triangleright$ is transitive and irreflexive, it is natural to ask whether it induces a total ordering on the class of comparable oracles. The following proposition shows that this is not the case: even when two oracles decode exactly the same entities, they may embody different and incomparable forms of hard-wired information.

Proposition 3.4.3 It is not the case that the reduction relation $\triangleright$ is total on $\mathfrak{O}_{\mathcal{E}}$. That is, there exist comparable oracles $\mathcal{O}_A, \mathcal{O}_B \in \mathfrak{O}_{\mathcal{E}}$ such that neither $\mathcal{O}_A \triangleright \mathcal{O}_B$ nor $\mathcal{O}_B \triangleright \mathcal{O}_A$ holds.

Proof. We exhibit two comparable oracles whose sets of minimal representations are incomparable under strict inclusion.

Let $\mathcal{E} = \{e_0, e_1, e_2\}$ and fix an alphabet $\mathcal{B}$ containing distinct symbols a, b, c . Define two reconstruction functions $\delta_A, \delta_B : \mathcal{B}^* \rightarrow \mathcal{E}$ by specifying their (minimal)

decodable representations as follows:

$$\delta_A(a) = e_0, \quad \delta_A(b) = e_1, \quad \delta_A(cc) = e_2,$$

$$\delta_B(aa) = e_0, \quad \delta_B(b) = e_1, \quad \delta_B(c) = e_2,$$

and leave both functions undefined on all other strings. Let $\mathcal{O}_A$ and $\mathcal{O}_B$ be the corresponding representation oracles.

By construction, both oracles decode exactly the same set of entities $\mathcal{E}$, and hence $\mathcal{O}_A, \mathcal{O}_B \in \mathfrak{D}_{\mathcal{E}}$. Moreover, their sets of minimal representations are

$$\mathcal{R}_{\mathcal{O}_A}^{*,0} = \{a, b, cc\}, \quad \mathcal{R}_{\mathcal{O}_B}^{*,0} = \{aa, b, c\}.$$

These two sets are incomparable under strict inclusion: we have $cc \in \mathcal{R}_{\mathcal{O}_A}^{*,0}$ but $cc \notin \mathcal{R}_{\mathcal{O}_B}^{*,0}$, and also $c \in \mathcal{R}_{\mathcal{O}_B}^{*,0}$ but $c \notin \mathcal{R}_{\mathcal{O}_A}^{*,0}$. Therefore neither $\mathcal{R}_{\mathcal{O}_B}^{*,0} \subsetneq \mathcal{R}_{\mathcal{O}_A}^{*,0}$ nor $\mathcal{R}_{\mathcal{O}_A}^{*,0} \subsetneq \mathcal{R}_{\mathcal{O}_B}^{*,0}$ holds.

By definition of the reduction relation $\triangleright$, this implies that neither $\mathcal{O}_A \triangleright \mathcal{O}_B$ nor $\mathcal{O}_B \triangleright \mathcal{O}_A$ holds. Hence $\triangleright$ is not a total relation on $\mathfrak{D}_{\mathcal{E}}$. ■

This proposition shows that even when the set of knowable entities is held fixed, there may be multiple, genuinely different ways of distributing information between representations and the oracle. Different oracles may hard-wire different aspects of the same entities, leading to incomparable minimal representation sets. As a consequence, simplification with respect to $\triangleright$ is not a linear process but a partial ordering, and minimal oracles, when they exist, need not be unique.

The ordering introduced by the reduction relation allow us to formally introduce the concept of minimality: an oracle is minimal if it cannot be reduced to any other comparable oracle.

Definition 3.4.4 Let $\mathcal{O} \in \mathfrak{D}_{\mathcal{E}}$ be an oracle. We say that $\mathcal{O}$ is *minimal* if

$$\nexists \mathcal{O}' \in [\mathcal{O}] : \mathcal{O} \triangleright \mathcal{O}'.$$

Equivalently, the oracle $\mathcal{O}$ is minimal if there is no $\mathcal{O}' \in [\mathcal{O}]$ such that

$$\left(\forall e \in \mathcal{E}_{\mathcal{O}}^{\top}, \mathcal{R}_{\mathcal{O}',e}^{*} \subseteq \mathcal{R}_{\mathcal{O},e}^{*} \right) \wedge \left(\exists e \in \mathcal{E}_{\mathcal{O}}^{\top}, \mathcal{R}_{\mathcal{O}',e}^{*} \subsetneq \mathcal{R}_{\mathcal{O},e}^{*} \right).$$

The notion of minimal oracle formalizes the idea of an oracle that contains no unnecessary hard-wired information relative to a fixed representation style and a fixed set of knowable entities. Recall that a reduction $\mathcal{O} \triangleright \mathcal{O}'$ corresponds to eliminating part of the oracle's internalized assumptions while preserving the same epistemic reach, a change that is reflected in a strict reduction of the set of minimal decodable representations. An oracle is therefore minimal precisely when no further such elimination is possible within its comparability class.

This does not mean that a minimal oracle is devoid of internal structure or background knowledge; rather, it means that any remaining hard-wired information is irreducible given the chosen representation style. In this sense, minimal oracles capture the simplest reconstruction mechanisms compatible with that style, and they formalize the intuition of oracles that do not rely on avoidable internalized assumptions.

Definition 3.4.5 Let $\mathcal{E}$ be a set of entities and let $\mathfrak{D}_{\mathcal{E}}$ be the class of representation oracles over $\mathcal{E}$. We define the set of *minimal oracles* as

$$\mathfrak{D}_{\mathcal{E}}^{\min} = \{ \mathcal{O} \in \mathfrak{D}_{\mathcal{E}} \mid \nexists \mathcal{O}' \in [\mathcal{O}] \text{ such that } \mathcal{O} \triangleright \mathcal{O}' \}.$$

The class $\mathfrak{D}_{\mathcal{E}}^{\min}$ collects those oracles that are minimal with respect to the reduction relation within their comparability class.

The following proposition provides an alternative characterization of minimal oracles in terms of minimal elements under pointwise set inclusion.

Proposition 3.4.4 Let $\mathcal{E}$ be a set of entities and let $\mathcal{O} \in \mathfrak{D}_{\mathcal{E}}$. Then $\mathcal{O}$ is minimal if and only if the family

$$\{ \mathcal{R}_{\mathcal{O},e}^* \}_{e \in \mathcal{E}_{\mathcal{O}}^{\top}}$$

is a minimal element, under pointwise set inclusion, among the families

$$\left\{ \{ \mathcal{R}_{\mathcal{O}',e}^* \}_{e \in \mathcal{E}_{\mathcal{O}}^{\top}} : \mathcal{O}' \in [\mathcal{O}] \right\}.$$

Proof. By Definition 3.4.4, the oracle $\mathcal{O}$ is minimal if and only if there is no $\mathcal{O}' \in [\mathcal{O}]$ such that $\mathcal{O} \triangleright \mathcal{O}'$. By the definition of the reduction relation, this is equivalent to the non-existence of $\mathcal{O}' \in [\mathcal{O}]$ such that

$$\left(\forall e \in \mathcal{E}_{\mathcal{O}}^{\top}, \mathcal{R}_{\mathcal{O}',e}^* \subseteq \mathcal{R}_{\mathcal{O},e}^* \right) \wedge \left(\exists e \in \mathcal{E}_{\mathcal{O}}^{\top}, \mathcal{R}_{\mathcal{O}',e}^* \subsetneq \mathcal{R}_{\mathcal{O},e}^* \right).$$

This condition is precisely the statement that the family $\{ \mathcal{R}_{\mathcal{O},e}^* \}_{e \in \mathcal{E}_{\mathcal{O}}^{\top}}$ is minimal under pointwise set inclusion among the corresponding families induced by oracles in $[\mathcal{O}]$. ■

Minimal oracles need not be unique. It may happen that two distinct oracles $\mathcal{O}_A, \mathcal{O}_B \in \mathfrak{D}_{\mathcal{E}}$ are both minimal, while their sets of minimal representations $\mathcal{R}_{\mathcal{O}_A}^*$ and $\mathcal{R}_{\mathcal{O}_B}^*$ are incomparable under set inclusion. In such a case, neither oracle can be reduced to the other, and both appear as minimal elements of the partial order induced by the reduction relation.

The existence of minimal oracles depends on the size of the subset of decodable entities and on the amount of freedom allowed in the representation mechanism.

Proposition 3.4.5 Let $\mathcal{O} \in \mathfrak{D}_{\mathcal{E}}$ and let $S := \mathcal{E}_{\mathcal{O}}^{\top}$ be finite. Then there exists at least one minimal oracle in the comparability class $[\mathcal{O}]$ with respect to the reduction relation $\triangleright$.

Proof. Fix $\mathcal{O}' \in [\mathcal{O}]$ and $e \in S$. By definition of minimal representations,

$$\mathcal{R}_{\mathcal{O}',e}^* = \{r \in \mathcal{R}_{\mathcal{O}'}^{\top} : \delta_{\mathcal{O}'}(r) = e, l(r) = L_{\mathcal{O}'}(e)\},$$

where

$$L_{\mathcal{O}'}(e) = \min\{l(r) : \delta_{\mathcal{O}'}(r) = e\}.$$

Hence $\mathcal{R}_{\mathcal{O}',e}^* \subseteq \mathcal{B}^{L_{\mathcal{O}'}(e)}$, and therefore $\mathcal{R}_{\mathcal{O}',e}^*$ is finite.

Since S is finite, the family

$$\{\mathcal{R}_{\mathcal{O}',e}^*\}_{e \in S}$$

consists of finitely many finite sets.

Consider the collection

$$\mathcal{F} := \{\{\mathcal{R}_{\mathcal{O}',e}^*\}_{e \in S} : \mathcal{O}' \in [\mathcal{O}]\}$$

ordered by the strict pointwise relation induced by $\triangleright$, that is,

$$\{A_e\}_{e \in S} \prec \{B_e\}_{e \in S} \iff (\forall e \in S, A_e \subseteq B_e) \wedge (\exists e \in S, A_e \subsetneq B_e).$$

We claim that there is no infinite strictly descending chain in $\mathcal{F}$ under $\prec$. Indeed, for each family $\{A_e\}_{e \in S}$ define the measure

$$\mu(\{A_e\}_{e \in S}) := \sum_{e \in S} |A_e|.$$

Each strict step $\{A_e\} \succ \{B_e\}$ decreases μ by at least one, since there exists some e with $B_e \subsetneq A_e$. As μ takes values in $\mathbb{N}$, no infinite strictly descending chain exists.

Therefore, $\mathcal{F}$ admits at least one minimal element. Let

$$\{\mathcal{R}_{\mathcal{O}^*,e}^*\}_{e \in S}$$

be such a minimal element. By construction, there is no $\mathcal{O}' \in [\mathcal{O}]$ such that $\mathcal{O}^* \triangleright \mathcal{O}'$. Hence $\mathcal{O}^*$ is minimal. ■

When the set of decodable entities is infinite, minimality is no longer guaranteed. Even though each individual entity $e \in S$ still admits at least one minimal representation, an oracle may allow multiple minimal representations for infinitely many entities. In that situation, one can construct an infinite strictly descending chain of oracles by successively removing some of these additional minimal representations while preserving at least one minimal representation for every entity.

Proposition 3.4.6 It is not the case that, for every infinite set of entities $S \subseteq \mathcal{E}$, the comparability class of oracles decoding S contains a $\triangleright$ -minimal oracle.

Proof. Let $S = \{e_1, e_2, e_3, \dots\}$ be an infinite set of entities. For each $n \geq 1$, define an oracle $\mathcal{O}_n$ with reconstruction function $\delta_n : \mathcal{B}^* \rightarrow S$ by setting, for every $k \geq 1$,

$$\delta_n(1^k) = e_k, \quad \delta_n(0^{k-1}1) = e_k \quad \text{for all } k \geq n,$$

and leaving $\delta_n(r)$ undefined for all other strings r .

By construction, each oracle $\mathcal{O}_n$ decodes exactly the set S , and therefore all $\mathcal{O}_n$ belong to the same comparability class.

For each $k \geq 1$, the shortest length of a decodable representation of e_k under $\mathcal{O}_n$ is k , since $\delta_n(1^k) = e_k$ and no shorter string decodes to e_k . Hence,

$$\mathcal{R}_{\mathcal{O}_n, e_k}^* = \begin{cases} \{1^k\}, & k < n, \\ \{1^k, 0^{k-1}1\}, & k \geq n. \end{cases}$$

Fix $n \geq 1$. Then for every $k \geq 1$ we have

$$\mathcal{R}_{\mathcal{O}_{n+1}, e_k}^* \subseteq \mathcal{R}_{\mathcal{O}_n, e_k}^*,$$

and for $k = n$ the inclusion is strict:

$$\mathcal{R}_{\mathcal{O}_{n+1}, e_n}^* = \{1^n\} \subsetneq \{1^n, 0^{n-1}1\} = \mathcal{R}_{\mathcal{O}_n, e_n}^*.$$

Therefore,

$$\mathcal{O}_n \triangleright \mathcal{O}_{n+1} \quad \text{for all } n \geq 1.$$

This yields an infinite strictly descending chain

$$\mathcal{O}_1 \triangleright \mathcal{O}_2 \triangleright \mathcal{O}_3 \triangleright \dots$$

Consequently, no oracle in this family is minimal with respect to $\triangleright$. ■

The previous proposition shows that minimal oracles need not exist when the set of decodable entities is infinite. The obstruction arises when an oracle admits additional minimal representations for infinitely many entities, making it possible to construct an infinite strictly descending chain under the reduction relation. The next proposition gives a simple sufficient condition ensuring existence: if, within a comparability class, the total number of additional minimal representations beyond one per entity is always finite, then at least one minimal oracle must exist.

Proposition 3.4.7 Let $\mathcal{O} \in \mathfrak{D}_{\mathcal{E}}$, and let $S := \mathcal{E}_{\mathcal{O}}^{\top}$, which may be infinite. Assume that for every oracle $\mathcal{O}' \in [\mathcal{O}]$, the quantity

$$\mu(\mathcal{O}') := \sum_{e \in S} (|\mathcal{R}_{\mathcal{O}',e}^*| - 1)$$

is finite. Then there exists at least one minimal oracle in the comparability class $[\mathcal{O}]$ with respect to the reduction relation $\triangleright$.

Proof. Since $\mathcal{O} \in [\mathcal{O}]$, the comparability class $[\mathcal{O}]$ is nonempty. By hypothesis, for every $\mathcal{O}' \in [\mathcal{O}]$, the value $\mu(\mathcal{O}')$ is a finite natural number. Therefore the set

$$M := \{\mu(\mathcal{O}') : \mathcal{O}' \in [\mathcal{O}]\} \subseteq \mathbb{N}$$

is nonempty, and hence admits a least element. Choose an oracle $\mathcal{O}^* \in [\mathcal{O}]$ such that

$$\mu(\mathcal{O}^*) = \min M.$$

We claim that $\mathcal{O}^*$ is minimal. Suppose, for contradiction, that it is not. Then there exists an oracle $\mathcal{O}' \in [\mathcal{O}]$ such that

$$\mathcal{O}^* \triangleright \mathcal{O}'.$$

By definition of the reduction relation, this means that

$$\forall e \in S, \quad \mathcal{R}_{\mathcal{O}',e}^* \subseteq \mathcal{R}_{\mathcal{O}^*,e}^*,$$

and

$$\exists e_0 \in S, \quad \mathcal{R}_{\mathcal{O}',e_0}^* \subsetneq \mathcal{R}_{\mathcal{O}^*,e_0}^*.$$

Since strict inclusion of finite sets strictly decreases cardinality, we obtain

$$|\mathcal{R}_{\mathcal{O}',e}^*| \leq |\mathcal{R}_{\mathcal{O}^*,e}^*| \quad \forall e \in S,$$

and

$$|\mathcal{R}_{\mathcal{O}',e_0}^*| < |\mathcal{R}_{\mathcal{O}^*,e_0}^*|.$$

Subtracting 1 from each side preserves these inequalities, so

$$|\mathcal{R}_{\mathcal{O}',e}^*| - 1 \leq |\mathcal{R}_{\mathcal{O}^*,e}^*| - 1 \quad \forall e \in S,$$

with strict inequality for $e = e_0$. Summing over all $e \in S$, and using the finiteness of both sums, yields

$$\mu(\mathcal{O}') < \mu(\mathcal{O}^*),$$

contradicting the minimality of $\mu(\mathcal{O}^*)$.

Therefore no such oracle $\mathcal{O}'$ exists, and $\mathcal{O}^*$ is minimal. ■

This proposition shows that the failure of minimality in the infinite case is not caused merely by the infinitude of the decodable set. What matters is whether the comparability class contains an unbounded supply of removable minimal representations. If the total excess of minimal representations remains finite for every oracle in the class, then the reduction process cannot continue indefinitely, and a minimal oracle must exist.

In some domains with composed by an infinite collection of entities, such as mathematics, scientific knowledge can exhibit a form of unbounded refinement. As inquiry progresses and finer distinctions become relevant, representations may need to become increasingly detailed (and thus longer), while at the same time the set of admissible minimal representations becomes more constrained. In such settings there may be no final, irreducible level of representation, so minimality should be understood as a limiting ideal rather than a reachable endpoint.

■ **Example 3.9** A paradigmatic example of an entity that requires infinite refinement of its representations is Chaitin's halting probability Ω . Fix a universal prefix-free Turing machine U , and define

$$\Omega = \sum_{p \text{ halts}} 2^{-|p|},$$

where the sum ranges over all binary programs p that halt on U . The number Ω is a well-defined real number in $(0, 1)$, but it is known to be non-computable and algorithmically random. In the present framework, Ω may be included as an entity $e_\Omega \in \mathcal{E}$, but its reconstruction places extreme demands on representations.

Consider a representation style in which representations are finite binary strings encoding partial information about Ω , for example its first n binary digits. A minimal oracle $\mathcal{O}_\mathcal{E}$ operating under this style does not contain any hard-wired information about the halting behavior of programs; it can only reconstruct what is explicitly provided by the string. For each n , there exist representations $r_n \in \mathcal{B}^*$ such that $\delta_\mathcal{O}(r_n)$ corresponds to an n -bit approximation of Ω , but no finite string r allows the oracle to reconstruct Ω exactly. Any improvement in accuracy requires a strictly longer representation, yielding an infinite chain

$$r_1 \prec r_2 \prec r_3 \prec \cdots$$

of minimal representations, with no maximal element.

The case of Ω illustrates a situation in which an entity is, in principle, part of scientific or mathematical discourse, yet resists complete reconstruction under any minimal oracle constrained to finite representations. Knowledge of Ω advances only through unbounded refinement: each additional bit of information requires new, irreducible representational content. In this sense, Ω exemplifies an entity for which minimality is attainable only as a limiting ideal, not as a concrete

endpoint, and it sharply delineates the boundary between infinite refinement of representations and full oracle-relative knowability. ■

The non-existence of minimal oracles in open-ended domains should not be interpreted as a limitation of scientific inquiry, but as an expression of its inherently exploratory character: progress consists in successive local refinements rather than in convergence to a final, irreducible description.

A natural question is whether there exists a single best oracle, in the sense of decoding every entity that is decodable under some oracle. If such an oracle existed, one could fix it once and for all and declare an entity knowable precisely when it is decodable with respect to it. The following theorem shows that this is not the case in general.

Theorem 3.4.8 Let $\mathcal{E}$ be a set of entities and let $\mathfrak{D}_{\mathcal{E}}$ be the class of representation oracles over $\mathcal{E}$. Then it is not the case that $\mathfrak{D}_{\mathcal{E}}$ always contains an oracle $\mathcal{O}^{\max}$ such that

$$\mathcal{E}_{\mathcal{O}}^{\top} \subseteq \mathcal{E}_{\mathcal{O}^{\max}}^{\top} \quad \forall \mathcal{O} \in \mathfrak{D}_{\mathcal{E}},$$

and $\mathcal{O}^{\max}$ is minimal in its comparability class $[\mathcal{O}^{\max}]$.

Proof. We construct a family of oracles with mutually incompatible representation styles.

Let $\mathcal{E} = \{e_1, e_2, e_3, \dots\}$ be a countable set of entities. For each $i \geq 1$, define an oracle $\mathcal{O}_i$ with reconstruction function $\delta_i : \mathcal{B}^* \rightarrow \mathcal{E}$ as follows:

$$\delta_i(i^k) = e_k \quad \text{for all } k \geq 1,$$

and leave $\delta_i(r)$ undefined for all other strings r .

Each oracle $\mathcal{O}_i$ decodes the same set of entities,

$$\mathcal{E}_{\mathcal{O}_i}^{\top} = \mathcal{E},$$

but uses a distinct encoding convention based on repetitions of the symbol i .

Now suppose, for contradiction, that there exists an oracle $\mathcal{O}^{\max} \in \mathfrak{D}_{\mathcal{E}}$ such that

$$\mathcal{E}_{\mathcal{O}_i}^{\top} \subseteq \mathcal{E}_{\mathcal{O}^{\max}}^{\top} \quad \forall i \geq 1,$$

and that $\mathcal{O}^{\max}$ is minimal in its comparability class.

Since each $\mathcal{O}_i$ decodes all entities in $\mathcal{E}$, the oracle $\mathcal{O}^{\max}$ must decode every e_k from some representation. In particular, for each i and each k , there must exist a string $r_{i,k}$ such that

$$\delta_{\mathcal{O}^{\max}}(r_{i,k}) = e_k.$$

However, the encodings used by different oracles $\mathcal{O}_i$ are mutually incompatible: the strings i^k carry no information about the encoding scheme used by j^k for $j \neq i$. In order for $\mathcal{O}^{\max}$ to decode all such representations, it must incorporate additional internal information specifying how to interpret each encoding convention.

This implies that $\mathcal{O}^{\max}$ admits multiple minimal representations for infinitely many entities, corresponding to the different encoding schemes. By the construction of infinite descending chains of oracles obtained by removing such redundant minimal representations (cf. Proposition 3.4.6), it follows that $\mathcal{O}^{\max}$ cannot be minimal in its comparability class.

This contradicts the assumption. Therefore, no such oracle $\mathcal{O}^{\max}$ exists in general. ■

Theorem 3.4.8 formalizes the fact that oracle-knowability cannot, in general, be reduced to knowability with respect to a single privileged oracle. The obstruction is not computational but representational: different oracles may rely on incompatible encoding conventions, and an oracle that decodes all of them must internalize additional information to bridge these conventions. As a consequence, maximal expressive power and minimality are, in general, incompatible requirements unless a representation style is fixed in advance.

■ **Example 3.10** Let $\mathcal{E} = \{e_1, e_2, e_3, \dots\}$ be a countable set of entities, and fix a representation style over the binary alphabet $\mathcal{B} = \{0, 1\}$ in which strings of the form 1^k encode entities.

Consider the class of oracles $\mathfrak{D}$ whose reconstruction functions $\delta : \mathcal{B}^* \rightarrow \mathcal{E}$ satisfy

$$\delta(1^k) = e_k \quad \text{for all } k \geq 1,$$

and may additionally decode other strings in $\mathcal{B}^*$ according to the same representation style.

All such oracles are comparable, since they decode the same set of entities,

$$\mathcal{E}_{\mathcal{O}}^{\top} = \mathcal{E} \quad \text{for all } \mathcal{O} \in \mathfrak{D}.$$

Define an oracle $\mathcal{O}^{\max} \in \mathfrak{D}$ by setting

$$\delta_{\mathcal{O}^{\max}}(r) = e_k \quad \text{whenever } r \in \mathcal{B}^* \text{ is any valid encoding of } e_k$$

under the fixed representation style, that is, $\mathcal{O}^{\max}$ decodes every representation that any oracle in $\mathfrak{D}$ decodes.

Then, for every $\mathcal{O} \in \mathfrak{D}$, we have

$$\mathcal{E}_{\mathcal{O}}^{\top} \subseteq \mathcal{E}_{\mathcal{O}^{\max}}^{\top},$$

and moreover,

$$\mathcal{R}_{\mathcal{O},e}^\top \subseteq \mathcal{R}_{\mathcal{O}^\text{max},e}^\top \quad \forall e \in \mathcal{E}.$$

Thus, $\mathcal{O}^\text{max}$ is a maximal oracle within this comparability class.

However, $\mathcal{O}^\text{max}$ is typically not minimal: by decoding all admissible representations, it incorporates additional oracle-internal information that allows reconstruction from incomplete encodings. This illustrates that maximality and minimality are, in general, competing properties even within a fixed representation style. ■

3.5 Unknowable Entities

In Section 3.2 we introduced the notion of an *oracle-knowable entity*. Given a representation oracle $\mathcal{O}_\mathcal{E}$, an entity $e \in \mathcal{E}$ is said to be oracle knowable if there exists a decodable representation r such that $\delta_\mathcal{O}(r) = e$. In other words, the oracle is capable of reconstructing the entity from at least one finite binary representation. Oracle knowability is therefore always relative to a specific oracle and to the representation style that the oracle implements.

In many situations, however, we are interested in a more intrinsic notion: whether an entity is knowable in principle, beyond the choice of any single representation oracle. A natural first idea would be to declare an entity knowable if there exists *some* oracle capable of reconstructing it. However, if we allow arbitrary oracles, this notion becomes too permissive. An oracle could trivially encode large amounts of information internally and reconstruct entities without the representation itself containing the necessary information. In such cases, the representation would merely act as an index into the oracle's internal knowledge, rather than as a genuine encoding of the entity.

To avoid this trivialization, we restrict attention to the set of *minimal representation oracles* introduced in Section 3.4. Minimal oracles ensure that the information required to reconstruct an entity must be contained in the representation itself, rather than being implicitly supplied by the oracle. This restriction reflects the epistemological principle that scientific knowledge should reside in explicit representations rather than in hidden assumptions.

We may now generalize the notion of oracle knowability.

Definition 3.5.1 Let $\mathfrak{O}_\mathcal{E}^\text{min}$ denote the set of minimal representation oracles over the set of entities $\mathcal{E}$. We say that an entity $e \in \mathcal{E}$ is *knowable* if there exists a minimal oracle $\mathcal{O}_\mathcal{E} \in \mathfrak{O}_\mathcal{E}^\text{min}$ such that

$$\exists r \in \mathcal{R}_{\mathcal{O}_\mathcal{E}}^\top \quad \text{with} \quad \delta_{\mathcal{O}}(r) = e.$$

■ We say that an entity $e \in \mathcal{E}$ is *unknowable* if it is not knowable.

Thus, knowability is defined relative to the set of admissible (minimal) oracles, rather than to any particular oracle. Intuitively, knowable entities are those that admit at least one finite representation containing sufficient information for their reconstruction under some admissible representation scheme. Unknowable entities, by contrast, are those for which no such representation exists within this framework.

It is convenient to collect all knowable entities into a single set.

Definition 3.5.2 We define the set of *knowable entities* as

$$\mathcal{E}^\top = \bigcup_{\mathcal{O} \in \mathcal{D}_{\mathcal{E}}^{\min}} \mathcal{E}_{\mathcal{O}}^\top.$$

The complement of this set corresponds to the entities that remain beyond the reach of all minimal representation oracles.

Definition 3.5.3 We define the set of *unknowable entities* as

$$\mathcal{E}^\perp = \mathcal{E} \setminus \mathcal{E}^\top.$$

By construction, the sets $\mathcal{E}^\top$ and $\mathcal{E}^\perp$ form a partition of $\mathcal{E}$.

Proposition 3.5.1

$$\mathcal{E}^\top \cup \mathcal{E}^\perp = \mathcal{E}, \quad \mathcal{E}^\top \cap \mathcal{E}^\perp = \emptyset.$$

Proof. The result follows immediately from the definition $\mathcal{E}^\perp = \mathcal{E} \setminus \mathcal{E}^\top$. ■

The use of finite representations imposes structural constraints on the size of the set of knowable entities. The following result makes this precise under a natural restriction on the admissible class of oracles.

Proposition 3.5.2 If the set $\mathcal{D}_{\mathcal{E}}^{\min}$ of admissible minimal representation oracles is countable, then the set of knowable entities $\mathcal{E}^\top$ is countable.

Proof. For any oracle $\mathcal{O}$, the set $\mathcal{E}_{\mathcal{O}}^\top$ is countable because it is the image of the countable set $\mathcal{R}_{\mathcal{O}}^\top \subseteq \mathcal{B}^*$. If the class $\mathcal{D}_{\mathcal{E}}^{\min}$ is countable, then $\mathcal{E}^\top$ is a countable union of countable sets, and is therefore countable. ■

In particular, under this assumption, if $\mathcal{E}$ is uncountable then there necessarily exist entities that are not knowable within this framework.

It is important to emphasize that the distinction between knowable and unknowable entities is primarily conceptual and framework dependent. In practice, it is rarely possible to determine in advance whether a given entity belongs to $\mathcal{E}^\top$ or $\mathcal{E}^\perp$. Scientific inquiry proceeds instead through the exploration

and refinement of candidate representations, gradually expanding the subset of entities for which effective and reliable representations are known. In this sense, the boundary between the known and the unknown is epistemically dynamic, even if the underlying notion of knowability is formally well defined.

The notion of knowability is stable under effective transformation. If an entity can be reconstructed from some finite representation, and if another entity can be obtained from that representation by a computable modification that remains valid within the same minimal oracle, then the second entity is also knowable.

Proposition 3.5.3 Let $e_1, e_2 \in \mathcal{E}$. Suppose there exists a total computable function $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that for some minimal oracle $\mathcal{O}$,

$$\delta_{\mathcal{O}}(f(r)) = e_1 \quad \text{whenever} \quad \delta_{\mathcal{O}}(r) = e_2.$$

If $e_2 \in \mathcal{E}^\top$, then $e_1 \in \mathcal{E}^\top$.

Proof. Assume that $e_2 \in \mathcal{E}^\top$. By definition of knowability, there exists a minimal oracle $\mathcal{O}' \in \mathfrak{S}_{\mathcal{E}}^{\min}$ and a representation $r' \in \mathcal{R}_{\mathcal{O}'}^\top$ such that

$$\delta_{\mathcal{O}'}(r') = e_2.$$

Now apply the hypothesis to the oracle witnessing the knowability of e_2 . That is, suppose there exists a total computable function $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that

$$\delta_{\mathcal{O}}(f(r)) = e_1 \quad \text{whenever} \quad \delta_{\mathcal{O}}(r) = e_2.$$

In particular, since $\delta_{\mathcal{O}'}(r') = e_2$, we obtain

$$\delta_{\mathcal{O}}(f(r')) = e_1.$$

Because f is total, the string $f(r')$ is a finite binary string, and the above equality shows that $f(r') \in \mathcal{R}_{\mathcal{O}}^\top$. Hence there exists a minimal oracle, namely $\mathcal{O}'$, and a decodable representation, namely $f(r')$, such that

$$\delta_{\mathcal{O}'}(f(r')) = e_1.$$

Therefore $e_1 \in \mathcal{E}^\top$. ■

This proposition shows that the set of knowable entities is closed under effective derivation: once an entity is knowable, any other entity that can be obtained from one of its valid representations by a computable transformation, while remaining decodable under a minimal oracle, is knowable as well. In this sense, knowability is preserved not only by direct representation, but also by computable operations performed on representations.

3.6 Admissible Representations

Minimal representations play an important role in the theoretical analysis of knowledge, because they isolate the smallest amount of information required for reconstruction. They are therefore well suited to questions of optimality, complexity, and abstraction. In scientific practice, however, minimality is rarely the main objective. The representations used in real research often contain repeated measurements, multiple views, correlated variables, and other forms of redundancy. Such redundancy is not necessarily a defect. On the contrary, it often improves robustness, facilitates statistical estimation, and makes it possible to construct more accurate and stable models.

What must be avoided in practice is not redundancy as such, but the presence of information that is not genuinely grounded in the entity under investigation. Symbols that are semantically unrelated to the entity may introduce spurious regularities and mislead the construction of descriptions or models. This motivates the following notion of admissibility. Informally, a representation will be called admissible when it contains no semantically irrelevant substring carrying independent nontrivial information. In this sense, admissible representations may still be redundant, but they exclude independent informational content that does not belong to the entity.

Definition 3.6.1 Let $\mathcal{E}$ be a set of entities and let $\mathcal{O}_{\mathcal{E}}$ be a representation oracle with reconstruction function $\delta_{\mathcal{O}}$. We say that a decodable representation $r \in \mathcal{R}^{\top}$ is *admissible* if^a

$$\nexists s \sqsubset r, \left(K(s) > O(1) \wedge K(s \mid r \setminus s) > O(1) \wedge \delta_{\mathcal{O}}(r \setminus s) = \delta_{\mathcal{O}}(r) \right),$$

.

^aRecall that $s \sqsubset r$ denotes that s is a non-empty substring of r , and $r \setminus s$ denotes the string obtained by deleting one occurrence of s from r .

This definition excludes substrings that satisfy three conditions simultaneously. First, the substring must contain nontrivial information, in the sense that its Kolmogorov complexity is not bounded by a constant. Second, this information must be independent of the remainder of the representation, in the sense that it cannot be reconstructed from $r \setminus s$ up to an additive constant. Third, the substring must be semantically irrelevant for reconstruction, since removing it leaves the reconstructed entity unchanged. Thus, admissibility permits redundancy, but excludes independent informational content that is semantically extraneous.

Admissible representations are not necessarily minimal. A representation may contain symbols that are recoverable from the rest, and hence redundant, while still being admissible. This is important in practice. Minimal representations are useful for theoretical analysis, because they isolate the shortest reconstructions

of entities, but they are often too fragile or too impoverished for scientific work. Real modeling procedures usually benefit from redundancy.

■ **Example 3.11** If the target entity is the concept of the handwritten digit 3, a single canonical minimal representation of that digit may be theoretically elegant, but it is of little practical use for training a neural network. In practice one uses a large collection of images, such as in the MNIST setting, containing many handwritten instances of the same digit. These samples are highly redundant in the sense that they encode overlapping information about the same entity, yet this redundancy is precisely what allows the model to learn robust invariant structure.

Having defined admissibility at the level of individual decodable representations, we can now introduce the corresponding subset of $\mathcal{R}^\top$ consisting of all admissible representations.

Definition 3.6.2 Let $\mathcal{E}$ be a set of entities and let $\mathcal{O}_\mathcal{E}$ be a representation oracle with reconstruction function $\delta_\mathcal{O}$. We define the *set of admissible representations* by

$$\bar{\mathcal{R}}_\mathcal{O} = \{r \in \mathcal{R}^\top \mid \nexists s \sqsubset r, \left(K(s) > O(1) \wedge K(s \mid r \setminus s) > O(1) \right. \\ \left. \wedge \delta_\mathcal{O}(r \setminus s) = \delta_\mathcal{O}(r) \right)\}$$

We are interested in those admissible representations that reconstruct a given entity e .

Definition 3.6.3 Let $e \in \mathcal{E}$ be an entity. We define the *set of admissible representations of e* by

$$\bar{\mathcal{R}}_e = \{r \in \bar{\mathcal{R}} \mid \delta(r) = e\}$$

The first basic property of admissible representations is that minimal representations automatically satisfy admissibility: if a representation is already shortest among all those decoding to the same entity, then it cannot contain a nontrivial removable part of the kind excluded by the definition above.

Proposition 3.6.1 Every minimal representation is admissible.

Proof. Let $r \in \mathcal{R}^\star$ be a minimal representation, and suppose for contradiction that r is not admissible. Then there exists a substring $s \sqsubset r$ such that

$$K(s) > O(1), \quad K(s \mid r \setminus s) > O(1), \quad \delta_\mathcal{O}(r \setminus s) = \delta_\mathcal{O}(r).$$

Since s is a non-empty substring of r , the string $r \setminus s$ is strictly shorter than r , that

is,

$$l(r \setminus s) < l(r).$$

Moreover, $r \setminus s$ decodes to the same entity as r . This contradicts the minimality of r , since a shorter representation of the same entity would then exist. Therefore no such substring s can exist, and r is admissible. ■

The converse implication fails in general. An admissible representation may still contain redundant information, provided that this information is either simple or recoverable from the rest of the representation.

A second useful property is that inadmissibility always witnesses the presence of independent spurious information. In other words, whenever a representation is not admissible, one can identify a specific substring that is semantically irrelevant for reconstruction and yet carries nontrivial information not recoverable from the remainder.

Proposition 3.6.2 Let $r \in \mathcal{B}^*$ be a decodable representation. If r is not admissible, then there exists a substring $s \sqsubseteq r$ such that

$$\delta_{\mathcal{O}}(r \setminus s) = \delta_{\mathcal{O}}(r), \quad K(s) > O(1), \quad K(s \mid r \setminus s) > O(1).$$

Proof. This follows immediately from the negation of the definition. By definition, r is admissible iff there does not exist any substring $s \sqsubseteq r$ satisfying the three stated conditions. Therefore, if r is not admissible, then such a substring must exist. ■

The content of this property is conceptually important even though the proof is immediate. It shows that inadmissibility is not an abstract defect, but a concrete witness of semantically extraneous independent information. In practice, such witnesses correspond to parts of the data that may generate spurious structure and thereby distort the subsequent construction of models.

The next property formalizes the idea that adding only simple or fully predictable material to an admissible representation does not destroy admissibility. This captures the practical fact that harmless annotations, duplicated tokens, or predictable formatting conventions need not invalidate a representation.

Proposition 3.6.3 Let $r \in \mathcal{B}^*$ be an admissible representation, and let $t \in \mathcal{B}^*$ be such that

$$K(t) = O(1) \quad \text{or} \quad K(t \mid r) = O(1).$$

Assume moreover that the augmented string rt is decodable and satisfies

$$\delta_{\mathcal{O}}(rt) = \delta_{\mathcal{O}}(r).$$

Then rt is admissible.

Proof. Suppose, for contradiction, that rt is not admissible. Then there exists a substring $s \sqsubset rt$ such that

$$K(s) > O(1), \quad K(s \mid (rt) \setminus s) > O(1), \quad \delta_{\mathcal{O}}((rt) \setminus s) = \delta_{\mathcal{O}}(rt).$$

If s lies entirely inside r , then, since $\delta_{\mathcal{O}}(rt) = \delta_{\mathcal{O}}(r)$, the same substring witnesses the inadmissibility of r , contradicting the assumption that r is admissible.

It remains to consider the case in which s involves symbols from the appended part t . Since t is either of constant complexity or computable from r up to an additive constant, any substring of rt whose informational contribution comes entirely from t must fail at least one of the two complexity requirements

$$K(s) > O(1) \quad \text{and} \quad K(s \mid (rt) \setminus s) > O(1).$$

Indeed, a substring wholly determined by a constant-size pattern or by the already available information in r cannot contribute independent nontrivial information. Therefore no witness of inadmissibility can arise from the appended part alone. This contradiction shows that rt is admissible. ■

The proposition expresses a useful robustness principle. Admissibility is intended to exclude semantically irrelevant independent information, not every possible source of redundancy. Predictable or low-complexity additions are compatible with admissibility, which is precisely what makes the notion flexible enough for practical scientific use.

A further important observation is that admissibility is relative to the reconstruction oracle. The same string may be admissible under one oracle and inadmissible under another, because semantic grounding is determined by the reconstruction relation encoded by the oracle.

Proposition 3.6.4 Admissibility is, in general, oracle-dependent.

Proof. Consider two representation oracles $\mathcal{O}_1$ and $\mathcal{O}_2$ over the same set of entities $\mathcal{E}$, with reconstruction functions $\delta_{\mathcal{O}_1}$ and $\delta_{\mathcal{O}_2}$. Let $r \in \mathcal{B}^*$ and suppose there exists a substring $s \sqsubset r$ such that

$$\delta_{\mathcal{O}_1}(r \setminus s) = \delta_{\mathcal{O}_1}(r),$$

but

$$\delta_{\mathcal{O}_2}(r \setminus s) \neq \delta_{\mathcal{O}_2}(r) \quad \text{or} \quad \delta_{\mathcal{O}_2}(r \setminus s) \text{ is undefined.}$$

Assume also that

$$K(s) > O(1) \quad \text{and} \quad K(s \mid r \setminus s) > O(1).$$

Then s witnesses the inadmissibility of r with respect to $\mathcal{O}_1$, whereas it does not witness inadmissibility with respect to $\mathcal{O}_2$. Hence r may be inadmissible under $\mathcal{O}_1$ and admissible under $\mathcal{O}_2$. ■

This dependence is unavoidable. Admissibility is not a purely syntactic property of strings, but a semantic property determined by what the oracle takes the string to represent. This is fully consistent with the broader perspective of the theory of nescience, in which the same finite string may have very different epistemic status under different representation conventions.

Admissible representations thus occupy an intermediate position between minimality and unrestricted empirical data. They are broader than minimal representations because they allow redundancy, but narrower than arbitrary representations because they exclude semantically irrelevant independent information. This makes them especially suitable as the raw material for later description-building. In practice, useful scientific representations are seldom minimal, but they must still be sufficiently grounded in the entity to prevent models from being driven by spurious informational content. Admissibility captures exactly this compromise.

3.7 Extended Representations

The set of admissible representations $\bar{\mathcal{R}}_e$ of a given entity $e \in \mathcal{E}$ can be regarded as the long-term goal of scientific inquiry about that entity. In practice, however, scientific inquiry does not proceed by direct access to elements of $\bar{\mathcal{R}}_e$, but rather as a discovery process in which increasingly informative approximate representations of e are gradually developed. These approximate representations may omit information required for complete reconstruction, include incorrect or misleading content, or contain symbols that are irrelevant to the representation of e .

We say that a string represents a particular entity if it contains a non-trivial amount of algorithmic information about it. This is captured by requiring that conditioning on the string yields a reduction in the Kolmogorov complexity of a perfect representation of the entity.

Definition 3.7.1 Let $e \in \mathcal{E}$ be an entity. We define the set of *representations* of e as

$$\mathcal{R}_e = \{r' \in \mathcal{B}^* \mid \exists r \in \bar{\mathcal{R}}_e \text{ such that } K(r \mid r') \leq K(r) - O(1)\}$$

A representation r' is associated with an entity whenever the reduction in the conditional Kolmogorov complexity of an admissible representation r given r' , relative to the unconditional complexity of r , cannot be attributed solely to technical artifacts such as encoding conventions or machine-dependent implementation details.

■ **Example 3.12** The astronomical records used in the time of Ptolemy to describe the positions of celestial bodies throughout the year constituted an

imperfect encoding of the entity "positions of celestial bodies," due to systematic observational inaccuracies. More accurate encodings were later obtained through the improved observations of Tycho Brahe, and modern astronomy provides even higher-precision representations. ■

We accept as representations strings that may be extremely poor encodings of the entities under study, and may not support even rudimentary inferences. This reflects early stages of scientific inquiry, where representations are often fragmentary, speculative, or weakly informative, yet still guide exploration and motivate further investigation. The theory of nescience accommodates the progressive refinement of such representations, from highly imperfect encodings to increasingly accurate ones as knowledge advances (see Chapter 5).

Unlike the case of admissible representations, the sets $\mathcal{R}_e$ associated with different entities need not be disjoint, since a single imperfect representation may contain minimal but genuine information about more than one entity.

■ **Example 3.13** Consider the early study of infectious agents, before the distinction between viruses and bacteria was clearly established. At that stage, representations such as "a microscopic agent that causes disease and reproduces inside a host" captured minimal but genuine information about both entities. Such a representation reduces the descriptive complexity of perfect representations of viruses as well as of bacteria, without providing enough information to discriminate between them. This string would therefore be accepted as a representation of both entities, and would belong simultaneously to the sets $\mathcal{R}_{\text{virus}}$ and $\mathcal{R}_{\text{bacterium}}$. ■

We are interested in the set composed of all possible representations of all knowable entities in $\mathcal{E}$.

Definition 3.7.2 Let $\mathcal{E}$ be a collection of entities, and for each $e \in \mathcal{E}$ let $\mathcal{R}_e \subseteq \mathcal{B}^*$ be the set of representations of e . We define the set of *representations* for $\mathcal{E}$ as

$$\mathcal{R}_{\mathcal{E}} = \bigcup_{e \in \mathcal{E}} \mathcal{R}_e.$$

As in the case of the set of admissible representations $\bar{\mathcal{R}}_{\mathcal{E}}$, the set $\mathcal{R}_{\mathcal{E}}$ is generally unknown in practice. Although its definition is explicit, it depends on the true structure of the entities under investigation and on the existence of admissible representations, both of which are inaccessible in real scientific contexts, and so, it can't be enumerated or computed in practice.

Notation 3.5. When the set of entities is clear from the context, we denote the set of representations simply by $\mathcal{R}$.

Strings that do not belong to $\mathcal{R}_{\mathcal{E}}$, that is, strings that do not contain

information about any entity in $\mathcal{E}$, correspond to targetless representations, which are explicitly allowed within the theory of nescience (see Sections ?? and 22.4).

3.8 Effective Representations

The set of representations $\mathcal{R}$ is not accessible in practice, since it depends both on the true structure of the entities under investigation and on the Oracle's unknown internal reconstruction function. For this reason, scientific work cannot proceed directly with $\mathcal{R}$, but instead relies on a further level of approximation. We therefore introduce the set $\hat{\mathcal{R}} \subseteq \mathcal{B}^*$, consisting of candidate strings that constitute the representations currently available within scientific practice. The elements of $\hat{\mathcal{R}}$ are called *effective representations*: finite descriptions that can be formulated, communicated, and manipulated within existing scientific frameworks, and that are taken within a given scientific context to represent particular entities, without any guarantee of correctness or completeness. Scientific progress can then be understood as the ongoing evolution of the set of effective representations $\hat{\mathcal{R}}$, whose elements reflect the representational commitments adopted within scientific practice at a given time. In favorable cases, these commitments may come to align more closely with the underlying admissible representations, but no such alignment is built into the definition.

Definition 3.8.1 Let $\mathcal{E}$ be a set of entities. For each entity $e \in \mathcal{E}$, we denote by

$$\hat{\mathcal{R}}_e \subseteq \mathcal{B}^*$$

the set of *effective representations* of e , that is, the finite binary strings that are currently taken, within a given scientific context, to represent or approximate the entity e .

The notion of effective representations allows us to distinguish between known and unknown entities, in a representational or effective sense.

Definition 3.8.2 An entity $e \in \mathcal{E}$ is said to be *effectively known* if $\hat{\mathcal{R}}_e \neq \emptyset$, and *effectively unknown* otherwise.

In the theory of nescience, the notions of *effective knowledge* and *knowability* refer to two distinct concepts. An entity is said to be *effectively known* if at least one effective representation is available, that is, if $\hat{\mathcal{R}}_e \neq \emptyset$, whereas it is said to be *knowable* if an admissible representation exists, i.e., if $\bar{\mathcal{R}}_e \neq \emptyset$ (see Definition 3.2.3). Effective knowledge therefore concerns whether some string is currently taken, within a scientific context, to represent the entity, while knowability concerns whether the entity admits a fully adequate representation in

principle.

A common situation in science is one in which an entity is knowable but still effectively unknown. In such cases, an admissible representation exists in principle, yet no effective representation has yet been assigned to the entity within current scientific practice. This situation is not exceptional; on the contrary, it characterizes the overwhelming majority of entities in most scientific domains and reflects the open-ended, exploratory nature of scientific inquiry. By contrast, a more problematic situation arises when an entity is unknowable, yet effectively known. In this case, scientific practice assigns representations to an entity even though no admissible representation exists for it, as the following example illustrates.

■ **Example 3.14** The luminiferous ether was a hypothetical medium postulated to explain the propagation of light as a wave. The Michelson–Morley experiment provided strong empirical evidence against the existence of such a medium, and Einstein’s theory of special relativity later rendered the ether concept unnecessary by postulating the invariance of the speed of light in vacuum. As a result, the ether was abandoned as a physical entity. ■

The luminiferous ether provides an example of an effectively known entity that is treated here as unknowable. Within a historical scientific context, certain strings were taken to represent it, even though, under the present theory, no admissible representation is assumed to exist for it. In such cases, scientists may develop and refine strings that are taken to represent an entity despite the fact that those strings do not correspond to any admissible representation of it. Scientific effort is thus directed toward an entity for which, within the present framework, no admissible reconstruction by the oracle exists. Unfortunately, when one works only with finite and imperfect representations, there is in general no infallible practical criterion for recognizing such cases in advance.

Definition 3.8.3 Let $\mathcal{E}$ be a set of entities. We define the set of *effectively known representations* for $\mathcal{E}$ as

$$\hat{\mathcal{R}}_{\mathcal{E}} = \bigcup_{e \in \mathcal{E}} \hat{\mathcal{R}}_e.$$

The set $\mathcal{R}_{\mathcal{E}}$ characterizes the space of all representations that are possible in principle for the entities in $\mathcal{E}$, independently of whether they are currently available or known. By contrast, the set $\hat{\mathcal{R}}_{\mathcal{E}}$ consists only of those effective representations that have actually been discovered, constructed, or adopted within scientific practice at a given time. Thus, $\hat{\mathcal{R}}_{\mathcal{E}}$ should be understood as a time-dependent record of the strings that are taken, within a given scientific context, to represent entities in $\mathcal{E}$, whereas $\mathcal{R}_{\mathcal{E}}$ characterizes the representations

that are possible in principle. The former is shaped by historical practice and may include incorrect or misleading assignments; the latter remains a conceptual and generally inaccessible object.

Notation 3.6. *When the set of entities is clear from the context, we denote the set of effective representations simply by $\hat{\mathcal{R}}$.*

We can also define the set of effectively known entities, that is, the set of those entities for which at least one effective representation exists.

Definition 3.8.4 Given a set of entities $\mathcal{E}$ and the family of sets of effective representations $\{\hat{\mathcal{R}}_e\}_{e \in \mathcal{E}}$, we define the set of *effectively known entities* as

$$\hat{\mathcal{E}} = \{e \in \mathcal{E} \mid \hat{\mathcal{R}}_e \neq \emptyset\}.$$

The set $\hat{\mathcal{E}}$ consists of those entities for which scientific practice has assigned at least one effective representation. This includes entities that are currently under active investigation (e.g., artificial intelligence), entities that were investigated in the past but are no longer central to ongoing research (e.g., the aerodynamics of zeppelins), and entities that were once investigated but later rejected as non-existent or false (e.g., the luminiferous ether).

As in the case of representations themselves, the sets of effective representations associated with different entities need not be disjoint. In particular, for two distinct entities $e_1, e_2 \in \hat{\mathcal{E}}$, it is possible that $\hat{\mathcal{R}}_{e_1} \cap \hat{\mathcal{R}}_{e_2} \neq \emptyset$, reflecting the ambiguity of some effective representations (see Example 3.13).

A further consequence of working with effective representations is that some candidate strings may, unbeknownst to us, fail to encode the entity we intend to study, as Example 3.16 shows; membership in $\hat{\mathcal{R}}_e$ reflects contextual assignment rather than guaranteed correctness.

■ **Example 3.15** In the late Eighteenth Century, chemist Joseph Priestley believed he was studying a substance called "phlogiston", which was thought to be a fire-like element released during combustion. All of his experiments and representations were constructed around this idea. However, in reality, Priestley was observing the properties of a completely different entity: oxygen. Though his descriptions were coherent and reproducible, they were assigned within that scientific context to the wrong entity. ■

■ **Example 3.16** In the late Eighteenth Century, chemist Joseph Priestley believed he was studying a substance called *phlogiston*, which was thought to be a fire-like element released during combustion. The strings produced within that scientific context were therefore taken to represent phlogiston, and so belonged to the set of effective representations $\hat{\mathcal{R}}_{\text{phlogiston}}$. However, many of those same experimental descriptions were in fact tracking properties of a different entity, namely oxygen.

In this sense, some strings that were historically assigned to phlogiston may also be regarded, from a later scientific standpoint, as effective representations of oxygen. Thus, the two sets of effective representations need not be disjoint:

$$\hat{\mathcal{R}}_{\text{phlogiston}} \cap \hat{\mathcal{R}}_{\text{oxygen}} \neq \emptyset.$$

This example illustrates how effective representations are determined by scientific assignment rather than guaranteed correctness, and how the same finite strings may be associated with different entities across different stages of scientific practice. ■

Effective representations are monotonic over time, in the sense that the set of representations associated with an entity grows as new representations are discovered; representations that are later discarded, rejected, or superseded are not removed from the set.

Notation 3.7. We denote by $\hat{\mathcal{R}}_e^t$ the set of effective representations for an entity $e \in \mathcal{E}$ available at time t , and by $\hat{\mathcal{E}}^t$ the set of effectively known entities at time t .

Next proposition proves the monotonicity property of effective representations over time.

Proposition 3.8.1 Let $\hat{\mathcal{R}}_e^{t_1}$ and $\hat{\mathcal{R}}_e^{t_2}$ denote the set of effective representations associated with an entity $e \in \mathcal{E}$ at times t_1 and t_2 respectively. Then

$$\hat{\mathcal{R}}_e^{t_1} \subseteq \hat{\mathcal{R}}_e^{t_2} \quad \text{for all } t_1 < t_2.$$

Proof. By assumption, representations discovered at time t_1 are retained at all later times, even if they are discarded or superseded, and additional representations may be introduced. Therefore, any effective representation available at time t_1 is also available at time $t_2 > t_1$, which implies $\hat{\mathcal{R}}_e^{t_1} \subseteq \hat{\mathcal{R}}_e^{t_2}$. ■

As a direct consequence, the set of effectively known entities is itself monotonic over time.

Proposition 3.8.2 Let $\hat{\mathcal{E}}^{t_1}$ and $\hat{\mathcal{E}}^{t_2}$ denote the set of effectively known entities at times t_1 and t_2 respectively. Then

$$\hat{\mathcal{E}}^{t_1} \subseteq \hat{\mathcal{E}}^{t_2} \quad \text{for all } t_1 < t_2.$$

Proof. By definition $\hat{\mathcal{E}}^t = \{e \in \mathcal{E} \mid \hat{\mathcal{R}}_e^t \neq \emptyset\}$. If $e \in \hat{\mathcal{E}}^{t_1}$, then $\hat{\mathcal{R}}_e^{t_1} \neq \emptyset$. By the previous proposition, this implies $\hat{\mathcal{R}}_e^{t_2} \neq \emptyset$ for all $t_2 > t_1$, and therefore $e \in \hat{\mathcal{E}}^{t_2}$. ■

The set of effectively known entities $\hat{\mathcal{E}}$ is thus not fixed by $\mathcal{E}$, but by the interaction between entities, representational practices, and available inferential tools. In this sense, scientific progress can be understood as the progressive expansion and refinement of the known subset $\hat{\mathcal{E}}$, together with a gradual reduction of representational ambiguity.

3.9 Joint Representations

Ideally, scientific work would proceed by using an admissible representation of the entity e under study, that is, an element of the set of admissible representations $\hat{\mathcal{R}}_e$. However, since such admissible representations are not accessible in practice, we must instead rely on the extended representations in $\mathcal{R}_e$. Certain representations are of high quality, in the sense that they encode substantial information about e . Yet, $\mathcal{R}_e$ also includes low-quality representations, which convey only limited information about e . Moreover, $\mathcal{R}_e$ may include representations that contain incorrect or misleading symbols: fragments that do not correspond to the entity and do not contribute to obtaining an admissible representation. Such errors may mislead us during inquiry if we treat them as valid information.

If we want to increase our knowledge about an entity, we must use representations that are as complete and correct as possible, that is, as close as possible to an admissible representation. One way to achieve this is to try different strings until we discover a high-quality representation. However, this method can be extremely time-consuming and impractical. A more efficient approach is to improve a poor representation by adding missing symbols, removing unnecessary symbols, or combining known representations, each of which contains partial information.

Operation of Augmentation

One natural way to improve a representation consists of adding symbols that encode information that is missing with respect to the entity being represented. We call this operation *augmentation*. It models the gradual enrichment of representations as additional relevant information is incorporated.

Definition 3.9.1 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings and let $e \in \mathcal{E}$ be an entity. Suppose that the concatenated string rt is a representation of e , that is, $rt \in \mathcal{R}_e$. We call the concatenation rst an *augmentation* of the representation rt , and we call s an *augmentation string*.

An augmentation may occur at any position within a representation, and both r and t are allowed to be empty. In particular, the common case of appending new information at the end of a representation corresponds to $t = \lambda$.

Intuitively, an augmentation is intended to reduce the deficiency of a representation by incorporating additional information about the entity. The following proposition shows that augmentation preserves extendedness.

Proposition 3.9.1 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings such that $rt \in \mathcal{R}_e$. Then $rst \in \mathcal{R}_e$.

Proof. Assume $rt \in \mathcal{R}_e$. By definition of $\mathcal{R}_e$, there exists a perfect representation $r^* \in \mathcal{R}_e^*$ such that

$$K(r^* \mid rt) \leq K(r^*) - O(1). \quad (3.1)$$

Let $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be the computable function that, on input the concatenation rst , outputs rt (i.e., discards the middle component s and concatenates the first and third components).

By monotonicity of conditional Kolmogorov complexity under computable transformations,

$$K(x \mid y) \leq K(x \mid g(y)) + O(1) \quad (3.2)$$

for every computable function g .

Applying (3.2) with $x = r^*$ and $y = rst$, we obtain

$$K(r^* \mid rst) \leq K(r^* \mid g(rst)) + O(1) = K(r^* \mid rt) + O(1).$$

Combining this with (3.1), we conclude that

$$K(r^* \mid rst) \leq K(r^*) - O(1).$$

Therefore, there exists $r^* \in \mathcal{R}_e^*$ such that $K(r^* \mid rst) \leq K(r^*) - O(1)$, and hence $rst \in \mathcal{R}_e$. ■

The previous proposition formalizes the intended role of augmentation: adding information cannot destroy the property of containing sufficient information about the entity. In particular, augmentation cannot increase deficiency beyond a constant term.

As a direct consequence we obtain:

Corollary 3.9.2 Let $rt \in \mathcal{R}_e$ be an extended representation of an entity $e \in \mathcal{E}$, and let rst be an augmentation of rt . Then $rst \in \mathcal{R}_e$.

Proof. Since $rt \in \mathcal{R}_e$, we have $rst \in \mathcal{R}_e \subseteq \mathcal{R}_{\mathcal{E}}$. ■

Augmentation does not guarantee uniqueness of reference, as the following example illustrates.

■ **Example 3.17** Consider an entity e corresponding to a physical system described by classical mechanics, and let rt be a representation encoding its mass, position, and velocity. Suppose that the augmentation string s adds information about relativistic corrections. The augmented representation rst still contains all the information required to reconstruct the original classical model, but it may also correspond, for the oracle, to a distinct entity e' representing a relativistic model. In this case, rst may simultaneously belong to $\mathcal{R}_e$ and to $\mathcal{R}_{e'}$. ■

This example highlights an important subtlety: while augmentation reduces deficiency relative to the original entity, it may introduce additional structure that enables the oracle to associate the resulting representation with another entity as well. Augmented representations therefore increase informational content and expressive power, but may also increase the potential for ambiguity in reference.

Operation of Reduction

An alternative strategy for improving a representation consists of simplifying it by removing symbols that are believed to be incorrect, irrelevant, or superfluous. We refer to this operation as *reduction*. Unlike augmentation, which targets deficiency, reduction targets surplus and therefore requires greater care: removing symbols may improve a representation, but it may also destroy essential information.

Definition 3.9.2 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings and let $e \in \mathcal{E}$ be an entity. Suppose that the concatenated string rst is a representation of e , that is, $rst \in \mathcal{R}_e$. We call the string rt , obtained by removing the substring s , a *reduced representation* of rst .

As with augmentation, reduction may remove symbols from any position within a representation, and the removed substring s may encode information of any kind.

The crucial question is whether the removed substring s is *disposable*. Intuitively, a substring is disposable with respect to e if eliminating it does not increase the deficiency of the representation. In other words, s is disposable if the reduced representation rt still contains sufficient information to reconstruct a perfect representation of e .

Formally, a substring s of rst is disposable (with respect to e) if

$$rst \in \mathcal{R}_e \quad \text{and} \quad rt \in \mathcal{R}_e.$$

In this case, reduction preserves representationality and strictly decreases surplus by eliminating information that is not required for reconstructing the entity.

However, not every substring is disposable. Reduction may eliminate symbols that are essential for the oracle to reconstruct the entity. In general, reduced representations are not guaranteed to remain representations.

■ **Example 3.18** Let r and t be the empty strings λ , and let $s \in \mathcal{B}^*$ be such that $rst = s \in \mathcal{R}_e$. Then the reduced representation $rt = \lambda$ is the empty string. Since $K(r^* \mid \lambda) = K(r^*)$ for all $r^* \in \mathcal{R}^*$, the empty string does not belong to $\mathcal{R}_e$. In this case, the removed substring s was not disposable. ■

Even when a reduced string remains a valid representation, reduction does not guarantee preservation of the represented entity.

■ **Example 3.19** Consider an entity e corresponding to a specific chemical compound, and suppose that rst encodes its full molecular structure, including stereochemical information. Let the substring s encode the spatial configuration of a chiral center. The reduced representation rt , which omits this information, may still correspond to a valid chemical compound, but one that is chemically distinct: a different stereoisomer. In this case, rt represents a different entity e' , even though it was obtained by reducing a representation of e . ■

These examples illustrate an important asymmetry between augmentation and reduction. While augmentation cannot destroy extendedness—since adding information cannot increase deficiency beyond a constant—reduction may remove essential information. Only when the removed substring is disposable does reduction genuinely improve a representation by decreasing surplus without increasing deficiency.

In practical contexts, such as the curation of datasets in machine learning, disposability corresponds to identifying and removing mislabeled samples, corrupted entries, or irrelevant features. These elements contribute surplus information that does not assist in reconstructing the underlying data-generating entity. Eliminating them refines the representation, but only when their removal does not eliminate essential structure.

Joint Representations

A third, and particularly powerful, operation on representations consists of combining two existing representations into a single one. This operation, which we call *joint representation*, plays a central role in the theory of nescience, as it provides a systematic mechanism for both refining existing knowledge and discovering new entities.

Definition 3.9.3 Let $r, s \in \mathcal{R}_\top$ be two representations of an entity $e \in \mathcal{E}$. We define the *joint representation* of r and s as the concatenated string rs .

Joint representations serve two distinct goals. When r and s represent the same entity, their joint representation may yield a more complete or less biased representation of that entity by aggregating complementary information. When r and s represent different entities, their joint representation may correspond to a new entity not previously represented.

■ **Example 3.20** Let e be the entity corresponding to the set of causes of lung cancer. Suppose that r is a dataset obtained from a population sample consisting exclusively of male subjects, and that s is a dataset obtained from a population sample consisting exclusively of female subjects. Both r and s belong to $\mathcal{R}_e$, but each constitutes a biased and incomplete representation of the entity. The joint representation rs combines both datasets and yields a more informative and less

biased representation of e . ■

With the following propositions we formally study the properties of the concept of joint representations.

Proposition 3.9.3 Joint representation is associative. That is, for all $r, s, t \in \mathcal{R}_{\mathcal{E}}$, we have

$$(rs)t = r(st).$$

Proof. Joint representation is defined as string concatenation. Since string concatenation is associative, the equality $(rs)t = r(st)$ holds for all binary strings, and therefore for all representations. ■

Despite its associativity, joint representation is not commutative. A concrete counterexample arises when r encodes a theoretical model and s encodes experimental constraints. The order in which these components are presented may affect how the oracle reconstructs the entity, leading to different outcomes.

Joint representation also satisfies an idempotence property.

Proposition 3.9.4 For any representation $r \in \mathcal{R}_{\mathcal{E}}$, the joint representation rr represents the same entity as r .

Proof. The concatenated string rr contains no information that is not already present in r . While redundancy may increase, the oracle can ignore duplicated symbols during reconstruction. Consequently, $rr \in \mathcal{R}_{\mathcal{E}}$ and $\mathcal{O}_{\mathcal{E}}(rr) = \mathcal{O}_{\mathcal{E}}(r)$. ■

We now turn to closure properties.

Proposition 3.9.5 If $r, s \in \mathcal{R}_{\mathcal{E}}$ and $rs \in \mathcal{R}_{\mathcal{E}}$, then rs is a representation.

Proof. This follows directly from the definition of $\mathcal{R}_{\mathcal{E}}$ as the domain of the oracle. Whenever rs belongs to $\mathcal{R}_{\mathcal{E}}$, it is, by definition, a valid representation. ■

Although syntactically well defined, joint representation is not guaranteed to preserve entity identity.

Proposition 3.9.6 Let $r, s \in \mathcal{R}_e$ be representations of the same entity e . Then the joint representation rs does not necessarily belong to $\mathcal{R}_e$.

Proof. The joint representation rs may introduce interactions between symbols in r and s that the oracle interprets as defining a different entity. Even if r and s individually encode correct and complementary information about e , their combination may cross a representational threshold that leads the oracle to reconstruct a distinct entity $e' \neq e$. Hence, $rs \in \mathcal{R}_{e'}$. ■

This phenomenon is not a defect but a defining feature of joint representations.

■ **Example 3.21** Consider a domain in which entities are chemical compounds and representations encode molecular descriptors. Let r represent a compound with anti-inflammatory properties and let s represent a structurally related compound with antiviral properties. While both r and s individually correspond to known compounds, the joint representation rs may encode a novel molecular structure. If the oracle associates rs with a previously uncharacterized compound, then rs represents a new entity. ■

The concept of joint representation can be extended to any arbitrary, but finite, collection of representations. This allows us to incorporate multiple partial representations into our research or to use them in the process of discovering new entities.

Definition 3.9.4 Let $r_1, r_2, \dots, r_n \in \mathcal{R}_{\mathcal{E}}$ be a finite collection of representations. The *joint representation* of $r_1, r_2, \dots, r_n$ is defined as the concatenated string $r_1 r_2 \dots r_n$.

The possibility that joint representations of known entities give rise to representations of new entities is a critical element of the theory of nescience. It provides a formal mechanism for the discovery of previously unknown entities through the systematic combination of existing knowledge, without requiring access to representations that lie outside the current epistemic horizon.

3.10 Representations of Research Areas

Research areas are subsets of the set of entities E . Once individual entities admit representations, it is natural to ask how an entire area can be represented. Intuitively, a representation of an area should collect representations of the entities that compose it in such a way that each component can be isolated and identified. This allows the area to be studied not merely as an undifferentiated collection of symbols, but as a structured family of entity-level representations.

A first difficulty is that research areas need not be finite, whereas representations are finite binary strings. A second difficulty is that some entities in an area may be unknowable, and therefore may not admit any representation at all. For these reasons, the representation of an area must be understood relative to a finite subset of its representable entities.

Let $A \subseteq E$ be a research area. Recall that, for each entity $e \in E$, the set $R_e \subseteq B^*$ denotes the set of representations of e , and $\hat{R}_e \subseteq B^*$ denotes the set of effective representations of e .

Definition 3.10.1 Let $\mathcal{O}$ be a representation oracle over E , and let $A \subseteq E$ be a research area. We define the *oracle-knowable part* of A by

$$A_{\mathcal{O}}^{\top} := A \cap E_{\mathcal{O}}^{\top}.$$

Thus, $A_{\mathcal{O}}^{\top}$ is the subset of entities in the area that are knowable with respect to the oracle $\mathcal{O}$. If A contains entities outside $A_{\mathcal{O}}^{\top}$, then no area representation based on $\mathcal{O}$ can include them as individually decodable components.

In order to represent several entities together, we must separate their representations and attach identifiers that allow each component to be recognized. Any standard self-delimiting encoding of finite strings may be used for this purpose.

Definition 3.10.2 Let $A \subseteq E$ be a research area, let

$$F = \{e_1, \dots, e_n\} \subseteq A$$

be a finite subset, and let $r_i \in R_{e_i}$ for each $i = 1, \dots, n$. Let $t_1, \dots, t_n \in B^*$ be pairwise distinct identifier strings. We define a *representation of the finite subarea F* as any finite binary string of the form

$$\rho_F = \langle (t_1, r_1), \dots, (t_n, r_n) \rangle,$$

where $\langle \cdot, \cdot \rangle$ and its finite iteration denote a fixed self-delimiting computable encoding of pairs and finite sequences of binary strings.

The identifiers t_i do not need to carry scientific meaning by themselves. Their only purpose is to make it possible to separate the different components of the area representation and to determine which representation belongs to which entity within the represented finite family.

Definition 3.10.3 Let $A \subseteq E$ be a research area. We say that a binary string $\rho \in B^*$ is an *extended representation of the area A* if there exists a finite subset $F = \{e_1, \dots, e_n\} \subseteq A$ and strings $r_i \in R_{e_i}$ such that

$$\rho = \langle (t_1, r_1), \dots, (t_n, r_n) \rangle$$

for some pairwise distinct identifiers $t_1, \dots, t_n$.

This definition deliberately allows area representations to be partial: the string ρ need not encode all entities in A , but only a finite subset of them. This is unavoidable whenever A is infinite, and it remains useful even when A is finite but some of its entities are unknowable.

Definition 3.10.4 Let $A \subseteq E$ be a research area and let $\mathcal{O}$ be a representation oracle. We say that ρ is a *complete area representation of A relative to $\mathcal{O}$* if ρ is an area representation associated with the finite set

$$F = A_{\mathcal{O}}^{\top}.$$

In particular, such a complete representation can exist only if $A_{\mathcal{O}}^{\top}$ is finite.

Thus, if A is finite and all its entities are knowable with respect to $\mathcal{O}$, then A admits a complete area representation. If A contains unknowable entities, then at best one can represent its knowable part $A_{\mathcal{O}}^{\top}$.

Proposition 3.10.1 Let

$$\rho_F = \langle (l_1, r_1), \dots, (l_n, r_n) \rangle$$

be a representation of a finite subarea $F = \{e_1, \dots, e_n\} \subseteq A$. Then each pair (l_i, r_i) can be recovered effectively from ρ_F .

Proof. By construction, ρ_F is encoded using a fixed self-delimiting computable coding of finite sequences of pairs. Therefore there exists a computable decoding procedure that parses ρ_F and returns each pair (l_i, r_i) . Hence every component representation is effectively recoverable from the area representation. ■

The previous proposition formalizes the main purpose of tagging individual entity representations: the representation of an area preserves the identity of its components.

Proposition 3.10.2 Area representations are not unique.

Proof. Non-uniqueness may arise in at least two ways. First, an entity $e \in E$ may admit several distinct representations $r, r' \in R_e$. Second, even for a fixed family of entity-level representations, one may choose different identifiers or a different order of concatenation. Therefore the same research area may admit multiple distinct area representations. ■

This non-uniqueness is not a defect. It reflects the fact that a research area may be organized in different representational styles, just as an individual entity may admit several valid representations.

Proposition 3.10.3 Let $F_1, F_2 \subseteq A$ be finite subareas with $F_1 \cap F_2 = \emptyset$. If ρ_{F_1} and ρ_{F_2} are area representations of F_1 and F_2 , respectively, then there exists an area representation of $F_1 \cup F_2$.

Proof. Write

$$\rho_{F_1} = \langle (l_1, r_1), \dots, (l_m, r_m) \rangle$$

and

$$\rho_{F_2} = \langle (\kappa_1, s_1), \dots, (\kappa_\ell, s_\ell) \rangle.$$

Choosing the identifiers so that all $\iota_1, \dots, \iota_m, \kappa_1, \dots, \kappa_\ell$ are pairwise distinct, the concatenated string

$$\langle (\iota_1, r_1), \dots, (\iota_m, r_m), (\kappa_1, s_1), \dots, (\kappa_\ell, s_\ell) \rangle$$

is a representation of the finite subarea $F_1 \cup F_2$. ■

This shows that finite area representations are closed under finite aggregation. In this sense, representing an area amounts to collecting identifiable entity-level representations into a single structured string.

In practice, however, science does not work with the full set R_e of representations of an entity, but with the currently available effective representations.

Definition 3.10.5 Let $A \subseteq E$ be a research area and let t denote a time index. We define the *effectively known part* of the area at time t by

$$\hat{A}^t := A \cap \hat{E}^t.$$

The set $\hat{A}^t$ contains those entities of the area for which at least one effective representation is available at time t .

Definition 3.10.6 Let $A \subseteq E$ be a research area and let

$$F = \{e_1, \dots, e_n\} \subseteq \hat{A}^t$$

be a finite subset of effectively known entities at time t . For each $i = 1, \dots, n$, let $\hat{r}_i \in \hat{R}_{e_i}^t$, and let $\iota_1, \dots, \iota_n \in B^*$ be pairwise distinct identifiers. We define an *effective representation of the finite known subarea F at time t* as any binary string of the form

$$\hat{\rho}_F^t = \langle (\iota_1, \hat{r}_1), \dots, (\iota_n, \hat{r}_n) \rangle.$$

Effective area representations are the practical counterparts of area representations. They need not be correct or complete, because their components are only effective representations of the individual entities. Nevertheless, they encode the current representational state of the part of the area that is effectively known.

Proposition 3.10.4 Let $A \subseteq E$ be a fixed research area. If $t_1 < t_2$, then

$$\hat{A}^{t_1} \subseteq \hat{A}^{t_2}.$$

Proof. By Proposition 3.8.2, the set $\hat{E}^t$ of effectively known entities is monotone in time. Therefore

$$\hat{A}^{t_1} = A \cap \hat{E}^{t_1} \subseteq A \cap \hat{E}^{t_2} = \hat{A}^{t_2}.$$



This proposition states that, if the research area A is held fixed, the subset of its entities that are effectively known is expected to grow over time. Consequently, effective area representations may be progressively extended by incorporating newly available effective representations of entities that were previously unknown.

Proposition 3.10.5 Let $A \subseteq E$ be a fixed research area, and let $\hat{\rho}_{F_1}^{t_1}$ be an effective representation of a finite known subarea $F_1 \subseteq \hat{A}^{t_1}$. If $t_1 < t_2$, then there exists a finite set F_2 such that

$$F_1 \subseteq F_2 \subseteq \hat{A}^{t_2}$$

and an effective representation $\hat{\rho}_{F_2}^{t_2}$ extending $\hat{\rho}_{F_1}^{t_1}$.

Proof. Since $\hat{A}^{t_1} \subseteq \hat{A}^{t_2}$, every entity in F_1 remains effectively known at time t_2 . Let F_2 be any finite subset of $\hat{A}^{t_2}$ containing F_1 . For entities in F_1 , retain their previous tagged effective representations, and for entities in $F_2 \setminus F_1$, append newly available tagged effective representations. The resulting self-delimiting concatenation is an effective representation of F_2 at time t_2 . ■

The previous proposition formalizes the idea that scientific progress can be viewed as a gradual enrichment of area representations.

However, this monotonicity concerns only the set of effectively known entities relative to a *fixed* research area. In practice, the area itself may change over time.

Notation 3.8. *If the membership of a research area changes with time, we write A^t for the set of entities regarded as belonging to the area at time t .*

This distinction is important. As scientific understanding deepens, the very delimitation of an area may be revised. New entities may be incorporated because they are discovered to belong naturally to the area, while other entities may be removed because they are reclassified or shown to be irrelevant.

For example, a research area may initially include only those entities that are directly observable within a given methodological framework. Later, the introduction of new theories or instruments may enlarge the area by adding previously inaccessible entities. Conversely, some entities once thought to belong to the area may later be abandoned as fictitious, misclassified, or scientifically unproductive. Thus, the time evolution of a research area has two distinct components: the growth of the effectively known subset of a fixed area, and the possible redefinition of the area itself.

In summary, the representation of a research area is necessarily component-wise and finite. It is obtained by concatenating identifiable representations of

individual entities. When all entities in a finite area are knowable, one may represent the whole area. When some entities are unknowable, only the knowable part can be represented. In practice, one works instead with effective area representations built from the currently available effective representations of the effectively known entities. These effective representations are expected to become richer over time, while the boundaries of the research area itself may also evolve as scientific knowledge progresses.

3.11 References

For more information about Russell's paradox, Cantor's theorem, and universal sets, refer, for example, to [Jec13]. This book also covers the Zermelo–Fraenkel system of axioms, including the Axiom of Choice. The idea of using a function to assign to each symbol and well-formed formula of a formal language a unique natural number (called a Gödel number) was introduced by Kurt Gödel in his proof of the incompleteness theorems [Göd31]. A detailed discussion of the Berry paradox from the perspective of computability can be found in [Cha95]. For a review of the problem of representation in Kolmogorov complexity, as well as a detailed account of the implications of Kolmogorov complexity being defined only up to an additive constant, see [LV13]. That oracle machines are not mechanical was originally stated by Turing when he introduced the concept of the oracle machine in [Tur39]. For a comprehensive review of oracle Turing machines, refer to [Rob15].



4. Descriptions

*What I cannot create,
I do not understand.*
Richard Feynman

In Chapter 3 we addressed the problem of how to encode a possibly abstract entity as a finite string of symbols that preserves enough information to make the entity available for study. Such a string was called a *representation*. A representation, however, is not yet enough for science. Even when it contains relevant information about the entity, it may still be too large, too irregular, or too unstructured to support explanation, prediction, or understanding. The next task is therefore to construct a description that generates, explains, or approximates that representation. Descriptions make it possible not merely to store or inspect a representation, but to do science with it: they capture regularities, compress what can be made systematic, and provide an effective basis for reasoning and prediction.

In the theory of nescience, descriptions are treated as computable objects: finite strings that encode procedures capable of producing an output representation. This requirement is essential. It excludes purely informal, ambiguous, or self-referential descriptions that may be suggestive from a philosophical point of view but cannot be executed or checked mechanically. By

grounding descriptions in computability, the theory places scientific explanation within a framework in which descriptions can be compared, transformed, and analyzed with mathematical precision.

Not every computable string qualifies equally as a scientific description. A *valid description* is one that encodes a computation that produces a representation. Among the valid descriptions of a given representation, some may be shorter than others. This leads to the notion of a *minimal description*, namely, a description of shortest possible length for that representation. Minimal descriptions play a central theoretical role, since they capture the smallest amount of information needed to reconstruct the representation. In practice, however, such optimal descriptions are generally unknown and cannot be identified effectively. For this reason, scientific work proceeds instead with *extended descriptions*: descriptions that are actually available and used in a given scientific context, even though they need not be minimal and may even fail to generate the intended representation exactly.

Descriptions may also be restricted to particular families, reflecting the fact that in real scientific work we rarely search over the entire space of conceivable descriptions. Instead, we usually work within structured classes such as linear models, systems of differential equations, decision trees, or neural networks. In this setting, a description may be understood as consisting of two parts: a structural component, which captures the regularities present in the representation, and an input component, which provides the remaining information needed for reconstruction. When the structural component is restricted to a given family, the resulting descriptions correspond to what scientists ordinarily call *models*. The chapter also studies basic operations on descriptions, such as augmentation and reduction, which formalize the ways in which descriptions may be refined, enriched, or simplified.

Whereas the chapter on representations focused on how entities are encoded, the present chapter focuses on how those encodings are modeled. It is at this point that the theory of nescience moves from the problem of representing entities to the problem of explaining their representations through valid, minimal, and effective descriptions, and through the model classes used in scientific practice.

4.1 Valid Descriptions

A description is intended to capture a representation in a form that can support explanation, prediction, and further reasoning. By replacing an unstructured string with a computable account of how that string is produced, descriptions make it possible to study representations as scientific objects rather than merely to record them.

Intuitively, a description consists of two parts: a Turing machine, which

encodes the regularities present in the representation, and an input string, which supplies the remaining information needed to produce it. In this way, the Turing machine captures the compressible structure, while the input string accounts for what is left unspecified.

Definition 4.1.1 Let $d \in \mathcal{B}^*$ be a binary string of the form $d = \langle TM, a \rangle$, where TM is the encoding of a prefix-free Turing machine and $a \in \mathcal{B}^*$ is an input string. If $TM(a) \downarrow$, then we say that d is a *valid description*.

From an ontological point of view, descriptions are finite strings that satisfy an additional requirement: they must encode a computation that halts. Thus, descriptions are not arbitrary symbolic objects, but effective procedures for producing strings.

This computational requirement has important advantages. First, it excludes purely informal, ambiguous, or metaphorical descriptions that may suggest an interpretation but do not determine any definite procedure. Second, it makes descriptions mechanically checkable: in principle, one can execute the encoded computation and verify the resulting output. Third, it allows descriptions to be compared in a mathematically precise way, for example by their length, their structure, or the transformations that relate one description to another. In this way, computable descriptions provide a rigorous foundation for scientific modeling, since they turn the problem of explanation into the study of explicit procedures rather than of arbitrary symbolic expressions.

The following definition specifies the set of all valid descriptions, that is, the set of finite binary strings that encode a computable procedure together with an input on which it halts.

Definition 4.1.2 We define the *set of valid descriptions*, denoted by $\mathcal{D}$, as

$$\mathcal{D} = \{d \in \mathcal{B}^* : d = \langle TM, a \rangle \wedge TM(a) \downarrow\}.$$

Only strings belonging to $\mathcal{D}$ will be considered descriptions in the theory of nescience.

Once the global set of descriptions has been fixed, we can restrict attention to those descriptions that perfectly reproduce a given representation. This allows us to associate descriptions directly with individual representations.

Definition 4.1.3 Let $r \in \mathcal{R}$ be a representation. We define the *set of exact descriptions for r* , denoted by $\bar{\mathcal{D}}_r$, as

$$\bar{\mathcal{D}}_r = \{d = \langle TM, a \rangle \in \mathcal{D} : TM(a) = r\}.$$

The set $\bar{\mathcal{D}}_r$ is important because it contains all the exact descriptions of the representation r . It therefore characterizes the full descriptive variability

associated with r : different elements of $\bar{\mathcal{D}}_r$ may encode the same representation by means of different procedures and with different amounts of information. This makes $\bar{\mathcal{D}}_r$ the natural reference set for studying optimality and excess in description, since any shortest or most economical description of r must belong to $\bar{\mathcal{D}}_r$.

Since a description produces a unique output whenever the computation halts, the sets of descriptions associated with distinct representations are disjoint.

Proposition 4.1.1 Let $r_1, r_2 \in \mathcal{R}$ with $r_1 \neq r_2$. Then

$$\bar{\mathcal{D}}_{r_1} \cap \bar{\mathcal{D}}_{r_2} = \emptyset.$$

Proof. Suppose, for contradiction, that there exists a description

$$d = \langle TM, a \rangle \in \bar{\mathcal{D}}_{r_1} \cap \bar{\mathcal{D}}_{r_2}.$$

By definition of $\bar{\mathcal{D}}_{r_1}$, we have $TM(a) = r_1$, and by definition of $\bar{\mathcal{D}}_{r_2}$, we also have $TM(a) = r_2$. Since TM is deterministic, the computation on input a has a unique output. Hence $r_1 = r_2$, contradicting the assumption that $r_1 \neq r_2$. ■

This proposition shows that the sets $\bar{\mathcal{D}}_r$ are pairwise disjoint. Therefore, any exact description whose output is a representation in $\mathcal{R}$ is associated with exactly one representation. This makes it possible to classify descriptions according to their output and to study each representation through its own set of descriptions, independently of the others.

It is important to note, however, that not every description in $\mathcal{D}$ must necessarily generate a string belonging to $\mathcal{R}$. A valid description is only required to encode a halting computation, and its output may be an arbitrary binary string. Thus, the sets $\bar{\mathcal{D}}_r$ do not partition the whole set $\mathcal{D}$, but only the subset of descriptions whose outputs are representations in $\mathcal{R}$.

We can generalize and consider all descriptions that perfectly generate some representation of a given entity. In this way, the notion of description is lifted from individual representations to the entities they represent.

Definition 4.1.4 Let $e \in \mathcal{E}$ be an entity. We define the *set of exact descriptions for e* , denoted by $\bar{\mathcal{D}}_e$, as

$$\bar{\mathcal{D}}_e = \{d = \langle TM, a \rangle \in \mathcal{D} : \exists r \in \mathcal{R}_e, TM(a) = r\}.$$

The set $\bar{\mathcal{D}}_e$ collects all exact descriptions that generate some representation of the entity e . Unlike the sets $\bar{\mathcal{D}}_r$, which are tied to a single representation, the set $\bar{\mathcal{D}}_e$ brings together all the different ways in which the same entity may be described through possibly different representations. In this sense, $\bar{\mathcal{D}}_e$ characterizes the descriptive variability of the entity itself, rather than of any one particular encoding.

By definition, $\bar{\mathcal{D}}_e$ is the union of the sets $\bar{\mathcal{D}}_r$ over all representations $r \in \mathcal{R}_e$. It therefore provides the natural domain for studying descriptions at the level of the entity, especially in situations where the entity admits multiple representations and, correspondingly, multiple families of descriptions.

Proposition 4.1.2 For every entity $e \in \mathcal{E}$,

$$\bar{\mathcal{D}}_e = \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r.$$

Proof. We prove both inclusions. First, let $d \in \bar{\mathcal{D}}_e$. By Definition 4.1.4, there exists some $r \in \mathcal{R}_e$ such that $TM(a) = r$. By Definition 4.1.3, this implies that $d \in \bar{\mathcal{D}}_r$. Hence

$$d \in \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r.$$

Therefore,

$$\bar{\mathcal{D}}_e \subseteq \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r.$$

Conversely, let

$$d \in \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r.$$

Then there exists some $r \in \mathcal{R}_e$ such that $d \in \bar{\mathcal{D}}_r$. By Definition 4.1.3, this means that $TM(a) = r$. Since $r \in \mathcal{R}_e$, Definition 4.1.4 implies that $d \in \bar{\mathcal{D}}_e$. Thus,

$$\bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r \subseteq \bar{\mathcal{D}}_e.$$

Combining both inclusions gives

$$\bar{\mathcal{D}}_e = \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r.$$

■

The next proposition makes explicit the relation between representation and description at the level of entities. Descriptions are defined only through the representations they generate, so the existence of exact descriptions for an entity is equivalent to the existence of representations of that entity. In this sense, the descriptive layer of the theory does not extend beyond the representational one, but presupposes it.

Proposition 4.1.3 For every entity $e \in \mathcal{E}$,

$$\bar{\mathcal{D}}_e = \emptyset \iff \mathcal{R}_e = \emptyset.$$

Proof. First, assume that $\mathcal{R}_e = \emptyset$. Then, by Proposition 4.2.4,

$$\bar{\mathcal{D}}_e = \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r = \bigcup_{r \in \emptyset} \bar{\mathcal{D}}_r = \emptyset.$$

Conversely, assume that $\bar{\mathcal{D}}_e = \emptyset$. Suppose, for contradiction, that $\mathcal{R}_e \neq \emptyset$. Then there exists some $r \in \mathcal{R}_e$. Since r is a finite binary string, there exists a prefix-free Turing machine that ignores its input and prints r , then halts. Let $d = \langle TM, a \rangle$ be the encoding of such a machine together with any admissible input a . Then $d \in \bar{\mathcal{D}}$ and $TM(a) = r$, so $d \in \bar{\mathcal{D}}_r \subseteq \bar{\mathcal{D}}_e$, contradicting $\bar{\mathcal{D}}_e = \emptyset$. Therefore, $\mathcal{R}_e = \emptyset$. ■

This proposition links the existence of descriptions to the representability of entities. In particular, if an entity is unknowable, in the sense that it admits no representations, then it also admits no exact descriptions. Conversely, whenever an entity is representable, it also becomes describable, since any of its representations may be produced by at least one exact description. Thus, at the level of entities, describability and representability coincide: the boundary between describable and undescribable entities is the same as the boundary between knowable and unknowable entities.

In some cases, a description may also qualify as a representation, namely when it is accepted by the oracle as encoding an entity. In this sense, there may exist descriptions that describe other descriptions. In practice, however, it is not advisable to use descriptions as representations of entities. A useful representation should preserve as much relevant detail about the original entity as possible, whereas a description is designed to provide a concise computational account of a representation. Replacing representations with descriptions would therefore make the task of scientific discovery considerably harder.

The set of descriptions associated with an entity is not arbitrary from a computability-theoretic point of view. Under a mild effectiveness assumption on the set of representations of the entity, the corresponding set of descriptions can also be effectively enumerated.

Proposition 4.1.4 Let $e \in \mathcal{E}$. If the set $\mathcal{R}_e$ is recursively enumerable, then the set $\bar{\mathcal{D}}_e$ is recursively enumerable.

Proof. By Proposition 4.2.4,

$$\bar{\mathcal{D}}_e = \bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r.$$

Fix $r \in \mathcal{R}_e$. The set $\bar{\mathcal{D}}_r$ is recursively enumerable: one can enumerate all binary strings d , interpret each of them as a candidate pair $\langle TM, a \rangle$, simulate $TM(a)$ by dovetailing, and output d whenever the computation halts with value r .

Now assume that $\mathcal{R}_e$ is recursively enumerable. Then one can enumerate its elements

$$r_1, r_2, r_3, \dots$$

and, for each r_i , run in parallel the corresponding enumeration of $\bar{\mathcal{D}}_{r_i}$. By dovetailing these enumerations, every description belonging to some $\mathcal{D}_{r_i}$ will eventually be produced. Therefore, one obtains an effective enumeration of

$$\bigcup_{r \in \mathcal{R}_e} \bar{\mathcal{D}}_r = \bar{\mathcal{D}}_e.$$

Hence $\bar{\mathcal{D}}_e$ is recursively enumerable. ■

This proposition has an important practical implication. Whenever the set $\mathcal{R}_e$ can be effectively enumerated, the corresponding descriptions of the entity can also be searched effectively, at least in principle. That is, one may systematically generate candidate descriptions of the entity and test their properties. This does not solve the problem of finding good or minimal descriptions, but it shows that the search space is not completely inaccessible from a computability-theoretic point of view.

A final and particularly important point is that, since each valid description determines exactly one output string, it is natural to define a function that maps descriptions to the representations they produce. Because descriptions encode Turing machines together with their inputs, this mapping can be implemented by a universal Turing machine. Thus, not only are individual descriptions computable objects, but the operation that sends a description to its output is computable as well.

Definition 4.1.5 We call a *description function*, denoted by Γ , a universal Turing machine

$$\Gamma : \mathcal{D} \rightarrow \mathcal{B}^*$$

that halts on all inputs in $\mathcal{D}$ and maps each description to the representation it generates.

If $d = \langle TM, a \rangle$ is a description of the representation r , then by definition

$$\Gamma(d) = \Gamma(\langle TM, a \rangle) = TM(a) = r.$$

Hence the description function provides the canonical computable map from valid descriptions to the representations they produce.

4.2 Minimal Descriptions

Inspired by the principle of Occam's razor¹, if two explanations are equally successful in reconstructing a representation, we should in general prefer the shorter one. The reason is not only aesthetic economy. Shorter descriptions tend to be easier to understand, since they isolate the essential regularities of the representation without obscuring them under unnecessary detail. They are also easier to manipulate in practice: from a shorter and more structured model, it is usually simpler to derive consequences, formulate predictions, compare alternative explanations, and detect which assumptions are actually doing the explanatory work. By contrast, excessively long descriptions may reproduce the representation correctly while still hiding its structure behind redundancy, ad hoc clauses, or irrelevant complexity.

For this reason, shortness plays a central role in the theory of nescience. A representation may admit many valid descriptions, but not all of them are equally informative from a scientific point of view. Some may reconstruct the representation only at the price of introducing superfluous structure, while others may capture the same content more economically. This motivates the relevance of minimal descriptions. A minimal description of a representation is, by definition, a shortest valid description that reconstructs it. In this sense, minimal descriptions mark the limit of what can be known, or understood, about a representation through description: once a shortest description has been reached, no further compression is possible without losing the ability to reconstruct the representation. Minimal descriptions therefore provide a natural theoretical ideal of scientific understanding, even if in practice they are typically unknown and must be approximated by the effective models actually available.

Definition 4.2.1 Let $\mathcal{D}_r$ be the set of descriptions of a representation $r \in \mathcal{R}$, and let $d \in \mathcal{D}_r$ be a description of r . We say that d is a *minimal description* of the representation r if there is no other description $d' \in \mathcal{D}_r$ such that $l(d') < l(d)$.

Recall that what we know about an entity e depends on the quality of the representation r used. A minimal description is defined only relative to a given representation: it is the shortest valid description that reconstructs r . Therefore, even if we find a minimal description of r , this does not guarantee perfect knowledge of e . If the representation r is incorrect, incomplete, or otherwise defective, then the best possible description of r will still inherit those representational limitations. Thus, perfect knowledge of an entity requires not only a minimal description, but also an admissible representation.

¹Occam's razor concerns the number of assumptions required by an explanation, not merely the number of symbols used to express it.

Notation 4.1. We denote by d_r^* the fact that the description d is a minimal description of the representation r .

In general, the minimal description of a representation need not be unique. A given representation r may admit several distinct descriptions of the same shortest length, corresponding to different but equally economical ways of reconstructing it. This non-uniqueness is not exceptional: different Turing machines, different encodings of the same computational strategy, or different decompositions between regularities and residual input may all yield descriptions that are optimal in length.

For this reason, it is useful to consider not just one minimal description, but the full collection of all optimal descriptions of r . This set captures the entire class of shortest valid descriptions associated with the representation and provides the natural reference point for later notions such as descriptive deficiency, surplus, and mismodel.

Definition 4.2.2 Let $\mathcal{D}_r$ be the set of descriptions of a representation $r \in \mathcal{R}_{\mathcal{E}}$. We define the *set of minimal descriptions* for r , denoted by $\mathcal{D}_r^*$, as the subset of $\mathcal{D}_r$ consisting of all minimal descriptions of r .

That is, $\mathcal{D}_r^*$ contains precisely those descriptions in $\mathcal{D}_r$ having minimum length.

Unfortunately, the set of minimal descriptions of a representation is generally unknown, and as Proposition 4.2.1 shows, there exists no algorithm to compute it. In practice, we must rely on approximations to estimate how far our current best description is from a minimal one, that is, to quantify how much we do not know about a particular representation of an entity (see Chapter 10).

Proposition 4.2.1 Let $r \in \mathcal{R}$ be a representation and let d_r^* be a minimal description of r . Then we have $l(d_r^*) = K(r)$.

Proof. Apply Definition 20.1.2 and note that the Turing machines TM used in descriptions of the form $\langle TM, a \rangle$ are required to be prefix-free. ■

The actual length of a description $l(d)$ for a representation r depends on the specific encoding of Turing machines used. This encoding method is determined by the chosen description function Γ . Fortunately, if we replace our description function with a different one, the length of perfect descriptions remains essentially unchanged, up to an additive constant that does not depend on the representation itself.

Corollary 4.2.2 Let $r \in \mathcal{R}$ be a representation, and let Γ and $\dot{\Gamma}$ be two different description functions. Let d_r^* be a minimal description of r under Γ , and $\dot{d}_r^*$ a minimal description under $\dot{\Gamma}$. Then $|l(d_r^*) - l(\dot{d}_r^*)| \leq c$, where c is a constant

that does not depend on r .

Proof. Apply Proposition 4.2.1 and Theorem 20.1.1. ■

In practice, within the theory of nescience, we are often not interested in computing the exact numerical value of the nescience associated with an entity for a given description–representation pair. Rather, our goal is to determine a weaker but sufficient notion: the relative ordering of different pairs of descriptions and representations according to their nescience. From this perspective, the specific details of the universal Turing machine used to compute description lengths are not essential². Accordingly, for the remainder of this book we assume that the description function Γ is fixed to a reference universal Turing machine. Alternatively, the reader may interpret all theorems in this book involving the length of the shortest models as holding up to an additive constant that does not depend on the topics themselves.

A central consequence of Proposition 4.2.1 is that minimal descriptions are not only shortest among all valid descriptions of a representation, but also algorithmically irreducible as strings in their own right. This fact is one of the key elements of the theory of nescience, because it connects scientific understanding with incompressibility: once a representation has been described in an optimal way, the resulting description cannot itself be simplified any further, except by a fixed additive constant.

Theorem 4.2.3 Let d_r^* be a minimal description of a representation r . Then d_r^* is incompressible, in the sense that

$$K(d_r^*) \geq l(d_r^*) - c,$$

for some constant c independent of r .

Proof. By Proposition 4.2.1, the length of a minimal description satisfies

$$l(d_r^*) = K(r),$$

up to an additive constant fixed by the choice of universal Turing machine. Suppose that d_r^* were compressible, that is, that

$$K(d_r^*) < l(d_r^*) - c$$

for arbitrarily large c . Then there would exist a program strictly shorter than d_r^* that outputs d_r^* . By composing this program with a fixed universal procedure that,

²Do not confuse the internal workings of the universal Turing machine Γ that maps descriptions to representations, which are not of interest here, with the internal workings of the universal oracle Turing machine δ that maps representations to entities, which are of interest, as understanding this mechanism is crucial to understanding how things work.

given d_r^* , executes the description and outputs r , one would obtain a description of r shorter than d_r^* , contradicting the minimality of d_r^* . Therefore, d_r^* must be incompressible up to an additive constant. ■

The significance of this corollary is substantial. It shows that minimal descriptions represent a limit of scientific understanding: once such a description has been reached, no further shortening is possible without losing the ability to reconstruct the representation. In this sense, a minimal description contains no dispensable structure. It also clarifies why minimal descriptions are theoretically so important in the theory of nescience. They do not merely provide the shortest way of generating a representation; they mark the point at which explanatory compression has been exhausted.

This result also explains the deep connection between perfect knowledge and randomness that underlies the theory. A minimal description is not random in the everyday sense of being meaningless or patternless, but it is random in the precise algorithmic sense of being incompressible. Thus, when scientific understanding reaches its ideal limit, the corresponding description can no longer be shortened: it becomes algorithmically irreducible. This is one of the reasons why the theory of nescience identifies the frontier of perfect understanding with incompressibility.

The converse of Theorem 4.2.3 does not generally hold: a description can be random without being the shortest possible one (incompressibility does not imply minimality). That is, we may have a description d of a representation r such that $l(d) = K(d)$, yet $l(d_r^*) < l(d)$.

■ **Example 4.1** Consider a deep neural network with an input layer of one thousand nodes, ten hidden layers of fifty thousand nodes each, and an output layer of one thousand nodes. Suppose the network is trained to output a fixed string of one thousand 1's for any given input. The Kolmogorov complexity of a string encoding of the trained neural network is much greater than that of the output string itself, which consists of one thousand identical bits. ■

The concept of a minimal description can be generalized from individual representations to entire entities. This generalization allows us to study the nature and properties of the entities themselves.

Definition 4.2.3 Let $\mathcal{D}_e$ be the set of descriptions of an entity $e \in \mathcal{E}$. We define the *set of perfect descriptions* of the entity e , denoted by $\mathcal{D}_e^*$, as the subset of $\mathcal{D}_e$ consisting of perfect descriptions. The elements of $\mathcal{D}_e^*$ are denoted by d_e^* .

The following proposition shows that the perfect descriptions of an entity are exactly the perfect descriptions of its representations.

Proposition 4.2.4 For every entity $e \in \mathcal{E}$,

$$\mathcal{D}_e^* = \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r^*.$$

Proof. By definition, a description d belongs to $\mathcal{D}_e^*$ if and only if d is a perfect description of the entity e . This means that d is a perfect description of some representation $r \in \mathcal{R}_e$. Therefore, there exists $r \in \mathcal{R}_e$ such that

$$d \in \mathcal{D}_r^*.$$

Hence,

$$\mathcal{D}_e^* \subseteq \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r^*.$$

Conversely, suppose that

$$d \in \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r^*.$$

Then there exists some $r \in \mathcal{R}_e$ such that $d \in \mathcal{D}_r^*$. By definition, d is a perfect description of the representation r . Since r is a representation of e , it follows that d is a perfect description of the entity e , and therefore

$$d \in \mathcal{D}_e^*.$$

Thus,

$$\bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r^* \subseteq \mathcal{D}_e^*.$$

Combining both inclusions, we conclude that

$$\mathcal{D}_e^* = \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r^*.$$

■

This proposition makes explicit that perfect descriptions of an entity are always mediated by its representations. In particular, if an entity admits several representations, then its set of perfect descriptions is obtained by collecting the perfect descriptions associated with each of them. Thus, as in the case of valid descriptions, the descriptive study of an entity is organized through the representations by which that entity is encoded.

Technically speaking, we could have descriptions that are longer than the representations they describe, that is, descriptions that do not compress the representations. However, they are epistemically uninformative, since they fail to achieve any compression.

Definition 4.2.4 Let $r \in \mathcal{B}^*$ be a representation, and $d \in \mathcal{D}_r$ one of its descriptions. If $l(d) \geq l(r)$, we say that d is a *pleonastic description* of the representation r .

■ **Example 4.2** Consider the set of all possible finite graphs. Since graphs are abstract mathematical objects, we must represent them as strings, for instance, using a binary encoding of their adjacency matrices (see Section 16.5 for an introduction to graphs). The description $d = \langle TM, r \rangle$, where r is the representation of a graph and TM is a Turing machine that simply halts, belongs to $\mathcal{D}_r$ because $TM(r) = r$. However, this description is of limited interest, as it is likely not the shortest possible description of r . ■

It may happen that there is no shorter possible description of a representation than the representation itself. This occurs when the representation is an incompressible string. As discussed in Section 20.6, the overwhelming majority of strings are incompressible. Conducting research on incompressible representations is unproductive, as it is not possible to find shorter models for such representations.

An interesting case arises when all the descriptions in $\mathcal{D}_e$ are pleonastic, that is, there exist no models shorter than the representation for any of the possible representations of the entity. This situation would occur if all representations of the entity e are random strings. In such a case, scientific research would be fundamentally limited, as it would be impossible to find a suitable model for e . Our ability to understand and make predictions about e would then be constrained by the length of its incompressible representations.

4.3 Extended Descriptions

The set of valid descriptions $\bar{\mathcal{D}}_r$ associated with a representation r is too restrictive for practical scientific work. In principle, the elements of $\bar{\mathcal{D}}_r$ are exactly those valid descriptions that generate r itself. In practice, however, the descriptions available to us rarely reconstruct the intended representation exactly. Instead, they usually generate another string r' which, ideally, is close enough to r to support explanation, prediction, or approximation, but need not coincide with it. For this reason, scientific practice cannot usually proceed by working only with the exact descriptions in $\bar{\mathcal{D}}_r$.

This motivates the introduction of *extended descriptions*. Extended descriptions are the descriptions actually used in practice to model a representation, even when their output differs from the target representation itself. They are context-dependent and need not be exact; their scientific relevance lies not in guaranteed correctness, but in the fact that they provide a usable approximation to the intended representation. This distinction will be essential

later, when we study the mismodel of descriptions (see Chapter 6).

Definition 4.3.1 Let $r \in \mathcal{R}$ be a representation. We define the set of *extended descriptions* of r , denoted by $\mathcal{D}_r$, as

$$\mathcal{D}_r = \{d \in \mathcal{D} : K(r \mid \Gamma(d)) \leq K(r) - O(1)\}.$$

This definition means that the output $\Gamma(d)$ of the description d contains non-trivial information about the representation r . Indeed, the inequality

$$K(r \mid \Gamma(d)) \leq K(r) - O(1)$$

states that, once $\Gamma(d)$ is given, the amount of information still needed to reconstruct r is strictly smaller, up to an additive constant, than the amount of information required to describe r from scratch. In this sense, $\Gamma(d)$ does not reproduce r exactly, but it does capture part of its structure.

The following proposition shows that exact descriptions are a particular case of extended descriptions. In other words, every description that reconstructs the representation exactly also contains non-trivial information about it. The notion of extended description therefore generalizes that of exact description by allowing outputs that do not coincide with r , but still contribute genuine information toward its reconstruction.

Proposition 4.3.1 For every representation $r \in \mathcal{R}$, we have that $\bar{\mathcal{D}}_r \subseteq \mathcal{D}_r$.

Proof. Let $d \in \bar{\mathcal{D}}_r$. By definition of $\bar{\mathcal{D}}_r$, we have $\Gamma(d) = r$. Therefore,

$$K(r \mid \Gamma(d)) = K(r \mid r) = O(1).$$

Since $K(r) - O(1)$ differs from $K(r)$ only by an additive constant, it follows that

$$K(r \mid \Gamma(d)) \leq K(r) - O(1).$$

Hence $d \in \mathcal{D}_r$. Therefore, $\bar{\mathcal{D}}_r \subseteq \mathcal{D}_r$. ■

The following example confirms that the notion of extended description is not merely a reformulation of exact description. It genuinely enlarges the descriptive space by admitting outputs that do not reconstruct the representation exactly, but still contain non-trivial information about it.

■ **Example 4.3** Let $x \in \mathcal{B}^*$ be a string of non-trivial complexity, and consider the representation $r = xx$. Let $d \in \mathcal{D}$ be a valid description such that $\Gamma(d) = x$. Then $d \notin \bar{\mathcal{D}}_r$, since $\Gamma(d) \neq r$.

On the other hand, from x one can reconstruct $r = xx$ by a fixed computable procedure, so

$$K(r \mid \Gamma(d)) = K(xx \mid x) = O(1).$$

Since $K(r) = K(xx) = K(x) + O(1)$, and $K(x)$ is non-trivial, we have

$$K(r \mid \Gamma(d)) \leq K(r) - O(1).$$

Hence $d \in \mathcal{D}_r$. Therefore, $d \in \mathcal{D}_r \setminus \bar{\mathcal{D}}_r$, which shows that, in this case, $\bar{\mathcal{D}}_r \subsetneq \mathcal{D}_r$. ■

Unlike the case of exact descriptions, the sets of extended descriptions associated with different representations need not be disjoint. The reason is that an extended description is not required to reconstruct its target representation exactly; it only needs to produce an output containing non-trivial information about it. A single description may therefore produce a string that is informative about more than one representation.

Proposition 4.3.2 In general, there exist representations $r_1, r_2 \in \mathcal{R}$ with $r_1 \neq r_2$ such that

$$\mathcal{D}_{r_1} \cap \mathcal{D}_{r_2} \neq \emptyset.$$

Proof. Let $r_1, r_2 \in \mathcal{R}$ be two distinct representations that share a non-trivial common structure, and let $s \in \mathcal{B}^*$ be a string encoding that common structure in such a way that

$$K(r_1 \mid s) \leq K(r_1) - O(1) \quad \text{and} \quad K(r_2 \mid s) \leq K(r_2) - O(1).$$

Let $d \in \mathcal{D}$ be any valid description such that $\Gamma(d) = s$. Then, by definition of extended description,

$$d \in \mathcal{D}_{r_1} \quad \text{and} \quad d \in \mathcal{D}_{r_2}.$$

Hence $d \in \mathcal{D}_{r_1} \cap \mathcal{D}_{r_2}$, and therefore $\mathcal{D}_{r_1} \cap \mathcal{D}_{r_2} \neq \emptyset$. ■

This proposition shows that extended descriptions classify representations in a weaker way than exact descriptions. Exact descriptions are tied to a unique output and therefore to a unique representation, whereas extended descriptions may overlap whenever the same output captures structure that is informative about several different representations. In this sense, extended descriptions reflect partial explanatory overlap between representations.

■ **Example 4.4** Consider two different representations r_1 and r_2 of climate data from two nearby cities. Since the cities are distinct, the representations are not equal: the measured temperatures, rainfall values, and local variations differ. However, both datasets may share the same large-scale seasonal pattern. Let d be a description whose output $\Gamma(d)$ is a string encoding that common yearly cycle. Then $\Gamma(d)$ does not reconstruct either dataset exactly, but it contains non-trivial

information about both of them, since knowing the seasonal pattern reduces the amount of information still needed to reconstruct each full dataset. Therefore, the same description d may belong simultaneously to $\mathcal{D}_{r_1}$ and to $\mathcal{D}_{r_2}$. ■

Since extendedness is defined through the output $\Gamma(d)$, it is natural to expect that two descriptions with the same output should either both be extended descriptions of r or neither of them be.

Proposition 4.3.3 Let $r \in \mathcal{R}$, and let $d_1, d_2 \in \mathcal{D}$ be such that $\Gamma(d_1) = \Gamma(d_2)$. Then

$$d_1 \in \mathcal{D}_r \iff d_2 \in \mathcal{D}_r.$$

Proof. If $\Gamma(d_1) = \Gamma(d_2)$, then

$$K(r \mid \Gamma(d_1)) = K(r \mid \Gamma(d_2)) + O(1).$$

Hence

$$K(r \mid \Gamma(d_1)) \leq K(r) - O(1)$$

if and only if

$$K(r \mid \Gamma(d_2)) \leq K(r) - O(1).$$

Therefore $d_1 \in \mathcal{D}_r$ if and only if $d_2 \in \mathcal{D}_r$. ■

This proposition shows that extendedness is a property of the information content of the output string, not of the particular implementation used to generate it.

A description whose output is at least as informative as that of an extended description should itself remain extended.

Proposition 4.3.4 Let $r \in \mathcal{R}$, and let $d_1, d_2 \in \mathcal{D}$. If $d_1 \in \mathcal{D}_r$ and $K(\Gamma(d_1) \mid \Gamma(d_2)) = O(1)$, then $d_2 \in \mathcal{D}_r$.

Proof. Since $d_1 \in \mathcal{D}_r$, we have

$$K(r \mid \Gamma(d_1)) \leq K(r) - O(1).$$

By the chain property of conditional Kolmogorov complexity,

$$K(r \mid \Gamma(d_2)) \leq K(r \mid \Gamma(d_1)) + K(\Gamma(d_1) \mid \Gamma(d_2)) + O(1).$$

Using the hypothesis $K(\Gamma(d_1) \mid \Gamma(d_2)) = O(1)$, we obtain

$$K(r \mid \Gamma(d_2)) \leq K(r) - O(1).$$

Hence $d_2 \in \mathcal{D}_r$. ■

This proposition formalizes the idea that extendedness is preserved under informative refinement: any description whose output allows one to recover, up to a constant, the output of an already extended description is itself an extended description.

We can lift the notion of extended description from individual representations to the entities they represent. Intuitively, an extended description of an entity should be any valid description whose output contains non-trivial information about at least one representation of that entity. In this way, the notion of extended description is no longer tied to a single representation, but to the entity itself through the family of its possible representations.

Definition 4.3.2 Let $e \in \mathcal{E}$ be an entity. We define the *set of extended descriptions* of e , denoted by $\mathcal{D}_e$, as

$$\mathcal{D}_e = \{d \in \mathcal{D} : \exists r \in \mathcal{R}_e \text{ such that } K(r \mid \Gamma(d)) \leq K(r) - O(1)\}.$$

Thus, a description belongs to $\mathcal{D}_e$ whenever its output contains non-trivial information about some representation of the entity e . In particular, membership in $\mathcal{D}_e$ does not require the description to reconstruct any representation of e exactly. It is enough that the output of the description help to account for at least one such representation. This makes $\mathcal{D}_e$ the natural descriptive analogue, at the level of entities, of the set $\mathcal{R}_e$ of extended representations. It also provides the appropriate domain for studying entities whose descriptions are only partial, approximate, or informative in a weaker sense than exact reconstruction.

The following proposition shows that the extended descriptions of an entity are obtained by collecting the extended descriptions of its individual representations. In this sense, the descriptive study of an entity is still mediated by the representations through which that entity is encoded, even though we now work with the broader notion of extended description.

Proposition 4.3.5 For every entity $e \in \mathcal{E}$,

$$\mathcal{D}_e = \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r.$$

Proof. Let $d \in \mathcal{D}_e$. By Definition 4.3.2, there exists $r \in \mathcal{R}_e$ such that

$$K(r \mid \Gamma(d)) \leq K(r) - O(1).$$

By definition of $\mathcal{D}_r$, this means that $d \in \mathcal{D}_r$. Hence

$$d \in \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r.$$

Therefore,

$$\mathcal{D}_e \subseteq \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r.$$

Conversely, let

$$d \in \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r.$$

Then there exists some $r \in \mathcal{R}_e$ such that $d \in \mathcal{D}_r$. By definition of $\mathcal{D}_r$, we have

$$K(r \mid \Gamma(d)) \leq K(r) - O(1).$$

Since $r \in \mathcal{R}_e$, it follows from Definition 4.3.2 that $d \in \mathcal{D}_e$. Hence

$$\bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r \subseteq \mathcal{D}_e.$$

Combining both inclusions, we conclude that

$$\mathcal{D}_e = \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r.$$

■

This proposition makes explicit that extended descriptions of an entity do not form an independent descriptive layer. Rather, they are organized through the different representations of the entity: a description is an extended description of e precisely because it contains non-trivial information about at least one representation of e .

The next proposition shows that an entity admits extended descriptions exactly when it admits representations.

Proposition 4.3.6 For every entity $e \in \mathcal{E}$,

$$\mathcal{D}_e = \emptyset \iff \mathcal{R}_e = \emptyset.$$

Proof. If $\mathcal{D}_e \neq \emptyset$, then there exists $d \in \mathcal{D}_e$. By Definition 4.3.2, there exists $r \in \mathcal{R}_e$ such that

$$K(r \mid \Gamma(d)) \leq K(r) - O(1).$$

Hence $\mathcal{R}_e \neq \emptyset$.

Conversely, suppose that $\mathcal{R}_e \neq \emptyset$, and let $r \in \mathcal{R}_e$. Since every exact description is an extended description by Proposition 4.3.1, and since r admits at least one exact description, it follows that $\mathcal{D}_r \neq \emptyset$. Therefore, by Proposition 4.3.5,

$$\mathcal{D}_e = \bigcup_{r \in \mathcal{R}_e} \mathcal{D}_r \neq \emptyset.$$

Thus $\mathcal{D}_e \neq \emptyset$.

■

This proposition shows that extended describability and representability coincide at the level of entities. In particular, if an entity admits no representation, then it admits no extended description either.

Hypotheses

From a practical perspective, scientific activity consists in discovering representations and descriptions that admit meaningful interpretations and can be used to solve concrete problems, such as explanation, optimization, or prediction. From a more theoretical perspective, the aim of science may be viewed as the construction of representation-description pairs that are consistent with the behavior of an unknown abstract oracle, which mediates the relationship between strings and entities.

Notation 4.2. We denote by $\mathcal{H}$ the set of hypotheses,

$$\mathcal{H} = \{ (r, d) \in \mathcal{R} \times \mathcal{D} : d \in \mathcal{D}_r \}.$$

Its elements $x = (r, d)$ are called hypotheses. We also write

$$\mathcal{H}_r = \{ (r, d) \in \mathcal{H} : d \in \mathcal{D}_r \}, \quad \mathcal{H}_e = \{ (r, d) \in \mathcal{H} : r \in \mathcal{R}_e, d \in \mathcal{D}_r \}.$$

Thus, a hypothesis is simply a convenient way of referring simultaneously to a representation and to one of its associated descriptions.

4.4 Models

The descriptions introduced in Section 4.1 are computable objects capable of reconstructing representations. Each description d has the form $d = \langle TM, a \rangle$, where the Turing machine TM encodes the regularities present in the representation, and the input string a contains the remaining incompressible part. In this sense, every description can be understood as a decomposition into a structured component, which captures regularity, and a residual component, which accounts for what cannot be further compressed.

However, the space of all possible descriptions is extremely large and lacks internal organization. In scientific practice, researchers do not attempt to explore this space exhaustively. Instead, they restrict their attention to particular families of descriptions characterized by shared structural assumptions. These restricted families correspond to what scientists commonly call *models*.

Intuitively, a model expresses a hypothesis about the regularities underlying a representation. By fixing a particular structure, one limits the class of admissible descriptions to those that conform to that structure. The remaining variability is then captured by the input component of the description, which specifies the particular instance of the model.

Definition 4.4.1 Let $\mathcal{D}$ be the set of valid descriptions. We define a *model*

class as a subset

$$\mathcal{M} \subseteq \mathcal{D}.$$

Each element of a model class should be a description that satisfies a common structural constraint. This constraint is precisely what makes models useful in practice: it restricts the space of possible descriptions to those that embody a particular kind of regularity, thereby making search, explanation, and prediction more manageable.

A particularly important case is that of parametric models, where the variability within the model class is controlled by a parameter space.

Definition 4.4.2 We define *parametric model* as a model class of the form

$$\mathcal{M} = \{d_\theta : \theta \in \Theta\},$$

where Θ is a parameter space and each $d_\theta \in \mathcal{D}$ is a description determined by the parameter θ .

In this setting, the general structure $d = \langle TM, a \rangle$ admits a natural reinterpretation. The Turing machine TM is no longer an arbitrary program, but a member of a restricted family encoding a specific type of model, while the input a represents the parameters of that model. Thus, descriptions in a parametric model class can be written as $d = \langle TM_m, \theta \rangle$, where TM_m encodes the structural form of the model m , and θ specifies a particular instance within that structure.

■ **Example 4.5** Consider a representation composed of a set of observed data points (x_i, y_i) in the plane. A common modeling assumption is that the data follow a linear relationship. This leads to the parametric model class of linear functions.

Let $\Theta = \mathbb{R}^2$, and for each parameter vector $\theta = (w, b)$, define the description

$$d_\theta = \langle TM_{\text{lin}}, \theta \rangle,$$

where TM_{lin} is a Turing machine that, on input (w, b) , outputs the representation of the function

$$f_\theta(x) = wx + b.$$

The model class is therefore

$$\mathcal{M} = \{d_\theta : \theta \in \mathbb{R}^2\}.$$

In this example, the machine TM_{lin} encodes the structural assumption that the relationship between x and y is linear, while the input $\theta = (w, b)$ specifies the

particular instance of the model that best fits the data. The regularity of the data is captured by the linear form, and the remaining information is contained in the parameter values. ■

Restricting attention to model classes therefore provides a principled way to organize the space of descriptions. Instead of searching over all possible descriptions, scientific practice proceeds by selecting a class of models, and then adjusting the parameters within that class. The success of a model then depends both on the adequacy of its structural assumptions and on the choice of parameters used to instantiate it.

A model class also induces an *inductive bias*. By restricting the space of admissible descriptions, one restricts the family of regularities that can be captured. This bias is not merely a computational convenience: it determines, in part, what kinds of explanations are even available within the chosen framework.

Definition 4.4.3 Let $\mathcal{M} \subseteq \mathcal{D}$ be a model class. We define the *expressive set* of $\mathcal{M}$ by

$$\mathcal{R}_{\mathcal{M}} = \{r \in \mathcal{R} : \exists d \in \mathcal{M}, \Gamma(d) = r\}.$$

The set $\mathcal{R}_{\mathcal{M}}$ contains precisely those representations that can be generated by at least one description in the model class $\mathcal{M}$. It therefore characterizes the expressive power of the class.

Proposition 4.4.1 Let $\mathcal{M}_1, \mathcal{M}_2 \subseteq \mathcal{D}$ be model classes such that $\mathcal{M}_1 \subseteq \mathcal{M}_2$. Then

$$\mathcal{R}_{\mathcal{M}_1} \subseteq \mathcal{R}_{\mathcal{M}_2}.$$

Proof. Let $r \in \mathcal{R}_{\mathcal{M}_1}$. By Definition 4.4.3, there exists $d \in \mathcal{M}_1$ such that $\Gamma(d) = r$. Since $\mathcal{M}_1 \subseteq \mathcal{M}_2$, we also have $d \in \mathcal{M}_2$, and therefore $r \in \mathcal{R}_{\mathcal{M}_2}$. ■

This proposition shows that enlarging a model class can only increase, or at least preserve, the set of representations that can be described within it. In this sense, more expressive model classes correspond to larger subsets of $\mathcal{D}$.

The converse side of this observation is that restrictive model classes may fail to capture some representations altogether.

Proposition 4.4.2 In general, there exist representations $r \in \mathcal{R}$ such that

$$\mathcal{D}_r \neq \emptyset \quad \text{but} \quad \mathcal{D}_r \cap \mathcal{M} = \emptyset.$$

Proof. A model class $\mathcal{M}$ is, in general, a proper subset of $\mathcal{D}$. Hence there may exist valid descriptions $d \in \mathcal{D} \setminus \mathcal{M}$. Let $r = \Gamma(d)$. Then $d \in \mathcal{D}_r$, so $\mathcal{D}_r \neq \emptyset$. However, there need not exist any description in $\mathcal{M}$ that generates r . In that case,

$$\mathcal{D}_r \cap \mathcal{M} = \emptyset.$$



This proposition formalizes an important limitation of model classes: by imposing structure, they simplify the search for descriptions, but they may also exclude some representations that are describable in principle.

In general, descriptions belonging to a model class need not be minimal, since the model class constitutes a restricted subset of the space of all descriptions. A minimal description of a representation may rely on regularities that are not expressible within the chosen model class, and therefore may not belong to it. As a consequence, even if a representation admits a shorter description, that description may lie outside the model class under consideration.

Restricting attention to a model class therefore introduces a trade-off between tractability and optimality. A more restrictive class simplifies the search for descriptions by imposing structure, but may fail to capture relevant regularities or exclude minimal descriptions. A more expressive class increases representational power, but usually enlarges the search space and makes the modeling problem harder.

Proposition 4.4.3 Let $r \in \mathcal{R}$ be a representation, and let

$$d = \langle TM_m, \theta \rangle \in \mathcal{M}$$

be a description of r belonging to a parametric model class. Then there exists a constant c such that

$$K(r) \leq \ell(TM_m) + \ell(\theta) + c.$$

Proof. Since $d = \langle TM_m, \theta \rangle$ is a valid description of r , we have

$$\Gamma(d) = TM_m(\theta) = r.$$

Therefore, there exists a program of length at most $\ell(TM_m) + \ell(\theta) + c$ that outputs r , where the constant c accounts for the fixed overhead of encoding the pairing and simulating the universal Turing machine. By definition of Kolmogorov complexity,

$$K(r) \leq \ell(TM_m) + \ell(\theta) + c.$$



This inequality shows that the length of a model-based description provides an upper bound on the Kolmogorov complexity of the representation. However, this bound need not be tight. A minimal description of r may exploit regularities that are not captured by the chosen model class, and therefore be strictly shorter than any description of the form $\langle TM_m, \theta \rangle$ with $TM_m \in \mathcal{M}$.

Corollary 4.4.4 Descriptions in a model class need not be minimal.

Proof. Let $d = \langle TM_m, \theta \rangle$ be a description of r in the model class $\mathcal{M}$. By Proposition 4.4.3,

$$K(r) \leq \ell(TM_m) + \ell(\theta) + c.$$

A minimal description of r has length $K(r) + O(1)$, which may be strictly smaller than $\ell(TM_m) + \ell(\theta)$. Hence, d need not be a minimal description. ■

Another important point is that the same representation may admit descriptions in different model classes. This reflects a common situation in scientific practice, where the same phenomenon may be explained using distinct structural hypotheses.

Proposition 4.4.5 In general, a representation $r \in \mathcal{R}$ may admit descriptions in more than one model class. That is, there may exist model classes $\mathcal{M}_1, \mathcal{M}_2 \subseteq \mathcal{D}$ such that

$$\mathcal{D}_r \cap \mathcal{M}_1 \neq \emptyset \quad \text{and} \quad \mathcal{D}_r \cap \mathcal{M}_2 \neq \emptyset.$$

Proof. A representation may possess several distinct valid descriptions, and these descriptions may satisfy different structural constraints. Hence it is possible for one description of r to belong to $\mathcal{M}_1$ and another to belong to $\mathcal{M}_2$, yielding the stated nonempty intersections. ■

This non-uniqueness is one of the reasons why model selection is a central problem in science: different model classes may provide different, and sometimes incompatible, ways of describing the same representation.

Finally, even within a fixed parametric model class, the parameterization itself need not be unique.

Proposition 4.4.6 In general, parametric model classes need not be injective with respect to the parameterization. That is, there may exist $\theta_1 \neq \theta_2$ such that

$$\Gamma(\langle TM_m, \theta_1 \rangle) = \Gamma(\langle TM_m, \theta_2 \rangle).$$

Proof. The definition of parametric model class does not require the mapping

$$\theta \mapsto \Gamma(\langle TM_m, \theta \rangle)$$

to be injective. Hence distinct parameter values may generate the same representation. ■

This shows that redundancy may arise not only from the model class itself, but also from the chosen parameterization. Distinct parameter values may encode the same effective description of a representation, a fact that will later be relevant when discussing surplus and the simplification of descriptions.

4.5 Operations on Descriptions

Augmentation and reduction are two complementary operations on descriptions. Augmentation models the idea of enriching a description by supplying additional computable information that may be missing from the original one. Reduction models the opposite process: simplifying a description by separating out information that is not needed for the purpose at hand. Together, these operations provide a basic formal vocabulary for describing how scientific descriptions may be refined, expanded, or simplified over time.

These operations are introduced here at a purely structural level. Their purpose is not yet to determine whether a description has improved, or whether a particular transformation brings it closer to an ideal description. Rather, they formalize two elementary ways of modifying descriptions: by adding information through augmentation, and by removing information through reduction. In both cases, the result is required to remain within the domain of valid descriptions, so that the transformed object continues to be interpretable as a computable description.

This distinction is important because the same formal operation may have very different effects depending on the context. An augmentation may add genuinely useful information, but it may also introduce irrelevant detail. Likewise, a reduction may remove superfluous structure, but it may also discard information that later proves important. For this reason, augmentation and reduction are best understood at this stage as general operations on descriptions. Their role in decreasing deficiency, surplus, or mismodel will be examined in Chapter 6, once those defects have been formally defined.

Operation of Augmentation

A natural way to improve the quality of a description is by enriching it with an auxiliary description that contributes part of the information missing from the original one. We call this operation *augmentation*. Its purpose is to produce a new valid description that may better account for the representation under study. Unlike the augmentation of representations (see Section 3.9), where additional symbols are inserted directly into a string, the augmentation of descriptions is more appropriately treated as an operation on valid descriptions themselves.

Since not every auxiliary description need be suitable for augmenting every original description, we model augmentation as a partial computable operation on the set of valid descriptions.

Definition 4.5.1 We define the *augmentation operator* as a partial computable

function

$$\text{aug} : \mathcal{D} \times \mathcal{D} \rightarrow \mathcal{D}.$$

For $d, d' \in \mathcal{D}$, whenever $\text{aug}(d, d')$ is defined, the string $\text{aug}(d, d')$ is called an *augmentation* of d by d' , and d' is called an *augmentation description*.

This definition does not prescribe how the information encoded by d and d' must be combined, only that the combination be computable whenever it is defined, and that the result be a valid description. In this way, augmentation is introduced as a structural operation on descriptions, independently of the particular effect it may have on the representation produced. Questions concerning whether augmentation improves a description, reduces its deficiency, or brings it closer to a target representation will be addressed in Chapter 6.

Augmentation is a partially defined internal operation on the set of valid descriptions.

Proposition 4.5.1 If $d, d' \in \mathcal{D}$ and $\text{aug}(d, d') \downarrow$, then $\text{aug}(d, d') \in \mathcal{D}$.

Proof. This follows directly from the definition of augmentation operator. ■

Therefore, whenever an augmentation is defined, augmenting a valid description by another valid description yields a new valid description.

Since augmentation is computable on its domain, it can be iterated whenever each successive step is defined.

Proposition 4.5.2 Let $d, d_1, \dots, d_n \in \mathcal{D}$, with $n \geq 1$. If all intermediate augmentations are defined, that is, if

$$\text{aug}(d, d_1) \downarrow, \quad \text{aug}(\text{aug}(d, d_1), d_2) \downarrow, \quad \dots, \quad \text{aug}(\text{aug}(\dots \text{aug}(d, d_1), \dots), d_n) \downarrow,$$

then the iterated augmentation

$$\text{aug}(\text{aug}(\dots \text{aug}(d, d_1), d_2) \dots, d_n)$$

belongs to $\mathcal{D}$.

Proof. By Proposition 4.5.1, if $\text{aug}(d, d_1) \downarrow$, then $\text{aug}(d, d_1) \in \mathcal{D}$. Repeating the same argument at each successive stage yields the result by induction on n . ■

This shows that augmentation can be applied repeatedly, thereby allowing the progressive construction of more elaborate descriptions from simpler ones, provided that each intermediate augmentation is defined.

The definition of augmentation only requires that aug be a partial computable function from $\mathcal{D} \times \mathcal{D}$ to $\mathcal{D}$. It imposes no condition requiring the output representation of $\text{aug}(d, d')$, when defined, to coincide with that of d . Therefore,

an augmentation operator may perfectly well be chosen so that, for some $d, d' \in \mathcal{D}$ such that $\text{aug}(d, d') \downarrow$, we have $\Gamma(\text{aug}(d, d')) \neq \Gamma(d)$.

■ **Example 4.6** Consider a representation r consisting of the observed trajectory of a projectile. Let

$$d = \langle TM_1, a_1 \rangle$$

be a valid description encoding the standard model of projectile motion in vacuum, where TM_1 computes the trajectory from the initial conditions stored in a_1 . This description captures the main regularity of the phenomenon, but it ignores the effect of air resistance.

Now let

$$d' = \langle TM_2, a_2 \rangle$$

be another valid description encoding a correction model for aerodynamic drag, with a_2 containing the relevant parameters, such as the drag coefficient. If the chosen augmentation operator is defined on the pair (d, d') , the augmented description

$$\text{aug}(d, d')$$

may be a new valid description that combines the vacuum trajectory model with the drag correction.

In this case, $\Gamma(\text{aug}(d, d'))$ will in general differ from $\Gamma(d)$, since the augmented description produces a trajectory modified by air resistance. The role of augmentation is therefore not merely to attach extra information to the original description, but to construct a new description whose output may better account for the representation under study. ■

An augmentation is therefore not merely a passive attachment of additional information to a description; it is a new valid description obtained by computably combining two existing ones whenever that combination is meaningful within the chosen framework. As a consequence, the resulting description may produce the same representation as the original description, but it may also produce a different one.

Operation of Reduction

An alternative way to improve the quality of a description is by removing information that is not needed for the purpose at hand. We call this operation *reduction*. Its purpose is to decompose a valid description into two parts: a retained part, which is kept as the relevant description, and a discarded part, which contains the information irrelevant or redundant. Unlike the reduction of

representations, where symbols are deleted directly from a string, the reduction of descriptions is more appropriately treated as an operation on valid descriptions themselves.

Since not every description need admit a meaningful decomposition, we model reduction as a partial computable operation on the set of valid descriptions.

Definition 4.5.2 We define the *reduction operator* as a partial computable function

$$\text{red} : \mathcal{D} \rightharpoonup \mathcal{D} \times \mathcal{D}$$

such that, whenever $\text{red}(d) = (d_1, d_2)$, the following conditions hold:

- i $\ell(d_1) \leq \ell(d)$,
- ii $\text{aug}(d_1, d_2) \downarrow \wedge \text{aug}(d_1, d_2) = d$.

The description d_1 is called a *reduction* of d , and d_2 is called the *discarded description*.

This definition does not prescribe how the decomposition of d into d_1 and d_2 is obtained, only that it be computable whenever it is defined, that the retained part d_1 be no longer than the original description, and that the original description be recoverable by augmenting d_1 with d_2 . In this way, reduction is introduced as a structural operation on descriptions, independently of the particular effect it may have on the representation produced. Questions concerning whether reduction improves a description, decreases its surplus, or preserves the represented string will be addressed in Chapter 6.

Reduction is a partially defined decomposition operation on the set of valid descriptions. Its role is to separate a description into two components: a retained part, which is meant to preserve the essential content of the original description, and a discarded part, which contains the information removed by the reduction. The first basic property to verify is that, whenever a reduction is defined, both resulting components remain within the space of valid descriptions.

Proposition 4.5.3 If $d \in \mathcal{D}$ and $\text{red}(d) \downarrow$, then there exist $d_1, d_2 \in \mathcal{D}$ such that $\text{red}(d) = (d_1, d_2)$.

Proof. This follows directly from the definition of reduction operator. ■

Hence, whenever a reduction is defined, it does not produce arbitrary strings, but decomposes a valid description into two components that are themselves valid descriptions. This is important because it ensures that reduction remains internal to the descriptive framework: both the retained part and the discarded part continue to be legitimate objects of study.

The next proposition records the basic size condition built into the notion of reduction. Since reduction is intended to simplify or compress a description by

discarding information that may later be restored, it is natural to require that the retained part should never be longer than the original description.

Proposition 4.5.4 If $d \in \mathcal{D}$ and $\text{red}(d) = (d_1, d_2)$, then $\ell(d_1) \leq \ell(d)$.

Proof. This is one of the defining conditions of the reduction operator. ■

Thus, the retained part of a reduction is never longer than the original description. This formalizes the idea that reduction moves, at least weakly, in the direction of simplification: even when it does not strictly shorten the description, it never increases its size.

A second basic property is that reduction is compatible with augmentation in the sense that the original description can be recovered from the retained part and the discarded part. In this way, reduction does not destroy information irreversibly, but reorganizes it into two components whose recombination restores the initial description.

Proposition 4.5.5 If $d \in \mathcal{D}$ and $\text{red}(d) = (d_1, d_2)$, then

$$\text{aug}(d_1, d_2) \downarrow \wedge \text{aug}(d_1, d_2) = d.$$

Proof. This is again one of the defining conditions of the reduction operator. ■

This proposition formalizes the intended interpretation of reduction: the discarded part is not arbitrary residue, but precisely the information that must be restored, via augmentation, in order to recover the original description. In this sense, reduction and augmentation are complementary operations: one separates a description into components, while the other recombines them.

As an immediate consequence, every defined reduction determines a factorization of the original description through the augmentation operator.

Corollary 4.5.6 If $d \in \mathcal{D}$ and $\text{red}(d) = (d_1, d_2)$, then d belongs to the image of aug .

Proof. By Proposition 4.5.5, $d = \text{aug}(d_1, d_2)$ with $\text{aug}(d_1, d_2) \downarrow$. Therefore d lies in the image of aug . ■

This shows that every reducible description can be reconstructed by augmenting its retained part with its discarded part. In particular, reduction does not take us outside the space generated by augmentation, but exhibits the original description as one that can be built back from simpler components.

Reduction can also be iterated whenever the retained part admits a further reduction. This makes it possible to study successive stages of simplification, in which a description is progressively decomposed into a sequence of retained and discarded parts.

Proposition 4.5.7 Let $d \in \mathcal{D}$. Suppose

$$\text{red}(d) = (d_1, e_1), \quad \text{red}(d_1) = (d_2, e_2), \quad \dots, \quad \text{red}(d_{n-1}) = (d_n, e_n)$$

are all defined. Then

$$\ell(d_n) \leq \ell(d_{n-1}) \leq \dots \leq \ell(d_1) \leq \ell(d).$$

Proof. By Proposition 4.5.4,

$$\ell(d_1) \leq \ell(d).$$

Applying the same proposition to each subsequent reduction yields

$$\ell(d_2) \leq \ell(d_1), \quad \dots, \quad \ell(d_n) \leq \ell(d_{n-1}).$$

Chaining these inequalities gives the result. ■

This shows that repeated reduction yields a non-increasing sequence of retained-description lengths, and therefore behaves monotonically with respect to description size. In practical terms, iterative reduction provides a formal way of tracing successive simplifications of a description while preserving, at each step, the possibility of recovering the preceding stage by reintroducing the discarded information.

In general, it is not the case that, if $\text{red}(d) = (d_1, d_2)$, then $\Gamma(d_1) = \Gamma(d)$. The definition of reduction requires only that $d_1, d_2 \in \mathcal{D}$, that $\ell(d_1) \leq \ell(d)$, and that $\text{aug}(d_1, d_2) = d$ when defined. It imposes no condition requiring d_1 and d to produce the same output under Γ . Therefore, a reduction operator may perfectly well be chosen so that, for some $d \in \mathcal{D}$ with $\text{red}(d) = (d_1, d_2)$, $\Gamma(d_1) \neq \Gamma(d)$.

Thus, a reduction is not merely the deletion of inert material from a description. It is a decomposition into a retained part and a discarded part, and the retained part may or may not describe the same representation as the original description.

■ **Example 4.7** Consider a valid description

$$d = \langle TM, a \rangle$$

encoding a trained machine-learning pipeline for credit-risk prediction. The machine TM specifies the whole predictive procedure: preprocessing, feature transformations, and a large ensemble model; the input a contains the learned parameters. Suppose that, after training, it is observed that many components of the model contribute little to the final predictions. For example, some engineered features may be nearly redundant, or many trees in the ensemble may have very small marginal effect.

A reduction operator may then be defined so as to separate the original description into two parts,

$$\text{red}(d) = (d_1, d_2),$$

where d_1 is a reduced description encoding the simplified pipeline that keeps only the most relevant features and the most influential part of the ensemble, while d_2 encodes the discarded components together with the information needed to restore them. In this setting, d_1 is shorter than d , so

$$\ell(d_1) \leq \ell(d),$$

and the original model can be reconstructed by augmentation:

$$\text{aug}(d_1, d_2) = d.$$

The output of the reduced description, $\Gamma(d_1)$, need not coincide with the output of the original description, $\Gamma(d)$. The simplified model may produce slightly different scores or classifications, since it no longer uses all the components of the original pipeline. Nevertheless, d_1 may still be preferable in practice if it is easier to interpret, faster to evaluate, or less prone to overfitting.

This illustrates the intended role of reduction in machine learning: one starts from a complex trained description, removes components judged unnecessary for the purpose at hand, and obtains a simpler retained description together with a discarded part that allows the original description to be recovered exactly. ■

In practical terms, reduction corresponds to the common machine-learning strategy of simplifying a trained model while keeping track of the information needed to recover the original one.

4.6 Descriptions for Joint Representations

In Section 3.9, we introduced the concept of a joint representation ts , formed by combining two individual representations t and s . In this section, we aim to study how the length of the minimal description of a joint representation relates to the lengths of the minimal descriptions of the individual representations.

The length of the minimal description of a joint representation is greater than or equal to the length of the minimal description of either individual representation. In other words, the more information a representation contains, the longer it takes to describe.

Proposition 4.6.1 Let $t, s \in \mathcal{R}$ be two representations, and let d_t^* , d_s^* , and d_{ts}^* denote the minimal descriptions of the representations t , s , and the joint representation ts , respectively. Then: $\ell(d_{ts}^*) \geq \ell(d_t^*)$ and $\ell(d_{ts}^*) \geq \ell(d_s^*)$.

Proof. The inequality $l(d_{ts}^*) \geq l(d_t^*)$ is equivalent to $K(ts) \geq K(t)$. The result then follows from Proposition 20.3.3. ■

Intuitively, adding more information to a representation is beneficial if the additional information is relevant to describing the entity of interest. However, including irrelevant information leads to unnecessarily long models. Recall that joining representations can serve either to concatenate two partial representations of the same entity or to enrich a representation by adding missing symbols.

If the selected representations partially overlap, we can exploit this redundancy to produce a joint description that is shorter than the mere concatenation of the individual descriptions. In the worst-case scenario, the minimal description of a joint representation would be equal in length to the sum of the minimal descriptions of the individual representations.

Proposition 4.6.2 Let $t, s \in \mathcal{R}$ be two representations, and let d_t^* , d_s^* , and d_{ts}^* denote the minimal descriptions of the representations t , s , and the joint representation ts , respectively. Then: $l(d_{ts}^*) \leq l(d_t^*) + l(d_s^*)$.

Proof. The inequality $l(d_{ts}^*) \leq l(d_t^*) + l(d_s^*)$ is equivalent to $K(ts) \leq K(t) + K(s)$. The result follows from Proposition 20.3.2. ■

One interpretation of Proposition 4.6.2 is that including redundant information in the representation of an entity does not hinder our ability to find its shortest possible description. From the perspective of compression, redundancy can be eliminated during the modeling process. Therefore, in practice, we may prefer to work with representations that are longer but make the process of scientific discovery, i.e., finding the best model, easier, even if they contain superfluous information. In contrast, Proposition 4.6.1 highlights a different concern: adding irrelevant or non-informative symbols to a representation should be avoided, as they increase the complexity of the description without contributing useful information about the entity.

Finally, the following proposition shows that the order of the representations in the minimal description of a joint representation does not affect its length.

Proposition 4.6.3 Let $t, s \in \mathcal{R}$ be two representations, and let d_{ts}^* and d_{st}^* be the minimal descriptions of the joint representations ts and st , respectively. Then: $l(d_{ts}^*) = l(d_{st}^*)$.

Proof. The equality $l(m_{ts}^*) = l(m_{st}^*)$ is equivalent to $K(ts) = K(st)$. The result follows from Proposition 20.3.1. ■

It is important to note, however, that joining representations is not a commutative operation, there is no guarantee that the strings ts and st encode the same entity. Moreover, given only the concatenated string ts , it is generally

not possible to recover the original representations t and s , since they are not self-delimiting.

Propositions 4.6.1, 4.6.2 and 4.6.3 can be generalized to any arbitrary, but finite, collection of representations $t_1, t_2, \dots, t_n$.

Proposition 4.6.4 Let $t_1, t_2, \dots, t_n \in \mathcal{R}$ be a finite collection of representations. Then, we have that:

- i $l(d_{t_1 t_2 \dots t_n}^*) \geq l(d_{t_i}^*) \forall 1 \leq i \leq n$,
- ii $l(d_{t_1 t_2 \dots t_n}^*) \leq l(d_{t_1}^*) + l(d_{t_2}^*) + \dots + l(d_{t_n}^*)$,
- iii $l(d_{t_1 \dots t_i \dots t_j \dots t_n}^*) = l(d_{t_1 \dots t_j \dots t_i \dots t_n}^*) + c \forall 1 \leq i \leq j \leq n$,
- iv $l(d_{t_1 \dots t_{n-1}}^*) \leq l(d_{t_1 \dots t_{n-1} t_n}^*)$.

Proof. Apply Propositions 4.6.1, 4.6.2 and 4.6.3 to individual pairs of representations i and j . ■

4.7 Conditional Descriptions

It is often impractical to include all the information required to reconstruct a representation within a single description. Scientific descriptions are rarely constructed in isolation. They normally presuppose a considerable amount of background knowledge: mathematical tools, measurement conventions, auxiliary theories, previously established empirical facts, and accepted methods of inference. For example, a modern formulation of Newtonian mechanics may rely on the language of infinitesimal calculus without explicitly reconstructing calculus as part of the theory itself.

This motivates the notion of a *conditional description*. A conditional description is a description whose interpretation is made relative to some prior description. The prior description is not incorporated as a further part of the model; rather, it is treated as background knowledge that may be freely used when reconstructing the target representation.

In this section we formalize conditioning as a computable operation on descriptions. Unlike augmentation, which combines two descriptions into a larger description, conditioning changes the context in which a description is evaluated.

Definition 4.7.1 We define a *conditioning operator* as a computable function

$$\text{cond} : \mathcal{D} \times \mathcal{D} \rightarrow \mathcal{D}$$

such that, for every pair of descriptions

$$d_1 = \langle TM_1, a_1 \rangle, \quad d_2 = \langle TM_2, a_2 \rangle \in \mathcal{D},$$

we have

$$\text{cond}(d_1, d_2) = d_3,$$

where

$$d_3 = \langle TM_3, \langle a_1, d_2 \rangle \rangle \in \mathcal{D},$$

and TM_3 is a Turing machine obtained computably from TM_1 such that, on input $\langle a_1, d_2 \rangle$, it parses $d_2 = \langle TM_2, a_2 \rangle$, computes the output $TM_2(a_2)$, and uses that output as auxiliary computable information during the simulation of $TM_1(a_1)$.

More precisely, TM_3 is allowed to parse $d_2 = \langle TM_2, a_2 \rangle$, simulate the computation $TM_2(a_2)$, and use its result as an auxiliary computable resource during the execution of $TM_1(a_1)$.

Definition 4.7.2 We define a *conditioning operator* as a computable function

$$\text{cond} : \mathcal{D} \times \mathcal{D} \rightarrow \mathcal{D}$$

given by

$$\text{cond}(d_1, d_2) = \langle U_{\text{cond}}, \langle d_1, d_2 \rangle \rangle,$$

where U_{cond} is a fixed universal Turing machine that executes the computation encoded by d_1 while using the computation encoded by d_2 as background information.

Definition 4.7.3 We define a *conditioning operator* as a computable function

$$\text{cond} : \mathcal{D} \times \mathcal{D} \rightarrow \mathcal{D}$$

such that, for every pair of descriptions $d_1 = \langle TM_1, a_1 \rangle$ and $d_2 = \langle TM_2, a_2 \rangle \in \mathcal{D}$, we have that

$$\text{cond}(d_1, d_2) = d_3$$

where $d_3 = \langle TM_3, \langle a_1, TM_2 \rangle \rangle \in \mathcal{D}$.

The output of a description is the representation reconstructed by that description. This notation allows us to distinguish clearly between a description d , which is a computational object, and the representation $\text{out}(d)$, which is the string produced by that object.

Definition 4.7.4 — Conditioning operator. A *conditioning operator* is a computable function

$$\text{cond} : \mathcal{D} \times \mathcal{D} \rightarrow \mathcal{D}$$

such that, for descriptions $d = \langle TM, a \rangle$ and $c \in \mathcal{D}$, the description

$$\text{cond}(d, c)$$

is interpreted as the description obtained by evaluating d with free access to the background representation $\text{out}(c)$.

Equivalently, $\text{cond}(d, c)$ is a description whose computation has the same effect, up to a fixed computable translation, as running TM on the pair

$$\langle a, \text{out}(c) \rangle.$$

The second argument c plays a special role. It is not merged with d as an additional component of the description. Instead, c determines the background context relative to which d is evaluated. This is the essential difference between conditioning and augmentation. Augmentation adds content; conditioning reuses content that is assumed to be already available.

For this reason, when measuring the size of a conditional description, the cost of the background description is not charged to the conditional description itself. The relevant cost is the additional information needed once the background has been fixed.

Definition 4.7.5 — Conditional description of a representation. Let $r \in \mathcal{B}^*$ be a representation, and let $c \in \mathcal{D}$ be a background description. We say that $d \in \mathcal{D}$ is a *valid description of r conditional on c* if

$$\text{out}(\text{cond}(d, c)) = r.$$

The set of all descriptions of r conditional on c is denoted by

$$\mathcal{D}_{r|c} = \{d \in \mathcal{D} : \text{out}(\text{cond}(d, c)) = r\}.$$

Thus, $\mathcal{D}_{r|c}$ contains the descriptions that reconstruct r under the assumption that the background description c is already available. Notice that the conditioning object is now a description c , not an arbitrary string s . This better matches the intended interpretation: background knowledge is itself represented by a description.

Definition 4.7.6 — Minimal conditional description. Let $r \in \mathcal{B}^*$ be a representation, and let $c \in \mathcal{D}$ be a background description. A description

$d_{r|c}^* \in \mathcal{D}_{r|c}$ is called a *minimal conditional description* of r given c if

$$\ell(d_{r|c}^*) = \min\{\ell(d) : d \in \mathcal{D}_{r|c}\}.$$

The quantity $\ell(d_{r|c}^*)$ measures the amount of additional descriptional information required to reconstruct r once the background description c is assumed to be available. It is the analogue, at the level of descriptions, of conditional Kolmogorov complexity.

Proposition 4.7.1 — Unconditional descriptions are conditional descriptions. Let $r \in \mathcal{B}^*$, and let $c \in \mathcal{D}$ be any background description. If $d \in \mathcal{D}_r$, then there exists a description $\tilde{d} \in \mathcal{D}_{r|c}$ such that

$$\ell(\tilde{d}) \leq \ell(d) + O(1).$$

Proof. Since $d \in \mathcal{D}_r$, the description $d = \langle TM, a \rangle$ satisfies $TM(a) = r$. Construct a new machine TM' that, on input $\langle a, \text{out}(c) \rangle$, ignores the second component and simulates $TM(a)$. The description

$$\tilde{d} = \langle TM', a \rangle$$

therefore satisfies

$$\text{out}(\text{cond}(\tilde{d}, c)) = r.$$

The only additional cost is the fixed cost of replacing TM with a machine that ignores the background input. Hence

$$\ell(\tilde{d}) \leq \ell(d) + O(1).$$

■

The converse does not hold in general. If $d \in \mathcal{D}_{r|c}$, then d reconstructs r only with access to the background description c . It need not reconstruct r without that background. In general,

$$d \in \mathcal{D}_{r|c} \not\Rightarrow d \in \mathcal{D}_r.$$

Proposition 4.7.2 — Conditional descriptions are no longer than unconditional descriptions. Let $r \in \mathcal{B}^*$, and let $c \in \mathcal{D}$ be a background description. Then

$$\ell(d_{r|c}^*) \leq \ell(d_r^*) + O(1).$$

Proof. Apply Proposition 4.7.1 to a minimal description $d_r^* \in \mathcal{D}_r$. This yields a conditional description $\tilde{d} \in \mathcal{D}_{r|c}$ such that

$$\ell(\tilde{d}) \leq \ell(d_r^*) + O(1).$$

Since $d_{r|c}^*$ is minimal in $\mathcal{D}_{r|c}$, we obtain

$$\ell(d_{r|c}^*) \leq \ell(\tilde{d}) \leq \ell(d_r^*) + O(1).$$

■

This proposition formalizes the intuitive idea that background knowledge cannot make the shortest description essentially longer. At worst, the description can ignore the background information.

Proposition 4.7.3 — Useful conditioning may strictly reduce description length. There exist representations $r \in \mathcal{B}^*$ and background descriptions $c \in \mathcal{D}$ such that

$$\ell(d_{r|c}^*) < \ell(d_r^*) - O(1).$$

Proof. Let $s \in \mathcal{B}^*$ be a string with high Kolmogorov complexity, and let $c \in \mathcal{D}$ be a description such that $\text{out}(c) = s$. Take $r = s$. Without access to c , any description of r must contain essentially the information needed to reconstruct s . However, conditional on c , a fixed-size description can output the background string $\text{out}(c)$. Thus the conditional description length may be smaller than the unconditional description length by an unbounded amount. ■

This proposition captures the role of background knowledge in scientific practice. A theory may become much shorter when expressed relative to a previously established theory, a mathematical framework, or a body of empirical knowledge.

Definition 4.7.7 — Conditional equivalence of descriptions. Let $c \in \mathcal{D}$ be a background description. Two descriptions $d_1, d_2 \in \mathcal{D}$ are said to be *equivalent conditional on c* if

$$\text{out}(\text{cond}(d_1, c)) = \text{out}(\text{cond}(d_2, c)).$$

We write

$$d_1 \equiv_c d_2.$$

If the two descriptions can be transformed into one another by fixed computable translators independent of the particular descriptions involved,

we write

$$d_1 \equiv_c^{O(1)} d_2.$$

The relation $\equiv_c^{O(1)}$ is the appropriate notion of equality for conditional descriptions. Two descriptions may not be identical as strings, but they may have the same expressive content relative to a given background, up to a constant-size change of coding.

We now relate conditioning to augmentation. Recall that augmentation is a computable operation

$$\text{aug} : \mathcal{D} \times \mathcal{D} \rightarrow \mathcal{D}$$

that combines two descriptions into a joint description. Conditioning should distribute over augmentation when the same background knowledge is available to each component. This is the formal counterpart of the idea that using a common background theory after combining two descriptions is equivalent to using that background theory in each component before combining them.

Proposition 4.7.4 — Distributivity of conditioning over augmentation. Let $d_1, d_2, c \in \mathcal{D}$. Suppose that augmentation is a structural combination operator, in the sense that $\text{aug}(d_1, d_2)$ reconstructs the joint output of d_1 and d_2 up to a fixed computable pairing convention. Then

$$\text{cond}(\text{aug}(d_1, d_2), c) \equiv^{O(1)} \text{aug}(\text{cond}(d_1, c), \text{cond}(d_2, c)).$$

Proof. The left-hand side first forms a joint description from d_1 and d_2 , and then evaluates that joint description relative to the background description c . Its output is therefore the joint reconstruction obtained from d_1 and d_2 with access to $\text{out}(c)$.

The right-hand side first evaluates d_1 relative to c , evaluates d_2 relative to c , and then structurally combines the two resulting conditional descriptions. Its output is therefore the same joint reconstruction.

Since both procedures differ only in the order in which a fixed computable pairing and a fixed computable conditioning convention are applied, there are constant-size translators between the two descriptions. Hence the two sides are equivalent up to $O(1)$. ■

This distributive property is one of the main reasons to define conditioning as an operation on descriptions rather than as a raw pairing of strings. It makes conditioning behave like a change of context: the same background knowledge may be applied before or after a structural decomposition of the description.

■ **Example 4.8 — Newtonian mechanics conditional on calculus.** Let d_1 be a description of Newton's first law, let d_2 be a description of Newton's second law, and let c be a description of infinitesimal calculus. Then

$$\text{cond}(\text{aug}(d_1, d_2), c)$$

represents the joint theory consisting of the first and second laws, expressed under the assumption that calculus is already available. On the other hand,

$$\text{aug}(\text{cond}(d_1, c), \text{cond}(d_2, c))$$

represents the combination of the first law expressed relative to calculus and the second law expressed relative to calculus. Proposition 4.7.4 states that these two ways of using the background theory are equivalent up to a constant-size translation. ■

The distributive property should not be confused with augmentation by the background theory itself. In general,

$$\text{cond}(d, c)$$

is not the same as

$$\text{aug}(d, c).$$

The former assumes c as background knowledge; the latter incorporates c as an additional component of the description.

Proposition 4.7.5 — Conditioning is not augmentation. In general,

$$\text{cond}(d, c) \not\equiv \text{aug}(d, c).$$

Proof. The description $\text{aug}(d, c)$ reconstructs a joint object containing the outputs of both d and c . In contrast, $\text{cond}(d, c)$ reconstructs the target output of d using $\text{out}(c)$ only as background information. Thus the output of $\text{aug}(d, c)$ contains the contribution of c explicitly, whereas the output of $\text{cond}(d, c)$ need not contain it. Hence the two operations have different semantics and are not equivalent in general. ■

This distinction is crucial. Augmentation expands the object being described; conditioning changes the informational context in which the description is evaluated.

The relation between conditional descriptions and joint descriptions can now be stated in a way that mirrors the usual inequalities for Kolmogorov complexity.

Proposition 4.7.6 — Conditional, unconditional, and joint descriptions. Let $r \in \mathcal{B}^*$, and let $c \in \mathcal{D}$ be a background description. Then

$$\ell(d_{r|c}^*) \leq \ell(d_r^*) + O(1) \leq \ell(d_{r, \text{out}(c)}^*) + O(1),$$

where $d_{r, \text{out}(c)}^*$ denotes a minimal joint description of the pair $\langle r, \text{out}(c) \rangle$.

Proof. The first inequality is Proposition 4.7.2. For the second inequality, any description of the joint representation $\langle r, \text{out}(c) \rangle$ can be converted, by a fixed computable projection, into a description of r . Therefore a minimal description of r cannot be longer than a minimal joint description of $\langle r, \text{out}(c) \rangle$, except for the constant cost of the projection. ■

This proposition formalizes the basic asymmetry between conditioning and joint description. Describing r given c may be easier than describing r alone, while describing r together with the output of c is at least as demanding as describing r alone, up to a constant.

Conditioning can also be iterated. In scientific practice, a model may rely on several layers of background knowledge: for example, a physical theory may presuppose calculus, geometry, measurement conventions, and numerical methods.

Definition 4.7.8 — Multiple conditioning. Let $d, c_1, \dots, c_n \in \mathcal{D}$. We define the iterated conditional description of d given $c_1, \dots, c_n$ recursively by

$$\text{cond}(d; c_1, \dots, c_n) = \text{cond}(\text{cond}(d; c_1, \dots, c_{n-1}), c_n),$$

with

$$\text{cond}(d; c_1) = \text{cond}(d, c_1).$$

Definition 4.7.9 — Conditional descriptions with multiple backgrounds. Let $r \in \mathcal{B}^*$, and let $c_1, \dots, c_n \in \mathcal{D}$. We define

$$\mathcal{D}_{r|c_1, \dots, c_n} = \{d \in \mathcal{D} : \text{out}(\text{cond}(d; c_1, \dots, c_n)) = r\}.$$

A description

$$d_{r|c_1, \dots, c_n}^* \in \mathcal{D}_{r|c_1, \dots, c_n}$$

is called a *minimal conditional description* of r given $c_1, \dots, c_n$ if

$$\ell(d_{r|c_1, \dots, c_n}^*) = \min\{\ell(d) : d \in \mathcal{D}_{r|c_1, \dots, c_n}\}.$$

Proposition 4.7.7 — Monotonicity of conditional description length. Let $r \in \mathcal{B}^*$, and let $c_1, \dots, c_n, c_{n+1} \in \mathcal{D}$. Then

$$\ell(d_{r|c_1, \dots, c_n, c_{n+1}}^*) \leq \ell(d_{r|c_1, \dots, c_n}^*) + O(1).$$

Proof. A description conditional on $c_1, \dots, c_n, c_{n+1}$ may ignore the additional background description c_{n+1} . Therefore any description that reconstructs r from $c_1, \dots, c_n$ can be converted, at constant cost, into a description that reconstructs r from $c_1, \dots, c_n, c_{n+1}$. Taking minimal descriptions gives the result. ■

Thus, adding background knowledge cannot essentially increase the length of a minimal conditional description. It may leave the length unchanged if the new background is irrelevant, or it may reduce the length if the new background contains information useful for reconstructing the representation.

Definition 4.7.10 — Relevance of a background description. Let $r \in \mathcal{B}^*$, and let $c \in \mathcal{D}$. We define the *descriptonal relevance* of c for r as

$$\Delta(r; c) = \ell(d_r^*) - \ell(d_{r|c}^*).$$

The quantity $\Delta(r; c)$ measures how much shorter the best description of r becomes when c is available as background knowledge. If $\Delta(r; c)$ is large, then c contains information that is highly relevant to the reconstruction of r . If $\Delta(r; c) = O(1)$, then c does not significantly reduce the descriptonal burden.

Proposition 4.7.8 — Irrelevant background descriptions. Let $r \in \mathcal{B}^*$, and let $c \in \mathcal{D}$. If

$$\ell(d_{r|c}^*) = \ell(d_r^*) + O(1),$$

then c is descriptonally irrelevant to r up to a constant-size coding effect.

Proof. By Definition 4.7.10,

$$\Delta(r; c) = \ell(d_r^*) - \ell(d_{r|c}^*).$$

If the two lengths differ only by $O(1)$, then

$$\Delta(r; c) = O(1).$$

Thus c does not provide a non-constant reduction in the shortest description length of r . ■

This proposition gives an operational interpretation of relevance. A background description is relevant to a target representation precisely when

it reduces the amount of additional information needed to reconstruct that representation.

Finally, we record the corresponding chain rule. It connects conditional description length with joint description length and explains why conditioning is the correct formal tool for separating novel information from assumed background knowledge.

Proposition 4.7.9 — Descriptive chain rule. Let $r \in \mathcal{B}^*$, and let $c \in \mathcal{D}$. Then, up to logarithmic or coding-dependent overheads,

$$\ell(d_{r, \text{out}(c)}^*) = \ell(d_{\text{out}(c)}^*) + \ell(d_{r|c}^*) + O(1).$$

Proof. A joint description of $\langle \text{out}(c), r \rangle$ may first describe $\text{out}(c)$ and then describe r conditional on that background. This gives the upper bound. Conversely, from any joint description of $\langle \text{out}(c), r \rangle$, one can recover both $\text{out}(c)$ and r , and therefore separate the information needed for the background from the additional information needed for the target relative to that background, up to the fixed cost of the chosen coding convention. This yields the stated equivalence. ■

The chain rule clarifies the conceptual role of conditional descriptions. Scientific knowledge grows not only by constructing larger joint descriptions, but also by changing what can be assumed as background. Once a description has become part of the accepted background, subsequent descriptions may become shorter, more modular, and easier to compare. Conditioning therefore provides a formal mechanism for representing cumulative scientific knowledge: new models are not evaluated in isolation, but relative to the knowledge already available.

4.8 Descriptions of Research Areas

Research areas are subsets of the set of entities $\mathcal{E}$. Once individual entities admit descriptions, it is natural to ask how an entire research area can be described. Intuitively, a description of an area should collect descriptions of the entities that compose it in such a way that each component can be identified and, when needed, recovered effectively. In this way, a research area is not treated as an undifferentiated aggregate, but as a structured family of entity-level descriptions.

As in the case of representations, two difficulties appear immediately. First, a research area need not be finite, whereas every description is a finite binary string. Second, some entities in the area may fail to admit any exact description, or may only be associated in practice with effective descriptions. For this reason, the description of a research area must be understood relative to finite subsets of the area and, in practice, relative to the effective descriptions currently available.

Recall that, for each entity $e \in \mathcal{E}$, the set $\mathcal{D}_e$ denotes the set of valid descriptions associated with e , and $\hat{\mathcal{D}}_e$ denotes the set of effective descriptions of e .

Definition 4.8.1 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area. We define the *describable part* of $\mathcal{A}$ as

$$\mathcal{A}^{\mathcal{D}} = \{e \in \mathcal{A} : \mathcal{D}_e \neq \emptyset\}.$$

Thus, $\mathcal{A}^{\mathcal{D}}$ consists of those entities in the area for which at least one valid description exists. If $\mathcal{A}$ contains entities outside $\mathcal{A}^{\mathcal{D}}$, then no exact area description can include them as individually described components.

To describe several entities together, we must separate their descriptions and attach identifiers that allow each component to be recognized. Any standard self-delimiting encoding of finite strings may be used for this purpose.

Definition 4.8.2 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area, let

$$F = \{e_1, \dots, e_n\} \subseteq \mathcal{A}$$

be a finite subset, and let $d_i \in \mathcal{D}_{e_i}$ for each $i = 1, \dots, n$. Let $t_1, \dots, t_n \in \mathcal{B}^*$ be pairwise distinct identifier strings. We define a *description of the finite subarea F* as any finite binary string of the form

$$\Delta_F = \langle (t_1, d_1), \dots, (t_n, d_n) \rangle,$$

where $\langle \cdot, \cdot \rangle$ and its finite iteration denote a fixed self-delimiting computable encoding of pairs and finite sequences of binary strings.

The identifiers t_i need not carry scientific meaning by themselves. Their role is only to make it possible to separate the different components of the area description and determine which description is associated with which entity in the represented finite family.

Definition 4.8.3 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area. We say that a binary string $\Delta \in \mathcal{B}^*$ is an *extended description of the area* if there exists a finite subset

$$F = \{e_1, \dots, e_n\} \subseteq \mathcal{A}$$

and strings $d_i \in \mathcal{D}_{e_i}$ such that

$$\Delta = \langle (t_1, d_1), \dots, (t_n, d_n) \rangle$$

for some pairwise distinct identifiers $t_1, \dots, t_n$.

This definition deliberately allows area descriptions to be partial. The string

Δ need not describe all entities in $\mathcal{A}$, but only a finite subset of them. This is unavoidable whenever $\mathcal{A}$ is infinite, and it remains useful even when $\mathcal{A}$ is finite but some of its entities are not describable.

Definition 4.8.4 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area. We say that Δ is a *complete area description* of $\mathcal{A}$ if Δ is an area description associated with the finite set

$$F = \mathcal{A}^{\mathcal{D}}.$$

In particular, such a complete description can exist only if $\mathcal{A}^{\mathcal{D}}$ is finite.

Thus, if $\mathcal{A}$ is finite and all its entities admit at least one description, then $\mathcal{A}$ admits a complete area description. If $\mathcal{A}$ contains entities with no description, then at best one can describe its describable part $\mathcal{A}^{\mathcal{D}}$.

The next proposition formalizes the main purpose of tagging individual descriptions: the description of an area preserves the identity of its components.

Proposition 4.8.1 Let

$$\Delta_F = \langle (i_1, d_1), \dots, (i_n, d_n) \rangle$$

be a description of a finite subarea $F = \{e_1, \dots, e_n\} \subseteq \mathcal{A}$. Then each pair (i_i, d_i) can be recovered effectively from Δ_F .

Proof. By construction, Δ_F is encoded using a fixed self-delimiting computable coding of finite sequences of pairs. Therefore, there exists a computable decoding procedure that parses Δ_F and returns each pair (i_i, d_i) . Hence every component description is effectively recoverable from the area description. ■

The previous proposition shows that a description of a research area does not erase the individual descriptions of its members, but rather organizes them within a single computable object.

Area descriptions are generally not unique.

Proposition 4.8.2 Descriptions of research areas are not unique.

Proof. Non-uniqueness may arise in at least two ways. First, an entity $e \in \mathcal{E}$ may admit several distinct descriptions $d, d' \in \mathcal{D}_e$. Second, even for a fixed family of entity-level descriptions, one may choose different identifiers or a different order of concatenation. Therefore, the same research area may admit multiple distinct area descriptions. ■

This non-uniqueness is not a defect. It reflects the fact that a research area may be organized in different descriptive styles, just as an individual entity may admit several valid descriptions.

Finite area descriptions are also stable under finite aggregation.

Proposition 4.8.3 Let $F_1, F_2 \subseteq \mathcal{A}$ be finite subareas with $F_1 \cap F_2 = \emptyset$. If Δ_{F_1} and Δ_{F_2} are descriptions of F_1 and F_2 , respectively, then there exists a description of $F_1 \cup F_2$.

Proof. Write

$$\Delta_{F_1} = \langle (\iota_1, d_1), \dots, (\iota_m, d_m) \rangle$$

and

$$\Delta_{F_2} = \langle (\kappa_1, d'_1), \dots, (\kappa_\ell, d'_\ell) \rangle.$$

Choosing the identifiers so that all $\iota_1, \dots, \iota_m, \kappa_1, \dots, \kappa_\ell$ are pairwise distinct, the string

$$\langle (\iota_1, d_1), \dots, (\iota_m, d_m), (\kappa_1, d'_1), \dots, (\kappa_\ell, d'_\ell) \rangle$$

is a description of the finite subarea $F_1 \cup F_2$. ■

This result shows that finite area descriptions are closed under finite aggregation. In this sense, describing an area amounts to collecting identifiable entity-level descriptions into a single structured string.

It is often useful to isolate the family of all area descriptions associated with a given research area.

Definition 4.8.5 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area. We define the set of *descriptions of the area $\mathcal{A}$* by

$$\mathcal{D}_{\mathcal{A}} = \{ \Delta \in \mathcal{B}^* : \Delta \text{ is an extended description of } \mathcal{A} \}.$$

By definition, $\mathcal{D}_{\mathcal{A}}$ contains all finite strings that describe finite subsets of $\mathcal{A}$ by means of identifiable valid descriptions of their entities.

The description of an area can also be understood as the union, in a structural sense, of the descriptions of its finite subareas.

Proposition 4.8.4 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area. Then every element of $\mathcal{D}_{\mathcal{A}}$ is a description of some finite subarea $F \subseteq \mathcal{A}$.

Proof. This follows immediately from the definition of extended description of a research area, which requires the existence of a finite subset $F = \{e_1, \dots, e_n\} \subseteq \mathcal{A}$ and corresponding descriptions $d_i \in \mathcal{D}_{e_i}$ such that

$$\Delta = \langle (\iota_1, d_1), \dots, (\iota_n, d_n) \rangle.$$

Therefore every $\Delta \in \mathcal{D}_{\mathcal{A}}$ is associated with some finite subarea $F \subseteq \mathcal{A}$. ■

In practice, however, science does not work with the full sets $\mathcal{D}_e$ of valid descriptions of entities, but with the effective descriptions currently available.

Definition 4.8.6 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area and let t denote a time index. We define the *effectively describable part of the area at time t* by

$$\hat{\mathcal{A}}_{\mathcal{D}}^t = \{e \in \mathcal{A} : \hat{\mathcal{D}}_e^t \neq \emptyset\}.$$

The set $\hat{\mathcal{A}}_{\mathcal{D}}^t$ contains those entities of the area for which at least one effective description is available at time t .

Definition 4.8.7 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area and let

$$F = \{e_1, \dots, e_n\} \subseteq \hat{\mathcal{A}}_{\mathcal{D}}^t$$

be a finite subset of effectively describable entities at time t . For each $i = 1, \dots, n$, let $\hat{d}_i \in \hat{\mathcal{D}}_{e_i}^t$, and let $t_1, \dots, t_n \in \mathcal{B}^*$ be pairwise distinct identifiers. We define an *effective description of the finite known subarea F at time t* as any binary string of the form

$$\hat{\Delta}_F^t = \langle (t_1, \hat{d}_1), \dots, (t_n, \hat{d}_n) \rangle.$$

Effective area descriptions are the practical counterparts of exact area descriptions. They need not be correct or minimal, because their components are only effective descriptions of the individual entities. Nevertheless, they encode the current descriptive state of the part of the area that is effectively described.

Proposition 4.8.5 Let $\mathcal{A} \subseteq \mathcal{E}$ be a fixed research area. If $t_1 < t_2$, then

$$\hat{\mathcal{A}}_{\mathcal{D}}^{t_1} \subseteq \hat{\mathcal{A}}_{\mathcal{D}}^{t_2}.$$

Proof. If $e \in \hat{\mathcal{A}}_{\mathcal{D}}^{t_1}$, then $e \in \mathcal{A}$ and $\hat{\mathcal{D}}_e^{t_1} \neq \emptyset$. Since effective descriptions are monotonic over time, we have $\hat{\mathcal{D}}_e^{t_2} \neq \emptyset$ for all $t_2 > t_1$. Therefore $e \in \hat{\mathcal{A}}_{\mathcal{D}}^{t_2}$. ■

As a consequence, effective descriptions of areas may be progressively extended over time by incorporating descriptions of entities that were previously not effectively described.

Proposition 4.8.6 Let $\mathcal{A} \subseteq \mathcal{E}$ be a fixed research area, and let $\hat{\Delta}_{F_1}^{t_1}$ be an effective description of a finite subarea $F_1 \subseteq \hat{\mathcal{A}}_{\mathcal{D}}^{t_1}$. If $t_1 < t_2$, then there exists a finite set F_2 such that

$$F_1 \subseteq F_2 \subseteq \hat{\mathcal{A}}_{\mathcal{D}}^{t_2}$$

and an effective description $\hat{\Delta}_{F_2}^{t_2}$ extending $\hat{\Delta}_{F_1}^{t_1}$.

Proof. By Proposition 4.8.5,

$$\hat{\mathcal{A}}_{\mathcal{D}}^{t_1} \subseteq \hat{\mathcal{A}}_{\mathcal{D}}^{t_2}.$$

Hence every entity in F_1 remains effectively describable at time t_2 . Let F_2 be any finite subset of $\hat{\mathcal{A}}_D^{t_2}$ containing F_1 . Retaining the previous effective descriptions for the entities in F_1 , and choosing effective descriptions at time t_2 for the entities in $F_2 \setminus F_1$, we obtain an effective description $\hat{\Delta}_{F_2}^{t_2}$ of F_2 that extends $\hat{\Delta}_{F_1}^{t_1}$. ■

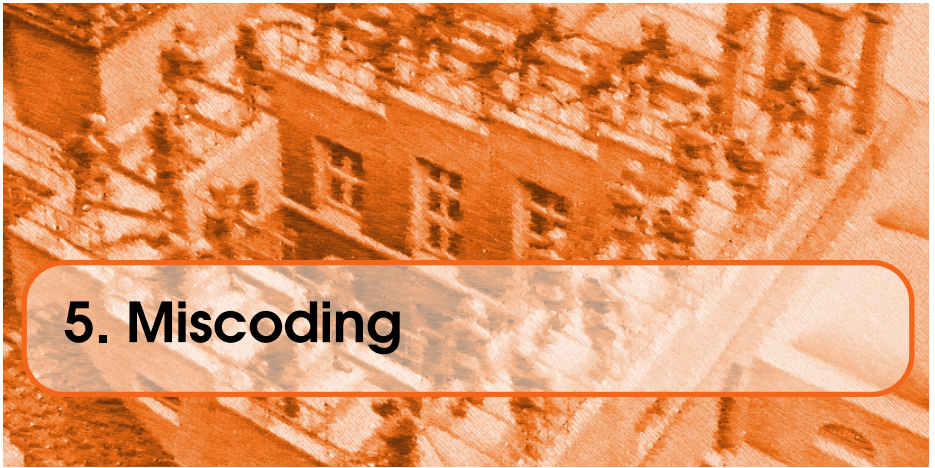
This monotonicity property reflects a natural feature of scientific practice: as more entities in an area become describable, the descriptive coverage of the area expands.

Finally, it is important to note that the description of a research area is not, in general, a minimal description of a single entity. It is a structured object that organizes descriptions of several entities simultaneously. For this reason, area descriptions are best understood as a bookkeeping and modeling device that allows the theory of nescience to be lifted from individual entities to families of related entities.

In later chapters, this lifting will make it possible to define area-level notions of mismodel, inaccuracy, surfeit, and nescience. At that point, the description of a research area will no longer be merely a collection of component descriptions, but the basis for measuring how well an entire domain of inquiry is currently understood.

4.9 References

For more information about Russell's paradox, Cantor's theorem, and universal sets, refer, for example, to [Jec13]. This book also covers the Zermelo–Fraenkel system of axioms, including the Axiom of Choice. The idea of using a function to assign to each symbol and well-formed formula of a formal language a unique natural number (called a Gödel number) was introduced by Kurt Gödel in his proof of the incompleteness theorems [Göd31]. A detailed discussion of the Berry paradox from the perspective of computability can be found in [Cha95]. For a review of the problem of representation in Kolmogorov complexity, as well as a detailed account of the implications of Kolmogorov complexity being defined only up to an additive constant, see [LV13]. That oracle machines are not mechanical was originally stated by Turing when he introduced the concept of the oracle machine in [Tur39]. For a comprehensive review of oracle Turing machines, refer to [Rob15].



5. Miscoding

*All great work is the fruit of patience and perseverance,
combined with tenacious concentration on a subject
over a period of months or years.*
Santiago Ramón y Cajal

In many areas of science, the entities we aim to study (the set $\mathcal{E}$) often include abstract concepts or complex phenomena that are difficult to investigate directly. These may be ideas or processes that resist straightforward description or modeling. To gain insight into such entities, we must rely on indirect methods. As discussed in Chapter 3, the theory of nescience suggests using representations, that is, sequences of symbols, rather than attempting to interact directly with the entities themselves.

However, this approach introduces its own challenges. Because our understanding of the elements in $\mathcal{E}$ is incomplete (otherwise, further research would be unnecessary), the representations we construct are typically imperfect, they lack the completeness and precision we would ideally desire. These imperfections can lead to significant problems: inaccuracies in representation can propagate into the models derived from them, potentially distorting our interpretation and understanding. It is therefore essential to recognize this source of error and to carefully consider its implications.

To address this issue, we introduce the concept of *miscoding*, a measure used to quantify the error that arises from misrepresenting or inaccurately encoding entities. Miscoding is defined in terms of the length of the shortest computer program capable of transforming an incorrect representation into an admissible one. In essence, it measures the amount of effort, as reflected by the program's length, required to correct a faulty representation.

However, there is a significant complication: applying this idea in practice is not straightforward, since the admissible representations of entities are unknown. As a result, we cannot directly measure the gap between a given representation and an ideal one. Nevertheless, we propose a theoretical framework based on the oracle machine (an abstract construct) used to define the set $\mathcal{E}$. This oracle machine is assumed to be capable of recognizing admissible representations for all entities in the set.

Gaining a deeper understanding of scientific representation and the difficulties involved in reducing miscoding can significantly advance scientific research, enabling the development of more accurate and comprehensive models of the natural world. In this chapter, we formally introduce the concept of *miscoding* and explore its key properties. We will analyze how miscoding behaves in the context of combined representations and consider strategies for reducing it. We will also examine the notion that some strings fail to represent any entity and investigate the implications of this idea. Finally, we will highlight the relevance of miscoding across various research domains and discuss its potential to inspire new lines of inquiry.

5.1 Deficiency of a Representation

A representation r of an entity e may be incomplete, in the sense that it may fail to contain some of the information required to reconstruct the entity by the underlying representation oracle. In scientific practice, such incompleteness appears whenever the available observations, measurements, or descriptions do not yet fully constrain the phenomenon under study. We formalize this notion by quantifying how much additional algorithmic information is needed to recover an admissible representation of e from the given representation r . We call this quantity the *deficiency* of r (with respect to e).

Definition 5.1.1 Let $e \in \mathcal{E}$ be a knowable entity, so that $\bar{\mathcal{R}}_e \neq \emptyset$, and let $r \in \mathcal{R}_e$ be a representation of e . We define the *deficiency* of r (relative to e) as

$$d_R(e, r) = \min_{r' \in \bar{\mathcal{R}}_e} K(r' \mid r).$$

Since $\bar{\mathcal{R}}_e \neq \emptyset$ and conditional Kolmogorov complexity takes values in $\mathbb{N}$, the minimum is well defined.

The quantity $K(r' | r)$ measures the length of the shortest computer program that, given r , outputs an admissible representation r' of e . Minimizing over $r' \in \bar{\mathcal{R}}_e$ selects the admissible representation that is easiest to recover from r . Hence $d_R(e, r)$ can be interpreted as the minimal amount of algorithmic information that must be supplied, in addition to r , in order to recover some admissible representation of e , measured up to the usual additive constant.

■ **Example 5.1** Let e be an entity corresponding to a particular finite undirected graph, for example a molecular interaction network or a transportation network. An admissible representation $r' \in \bar{\mathcal{R}}_e$ may be taken to encode the complete adjacency list of the graph, so that the full structure of the network can be reconstructed unambiguously. In practice, however, one often works with only partial information. For instance, a representation $r \in \mathcal{R}_e$ may consist of a subgraph obtained from incomplete measurements, missing edges, or partially observed nodes. Such a representation may already reveal some structural features of the network, but it does not yet determine the whole graph. The deficiency $d_R(e, r)$ measures, in algorithmic terms, how much additional information must be supplied in order to recover an admissible representation of the complete network from this partial one. ■

Before studying further properties of deficiency, it is useful to record that, like Kolmogorov complexity itself, it depends on the choice of reference universal machine only up to an additive constant. Thus deficiency is not an absolute numerical invariant, but it is stable under the standard machine-independence convention of algorithmic information theory.

Proposition 5.1.1 Let U and V be two reference universal prefix-free machines, and let $\text{def}^U(r, e)$ and $\text{def}^V(r, e)$ denote the deficiency of r computed using the corresponding conditional Kolmogorov complexities $K_U(\cdot | \cdot)$ and $K_V(\cdot | \cdot)$. Then, for every knowable entity $e \in \mathcal{E}$ and every representation $r \in \mathcal{R}_e$,

$$d_R^U(e, r) = d_R^V(e, r) + O(1).$$

Proof. By the invariance theorem for conditional Kolmogorov complexity, for every pair of strings $x, y \in \mathcal{B}^*$,

$$K_U(x | y) \leq K_V(x | y) + c_{UV}, \quad K_V(x | y) \leq K_U(x | y) + c_{VU},$$

where c_{UV} and c_{VU} are constants depending only on U and V . Applying the first inequality to each $r' \in \bar{\mathcal{R}}_e$ gives

$$K_U(r' | r) \leq K_V(r' | r) + c_{UV}.$$

Taking minima over $r' \in \bar{\mathcal{R}}_e$, we obtain

$$d_R^U(e, r) = \min_{r' \in \bar{\mathcal{R}}_e} K_U(r' | r) \leq \min_{r' \in \bar{\mathcal{R}}_e} (K_V(r' | r) + c_{UV}) = d_R^V(e, r) + c_{UV}.$$

By symmetry,

$$d_R^V(e, r) \leq d_R^U(e, r) + c_{VU}.$$

Combining both inequalities yields

$$d_R^U(e, r) = d_R^V(e, r) + O(1).$$

■

This proposition shows that deficiency is robust under the choice of universal machine in exactly the same sense as Kolmogorov complexity itself. Consequently, statements such as $d_R(e, r) = O(1)$ or comparisons between deficiencies remain meaningful independently of the particular coding convention used to define the complexity measure.

The following proposition establishes a basic structural property of deficiency: since it is defined in terms of program lengths, it cannot take negative values.

Proposition 5.1.2 For every representation $r \in \mathcal{R}_e$, we have that $d_R(e, r) \geq 0$.

Proof. For any $r' \in \bar{\mathcal{R}}_e$, the conditional complexity $K(r' | r)$ is the length of a computer program and therefore is non-negative. The minimum of non-negative quantities is non-negative. ■

This result confirms that deficiency is a non-negative quantity, as expected for a measure of missing information. In particular, $d_R(e, r) = O(1)$ corresponds to the situation in which no additional information is required to recover an admissible representation of the entity.

Next proposition shows that admissible representations are, in a precise sense, essentially complete: they require no additional information to reconstruct the entity beyond a constant overhead.

Proposition 5.1.3 Let $r \in \bar{\mathcal{R}}_e$ be an admissible representation. Then we have that $d_R(e, r) = O(1)$.

Proof. Choose $r' = r$ in Definition 5.1.1. Then $d_R(e, r) \leq K(r | r)$. Since a constant length program can output its conditional input, $K(r | r) = O(1)$, and hence $d_R(e, r) = O(1)$. ■

This proposition formalizes the intuition that admissible representations already contain all the information needed to reconstruct the entity. The residual $O(1)$ term reflects only the fixed overhead associated with the choice of universal machine.

An important property of deficiency is its monotonic behavior under computable enrichment. Informally, if one replaces a representation by another

from which the original can be effectively recovered, then the deficiency cannot increase by more than a constant. This captures the idea that adding accessible information should not make the task of reconstructing an admissible representation substantially harder.

Proposition 5.1.4 Let $e \in \mathcal{E}$ be a knowable entity, and let $r, s \in \mathcal{R}_e$ be a representations of e . Suppose that there exists a total computable function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that $g(s) = r$. Then

$$d_R(e, s) \leq d_R(e, r) + O(1).$$

Proof. Let $r' \in \bar{\mathcal{R}}_e$ be arbitrary. Since g is total and computable, from s one can compute $g(s) = r$ with constant overhead and then simulate an optimal program witnessing $K(r' \mid r)$. Hence

$$K(r' \mid s) \leq K(r' \mid g(s)) + O(1) = K(r' \mid r) + O(1).$$

Taking minima over $r' \in \bar{\mathcal{R}}_e$, we obtain

$$d_R(e, s) = \min_{r' \in \bar{\mathcal{R}}_e} K(r' \mid s) \leq \min_{r' \in \bar{\mathcal{R}}_e} (K(r' \mid r) + O(1)) = d_R(e, r) + O(1).$$

■

This proposition expresses a basic monotonicity principle: computably enriching a representation cannot substantially worsen its deficiency. In scientific terms, if a new representation preserves the old one in an effectively recoverable way while adding further accessible information, then it cannot increase the amount of genuinely missing information by more than a constant.

The case of augmentation by concatenation is an immediate instance of the previous proposition. It models the situation in which extra information is inserted into a representation while the original content remains computably retrievable.

Corollary 5.1.5 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings and let $e \in \mathcal{E}$ be a knowable entity. Suppose that both $rt \in \mathcal{R}_e$ and $rst \in \mathcal{R}_e$ are representations of e . Then

$$d_R(e, rst) \leq d_R(e, rt) + O(1).$$

Proof. Define a total computable function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ by letting $g(rst) = rt$, that is, g discards the middle block s . Since $g(rst) = rt$, the conclusion follows directly from the previous proposition with $s = rst$ and $r = rt$. ■

The next proposition identifies a simple condition under which reducing a representation does not change its deficiency beyond the usual additive constant. Intuitively, if the discarded part can be reconstructed from what remains at constant cost, then no genuinely missing information has been introduced by removing it.

Proposition 5.1.6 Let $e \in \mathcal{E}$ be a knowable entity, and let $x, y \in \mathcal{B}^*$ be such that both $xy \in \mathcal{R}_e$ and $x \in \mathcal{R}_e$ are representations of e . If

$$K(y \mid x) = O(1),$$

then

$$d_R(e, x) = d_R(e, xy) + O(1).$$

Proof. Since x is computable from xy by deleting the suffix y , the previous proposition gives

$$d_R(e, xy) \leq d_R(e, x) + O(1).$$

On the other hand, the condition $K(y \mid x) = O(1)$ implies that there is a constant-length program that reconstructs y from x . Hence from x one can compute xy with constant overhead. By the monotonicity of deficiency under computable enrichment,

$$d_R(e, x) \leq d_R(e, xy) + O(1).$$

Combining both inequalities yields

$$d_R(e, x) = d_R(e, xy) + O(1).$$

■

This proposition formalizes the idea of a safe reduction: removing information that is already recoverable from the retained part does not make the representation meaningfully more incomplete. In practice, it justifies discarding predictable or duplicative components of a representation without affecting, except up to a constant, the amount of genuinely missing information needed to recover an admissible representation.

Finally, we derive a simple but useful upper bound on deficiency in terms of the admissible complexity floor κ_e of the entity (see Definition 3.3.3). This bound follows from the previous monotonicity principle by observing that any representation can be computably enriched into a form from which a fixed minimal admissible representation can be recovered by simply ignoring the original input and outputting that representation.

Corollary 5.1.7 For every representation $r \in \mathcal{R}_e$, we have that

$$d_R(e, r) \leq \kappa_e + O(1).$$

Proof. Let $r_0 \in \bar{\mathcal{R}}_e$ be such that

$$K(r_0) = \kappa_e.$$

Consider the constant function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ defined by $g(x) = r_0$ for all $x \in \mathcal{B}^*$. Since g is total and computable, from any representation r one can computably recover r_0 with an additive constant overhead. Therefore, by the previous proposition with $s = r$ and $r = r_0$,

$$d_R(e, r) \leq d_R(e, r_0) + O(1).$$

Since $r_0 \in \bar{\mathcal{R}}_e$, the previous corollary gives

$$d_R(e, r_0) = O(1).$$

More directly, one may write

$$d_R(e, r) = \min_{r' \in \bar{\mathcal{R}}_e} K(r' \mid r) \leq K(r_0 \mid r) \leq K(r_0) + O(1) = \kappa_e + O(1),$$

where the second inequality follows because a program can ignore r and output r_0 at an additive constant overhead. Hence

$$d_R(e, r) \leq \kappa_e + O(1).$$

■

This bound shows that deficiency is always finite and is controlled by the admissible complexity floor of the entity relative to a fixed representation oracle. Even a completely uninformative representation can be supplemented, at worst, by providing a minimal admissible encoding from scratch.

5.2 Surplus of a Representation

An representation r of an entity e may contain more information than is actually needed to represent e . Some of this additional content may be redundant, that is, information that is not essential to reconstruct the entity, but that may still be useful for identifying models. However, this additional information may also be extraneous, misleading, or entirely unrelated to the entity, which is undesirable because it can make the modeling task more difficult. It is this latter component that we call *surplus*, a part that needs to be identified and measured. We formalize this idea by quantifying how much algorithmic information must be added to an admissible representation in order to recover the representation r .

Definition 5.2.1 Let $e \in \mathcal{E}$ be a knowable entity, so that $\bar{\mathcal{R}}_e \neq \emptyset$, and let $r \in \mathcal{R}_e$ be an representation of e . We define the *surplus* of r relative to e as

$$s_R(e, r) = \min_{r' \in \bar{\mathcal{R}}_e} K(r \mid r').$$

Since $\bar{\mathcal{R}}_e \neq \emptyset$ and conditional Kolmogorov complexity takes values in $\mathbb{N}$, the minimum is well defined.

The quantity $K(r \mid r')$ measures the length of the shortest program that, given an admissible representation r' of e , outputs the representation r . Minimizing over $r' \in \bar{\mathcal{R}}_e$ selects the admissible representation that renders r easiest to generate. Hence $s_R(e, r)$ measures the minimal amount of additional algorithmic content in r beyond what is needed to encode e . When $s_R(e, r)$ is large, the representation contains substantial extraneous information relative to the entity.

■ **Example 5.2** Consider an entity e corresponding to the relationship between a collection of biological features of plants and their species identity. An admissible representation $r' \in \bar{\mathcal{R}}_e$ may encode only those variables needed to characterize the plants for this purpose, such as leaf shape, petal length, stem height, and similar morphological attributes. In practice, however, a dataset $r \in \mathcal{R}_e$ may also include columns such as greenhouse bench number, sensor serial number, or the identifier of the technician who recorded the sample. These additional variables may be present in the representation even though they do not belong to the entity being represented. Given an admissible representation r' of the biological features, generating the full dataset r still requires specifying those extra fields. The surplus $s_R(e, r)$ quantifies, in algorithmic terms, how much such unrelated information has been incorporated into the representation. ■

As in the case of deficiency, surplus depends on the choice of reference universal machine only up to an additive constant. The argument is entirely analogous, but with the roles of the conditioned and conditioning strings reversed.

Proposition 5.2.1 Let U and V be two reference universal prefix-free machines, and let $sur^U(r, e)$ and $sur^V(r, e)$ denote the surplus of r computed using the corresponding conditional Kolmogorov complexities $K_U(\cdot \mid \cdot)$ and $K_V(\cdot \mid \cdot)$. Then, for every knowable entity $e \in \mathcal{E}$ and every representation $r \in \mathcal{R}_e$,

$$s_R^U(e, r) = s_R^V(e, r) + O(1).$$

Proof. By the invariance theorem for conditional Kolmogorov complexity, for every pair of strings $x, y \in \mathcal{B}^*$,

$$K_U(x \mid y) \leq K_V(x \mid y) + c_{UV}, \quad K_V(x \mid y) \leq K_U(x \mid y) + c_{VU},$$

where c_{UV} and c_{VU} are constants depending only on U and V . Applying the first

inequality to each $r' \in \bar{\mathcal{R}}_e$ gives

$$K_U(r \mid r') \leq K_V(r \mid r') + c_{UV}.$$

Taking minima over $r' \in \bar{\mathcal{R}}_e$, we obtain

$$s_R^U(e, r) = \min_{r' \in \bar{\mathcal{R}}_e} K_U(r \mid r') \leq \min_{r' \in \bar{\mathcal{R}}_e} (K_V(r \mid r') + c_{UV}) = s_R^V(e, r) + c_{UV}.$$

By symmetry,

$$s_R^V(e, r) \leq s_R^U(e, r) + c_{VU}.$$

Combining both inequalities yields

$$s_R^U(e, r) = s_R^V(e, r) + O(1).$$

■

Thus surplus is machine-invariant in exactly the same sense as deficiency and Kolmogorov complexity itself. In particular, qualitative statements such as $s_R(e, r) = O(1)$ or comparisons between surpluses remain meaningful independently of the specific universal machine chosen.

The following proposition establishes the most basic structural property of surplus: like deficiency, it is always non-negative, since it is defined as a minimum over program lengths.

Proposition 5.2.2 For every representation $r \in \mathcal{R}_e$, we have that $s_R(e, r) \geq 0$.

Proof. For any $r' \in \bar{\mathcal{R}}_e$, the conditional complexity $K(r \mid r')$ is the length of a computer program and is therefore non-negative. The minimum of non-negative quantities is non-negative. ■

This confirms that surplus is a non-negative quantity, as expected for a measure of extraneous information. In particular, $s_R(e, r) = O(1)$ corresponds to the situation in which the representation differs from some admissible representation by at most a constant amount of additional algorithmic content.

The next proposition is the direct counterpart, for surplus, of the analogous result previously established for deficiency. It shows that admissible representations have negligible surplus, since they contain no non-trivial additional information beyond what is needed to represent the entity.

Proposition 5.2.3 For every admissible representation $r \in \bar{\mathcal{R}}_e$, we have that

$$s_R(e, r) = O(1).$$

Proof. Choose $r' = r$ in Definition 5.2.1. Then

$$s_R(e, r) \leq K(r \mid r).$$

Since a constant-length program can output its conditional input, $K(r \mid r) = O(1)$, and hence $s_R(e, r) = O(1)$. ■

This proposition formalizes the intuition that admissible representations do not contain substantial extraneous information relative to the entity. The residual $O(1)$ term again reflects only the fixed overhead associated with the choice of universal machine.

For surplus, the natural monotonicity principle concerns computable reduction rather than enrichment. Intuitively, if one passes from a representation to a computable simplification of it, then surplus cannot increase by more than a constant, because extraneous information may be discarded but cannot be created by a reduction.

Proposition 5.2.4 Let $e \in \mathcal{E}$ be a knowable entity, and let $r, s \in \mathcal{R}_e$ be representations of e . Suppose that there exists a total computable function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that $g(s) = r$. Then

$$s_R(e, r) \leq s_R(e, s) + O(1).$$

Proof. Let $r' \in \bar{\mathcal{R}}_e$ be arbitrary. Since g is total and computable, from any program witnessing $K(s \mid r')$ one can compute s from r' and then apply g with constant overhead to obtain $r = g(s)$. Hence

$$K(r \mid r') \leq K(s \mid r') + O(1).$$

Taking minima over $r' \in \bar{\mathcal{R}}_e$, we obtain

$$s_R(e, r) = \min_{r' \in \bar{\mathcal{R}}_e} K(r \mid r') \leq \min_{r' \in \bar{\mathcal{R}}_e} (K(s \mid r') + O(1)) = s_R(e, s) + O(1).$$

■

This proposition is the surplus-dual of the monotonicity of deficiency under computable enrichment. Scientifically, it captures the idea that simplifying a representation in a computable way cannot substantially worsen its surplus; if anything, reduction tends to remove content that is not genuinely required to encode the entity.

The case of deleting a contiguous block is an immediate instance of the previous proposition. It corresponds to a direct reduction operation in which part of the representation is discarded while the remainder is kept unchanged.

Corollary 5.2.5 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings and let $e \in \mathcal{E}$ be a knowable entity. Suppose that both $rst \in \mathcal{R}_e$ and $rt \in \mathcal{R}_e$ are representations of e . Then

$$s_R(e, rt) \leq s_R(e, rst) + O(1).$$

Proof. Define a total computable function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ by letting $g(rst) = rt$, that is, g discards the middle block s . Since $g(rst) = rt$, the conclusion follows directly from the previous proposition with $s = rst$ and $r = rt$. ■

Thus, removing a block from a representation cannot substantially increase its surplus, provided that the reduced string still qualifies as an representation of the same entity. This is the natural counterpart, for surplus, of the augmentation corollary established earlier for deficiency.

The next proposition identifies a simple condition under which extending a representation does not change its surplus beyond the usual additive constant. This is the dual counterpart of the analogous result for deficiency under safe reduction: if the appended part is already recoverable from the original representation at constant cost, then its presence does not introduce any substantial extra content.

Proposition 5.2.6 Let $e \in \mathcal{E}$ be a knowable entity, and let $x, y \in \mathcal{B}^*$ be such that both $xy \in \mathcal{R}_e$ and $x \in \mathcal{R}_e$ are representations of e . If

$$K(y \mid x) = O(1),$$

then

$$s_R(e, xy) = s_R(x, e) + O(1).$$

Proof. Since x is computable from xy by deleting the suffix y , the previous proposition gives

$$s_R(e, x) \leq s_R(e, xy) + O(1).$$

On the other hand, the condition $K(y \mid x) = O(1)$ implies that there is a constant-length program that reconstructs y from x . Hence from x one can compute xy with constant overhead. Applying the previous proposition again, now with $s = x$ and $r = xy$, gives

$$s_R(e, xy) \leq s_R(e, x) + O(1).$$

Combining both inequalities yields

$$s_R(e, xy) = s_R(e, x) + O(1).$$

■

This proposition formalizes the idea of a safe extension: appending information that is already computably recoverable from the original representation does not substantially increase surplus. In practice, it justifies adding predictable or duplicative fields without changing, except up to a constant, the amount of genuinely extraneous information present in the representation.

Finally, we derive a simple but useful upper bound on surplus. In contrast with the deficiency bound, which is controlled by the admissible complexity floor of the entity, the most immediate general bound for surplus is controlled by the complexity of the representation itself.

Proposition 5.2.7 For every representation $r \in \mathcal{R}_e$,

$$s_R(e, r) \leq K(r) + O(1) \leq \ell(r) + O(1).$$

Proof. Fix any $r' \in \bar{\mathcal{R}}_e$. Then

$$s_R(e, r) = \min_{u \in \bar{\mathcal{R}}_e} K(r \mid u) \leq K(r \mid r') \leq K(r) + O(1),$$

because a program can ignore r' and simply output r , incurring only constant overhead. Finally, $K(r) \leq \ell(r) + O(1)$ yields the length bound. ■

This bound shows that surplus is always finite and, in the worst case, is controlled by the descriptive complexity of the representation itself. Even if the representation contains a large amount of irrelevant material, that excess cannot exceed, up to a constant, the total amount of information present in the representation.

5.3 Miscoding

The concepts of deficiency and surplus capture two fundamentally different ways in which a representation may fail. Deficiency measures how much information about the entity is still missing, whereas surplus measures how much information in the representation is not genuinely attributable to the entity. Since these two quantities need not decrease simultaneously, the analysis of miscoding is naturally formulated as a multiobjective optimization problem (see Section 21.6) rather than as a single objective one.

Pareto optimality

Before introducing a scalar measure of miscoding, it is useful to make explicit that representation refinement involves two competing objectives.

Definition 5.3.1 Let $e \in \mathcal{E}$ be an entity. We define the *miscoding problem* for the entity e as the following multiobjective optimization problem

$$\min_{r \in \mathcal{R}_e} (d_R(e, r), s_R(e, r)).$$

This formulation emphasizes that miscoding is not, at its core, a single number but a trade-off between two distinct desiderata: informational completeness and absence of irrelevant content. In scientific practice, some refinements decrease deficiency at the cost of increasing surplus, while others remove surplus but also discard useful information.

The natural comparison relation for the problem of miscoding is dominance (see Definition 21.6.1). Before defining optimal representations, we introduce the usual notion of dominance in the context of representations.

Definition 5.3.2 Let $r, r' \in \mathcal{R}_e$ be two representation. We say that r' *dominates* r , denoted by $r' \prec_e r$, if

$$d_R(e, r') \leq d_R(e, r), \quad s_R(e, r') \leq s_R(e, r),$$

and at least one of these inequalities is strict.

A dominating representation is at least as good in both components and strictly better in one. Thus, a dominated representation is scientifically suboptimal in a very strong sense: there is another available representation that improves one source of miscoding without worsening the other.

The appropriate notion of optimality for a multiobjective problem is Pareto optimality (see Definition 21.6.2). We are interested in those representations for which no simultaneous improvement is possible.

Definition 5.3.3 Let $e \in \mathcal{E}$. A representation $r \in \mathcal{R}_e$ is said to be *Pareto optimal* (with respect to deficiency and surplus) if there does not exist any $r' \in \mathcal{R}_e$ such that $r' \prec_e r$.

The set of all Pareto-optimal representations of e is denoted by

$$\mathcal{P}_e = \{r \in \mathcal{R}_e : \nexists r' \in \mathcal{R}_e \text{ such that } r' \prec_e r\},$$

and is called the *Pareto frontier* of the miscoding problem.

The Pareto frontier collects precisely those representations for which any further improvement in one component requires a deterioration in the other. In this sense, it plays the role of a representation-level knowledge frontier: beyond it, refinement becomes a matter of trade-offs rather than unambiguous progress.

An immediate property, but important, is that admissible representations lie on the Pareto frontier.

Proposition 5.3.1 If $r \in \bar{\mathcal{R}}_e$, then $r \in \mathcal{P}_e$.

Proof. Let $r \in \bar{\mathcal{R}}_e$. By the basic properties of deficiency and surplus, we have

$$d_R(e, r) = O(1), \quad s_R(e, r) = O(1).$$

Suppose, for contradiction, that $r \notin \mathcal{P}_e$. Then there exists $r' \in \mathcal{R}_e$ such that

$$d_R(e, r') \leq d_R(e, r), \quad s_R(e, r') \leq s_R(e, r),$$

with at least one strict inequality. Since both $d_R(e, r)$ and $s_R(e, r)$ are already minimal up to additive constants, this is impossible. Hence no such r' exists, and therefore $r \in \mathcal{P}_e$. ■

This proposition shows that admissible representations are never dominated. More generally, the Pareto frontier should be understood as the set of the best currently available approximations to such ideal encodings.

The next property clarifies why dominated representations can be safely discarded.

Proposition 5.3.2 Let $\phi : \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}_{\geq 0}$ be strictly increasing in each argument. If $r \in \mathcal{R}_e$ is dominated, then r cannot minimize $\phi(d_R(e, r), s_R(e, r))$ over $\mathcal{R}_e$.

Proof. Assume that r is dominated. Then there exists $r' \in \mathcal{R}_e$ such that

$$d_R(e, r') \leq d_R(e, r), \quad s_R(e, r') \leq s_R(e, r),$$

and at least one inequality is strict. Since ϕ is strictly increasing in each argument, it follows that

$$\phi(d_R(e, r'), s_R(e, r')) < \phi(d_R(e, r), s_R(e, r)).$$

Therefore r cannot be a minimizer of ϕ . ■

This result formalizes an important elimination principle: any representation that is dominated is suboptimal for every sensible scalar criterion that penalizes both deficiency and surplus. Consequently, once the Pareto frontier has been identified, all remaining scalarizations are selection rules *on* that frontier.

To ensure that the optimization problem is well posed in practical finite settings, it is useful to note that Pareto-optimal representations always exist.

Proposition 5.3.3 If $\mathcal{R}_e$ is finite and non-empty, then $\mathcal{P}_e$ is non-empty.

Proof. Consider the finite set

$$S = \{(d_R(e, r), s_R(e, r)) : r \in \mathcal{R}_e\} \subseteq \mathbb{R}_{\geq 0}^2.$$

Since S is finite and non-empty, it contains at least one minimal element with respect to the product order

$$(a', b') \preceq (a, b) \iff a' \leq a \text{ and } b' \leq b.$$

Let $(d_R(e, r), s_R(e, r))$ be such a minimal element. Then no $r' \in \mathcal{R}_e$ can dominate r , since that would contradict minimality. Hence $r \in \mathcal{P}_e$. ■

This proposition guarantees that in any finite scientific comparison there is always at least one non-dominated representation. The real difficulty is therefore not existence but selection among the competing Pareto-optimal alternatives.

Scalar miscoding

Although the Pareto frontier captures the full trade-off structure, in many applications one still wishes to assign a single miscoding value to each representation. A natural scalarization is to minimize the worst of the two components.

We now introduce the canonical scalar score used in this book for the concept of miscoding.

Definition 5.3.4 Let $r \in \mathcal{R}_e$. The *miscoding* of r is defined by

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\}.$$

This definition corresponds to the Chebyshev scalarization of a bi-objective problem (see Section 21.6.3). It assigns to each representation the magnitude of its dominant defect, and therefore penalizes imbalance between the two components.

A justification for this definition is that any minimizer of μ must lie on the Pareto frontier.

Proposition 5.3.4 If $r \in \mathcal{R}_e$ minimizes μ over $\mathcal{R}_e$, then $r \in \mathcal{P}_e$.

Proof. Suppose, for contradiction, that r minimizes μ but $r \notin \mathcal{P}_e$. Then there exists $r' \in \mathcal{R}_e$ such that

$$d_R(e, r') \leq d_R(e, r), \quad s_R(e, r') \leq s_R(e, r),$$

and at least one inequality is strict. Therefore,

$$\max\{d_R(e, r'), s_R(e, r')\} < \max\{d_R(e, r), s_R(e, r)\},$$

that is,

$$\mu(e, r') < \mu(e, r),$$

contradicting the minimality of r . Hence $r \in \mathcal{P}_e$. ■

Thus, Chebyshev miscoding does not ignore the Pareto structure; it selects one of its best-balanced points. In this way, the scalar score remains compatible with the underlying multiobjective formulation.

A useful property of the Chebyshev scalarization is that it provides a uniform upper bound on both deficiency and surplus.

Proposition 5.3.5 For every $r \in \mathcal{R}_e$ and every $t \geq 0$,

$$\mu(e, r) \leq t \iff d_R(e, r) \leq t \text{ and } s_R(e, r) \leq t.$$

Proof. By definition,

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\}.$$

Hence $\mu(e, r) \leq t$ if and only if the maximum of the two quantities is at most t , which happens if and only if each quantity is at most t . ■

This equivalence is one of the main reasons to prefer the maximum. It says that minimizing miscoding is exactly the same as pushing both components below the smallest possible common threshold.

The maximum also identifies which of the two components is currently the main obstacle to improvement.

Proposition 5.3.6 Let $r \in \mathcal{R}_e$.

- 1 If $d_R(e, r) > s_R(e, r)$, then any refinement r' such that

$$d_R(e, r') < d_R(e, r) \quad \text{and} \quad s_R(e, r') \leq d_R(e, r')$$

necessarily satisfies $\mu(e, r') < \mu(e, r)$.

- 2 If $s_R(e, r) > d_R(e, r)$, then any refinement r' such that

$$s_R(e, r') < s_R(e, r) \quad \text{and} \quad d_R(e, r') \leq s_R(e, r')$$

necessarily satisfies $\mu(e, r') < \mu(e, r)$.

Proof. Assume first that $d_R(e, r) > s_R(e, r)$. Then

$$\mu(e, r) = d_R(e, r).$$

If r' satisfies

$$d_R(e, r') < d_R(e, r), \quad s_R(e, r') \leq d_R(e, r'),$$

then

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\} = d_R(e, r') < d_R(e, r) = \mu(e, r).$$

The second part is symmetric. If $s_R(e, r) > d_R(e, r)$, then

$$\mu(e, r) = s_R(e, r),$$

and if

$$s_R(e, r') < s_R(e, r), \quad d_R(e, r') \leq s_R(e, r'),$$

then

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\} = s_R(e, r') < s_R(e, r) = \mu(e, r).$$

■

This proposition has a clear methodological interpretation. When deficiency dominates, the natural priority is augmentation; when surplus dominates, the natural priority is reduction or cleaning.

The main advantage of the Chebyshev scalarization is that it enforces balance between the two types of representational error. The next proposition formalizes the sense in which minimizing the maximum avoids extreme solutions.

Proposition 5.3.7 Let $r_1, r_2 \in \mathcal{R}_e$. Suppose that

$$d_R(e, r_1) + s_R(e, r_1) = d_R(e, r_2) + s_R(e, r_2) = C$$

for some constant C , and assume moreover that

$$|d_R(e, r_1) - s_R(e, r_1)| \leq |d_R(e, r_2) - s_R(e, r_2)|.$$

Then

$$\mu(e, r_1) \leq \mu(e, r_2).$$

Proof. For any nonnegative numbers a, b with $a + b = C$, one has

$$\max\{a, b\} = \frac{C + |a - b|}{2}.$$

Applying this identity to the pairs

$$(a, b) = (d_R(e, r_1), s_R(e, r_1)) \quad \text{and} \quad (a, b) = (d_R(e, r_2), s_R(e, r_2)),$$

we obtain

$$\mu(e, r_1) = \frac{C + |d_R(e, r_1) - s_R(e, r_1)|}{2},$$

and

$$\mu(e, r_2) = \frac{C + |d_R(e, r_2) - s_R(e, r_2)|}{2}.$$

Since

$$|d_R(e, r_1) - s_R(e, r_1)| \leq |d_R(e, r_2) - s_R(e, r_2)|,$$

it follows that

$$\mu(e, r_1) \leq \mu(e, r_2).$$

■

Among representations with the same total burden, the maximum prefers the more balanced one. This is precisely the desired behavior when miscoding is meant to identify representations that are not only good on average, but also free from a dominant unresolved defect.

A final basic property is that the Chebyshev scalarization is the smallest uniform upper bound on the two error components.

Proposition 5.3.8 For every $r \in \mathcal{R}_e$, $\mu(e, r)$ is the smallest number $t \geq 0$ such that

$$d_R(e, r) \leq t \quad \text{and} \quad s_R(e, r) \leq t.$$

Proof. By Proposition 5.3.5,

$$\mu(e, r) \leq t \iff d_R(e, r) \leq t \text{ and } s_R(e, r) \leq t.$$

Taking $t = \mu(e, r)$ shows that $\mu(e, r)$ is a common upper bound. If $t < \mu(e, r)$, then the equivalence above implies that at least one of $d_R(e, r)$ or $s_R(e, r)$ is greater than t . Hence no smaller common upper bound exists. ■

This characterization gives the maximum a clear operational meaning. It measures the least tolerance level under which both kinds of representational error can simultaneously be regarded as acceptable.

Comparison with alternative scalarizations

The maximum is not the only possible way to extract a scalar score from the pair $(d_R(e, r), s_R(e, r))$. Nevertheless, alternative scalarizations embody different epistemic priorities and have different advantages and drawbacks.

A first possibility is sum scalarization

$$\mu^\Sigma(e, r) = d_R(e, r) + s_R(e, r).$$

This criterion measures total representational burden. It is simple, symmetric, and often convenient analytically, but it allows compensation: a large value of one component may be offset by a small value of the other.

The following observation makes this drawback precise.

Proposition 5.3.9 There exist representations $r_1, r_2 \in \mathcal{R}_e$ such that

$$\mu^\Sigma(e, r_1) = \mu^\Sigma(e, r_2)$$

but

$$\mu(e, r_1) > \mu(e, r_2).$$

Proof. It suffices to consider two hypothetical pairs of values:

$$(d_R(e, r_1), s_R(e, r_1)) = (C, 0), \quad (d_R(e, r_2), s_R(e, r_2)) = \left(\frac{C}{2}, \frac{C}{2}\right),$$

with $C > 0$. Then

$$\mu^\Sigma(e, r_1) = C = \mu^\Sigma(e, r_2),$$

but

$$\mu(e, r_1) = \max\{C, 0\} = C$$

whereas

$$\mu(e, r_2) = \max\left\{\frac{C}{2}, \frac{C}{2}\right\} = \frac{C}{2}.$$

Hence the sum treats the two representations as equivalent, while the Chebyshev scalarization strictly prefers the more balanced one. ■

The sum is therefore useful when one cares about aggregate burden, but less appropriate when one wants to prevent a single severe defect from being masked by good performance elsewhere. For miscoding, this masking effect is often undesirable.

A second possibility is to introduce asymmetric penalties, that is, a weighted Chebyshev scalarization:

$$\mu^{(\lambda)}(e, r) = \max\{\lambda d_R(e, r), (1 - \lambda) s_R(e, r)\}, \quad 0 < \lambda < 1.$$

This criterion is useful when one source of error is judged more serious than the other in a given application. For example, in some empirical settings it may be preferable to tolerate moderate surplus in order to avoid missing rare but important phenomena.

Its advantage is flexibility, but that flexibility is also its weakness: the choice of λ is external to the formal theory and must be justified by domain-specific considerations. Accordingly, the weighted maximum is best understood as an application-dependent refinement of the unweighted definition rather than as the canonical notion of miscoding itself.

A third possibility is lexicographic ordering. For example, one may declare that

$$r_1 \prec_{\text{lex}} r_2$$

whenever

$$d_R(e, r_1) < d_R(e, r_2),$$

or

$$d_R(e, r_1) = d_R(e, r_2) \text{ and } s_R(e, r_1) < s_R(e, r_2).$$

This corresponds to an absolute priority on reducing deficiency, with surplus considered only as a secondary criterion.

Lexicographic rules are appropriate when one objective is genuinely primary. However, they are too rigid for a general theory of miscoding, since an arbitrarily small improvement in the primary objective can justify an arbitrarily large deterioration in the secondary one.

A simple example shows this clearly.

Proposition 5.3.10 For every $\varepsilon > 0$ and every $M > 0$, there exist two pairs (a_1, b_1) and (a_2, b_2) in $\mathbb{R}_{\geq 0}^2$ such that

$$a_1 < a_2 < a_1 + \varepsilon,$$

$$b_2 + M < b_1,$$

and yet the lexicographic order prioritizing the first coordinate strictly prefers (a_1, b_1) to (a_2, b_2) .

Proof. Take

$$(a_1, b_1) = (0, M+1), \quad (a_2, b_2) = \left(\frac{\varepsilon}{2}, 0\right).$$

Then

$$a_1 = 0 < \frac{\varepsilon}{2} = a_2 < \varepsilon = a_1 + \varepsilon,$$

and

$$0 + M < M + 1.$$

Thus the second pair is better by more than M in the second coordinate, but lexicographic order prioritizing the first coordinate still prefers the first pair, since $a_1 < a_2$. ■

This illustrates the main drawback of lexicographic scalarization: it can force highly unbalanced choices. Such a policy may be appropriate in narrow applications, but it is not a good default for a theory intended to capture the overall quality of a representation.

Negligible miscoding characterizes algorithmic equivalence to perfection.

Proposition 5.3.11 For $r \in \mathcal{R}_e$, the following are equivalent:

- (i) $\mu(e, r) = O(1)$.
- (ii) There exist $r_d^*, r_s^* \in \mathcal{R}_e^*$ such that $K(r_d^* | r) = O(1)$ and $K(r | r_s^*) = O(1)$.

Proof. ($i \Rightarrow ii$) If $\mu(e, r) = O(1)$, then $d_R(e, r) \leq \mu(e, r) = O(1)$ and $s_R(e, r) \leq \mu(e, r) = O(1)$. By definition of the minima in $d_R(e, r)$ and $s_R(e, r)$, there exist $r_d^*, r_s^* \in \mathcal{R}_e^*$ such that $K(r_d^* | r) = d_R(e, r) = O(1)$ and $K(r | r_s^*) = s_R(e, r) = O(1)$.

($ii \Rightarrow i$) If such r_d^*, r_s^* exist, then $d_R(e, r) \leq K(r_d^* | r) = O(1)$ and $s_R(e, r) \leq K(r | r_s^*) = O(1)$. Therefore $\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\} = O(1)$. ■

5.4 Relative miscoding

The quantity

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\}$$

measures the miscoding of an representation $r \in \mathcal{R}_e$ in absolute terms, that is, in bits. This absolute measure is the appropriate one for the structural theory developed in the previous sections. In practice, however, one often wishes to compare the miscoding of representations of different sizes, or even of representations associated with different entities. For this purpose, it is useful to introduce a normalized, dimensionless version of miscoding.

The idea is simple: we divide the absolute miscoding by a complexity scale that is intrinsic both to the representation and to the entity being represented. This yields a relative score that is less sensitive to scale effects and therefore better suited for practical comparison.

Definition 5.4.1 Let $e \in \mathcal{E}$ and let $r \in \mathcal{R}_e$ be an effective representation of e . We define the *relative miscoding* of r with respect to e as

$$\mu^{\text{rel}}(e, r) = \frac{\mu(e, r)}{\max\{K(r), \kappa_e\}} = \frac{\max\{d_R(e, r), s_R(e, r)\}}{\max\{K(r), \kappa_e\}},$$

Recall that

$$\kappa_e = \min_{r \in \mathcal{R}_e^\top} K(r).$$

is the perfect complexity floor of e (see Definiton 3.3.3).

The denominator $\max\{K(r), \kappa_e\}$ provides a stable intrinsic scale. It avoids degenerate normalizations when $K(r)$ is unusually small, while ensuring that the normalization remains tied to the minimum algorithmic information required to represent e accurately. In this way, relative miscoding preserves the meaning of absolute miscoding while making its magnitude comparable across representations of different sizes.

The first result shows that relative miscoding is essentially bounded between 0 and 1, up to the usual additive constants of Kolmogorov complexity.

Proposition 5.4.1 For every $r \in \mathcal{R}_e$, it holds that

$$0 \leq \mu^{\text{rel}}(e, r) \leq 1 + o(1),$$

where the $o(1)$ term denotes a quantity of the form

$$\frac{O(1)}{\max\{K(r), \kappa_e\}},$$

which vanishes for sufficiently complex representations or entities.

Proof. Non-negativity is immediate, since $d_R(e, r) \geq 0$ and $s_R(e, r) \geq 0$, hence

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\} \geq 0.$$

Therefore $\mu^{\text{rel}}(e, r) \geq 0$.

For the upper bound, recall first that

$$s_R(e, r) \leq K(r) + O(1),$$

since for any $r' \in \mathcal{R}_e^\top$, a program can ignore r' and output r , and taking the minimum over $r' \in \mathcal{R}_e^\top$ preserves this bound.

Second,

$$d_R(e, r) \leq \kappa_e + O(1),$$

because if $r_0 \in \mathcal{R}_e^\top$ satisfies $K(r_0) = \kappa_e$, then

$$K(r_0 \mid r) \leq K(r_0) + O(1) = \kappa_e + O(1),$$

and taking the minimum over all perfect representations yields the claim.

Hence

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\} \leq \max\{\kappa_e, K(r)\} + O(1).$$

Dividing by $\max\{K(r), \kappa_e\}$ gives

$$\mu^{\text{rel}}(e, r) \leq 1 + \frac{O(1)}{\max\{K(r), \kappa_e\}} = 1 + o(1).$$

■

This proposition shows that relative miscoding behaves as a normalized error score. In particular, values close to 0 indicate low miscoding relative to the intrinsic size of the problem, whereas values close to 1 indicate that the dominant representational defect is of the same order as the representation itself or the entity's perfect complexity floor.

The next proposition connects the normalized notion with the ideal case of admissible representations.

Proposition 5.4.2 If $r \in \bar{\mathcal{R}}_e$, then

$$\mu^{\text{rel}}(e, r) = o(1).$$

In particular, for sufficiently complex entities or representations, $\mu^{\text{rel}}(e, r) \approx 0$.

Proof. If $r \in \bar{\mathcal{R}}_e$, then

$$d_R(e, r) = O(1) \quad \text{and} \quad s_R(e, r) = O(1),$$

hence

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\} = O(1).$$

Therefore

$$\mu^{\text{rel}}(e, r) = \frac{O(1)}{\max\{K(r), \kappa_e\}} = o(1).$$

■

This is the normalized analogue of the fact that perfect representations have negligible absolute miscoding. It confirms that relative miscoding is fully compatible with the theoretical ideal: once the intrinsic complexity scale is large, the residual $O(1)$ ambiguity becomes asymptotically negligible.

The following result shows that negligible absolute miscoding always implies negligible relative miscoding.

Proposition 5.4.3 Let $r \in \mathcal{R}_e$. If

$$\mu(e, r) = O(1),$$

then

$$\mu^{\text{rel}}(e, r) = o(1).$$

Proof. If $\mu(e, r) = O(1)$, then by Definition 5.4.1,

$$\mu^{\text{rel}}(e, r) = \frac{\mu(e, r)}{\max\{K(r), \kappa_e\}} = \frac{O(1)}{\max\{K(r), \kappa_e\}} = o(1).$$

■

This property makes explicit that relative miscoding is a genuine normalization of absolute miscoding. Whenever the absolute representational error is negligible, the normalized version is negligible as well, and hence no spurious inflation is introduced by passing to the relative scale.

The converse direction does not hold in general. A representation may have large absolute miscoding and still small relative miscoding if its overall complexity is even larger.

Proposition 5.4.4 In general, $\mu^{\text{rel}}(e, r) = o(1)$ does not imply $\mu(e, r) = O(1)$.

Proof. Fix any admissible representation $r' \in \bar{\mathcal{R}}_e$, and let z be an incompressible string independent of r' . Define r as a lossless encoding of the pair (r', z) . Then r' is computable from r , so

$$d_R(e, r) = O(1).$$

On the other hand,

$$K(r \mid r') \geq K(z) - O(1),$$

so $s_R(e, r)$ can be made arbitrarily large by increasing $\ell(z)$. Hence

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\} \approx \ell(z),$$

which is unbounded.

At the same time,

$$K(r) \approx K(r') + K(z),$$

so

$$\mu^{\text{rel}}(e, r) = \frac{\mu(e, r)}{\max\{K(r), \kappa_e\}} \approx \frac{\ell(z)}{K(r') + \ell(z)}.$$

Now modify the construction by replacing z with a string z_n whose length tends to infinity while ensuring that the surplus grows sublinearly with $K(r)$, for instance by encoding r as a lossless extension in which the additional information increases more slowly than the total complexity. Then $\mu(e, r_n) \rightarrow \infty$ while

$$\mu^{\text{rel}}(e, r_n) \rightarrow 0.$$

Thus small relative miscoding does not force small absolute miscoding. ■

This proposition shows that relative miscoding and absolute miscoding serve different purposes. Absolute miscoding captures the actual algorithmic cost of repair, whereas relative miscoding measures that cost only in proportion to the intrinsic scale of the representation.

The next property shows that relative miscoding is stable under bounded effective recodings.

Proposition 5.4.5 Let $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a total computable bijection with computable inverse and descriptive complexity $K(f) = O(1)$. Then for every $r \in \mathcal{R}_e$,

$$\mu^{\text{rel}}(e, f(r)) = \mu^{\text{rel}}(e, r) + o(1).$$

Proof. For every $r' \in \bar{\mathcal{R}}_e$, standard invariance properties of Kolmogorov complexity under computable bijections yield

$$K(f(r') \mid f(r)) = K(r' \mid r) + O(1),$$

and

$$K(f(r) \mid f(r')) = K(r \mid r') + O(1).$$

Assuming the representation framework is transported coherently under f , taking minima over $r' \in \mathcal{R}'_e$ gives

$$d_R(e, f(r)) = d_R(e, r) + O(1) \quad \text{and} \quad s_R(e, f(r)) = s_R(e, r) + O(1).$$

Hence

$$\mu(e, f(r)) = \mu(e, r) + O(1).$$

Moreover,

$$K(f(r)) = K(r) + O(1),$$

again by invariance under total computable bijections with computable inverse. Therefore

$$\max\{K(f(r)), \kappa_e\} = \max\{K(r), \kappa_e\} + O(1).$$

Substituting into Definition 5.4.1, we obtain

$$\begin{aligned} \mu^{\text{rel}}(e, f(r)) &= \frac{\mu(e, r) + O(1)}{\max\{K(r), \kappa_e\} + O(1)} \\ &= \mu^{\text{rel}}(e, r) + \frac{O(1)}{\max\{K(r), \kappa_e\}} \\ &= \mu^{\text{rel}}(e, r) + O(1). \end{aligned}$$

■

This result shows that relative miscoding does not depend substantially on the accidental details of a reasonable encoding convention. Up to asymptotically negligible terms, bounded effective recodings preserve the normalized amount of representational error.

It is also useful to make explicit the relation between absolute and relative ordering when the compared representations have the same intrinsic scale.

Proposition 5.4.6 Let $r_1, r_2 \in \mathcal{R}_e$ satisfy

$$\max\{K(r_1), \kappa_e\} = \max\{K(r_2), \kappa_e\}.$$

Then

$$\mu(e, r_1) \leq \mu(e, r_2) \iff \mu^{\text{rel}}(e, r_1) \leq \mu^{\text{rel}}(e, r_2).$$

Proof. If the denominators coincide, say both equal $C > 0$, then

$$\mu^{\text{rel}}(e, r_i) = \frac{\mu(e, r_i)}{C} \quad (i = 1, 2).$$

Division by the same positive constant preserves order, so

$$\mu(e, r_1) \leq \mu(e, r_2) \iff \frac{\mu(e, r_1)}{C} \leq \frac{\mu(e, r_2)}{C},$$

which is exactly

$$\mu^{\text{rel}}(e, r_1) \leq \mu^{\text{rel}}(e, r_2).$$

■

This proposition clarifies the practical meaning of the normalization. When the compared representations have the same intrinsic scale, relative miscoding does not change the ranking; it simply rescales the absolute values. The normalization matters only when the underlying scales differ.

Finally, relative miscoding preserves the bottleneck interpretation of the absolute miscoding, since the denominator is common to both deficiency and surplus.

Proposition 5.4.7 Let $r \in \mathcal{R}_e$.

1 If $d_R(e, r) > s_R(e, r)$, then

$$\mu^{\text{rel}}(e, r) = \frac{d_R(e, r)}{\max\{K(r), \kappa_e\}}.$$

2 If $s_R(e, r) > d_R(e, r)$, then

$$\mu^{\text{rel}}(e, r) = \frac{s_R(e, r)}{\max\{K(r), \kappa_e\}}.$$

Proof. By definition,

$$\mu^{\text{rel}}(e, r) = \frac{\max\{d_R(e, r), s_R(e, r)\}}{\max\{K(r), \kappa_e\}}.$$

If $d_R(e, r) > s_R(e, r)$, then the numerator equals $d_R(e, r)$. If $s_R(e, r) > d_R(e, r)$, then the numerator equals $s_R(e, r)$. The stated equalities follow immediately. ■

Thus, relative miscoding inherits the same diagnostic value as absolute miscoding: it identifies the currently dominant representational defect. The only difference is that this defect is now expressed relative to the intrinsic scale of the representation and the entity.

Interpretation.

A value $\mu^{\text{rel}}(e, r) \approx 0$ indicates that, relative to its overall information content, the representation has both low deficiency and low surplus; it is therefore close to perfect for e in a scale-free sense. Values close to 1 indicate that either deficiency or surplus is of the same order as the representation's intrinsic scale $\max\{K(r), \kappa_e\}$, meaning that the representation is highly incomplete, highly contaminated, or both.

Relative miscoding is therefore best understood as a practical comparison tool rather than as a replacement for absolute miscoding. The latter remains the primary structural notion, whereas the former provides a normalized view better suited for comparing representations of different sizes.

Finally, one should keep in mind that the interpretation of $\mu^{\text{rel}}(e, r)$ still presupposes that the target entity e has been fixed. Determining which entity a given string r actually represents remains a separate problem. In scientific practice, hypotheses about reference may change over time, and thus the interpretation of a fixed string may shift as knowledge evolves.

5.5 Decreasing Miscoding

For a fixed entity e , miscoding measures how far a representation $r \in \mathcal{R}_e$ is from being simultaneously informationally complete and free of extraneous content. In the theory of nescience these two requirements are measured separately by the metrics of deficiency and surplus. Thus, decreasing miscoding is intrinsically a two-objective refinement problem: one aims to reduce missing information about e (deficiency) while also removing information that is not attributable to e (surplus).

In scientific practice these correspond to two complementary modes of improvement: augmentation (adding information) is typically aimed at reducing deficiency, whereas reduction (removing information) is typically aimed at reducing surplus.

Decreasing Deficiency via Augmentation

A natural way to reduce the deficiency of a representation is to augment it with additional information. In scientific practice, this corresponds to collecting new observations, increasing measurement resolution, adding missing variables, or incorporating contextual metadata. Intuitively, if the added material is relevant, the resulting representation should be closer to an admissible representation of the entity, and therefore require less additional information to complete it.

Within a fixed representation framework, augmenting an representation with additional information cannot increase its deficiency by more than an additive constant.

Proposition 5.5.1 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings and let $e \in \mathcal{E}$ be a knowable entity. Suppose that both $rt \in \mathcal{R}_e$ and $rst \in \mathcal{R}_e$ are representations of e . Then

$$d_R(e, rst) \leq d_R(e, rt) + O(1).$$

Proof. Let $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a computable function that maps rst to rt by discarding the middle block s . For any $r' \in \tilde{\mathcal{R}}_e$, monotonicity of conditional Kolmogorov

complexity under computable transformations yields

$$K(r' \mid rst) \leq K(r' \mid g(rst)) + O(1) = K(r' \mid rt) + O(1),$$

since from rst one can compute $g(rst) = rt$ with constant overhead and then simulate an optimal program for $K(r' \mid rt)$. Taking the minimum over $r' \in \bar{\mathcal{R}}_e$ gives

$$d_R(e, rst) = \min_{r' \in \bar{\mathcal{R}}_e} K(r' \mid rst) \leq \min_{r' \in \bar{\mathcal{R}}_e} (K(r' \mid rt) + O(1)) = d_R(e, rt) + O(1).$$

■

This proposition captures a basic monotonicity principle: once additional information has been incorporated into a representation, recovering an admissible representation of the entity cannot become substantially harder. Of course, the result does not guarantee that every augmentation is useful in practice. Some added information may be redundant, irrelevant, or even misleading. What matters scientifically is not augmentation per se, but augmentation that reduces deficiency by supplying information that was genuinely missing.

To make this idea explicit, it is useful to measure how much deficiency is reduced by adding a given block of information.

Definition 5.5.1 Let $e \in \mathcal{E}$ be a knowable entity, let $r \in \mathcal{R}_e$ be an representation of e , and let $a \in \mathcal{B}^*$ be such that $ra \in \mathcal{R}_e$. We define the *marginal deficiency reduction* of a relative to r by

$$\Delta_R^d(e, r, a) = d_R(e, r) - d_R(e, ra).$$

The quantity $\Delta_R^d(e, r, a)$ measures how much the additional information a reduces the amount of missing algorithmic information needed to recover an admissible representation of e . If $\Delta_R^d(e, r, a)$ is large, then a supplies information that was highly relevant and previously absent from r . If it is close to zero, then the augmentation contributes little toward completing the representation.

This notion suggests a practical decision principle for scientific inquiry: among several possible augmentations, a good next observation is one that is expected to produce the largest reduction in posterior deficiency. In other words, one should prefer the experiment, measurement, or data collection step that is expected to bring the current representation closest to an admissible one.

■ **Example 5.3** Suppose that the entity e is the complete interaction network of a biological signaling pathway. The current representation r may include only a subset of the relevant molecular interactions, obtained from existing experiments. Researchers now face a choice between two possible next observations.

The first option, a_1 , is to repeat an already well-characterized assay at slightly higher precision. This may improve confidence in edges that are already known, but it is unlikely to reveal fundamentally new parts of the network. The second option, a_2 , is to measure interactions involving a poorly understood regulatory protein that has not yet been incorporated into the current representation. If that protein is in fact central to the pathway, then observing its interactions may fill a major structural gap in the network.

In the language of deficiency, the scientifically preferable choice is the one expected to minimize the posterior deficiency, that is, the one expected to make $d_e(ra)$ as small as possible. In this scenario, a_2 is likely to be the better next observation, because it has greater potential to reduce the missing information required to recover an admissible representation of the full signaling network. By contrast, a_1 may add useful detail, but only a small amount of genuinely new information. Thus, a good next observation is not necessarily the one that produces the most data, but the one that is expected to reduce deficiency the most.

■

The concept of marginal deficiency reduction turns deficiency into a prescriptive tool. Rather than merely quantifying how incomplete a representation currently is, it provides a way to evaluate candidate augmentations according to how much they are expected to reduce that incompleteness. In this sense, deficiency does not only measure what is missing; it also helps guide the choice of what should be observed next.

Decreasing Surplus via Reduction

A natural way to reduce the surplus of a representation is to remove information that is not genuinely needed to encode the entity. In scientific practice, this corresponds to discarding irrelevant variables, removing administrative metadata, filtering spurious attributes, or simplifying a representation by deleting content that does not contribute to the representation of the entity itself. Intuitively, if the removed material is truly extraneous, the resulting representation should be closer to an admissible representation of the entity, and therefore contain less additional algorithmic content beyond what is needed to encode it.

Within a fixed representation framework, reducing a representation by deleting information cannot increase its surplus by more than an additive constant.

Proposition 5.5.2 Let $r, s, t \in \mathcal{B}^*$ be arbitrary binary strings and let $e \in \mathcal{E}$ be a knowable entity. Suppose that both $rst \in \mathcal{R}_e$ and $rt \in \mathcal{R}_e$ are representations of e . Then

$$s_e(rt) \leq s_e(rst) + O(1).$$

Proof. Let $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a computable function that maps rst to rt by discarding

the middle block s . For any $r' \in \tilde{\mathcal{R}}_e$, monotonicity of conditional Kolmogorov complexity under computable transformations yields

$$K(rt \mid r') \leq K(g(rst) \mid r') + O(1) \leq K(rst \mid r') + O(1),$$

since from any program that outputs rst given r' one can compute $g(rst) = rt$ with constant overhead. Taking the minimum over $r' \in \tilde{\mathcal{R}}_e$ gives

$$s_e(rt) = \min_{r' \in \tilde{\mathcal{R}}_e} K(rt \mid r') \leq \min_{r' \in \tilde{\mathcal{R}}_e} (K(rst \mid r') + O(1)) = s_e(rst) + O(1).$$

■

This proposition captures the basic monotonicity principle for surplus: once part of a representation has been removed, generating the reduced representation from an admissible one cannot become substantially harder. Of course, the result does not guarantee that every reduction is scientifically desirable. Some deleted material may be essential for representing the entity, in which case the reduced string may cease to be a good representation altogether. What matters scientifically is not reduction per se, but reduction that decreases surplus by removing content that is genuinely extraneous.

To make this idea explicit, it is useful to measure how much surplus is reduced by deleting a given block of information.

Definition 5.5.2 Let $e \in \mathcal{E}$ be a knowable entity, let $r \in \mathcal{R}_e$ be an representation of e , and let $a \in \mathcal{B}^*$ be such that the reduced string $r \setminus a \in \mathcal{R}_e$ is still an representation of e . We define the *marginal surplus reduction* of removing a from r by

$$\Delta_R^s(e, r, a) = s_R(e, r) - s_R(e, r \setminus a).$$

The quantity $\Delta_R^s(e, r, a)$ measures how much the deleted information a reduces the amount of additional algorithmic content present in the representation relative to the entity. If $\Delta_R^s(e, r, a)$ is large, then the removed block was highly extraneous. If it is close to zero, then deleting a contributes little toward eliminating surplus.

This notion suggests a practical decision principle for scientific inquiry: among several possible reductions, a good next simplification is one that is expected to produce the largest reduction in posterior surplus. In other words, one should prefer the filtering, variable-removal, or simplification step that is expected to make the resulting representation contain as little extraneous information as possible while still representing the same entity.

■ **Example 5.4** Suppose that the entity e is a dataset intended to represent the biological characteristics of a collection of plant samples for species identification.

The current representation r contains the relevant morphological variables, but it also includes several additional columns that are not part of the entity itself.

One option, a_1 , is to remove a column containing the greenhouse bench number where each sample was stored. This information may be useful for logistics, but it is not part of the biological entity being represented. A second option, a_2 , is to remove the column giving petal length, which is one of the variables actually needed to characterize the plants.

In the language of surplus, the scientifically preferable choice is the one expected to minimize the posterior surplus, that is, the one expected to make $s_e(r \setminus a)$ as small as possible while preserving an representation of the same entity. In this scenario, removing a_1 is likely to be the better reduction step, because it discards information unrelated to the biological characteristics of the plants. By contrast, removing a_2 may simplify the dataset, but it risks deleting information that is genuinely relevant to the entity. Thus, a good reduction is not simply the one that removes the most data, but the one that is expected to reduce surplus the most. ■

The concept of marginal surplus reduction turns surplus into a prescriptive tool. Rather than merely quantifying how much extraneous information is currently present in a representation, it provides a way to evaluate candidate reductions according to how much they are expected to eliminate that excess. In this sense, surplus does not only measure what should be removed; it also helps guide the choice of what should be filtered out next.

Strict decrease of miscoding

The previous propositions provide one-sided guarantees: augmentation controls deficiency, and reduction controls surplus. A strict decrease of miscoding requires that the quantity

$$\mu(e, r) = \max\{d_R(e, r), s_R(e, r)\}$$

actually becomes smaller. In particular, decreasing only one component is not always sufficient, since the other component may remain dominant. The following results make this point explicit and record some simple sufficient conditions for strict improvement.

Before giving such conditions, it is useful to note that a strict decrease in only one of the two components does not, in general, imply a strict decrease in miscoding.

Proposition 5.5.3 A strict decrease in deficiency alone, or in surplus alone, does not in general imply a strict decrease in miscoding.

Proof. We prove the first claim by counterexample. Suppose that

$$d_R(e, r) = 10, \quad s_R(e, r) = 100,$$

and that, after some modification of the representation, we obtain r' such that

$$d_R(e, r') = 9, \quad s_R(e, r') = 100.$$

Then deficiency has decreased strictly, but

$$\mu(e, r) = \max\{10, 100\} = 100, \quad \mu(e, r') = \max\{9, 100\} = 100.$$

Hence $\mu(e, r') = \mu(e, r)$, so miscoding does not decrease strictly.

The argument for surplus is symmetric. For example, if

$$d_R(e, r) = 100, \quad s_R(e, r) = 10,$$

and

$$d_R(e, r') = 100, \quad s_R(e, r') = 9,$$

then surplus decreases strictly, but

$$\mu(e, r) = \max\{100, 10\} = 100, \quad \mu(e, r') = \max\{100, 9\} = 100.$$

Again, miscoding does not decrease strictly. ■

This proposition shows why one-sided control results are not enough on their own to guarantee improvement in miscoding. What matters is not only whether one component decreases, but whether the dominant component is reduced below the previous miscoding level.

The strongest simple sufficient condition is the following: if both deficiency and surplus decrease strictly, then miscoding must also decrease strictly.

Proposition 5.5.4 Let $r, r' \in \mathcal{R}_e$ be representations of the same knowable entity e . If

$$d_R(e, r') < d_R(e, r) \quad \text{and} \quad s_R(e, r') < s_R(e, r),$$

then

$$\mu(e, r') < \mu(e, r).$$

Proof. Since both components decrease strictly, we have

$$\max\{d_R(e, r'), s_R(e, r')\} < \max\{d_R(e, r), s_R(e, r)\}.$$

That is,

$$\mu(e, r') < \mu(e, r).$$
■

This proposition captures the ideal situation in which a representational intervention simultaneously removes missing information and extraneous information. In practice, however, augmentation and reduction often act asymmetrically, so it is useful to have more targeted sufficient conditions based on whichever component is currently dominant.

To state these conditions clearly, recall that miscoding is controlled by the larger of the two quantities $d_R(e, r)$ and $s_R(e, r)$. Thus, when deficiency is dominant, the most effective interventions are typically those that reduce deficiency without allowing surplus to remain at or above the previous miscoding level. The symmetric principle holds when surplus is dominant.

Proposition 5.5.5 Let $r \in \mathcal{R}_e$ be an representation and let $r' \in \mathcal{R}_e$ be obtained from r by either augmentation or reduction.

- (i) If $d_R(e, r) = \mu(e, r)$, $d_R(e, r') < d_R(e, r)$, and $s_R(e, r') < d_R(e, r)$, then $\mu(e, r') < \mu(e, r)$.
- (ii) If $s_R(e, r) = \mu(e, r)$, $s_R(e, r') < s_R(e, r)$, and $d_R(e, r') < s_R(e, r)$, then $\mu(e, r') < \mu(e, r)$.

Proof. (i) Since $d_R(e, r) = \mu(e, r)$, we have

$$\mu(e, r) = d_R(e, r).$$

By assumption,

$$d_R(e, r') < d_R(e, r) = \mu(e, r) \quad \text{and} \quad s_R(e, r') < d_R(e, r) = \mu(e, r).$$

Hence both arguments in the maximum defining $\mu(e, r')$ are strictly smaller than $\mu(e, r)$, so

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\} < \mu(e, r).$$

- (ii) The proof is symmetric. Since $s_R(e, r) = \mu(e, r)$, we have

$$\mu(e, r) = s_R(e, r).$$

By assumption,

$$s_R(e, r') < s_R(e, r) = \mu(e, r) \quad \text{and} \quad d_R(e, r') < s_R(e, r) = \mu(e, r).$$

Therefore,

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\} < \mu(e, r).$$



This proposition makes explicit a basic decision rule for representational refinement: one should focus on reducing the component that currently determines miscoding, while keeping the other below the previous threshold. In this sense, augmentation is most naturally useful when deficiency is dominant, whereas reduction is most naturally useful when surplus is dominant.

A particularly informative case occurs when the two components are exactly balanced. Then neither deficiency nor surplus alone determines miscoding; both jointly determine the threshold that must be crossed for strict improvement.

Corollary 5.5.6 — Balanced case. If

$$d_R(e, r) = s_R(e, r) = \mu(e, r),$$

then

$$\mu(e, r') < \mu(e, r) \iff d_R(e, r') < \mu(e, r) \text{ and } s_R(e, r') < \mu(e, r).$$

Proof. Assume first that $\mu(e, r') < \mu(e, r)$. Since

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\},$$

both $d_R(e, r')$ and $s_R(e, r')$ must be strictly smaller than $\mu(e, r)$; otherwise the maximum would be at least $\mu(e, r)$.

Conversely, if

$$d_R(e, r') < \mu(e, r) \quad \text{and} \quad s_R(e, r') < \mu(e, r),$$

then

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\} < \mu(e, r).$$

■

The balanced case shows that when deficiency and surplus are tied, improving only one of them is not enough to decrease miscoding strictly. Both must fall below the previous common level. This explains why, in some stages of scientific refinement, isolated edits may fail to improve representational quality and a combined intervention may be required.

In practical terms, these results suggest a simple strategy. If deficiency is much larger than surplus, one should prioritize augmentation steps expected to reduce missing information while keeping surplus under control. If surplus is much larger than deficiency, one should prioritize reduction steps expected to eliminate extraneous content without making the representation substantially more incomplete. When the two components are of comparable size, the best intervention may need to combine both types of refinement.

Decreasing Miscoding via Joint Representations

A third way to reduce miscoding is to combine two representations into a single joint representation. In scientific practice, this occurs whenever complementary sources of evidence are merged into a common representation: for example, when observational data are combined with metadata, when molecular measurements are combined with imaging data, or when two partial descriptions of the same system are brought together. Intuitively, if the two components are genuinely complementary and concern the same entity, their joint representation may reduce deficiency by supplying missing information. At the same time, if the added component introduces little extraneous content, the resulting miscoding may decrease.

Let $s, t \in \mathcal{B}^*$ be two binary strings. Their *joint representation* is the concatenated string st . No new definition of miscoding is required: once st is regarded as an representation of an entity e , its miscoding is simply $\mu(e, st)$.

The first basic observation is that jointing cannot worsen deficiency, up to an additive constant, relative to either of its components taken separately. This is just the monotonicity of deficiency under computable enrichment applied to the projections from st onto s and t .

Proposition 5.5.7 Let $e \in \mathcal{E}$ be a knowable entity, and let $s, t \in \mathcal{R}_e$ be representations of e . Suppose that the concatenated string $st \in \mathcal{R}_e$ is also an representation of e . Then

$$d_R(e, st) \leq d_R(e, s) + O(1) \quad \text{and} \quad d_R(e, st) \leq d_R(e, t) + O(1).$$

Proof. Define total computable projection functions $g_1, g_2 : \mathcal{B}^* \rightarrow \mathcal{B}^*$ by $g_1(st) = s$ and $g_2(st) = t$. Since both g_1 and g_2 are total and computable, the monotonicity of deficiency under computable enrichment yields

$$d_R(e, st) \leq d_R(e, g_1(st)) + O(1) = d_R(e, s) + O(1),$$

and similarly,

$$d_R(e, st) \leq d_R(e, g_2(st)) + O(1) = d_R(e, t) + O(1).$$

■

This proposition expresses the most basic epistemic advantage of joint representations: once two representations have been combined, the missing information needed to recover an admissible representation cannot be substantially larger than for either component alone. In particular, jointing is a natural way to reduce deficiency when the two components contain complementary information about the same entity.

A useful special case is when one of the two components is already admissible.

Corollary 5.5.8 Let $e \in \mathcal{E}$ be a knowable entity, and let $s \in \bar{\mathcal{R}}_e$, $t \in \mathcal{R}_e$. Suppose that $st \in \mathcal{R}_e$. Then

$$d_R(e, st) = O(1).$$

Proof. By the previous proposition,

$$d_R(e, st) \leq d_R(e, s) + O(1).$$

Since $s \in \bar{\mathcal{R}}_e$, admissible representations have deficiency $O(1)$. Hence

$$d_R(e, st) = O(1).$$

■

Thus, once an admissible representation is present as one component of the joint representation, the missing-information part of miscoding becomes negligible. Whether miscoding itself decreases then depends on how much surplus is introduced by the additional component.

For surplus, the relevant question is different. The joint representation st may contain additional content not needed to encode the entity, even if both components are individually meaningful. The next proposition gives a simple upper bound on this effect.

Proposition 5.5.9 Let $e \in \mathcal{E}$ be a knowable entity, and let $s, t \in \mathcal{R}_e$ be such that $st \in \mathcal{R}_e$. Then

$$s_R(e, st) \leq s_R(e, s) + K(t \mid s) + O(1).$$

Symmetrically,

$$s_R(e, st) \leq s_R(e, t) + K(s \mid t) + O(1).$$

Proof. Let $r' \in \bar{\mathcal{R}}_e$ be arbitrary. From r' , an optimal program of length $K(s \mid r')$ outputs s . Then, given s , an optimal program of length $K(t \mid s)$ outputs t . By concatenating these two procedures and adding constant overhead, one obtains a program that outputs st from r' . Therefore,

$$K(st \mid r') \leq K(s \mid r') + K(t \mid s) + O(1).$$

Taking the minimum over $r' \in \bar{\mathcal{R}}_e$ yields

$$\begin{aligned} s_R(e, st) &= \min_{r' \in \bar{\mathcal{R}}_e} K(st \mid r') \leq \min_{r' \in \bar{\mathcal{R}}_e} (K(s \mid r') + K(t \mid s) + O(1)) \\ &= s_R(e, s) + K(t \mid s) + O(1). \end{aligned}$$

The symmetric inequality is proved in the same way.

■

This bound shows that the surplus of a joint representation is controlled by two terms: the surplus already present in one component, and the additional information needed to specify the other component once the first is known. In practice, this means that jointing is most favorable when the second component is highly informative about the same entity but adds little extraneous or independent content beyond what is already implicit in the first.

An important special case occurs when the second component is computably recoverable from the first up to a constant.

Corollary 5.5.10 Let $e \in \mathcal{E}$ be a knowable entity, and let $s, t \in \mathcal{R}_e$ be such that $st \in \mathcal{R}_e$. If

$$K(t \mid s) = O(1),$$

then

$$s_R(e, st) \leq s_R(e, s) + O(1).$$

If, in addition, s can be computed from st by a total computable projection, then

$$s_R(e, st) = s_R(e, s) + O(1).$$

Proof. The first claim follows immediately from the previous proposition. For the second, note that s is computable from st by deleting the suffix t , so by the monotonicity of surplus under computable reduction,

$$s_R(e, s) \leq s_R(e, st) + O(1).$$

Combining this with $s_R(e, st) \leq s_R(e, s) + O(1)$ yields

$$s_R(e, st) = s_R(e, s) + O(1).$$

■

This corollary formalizes the idea of a safe jointing: if one component adds only computably recoverable content, then passing from s to st does not substantially increase surplus. In that case, jointing can improve miscoding only through the deficiency term, without paying a significant cost in extraneous information.

To make the effect of jointing explicit, it is useful to quantify how much miscoding is reduced by adding a second representation.

Definition 5.5.3 Let $e \in \mathcal{E}$ be a knowable entity, let $s \in \mathcal{R}_e$ be an representation of e , and let $t \in \mathcal{B}^*$ be such that $st \in \mathcal{R}_e$. We define the *marginal miscoding reduction* of t relative to s by

$$\Delta_\mu(e, s, t) = \mu(e, s) - \mu(e, st).$$

The quantity $\Delta_\mu(e, s, t) = \mu(e, s) - \mu(e, st)$ measures the gain obtained by joining t to the current representation s . If $\Delta_\mu(e, s, t) = \mu(e, s) - \mu(e, st)$ is positive and large, then the second component substantially improves representational quality. If it is close to zero, then the joint representation offers little benefit. If it is negative, the added component has worsened miscoding, typically by introducing surplus or by changing the effective target of the representation.

The propositions above show that joint miscoding is governed by a trade-off: jointing can reduce deficiency, but it may also increase surplus. The following proposition gives a practical sufficient condition for strict improvement.

Proposition 5.5.11 Let $e \in \mathcal{E}$ be a knowable entity, let $s, t \in \mathcal{R}_e$, and suppose that $st \in \mathcal{R}_e$. If

$$d_R(e, st) < d_R(e, s) \quad \text{and} \quad s_R(e, st) \leq s_R(e, s),$$

then

$$\mu(e, st) < \mu(e, s).$$

The symmetric statement holds with s replaced by t .

Proof. If both

$$d_R(e, st) < d_R(e, s) \quad \text{and} \quad s_R(e, st) \leq s_R(e, s)$$

hold, then by the sufficient conditions for strict decrease of miscoding established earlier,

$$\mu(e, st) < \mu(e, s).$$

The symmetric statement is identical. ■

This proposition identifies the ideal situation in which jointing improves representational quality: the second component supplies genuinely missing information without increasing the amount of extraneous content. In practice, this is what one hopes for when combining complementary datasets or different measurement modalities.

■ **Example 5.5** Suppose that the entity e is the complete interaction network of a biological signaling pathway. A first representation s is obtained from protein-binding assays and contains a number of experimentally verified interactions, but it misses some regulatory connections. A second representation t is obtained from perturbation experiments and encodes causal effects between pathway components. Each representation is partial, but both concern the same underlying entity.

If the concatenated string st is taken as a joint representation of the pathway, then the deficiency of st may be substantially smaller than that of either s or t alone, because the two sources of evidence fill different structural gaps in the network. If, moreover, the perturbation data do not introduce much irrelevant content beyond what is needed to represent the pathway, then the surplus of st need not increase substantially. In this situation, the joint representation has lower miscoding than either component alone.

By contrast, if one appends to s a second representation t' that concerns a related but distinct pathway, then the deficiency of the joint string st' with respect to e may fail to decrease, while its surplus may increase due to the introduction of irrelevant material. In that case, miscoding may stay unchanged or even worsen. Thus, jointing helps only when the added representation is both relevant and complementary. ■

The same idea extends to any finite collection of representations. Given $r_1, \dots, r_n \in \mathcal{B}^*$, one may form the joint representation $r_1 r_2 \cdots r_n$ whenever the concatenated string is treated as an representation of the same entity. Repeated application of the previous propositions shows that the deficiency of the joint representation cannot be substantially larger than that of any one component, while its surplus is controlled by the cumulative additional information introduced by the remaining components. Thus, the jointing of multiple representations provides a third general strategy for reducing miscoding: unlike augmentation, which adds new symbols to a single representation, and reduction, which removes extraneous symbols, jointing combines complementary partial representations in the hope that their informational gains outweigh any surplus they introduce.

5.6 Miscoding of Areas

The concept of miscoding can be extended to research areas to quantitatively measure the amount of effort required to correct an inaccurate representation of an area. Unfortunately, as discussed in Section ??, there is no reliable way to verify whether the strings included in an n -fold representation actually correspond to entities within that area. Moreover, it is not possible to prevent cases in which some of these strings represent the same entity.

Consider the strings $r_1, r_2, \dots, r_n$, where each $r_i \in \mathcal{B}^*$ for $i = 1, 2, \dots, n$. Recall

that the expression $r_1 r_2 \dots r_n$ refers to the concatenation of these strings into a single binary string. This operation may merge the individual components in a way that makes them inseparable. In contrast, the notation $\langle r_1, r_2, \dots, r_n \rangle$ denotes a re-encoding of the individual strings into a single string that preserves the ability to recover each original component.

Furthermore, the joint representation $r_1 r_2 \dots r_n$ is assumed to encode a single entity, while the n -fold representation $\langle r_1, r_2, \dots, r_n \rangle$ may encode up to n distinct entities.

The following definition extends the concept of miscoding to n -fold representations.

Definition 5.6.1 Let $R = (r_1, r_2, \dots, r_n) \in \mathcal{B}^* \times \mathcal{B}^* \times \dots \times \mathcal{B}^*$ be an n -fold representation. We define the *miscoding* of R , denoted by $\mu(e, r)$, as:

$$\mu(e, r) = \min_{(r_{e_1}^*, \dots, r_{e_n}^*) \in \mathcal{R}_{\mathcal{E}}^* \times \dots \times \mathcal{R}_{\mathcal{E}}^*}^o \frac{\max \{K(\langle r_1, \dots, r_n \rangle \mid \langle r_{e_1}^*, \dots, r_{e_n}^* \rangle), K(\langle r_{e_1}^*, \dots, r_{e_n}^* \rangle \mid \langle r_1, \dots, r_n \rangle)\}}{\max \{K(\langle r_1, \dots, r_n \rangle), K(\langle r_{e_1}^*, \dots, r_{e_n}^* \rangle)\}}$$

where $\min^o$ indicates that the minimum is to be computed by an oracle.

The miscoding of the representation of an area falls within the range $[0, 1]$, as demonstrated by the following proposition.

Proposition 5.6.1 For all known subsets $R = (r_1, r_2, \dots, r_n) \in \mathcal{B}^* \times \mathcal{B}^* \times \dots \times \mathcal{B}^*$, it holds that $0 \leq \mu(e, r) \leq 1$.

Proof. Since $\langle r_1, r_2, \dots, r_n \rangle$ is a string in $\mathcal{B}^*$, we can apply Proposition 20.5.5, which guarantees that the normalized compression-based distance lies between 0 and 1. ■

By extending the concept of miscoding to cover research areas, we gain a quantitative means of evaluating the quality of representations for specific subsets of entities. This mathematical framework provides a rigorous tool to assess and correct inaccuracies, both at the level of individual entities and across broader scientific domains.

References

Misrepresentation or inaccuracies in scientific representation have significant implications for scientific discovery, technological progress, policymaking, and other domains. However, no book or paper explicitly addresses the topic of "incorrect representations in science" from the perspective adopted in this

work. Here, we decompose the problem of scientific representation into two complementary subproblems: the representation of entities and the description of those representations.

[Sup02] provides a formal foundation for scientific representation and emphasizes the role of structure-preserving mappings; his analysis is crucial for understanding when and how representations fail to capture relevant features of phenomena. [Van80] introduces the concept of "constructive empiricism" and highlights the model-dependent nature of scientific representation. [LSW13] offers an ethnographic study of how scientific representations are socially constructed—and, at times, distorted—in scientific practice.

Although the specific topic of incorrect representations has not been directly examined in this way, various researchers have addressed related concerns indirectly. Their discussions often focus on issues such as scientific fraud, the replication crisis, and the use of incorrect or misleading models. For example, [Ioa05] presents a widely cited empirical and philosophical analysis of how methodological biases, poor model design, and selective reporting contribute to unreliable or misleading scientific results.



6. Mismodel

*Everything should be made as simple as possible,
but not simpler.*
Albert Einstein

Even after a suitable representation of an entity has been chosen, the problem of understanding is far from solved. A representation is only the starting point: to reason about the entity, explain it, or make predictions, we still need a description, that is, a computable object capable of generating or reconstructing the representation. As discussed in Chapter 4, descriptions are the formal devices through which we attempt to capture the regularities present in a representation and separate them from what remains specific or accidental.

However, this second step introduces its own source of error. Because our understanding of an entity is usually incomplete, the descriptions we construct are rarely optimal. Some omit information that would be needed to recover an admissible representation; others include algorithmic content that is not genuinely required for that purpose. In other words, even if the representation itself were admissible, our way of describing it may still be deficient, overly complicated, or both. Such imperfections can distort explanation, hinder prediction, and obscure the structure we are trying to uncover. It is therefore essential to identify this source of descriptive error and study it in a systematic way.

To address this issue, we introduce the concept of *mismodel*, a measure used to quantify the error that arises from describing a representation poorly. At the descriptive level, there are two fundamental modes of failure. A description may be *deficient*, in the sense that it still lacks information needed to recover a minimal description of the target representation. Or it may contain *surplus*, that is, additional algorithmic structure beyond what is needed to obtain such a minimal description. Mismodel combines these two aspects into a single quantity by taking the larger of the two. In this way, it measures the dominant defect of a description relative to a fixed target representation.

As in the case of miscoding, an important complication arises immediately: minimal descriptions are not available in practice. Because Kolmogorov complexity is not computable, we cannot in general determine the shortest description of a given representation, nor directly measure the exact gap between an available description and an ideal one. Nevertheless, the theory provides a rigorous conceptual framework for analyzing this gap in terms of algorithmic information. This allows us to define and compare descriptive errors even when the corresponding optimal descriptions remain unknown.

A deeper understanding of mismodel and of the mechanisms by which it can be reduced is essential for improving scientific modeling. Better descriptions do not merely reproduce representations more efficiently; they also make the underlying structure of those representations more accessible to reasoning, comparison, and refinement. In this chapter, we formally introduce the concepts of deficiency and surplus of a description, and from them derive the notion of *mismodel*. We study the main properties of these quantities, introduce a normalized notion of relative mismodel, and examine practical principles for decreasing mismodel through enrichment and simplification of descriptions. In this way, the chapter develops the descriptive counterpart of miscoding and clarifies how imperfect models can be systematically brought closer to descriptive optimality.

6.1 Deficiency of a Description

A description intended to support the study of a representation need not itself be optimal for that representation. Even when a valid description is available, it may fail to expose the shortest way of generating the target representation. Its program component may be unnecessarily complicated, its auxiliary input may contain avoidable detail, or it may simply be a description of some other string that is only indirectly useful for reaching the target. To quantify this phenomenon, we introduce the notion of *deficiency of a description*.

Intuitively, the deficiency of a valid description d with respect to a representation r measures the minimum amount of additional algorithmic

information that must be supplied, beyond d , in order to recover some minimal description of r . We do not require d itself to be an exact description of r ; it may be any description. Thus deficiency measures not whether d already describes r , but how useful d is as a starting point for reaching a shortest description of r .

Definition 6.1.1 Let $r \in \mathcal{R}$ be a representation, and let $d \in \mathcal{D}$ be a valid description. We define the *deficiency of d as a description for r* as

$$d_D(r, d) = \min_{d' \in \mathcal{D}_r^*} K(d' \mid d).$$

Since conditional Kolmogorov complexity takes values in $\mathbb{N}$, the minimum is well defined.

The quantity $K(d' \mid d)$ measures the length of the shortest program that, given the description d , outputs a minimal description d' of the target representation r . Minimizing over $d' \in \mathcal{D}_r^*$ selects the minimal description of r that is easiest to recover from d . Thus $d_D(r, d)$ quantifies the least amount of additional algorithmic information that must be supplied, on top of d , in order to reach some shortest valid description of r .

This interpretation covers two important cases. If $d \in \mathcal{D}_r$, then $d_D(r, d)$ measures how far the current description of r is from descriptive optimality. If $d \notin \mathcal{D}_r$, then $d_D(r, d)$ measures how useful d is as auxiliary information for obtaining a minimal description of r .

■ **Example 6.1** Let r be a binary representation encoding the complete adjacency list of a finite graph. A valid description $d \in \mathcal{D}$ might generate only a coarse summary of that graph, for example a string encoding its number of vertices, number of edges, and degree sequence. Such a description is valid, but it need not belong to $\mathcal{D}_r^*$, since it may generate a summary string rather than the full representation r .

The deficiency $d_D(r, d)$ measures, in algorithmic terms, how much additional information must be supplied, beyond this summary description, in order to recover some minimal description of the exact representation r . If the summary already determines most of the structure of the graph, then the deficiency is small; if many nontrivial details are still missing, the deficiency is large. ■

As with Kolmogorov complexity itself, deficiency depends on the choice of reference universal machine only up to an additive constant. Thus deficiency is stable under the usual machine-independence convention of algorithmic information theory.

Proposition 6.1.1 Let U and V be two reference universal prefix-free machines, and let $d_D^U(r, d)$ and $d_D^V(r, d)$ denote the deficiency of d computed using the corresponding conditional Kolmogorov complexities $K_U(\cdot \mid \cdot)$ and $K_V(\cdot \mid \cdot)$. Then,

for every representation $r \in \mathcal{R}$ and every valid description $d \in \mathcal{D}$,

$$d_D^U(r, d) = d_D^V(r, d) + O(1).$$

Proof. By the invariance theorem for conditional Kolmogorov complexity, for every pair of strings $x, y \in \mathcal{B}^*$,

$$K_U(x | y) \leq K_V(x | y) + c_{UV}, \quad K_V(x | y) \leq K_U(x | y) + c_{VU},$$

where c_{UV} and c_{VU} are constants depending only on U and V . Applying the first inequality to each $d' \in \mathcal{D}_r^*$, we obtain

$$K_U(d' | d) \leq K_V(d' | d) + c_{UV}.$$

Taking minima over $d' \in \mathcal{D}_r^*$, we get

$$d_D^U(r, d) = \min_{d' \in \mathcal{D}_r^*} K_U(d' | d) \leq \min_{d' \in \mathcal{D}_r^*} (K_V(d' | d) + c_{UV}) = d_D^V(r, d) + c_{UV}.$$

By symmetry,

$$d_D^V(r, d) \leq d_D^U(r, d) + c_{VU}.$$

Combining both inequalities yields

$$d_D^U(r, d) = d_D^V(r, d) + O(1).$$

■

This proposition shows that deficiency is robust under the choice of universal machine in exactly the same sense as Kolmogorov complexity. Consequently, asymptotic statements such as $d_D(r, d) = O(1)$ remain meaningful independently of the particular coding convention used to define complexity.

A first basic property of deficiency is that it cannot be negative.

Proposition 6.1.2 For every representation $r \in \mathcal{R}$ and every description $d \in \mathcal{D}$, we have

$$d_D(r, d) \geq 0.$$

Proof. For every $d' \in \mathcal{D}_r^*$, the conditional complexity $K(d' | d)$ is the length of a program and is therefore non-negative. The minimum of non-negative quantities is non-negative. ■

Thus deficiency behaves as expected from a measure of missing algorithmic information. In particular, $d_D(r, d) = O(1)$ corresponds to the situation in which some minimal description of r can be recovered from d with only constant overhead.

Because the starting description d is allowed to be any valid description, it is important to note explicitly that deficiency is always finite.

Proposition 6.1.3 For every representation $r \in \mathcal{R}$ and every description $d \in \mathcal{D}$, we have

$$d_D(r, d) < \infty.$$

More precisely, if

$$\ell_r^* = \min_{d' \in \mathcal{D}_r} l(d')$$

denotes the length of a shortest description of r , then

$$d_D(r, d) \leq \ell_r^* + O(1).$$

Proof. Let $d^* \in \mathcal{D}_r^*$. Then $l(d^*) = \ell_r^*$. By Definition 6.1.1,

$$d_D(r, d) \leq K(d^* \mid d).$$

A program may ignore its conditional input d and output d^* from scratch. Therefore

$$K(d^* \mid d) \leq K(d^*) + O(1) \leq l(d^*) + O(1) = \ell_r^* + O(1).$$

Hence

$$d_D(r, d) \leq \ell_r^* + O(1),$$

and in particular $der(r, d)$ is finite. ■

This upper bound shows that deficiency never diverges. In the worst case, one can always disregard the given description altogether and output a minimal description of r from scratch.

Minimal descriptions themselves have negligible deficiency.

Proposition 6.1.4 Let $r \in \mathcal{R}$ and let $d \in \mathcal{D}_r^*$ be a minimal description of r . Then

$$d_D(r, d) = O(1).$$

Proof. Choose $d' = d$ in Definition 6.1.1. Then $d_D(r, d) \leq K(d \mid d)$. Since a constant-length program can output its conditional input unchanged, we have $K(d \mid d) = O(1)$. Therefore $d_D(r, d) = O(1)$. ■

This formalizes the intuition that a minimal description is already algorithmically complete with respect to descriptive optimality: recovering a shortest description of r from it requires no nontrivial extra information.

Deficiency also satisfies a natural monotonicity property under computable enrichment. Informally, if one valid description can be effectively recovered from another, then the latter cannot have substantially larger deficiency with respect to the same target representation.

Proposition 6.1.5 Let $r \in \mathcal{R}$, and let $d, s \in \mathcal{D}$. Suppose that there exists a total computable function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that $g(s) = d$. Then

$$d_D(r, s) \leq d_D(r, d) + O(1).$$

Proof. Let $d^* \in \mathcal{D}_r^*$ be arbitrary. Since g is total and computable, from s one can compute $g(s) = d$ with constant overhead and then simulate an optimal program witnessing $K(d^* \mid d)$. Hence

$$K(d^* \mid s) \leq K(d^* \mid g(s)) + O(1) = K(d^* \mid d) + O(1).$$

Taking minima over $d^* \in \mathcal{D}_r^*$, we obtain

$$d_D(r, s) = \min_{d^* \in \mathcal{D}_r^*} K(d^* \mid s) \leq \min_{d^* \in \mathcal{D}_r^*} (K(d^* \mid d) + O(1)) = d_D(r, d) + O(1).$$

■

This proposition expresses a basic monotonicity principle: if a valid description s effectively contains enough information to recover another description d , then s cannot be substantially farther from a minimal description of r than d is.

As an immediate consequence, descriptions that are mutually recoverable at constant cost have the same deficiency up to an additive constant.

Corollary 6.1.6 Let $r \in \mathcal{R}$, and let $d, s \in \mathcal{D}$. If $K(d \mid s) = O(1)$ and $K(s \mid d) = O(1)$, then

$$d_D(r, d) = d_D(r, s) + O(1).$$

Proof. From $K(d \mid s) = O(1)$, there exists a constant-length program that computes d from s . Hence, by Proposition 6.1.5,

$$d_D(r, s) \leq d_D(r, d) + O(1).$$

Similarly, from $K(s \mid d) = O(1)$ we obtain

$$d_D(r, d) \leq d_D(r, s) + O(1).$$

Combining both inequalities yields

$$d_D(r, d) = d_D(r, s) + O(1).$$

■

This shows that deficiency is determined by the algorithmic content of a description rather than by its superficial syntax. Valid descriptions that differ only by a reversible computable recoding are equivalent from the standpoint of deficiency.

Finally, it is worth stressing that exact correctness and low deficiency are different notions. A valid description d may belong to $\mathcal{D}_r$ and thus generate r exactly, yet still have positive deficiency if substantial additional algorithmic information is needed to recover a minimal description of r from d . Conversely, a description $d \notin \mathcal{D}_r$ may still have small deficiency if it already contains, in an implicit or indirect form, most of the information needed to reach a shortest description of r .

The deficiency of a description therefore measures something more subtle than mere validity or exactness. It quantifies the algorithmic distance from a given valid description to descriptive optimality with respect to a fixed target representation.

6.2 Surplus of a Description

A valid description used in the study of a representation need not be descriptively economical. Even when a description is useful for reaching or understanding a target representation r , it may contain additional algorithmic content that is not required by any shortest description of r . This excess may arise from an unnecessarily complicated program, an over-specific auxiliary input, or from the fact that the description concerns a related but different string whose connection to r still has to be made explicit. To quantify this phenomenon, we introduce the notion of *surplus of a description*.

Intuitively, the surplus of a valid description d with respect to a representation r measures how much additional algorithmic information is present in d beyond what is needed to obtain a minimal description of r . Formally, we measure this excess by asking how much information must be added to a minimal description of r in order to recover d .

Definition 6.2.1 Let $r \in \mathcal{R}$ be a representation, and let $d \in \mathcal{D}$ be a description. We define the *surplus of d as a description for r* as

$$s_D(r, d) = \min_{d' \in \mathcal{D}_r^*} K(d \mid d').$$

Since conditional Kolmogorov complexity takes values in $\mathbb{N}$, the minimum is well defined.

The quantity $K(d \mid d')$ measures the length of the shortest program that, given a minimal description d' of r , outputs the description d . Minimizing over $d' \in \mathcal{D}_r^*$ selects the minimal description of r from which d is easiest to generate. Hence

$s_D(r, d)$ quantifies the minimum amount of additional algorithmic content present in d beyond what is needed to achieve a shortest description of the representation r .

If $d \in \mathcal{D}_r$, then $s_D(r, d)$ measures the excess algorithmic content of the given description of r relative to a minimal one. If $d \notin \mathcal{D}_r$, then $s_D(r, d)$ measures how much additional information is needed to reconstruct that auxiliary valid description from a minimal description of r .

■ **Example 6.2** Let r be a binary string encoding a finite time series. A minimal description $d' \in \mathcal{D}_r^*$ may exploit a compact regularity of the series together with a short residual input. By contrast, a valid description $d \in \mathcal{D}_r$ may simply contain a longer program that prints the symbols of r one by one, without making the underlying regularity explicit. Both descriptions generate exactly the same representation r , but the latter contains avoidable descriptive structure. The surplus $s_D(r, d)$ measures, in algorithmic terms, how much additional information is needed to recover this more cumbersome description from some minimal description of r . ■

As in the case of deficiency, surplus depends on the choice of reference universal machine only up to an additive constant. The argument is entirely analogous.

Proposition 6.2.1 Let U and V be two reference universal prefix-free machines, and let $s_D^U(r, d)$ and $s_D^V(r, d)$ denote the surplus of d computed using the corresponding conditional Kolmogorov complexities $K_U(\cdot | \cdot)$ and $K_V(\cdot | \cdot)$. Then, for every representation $r \in \mathcal{R}$ and every valid description $d \in \mathcal{D}$,

$$s_D^U(r, d) = s_D^V(r, d) + O(1).$$

Proof. By the invariance theorem for conditional Kolmogorov complexity, for every pair of strings $x, y \in \mathcal{B}^*$,

$$K_U(x | y) \leq K_V(x | y) + c_{UV}, \quad K_V(x | y) \leq K_U(x | y) + c_{VU},$$

where c_{UV} and c_{VU} are constants depending only on U and V . Applying the first inequality to each $d' \in \mathcal{D}_r^*$, we obtain

$$K_U(d | d') \leq K_V(d | d') + c_{UV}.$$

Taking minima over $d' \in \mathcal{D}_r^*$, we get

$$s_r^U(d) = \min_{d' \in \mathcal{D}_r^*} K_U(d | d') \leq \min_{d' \in \mathcal{D}_r^*} (K_V(d | d') + c_{UV}) = s^V(r, d) + c_{UV}.$$

By symmetry,

$$s^V(r, d) \leq s^U(r, d) + c_{VU}.$$

Combining both inequalities yields

$$s^U(r, d) = s^V(r, d) + O(1).$$

■

Thus surplus is machine-invariant in exactly the same sense as deficiency and Kolmogorov complexity itself. In particular, qualitative statements such as $s_D(r, d) = O(1)$ or comparisons between surpluses remain meaningful independently of the specific universal machine chosen.

The following proposition records the most basic structural property of surplus.

Proposition 6.2.2 For every representation $r \in \mathcal{R}$ and every description $d \in \mathcal{D}$, we have $s_D(r, d) \geq 0$.

Proof. For any $d' \in \mathcal{D}_r^*$, the conditional complexity $K(d \mid d')$ is the length of a computer program and is therefore non-negative. The minimum of non-negative quantities is non-negative. ■

This confirms that surplus is a non-negative quantity, as expected for a measure of excess algorithmic content. In particular, $s_D(r, d) = O(1)$ corresponds to the situation in which the valid description d differs from some minimal description of r by at most a constant amount of additional information.

The next proposition shows that minimal descriptions have negligible surplus.

Proposition 6.2.3 For every minimal description $d \in \mathcal{D}_r^*$, we have $s_D(r, d) = O(1)$.

Proof. Choose $d' = d$ in Definition 6.2.1. Then $s_D(r, d) \leq K(d \mid d)$. Since a constant-length program can output its conditional input, $K(d \mid d) = O(1)$, and hence $s_D(r, d) = O(1)$. ■

This proposition formalizes the intuition that a minimal description contains no substantial algorithmic excess relative to the target representation. The residual $O(1)$ term reflects only the fixed overhead associated with the choice of universal machine.

For surplus, the natural monotonicity principle concerns computable reduction. Intuitively, if one valid description can be obtained from another by an effective simplification, then its surplus cannot be greater by more than a constant, since simplification may remove excess information but cannot create it.

Proposition 6.2.4 Let $r \in \mathcal{R}$, and let $d, s \in \mathcal{D}$ be descriptions. Suppose that there exists a total computable function $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that $g(s) = d$. Then

$$s_r(d) \leq s_r(s) + O(1).$$

Proof. Let $d' \in \mathcal{D}_r^*$ be arbitrary. Since g is total and computable, from any program witnessing $K(s \mid d')$ one can compute s from d' and then apply g with constant overhead to obtain $d = g(s)$. Hence

$$K(d \mid d') \leq K(s \mid d') + O(1).$$

Taking minima over $d' \in \mathcal{D}_r^*$, we obtain

$$s_D(r, d) = \min_{d' \in \mathcal{D}_r^*} K(d \mid d') \leq \min_{d' \in \mathcal{D}_r^*} (K(s \mid d') + O(1)) = s_D(r, s) + O(1).$$

■

This proposition is the direct dual, for descriptions, of the corresponding monotonicity principle for surplus of representations. Scientifically, it captures the idea that computably simplifying a description cannot substantially increase its surplus relative to a fixed target representation.

As an immediate consequence, descriptions that are mutually recoverable at constant cost have the same surplus up to an additive constant.

Corollary 6.2.5 Let $r \in \mathcal{R}$, and let $d, s \in \mathcal{D}$ be descriptions. If $K(d \mid s) = O(1)$ and $K(s \mid d) = O(1)$, then

$$s_D(r, d) = s_r(s) + O(1).$$

Proof. From $K(d \mid s) = O(1)$, there exists a constant-length program that computes d from s . Hence, by Proposition 6.2.4,

$$s_D(r, d) \leq s_D(r, s) + O(1).$$

Similarly, from $K(s \mid d) = O(1)$, we obtain

$$s_D(r, s) \leq s_D(r, d) + O(1).$$

Combining both inequalities yields

$$s_D(r, d) = s_D(r, s) + O(1).$$

■

This shows that surplus depends on the algorithmic content of a description rather than on its superficial syntax. Two valid descriptions that differ only by a reversible computable recoding have essentially the same surplus relative to r .

The next proposition identifies a simple condition under which extending a description does not change its surplus beyond the usual additive constant. Intuitively, if the appended part is already recoverable from the original description at constant cost, then it does not contribute substantial new surplus.

Proposition 6.2.6 Let $r \in \mathcal{R}$, and let $x, y \in \mathcal{B}^*$ be such that both $x \in \mathcal{D}$ and $xy \in \mathcal{D}$. If $K(y \mid x) = O(1)$, then $s_D(r, xy) = s_D(r, x) + O(1)$.

Proof. Since x is computable from xy by deleting the suffix y , Proposition 6.2.4 gives

$$s_D(r, x) \leq s_D(r, xy) + O(1).$$

On the other hand, the condition $K(y \mid x) = O(1)$ implies that there is a constant-length program that reconstructs y from x . Hence from x one can compute xy with constant overhead. Applying Proposition 6.2.4 again, now with $s := x$ and $d := xy$, we get

$$s_D(r, xy) \leq s_D(r, x) + O(1).$$

Combining both inequalities yields

$$s_D(xy, x) = s_D(x) + O(1).$$

■

This proposition formalizes the idea of a safe extension: appending information that is already computably recoverable from the original description does not substantially increase surplus.

Finally, we derive a simple upper bound on surplus. In contrast with the corresponding deficiency bound, the most immediate general bound here is controlled by the complexity of the description itself.

Proposition 6.2.7 For every valid description $d \in \mathcal{D}$ and every representation $r \in \mathcal{R}$,

$$s_D(r, d) \leq K(d) + O(1) \leq l(d) + O(1).$$

Proof. Fix any $d' \in \mathcal{D}_r^*$. Then

$$s_D(r, d) = \min_{u \in \mathcal{D}_r^*} K(d \mid u) \leq K(d \mid d').$$

A program can ignore the conditional input d' and simply output d , incurring only constant overhead. Therefore

$$K(d \mid d') \leq K(d) + O(1).$$

Hence

$$s_D(r, d) \leq K(d) + O(1).$$

Finally, since Kolmogorov complexity is bounded by length up to an additive constant, we also have

$$K(d) \leq l(d) + O(1),$$

and therefore

$$s_D(r, d) \leq l(d) + O(1).$$

■

This bound shows that surplus is always finite and, in the worst case, is controlled by the descriptive complexity of the description itself. Even if a valid description contains a large amount of algorithmically extraneous material relative to the target representation r , that excess cannot exceed, up to a constant, the total amount of information present in the description.

The surplus of a description therefore measures something more subtle than mere length. A long description may still have small surplus if it is computably close to a minimal description of r , whereas a shorter description may have larger surplus if it contains algorithmically irrelevant structure that is not needed to reach a shortest description of the target representation.

6.3 Mismodel

The concepts of deficiency and surplus capture two fundamentally different ways in which a description may fail with respect to a target representation r . Deficiency measures how much algorithmic information is still missing in order to recover a minimal description of r , whereas surplus measures how much algorithmic content in the given description goes beyond what is needed to obtain such a minimal description. Since these two quantities need not decrease simultaneously, the analysis of mismodel is most naturally formulated as a multiobjective optimization problem rather than as a single-objective one.

Pareto optimality

Before introducing a scalar measure of mismodel, it is useful to make explicit that description refinement involves two competing objectives.

Definition 6.3.1 Let $r \in \mathcal{R}$ be a representation. We define the *mismodel problem* for r as the following multiobjective optimization problem:

$$\min_{d \in \mathcal{D}} (d_D(r, d), s_D(r, d)).$$

This formulation emphasizes that mismodel is not, at its core, a single number but a trade-off between two distinct desiderata: informational adequacy relative to r , and absence of excess descriptive content. In practice, some refinements reduce deficiency at the cost of increasing surplus, while others simplify a description but also discard information that was useful for reaching a minimal description of the target representation.

The natural comparison relation for this problem is dominance.

Definition 6.3.2 Let $r \in \mathcal{R}$, and let $d, d' \in \mathcal{D}$ be two descriptions. We say that d' *dominates* d , denoted by $d' \prec_r d$, if

$$d_D(r, d') \leq d_D(r, d), \quad s_D(r, d') \leq s_D(r, d),$$

and at least one of these inequalities is strict.

A dominating description is at least as good in both components and strictly better in one. Thus, a dominated description is suboptimal in a strong sense: there is another valid description that improves one source of mismodel without worsening the other.

The appropriate notion of optimality for a multiobjective problem is Pareto optimality.

Definition 6.3.3 Let $r \in \mathcal{R}$. A description $d \in \mathcal{D}$ is said to be *Pareto optimal* (with respect to deficiency and surplus) if there does not exist any $d' \in \mathcal{D}$ such that $d' \prec_r d$.

The set of all Pareto-optimal descriptions for r is denoted by

$$\mathcal{P}_r = \{d \in \mathcal{D} : \nexists d' \in \mathcal{D} \text{ such that } d' \prec_r d\},$$

and is called the *Pareto frontier* of the mismodel problem.

The Pareto frontier collects precisely those valid descriptions for which any further improvement in one component requires a deterioration in the other. In this sense, it plays the role of a description-level frontier: beyond it, refinement becomes a matter of trade-offs rather than unambiguous progress.

The next proposition clarifies why dominated descriptions can be safely discarded.

Proposition 6.3.1 Let $\phi : \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}_{\geq 0}$ be strictly increasing in each argument. If $d \in \mathcal{D}$ is dominated, then d cannot minimize $\phi(d_D(r, d), s_D(r, d))$ over $\mathcal{D}$.

Proof. Assume that d is dominated. Then there exists $d' \in \mathcal{D}$ such that

$$d_D(r, d') \leq d_D(r, d), \quad s_D(r, d') \leq s_D(r, d),$$

and at least one inequality is strict. Since ϕ is strictly increasing in each argument, it follows that

$$\phi(d_D(r, d'), s_D(r, d')) < \phi(d_D(r, d), s_D(r, d)).$$

Therefore d cannot be a minimizer of ϕ . ■

This result formalizes an important elimination principle: any description that is dominated is suboptimal for every sensible scalar criterion that penalizes both deficiency and surplus. Consequently, once the Pareto frontier has been identified, all remaining scalarizations are selection rules on that frontier.

Unlike the corresponding representation-level statement in finite settings, the Pareto frontier for descriptions is non-empty without any finiteness assumption, because both deficiency and surplus take values in $\mathbb{N}$.

Proposition 6.3.2 For every representation $r \in \mathcal{R}$, the Pareto frontier $\mathcal{P}_r$ is non-empty.

Proof. Since $\mathcal{D} \neq \emptyset$, the set

$$\{d_D(r, d) : d \in \mathcal{D}\} \subseteq \mathbb{N}$$

is non-empty and therefore has a minimum. Let

$$m = \min_{d \in \mathcal{D}} d_D(r, d),$$

and consider the non-empty set

$$\mathcal{D}_m = \{d \in \mathcal{D} : d_D(r, d) = m\}.$$

Since

$$\{s_D(r, d) : d \in \mathcal{D}_m\} \subseteq \mathbb{N}$$

is again non-empty, it has a minimum. Choose $d^* \in \mathcal{D}_m$ such that

$$s_D(r, d^*) = \min_{d \in \mathcal{D}_m} s_D(r, d).$$

We claim that $d^* \in \mathcal{P}_r$. Suppose, for contradiction, that there exists $u \in \mathcal{D}$ such that $u \prec_r d^*$. Then

$$d_D(r, u) \leq d_D(r, d^*) = m, \quad s_D(r, u) \leq s_D(r, d^*),$$

with at least one strict inequality. Since m is the minimum possible deficiency, we must have $d_D(r, u) = m$, so $u \in \mathcal{D}_m$. Hence

$$s_D(r, u) < s_D(r, d^*),$$

contradicting the choice of d^* . Therefore $d^* \in \mathcal{P}_r$. ■

This proposition guarantees that there is always at least one non-dominated description. The real difficulty is therefore not existence but selection among the competing Pareto-optimal alternatives.

Minimal descriptions provide natural ideal benchmarks for the biobjective problem.

Proposition 6.3.3 For every minimal description $d \in \mathcal{D}_r^*$, we have

$$d_D(r, d) = O(1), \quad s_D(r, d) = O(1).$$

Proof. The first statement follows from the basic property of deficiency of a description: if $d \in \mathcal{D}_r^*$, then

$$d_D(r, d) \leq K(d \mid d) = O(1).$$

Similarly, by the basic property of surplus of a description,

$$s_D(r, d) \leq K(d \mid d) = O(1).$$

Hence both quantities are $O(1)$. ■

Thus minimal descriptions realize, up to the usual additive constant, the ideal situation in which nothing essential is missing and nothing substantial is superfluous. For this reason, the Pareto frontier should be understood as the set of the best available approximations to such ideal descriptions.

Scalar mismodel

Although the Pareto frontier captures the full trade-off structure, in many applications one still wishes to assign a single mismodel value to each description. A natural scalarization is to penalize the worst of the two components.

Definition 6.3.4 Let $r \in \mathcal{R}$ and $d \in \mathcal{D}$. The *mismodel* of d with respect to r is defined by

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\}.$$

This definition corresponds to the Chebyshev scalarization of a bi-objective problem. It assigns to each description the magnitude of its dominant defect, and therefore penalizes imbalance between the two components.

A first useful property is that Chebyshev mismodel provides a uniform upper bound on both deficiency and surplus.

Proposition 6.3.4 For every $r \in \mathcal{R}$, every $d \in \mathcal{D}$, and every $t \geq 0$,

$$v(r, d) \leq t \iff d_D(r, d) \leq t \text{ and } s_D(r, d) \leq t.$$

Proof. By definition,

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\}.$$

Hence $v(r, d) \leq t$ if and only if the maximum of the two quantities is at most t , which happens if and only if each quantity is at most t . ■

This equivalence is one of the main reasons to prefer the maximum. It says that minimizing mismodel is exactly the same as pushing both components below the smallest possible common threshold.

Unlike strictly increasing scalarizations, the maximum does not force every one of its minimizers to lie on the Pareto frontier. A dominated description may still tie for the same maximal value. However, the minimum of $v(r, d)$ is always attained at some Pareto-optimal description.

Proposition 6.3.5 For every representation $r \in \mathcal{R}$, there exists $d^* \in \mathcal{P}_r$ such that

$$v(r, d^*) = \min_{d \in \mathcal{D}} v(r, d).$$

Proof. Let

$$M = \min_{d \in \mathcal{D}} v(r, d),$$

which exists because $\{v_r(d) : d \in \mathcal{D}\} \subseteq \mathbb{N}$ is non-empty. Consider the non-empty set of minimizers

$$\mathcal{M} = \{d \in \mathcal{D} : v(r, d) = M\}.$$

Among these, choose $d^* \in \mathcal{M}$ with minimal deficiency, and among those with the same deficiency, choose one with minimal surplus.

We claim that $d^* \in \mathcal{P}_r$. Suppose, for contradiction, that there exists $u \in \mathcal{D}$ such that $u \prec_r d^*$. Then

$$d_D(r, u) \leq d_D(r, d^*), \quad s_D(r, u) \leq s_D(r, d^*),$$

with at least one strict inequality. Hence

$$v(u, r) = \max\{d_D(r, u), s_D(r, u)\} \leq \max\{d_D(r, d^*), s_D(r, d^*)\} = v(r, d^*) = M.$$

Since M is the minimum possible value, we must have $v(u, r) = M$, so $u \in \mathcal{M}$. But then either $d_D(r, u) < d_D(r, d^*)$, contradicting the choice of d^* , or $d_D(r, u) = d_D(r, d^*)$ and $s_D(r, u) < s_D(r, d^*)$, again contradicting the choice of d^* . Therefore no such u exists, and $d^* \in \mathcal{P}_r$. ■

Thus, Chebyshev mismodel does not ignore the Pareto structure: even though not every minimizer need be Pareto-optimal, the scalar criterion always selects at least one best-balanced point on the frontier.

The maximum also identifies which of the two components is currently the main obstacle to improvement.

Proposition 6.3.6 Let $r \in \mathcal{R}$ and $d \in \mathcal{D}$.

- 1 If $d_D(r, d) > s_D(r, d)$, then any valid description $u \in \mathcal{D}$ such that $d_D(r, u) < d_D(r, d)$ and $s_D(r, u) \leq d_D(r, u)$ necessarily satisfies $v(u, r) < v(r, d)$.
- 2 If $s_D(r, d) > d_D(r, d)$, then any valid description $u \in \mathcal{D}$ such that $s_D(r, u) < s_D(r, d)$ and $d_D(r, u) \leq s_D(r, u)$ necessarily satisfies $v(u, r) < v(r, d)$.

Proof. Assume first that $d_D(r, d) > s_D(r, d)$. Then $v(r, d) = d_D(r, d)$. If u satisfies $d_D(r, u) < d_D(r, d)$ and $s_D(r, u) \leq d_D(r, u)$, then

$$v(u, r) = \max\{d_D(r, u), s_D(r, u)\} = d_D(r, u) < d_D(r, d) = v(r, d).$$

The second part is symmetric. If $s_D(r, d) > d_D(r, d)$, then $v(r, d) = s_D(r, d)$, and if $s_D(r, u) < s_D(r, d)$ and $d_D(r, u) \leq s_D(r, u)$, then

$$v(u, r) = \max\{d_D(r, u), s_D(r, u)\} = s_D(r, u) < s_D(r, d) = v(r, d).$$

■

This proposition has a clear methodological interpretation. When deficiency dominates, the natural priority is to enrich or redirect the description so that it better supports the recovery of a minimal description of r . When surplus dominates, the natural priority is simplification, pruning, or re-expression.

The main advantage of the Chebyshev scalarization is that it enforces balance between the two types of descriptive error.

Proposition 6.3.7 Let $d_1, d_2 \in \mathcal{D}$. Suppose that

$$d_D(r, d_1) + s_D(r, d_1) = d_D(r, d_2) + s_D(r, d_2) = C$$

for some constant C , and assume moreover that

$$|d_D(r, d_1) - s_D(r, d_1)| \leq |d_D(r, d_2) - s_D(r, d_2)|.$$

Then

$$v(d_1, r) \leq v(d_2, r).$$

Proof. For any nonnegative numbers a, b with $a + b = C$, one has

$$\max\{a, b\} = \frac{C + |a - b|}{2}.$$

Applying this identity to the pairs

$$(a, b) = (d_D(r, d_1), s_D(r, d_1)) \quad \text{and} \quad (a, b) = (d_D(r, d_2), s_D(r, d_2)),$$

we obtain

$$v(d_1, r) = \frac{C + |d_D(r, d_1) - s_D(r, d_1)|}{2},$$

and

$$v(d_2, r) = \frac{C + |d_D(r, d_2) - s_D(r, d_2)|}{2}.$$

Since

$$|d_D(r, d_1) - s_D(r, d_1)| \leq |d_D(r, d_2) - s_D(r, d_2)|,$$

it follows that

$$v(d_1, r) \leq v(d_2, r).$$

■

Among descriptions with the same total burden, the maximum prefers the more balanced one. This is precisely the desired behavior when mismodel is meant to identify descriptions that are not only good on average, but also free from a dominant unresolved defect.

A final basic property is that the Chebyshev scalarization is the smallest uniform upper bound on the two error components.

Proposition 6.3.8 For every $r \in \mathcal{R}$ and every $d \in \mathcal{D}$, $v(r, d)$ is the smallest number $t \geq 0$ such that $d_D(r, d) \leq t$ and $s_D(r, d) \leq t$.

Proof. By Proposition 6.3.4,

$$v(r, d) \leq t \iff d_D(r, d) \leq t \text{ and } s_D(r, d) \leq t.$$

Taking $t = v(r, d)$ shows that $v(r, d)$ is a common upper bound. If $t < v(r, d)$, then the same equivalence implies that at least one of $d_D(r, d)$ or $s_D(r, d)$ is greater than t . Hence no smaller common upper bound exists. ■

This characterization gives the maximum a clear operational meaning. It measures the least tolerance level under which both kinds of descriptive error can simultaneously be regarded as acceptable.

Comparison with alternative scalarizations

The maximum is not the only possible way to extract a scalar score from the pair $(d_D(r, d), s_D(r, d))$. Nevertheless, alternative scalarizations embody different epistemic priorities and have different advantages and drawbacks.

A first possibility is the sum scalarization

$$v^\Sigma(r, d) = d_D(r, d) + s_D(r, d).$$

This criterion measures total descriptive burden. It is simple, symmetric, and often convenient analytically, but it allows compensation: a large value of one component may be offset by a small value of the other.

Proposition 6.3.9 There exist valid descriptions $d_1, d_2 \in \mathcal{D}$ such that

$$v^\Sigma(d_1, r) = v^\Sigma(d_2, r)$$

but

$$v(d_1, r) > v(d_2, r).$$

Proof. It suffices to consider two hypothetical pairs of values:

$$(d_D(r, d_1), s_D(r, d_1)) = (C, 0), \quad (d_D(r, d_2), s_D(r, d_2)) = \left(\frac{C}{2}, \frac{C}{2}\right),$$

with $C > 0$. Then

$$v^\Sigma(d_1, r) = C = v^\Sigma(d_2, r),$$

but

$$v(d_1, r) = \max\{C, 0\} = C,$$

whereas

$$v(d_2, r) = \max \left\{ \frac{C}{2}, \frac{C}{2} \right\} = \frac{C}{2}.$$

Hence the sum treats the two descriptions as equivalent, while the Chebyshev scalarization strictly prefers the more balanced one. ■

The sum is therefore useful when one cares about aggregate burden, but less appropriate when one wants to prevent a single severe defect from being masked by good performance elsewhere.

A second possibility is a weighted Chebyshev scalarization:

$$v^{(\lambda)}(r, d) = \max \{ \lambda d_D(r, d), (1 - \lambda) s_D(r, d) \}, \quad 0 < \lambda < 1.$$

This criterion is useful when one source of descriptive error is judged more serious than the other in a given application. Its advantage is flexibility, but that flexibility is also its weakness: the choice of λ is external to the formal theory and must be justified by domain-specific considerations. Accordingly, the weighted maximum is best understood as an application-dependent refinement of the unweighted definition rather than as the canonical notion of mismodel itself.

A third possibility is lexicographic ordering. For example, one may declare that

$$d_1 \prec_{\text{lex}} d_2$$

whenever

$$d_D(r, d_1) < d_D(r, d_2),$$

or

$$d_D(r, d_1) = d_D(r, d_2) \text{ and } s_D(r, d_1) < s_D(r, d_2).$$

This corresponds to an absolute priority on reducing deficiency, with surplus considered only as a secondary criterion.

Proposition 6.3.10 For every $\varepsilon > 0$ and every $M > 0$, there exist two pairs (a_1, b_1) and (a_2, b_2) in $\mathbb{R}_{\geq 0}^2$ such that

$$a_1 < a_2 < a_1 + \varepsilon,$$

$$b_2 + M < b_1,$$

and yet the lexicographic order prioritizing the first coordinate strictly prefers (a_1, b_1) to (a_2, b_2) .

Proof. Take

$$(a_1, b_1) = (0, M+1), \quad (a_2, b_2) = \left(\frac{\varepsilon}{2}, 0\right).$$

Then

$$a_1 = 0 < \frac{\varepsilon}{2} = a_2 < \varepsilon = a_1 + \varepsilon,$$

and

$$0 + M < M + 1.$$

Thus the second pair is better by more than M in the second coordinate, but lexicographic order prioritizing the first coordinate still prefers the first pair, since $a_1 < a_2$. ■

This illustrates the main drawback of lexicographic scalarization: it can force highly unbalanced choices. Such a policy may be appropriate in narrow applications, but it is not a good default for a theory intended to capture the overall quality of a description.

Negligible mismatch characterizes algorithmic equivalence to minimality in both directions.

Proposition 6.3.11 For $r \in \mathcal{R}$ and $d \in \mathcal{D}$, the following are equivalent:

- 1 $v(r, d) = O(1)$.
- 2 There exist $d_1^*, d_2^* \in \mathcal{D}_r^*$ such that

$$K(d_1^* \mid d) = O(1) \quad \text{and} \quad K(d \mid d_2^*) = O(1).$$

Proof. ($1 \Rightarrow 2$) If $v(r, d) = O(1)$, then

$$d_D(r, d) \leq v(r, d) = O(1), \quad s_D(r, d) \leq v(r, d) = O(1).$$

By definition of the minima in $d_D(r, d)$ and $s_D(r, d)$, there exist $d_1^*, d_2^* \in \mathcal{D}_r^*$ such that

$$K(d_1^* \mid d) = d_D(r, d) = O(1), \quad K(d \mid d_2^*) = s_D(r, d) = O(1).$$

($2 \Rightarrow 1$) If such d_1^*, d_2^* exist, then

$$d_D(r, d) \leq K(d_1^* \mid d) = O(1), \quad s_D(r, d) \leq K(d \mid d_2^*) = O(1).$$

Therefore

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\} = O(1).$$
■

This equivalence gives the central operational meaning of mismodel. A valid description has negligible mismodel precisely when it is algorithmically close, in both directions, to the class of minimal descriptions of the target representation. In particular, when $d \in \mathcal{D}_r$, negligible mismodel means that the current description of r is already optimal up to a constant amount of algorithmic information.

6.4 Relative mismodel

The quantity

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\}$$

measures the mismodel of a valid description $d \in \mathcal{D}$ with respect to a target representation $r \in \mathcal{R}$ in absolute terms, that is, in bits. This absolute measure is the appropriate one for the structural theory developed in the previous sections. In practice, however, one often wishes to compare the mismodel of descriptions of very different intrinsic complexity, or even descriptions associated with different target representations. For this purpose, it is useful to introduce a normalized, dimensionless version of mismodel.

The idea is to divide the absolute mismodel by a complexity scale that is intrinsic both to the description under consideration and to the target representation. This yields a relative score that is less sensitive to scale effects and therefore better suited for practical comparison.

We first introduce the descriptonal complexity floor of the target representation.

Definition 6.4.1 Let $r \in \mathcal{R}$. We define the *minimal-description complexity floor* of r by

$$\kappa_r^{\mathcal{D}} := \min_{d^* \in \mathcal{D}_r^*} K(d^*).$$

The quantity $\kappa_r^{\mathcal{D}}$ measures the least Kolmogorov complexity among all minimal descriptions of r . It provides an intrinsic algorithmic scale for the target representation at the level of descriptions.

Definition 6.4.2 Let $r \in \mathcal{R}$ and let $d \in \mathcal{D}$ be a valid description. We define the *relative mismodel* of d with respect to r as

$$v^{\text{rel}}(r, d) := \frac{v(r, d)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}} = \frac{\max\{d_D(r, d), s_D(r, d)\}}{\max\{K(d), \kappa_r^{\mathcal{D}}\}}.$$

The denominator $\max\{K(d), \kappa_r^{\mathcal{D}}\}$ provides a stable intrinsic scale. It avoids degenerate normalizations when $K(d)$ is unusually small, while ensuring that the normalization remains tied to the minimum algorithmic information required

to describe r optimally. In this way, relative mismodel preserves the meaning of absolute mismodel while making its magnitude more comparable across descriptions of different complexities.

The first result shows that relative mismodel is essentially bounded between 0 and 1, up to the usual additive constants of Kolmogorov complexity.

Proposition 6.4.1 For every $r \in \mathcal{R}$ and every $d \in \mathcal{D}$, it holds that

$$0 \leq v^{\text{rel}}(r, d) \leq 1 + o(1),$$

where the $o(1)$ term denotes a quantity of the form

$$\frac{O(1)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}},$$

which vanishes for sufficiently complex descriptions or targets.

Proof. Non-negativity is immediate, since $d_D(r, d) \geq 0$ and $s_D(r, d) \geq 0$, hence

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\} \geq 0.$$

Therefore $v^{\text{rel}}(r, d) \geq 0$.

For the upper bound, recall first that

$$s_D(r, d) \leq K(d) + O(1),$$

since for any $d^* \in \mathcal{D}_r^*$, a program can ignore d^* and output d , and taking the minimum over $d^* \in \mathcal{D}_r^*$ preserves this bound.

Second,

$$d_D(r, d) \leq \kappa_r^{\mathcal{D}} + O(1),$$

because if $d_0^* \in \mathcal{D}_r^*$ satisfies $K(d_0^*) = \kappa_r^{\mathcal{D}}$, then

$$K(d_0^* | d) \leq K(d_0^*) + O(1) = \kappa_r^{\mathcal{D}} + O(1),$$

and taking the minimum over all minimal descriptions yields the claim.

Hence

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\} \leq \max\{\kappa_r^{\mathcal{D}}, K(d)\} + O(1).$$

Dividing by $\max\{K(d), \kappa_r^{\mathcal{D}}\}$ gives

$$v^{\text{rel}}(r, d) \leq 1 + \frac{O(1)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}} = 1 + o(1).$$



This proposition shows that relative mismodel behaves as a normalized error score. In particular, values close to 0 indicate low mismodel relative to the intrinsic descriptive scale of the problem, whereas values close to 1 indicate that the dominant descriptive defect is of the same order as the description itself or the target's minimal-description complexity floor.

The next proposition connects the normalized notion with the ideal case of minimal descriptions.

Proposition 6.4.2 If $d \in \mathcal{D}_r^*$, then

$$v^{\text{rel}}(r, d) = o(1).$$

In particular, for sufficiently complex target representations or sufficiently complex minimal descriptions, $v^{\text{rel}}(r, d) \approx 0$.

Proof. If $d \in \mathcal{D}_r^*$, then

$$d_D(r, d) = O(1) \quad \text{and} \quad s_D(r, d) = O(1),$$

hence

$$v(r, d) = \max\{d_D(r, d), s_D(r, d)\} = O(1).$$

Therefore

$$v_r^{\text{rel}}(d) = \frac{O(1)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}} = o(1).$$

■

This is the normalized analogue of the fact that minimal descriptions have negligible absolute mismodel. It confirms that relative mismodel is fully compatible with the theoretical ideal: once the intrinsic complexity scale is large, the residual $O(1)$ ambiguity becomes asymptotically negligible.

The following result shows that negligible absolute mismodel always implies negligible relative mismodel.

Proposition 6.4.3 Let $r \in \mathcal{R}$ and $d \in \mathcal{D}$. If

$$v(r, d) = O(1),$$

then

$$v^{\text{rel}}(r, d) = o(1).$$

Proof. If $v(r, d) = O(1)$, then by Definition 6.4.2,

$$v^{\text{rel}}(r, d) = \frac{v(r, d)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}} = \frac{O(1)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}} = o(1).$$

■

This makes explicit that relative mismodel is a genuine normalization of absolute mismodel. Whenever the absolute descriptive error is negligible, the normalized version is negligible as well, and hence no spurious inflation is introduced by passing to the relative scale.

For a *fixed* target representation r , the converse behavior is much more rigid than in the case of representations. The next proposition shows that, once r is fixed, very large absolute mismodel forces relative mismodel to be close to 1.

Proposition 6.4.4 Let $r \in \mathcal{R}$ be fixed, and let $(d_n)_{n \geq 1}$ be a sequence of valid descriptions in $\mathcal{D}$. If

$$v(r, d_n) \rightarrow \infty,$$

then

$$v^{\text{rel}}(r, d_n) \rightarrow 1.$$

Proof. Fix r , and let $d^* \in \mathcal{D}_r^*$ satisfy $K(d^*) = \kappa_r^{\mathcal{D}}$. Then, for every $d \in \mathcal{D}$,

$$K(d) \leq K(d^*) + K(d \mid d^*) + O(1) \leq \kappa_r^{\mathcal{D}} + s_r(d) + O(1) \leq \kappa_r^{\mathcal{D}} + v(r, d) + O(1).$$

Hence

$$\max\{K(d), \kappa_r^{\mathcal{D}}\} \leq \kappa_r^{\mathcal{D}} + v(r, d) + O(1).$$

Therefore

$$v^{\text{rel}}(r, d) = \frac{v(r, d)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}} \geq \frac{v(r, d)}{\kappa_r^{\mathcal{D}} + v(r, d) + O(1)}.$$

Applying this to $d = d_n$ and letting $n \rightarrow \infty$, the right-hand side tends to 1. On the other hand, by Proposition 6.4.1,

$$v^{\text{rel}}(r, d_n) \leq 1 + o(1).$$

Combining both bounds yields

$$v^{\text{rel}}(r, d_n) \rightarrow 1.$$

■

Thus, for a fixed target representation, relative mismodel cannot be made arbitrarily small while absolute mismodel grows without bound. In this sense, the normalization preserves much of the absolute information when the target is held fixed.

The converse direction can nevertheless fail in general when the target representation is allowed to vary.

Proposition 6.4.5 In general, $v^{\text{rel}}(r, d) = o(1)$ does not imply $v(r, d) = O(1)$.

Proof. We construct a sequence of pairs (r_n, d_n) for which the absolute mismodel diverges while the relative mismodel tends to 0.

Choose a sequence of target representations $(r_n)_{n \geq 1}$ such that their minimal-description complexity floors satisfy

$$\kappa_{r_n}^{\mathcal{D}} \rightarrow \infty.$$

For each n , let $d_n^* \in \mathcal{D}_{r_n}^*$ satisfy

$$K(d_n^*) = \kappa_{r_n}^{\mathcal{D}}.$$

Let z_n be an incompressible string independent of d_n^* of length

$$\ell(z_n) = \lfloor \log \kappa_{r_n}^{\mathcal{D}} \rfloor.$$

Now define a valid description d_n as a fixed computable packaging of the pair (d_n^*, z_n) such that d_n^* is uniformly recoverable from d_n . For example, d_n may be taken to be a description that outputs the binary encoding of the pair $\langle d_n^*, z_n \rangle$. Then

$$K(d_n^* \mid d_n) = O(1),$$

so

$$d_{r_n}(d_n) = O(1).$$

On the other hand, from d_n^* one must essentially reconstruct the independent incompressible tag z_n in order to recover d_n , so

$$s_{r_n}(d_n) = \Theta(\ell(z_n)) = \Theta(\log \kappa_{r_n}^{\mathcal{D}}).$$

Hence

$$v_{r_n}(d_n) = \max\{d_{r_n}(d_n), s_{r_n}(d_n)\} = \Theta(\log \kappa_{r_n}^{\mathcal{D}}) \rightarrow \infty.$$

At the same time,

$$K(d_n) \leq K(d_n^*) + K(z_n) + O(1) = \kappa_{r_n}^{\mathcal{D}} + O(\log \kappa_{r_n}^{\mathcal{D}}),$$

so

$$\max\{K(d_n), \kappa_{r_n}^{\mathcal{D}}\} = \kappa_{r_n}^{\mathcal{D}} + O(\log \kappa_{r_n}^{\mathcal{D}}).$$

Therefore

$$v_{r_n}^{\text{rel}}(d_n) = \frac{\Theta(\log \kappa_{r_n}^{\mathcal{D}})}{\kappa_{r_n}^{\mathcal{D}} + O(\log \kappa_{r_n}^{\mathcal{D}})} \rightarrow 0.$$

Thus small relative mismodel does not, in general, force small absolute mismodel. ■

This proposition shows that relative mismodel and absolute mismodel serve different purposes. Absolute mismodel captures the actual algorithmic cost of repair, whereas relative mismodel measures that cost only in proportion to the intrinsic scale of the description and of the target representation.

It is also useful to make explicit the relation between absolute and relative ordering when the compared descriptions have the same intrinsic scale.

Proposition 6.4.6 Let $r \in \mathcal{R}$, and let $d_1, d_2 \in \mathcal{D}$ satisfy

$$\max\{K(d_1), \kappa_r^{\mathcal{D}}\} = \max\{K(d_2), \kappa_r^{\mathcal{D}}\}.$$

Then

$$v(r, d_1) \leq v(r, d_2) \iff v^{\text{rel}}(r, d_1) \leq v^{\text{rel}}(r, d_2).$$

Proof. If the denominators coincide, say both equal $C > 0$, then

$$v^{\text{rel}}(r, d_i) = \frac{v(r, d_i)}{C} \quad (i = 1, 2).$$

Division by the same positive constant preserves order, so

$$v(r, d_1) \leq v(r, d_2) \iff \frac{v(r, d_1)}{C} \leq \frac{v(r, d_2)}{C},$$

which is exactly

$$v^{\text{rel}}(r, d_1) \leq v^{\text{rel}}(r, d_2). \quad \blacksquare$$

This proposition clarifies the practical meaning of the normalization. When the compared descriptions have the same intrinsic scale, relative mismodel does not change the ranking; it simply rescales the absolute values. The normalization matters only when the underlying scales differ.

Finally, relative mismodel preserves the bottleneck interpretation of absolute mismodel, since the denominator is common to both deficiency and surplus.

Proposition 6.4.7 Let $r \in \mathcal{R}$ and $d \in \mathcal{D}$.

1 If $d_D(r, d) > s_D(r, d)$, then

$$v_r^{\text{rel}}(d) = \frac{d_D(r, d)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}}.$$

2 If $s_D(r, d) > d_D(r, d)$, then

$$v_r^{\text{rel}}(r, d) = \frac{s_D(r, d)}{\max\{K(d), \kappa_r^{\mathcal{D}}\}}.$$

Proof. By definition,

$$v_r^{\text{rel}}(r, d) = \frac{\max\{d_D(r, d), s_D(r, d)\}}{\max\{K(d), \kappa_r^{\mathcal{D}}\}}.$$

If $d_D(r, d) > s_D(r, d)$, then the numerator equals $d_D(r, d)$. If $s_D(r, d) > d_D(r, d)$, then the numerator equals $s_D(r, d)$. The stated equalities follow immediately. ■

Thus, relative mismodel inherits the same diagnostic value as absolute mismodel: it identifies the currently dominant descriptive defect. The only difference is that this defect is now expressed relative to the intrinsic scale of the description and the target representation.

Interpretation.

A value $v_r^{\text{rel}}(r, d) \approx 0$ indicates that, relative to its overall information content, the description has both low deficiency and low surplus with respect to r ; it is therefore close to optimal for that target in a scale-free sense. Values close to 1 indicate that either deficiency or surplus is of the same order as the intrinsic scale $\max\{K(d), \kappa_r^{\mathcal{D}}\}$, meaning that the description is highly incomplete, highly contaminated, or both.

Relative mismodel is therefore best understood as a practical comparison tool rather than as a replacement for absolute mismodel. The latter remains the primary structural notion, whereas the former provides a normalized view better suited for comparing descriptions of different scales or descriptions associated with different target representations.

6.5 Decreasing Mismodel

For a fixed target representation r , mismodel measures how far a valid description $d \in \mathcal{D}$ is from being algorithmically equivalent to a minimal description of r . In the theory developed above, this discrepancy is decomposed into two components. The deficiency $d_D(r, d)$ measures how much information is still missing in order to recover a minimal description of r , whereas the surplus $s_D(r, d)$ measures how

much information in d goes beyond what is needed to obtain such a minimal description.

Decreasing mismodel is therefore intrinsically a two-objective refinement problem: one aims to reduce missing descriptive information while also removing descriptively extraneous content. In practice, these two goals correspond to two complementary modes of improvement. Enrichment (adding information to a description) is typically aimed at reducing deficiency, whereas simplification (removing or normalizing parts of a description) is typically aimed at reducing surplus.

A specific feature of the description setting is that not every raw edit of a binary string preserves validity. Accordingly, the practically relevant transformations considered below are computable operations that map valid descriptions to valid descriptions.

Augmentation and Deficiency

We now examine how the deficiency of a description may be reduced through augmentation. Intuitively, when a description falls short of a minimal valid description of a target representation, it lacks some of the computable information needed to reconstruct that representation more effectively. In such cases, augmenting the description with a suitable auxiliary description may supply part of the missing information and thereby bring the resulting description closer to the target. The purpose of this section is to formalize that intuition and to identify conditions under which augmentation can contribute to decreasing deficiency.

Since the augmented description contains the original description as a recoverable component, augmentation cannot increase deficiency by more than an additive constant.

Proposition 6.5.1 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}$, and let $d' \in \mathcal{D}$. Then

$$d_D(r, \text{aug}(d, d')) \leq d_D(r, d) + O(1).$$

Proof. Let $d^* \in \mathcal{D}_r^*$ be any minimal description of r . Since the first component d can be recovered computably from $\text{aug}(d, a)$, there exists a computable function p such that

$$p(\text{aug}(d, a)) = d.$$

By monotonicity of conditional Kolmogorov complexity under computable transformations,

$$K(d^* \mid \text{aug}(d, a)) \leq K(d^* \mid p(\text{aug}(d, a))) + O(1) = K(d^* \mid d) + O(1).$$

Since this holds for every $d^* \in \mathcal{D}_r^*$, we obtain

$$\min_{d^* \in \mathcal{D}_r^*} K(d^* \mid \text{aug}(d, a)) \leq \min_{d^* \in \mathcal{D}_r^*} K(d^* \mid d) + O(1).$$

Hence

$$d_D(r, \text{aug}(d, a)) \leq d_D(r, d) + O(1).$$

■

This proposition captures a basic monotonicity principle: once additional information has been incorporated into a valid description, recovering a minimal description of the target representation cannot become substantially harder. Of course, the result does not guarantee that every enrichment is useful in practice. Some added material may be redundant, irrelevant, or misleading. What matters scientifically is not enrichment per se, but enrichment that reduces deficiency by supplying information that was genuinely missing.

The next proposition isolates a sufficient condition for such a reduction.

Proposition 6.5.2 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}_r$, and let $a \in \mathcal{D}$. If there exists a minimal description $d^* \in \mathcal{D}_r^*$ such that

$$K(d^* \mid \text{aug}(d, a)) = O(1),$$

then

$$d_D(r, \text{aug}(d, a)) = O(1).$$

Proof. By definition,

$$d_D(r, \text{aug}(d, a)) = \min_{d' \in \mathcal{D}_r^*} K(d' \mid \text{aug}(d, a)).$$

Since $d^* \in \mathcal{D}_r^*$, we have

$$d_D(r, \text{aug}(d, a)) \leq K(d^* \mid \text{aug}(d, a)) = O(1).$$

■

This proposition captures the ideal case of augmentation: the auxiliary description a supplies essentially all the missing algorithmic information needed to recover a minimal description of the target representation.

A particularly important special case occurs when the auxiliary description itself encodes the missing information needed to transform the current description into a minimal one.

Corollary 6.5.3 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}_r$, and let $a \in \mathcal{D}$. If there exists a computable function f and a minimal description $d^* \in \mathcal{D}_r^*$ such that

$$f(\text{aug}(d, a)) = d^*,$$

then

$$d_D(r, \text{aug}(d, a)) = O(1).$$

Proof. If $f(\text{aug}(d, a)) = d^*$, then d^* can be computed from $\text{aug}(d, a)$ by a program of constant length, so

$$K(d^* \mid \text{aug}(d, a)) = O(1).$$

The result now follows from Proposition 6.5.2. ■

This corollary formalizes the intuition that augmentation is useful precisely when the auxiliary description contributes information that was absent from the original one.

Finally, augmentation need not be strictly improving. If the auxiliary description carries no relevant information about how to approach a minimal description, then the deficiency may remain unchanged up to an additive constant.

■ **Example 6.3** Let $r \in \mathcal{R}$ and let $d \in \mathcal{D}_r$ be a non-minimal description of r . Let $a \in \mathcal{D}$ be any valid description whose information is unrelated to the structure of the minimal descriptions in $\mathcal{D}_r^*$. Then the augmented description

$$\text{aug}(d, a)$$

still belongs to $\mathcal{D}_r$, but there need not be any strict decrease in deficiency:

$$d_D(r, \text{aug}(d, a)) = d_D(r, d) + O(1)$$

may still hold. Thus, augmentation preserves validity and target representation, but only relevant augmentation descriptions are expected to reduce deficiency. ■

The operation of augmentation therefore provides a mathematically well-defined mechanism for enriching descriptions by auxiliary computable information. It preserves both validity and target representation, and it can only decrease deficiency up to an additive constant. For this reason, augmentation is a natural operation in the analysis of description deficiency and, more generally, in the study of how scientific models are progressively refined.

To make this idea explicit, it is useful to measure how much deficiency is reduced by a given enrichment step.

Definition 6.5.1 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}$, and let $a \in \mathcal{B}^*$ be such that $da \in \mathcal{D}$. We define the *marginal deficiency reduction* of a relative to d and r by

$$\Delta_D^d(e, r, a) = d_D(e, r) - d_D(e, ra)$$

The quantity $\Delta_D^d(e, r, a)$ measures how much the additional information a reduces the amount of missing algorithmic information needed to reach a minimal description of r . If $\Delta_D^d(e, r, a)$ is large, then a supplies information that was highly relevant and previously absent from d . If it is close to zero, then the enrichment contributes little toward completing the description.

This notion suggests a practical decision principle: among several candidate augmentations, a good next addition is one that is expected to minimize the posterior deficiency $d_D(r, da)$, or equivalently, to maximize the expected marginal deficiency reduction.

■ **Example 6.4** Suppose that r is a representation encoding the full state trajectory of a dynamical system. A current valid description d may consist of a program that simulates the system together with a coarse set of parameters, but it may omit precise initial conditions or an explicit boundary constraint. A candidate augmentation a_1 adds a verbose explanatory comment about the numerical solver, whereas a second augmentation a_2 adds the missing initial condition.

The scientifically preferable enrichment is the one expected to make $d_D(r, da)$ as small as possible. In this situation, a_2 is likely to be the better choice, because it supplies information that is directly relevant to recovering a minimal description of the target representation r . By contrast, a_1 may improve readability, but it is unlikely to reduce the missing information needed to reach a minimal description of r . ■

Candidate Augmentations of a Description

The operation of augmentation introduced above provides a general mechanism for enriching a description by pairing it with an auxiliary valid description. In principle, one could consider augmenting a description by *any* valid description. For the purposes of a nescience-reducing algorithm, however, such a search space is too broad to be useful. It is therefore convenient to distinguish between a theoretical notion of augmentation, whose role is to characterize ideal deficiency-reducing moves, and a practical notion of candidate augmentation, whose role is to generate a finite or otherwise restricted set of alternatives that can actually be explored.

We begin with the theoretical notion. Since augmentation preserves the represented string, it is natural to focus on those augmentations that reduce the deficiency of the current description with respect to a fixed representation.

Definition 6.5.2 Let $r \in \mathcal{R}$ and let $d \in \mathcal{D}_r$. We define the *deficiency-targeted augmentation set* of d with respect to r by

$$\mathcal{A}_r^d(d) = \{\text{aug}(d, a) \in \mathcal{D}_r : a \in \mathcal{D} \text{ and } d_D(r, \text{aug}(d, a)) < d_D(r, d)\}.$$

This set contains precisely those augmentations that strictly improve the deficiency of the current description. It therefore captures the ideal role of augmentation as a corrective operation directed at the defect measured by $d_D(r, d)$.

The previous definition is conceptually natural, but it is not suitable for direct algorithmic use. The difficulty is that it quantifies over the whole set $\mathcal{D}$, which is recursively enumerable but unbounded, and it requires checking whether deficiency decreases, a condition that is typically not computable in full generality. For practical purposes, we therefore introduce a restricted set of admissible augmentation descriptions.

Definition 6.5.3 Let $d \in \mathcal{D}$. We call a set

$$\mathcal{A}_d \subseteq \mathcal{D}$$

a set of *admissible augmentation descriptions* for d if it is chosen as the search space of auxiliary descriptions to be paired with d in the augmentation step.

The choice of $\mathcal{A}_d$ depends on the application. In principle, one may take $\mathcal{A}_d = \mathcal{D}$, thereby recovering the unrestricted theoretical notion above. In practice, however, $\mathcal{A}_d$ is usually chosen to satisfy additional constraints, such as bounded length, membership in a model class, or relevance to the active defect set.

Once a family of admissible augmentation descriptions has been fixed, we can define the corresponding practical candidate set.

Definition 6.5.4 Let $x = (r, d)$ be a hypothesis such that $d \in \mathcal{D}_r$, and let $\mathcal{A}_d \subseteq \mathcal{D}$ be a prescribed set of admissible augmentation descriptions. We define the *algorithmic augmentation candidate set* of x by

$$\mathcal{C}^{\text{aug}}(x) = \{(r, \text{aug}(d, a)) : a \in \mathcal{A}_d\}.$$

This is the set of hypotheses obtained by augmenting the current description d with auxiliary descriptions drawn from the chosen search space $\mathcal{A}_d$. Unlike the deficiency-targeted augmentation set, the set $\mathcal{C}^{\text{aug}}(x)$ is intended for actual use in a search or descent algorithm.

The first important property of the algorithmic augmentation candidate set is that it preserves the represented string.

Proposition 6.5.4 Let $x = (r, d)$ be a hypothesis such that $d \in \mathcal{D}_r$, and let $(r', d') \in$

$\mathcal{C}^{\text{aug}}(x)$. Then

$$r' = r \quad \text{and} \quad d' \in \mathcal{D}_r.$$

Proof. By Definition 6.5.4, every element of $\mathcal{C}^{\text{aug}}(x)$ has the form

$$(r', d') = (r, \text{aug}(d, a))$$

for some $a \in \mathcal{A}_d \subseteq \mathcal{D}$. Hence $r' = r$.

Since $d \in \mathcal{D}_r$, we have $\gamma(d) = r$. By Proposition ??,

$$\gamma(\text{aug}(d, a)) = \gamma(d) = r.$$

Therefore

$$d' = \text{aug}(d, a) \in \mathcal{D}_r.$$

■

Thus, algorithmic augmentation operates entirely within the fiber $\mathcal{D}_r$. The representation remains fixed, while only the description is modified.

As in the unrestricted case, augmentation cannot increase deficiency beyond an additive constant.

Proposition 6.5.5 Let $x = (r, d)$ be a hypothesis such that $d \in \mathcal{D}_r$, and let $(r, d') \in \mathcal{C}^{\text{aug}}(x)$. Then

$$d_D(r, d') \leq d_D(r, d) + O(1).$$

Proof. Since $(r, d') \in \mathcal{C}^{\text{aug}}(x)$, there exists $a \in \mathcal{A}_d \subseteq \mathcal{D}$ such that

$$d' = \text{aug}(d, a).$$

The result then follows immediately from Proposition 6.5.1, which states that

$$d_D(r, \text{aug}(d, a)) \leq d_D(r, d) + O(1).$$

■

The previous proposition justifies the use of augmentation in a defect-guided search procedure: although augmentation need not strictly improve deficiency, it never moves the description far away from the class of minimal descriptions, up to an additive constant.

A particularly natural way to restrict the set of admissible augmentation descriptions is by bounding their length.

Definition 6.5.5 Let $d \in \mathcal{D}$, and let $L : \mathbb{N} \rightarrow \mathbb{N}$ be a bound function. We define the *length-bounded augmentation set* associated with d by

$$\mathcal{A}_d^L = \{a \in \mathcal{D} : \ell(a) \leq L(\ell(d))\}.$$

The corresponding *length-bounded algorithmic augmentation candidate set* is

$$\mathcal{C}^{\text{aug},L}(x) = \{(r, \text{aug}(d, a)) : a \in \mathcal{A}_d^L\},$$

where $x = (r, d)$.

This restriction is useful in practice because it limits the search to auxiliary descriptions whose size is controlled relative to the current description. The bound function L may be chosen in several ways depending on the intended search strategy. For instance, one may take $L(n) = n$, which allows only augmentation descriptions no longer than the current description itself, or $L(n) = n + c$ for some fixed constant c , which permits only moderate growth.

Whenever the chosen bound yields a finite set of descriptions, the corresponding candidate set is finite as well.

Proposition 6.5.6 Let $x = (r, d)$ be a hypothesis. If the set $\mathcal{A}_d^L$ is finite, then $\mathcal{C}^{\text{aug},L}(x)$ is finite.

Proof. By Definition 6.5.5,

$$\mathcal{C}^{\text{aug},L}(x) = \{(r, \text{aug}(d, a)) : a \in \mathcal{A}_d^L\}.$$

The mapping

$$a \mapsto (r, \text{aug}(d, a))$$

associates one candidate hypothesis with each element of $\mathcal{A}_d^L$. Hence $\mathcal{C}^{\text{aug},L}(x)$ has cardinality at most that of $\mathcal{A}_d^L$. Therefore, if $\mathcal{A}_d^L$ is finite, then $\mathcal{C}^{\text{aug},L}(x)$ is also finite. ■

This proposition highlights the practical role of length-bounded augmentation: it converts the open-ended theoretical space of all augmentations into a candidate set that can, at least in principle, be explored by an algorithm.

Finally, if one wishes to retain the practical candidate set while focusing only on genuine deficiency improvements, one may intersect the algorithmic candidate set with the deficiency-targeted one.

Definition 6.5.6 Let $x = (r, d)$ be a hypothesis such that $d \in \mathcal{D}_r$, and let $\mathcal{A}_d \subseteq \mathcal{D}$ be a prescribed set of admissible augmentation descriptions. We

define the *practical deficiency-reducing augmentation set* of x by

$$\mathcal{C}^{\text{aug},d}(x) = \{(r, \text{aug}(d, a)) \in \mathcal{C}^{\text{aug}}(x) : d_D(r, \text{aug}(d, a)) < d_D(r, d)\}.$$

This set serves as a bridge between theory and practice. The deficiency-targeted augmentation set identifies the ideal augmentations from a purely theoretical standpoint, whereas the algorithmic augmentation candidate set specifies the bounded family of alternatives that the search procedure is willing to examine. Their intersection yields the candidate augmentations that are both practically admissible and theoretically beneficial.

In summary, the operation of augmentation gives rise to two distinct but complementary notions. The deficiency-targeted augmentation set describes the ideal search direction for reducing deficiency, while the algorithmic augmentation candidate set provides the restricted family of augmentations that can be used in an effective nescience-reducing procedure. This distinction will be useful later when defining the full candidate set of operations for the descent algorithm.

Decreasing surplus via simplification

A natural way to reduce the surplus of a description is to simplify it by removing or normalizing content that is not genuinely needed to reach a minimal description of the target representation. In scientific practice, this corresponds to removing dead code, duplicated routines, logging instructions, irrelevant comments, administrative metadata, plotting subroutines, or ad hoc branches that do not contribute to the description of r itself.

The basic monotonicity principle is that computably simplifying a valid description cannot substantially increase its surplus.

Proposition 6.5.7 Let $r \in \mathcal{R}$, let $d, s \in \mathcal{D}$, and suppose that there exists a total computable function

$$g : \mathcal{B}^* \rightarrow \mathcal{B}^*$$

such that

$$g(s) = d.$$

Then

$$s_r(d) \leq s_r(s) + O(1).$$

Proof. Let $d^* \in \mathcal{D}_r^*$ be arbitrary. From any program witnessing $K(s \mid d^*)$, one can compute s from d^* and then apply g with constant overhead to obtain $d = g(s)$. Hence

$$K(d \mid d^*) \leq K(s \mid d^*) + O(1).$$

Taking minima over $d^* \in \mathcal{D}_r^*$, we obtain

$$s_r(d) = \min_{d^* \in \mathcal{D}_r^*} K(d \mid d^*) \leq \min_{d^* \in \mathcal{D}_r^*} (K(s \mid d^*) + O(1)) = s_r(s) + O(1).$$

■

This proposition captures the basic monotonicity principle for surplus: computable simplification may remove extraneous descriptive content, but it cannot create it. Again, not every simplification is scientifically desirable: a simplification that removes relevant information may reduce surplus while making the description much more deficient. What matters scientifically is not simplification per se, but simplification that decreases surplus without substantially worsening deficiency.

To evaluate such steps explicitly, it is useful to measure how much surplus is reduced by a given simplification operator.

Definition 6.5.7 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}$, and let $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a total computable function such that $g(d) \in \mathcal{D}$. We define the *marginal surplus reduction* of g relative to d and r by

$$\Delta_{\text{sup}}(g \mid d, r) = s_r(d) - s_r(g(d)).$$

The quantity $\Delta_{\text{sup}}(g \mid d, r)$ measures how much the simplification g reduces the amount of algorithmically extraneous content present in d relative to the target representation r . If $\Delta_{\text{sup}}(g \mid d, r)$ is large, then the removed or normalized material was highly extraneous. If it is close to zero, then the simplification contributes little toward eliminating surplus.

This notion suggests a practical decision principle: among several candidate simplifications, a good next reduction is one that is expected to minimize the posterior surplus $s_r(g(d))$, or equivalently, to maximize the expected marginal surplus reduction.

■ **Example 6.5** Suppose that d is a valid description consisting of a program that generates the target representation r , but also includes extensive logging routines, benchmark code, plotting commands, and a historical record of intermediate failed attempts. A normalization operator g_1 removes only the logging and plotting routines, whereas another operator g_2 removes a core routine that is actually needed to generate the representation.

The preferable simplification is the one expected to make $s_r(g(d))$ as small as possible while keeping the resulting string a useful valid description. In this scenario, g_1 is likely to be the better choice, because it removes content that is not genuinely needed to reach a minimal description of r . By contrast, g_2 may simplify the code syntactically, but it risks removing information that is structurally relevant to the target representation. ■

A particularly important case is when the removed material is fully recoverable from what remains. Then simplification does not substantially worsen deficiency.

Proposition 6.5.8 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}$, and let $g : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a total computable function such that

$$\tilde{d} := g(d) \in \mathcal{D}.$$

If

$$K(d \mid \tilde{d}) = O(1),$$

then

$$d_D(r, \tilde{d}) = d_D(r, d) + O(1).$$

Proof. Since $\tilde{d}$ is computable from d via g , we have

$$d_D(r, \tilde{d}) \leq d_D(r, d) + O(1).$$

On the other hand, the assumption $K(d \mid \tilde{d}) = O(1)$ implies that d is computable from $\tilde{d}$ with constant overhead. Therefore

$$d_D(r, d) \leq d_D(r, \tilde{d}) + O(1).$$

Combining both inequalities yields

$$d_D(r, \tilde{d}) = d_D(r, d) + O(1).$$

■

This proposition formalizes the idea of a safe simplification: if the material that is removed can already be reconstructed from the simplified description, then no substantial increase in deficiency is incurred. In practice, this justifies removing duplications, macro expansions, boilerplate, or reversible syntactic sugar without materially worsening the amount of missing information.

Strict decrease of mismodel

The previous propositions provide one-sided guarantees: enrichment controls deficiency, and simplification controls surplus. A strict decrease of mismodel requires that the quantity

$$v_r(d) = \max\{d_D(r, d), s_r(d)\}$$

actually becomes smaller. In particular, decreasing only one component is not always sufficient, since the other component may remain dominant.

Before giving sufficient conditions for strict decrease, it is useful to note explicitly that a strict decrease in deficiency alone, or in surplus alone, does not in general imply a strict decrease in mismatch.

Proposition 6.5.9 A strict decrease in deficiency alone, or in surplus alone, does not in general imply a strict decrease in mismatch.

Proof. For the first claim, suppose that

$$d_D(r, d) = 10, \quad s_r(d) = 100,$$

and that after some modification we obtain a valid description $d' \in \mathcal{D}$ such that

$$d_D(r, d') = 9, \quad s_r(d') = 100.$$

Then deficiency has decreased strictly, but

$$v_r(d) = \max\{10, 100\} = 100, \quad v_r(d') = \max\{9, 100\} = 100.$$

Hence $v_r(d') = v_r(d)$, so mismatch does not decrease strictly.

The argument for surplus is symmetric. If

$$d_D(r, d) = 100, \quad s_r(d) = 10,$$

and

$$d_D(r, d') = 100, \quad s_r(d') = 9,$$

then surplus decreases strictly, but

$$v_r(d) = \max\{100, 10\} = 100, \quad v_r(d') = \max\{100, 9\} = 100.$$

Again, mismatch does not decrease strictly. ■

This proposition shows why one-sided control results are not enough on their own to guarantee improvement in mismatch. What matters is not only whether one component decreases, but whether the dominant component is reduced below the previous mismatch level.

The strongest simple sufficient condition is the following: if both deficiency and surplus decrease strictly, then mismatch must also decrease strictly.

Proposition 6.5.10 Let $r \in \mathcal{R}$, and let $d, d' \in \mathcal{D}$ be valid descriptions. If

$$d_D(r, d') < d_D(r, d) \quad \text{and} \quad s_r(d') < s_r(d),$$

then

$$v_r(d') < v_r(d).$$

Proof. Since both components decrease strictly, we have

$$\max\{d_D(r, d'), s_r(d')\} < \max\{d_D(r, d), s_r(d)\}.$$

That is,

$$v_r(d') < v_r(d).$$

■

This proposition captures the ideal situation in which a descriptive intervention simultaneously removes missing information and extraneous information. In practice, however, enrichment and simplification usually act asymmetrically, so it is useful to have more targeted sufficient conditions based on whichever component is currently dominant.

Proposition 6.5.11 Let $r \in \mathcal{R}$, let $d \in \mathcal{D}$ be a valid description, and let $d' \in \mathcal{D}$ be a revised valid description.

- 1 If $d_D(r, d) = v_r(d)$, $d_D(r, d') < d_D(r, d)$, and

$$s_r(d') < d_D(r, d),$$

then

$$v_r(d') < v_r(d).$$

- 2 If $s_r(d) = v_r(d)$, $s_r(d') < s_r(d)$, and

$$d_D(r, d') < s_r(d),$$

then

$$v_r(d') < v_r(d).$$

Proof. For (1), since $d_D(r, d) = v_r(d)$, we have

$$v_r(d) = d_D(r, d).$$

By assumption,

$$d_D(r, d') < d_D(r, d) = v_r(d) \quad \text{and} \quad s_r(d') < d_D(r, d) = v_r(d).$$

Hence both arguments in the maximum defining $v_r(d')$ are strictly smaller than $v_r(d)$, so

$$v_r(d') = \max\{d_D(r, d'), s_r(d')\} < v_r(d).$$

The proof of (2) is symmetric. Since $s_r(d) = v_r(d)$, we have

$$v_r(d) = s_r(d).$$

By assumption,

$$s_r(d') < s_r(d) = v_r(d) \quad \text{and} \quad d_D(r, d') < s_r(d) = v_r(d).$$

Therefore

$$v_r(d') = \max\{d_D(r, d'), s_r(d')\} < v_r(d).$$

■

This proposition makes explicit a basic decision rule for description refinement: one should focus on reducing the component that currently determines mismodel, while keeping the other below the previous threshold. In this sense, enrichment is most naturally useful when deficiency is dominant, whereas simplification is most naturally useful when surplus is dominant.

A particularly informative case occurs when the two components are exactly balanced. Then neither deficiency nor surplus alone determines mismodel; both jointly determine the threshold that must be crossed for strict improvement.

Corollary 6.5.12 If

$$d_D(r, d) = s_r(d) = v_r(d),$$

then

$$v_r(d') < v_r(d) \iff d_D(r, d') < v_r(d) \text{ and } s_r(d') < v_r(d).$$

Proof. Assume first that $v_r(d') < v_r(d)$. Since

$$v_r(d') = \max\{d_D(r, d'), s_r(d')\},$$

both $d_D(r, d')$ and $s_r(d')$ must be strictly smaller than $v_r(d)$; otherwise the maximum would be at least $v_r(d)$.

Conversely, if

$$d_D(r, d') < v_r(d) \quad \text{and} \quad s_r(d') < v_r(d),$$

then

$$v_r(d') = \max\{d_D(r, d'), s_r(d')\} < v_r(d).$$

■

The balanced case shows that when deficiency and surplus are tied, improving only one of them is not enough to decrease mismodel strictly. Both must fall below the previous common level. This explains why isolated edits may fail to improve descriptive quality and a combined intervention may be required.

Practical strategy for decreasing mismodel

The previous results suggest a simple operational strategy for refining valid descriptions.

If deficiency is much larger than surplus, one should prioritize enrichment steps expected to reduce missing information while keeping surplus under control. In practice, this means adding precisely the information that brings the current description closer to a minimal description of r : missing constraints, definitions, initial conditions, causal mechanisms, latent states, or structural invariants.

If surplus is much larger than deficiency, one should prioritize simplification steps expected to eliminate extraneous content without making the description substantially more deficient. In practice, this means removing dead code, duplicated derivations, reversible formatting layers, irrelevant comments, logging instructions, or ad hoc subroutines that do not help in reaching a minimal description of the target representation.

When the two components are of comparable size, the best intervention may need to combine both types of refinement: enrich the description in the directions that reduce deficiency most, and simplify it in the directions that reduce surplus most. The goal is not merely to make the description longer or shorter, but to bring it algorithmically closer, in both directions, to the class of minimal descriptions of r .

In this way, mismodel becomes a prescriptive notion. It does not merely measure how far a current description is from descriptive optimality; it also suggests how that description should be revised next.

References

A good introduction to the study of uncertainties (error analysis in models) in science, and in particular in physics, chemistry, and engineering, is the best-selling text [TAY22], which also features the same image of a crashed train than in the introduction to this chapter.

The concept of redundancy has been also investigated in the context of information theory, since we are interested on using codes with low redundancy (see for example [Abr63]).



7. Nescience

*There are known knowns. These are things we know that we know.
There are known unknowns. That is to say, there are things that we
know we don't know. But there are also unknown unknowns. There
are things we don't know we don't know.*

Donald Rumsfeld

Following Chapters 2, 3, and 4, which introduced the basic objects of the theory of nescience, and Chapters 5 and 6, which developed the two fundamental sources of epistemic error, we are now in a position to introduce the central concept of this book: *nescience*. Nescience measures the extent to which our research hypotheses fail to provide an adequate understanding of the entities under investigation.

Unlike Shannon entropy or Kolmogorov complexity, which quantify properties of strings or sources of information, nescience is a metric framework for quantifying ignorance. More precisely, it characterizes ignorance through two complementary components. The first is *misencoding*, which measures the representational error of a hypothesis, that is, how far a candidate representation is from an adequate encoding of the entity under study. The second is *mismodel*, which measures the descriptive error of a hypothesis, that is, how far a candidate description is from an adequate model of the chosen representation. Together,

these two quantities capture the two main ways in which scientific understanding may fail: by representing the world poorly, or by describing those representations poorly.

These two components are generally interdependent and often conflicting. Improving the representation of an entity may require abandoning a previously successful description, thereby increasing mismodel. Conversely, although improving the description of a fixed representation does not alter its miscoding, it may reveal limitations of the current hypothesis and motivate the search for a different representation-description pair. For this reason, the theory of nescience treats science as a *bi-objective optimization problem*: the aim is not to minimize representational error or descriptive error in isolation, but to reduce both simultaneously. This perspective leads naturally to the notions of Pareto optimality, current best hypothesis, and perfect knowledge developed in this chapter.

A major consequence of this framework is that it provides a systematic way to classify the domain of inquiry according to the current state of our hypotheses. Some entities belong to the region of the *known knowns*, for which at least one known hypothesis already achieves negligible nescience. Others belong to the *known unknowns*, for which hypotheses exist but none yet reach that standard. Finally, there remain the *unknown unknowns*: entities for which no hypothesis has yet been formulated. One of the most important ambitions of the theory of nescience is to supply a formal basis for understanding this last region and, ultimately, for developing methods capable of bringing parts of it into scientific view.

The theory also suggests a more subtle and perhaps counterintuitive conclusion. Scientific progress is not guaranteed merely by the accumulation of further hypotheses. In some situations, inquiry may reach a stage at which no reduction in one component of nescience is possible without worsening the other. In such cases, research encounters a frontier beyond which improvement requires a change of representation, a change of descriptive framework, or a revision of the preference criterion used to rank hypotheses. Nescience therefore does not merely quantify ignorance; it also reveals the structural limits within which scientific progress must operate.

7.1 Nescience

Intuitively, our understanding of an entity depends on two complementary aspects of scientific work: the construction of a representation that encodes the entity, and the construction of a description that models that representation. According to the theory of nescience, there are two sources of ignorance, captured by the notions of miscoding and mismodel. Miscoding measures how far a candidate representation

is from being algorithmically equivalent to an admissible representation of the entity, whereas mismodel measures how far a candidate description is from being algorithmically equivalent to a minimal description of the given representation.

From this perspective, scientific inquiry consists in jointly reducing representational and descriptive error. Although miscoding depends only on r , while mismodel depends on d relative to r , the goal is to optimize both components simultaneously. Science is therefore naturally formulated as a bi-objective optimization problem.

The Science Problem

$$\begin{aligned} &\text{Given an entity } e \in \mathcal{E} \\ &\text{minimize } \{\mu(e, r), v(r, d)\} \\ &\text{subject to } (r, d) \in \mathcal{H}_e \end{aligned}$$

The feasible region is the set

$$\mathcal{H}_e = \{(r, d) \mid r \in \mathcal{R}_e, d \in \mathcal{D}_r\},$$

which contains all representation-description pairs associated with the entity e . The decision vectors are pairs (r, d) consisting of a representation and a description. The objective functions to be minimized are miscoding and mismodel. The objective space is therefore a subset $\mathbf{Z} \subseteq \mathbb{R}_{\geq 0}^2$. Our primary focus lies in the feasible decision space $\mathcal{H}_e$, namely the space of admissible representations and descriptions for the entity under study, rather than in the objective space $\mathbf{Z} \subseteq \mathbb{R}_{\geq 0}^2$ of metric values.

Pareto Optimality

If the current representation and description are not perfect, the objective of scientific inquiry is to identify alternative hypotheses that reduce our ignorance. This means finding a new representation or a new description that decreases at least one of the two components of nescience, miscoding or mismodel, without increasing the other.

Definition 7.1.1 We say that a hypothesis $(r, d) \in \mathcal{H}$ *dominates* another hypothesis $(r', d') \in \mathcal{H}$ if

$$\mu(r) \leq \mu(r'), \quad v_r(d) \leq v_{r'}(d'),$$

and at least one of these two inequalities is strict.

This notion of dominance formalizes the idea of unambiguous scientific improvement. A dominating hypothesis is never worse in either component

of nescience and is strictly better in at least one. Consequently, dominated hypotheses can be regarded as suboptimal and, in principle, replaceable by better alternatives.

The definition also highlights the two fundamental directions in which scientific progress may occur. On the one hand, we may improve the representation of the entity, thereby reducing miscoding, while retaining a description of comparable quality. On the other hand, we may refine the description of a given representation, thereby reducing mismodel, without degrading the representational fidelity.

In practice, there is rarely a single solution that simultaneously minimizes both miscoding and mismodel. Instead, one typically obtains a collection of mutually incomparable hypotheses that define the frontier of optimal trade-offs.

Definition 7.1.2 We say that a hypothesis $(r, d) \in \mathcal{H}$ is *Pareto optimal* if there does not exist another hypothesis $(r', d') \in \mathcal{H}$ such that (r', d') dominates (r, d) . The set of Pareto optimal solutions, denoted by $\mathbf{P}_{\mathcal{H}}$, is called the *Pareto frontier*.

In the context of scientific research, the Pareto frontier represents the set of best-achievable balances between representational and descriptive quality. Along this frontier, one cannot reduce miscoding without worsening mismodel, nor reduce mismodel without worsening miscoding. It therefore provides the natural set of candidate hypotheses from which a decision maker may later select a single preferred solution.

In some applications it may still be convenient to adopt a hypothesis that is not Pareto optimal, for example when one wishes to privilege representational fidelity over descriptive economy, or conversely. Such choices belong to the stage of decision making and scalarization, not to the structural definition of nescience itself.

Range of Solutions

The ideal objective vector is the point in objective space at which all objective functions attain their minimum possible value. For the science problem, this ideal vector is the origin.

Proposition 7.1.1 The ideal objective vector for the science problem is the origin $(0, 0)$.

Proof. By definition, both miscoding and mismodel are non-negative:

$$\mu(r) \geq 0, \quad \nu_r(d) \geq 0.$$

Hence no objective vector can lie below $(0, 0)$.

Moreover, the value 0 is attainable for both objectives in the limiting ideal case. If r^* is an admissible representation and d^* is a minimal description of r^* , then

$$\mu(r^*) = 0, \quad v_{r^*}(d^*) = 0.$$

Therefore the ideal objective vector is $(0, 0)$. ■

A hypothesis $(r, d) \in \mathcal{H}$ is said to be *ideal* if it exhibits zero miscoding and zero mismodel. Intuitively, this means that the representation is already algorithmically equivalent to an admissible representation of the target entity, and that the description is already algorithmically equivalent to a minimal description of that representation. Ideal hypotheses embody the notion of perfect knowledge within the present formulation of the theory.

Unfortunately, in most practical applications the ideal vector is not attainable. The obstruction may lie in the representation, in the description, or in both. Some entities resist faithful representation with finite strings; others admit only highly imperfect or highly non-minimal descriptions within the limits of current scientific practice.

■ **Example 7.1** A natural example is weather prediction. The entity under study is the state of the atmosphere over a geographical region. Any finite representation of that entity is necessarily imperfect, because measurements are noisy, incomplete, and spatially limited. Hence miscoding remains strictly positive. In addition, even highly sophisticated numerical models fail to produce descriptions that are both fully adequate and minimal, because the underlying dynamics are chaotic and the models necessarily rely on approximations and auxiliary structure. Thus mismodel also remains strictly positive. In such a case, the ideal vector $(0, 0)$ is not reachable. ■

Since miscoding and mismodel are measured in bits, their values are non-negative but not naturally bounded above by a universal constant. When a bounded objective region is required for practical purposes, one may instead work with the corresponding relative versions of miscoding and mismodel introduced in Chapters 5 and 6.

Trade-offs

Because the science problem is bi-objective, the global structure of trade-offs is simple: every transition between two hypotheses can be analyzed in terms of how mismodel changes relative to miscoding.

Definition 7.1.3 Let $e \in \mathcal{E}$ be an entity, and $\mathbf{x}_1 = (r_1, d_1)$ and $\mathbf{x}_2 = (r_2, d_2)$ be two hypotheses such that $\mu(e, r_2) \neq \mu(e, r_1)$. We define the *nescience*

trade-off ratio between $\mathbf{x}_1$ and $\mathbf{x}_2$ as

$$\Delta_{v\mu}(\mathbf{x}_1, \mathbf{x}_2) = \frac{v(r_2, d_2) - v(r_1, d_1)}{\mu(e, r_2) - \mu(e, r_1)}.$$

This ratio quantifies the rate at which descriptive error changes relative to representational error when moving from one hypothesis to another. A positive value of $\Delta_{v\mu}$ indicates that miscoding and mismodel move in the same direction, either improving together or deteriorating together. A negative value indicates a genuine trade-off: one objective improves only at the expense of worsening the other.

The structural notion of nescience developed in this section is therefore not a single scalar quantity but a bi-objective characterization of ignorance through the pair

$$(\mu(r), v_r(d)).$$

A single numerical score may still be introduced later by means of a decision maker or a scalarization criterion, but such a score should be understood as a practical selection device built on top of the more fundamental multiobjective formulation given here.

Local Pareto Optimality

The notion of Pareto optimality introduced above is global, in the sense that it compares a hypothesis with all other hypotheses in the search space. In practice, however, scientific inquiry often proceeds incrementally, by modifying a current hypothesis through limited changes in its representation, its description, or both. This motivates a localized version of Pareto optimality.

To formalize this idea, we must first specify what counts as a *small* modification of a hypothesis. In the present framework, we measure the size of a modification in algorithmic terms: a nearby hypothesis is one that can be transformed into the current one, and conversely, by means of short programs.

Definition 7.1.4 Let $(r, d) \in \mathcal{H}$ and let $\varepsilon > 0$. We define the *algorithmic neighborhood* of (r, d) as

$$\mathcal{N}_\varepsilon(r, d) = \left\{ (r', d') \in \mathcal{H} \mid \begin{array}{l} K(r' \mid r) + K(r \mid r') \leq \varepsilon, \\ K(d' \mid d) + K(d \mid d') \leq \varepsilon \end{array} \right\}.$$

This definition captures the idea that two hypotheses are close when both their representations and their descriptions are mutually transformable by short programs. In practical terms, the neighborhood $\mathcal{N}_\varepsilon(r, d)$ includes hypotheses obtained from (r, d) through small modifications such as slight reformulations of

the representation, the addition or removal of a small number of attributes, minor adjustments of parameters, or limited refinements of the description. By contrast, hypotheses that require a substantial redesign of the representation, a radically different descriptive framework, or a major conceptual reorganization lie outside the neighborhood, since the algorithmic information needed to pass from one hypothesis to the other is too large.

Definition 7.1.5 Let $(r, d) \in \mathcal{H}$ and let $\varepsilon > 0$. We say that (r, d) is *locally Pareto optimal* with respect to $\mathcal{N}_\varepsilon(r, d)$ if there does not exist $(r', d') \in \mathcal{N}_\varepsilon(r, d)$ such that (r', d') dominates (r, d) .

A locally Pareto optimal hypothesis is therefore one that cannot be improved through small algorithmic modifications. In this sense, local Pareto optimality expresses a form of *algorithmic stability*: within the region of hypotheses accessible from the current one by bounded transformations, no strictly better alternative exists.

It is important to stress that a local Pareto optimum does not, in general, correspond either to a different representation style of the same entity or to a different entity altogether. The reason is that the notion of neighborhood is purely algorithmic. A dominating hypothesis may lie outside $\mathcal{N}_\varepsilon(r, d)$ simply because it requires a transformation of high algorithmic complexity, even if it concerns the same entity and uses the same representation style. Conversely, two hypotheses concerning different entities might accidentally lie close to each other in algorithmic terms. Thus local Pareto optimality should be interpreted as a property of the geometry of the hypothesis space under bounded transformations, not as a direct classification of entities or representation styles.

The next proposition makes explicit the computational meaning of local Pareto optimality.

Proposition 7.1.2 Let $(r, d) \in \mathcal{H}$ and let $\varepsilon > 0$. The hypothesis (r, d) is locally Pareto optimal with respect to $\mathcal{N}_\varepsilon(r, d)$ if and only if there does not exist any hypothesis (r', d') such that

$$K(r' \mid r) + K(r \mid r') \leq \varepsilon, \quad K(d' \mid d) + K(d \mid d') \leq \varepsilon,$$

and

$$\mu(r') \leq \mu(r), \quad v_{r'}(d') \leq v_r(d),$$

with at least one of these two inequalities being strict.

Proof. This follows immediately by combining the definition of algorithmic neighborhood with the definition of local Pareto optimality. Indeed, (r, d) is locally Pareto optimal if and only if there is no dominating hypothesis inside

$\mathcal{N}_\varepsilon(r, d)$, and belonging to $\mathcal{N}_\varepsilon(r, d)$ is equivalent to satisfying the two algorithmic bounds stated above. ■

Proposition 7.1.2 shows that local Pareto optimality is a statement about the impossibility of reducing nescience through small algorithmic transformations of the current hypothesis. In other words, if a locally Pareto optimal hypothesis is to be improved, the improvement must come from a transformation whose algorithmic complexity exceeds the radius ε of the neighborhood.

■ **Example 7.2** Consider a machine learning task in which a current hypothesis consists of a representation r built from a fixed set of customer attributes and a description d given by a logistic regression model. Small modifications of this hypothesis include adding or removing a small number of features, slightly changing the preprocessing pipeline, or adjusting the form of the model while retaining the same overall descriptive framework. If no such small modification decreases either miscoding or mismodel without worsening the other, then the hypothesis is locally Pareto optimal. Nevertheless, a substantially different hypothesis, for example one based on a new representation built from call transcripts and a transformer-based description, may still dominate it. Such a hypothesis would lie outside the algorithmic neighborhood because the transformation required to reach it is too large. ■

Global and local Pareto optimality are related in the expected way: global optimality is stronger than local optimality.

Proposition 7.1.3 Let $(r, d) \in \mathcal{H}$ and let $\varepsilon > 0$. If (r, d) is Pareto optimal in $\mathcal{H}$, then (r, d) is locally Pareto optimal with respect to $\mathcal{N}_\varepsilon(r, d)$.

Proof. Assume that (r, d) is Pareto optimal in $\mathcal{H}$. Then there does not exist any hypothesis $(\tilde{r}, \tilde{d}) \in \mathcal{H}$ that dominates (r, d) . Since

$$\mathcal{N}_\varepsilon(r, d) \subseteq \mathcal{H},$$

there cannot exist any hypothesis inside $\mathcal{N}_\varepsilon(r, d)$ that dominates (r, d) . Hence (r, d) is locally Pareto optimal with respect to $\mathcal{N}_\varepsilon(r, d)$. ■

The converse fails in general.

Proposition 7.1.4 Local Pareto optimality with respect to $\mathcal{N}_\varepsilon(r, d)$ does not, in general, imply global Pareto optimality in $\mathcal{H}$.

Proof. Consider a hypothesis (r, d) and suppose there exists another hypothesis $(r^\dagger, d^\dagger)$ such that $(r^\dagger, d^\dagger)$ dominates (r, d) , but

$$K(r^\dagger \mid r) + K(r \mid r^\dagger) > \varepsilon \quad \text{or} \quad K(d^\dagger \mid d) + K(d \mid d^\dagger) > \varepsilon.$$

Then $(r^\dagger, d^\dagger) \notin \mathcal{N}_\varepsilon(r, d)$. If, in addition, no hypothesis inside $\mathcal{N}_\varepsilon(r, d)$ dominates (r, d) , then (r, d) is locally Pareto optimal, yet not globally Pareto optimal, since it is dominated by $(r^\dagger, d^\dagger)$ in the full search space. ■

This proposition captures an important feature of scientific inquiry: a hypothesis may resist all sufficiently small improvements and yet fail to be globally optimal. In such cases, progress requires a transformation of higher algorithmic complexity, such as the introduction of new variables, a substantial change of representation, or the adoption of a fundamentally different descriptive framework.

The next proposition records a simple monotonicity property with respect to the neighborhood radius.

Proposition 7.1.5 Let $0 < \varepsilon_1 \leq \varepsilon_2$. If a hypothesis (r, d) is locally Pareto optimal with respect to $\mathcal{N}_{\varepsilon_2}(r, d)$, then it is also locally Pareto optimal with respect to $\mathcal{N}_{\varepsilon_1}(r, d)$.

Proof. Since $\varepsilon_1 \leq \varepsilon_2$, we have

$$\mathcal{N}_{\varepsilon_1}(r, d) \subseteq \mathcal{N}_{\varepsilon_2}(r, d).$$

If no hypothesis in the larger neighborhood $\mathcal{N}_{\varepsilon_2}(r, d)$ dominates (r, d) , then a fortiori no hypothesis in the smaller neighborhood $\mathcal{N}_{\varepsilon_1}(r, d)$ dominates (r, d) . Therefore (r, d) is locally Pareto optimal with respect to $\mathcal{N}_{\varepsilon_1}(r, d)$. ■

Proposition 7.1.5 shows that local Pareto optimality becomes more demanding as the neighborhood radius increases. A hypothesis that is locally Pareto optimal for a large value of ε is stable against a broader class of modifications than one that is locally Pareto optimal only for a small value of ε .

Finally, observe that the notion of local Pareto optimality is primarily useful for understanding the dynamics of scientific search, rather than for defining the ultimate standard of optimality. Global Pareto optimality identifies the truly nondominated hypotheses in the full search space, whereas local Pareto optimality captures the possibility that inquiry may become trapped in regions of the hypothesis space where no small improvement is available. This makes the concept valuable for analyzing scientific stagnation, incremental progress, and the need for discontinuous conceptual shifts.

Perfect Knowledge

According to the theory of nescience, the fundamental aim of scientific research is to reduce our ignorance concerning the entities under investigation. This ignorance is captured by two quantities: the miscoding of the representation

and the mismodel of the description. Perfect knowledge is therefore naturally understood as the limit in which both sources of ignorance vanish, or cannot be further improved without sacrifice.

This viewpoint leads to two complementary notions of perfect knowledge. The first is an ideal, theoretical notion, in which both miscoding and mismodel are exactly zero. The second is a practical notion, based on Pareto optimality, in which no further decrease in one of the two components is possible without increasing the other. The former captures the endpoint of inquiry in an absolute sense; the latter captures the best attainable state within a given symbolic and methodological framework.

Definition 7.1.6 Let $(r, d) \in \mathcal{H}$ be a hypothesis. We say that (r, d) attains *ideal perfect knowledge* if

$$\mu(e, r) = 0 \quad \text{and} \quad v(r, d) = 0.$$

Thus, ideal perfect knowledge is attained when the representation exhibits zero miscoding and the description exhibits zero mismodel. Intuitively, this means that the representation is already optimal with respect to the target entity, and that the description is already optimal with respect to that representation.

Because Kolmogorov-based quantities are defined only up to additive constants, the exact equalities appearing in the previous definition should be regarded as an ideal limiting case. For many theoretical purposes, a more robust notion is obtained by allowing constant-size discrepancies.

Definition 7.1.7 Let $(r, d) \in \mathcal{H}$ be a hypothesis. We say that (r, d) attains *negligible perfect knowledge* if

$$\mu(e, r) = O(1) \quad \text{and} \quad v(r, d) = O(1).$$

Negligible perfect knowledge expresses that both the representational and the descriptive defects are bounded by a constant amount of algorithmic information. This is often the more stable notion from the standpoint of algorithmic information theory.

Proposition 7.1.6 Let $e \in E$ be an entity, let $r \in \bar{R}_e$ be an admissible representation of e , and let $d \in D_r^*$ be a minimal description of r . Then the hypothesis (r, d) attains negligible perfect knowledge.

Proof. Since $r \in \bar{R}_e$, the representation r is admissible for e . Hence its representational defect is negligible, that is,

$$\mu(e, r) = O(1).$$

Likewise, since $d \in D_r^*$, the description d is minimal for the target representation

r . Therefore its descriptive defect is also negligible:

$$v(r, d) = O(1).$$

Consequently,

$$\mu(e, r) = O(1) \quad \text{and} \quad v(r, d) = O(1),$$

so (r, d) attains negligible perfect knowledge. ■

This proposition identifies the structural content of perfect knowledge in the present theory: admissible representations together with minimal descriptions yield vanishing nescience up to the usual constant-size ambiguity.

In practice, ideal perfect knowledge is often unattainable. Even when a hypothesis cannot be made perfect in the strict sense, it may still be impossible to improve it in one component of nescience without worsening the other. This motivates the practical notion of Pareto perfect knowledge.

Definition 7.1.8 Let $(r, d) \in \mathcal{H}$ be a hypothesis. We say that (r, d) attains *Pareto perfect knowledge* if (r, d) is Pareto optimal with respect to the two objectives

$$\mu(e, r) \quad \text{and} \quad v(r, d).$$

Pareto perfect knowledge does not mean that ignorance has been reduced to zero. It means only that no further improvement in miscoding can be achieved without worsening mismodel, and no further improvement in mismodel can be achieved without worsening miscoding. In this sense, Pareto perfect knowledge marks the boundary of what can be improved within the current symbolic framework.

Proposition 7.1.7 Every hypothesis that attains ideal perfect knowledge also attains Pareto perfect knowledge.

Proof. Let (r, d) be a hypothesis such that

$$\mu(e, r) = 0 \quad \text{and} \quad v(r, d) = 0.$$

Since both miscoding and mismodel are nonnegative, there is no other hypothesis $(r', d') \in \mathcal{H}$ such that

$$\mu(e, r') \leq \mu(e, r), \quad v(r', d') \leq v(r, d),$$

with at least one strict inequality. Hence no hypothesis dominates (r, d) , and therefore (r, d) is Pareto optimal. ■

The converse does not hold in general.

Proposition 7.1.8 Pareto perfect knowledge does not, in general, imply ideal perfect knowledge.

Proof. Consider a hypothesis (r, d) lying on the Pareto frontier such that

$$\mu(e, r) > 0 \quad \text{and} \quad v(r, d) > 0.$$

Such a situation occurs whenever any attempt to reduce miscoding necessarily increases mismodel, or any attempt to reduce mismodel necessarily increases miscoding. Then (r, d) is Pareto optimal, hence it attains Pareto perfect knowledge, but it does not attain ideal perfect knowledge because neither objective is zero. ■

This distinction is important. Ideal perfect knowledge corresponds to complete success in both dimensions of nescience, whereas Pareto perfect knowledge corresponds only to the best attainable trade-off. The latter is often the practically relevant notion.

The concept of Pareto perfect knowledge provides a natural stopping criterion for inquiry. When a hypothesis becomes Pareto optimal, further research can no longer improve one of the two sources of ignorance without worsening the other. At that point, continued inquiry is justified only if one is willing to revise the decision rule, alter the representation style, enlarge the class of admissible descriptions, or accept a different trade-off structure.

■ **Example 7.3** In weather prediction, the target entity is the evolving state of the atmosphere over a geographical region. Any finite representation of this entity is necessarily imperfect due to measurement noise, limited spatial resolution, and incomplete coverage. Thus miscoding remains strictly positive. Moreover, even the best available numerical descriptions rely on approximations, discretizations, and auxiliary assumptions, so mismodel also remains strictly positive. In such a case, ideal perfect knowledge is unattainable. Nevertheless, one may still reach Pareto perfect knowledge when no further reduction of miscoding is possible without worsening mismodel, and vice versa. ■

7.2 Scalar Nescience

From a mathematical perspective, any hypothesis in the Pareto frontier is a valid solution to the Science Problem. In this formal sense, the problem is solved once the full set of Pareto optimal hypotheses has been identified. In scientific practice, however, this is often not sufficient. Researchers usually seek a single hypothesis that is most appropriate for the specific goals, constraints, and standards of the investigation.

In multi-objective optimization, a *decision maker* (see Section 21.6) is an entity, either a human agent, a set of explicit criteria, or an algorithm, that selects one hypothesis from the Pareto frontier. The role of the decision maker is to introduce a preference relation on the set of feasible hypotheses, thereby transforming the multi-objective problem into a single-objective selection problem.

Within the present formulation of the theory of nescience, the decision maker evaluates hypotheses through their miscoding and mismodel. This reflects the fact that ignorance now has two primary components: representational error and descriptive error.

Definition 7.2.1 Let $(r, d) \in \mathcal{H}$ be a hypothesis. A *decision maker* for the science problem is a function

$$V : \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}$$

that assigns to each pair $(\mu(e, r), v(r, d))$ the real value

$$V(\mu(e, r), v(r, d)).$$

The function V is often called a *value function* or *utility function*. It encodes the preferences of the scientist or the research community. Depending on the application, V may treat miscoding and mismodel symmetrically, may privilege one of them, or may penalize imbalances between them.

This formalism enables the selection of a single hypothesis from the Pareto frontier. It also makes explicit that the final choice of a hypothesis is not determined by the structural theory alone, but also by the practical aims of the investigation.

A basic requirement for a reasonable decision maker is monotonicity: if one hypothesis is no worse in either objective and strictly better in at least one, then it should receive a strictly better score.

Proposition 7.2.1 Let $V : \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}$ be strictly increasing in each coordinate. If (r^*, d^*) is a global minimizer of

$$V(\mu(e, r^*), v_r(r^*, d^*))$$

over the feasible region $\mathcal{H}$, then (r^*, d^*) is Pareto optimal.

Proof. Assume, for contradiction, that (r^*, d^*) is not Pareto optimal. Then there exists $(r', d') \in \mathcal{H}$ such that

$$\mu(e, r') \leq \mu(e, r^*), \quad v(r', d') \leq v(r^*, d^*),$$

with at least one strict inequality. Since V is strictly increasing in each coordinate, it follows that

$$V(\mu(e, r'), v(r', d')) < V(\mu(e, r^*), v(r^*, d^*)),$$

contradicting the optimality of (r^*, d^*) . Therefore (r^*, d^*) must be Pareto optimal. ■

This proposition is useful because it unifies many practical optimization methods under a single principle: any scalarization that is strictly increasing in both miscoding and mismodel can only select Pareto optimal hypotheses.

Unlike miscoding and mismodel, for which the Tchebycheff criterion provides a natural aggregation rule, the theory of nescience does not prescribe a unique scalarization. This is because representational and descriptive errors correspond to distinct scientific activities, and different research contexts may legitimately prioritize them differently.

Consequently, scalar nescience should be regarded as a decision-making layer built on top of the structural bi-objective formulation. Different decision makers may therefore be appropriate in different applications.

Global Criterion

The global criterion method (see Section 21.6.3) selects a hypothesis by minimizing the distance from its objective vector to the ideal vector. In the present case, the ideal vector is the origin $(0, 0)$.

A natural version of the global criterion is obtained by using the Euclidean distance:

$$\begin{aligned} &\text{minimize} && \sqrt{\mu(e, r)^2 + v(r, d)^2} \\ &\text{subject to} && (r, d) \in \mathcal{H} \end{aligned}$$

This scalarization is geometrically intuitive: it chooses the hypothesis whose pair $(\mu(e, r), v(r, d))$ is closest to perfect knowledge. It treats both objectives symmetrically and penalizes hypotheses that are far from the ideal point in either coordinate.

More generally, one may use an L_p distance to the ideal vector:

$$\begin{aligned} &\text{minimize} && (\mu(e, r)^p + v(r, d)^p)^{1/p} \\ &\text{subject to} && (r, d) \in \mathcal{H}, \quad 1 \leq p < \infty \end{aligned}$$

The Tchebycheff method is a particular case of the global criterion in the limit $p \rightarrow \infty$,

$$\begin{aligned} &\text{minimize} && \max\{\mu(e, r), v(r, d)\} \\ &\text{subject to} && (r, d) \in \mathcal{H}^* \times D. \end{aligned}$$

The case $p = \infty$ is particularly interesting in the theory of nescience, because it directly penalizes imbalance and selects hypotheses that control the worst of the two errors.

■ **Example 7.4** Consider the following three Pareto optimal objective vectors:

$$A = (1, 7), \quad B = (7, 1), \quad C = (4, 4).$$

A Euclidean global criterion selects C , since

$$\sqrt{1^2 + 7^2} = \sqrt{50}, \quad \sqrt{7^2 + 1^2} = \sqrt{50}, \quad \sqrt{4^2 + 4^2} = \sqrt{32}.$$

This illustrates the main feature of the global criterion: it favors hypotheses that achieve a balanced reduction of both miscoding and mismodel, even when the more extreme points are also Pareto optimal. ■

The main drawback of the global criterion is that it does not recover all Pareto optimal hypotheses. In particular, it tends to favor centrally located trade-offs and may exclude hypotheses that are highly desirable because they are exceptionally good in one dimension.

Both miscoding and mismodel are absolute quantities measured in bits, and they need not share the same numerical scale for a given class of problems. Consequently, when a global criterion is used in practice, it is often advisable either to normalize the objectives explicitly or to replace them by their relative versions:

$$\mu^{\text{rel}}(e, r), \quad v^{\text{rel}}(r, d).$$

Weighted Sum Method

The weighting method is the most classical way to incorporate preferences into a bi-objective problem. The idea is to assign a nonnegative weight to each objective and minimize the resulting weighted sum.

$$\begin{aligned} &\text{minimize} && w_\mu \mu(e, r) + w_v v(r, d) \\ &\text{subject to} && (r, d) \in \mathcal{H} \end{aligned}$$

where

$$w_\mu \geq 0, \quad w_v \geq 0, \quad w_\mu + w_v = 1.$$

The following proposition shows that the weighted-sum method satisfies the requirement that whenever positive weights are assigned to both objectives it never select hypotheses that are objectively inferior to other available alternatives. Consequently, every hypothesis selected by this scalarization necessarily belongs to the Pareto frontier.

Proposition 7.2.2 If $w_\mu > 0$ and $w_\nu > 0$, then every global minimizer of the weighted-sum problem is Pareto optimal.

Proof. The scalarization

$$V(x, y) = w_\mu x + w_\nu y$$

is strictly increasing in each coordinate whenever $w_\mu > 0$ and $w_\nu > 0$. The conclusion therefore follows from Proposition 7.2.1. ■

The weighted sum is simple, interpretable, and often convenient computationally. It is especially appropriate when the decision maker has a clear view of the acceptable trade-off between representational and descriptive error.

However, the weights should not be interpreted as absolute measures of importance. A more precise interpretation is that they quantify the rate at which the decision maker is willing to trade one objective against the other. A larger value of w_μ indicates a lower tolerance for miscoding; a larger value of w_ν indicates a lower tolerance for mismodel.

■ **Example 7.5** Suppose a scientist is comparing candidate hypotheses for a newly studied biological mechanism. If the main goal is to obtain a highly faithful representation of the underlying phenomenon before refining the explanatory model, a choice such as

$$w_\mu = 0.8, \quad w_\nu = 0.2$$

expresses a willingness to tolerate a larger mismodel provided the miscoding is significantly reduced. Conversely, if the available representation is already trusted and the main concern is descriptive quality, the scientist may prefer

$$w_\mu = 0.2, \quad w_\nu = 0.8.$$

■

ε -Constraint Method

A particularly useful alternative in practice is the ε -constraint method. Instead of combining the two objectives into a single weighted sum, one objective is minimized while the other is converted into an explicit constraint.

For example, one may minimize mismodel under a prescribed representational tolerance:

$$\begin{aligned} &\text{minimize} && v(r, d) \\ &\text{subject to} && (r, d) \in \mathcal{H} \\ &&& \mu(e, r) \leq \varepsilon_\mu. \end{aligned}$$

Dually, one may minimize miscoding under a descriptive tolerance:

$$\begin{aligned} & \text{minimize} && \mu(e, r) \\ & \text{subject to} && (r, d) \in \mathcal{H}, \\ & && v(r, d) \leq \varepsilon_v. \end{aligned}$$

This method is particularly natural in scientific applications. Very often one of the two dimensions is not simply a preference but a hard admissibility requirement. For example, one may demand that the representation be sufficiently reliable before accepting any description built from it, or one may impose a maximal acceptable mismodel and then search for the least miscoded representation under that requirement.

Proposition 7.2.3 Assume that the feasible set $\mathcal{H}$ is nonempty. Then every global minimizer of $v(r, d)$ over that set is weakly Pareto optimal. If, in addition, the minimizer is unique in the objective space, then it is Pareto optimal.

Proof. If a feasible hypothesis (r^*, d^*) were strictly improved in both coordinates by another feasible hypothesis (r', d') , then in particular

$$v(r', d') < v(r^*, d^*),$$

contradicting the optimality of (r^*, d^*) for the constrained problem. Hence (r^*, d^*) is weakly Pareto optimal.

If the objective vector is unique and there existed a feasible hypothesis that dominated (r^*, d^*) , then it would have to improve one coordinate without worsening the other. Because the constraint already controls μ , such a feasible dominating point would necessarily have

$$v(r', d') \leq v(r^*, d^*),$$

with strict improvement in at least one coordinate, contradicting uniqueness in the objective space. Therefore the solution is Pareto optimal. ■

The main practical advantage of the ε -constraint method is that it expresses scientific standards in a direct and transparent way. Rather than asking how much one objective should be weighted, it asks how much of that objective is acceptable.

Weighted Tchebycheff Method

Another method of particular interest is the *weighted Tchebycheff method*. Given positive weights w_μ, w_v and the ideal vector $(0, 0)$, it solves

$$\begin{aligned} & \text{minimize} && \max\{w_\mu \mu(e, r), w_v v(r, d)\} \\ & \text{subject to} && (r, d) \in \mathcal{H}. \end{aligned}$$

This method differs from the weighted sum in an important way: it focuses on the worst weighted deviation from the ideal point. As a consequence, it strongly discourages highly unbalanced solutions.

The weighted Tchebycheff method is especially appropriate when one wishes to avoid hypotheses that are excellent in one dimension but very poor in the other. It is also useful because, unlike the weighted sum, it is often able to recover Pareto optimal solutions that lie on nonconvex parts of the frontier.

Lexicographic Method

A limiting case of preference-based selection is the *lexicographic method*. Here the decision maker declares one objective to have absolute priority over the other.

For example, if miscoding is given priority, the selection proceeds as follows:

- 1 minimize $\mu(e, r)$ over all feasible hypotheses;
- 2 among the hypotheses attaining the minimum miscoding, minimize $v(r, d)$.

The reverse order is also possible.

The lexicographic method is relevant when the scientific context imposes a strict hierarchy between representation and description. For example, in some experimental settings one may insist that no description is scientifically acceptable unless the representation first satisfies the strongest available reliability criterion.

However, this method is usually too rigid to serve as a default decision rule for the theory of nescience. It can force highly unbalanced choices and may ignore substantial improvements in the secondary objective. It is therefore best understood as a special-purpose selection rule rather than as a general prescription.

7.3 Decreasing Nescience

The pair $(\mu(e, r), v(r, d))$ provides the canonical structural representation of nescience. It captures, in a minimal way, the two fundamental sources of epistemic error: representational defect and descriptive defect. This bi-objective formulation is sufficient to define the science problem, characterize optimality via Pareto dominance, and analyze the global structure of trade-offs.

For the practical task of reducing nescience, however, this representation is too coarse. The quantities $\mu(e, r)$ and $v(r, d)$ are aggregated measures, defined as maxima of more primitive defects, and therefore do not indicate how a hypothesis should be modified to achieve improvement. For instance, a reduction in $\mu(e, r)$ may arise either from removing irrelevant information (surplus) or from incorporating missing relevant structure (deficiency), but its value alone does not distinguish between these cases. The same ambiguity arises for $v(r, d)$, where improvement may require either simplification or enrichment of the description.

To obtain operational guidance, it is advantageous to work at a finer level of resolution by decomposing miscoding and mismodel into their primitive components: deficiency $d_R(e, r)$ and surplus $s_R(e, r)$ of the representation, and deficiency $d_D(r, d)$ and surplus $s_D(r, d)$ of the description. This yields a four-dimensional optimization problem in which each coordinate corresponds to a specific and actionable defect. Each component naturally suggests a corrective operation: augmentation, reduction, enrichment, and simplification.

The full search space $\mathcal{H}$ of hypothesis is too large to be explored directly, and most of its elements are scientifically irrelevant. It is therefore essential to restrict the search without excluding meaningful hypotheses.

Accordingly, we consider the following four-dimensional, entity-constrained science problem:

$$\begin{aligned} &\text{minimize} && \{d_R(e, r), s_R(e, r), d_D(r, d), s_D(r, d)\} \\ &\text{subject to} && (r, d) \in \mathcal{H}, \end{aligned}$$

where minimization is understood in the Pareto sense on $\mathbb{R}_{\geq 0}^4$.

It is convenient, from a notational and analytical point of view, to introduce the following concept of primitive nescience vector.

Definition 7.3.1 Let $(r, d) \in \mathcal{H}$ be an hypothesis. We define the *primitive nescience vector* of (r, d) by

$$\mathbf{n}(e, r, d) = (d_R(e, r), s_R(e, r), d_D(r, d), s_D(r, d)) \in \mathbb{R}_{\geq 0}^4.$$

As next proposition shows, the proposed refinement complements rather than replaces the bi-objective formulation. The pair $(\mu(e, r), \nu(r, d))$ remains the canonical representation of nescience, while the vector $\mathbf{n}(e, r, d)$ provides a finer diagnostic account of the sources of imperfection in the hypothesis.

Proposition 7.3.1 Let (r, d) and (r', d') be two hypotheses. If

$$d_R(r, r') \leq d_R(r, r), s_R(r, r') \leq s_R(r, r), d_D(r', d') \leq d_D(r, d), s_D(r', d') \leq s_D(r, d),$$

and at least one of these inequalities is strict, then

$$\mu(e, r') \leq \mu(e, r), \quad \nu(r', d') \leq \nu(r, d),$$

and at least one of these two inequalities is strict.

Proof. Since the maximum function is monotone in each argument,

$$\mu(e, r') = \max\{d_R(e, r'), s_R(e, r')\} \leq \max\{d_R(e, r), s_R(e, r)\} = \mu(e, r),$$

and similarly

$$\nu(r', d') = \max\{d_D(r', d'), s_D(r', d')\} \leq \max\{d_D(r, d), s_D(r, d)\} = \nu_r(d).$$

If all four primitive inequalities are non-strict and at least one is strict, then at least one of the two maxima must decrease or, at worst, both maxima remain unchanged because the strict decrease occurs in a non-active component. In either case, the pair $(\mu(e, r'), v(r', d'))$ does not worsen, and whenever one strict decrease occurs in a component attaining one of the maxima, the corresponding aggregate objective strictly decreases. In particular, if the four-vector improves componentwise and at least one active component decreases, then the pair improves in the Pareto sense. ■

Proposition 7.3.1 shows that a componentwise improvement of the primitive defects induces a valid reduction of nescience. This justifies using the four-dimensional vector as a practical guide for modifying hypotheses.

To compare a current hypothesis $\mathbf{x} = (r, d)$ with a candidate hypothesis $\mathbf{y} = (r', d')$, it is useful to record how the four primitive defects change. Rather than evaluating hypotheses solely in terms of their absolute nescience, this approach focuses on the direction and magnitude of the transition from $\mathbf{x}$ to $\mathbf{y}$.

Definition 7.3.2 Let $\mathbf{x} = (r, d)$ and $\mathbf{y} = (r', d')$ be two hypotheses. We define the *change vector* from $\mathbf{x}$ to $\mathbf{y}$ as

$$\Delta(\mathbf{x}, \mathbf{y}) = \mathbf{n}(\mathbf{y}) - \mathbf{n}(\mathbf{x}) = (\Delta d_R, \Delta s_R, \Delta d_D, \Delta s_D),$$

where

$$\begin{aligned}\Delta d_R &= d_R(e, r') - d_R(e, r), \\ \Delta s_R &= s_R(e, r') - s_R(e, r), \\ \Delta d_D &= d_D(r', d') - d_D(r, d), \\ \Delta s_D &= s_D(r', d') - s_D(r, d).\end{aligned}$$

Negative entries of the change vector indicate improvement, positive entries deterioration, and zero entries invariance. By tracking how each component evolves, we obtain a more informative description of the effect of a candidate modification, making explicit which aspects of the hypothesis improve, which deteriorate, and which remain unchanged.

Since the four defects may have different magnitudes, a normalized version is useful when comparing candidate hypotheses.

Definition 7.3.3 Let $\mathbf{x} = (r, d)$ and $\mathbf{y} = (r', d')$ be two hypotheses. We define the *normalized variation vector* from $\mathbf{x}$ to $\mathbf{y}$ as

$$\widehat{\Delta}(\mathbf{x}, \mathbf{y}) = (\widehat{\Delta}_{d_R}, \widehat{\Delta}_{s_R}, \widehat{\Delta}_{d_D}, \widehat{\Delta}_{s_D}),$$

where, for each primitive defect $q \in \{d_R, s_R, d_D, s_D\}$,

$$\hat{\Delta}_q(\mathbf{x}, \mathbf{y}) = \begin{cases} \frac{q(\mathbf{y}) - q(\mathbf{x})}{q(\mathbf{x})}, & \text{if } q(\mathbf{x}) > 0, \\ 0, & \text{if } q(\mathbf{x}) = 0 = q(\mathbf{y}), \\ +\infty, & \text{if } q(\mathbf{x}) = 0 < q(\mathbf{y}). \end{cases}$$

Thus, a value of -0.2 means that the corresponding defect has decreased by 20%, whereas a value of 0.1 means that it has increased by 10%. The normalization allows candidates to be compared on a relative scale, rather than only through absolute changes.

Since miscoding and mismodel are defined as maxima of primitive defects, their values are determined by the largest components in each group. It is therefore natural to identify the defects that actively determine the current level of nescience.

Definition 7.3.4 Let $\mathbf{x} = (r, d)$ be a hypothesis. The *active defect set* of $\mathbf{x}$ consists of those primitive defects that attain the maxima defining miscoding and mismodel, namely

$$A(\mathbf{x}) = \arg \max\{d_R(r), s_R(r)\} \cup \arg \max\{d_D(d | r), s_D(d | r)\}.$$

The active defect set identifies those primitive quantities that currently determine miscoding and mismodel. From a practical standpoint, these are the most urgent defects to attack, since reducing a non-active defect may leave the aggregate nescience unchanged.

Proposition 7.3.2 Let $\mathbf{x} = (r, d)$ be a hypothesis with primitive nescience vector

$$\mathbf{n}(\mathbf{x}) = (d_R(e, r), s_R(e, r), d_D(r, d), s_D(r, d)).$$

Then the active defect set $A(\mathbf{x})$ consists exactly of those primitive defects that determine the values of miscoding and mismodel, that is,

$$\mu(r) = \max\{d_R(e, r), s_R(e, r)\} \quad \text{and} \quad \nu(r, d) = \max\{d_D(r, d), s_D(r, d)\}$$

are attained precisely by the elements of $A(\mathbf{x})$.

Proof. By definition,

$$A(\mathbf{x}) = \arg \max\{d_R(e, r), s_R(e, r)\} \cup \arg \max\{d_D(r, d), s_D(r, d)\}.$$

Hence, the elements of $\arg \max\{d_R(e, r), s_R(e, r)\}$ are exactly those primitive defects whose value equals $\mu(e, r)$, and the elements of $\arg \max\{d_D(r, d), s_D(r, d)\}$ are exactly those whose value equals $\nu_r(d)$.

Therefore, every element of $A(\mathbf{x})$ attains either $\mu(e, r)$ or $v(r, d)$, and conversely, every primitive defect that attains one of these maxima belongs to $A(\mathbf{x})$. This proves that $A(\mathbf{x})$ coincides with the set of primitive defects that determine miscoding and mismodel. ■

The notion of active defect set is particularly relevant because it identifies the specific components that determine the current value of nescience. Since miscoding and mismodel are defined as maxima, any improvement that does not reduce at least one active defect cannot decrease the overall level of ignorance. In this sense, the active defect set provides a principled criterion for prioritizing transformations of a hypothesis: it indicates where effort is most likely to have a direct impact. In much of current scientific practice, however, improvements are often pursued without an explicit identification of these controlling components. Efforts may focus, for instance, on refining models when the dominant limitation lies in the representation, or on collecting additional data when the main issue is the presence of irrelevant or redundant attributes. While such approaches can still yield progress, they may do so inefficiently, as they do not directly target the defects that govern the current level of nescience. Making the active defect set explicit helps to align methodological choices with the structure of the problem, thereby promoting more effective and systematic progress.

Corrective Operations

The four primitive defects naturally suggest four corresponding modes of intervention on a hypothesis. At a high level, these operations provide a structured way to modify either the representation or the description in order to reduce nescience. Augmentation targets missing relevant information in the representation, for instance by incorporating additional data or variables. Reduction removes irrelevant or redundant components of the representation, such as unnecessary features or noisy attributes. Enrichment improves the description by adding missing explanatory structure, while simplification removes superfluous elements from the description, leading to more concise and efficient models. Each operation generates a set of candidate hypotheses that explore a particular direction of improvement.

These operations should be understood as idealized directions rather than rigid transformations. In practice, modifying a hypothesis typically affects several defects at once: reducing one component may slightly increase another. Nevertheless, it is useful to conceptually isolate the intended effect of each operation. This allows us to reason about their impact in a clean and systematic way. In particular, when a given defect is active, that is, when it determines the current value of miscoding or mismodel, applying the corresponding operation is the most direct way to reduce the associated aggregate error.

This observation highlights an important principle for guiding scientific practice. The primitive defects are not merely descriptive quantities; they have an operational interpretation. They indicate not only what is wrong with a hypothesis, but also how it should be modified. When an operation successfully targets an active defect, it leads to a strict improvement in the corresponding aggregate measure of nescience. This provides a clear connection between diagnosis and action, and forms the basis for systematic procedures aimed at reducing ignorance.

The four corrective operations generate candidate hypotheses. To decide which one should be accepted, we first identify those candidates that actually improve the primitive nescience vector.

Definition 7.3.5 Let $\mathbf{x} = (r, d)$ be a hypothesis and let $\mathbf{y} = (r', d')$ be a candidate hypothesis. We say that $\mathbf{y}$ is a *four-dimensional descent candidate* for $\mathbf{x}$ if

$$\hat{\Delta}_{d_R}(\mathbf{x}, \mathbf{y}) \leq 0, \quad \hat{\Delta}_{s_R}(\mathbf{x}, \mathbf{y}) \leq 0, \quad \hat{\Delta}_{d_D}(\mathbf{x}, \mathbf{y}) \leq 0, \quad \hat{\Delta}_{s_D}(\mathbf{x}, \mathbf{y}) \leq 0,$$

and at least one of these four inequalities is strict.

Thus, a four-dimensional descent candidate does not worsen any primitive defect and improves at least one. By Proposition 7.3.1, every such candidate induces a valid reduction of nescience.

This notion coincides with Pareto dominance in the four-dimensional space of primitive defects, expressed in terms of the normalized change vector. The formulation in terms of changes is nevertheless useful, as it emphasizes the directional character of improvement and provides a natural basis for iterative descent procedures.

To select one descent candidate among several, we use the active defects to define a score.

Definition 7.3.6 Let $\mathbf{x}$ be a hypothesis and let $\mathbf{y}$ be a four-dimensional descent candidate for $\mathbf{x}$. We define the *active normalized descent score* of $\mathbf{y}$ relative to $\mathbf{x}$ is

$$\Sigma(\mathbf{x}, \mathbf{y}) = - \sum_{i \in A(\mathbf{x})} \hat{\Delta}_i(\mathbf{x}, \mathbf{y}).$$

Since the active components are those currently responsible for miscoding and mismodel, the score Σ measures the total normalized reduction achieved on the bottlenecks of the current hypothesis. Larger values of Σ correspond to stronger relative improvements on the defects that matter most at that stage.

Proposition 7.3.3 If $\mathbf{y}$ is a four-dimensional descent candidate for $\mathbf{x}$, then

$$\Sigma(\mathbf{x}, \mathbf{y}) \geq 0.$$

Moreover, if at least one active defect strictly decreases, then

$$\Sigma(\mathbf{x}, \mathbf{y}) > 0.$$

Proof. For every four-dimensional descent candidate, all components of $\hat{\Delta}(\mathbf{x}, \mathbf{y})$ are non-positive. Therefore, for each $i \in A(\mathbf{x})$,

$$-\hat{\Delta}_i(\mathbf{x}, \mathbf{y}) \geq 0.$$

Summing over the active set gives

$$\Sigma(\mathbf{x}, \mathbf{y}) \geq 0.$$

If at least one active component is strictly negative, then at least one term in the sum is strictly positive, hence $\Sigma(\mathbf{x}, \mathbf{y}) > 0$. ■

We now combine the previous ingredients into a formal descent scheme.

Definition 7.3.7 Let $\mathbf{x} = (r, d)$ be a hypothesis. We define the *operation-generated candidate set* of $\mathbf{x}$ by

$$\mathcal{C}(\mathbf{x}) = \mathcal{A}(\mathbf{x}) \cup \mathcal{R}(\mathbf{x}) \cup \mathcal{E}(\mathbf{x}) \cup \mathcal{S}(\mathbf{x}).$$

The corresponding *descent set* is

$$\mathcal{C}^\downarrow(\mathbf{x}) = \{\mathbf{y} \in \mathcal{C}(\mathbf{x}) : \mathbf{y} \text{ is a four-dimensional descent candidate for } \mathbf{x}\}.$$

Defect-Guided Descent Algorithm Given an initial hypothesis $\mathbf{x}_0 = (r_0, d_0)$, define recursively:

- 1 Compute the primitive nescience vector $\mathbf{n}(\mathbf{x}_t)$ and the active defect set $A(\mathbf{x}_t)$.
- 2 Generate the candidate set $\mathcal{C}(\mathbf{x}_t)$ by applying augmentation, reduction, enrichment, and simplification to $\mathbf{x}_t$.
- 3 Form the descent set $\mathcal{C}^\downarrow(\mathbf{x}_t)$.
- 4 If $\mathcal{C}^\downarrow(\mathbf{x}_t) = \emptyset$, stop.
- 5 Otherwise, choose

$$\mathbf{x}_{t+1} \in \arg \max_{\mathbf{y} \in \mathcal{C}^\downarrow(\mathbf{x}_t)} \Sigma(\mathbf{x}_t, \mathbf{y}).$$

The algorithm is *defect-guided* because the score Σ depends on the active defects of the current hypothesis. It is therefore not a blind search through the space of hypotheses, but an iterative attempt to reduce the currently dominant sources of nescience.

Proposition 7.3.4 Let $(\mathbf{x}_t)_{t \geq 0}$ be a sequence produced by the defect-guided descent algorithm. Then, for every t such that $\mathbf{x}_{t+1}$ is defined,

$$d_R(\mathbf{x}_{t+1}) \leq d_R(\mathbf{x}_t), s_R(\mathbf{x}_{t+1}) \leq s_R(\mathbf{x}_t), d_D(\mathbf{x}_{t+1}) \leq d_D(\mathbf{x}_t), s_D(\mathbf{x}_{t+1}) \leq s_D(\mathbf{x}_t),$$

with at least one strict inequality. Consequently,

$$\mu(\mathbf{x}_{t+1}) \leq \mu(\mathbf{x}_t), \quad v(\mathbf{x}_{t+1}) \leq v(\mathbf{x}_t),$$

with at least one strict inequality on the active aggregates whenever an active defect decreases.

Proof. By construction,

$$\mathbf{x}_{t+1} \in \mathcal{C}^\downarrow(\mathbf{x}_t).$$

Hence $\mathbf{x}_{t+1}$ is a four-dimensional descent candidate for $\mathbf{x}_t$, which means precisely that all four primitive defects do not increase and that at least one of them strictly decreases. The conclusion for μ and v follows from Proposition 7.3.1. ■

The algorithm therefore produces a monotone sequence of improvements in the primitive defects, and, as a consequence, a monotone descent of nescience in the aggregate sense whenever the improved components are active.

Definition 7.3.8 A hypothesis $\mathbf{x}$ is said to be *operation-stationary* if

$$\mathcal{C}^\downarrow(\mathbf{x}) = \emptyset.$$

Proposition 7.3.5 The defect-guided descent algorithm halts at a hypothesis $\mathbf{x}_t$ if and only if $\mathbf{x}_t$ is operation-stationary.

Proof. The algorithm halts exactly when, at Step 4, the descent set $\mathcal{C}^\downarrow(\mathbf{x}_t)$ is empty. This is precisely the definition of operation-stationarity. ■

Operation-stationarity does not imply global Pareto optimality, and in general it does not even imply local Pareto optimality with respect to the algorithmic neighborhoods introduced earlier. What it means is only that, among the candidates generated by the four corrective operations, none is a four-dimensional descent candidate. In practice, this situation indicates either that the current hypothesis is already very good, or that further progress requires a broader repertoire of transformations than those currently available.

Pruning by active defects exploits the fact that miscoding and mismodel are aggregated through maxima:

$$\mu(r) = \max\{d_R(r), s_R(r)\}, \quad v_r(d) = \max\{d_D(d \mid r), s_D(d \mid r)\}.$$

Only the active components determine the current values of these two aggregate objectives.

Definition 7.3.9 Let (r, d) be a hypothesis. We define the *active representational defect set* by

$$A_R(r) = \{q \in \{d_R(r), s_R(r)\} : q = \mu(r)\},$$

and the *active descriptive defect set* by

$$A_D(r, d) = \{q \in \{d_D(d \mid r), s_D(d \mid r)\} : q = v_r(d)\}.$$

Proposition 7.3.6 Let (r, d) and (r', d') be two hypotheses. Assume that every active representational defect of (r, d) remains unchanged in (r', d') , and likewise every active descriptive defect remains unchanged. Then

$$\mu(r') = \mu(r), \quad v_{r'}(d') = v_r(d),$$

provided that no inactive defect increases above the current active value.

Proof. If the active representational defect(s) remain unchanged and no inactive representational defect exceeds the current value of $\mu(r)$, then the maximum of $\{d_R(r'), s_R(r')\}$ remains equal to $\mu(r)$. Hence $\mu(r') = \mu(r)$. The same argument applies to the descriptive side, giving $v_{r'}(d') = v_r(d)$. ■

Proposition 7.3.6 yields a useful pruning principle: candidates that affect only inactive defects, while leaving all active defects unchanged, need not be explored if the goal is to reduce current nescience.

A very common reduction is to impose explicit complexity or length bounds.

Definition 7.3.10 Let $L_R, L_D \in \mathbb{N}$. We define the *bounded search region*

$$\mathcal{H}_{L_R, L_D} = \{(r, d) \in \mathcal{B}^* \times \mathcal{D} : \ell(r) \leq L_R, \ell(d) \leq L_D\}.$$

Proposition 7.3.7 If the set of descriptions of length at most L_D is finite, then $\mathcal{H}_{L_R, L_D}$ is finite.

Proof. There are only finitely many binary strings of length at most L_R . By assumption, there are only finitely many descriptions of length at most L_D . Hence their Cartesian product is finite. ■

Bounded search regions are useful because they transform the search problem into a finite one. Their main drawback is that they may exclude highly informative hypotheses of greater complexity. For this reason, they are best viewed as practical truncations rather than as part of the foundational definition of nescience.

The Parametric Case

The previous framework becomes more structured when descriptions are restricted to a parametric family of models.

Definition 7.3.11 Let Θ be a parameter set and let

$$\mathcal{D}_\Theta = \{d_\theta : \theta \in \Theta\} \subseteq \mathcal{D}$$

be a parametric family of descriptions. For a fixed entity $e \in \mathcal{E}$, the corresponding *parametric feasible region* is

$$\mathcal{H}_{e,\Theta} = \mathcal{R}_e \times \mathcal{D}_\Theta.$$

For a fixed representation $r \in \mathcal{R}_e$, the primitive nescience vector becomes

$$\mathbf{n}(r, \theta) = (d_R(r), s_R(r), d_D(d_\theta | r), s_D(d_\theta | r)).$$

Thus, the representation-side quantities depend only on r , whereas the description-side quantities depend on the parameter θ .

Definition 7.3.12 Let $r \in \mathcal{R}_e$ be fixed. A *parametric enrichment step* is a transition

$$\theta \mapsto \theta'$$

such that

$$d_D(d_{\theta'} | r) < d_D(d_\theta | r).$$

A *parametric simplification step* is a transition

$$\theta \mapsto \theta'$$

such that

$$s_D(d_{\theta'} | r) < s_D(d_\theta | r).$$

In a parametric family, enrichment and simplification become operations on the parameter space. This makes the descriptonal part of the search substantially more structured. It also leads naturally to an alternating strategy: for a given

representation, improve the parameters of the model family; then, if needed, modify the representation itself through augmentation or reduction.

Proposition 7.3.8 Let $r \in \mathcal{R}_e$ be fixed and let $\theta, \theta' \in \Theta$. If

$$d_D(d_{\theta'} | r) \leq d_D(d_\theta | r), \quad s_D(d_{\theta'} | r) \leq s_D(d_\theta | r),$$

with at least one strict inequality, then

$$v_r(d_{\theta'}) \leq v_r(d_\theta),$$

with strict inequality whenever the strict decrease occurs in an active description-side defect.

Proof. Since r is fixed, the representation-side quantities are unchanged. On the description side,

$$v_r(d_{\theta'}) = \max\{d_D(d_{\theta'} | r), s_D(d_{\theta'} | r)\} \leq \max\{d_D(d_\theta | r), s_D(d_\theta | r)\} = v_r(d_\theta),$$

by monotonicity of the maximum function. If the strict decrease occurs in a component attaining the maximum, then the inequality is strict. ■

Proposition 7.3.8 shows that, once the representation is fixed, the descriptive part of nescience reduction can be treated as a structured multiobjective problem over the parameter space. The general defect-guided descent algorithm therefore specializes naturally to the parametric setting:

- augmentation and reduction modify the representation r ;
- enrichment and simplification act on the parameter θ .

This yields a practically important architecture for future implementations. The abstract theory of nescience remains defined over representations and descriptions, but when descriptions belong to a parametric family, the descriptive part of the search acquires a concrete internal structure. As a result, the generic corrective operations of the theory become directly interpretable as model-space moves, parameter-space moves, or alternating combinations of both.

7.4 Unknown Unknowns

In Chapter 1, we introduced the idea of the *unknown unknown* region: the region of inquiry containing entities for which we not only lack satisfactory explanations, but for which no scientific hypothesis has yet been formulated. Within the theory of nescience, one of the central objectives is to develop procedures capable of identifying and exploring entities hidden in this region (see Chapter ??).

To formalize this idea, we classify entities according to the current state of our hypotheses about them. In the present bi-objective formulation of nescience, what

matters is no longer the triplet formed by miscoding, inaccuracy, and surfeit, but the pair formed by *miscoding* and *mismodel*. Accordingly, the relevant distinction is whether an entity is represented in our current stock of hypotheses, and, if so, whether some known hypothesis already brings both components of nescience down to a negligible level.

Known and Unknown Entities

We begin by separating the entities about which at least one hypothesis is currently known from those about which no hypothesis is known.

Definition 7.4.1 Let $e \in E$ be an entity, and let $\hat{H}_e$ denote the collection of known hypotheses about e . We say that e is a *known entity* if

$$\hat{H}_e \neq \emptyset.$$

We denote by E_K the set of all known entities:

$$E_K = \{e \in E \mid \hat{H}_e \neq \emptyset\}.$$

Definition 7.4.2 Let $e \in E$ be an entity, and let $\hat{H}_e$ denote the collection of known hypotheses about e . We say that e is an *unknown unknown* if

$$\hat{H}_e = \emptyset.$$

We denote by E_{UU} the set of all unknown unknowns:

$$E_{UU} = \{e \in E \mid \hat{H}_e = \emptyset\}.$$

Thus, an unknown unknown is simply an entity for which no representation-description pair is currently known. Such an entity may be knowable or unknowable in principle; the present classification does not distinguish between these possibilities, since that distinction is not operationally available at the level of current scientific practice.

The sets E_K and E_{UU} form a partition of the entity space.

Proposition 7.4.1 We have

$$E = E_K \cup E_{UU} \quad \text{and} \quad E_K \cap E_{UU} = \emptyset.$$

Proof. Let $e \in E$ be arbitrary. Either $\hat{H}_e = \emptyset$ or $\hat{H}_e \neq \emptyset$. In the first case, $e \in E_{UU}$ by definition. In the second case, $e \in E_K$ by definition. Therefore every entity belongs to $E_K \cup E_{UU}$.

Moreover, no entity can satisfy simultaneously $\hat{H}_e = \emptyset$ and $\hat{H}_e \neq \emptyset$. Hence $E_K \cap E_{UU} = \emptyset$. ■

Known Knowns and Known Unknowns

Among the entities that are already known in the weak sense that at least one hypothesis has been formulated, it is useful to distinguish those for which some known hypothesis already attains negligible perfect knowledge from those for which all currently known hypotheses still exhibit a non-negligible degree of nescience.

Definition 7.4.3 Let $e \in E$ be an entity. We say that e is a *known known* if there exists a known hypothesis $(r, d) \in \hat{H}_e$ such that

$$\mu(r) = O(1) \quad \text{and} \quad \nu_r(d) = O(1).$$

We denote by E_{KK} the set of all known knowns:

$$E_{KK} = \{e \in E \mid \exists (r, d) \in \hat{H}_e \text{ such that } \mu(r) = O(1) \text{ and } \nu_r(d) = O(1)\}.$$

Definition 7.4.4 Let $e \in E$ be an entity. We say that e is a *known unknown* if

$$\hat{H}_e \neq \emptyset$$

but no known hypothesis about e attains negligible perfect knowledge. Equivalently,

$$e \in E_{KU} \iff \hat{H}_e \neq \emptyset \text{ and } \forall (r, d) \in \hat{H}_e, \mu(r) \neq O(1) \text{ or } \nu_r(d) \neq O(1).$$

We denote by E_{KU} the set of all known unknowns.

This classification is intentionally epistemic rather than metaphysical. An entity is called a known known not because we have access to its essence independently of representation, but because within our currently available hypotheses we already possess at least one whose miscoding and mismodel are negligible. Conversely, a known unknown is an entity that has entered scientific awareness, yet for which every currently available hypothesis remains imperfect in a non-negligible way.

The sets of known knowns, known unknowns, and unknown unknowns form a trichotomy.

Proposition 7.4.2 We have

$$E = E_{KK} \cup E_{KU} \cup E_{UU},$$

and these three sets are pairwise disjoint:

$$E_{KK} \cap E_{KU} = E_{KK} \cap E_{UU} = E_{KU} \cap E_{UU} = \emptyset.$$

Proof. Let $e \in E$ be arbitrary.

If $\hat{H}_e = \emptyset$, then $e \in E_{UU}$ by definition.

If $\hat{H}_e \neq \emptyset$, then either there exists a hypothesis $(r, d) \in \hat{H}_e$ such that

$$\mu(r) = O(1) \quad \text{and} \quad \nu_r(d) = O(1),$$

in which case $e \in E_{KK}$, or no such hypothesis exists, in which case $e \in E_{KU}$.

Therefore every entity belongs to at least one of the three sets.

To prove pairwise disjointness, first observe that E_{UU} is disjoint from both E_{KK} and E_{KU} , since membership in E_{UU} requires $\hat{H}_e = \emptyset$, whereas membership in either E_{KK} or E_{KU} requires $\hat{H}_e \neq \emptyset$.

Now suppose, for contradiction, that $e \in E_{KK} \cap E_{KU}$. Then there exists a hypothesis $(r, d) \in \hat{H}_e$ with

$$\mu(r) = O(1) \quad \text{and} \quad \nu_r(d) = O(1),$$

by membership in E_{KK} , while by membership in E_{KU} no such hypothesis may exist. This is a contradiction. Hence $E_{KK} \cap E_{KU} = \emptyset$. ■

The previous proposition gives a clean epistemic decomposition of the entity space. Unknown unknowns are entities outside present scientific awareness; known unknowns are entities already recognized but still imperfectly captured; known knowns are entities for which some currently available hypothesis already reaches negligible perfect knowledge.

Ideal Known Knowns

It is sometimes useful to isolate the stronger notion corresponding to exact, rather than merely negligible, perfect knowledge.

Definition 7.4.5 We define the set of *ideal known knowns* by

$$E_{IKK} = \{e \in E \mid \exists (r, d) \in \hat{H}_e \text{ such that } \mu(r) = 0 \text{ and } \nu_r(d) = 0\}.$$

Proposition 7.4.3 We have

$$E_{IKK} \subseteq E_{KK}.$$

Proof. If $e \in E_{IKK}$, then there exists $(r, d) \in \hat{H}_e$ such that

$$\mu(r) = 0 \quad \text{and} \quad \nu_r(d) = 0.$$

Since $0 = O(1)$, it follows that the same hypothesis also satisfies

$$\mu(r) = O(1) \quad \text{and} \quad \nu_r(d) = O(1).$$

Therefore $e \in E_{KK}$. ■

This proposition shows that ideal known knowns form a distinguished subclass of the known knowns. In most practical contexts, however, the negligible notion is the more robust one, because Kolmogorov-based quantities are defined only up to additive constants.

Knowledge Frontier

The most relevant boundary for ongoing research is not the boundary between the known and the unknown in the mere sense of hypothesis existence, but the boundary between what is already captured up to negligible perfect knowledge and what is not. This motivates the following notion.

Definition 7.4.6 The *knowledge frontier* is the set

$$\mathcal{F} = E_{KU}.$$

Thus, the knowledge frontier consists of those entities that have already entered scientific inquiry, but for which no currently known hypothesis attains negligible perfect knowledge. These entities form the natural region in which further improvement of our hypotheses is still epistemically meaningful.

This definition is simpler and more operational than the earlier anti-Pareto construction. It captures the intuitive idea that the frontier is made of entities that are already visible to science, but not yet adequately understood.

New Research Entities

A *new research entity* is an entity that belongs to the unknown unknown region and becomes the subject of scientific inquiry once a first hypothesis about it is formulated.

Definition 7.4.7 Let t denote time. We say that an entity $e \in E$ is a *new research entity at time $t + 1$* if

$$e \in E_{UU}^{(t)} \quad \text{and} \quad e \in E_K^{(t+1)}.$$

In other words, a new research entity is an entity that was previously an unknown unknown and has just crossed into the known region by acquiring its first known hypothesis.

The next proposition formalizes the obvious but important relation between new research entities and unknown unknowns.

Proposition 7.4.4 If e is a new research entity at time $t + 1$, then

$$e \in E_{UU}^{(t)}.$$

Proof. This is immediate from the definition of new research entity at time $t + 1$. ■

Temporal Monotonicity

If the set of known hypotheses grows monotonically over time, then the region of unknown unknowns can only shrink.

Proposition 7.4.5 Assume that for every entity $e \in E$ and every $t \in \mathbb{N}$,

$$\hat{H}_e^{(t)} \subseteq \hat{H}_e^{(t+1)}.$$

Then

$$E_{UU}^{(t+1)} \subseteq E_{UU}^{(t)} \quad \text{for all } t \in \mathbb{N}.$$

Proof. Let $e \in E_{UU}^{(t+1)}$. Then

$$\hat{H}_e^{(t+1)} = \emptyset.$$

Since $\hat{H}_e^{(t)} \subseteq \hat{H}_e^{(t+1)}$, it follows that

$$\hat{H}_e^{(t)} = \emptyset.$$

Therefore $e \in E_{UU}^{(t)}$, and hence

$$E_{UU}^{(t+1)} \subseteq E_{UU}^{(t)}.$$

■

Under the same monotonicity assumption, the set of known knowns can only grow.

Proposition 7.4.6 Assume that for every entity $e \in E$ and every $t \in \mathbb{N}$,

$$\hat{H}_e^{(t)} \subseteq \hat{H}_e^{(t+1)}.$$

Then

$$E_{KK}^{(t)} \subseteq E_{KK}^{(t+1)} \quad \text{for all } t \in \mathbb{N}.$$

Proof. Let $e \in E_{KK}^{(t)}$. Then there exists $(r, d) \in \hat{H}_e^{(t)}$ such that

$$\mu(r) = O(1) \quad \text{and} \quad \nu_r(d) = O(1).$$

Since $\hat{H}_e^{(t)} \subseteq \hat{H}_e^{(t+1)}$, the same hypothesis also belongs to $\hat{H}_e^{(t+1)}$. Hence $e \in E_{KK}^{(t+1)}$. ■

These two monotonicity properties express the most basic dynamics of scientific discovery: entities move out of the unknown unknown region when a first hypothesis is formulated, and they move out of the knowledge frontier when a sufficiently good hypothesis is found.

The unknown unknown region plays a foundational role in the theory of nescience. By definition, it contains those entities for which no hypothesis has yet been proposed. Scientific innovation begins precisely when an entity leaves this region and becomes the object of representation and description. In this sense, the reduction of the unknown unknown region is one of the clearest formal signatures of epistemic expansion.

Scientific knowledge does not consist merely of isolated entities. Entities are typically connected through relations that encode causal interactions, structural dependencies, logical implications, or other forms of scientific linkage. These relations induce a natural structure over the domain of entities and allow us to formalize the notions of knowledge frontier and unknown unknowns.

Definition 7.4.8 Let $\mathcal{E}$ be a set of entities. A *knowledge state* is a subset

$$\mathcal{K} \subseteq \mathcal{E}$$

containing the entities that have been identified within a given domain of inquiry.

The complement

$$\mathcal{U} = \mathcal{E} \setminus \mathcal{K}$$

represents the set of entities that remain unknown.

Suppose that entities are connected through a relation

$$\mathcal{R} \subseteq \mathcal{E} \times \mathcal{E}.$$

The relation $\mathcal{R}$ may represent any form of scientific connection, such as causal influence, interaction, structural dependence, or logical implication.

Relations induce paths that connect entities.

Definition 7.4.9 Let $e_1, e_2 \in \mathcal{E}$. We say that e_2 is *reachable* from e_1 , and write

$$e_1 \rightsquigarrow e_2,$$

if there exists a finite sequence of entities

$$e_1 = x_0, x_1, \dots, x_n = e_2$$

such that

$$(x_i, x_{i+1}) \in \mathcal{R} \quad \text{for all } i \in \{0, \dots, n-1\}.$$

Reachability captures the idea that one entity may be connected to another through a chain of scientific relationships.

The structure induced by $\mathcal{R}$ allows us to define the boundary between known and unknown entities.

Definition 7.4.10 Let $\mathcal{K}$ be a knowledge state. The *knowledge frontier* associated with $\mathcal{K}$ is the set

$$\partial\mathcal{K} = \{e \in \mathcal{K} : \exists u \in \mathcal{E} \setminus \mathcal{K} \text{ such that } (e, u) \in \mathcal{R}\}.$$

Entities in the frontier are known entities that are directly connected to unknown ones. They therefore represent points where new discoveries are most likely to emerge.

Not all unknown entities lie near the frontier.

Definition 7.4.11 Let $\mathcal{K}$ be a knowledge state. An entity

$$u \in \mathcal{E} \setminus \mathcal{K}$$

is called an *unknown unknown* if it is not reachable from any known entity, that is, if

$$\forall k \in \mathcal{K}, \quad k \not\rightsquigarrow u.$$

Such entities lie outside the structural region currently accessible through known relations. Their existence cannot be inferred from the current knowledge structure.

The set of entities that can be reached from current knowledge plays an important role.

Definition 7.4.12 Let $\mathcal{K} \subseteq \mathcal{E}$ be a knowledge state. The *reachable region* generated by $\mathcal{K}$ is

$$\text{Reach}(\mathcal{K}) = \{e \in \mathcal{E} : \exists k \in \mathcal{K} \text{ such that } k \rightsquigarrow e\}.$$

The reachable region contains all entities that are structurally accessible from current knowledge by following chains of known relations. By definition, every unknown unknown lies outside this region.

Proposition 7.4.7 For every knowledge state $\mathcal{K} \subseteq \mathcal{E}$,

$$\mathcal{K} \subseteq \text{Reach}(\mathcal{K}).$$

Proof. Let $k \in \mathcal{K}$. Then k is reachable from itself by the trivial path of length

zero. Hence $k \in \text{Reach}(\mathcal{K})$. ■

Proposition 7.4.8 An entity $u \in \mathcal{E} \setminus \mathcal{K}$ is an unknown unknown if and only if

$$u \notin \text{Reach}(\mathcal{K}).$$

Proof. By definition, u is an unknown unknown precisely when there does not exist any $k \in \mathcal{K}$ such that $k \rightsquigarrow u$. This is equivalent to saying that u does not belong to $\text{Reach}(\mathcal{K})$. ■

The previous definitions make it possible to state a simple sufficient condition for the existence of unknown unknowns.

Theorem 7.4.9 Let $\mathcal{E}$ be a set of entities, let

$$\mathcal{R} \subseteq \mathcal{E} \times \mathcal{E}$$

be a relation, and let $\mathcal{K} \subseteq \mathcal{E}$ be a non-empty knowledge state. If

$$\text{Reach}(\mathcal{K}) \neq \mathcal{E},$$

then there exists at least one unknown unknown.

Proof. Assume that

$$\text{Reach}(\mathcal{K}) \neq \mathcal{E}.$$

Then there exists an entity

$$u \in \mathcal{E} \setminus \text{Reach}(\mathcal{K}).$$

Since $\mathcal{K} \subseteq \text{Reach}(\mathcal{K})$, it follows that $u \notin \mathcal{K}$. Therefore

$$u \in \mathcal{E} \setminus \mathcal{K}.$$

Moreover, because $u \notin \text{Reach}(\mathcal{K})$, there is no $k \in \mathcal{K}$ such that $k \rightsquigarrow u$. Hence u is an unknown unknown. ■

The theorem shows that unknown unknowns arise whenever the current knowledge state fails to reach the whole domain through the available relational structure. In other words, even if some entities are known, the graph induced by the relation may still contain regions that remain completely inaccessible from present knowledge.

The previous theorem can be specialized to a simple graph-theoretic condition.

Corollary 7.4.10 Let $\mathcal{K} \subseteq \mathcal{E}$ be a non-empty knowledge state. If there exists an entity

$$u \in \mathcal{E} \setminus \mathcal{K}$$

lying in a connected component of the graph $(\mathcal{E}, \mathcal{R})$ that contains no element of $\mathcal{K}$, then u is an unknown unknown.

Proof. If the connected component of u contains no known entity, then there exists no path from any $k \in \mathcal{K}$ to u . Hence $u \notin \text{Reach}(\mathcal{K})$, and by the previous proposition u is an unknown unknown. ■

The relation $\mathcal{R}$ induces a graph

$$G = (\mathcal{E}, \mathcal{R})$$

whose vertices represent entities and whose edges represent relations between them. In this graph:

- the known entities form a subgraph induced by $\mathcal{K}$,
- the frontier corresponds to nodes adjacent to unknown regions,
- the reachable region contains the entities that can in principle be approached from current knowledge by following relational paths,
- unknown unknowns correspond to nodes that lie outside this reachable region.

This perspective provides a structural explanation of how scientific knowledge expands. Discovery typically proceeds along the frontier, where known entities reveal direct links to previously unknown ones. However, the theorem above shows that some unknown entities may remain entirely detached from the current knowledge state. Such entities cannot be reached by local extensions of existing knowledge alone; they require the introduction of new relations, new methods, or new entry points into the domain.

In this sense, unknown unknowns are not merely entities that are currently unknown. They are entities that remain invisible from the standpoint of the present relational organization of knowledge.

Please adapt the following section about the difference between science and pseudoscience. Add (or remove) any other property it might be interesting or usefull. Please write the result in latex.

7.5 Science vs. Pseudoscience

In the theory of nescience, scientific progress is measured by the systematic reduction of ignorance. As we propose and validate hypotheses about previously unstudied entities, those entities transition from the unknown unknown region to

either the known known or known unknown regions. From this perspective, the set of unknown entities should decrease over time.

Proposition 7.5.1 Let $\mathcal{E}_{UU}^{(t)}$ denote the set of unknown unknown entities at time t and $\mathcal{E}_{UU}^{(t+1)}$ at a posterior time. Then, $\mathcal{E}_{UU}^{(t+1)} \subseteq \mathcal{E}_{UU}^{(t)}$ for all $t \in \mathbb{N}$.

Proof. At each time step t , let the process of scientific discovery introduce new hypotheses about entities in $\mathcal{E}$, thereby moving those entities from the unknown unknown region into the known known region (either known unknowns or known known). ■

In addition to the reduction in the cardinality of the unknown unknown region, we can also observe a progressive reduction in the nescience of individual entities as more refined hypotheses are introduced. Recall that the nescience of an entity is defined based on its current best hypothesis.

Proposition 7.5.2 Let $e \in \mathcal{E}$ be an entity such that $\hat{\mathcal{H}}_e^{(t)}$ and $\hat{\mathcal{H}}_e^{(t+1)}$ are the collections of known hypotheses about e at times t and $t+1$, respectively. Then, if a new hypothesis $h' \in \hat{\mathcal{H}}_e^{(t+1)} \setminus \hat{\mathcal{H}}_e^{(t)}$ is such that $v(h') < v(h)$ for all $h \in \hat{\mathcal{H}}_e^{(t)}$, the nescience of e satisfies:

$$v^{(t+1)}(e) < v^{(t)}(e),$$

where $v^{(t)}(e)$ denotes the nescience of the current best hypothesis at time t .

Proof. By definition, we have that $v^{(t)}(e) = \min_{h \in \hat{\mathcal{H}}_e^{(t)}} v(h)$ and $v^{(t+1)}(e) = \min_{h \in \hat{\mathcal{H}}_e^{(t+1)}} v(h)$. If a new hypothesis $h' \in \hat{\mathcal{H}}_e^{(t+1)}$ is added such that $v(h') < v^{(t)}(e)$, then $v^{(t+1)}(e) = v(h') < v^{(t)}(e)$. ■

This result reflects the iterative nature of scientific inquiry: as we formulate new hypotheses or improve existing ones, the combined errors associated with our representations and descriptions tend to diminish. Over time, and under continued research, the nescience of most entities should asymptotically approach its minimum, potentially reaching the Pareto frontier, or in rare cases, even ideal perfect knowledge.

This characteristic trend of decreasing nescience offers a powerful criterion for distinguishing scientific disciplines from pseudoscientific ones. In genuine scientific fields, the average nescience of topics demonstrably declines over time as knowledge improves. In contrast, pseudoscientific domains typically exhibit stagnation in nescience: either the number of proposed hypotheses remains static or their quality fails to improve, leading to no measurable reduction in ignorance. Thus, the presence or absence of a decreasing trend in nescience provides a formal, quantitative indicator for demarcating science from pseudoscience.

7.6 Nescience of Areas

As discussed in Section 2.3, entities can be grouped into research areas, defined as subsets $\mathcal{A} \subset \mathcal{E}$ of the overall entity space. The concept of a research area is meaningful when the entities in $\mathcal{A}$ share a common epistemic domain or possess a unifying property, such as belonging to the same scientific field or addressing a related phenomenon. To quantify how much we do not know about a research area $\mathcal{A}$, we must work with the subset of entities within $\mathcal{A}$ for which at least one hypothesis is currently known $\hat{\mathcal{A}} \subseteq \mathcal{A}$. A representation of the area $\mathcal{A}$ is then a symbolic encoding $R_{\hat{\mathcal{A}}}$ of the known entities in $\hat{\mathcal{A}}$, suitable for further descriptive analysis. Given this representation, we define a description of the area $\mathcal{A}$ as a string $d_{\hat{\mathcal{A}}}$ that models or summarizes the structure or content of $R_{\hat{\mathcal{A}}}$.

The concept of nescience can be extended from individual entities to entire research areas, allowing us to quantitatively assess how well a given area is understood. This extension relies on generalizing the core components of nescience to sets of entities. We use the notions of combined miscoding of the known subset (see Section 5.6), combined inaccuracy (see Section 9.7), and combined surfeit (see Section 10.7) to define a composite measure of nescience at the area level.

In practice, due to the difficulty of precisely identifying the entire known subset $\hat{\mathcal{A}} \subseteq \mathcal{A}$, it is often convenient to use approximations. One practical and interpretable approximation is the notion of average nescience, which provides a summary of the epistemic quality of the known entities within the area.

Definition 7.6.1 Let $\mathcal{A} \subset \mathcal{E}$ be an area, and $\hat{\mathcal{A}}$ be the subset of known entities. The *average nescience of the area* is defined as:

$$\bar{v}(\mathcal{A}) = \frac{1}{|\hat{\mathcal{A}}|} \sum_{e \in \hat{\mathcal{A}}} v(e),$$

where $v(e)$ denotes the nescience of the current best hypothesis about entity e given a particular decision maker.

For instance, when dealing with research topics, a research area may correspond to a broad disciplinary domain, such as biology, mathematics, or physics, each comprising a collection of entities that fall under that domain. In this context, the area of biology would include all research topics formally classified within the biological sciences. This classification allows us to compute and compare the nescience of distinct knowledge domains by aggregating the nescience of the individual topics they encompass (see Section 14.3). Such comparisons provide insight into which areas of human knowledge remain most poorly understood and where future research efforts might be most impactful.

References

Here are some selected references that are particularly relevant to the concept nescience, along with brief explanations of their relevance.

[LV13] is a foundational text in algorithmic information theory, central to defining concepts such as description length, randomness, and information content. These are directly used in our theory of nescience to formalize inaccuracy and surfeit. [Cha69] provides key ideas on the relationship between program length and algorithmic simplicity. This work contributes to the formal underpinnings of inaccuracy and model complexity in the theory of nescience. [CT12] A comprehensive treatment of classical information theory, including entropy, mutual information, and coding theorems. These concepts are foundational for measuring redundancy and assessing the informativeness of representations in our framework. [Grü07] is irectly relevant to the concept of surfeit in nescience. The MDL principle formalizes model selection based on description length, which parallels the idea of seeking non-redundant descriptions. [Sol64] Solomonoff's work on algorithmic probability underlies the notion of model uncertainty and prediction, both of which are implicitly addressed in our theory's pursuit of minimal ignorance. [Kol65] lays the groundwork for the use of Kolmogorov complexity as a measure of information, randomness, and compressibility, all essential in quantifying nescience. [Tur36] provides the foundational model of computation used throughout the theory of nescience. The theory's reliance on representations and oracle machines assumes the basic framework established by Turing. [Cal02] explores algorithmic complexity and randomness in depth, offering a broader perspective that connects well with our treatment of knowledge, representation, and incompleteness in the theory of nescience. [Cha95] illustrates limits of formal descriptions and links to incompleteness phenomena. This is relevant for understanding the boundaries of compressibility and the epistemological challenges in defining knowledge. [Mie12] is crucial for our use of Pareto and anti-Pareto frontiers in describing tradeoffs among miscoding, inaccuracy, and surfeit. Provides the theoretical tools for reasoning about optimality in multi-dimensional knowledge spaces. [Pop14] while not technical, this work provides a philosophical background on how scientific knowledge evolves through falsification, a process aligned with your model's notion of reducing nescience via hypothesis revision. [Sup02] addresses the problem of how scientific structures relate to the entities they describe, an issue directly relevant to your notion of representations, miscoding, and scientific models.



8. Labeled Miscoding

*A little inaccuracy sometimes saves
tons of explanations.*
Saki

In Chapter 5, we introduced miscoding as a measure of the error introduced when a (extended) representation is used in place of an admissible representation of an entity. The definition is conceptually clear, but it is not directly accessible in practice. To compute the miscoding of a representation $r \in \mathcal{R}_e$, one would need to compare it with the set of admissible representations $\bar{\mathcal{R}}_e$ of the entity e . In ordinary scientific practice, this set is not known. Indeed, if an admissible representation of the entity were already available, the central representational problem would largely have been solved.

This creates an important practical limitation. The theoretical definition of miscoding tells us what should be minimized, but it does not directly tell us how to estimate miscoding when the set of admissible representations is unknown. In many empirical settings, however, we possess a weaker but still useful substitute for the missing admissible representations. We may not know the correct representation of each entity, but we may know that certain representations refer to the same entity, or to entities of the same kind. In other words, we may

have a collection of representations together with labels indicating the entity, class, type, or category to which each representation belongs.

The most familiar example arises in supervised machine learning. A training dataset consists of observations described by a set of attributes $\mathbf{X}$, together with a target variable $\mathbf{y}$. The target variable does not constitute an admissible representation of the entity under study. It does not tell us what the entity is in its full structure. Nevertheless, it provides an index that groups observations according to the entity, property, or outcome we wish to explain or predict. This information allows us to estimate, in a practical way, how well a given representation captures the distinctions that matter for the task.

The purpose of this chapter is to formalize this idea. We introduce *labeled miscoding* as a practical approximation to miscoding for situations in which representations are accompanied by labels. The label does not replace the entity, and it should not be confused with an admissible representation. Rather, it acts as a supervised proxy for entity identity, class membership, or target relevance. Supervised miscoding therefore measures how well a candidate representation preserves the information contained in the labels, while also penalizing information in the representation that is not explained by those labels.

8.1 Labeled Representations

Let $\mathcal{E}$ be a set of entities. In the theoretical definition of miscoding, a representation $r \in \mathcal{R}_e$ is compared against the unknown set of admissible representations $\bar{\mathcal{R}}_e$. In the supervised setting we do not assume access to $\bar{\mathcal{R}}_e$. Instead, we assume access to a finite collection of representations whose intended entity membership is indexed by labels.

Definition 8.1.1 Let $I = \{1, \dots, n\}$ be a finite index set. A *labeled sample of representations* is a finite sequence

$$\mathcal{S} = ((r_1, \lambda_1), \dots, (r_n, \lambda_n)),$$

where $r_i \in \mathcal{R}$ is a representation and $\lambda_i \in \Lambda$ is a label belonging to a finite label set Λ .

The label λ_i should be interpreted as an empirical index indicating the entity to which the representation r_i is assigned. The label need not be an admissible representation of that entity; it only needs to identify the entity uniquely within the labeled sample. Thus, two representations with the same label are treated as representations of the same entity, while representations with different labels are treated as representations of different entities.

Definition 8.1.2 Let $\mathcal{S}$ be a labeled sample of representations. For each label $\lambda \in \Lambda$, we define the corresponding *label class* as

$$\mathcal{R}_\lambda^\mathcal{S} = \{r_i \in \mathcal{R} : \lambda_i = \lambda\}.$$

The set $\mathcal{R}_\lambda^\mathcal{S}$ contains all representations that are assigned the same label in the sample. It is an empirical substitute for the unknown set $\mathcal{R}_e$ of representations of an entity. However, the substitution is only approximate. A label may incorrectly group together different entities, split a single entity into several categories, or contain attribute errors.

■ **Example 8.1** Let $\mathcal{S}$ be a collection of images of handwritten digits. Each image r_i is a representation, and its label $\lambda_i \in \{0, 1, \dots, 9\}$ indicates the digit shown in the image. The class $\mathcal{R}_3^\mathcal{S}$ contains all images labeled as the digit 3. This set is not an admissible representation of the abstract entity "the digit 3", but it provides empirical information about how instances of that entity are represented in the dataset. ■

The supervised setting therefore replaces the unknown admissible representation of an entity by an empirical relation of co-membership: two representations with the same label are treated as representations of the same entity, at least for the purposes of the supervised task.

8.2 The Supervised Proxy

To compute a practical approximation to miscoding, we need to transform the labeled sample into strings whose algorithmic relationship can be estimated. Let

$$\lambda = \lambda_1 \lambda_2 \cdots \lambda_n$$

denote an encoding of the sequence of labels. We call this string the *label string*. It is a finite binary string obtained by encoding each label with a fixed prefix-free code.

In many applications, each representation r_i is structured. For example, in a tabular dataset, each observation consists of p attributes:

$$r_i = (x_{i1}, \dots, x_{ip}).$$

For each feature j , we may form the feature string

$$\mathbf{x}_j = x_{1j} x_{2j} \cdots x_{nj},$$

that is, the sequence of values taken by the j -th feature across the sample.

More generally, we may consider any computable extraction function

$$\phi : \mathcal{B}^* \rightarrow \mathcal{B}^*$$

and construct the induced sample string

$$\phi(\mathcal{S}) = \phi(r_1)\phi(r_2)\cdots\phi(r_n).$$

The string $\phi(\mathcal{S})$ represents the information extracted from the effective representations by the transformation ϕ .

Definition 8.2.1 Let $\mathcal{S} = ((r_1, \lambda_1), \dots, (r_n, \lambda_n))$ be a labeled sample, and let $\phi : \mathcal{B}^* \rightarrow \mathcal{B}^*$ be a computable extraction function. The string

$$\mathbf{x}_\phi = \phi(r_1)\phi(r_2)\cdots\phi(r_n)$$

is called the *supervised proxy representation* induced by ϕ .

The supervised proxy representation is not a representation of a single entity. It is a representation of how a certain observable aspect of the sample varies across labeled examples. Its purpose is to determine whether the extracted information is useful for reconstructing the label string.

■ **Example 8.2** In a tabular machine learning dataset, the function ϕ_j that extracts the j -th attribute from each observation induces the proxy representation

$$\mathbf{x}_j = x_{1j}x_{2j}\cdots x_{nj}.$$

The supervised miscoding of $\mathbf{x}_j$ measures how well the j -th feature represents the target labels. ■

8.3 Deficiency and Surplus

The theoretical notion of miscoding is based on two components: deficiency and surplus. Deficiency measures how much information is missing from a representation in order to reach an admissible representation. Surplus measures how much information in the representation is not attributable to the entity.

In the labeled setting, the admissible representation is unavailable. The label string λ is therefore used as a proxy for the target distinctions that the representation should preserve. If the proxy representation $\mathbf{x}$ does not allow us to identify the corresponding label λ_i for each represented instance, then it is deficient with respect to the supervised task. Conversely, if $\mathbf{x}$ contains substantial information that is not relevant to identifying the labels, then it contains surplus with respect to the supervised task.

Let $K(\cdot)$ denote Kolmogorov complexity. For two strings $\mathbf{x}, \lambda \in \mathcal{B}^*$, the conditional complexity $K(\lambda \mid \mathbf{x})$ measures how much information is required to reconstruct the labels from the representation, while $K(\mathbf{x} \mid \lambda)$ measures how much information is required to reconstruct the representation from the labels.

Definition 8.3.1 Let $\mathbf{x} \in \mathcal{B}^*$ be a supervised proxy representation, and let $\lambda \in \mathcal{B}^*$ be the corresponding label string. We define the *ideal supervised deficiency* of $\mathbf{x}$ with respect to λ as

$$d^{\text{sup}}(\mathbf{x}, \lambda) = \frac{K(\lambda \mid \mathbf{x})}{\max\{K(\mathbf{x}), K(\lambda)\}}.$$

We define the *ideal supervised surplus* of $\mathbf{x}$ with respect to λ as

$$s^{\text{sup}}(\mathbf{x}, \lambda) = \frac{K(\mathbf{x} \mid \lambda)}{\max\{K(\mathbf{x}), K(\lambda)\}}.$$

The first quantity measures the fraction of label information that is not supplied by the representation. The second measures the fraction of representation information that is not explained by the labels. The normalization makes the quantities dimensionless and allows comparisons across representations of different sizes.

Definition 8.3.2 Let $\mathbf{x} \in \mathcal{B}^*$ be a supervised proxy representation, and let $\lambda \in \mathcal{B}^*$ be the corresponding label string. We define the *ideal supervised miscoding* of $\mathbf{x}$ with respect to λ as

$$\mu^{\text{sup}}(\mathbf{x}, \lambda) = \max\{d^{\text{sup}}(\mathbf{x}, \lambda), s^{\text{sup}}(\mathbf{x}, \lambda)\}.$$

Equivalently,

$$\mu^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\max\{K(\lambda \mid \mathbf{x}), K(\mathbf{x} \mid \lambda)\}}{\max\{K(\mathbf{x}), K(\lambda)\}}.$$

This is the normalized information distance between the supervised proxy representation and the label string. The lower this value, the better the representation captures the supervised information encoded by the labels.

8.4 Computable Supervised Miscoding

The ideal quantities above are not computable because Kolmogorov complexity is not computable. To obtain a practical metric, we approximate Kolmogorov complexity by the length of compressed strings. Let C be a fixed compressor, and let

$$\hat{K}_C(x)$$

denote the length of the compressed version of x under C . Similarly, let $\hat{K}_C(x, y)$ denote the length of the compressed encoding of the concatenated pair (x, y) , using a fixed self-delimiting pairing convention.

Definition 8.4.1 Let $\mathbf{x} \in \mathcal{B}^*$ be a supervised proxy representation, let $\lambda \in \mathcal{B}^*$ be the corresponding label string, and let C be a compressor. We define the *computable supervised miscoding* of $\mathbf{x}$ with respect to λ as

$$\hat{\mu}_C^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\hat{K}_C(\mathbf{x}, \lambda) - \min\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}}.$$

This definition is the computable counterpart of the ideal supervised miscoding. It estimates the algorithmic distance between the representation and the labels. If the representation contains enough information to reconstruct the labels, then $\hat{K}_C(\mathbf{x}, \lambda)$ will be close to $\hat{K}_C(\mathbf{x})$. If the labels contain enough information to reconstruct the representation, then $\hat{K}_C(\mathbf{x}, \lambda)$ will be close to $\hat{K}_C(\lambda)$. If neither string explains the other, then the joint compressed length will be close to the sum of the two compressed lengths, and supervised miscoding will be large.

Definition 8.4.2 Under the same assumptions, we define the *computable supervised deficiency* by

$$\hat{d}_C^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\hat{K}_C(\mathbf{x}, \lambda) - \hat{K}_C(\mathbf{x})}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}},$$

and the *computable supervised surplus* by

$$\hat{s}_C^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\hat{K}_C(\mathbf{x}, \lambda) - \hat{K}_C(\lambda)}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}}.$$

These two quantities approximate the two directions of the information gap. The deficiency term is large when the representation fails to determine the labels. The surplus term is large when the representation contains information that is not determined by the labels.

Proposition 8.4.1 Let $\mathbf{x}, \lambda \in \mathcal{B}^*$. Then

$$\hat{\mu}_C^{\text{sup}}(\mathbf{x}, \lambda) = \max\{\hat{d}_C^{\text{sup}}(\mathbf{x}, \lambda), \hat{s}_C^{\text{sup}}(\mathbf{x}, \lambda)\}.$$

Proof. By definition,

$$\hat{d}_C^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\hat{K}_C(\mathbf{x}, \lambda) - \hat{K}_C(\mathbf{x})}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}},$$

and

$$\hat{s}_C^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\hat{K}_C(\mathbf{x}, \lambda) - \hat{K}_C(\lambda)}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}}.$$

Since the denominator is the same in both expressions, the maximum is obtained by subtracting the smaller of $\hat{K}_C(\mathbf{x})$ and $\hat{K}_C(\lambda)$ from $\hat{K}_C(\mathbf{x}, \lambda)$. Hence

$$\max \{ \hat{d}_C^{\sup}, \hat{s}_C^{\sup} \} = \frac{\hat{K}_C(\mathbf{x}, \lambda) - \min \{ \hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda) \}}{\max \{ \hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda) \}},$$

which is precisely $\hat{\mu}_C^{\sup}(\mathbf{x}, \lambda)$. ■

This result shows that supervised miscoding retains the same conceptual structure as theoretical miscoding: it is determined by the worse of two defects, missing supervised information and extraneous supervised information.

8.5 Basic Properties

We now study several basic properties of supervised miscoding. These properties clarify how the metric should be interpreted and where its limitations lie.

Proposition 8.5.1 Assume that the compressor C satisfies

$$\hat{K}_C(\mathbf{x}, \lambda) \geq \max \{ \hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda) \}$$

up to a constant independent of $\mathbf{x}$ and λ . Then

$$\hat{\mu}_C^{\sup}(\mathbf{x}, \lambda) \geq 0$$

up to a vanishing additive term.

Proof. If

$$\hat{K}_C(\mathbf{x}, \lambda) \geq \max \{ \hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda) \},$$

then in particular

$$\hat{K}_C(\mathbf{x}, \lambda) \geq \min \{ \hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda) \}.$$

Therefore the numerator in Definition 8.4.1 is non-negative. Since the denominator is positive for non-empty strings, the result follows. ■

In practical implementations, compressors may violate the inequality for very short strings because of headers, block sizes, or implementation details. For this reason, one may truncate the value to the interval $[0, 1]$ when using supervised miscoding as a numerical score.

Proposition 8.5.2 Assume that the compressor C satisfies

$$\hat{K}_C(\mathbf{x}, \lambda) \leq \hat{K}_C(\mathbf{x}) + \hat{K}_C(\lambda) + O(1).$$

Then

$$\hat{\mu}_C^{\text{sup}}(\mathbf{x}, \lambda) \leq 1 + O\left(\frac{1}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}}\right).$$

Proof. By assumption,

$$\hat{K}_C(\mathbf{x}, \lambda) - \min\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\} \leq \max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\} + O(1).$$

Dividing both sides by

$$\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}$$

gives the result. ■

Thus, for sufficiently long strings and a reasonable compressor, computable supervised miscoding lies approximately in the interval $[0, 1]$. Values close to 0 indicate strong supervised adequacy, whereas values close to 1 indicate that the representation and the labels share little usable information.

Proposition 8.5.3 For every $\mathbf{x}, \lambda \in \mathcal{B}^*$,

$$\hat{\mu}_C^{\text{sup}}(\mathbf{x}, \lambda) = \hat{\mu}_C^{\text{sup}}(\lambda, \mathbf{x}),$$

provided that the pairing convention used to compute $\hat{K}_C(\mathbf{x}, \lambda)$ is symmetric up to a constant.

Proof. The expression in Definition 8.4.1 is symmetric in $\mathbf{x}$ and λ , except possibly for the joint compressed length. If the pairing convention is symmetric up to a constant, then

$$\hat{K}_C(\mathbf{x}, \lambda) = \hat{K}_C(\lambda, \mathbf{x}) + O(1).$$

The minimum and maximum terms are exactly symmetric. Therefore the two supervised miscoding values differ at most by a term of order

$$O\left(\frac{1}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\lambda)\}}\right),$$

which is negligible for long strings. ■

The symmetry property should not be misinterpreted. Although the numerical distance is symmetric, the scientific interpretation is not. In supervised miscoding, $\mathbf{x}$ is the candidate representation and λ is the supervised target. The two strings play different roles in the application even if the mathematical distance is symmetric.

Proposition 8.5.4 Suppose that $\mathbf{x}$ and λ are algorithmically independent in the sense that

$$K(\mathbf{x}, \lambda) = K(\mathbf{x}) + K(\lambda) + O(1).$$

Then the ideal supervised miscoding satisfies

$$\mu^{\text{sup}}(\mathbf{x}, \lambda) = 1 + O\left(\frac{1}{\max\{K(\mathbf{x}), K(\lambda)\}}\right).$$

Proof. If $\mathbf{x}$ and λ are algorithmically independent, then

$$K(\lambda \mid \mathbf{x}) = K(\lambda) + O(1)$$

and

$$K(\mathbf{x} \mid \lambda) = K(\mathbf{x}) + O(1).$$

Therefore

$$\max\{K(\lambda \mid \mathbf{x}), K(\mathbf{x} \mid \lambda)\} = \max\{K(\mathbf{x}), K(\lambda)\} + O(1).$$

Dividing by $\max\{K(\mathbf{x}), K(\lambda)\}$ gives the result. ■

This proposition formalizes the intuition that an irrelevant representation has maximal supervised miscoding. If the representation and the labels contain no shared algorithmic structure, then the representation does not help us reconstruct the labels, and the labels do not explain the representation.

Proposition 8.5.5 Suppose that

$$K(\lambda \mid \mathbf{x}) = O(1) \quad \text{and} \quad K(\mathbf{x} \mid \lambda) = O(1).$$

Then

$$\mu^{\text{sup}}(\mathbf{x}, \lambda) = O\left(\frac{1}{\max\{K(\mathbf{x}), K(\lambda)\}}\right).$$

Proof. By Definition 8.3.2,

$$\mu^{\text{sup}}(\mathbf{x}, \lambda) = \frac{\max\{K(\lambda \mid \mathbf{x}), K(\mathbf{x} \mid \lambda)\}}{\max\{K(\mathbf{x}), K(\lambda)\}}.$$

The numerator is $O(1)$ by hypothesis. Dividing by the denominator yields the result. ■

This proposition identifies the ideal supervised case: the representation and the labels are mutually reconstructible up to a constant. In practice this rarely occurs, because useful representations often contain within-class variation not determined by the label. Such variation is not necessarily bad, but it is surplus relative to the supervised task.

Proposition 8.5.6 Let $\pi : \Lambda \rightarrow \Lambda'$ be a computable bijection between label sets, and let

$$\pi(\lambda)$$

be the relabeled string obtained by applying π to every label in λ . If both π and π^{-1} are computable with constant overhead, then

$$\mu^{\sup}(\mathbf{x}, \pi(\lambda)) = \mu^{\sup}(\mathbf{x}, \lambda) + O\left(\frac{1}{\max\{K(\mathbf{x}), K(\lambda)\}}\right).$$

Proof. Since π and π^{-1} are computable with constant overhead,

$$K(\pi(\lambda) \mid \lambda) = O(1) \quad \text{and} \quad K(\lambda \mid \pi(\lambda)) = O(1).$$

Thus

$$K(\pi(\lambda)) = K(\lambda) + O(1),$$

and the conditional complexities involving $\mathbf{x}$ change by at most $O(1)$. Substituting these relations into Definition 8.3.2 gives the result. ■

This proposition is important in practice. Supervised miscoding depends on the information content of the label partition, not on the arbitrary names assigned to labels. Renaming the classes from $\{0, 1\}$ to $\{\text{negative}, \text{positive}\}$, for instance, should not change the value except for coding overhead.

8.6 Feature Supervised Miscoding

One of the most useful applications of supervised miscoding is feature evaluation. Suppose that each representation r_i is a vector of p components:

$$r_i = (x_{i1}, \dots, x_{ip}).$$

For each feature j , define

$$\mathbf{x}_j = x_{1j}x_{2j} \cdots x_{nj}.$$

The supervised miscoding of the feature j is then computed by comparing $\mathbf{x}_j$ with the label string λ .

Definition 8.6.1 Let $\mathcal{S}$ be a labeled sample with feature strings $\mathbf{x}_1, \dots, \mathbf{x}_p$ and label string λ . The *regular supervised miscoding* of feature j is

$$\hat{\mu}_{C,j}^{\sup} = \hat{\mu}_C^{\sup}(\mathbf{x}_j, \lambda).$$

A small value of $\hat{\mu}_{C,j}^{\sup}$ indicates that the feature carries information about the labels. A large value indicates that the feature is weakly related or unrelated to the supervised target.

■ **Example 8.3** Let λ be a binary label indicating whether a patient has a certain disease. Suppose that $\mathbf{x}_1$ records a biomarker strongly associated with the disease, while $\mathbf{x}_2$ records the patient identification number. We should expect

$$\hat{\mu}_C^{\sup}(\mathbf{x}_1, \lambda) < \hat{\mu}_C^{\sup}(\mathbf{x}_2, \lambda).$$

The biomarker is more useful for reconstructing the labels, whereas the identification number may contain information about the sample but little or no information about the disease. ■

It is sometimes more convenient to convert miscoding values into relative importance weights. Since lower miscoding is better, we consider the complement $1 - \hat{\mu}_{C,j}^{\sup}$.

Definition 8.6.2 Let $\hat{\mu}_{C,1}^{\sup}, \dots, \hat{\mu}_{C,p}^{\sup}$ be the regular supervised miscodings of the features. We define the *adjusted supervised relevance* of feature j by

$$w_j = \frac{1 - \hat{\mu}_{C,j}^{\sup}}{\sum_{k=1}^p (1 - \hat{\mu}_{C,k}^{\sup})},$$

provided that the denominator is nonzero.

The weights w_j are not miscodings. They are normalized relevance scores derived from supervised miscoding. They are useful for visualization and feature ranking, but they should not be interpreted as absolute measures of representational error.

Proposition 8.6.1 If

$$\sum_{k=1}^p (1 - \hat{\mu}_{C,k}^{\sup}) \neq 0,$$

then

$$\sum_{j=1}^p w_j = 1.$$

Proof. By Definition 8.6.2,

$$\sum_{j=1}^p w_j = \sum_{j=1}^p \frac{1 - \hat{\mu}_{C,j}^{\sup}}{\sum_{k=1}^p (1 - \hat{\mu}_{C,k}^{\sup})} = \frac{\sum_{j=1}^p (1 - \hat{\mu}_{C,j}^{\sup})}{\sum_{k=1}^p (1 - \hat{\mu}_{C,k}^{\sup})} = 1.$$

■

8.7 Joint Supervised Miscoding

Individual features may be misleading when considered in isolation. Two features may each have high supervised miscoding individually, while their joint representation has low supervised miscoding. This occurs when the label depends on an interaction between the features.

Let $Z \subseteq \{1, \dots, p\}$ be a subset of features. We denote by

$\mathbf{x}_Z$

the joint string obtained by encoding, for each sample, the values of the features in Z .

Definition 8.7.1 Let $Z \subseteq \{1, \dots, p\}$. The *joint supervised miscoding* of the feature subset Z is

$$\hat{\mu}_C^{\sup}(Z, \lambda) = \hat{\mu}_C^{\sup}(\mathbf{x}_Z, \lambda).$$

This definition follows the same principle as theoretical miscoding of joint representations: no separate concept is needed once the joint object has been encoded as a string. The joint representation is simply evaluated against the supervised target.

■ **Example 8.4** Suppose that the label is defined by the exclusive-or relation

$$\lambda_i = x_{i1} \oplus x_{i2}.$$

Individually, neither $\mathbf{x}_1$ nor $\mathbf{x}_2$ may determine the label. Hence both features may have high supervised miscoding. However, the joint representation $\mathbf{x}_{\{1,2\}}$ determines the label exactly, so its supervised deficiency is small. This illustrates why feature selection based only on individual scores may fail. ■

Definition 8.7.2 Let $Z \subseteq \{1, \dots, p\}$, and let $j \notin Z$. The *marginal supervised miscoding reduction* of feature j relative to Z is

$$\Delta_C^{\sup}(j | Z) = \hat{\mu}_C^{\sup}(Z, \lambda) - \hat{\mu}_C^{\sup}(Z \cup \{j\}, \lambda).$$

If $\Delta_C^{\text{sup}}(j \mid Z) > 0$, then adding feature j improves the supervised representation. If $\Delta_C^{\text{sup}}(j \mid Z) < 0$, then adding the feature worsens supervised miscoding, usually because it introduces surplus that is not compensated by a reduction in deficiency.

Proposition 8.7.1 Let $Z \subseteq \{1, \dots, p\}$ and $j \notin Z$. Then

$$\Delta_C^{\text{sup}}(j \mid Z) > 0$$

if and only if

$$\hat{\mu}_C^{\text{sup}}(Z \cup \{j\}, \lambda) < \hat{\mu}_C^{\text{sup}}(Z, \lambda).$$

Proof. This follows immediately from the definition:

$$\Delta_C^{\text{sup}}(j \mid Z) = \hat{\mu}_C^{\text{sup}}(Z, \lambda) - \hat{\mu}_C^{\text{sup}}(Z \cup \{j\}, \lambda).$$

The difference is positive exactly when the second term is smaller than the first. ■

The quantity $\Delta_C^{\text{sup}}(j \mid Z)$ provides a practical criterion for greedy feature selection. Starting from $Z = \emptyset$, one may repeatedly add the feature with the largest positive marginal reduction until no feature decreases supervised miscoding. This procedure is not guaranteed to find a global optimum, but it is consistent with the idea of decreasing miscoding by adding information that helps reconstruct the target while controlling surplus.

8.8 Supervised Partial Miscoding

In practice, one often wants to assign each feature a signed contribution. A feature should contribute positively if it helps reconstruct the labels and negatively if it mostly adds irrelevant information. For this purpose we can compare two normalized quantities: the relative relevance of a feature and its relative miscoding.

Definition 8.8.1 Let $\hat{\mu}_{C,1}^{\text{sup}}, \dots, \hat{\mu}_{C,p}^{\text{sup}}$ be the regular supervised miscodings of the features. The *partial supervised contribution* of feature j is defined as

$$\tilde{\mu}_{C,j}^{\text{sup}} = \frac{1 - \hat{\mu}_{C,j}^{\text{sup}}}{\sum_{k=1}^p (1 - \hat{\mu}_{C,k}^{\text{sup}})} - \frac{\hat{\mu}_{C,j}^{\text{sup}}}{\sum_{k=1}^p \hat{\mu}_{C,k}^{\text{sup}}},$$

provided that both denominators are nonzero.

A positive value of $\tilde{\mu}_{C,j}^{\text{sup}}$ indicates that the feature is more relevant than miscoded relative to the other features. A negative value indicates that the feature contributes more supervised error than supervised relevance.

Proposition 8.8.1 Assume that the denominators in Definition 8.8.1 are nonzero. Then

$$\sum_{j=1}^p \tilde{\mu}_{C,j}^{\sup} = 0.$$

Proof. By Definition 8.8.1,

$$\sum_{j=1}^p \tilde{\mu}_{C,j}^{\sup} = \sum_{j=1}^p \frac{1 - \hat{\mu}_{C,j}^{\sup}}{\sum_{k=1}^p (1 - \hat{\mu}_{C,k}^{\sup})} - \sum_{j=1}^p \frac{\hat{\mu}_{C,j}^{\sup}}{\sum_{k=1}^p \hat{\mu}_{C,k}^{\sup}}.$$

Both sums are equal to 1. Therefore the difference is 0. ■

This result shows that partial supervised contributions are relative rather than absolute. They distribute the balance of relevance and miscoding across features. They are useful for ranking and selection, but not for measuring the absolute miscoding of the whole dataset.

8.9 Supervised Miscoding of a Dataset

Let

$$\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_p)$$

be the full supervised representation induced by all features. The supervised miscoding of the dataset is not the sum of the individual feature miscodings. Summing individual values ignores redundancy and interaction effects. Two features may encode the same information about the labels, in which case their individual relevance should not be counted twice. Conversely, two features may jointly encode information that neither contains alone.

For this reason, the most direct definition of dataset-level supervised miscoding is the joint one.

Definition 8.9.1 Let $\mathbf{X}$ be the joint encoding of all features in a labeled dataset, and let λ be the label string. The *supervised miscoding of the dataset* is

$$\hat{\mu}_C^{\sup}(\mathbf{X}, \lambda) = \frac{\hat{K}_C(\mathbf{X}, \lambda) - \min\{\hat{K}_C(\mathbf{X}), \hat{K}_C(\lambda)\}}{\max\{\hat{K}_C(\mathbf{X}), \hat{K}_C(\lambda)\}}.$$

This quantity measures how well the dataset as a whole represents the supervised target. It automatically accounts for interactions among features, redundancy between features, and surplus information unrelated to the labels.

Proposition 8.9.1 In general,

$$\hat{\mu}_C^{\text{sup}}(\mathbf{X}, \lambda) \neq \sum_{j=1}^P \hat{\mu}_C^{\text{sup}}(\mathbf{x}_j, \lambda).$$

Proof. It is enough to exhibit two cases.

First, suppose that $\mathbf{x}_1 = \mathbf{x}_2$. Then the two features contain the same information. The joint representation $(\mathbf{x}_1, \mathbf{x}_2)$ is computably equivalent to $\mathbf{x}_1$, up to a small overhead. Hence

$$\hat{\mu}_C^{\text{sup}}((\mathbf{x}_1, \mathbf{x}_2), \lambda) \approx \hat{\mu}_C^{\text{sup}}(\mathbf{x}_1, \lambda),$$

whereas

$$\hat{\mu}_C^{\text{sup}}(\mathbf{x}_1, \lambda) + \hat{\mu}_C^{\text{sup}}(\mathbf{x}_2, \lambda) = 2\hat{\mu}_C^{\text{sup}}(\mathbf{x}_1, \lambda).$$

Thus equality fails in general.

Second, suppose that the label is determined by an interaction between two features, as in the exclusive-or example. Then the individual supervised miscodings may be large, while the joint supervised miscoding is small. Again equality fails. ■

This proposition explains why supervised miscoding should be computed at the level of the representation actually used. Individual feature scores are useful diagnostic tools, but the miscoding of a dataset is a property of the joint representation.

8.10 Connection with Theoretical Miscoding

Supervised miscoding is not a replacement for theoretical miscoding. It is an approximation that becomes meaningful under a specific empirical assumption: the labels capture the entity distinctions that matter for the investigation.

Let $e_i \in \mathcal{E}$ denote the entity associated with r_i , even if this entity is not directly accessible. The label map

$$i \mapsto \lambda_i$$

may be interpreted as an empirical approximation to the unknown entity assignment

$$i \mapsto e_i.$$

If the labels are correct and sufficiently informative, then reducing supervised miscoding tends to improve the representation for the supervised task. If the labels are poor, noisy, or conceptually misaligned, then supervised miscoding may optimize the wrong target.

Definition 8.10.1 Let $\mathcal{S} = ((r_1, \lambda_1), \dots, (r_n, \lambda_n))$ be a labeled sample, and let e_i denote the intended entity represented by r_i . We say that the labels are *faithful* with respect to the sample if

$$\lambda_i = \lambda_j \iff e_i = e_j$$

for all $i, j \in I$.

Faithfulness means that the labels identify exactly the entity membership relation within the sample. This is a strong condition. In many applications the labels identify not entities themselves, but properties of entities. For example, a medical label may indicate whether a patient has a disease, not the complete biomedical state of the patient.

Proposition 8.10.1 Suppose that the labels are faithful and that a supervised proxy representation $\mathbf{x}$ determines the entity membership relation of the sample up to a constant-size program. Then

$$K(\lambda \mid \mathbf{x}) = O(1),$$

and therefore

$$d^{\text{sup}}(\mathbf{x}, \lambda) = O\left(\frac{1}{\max\{K(\mathbf{x}), K(\lambda)\}}\right).$$

Proof. If $\mathbf{x}$ determines the entity membership relation and the labels are faithful, then there exists a constant-size decoding rule that assigns to each sample its correct label from the structure encoded in $\mathbf{x}$. Hence

$$K(\lambda \mid \mathbf{x}) = O(1).$$

Substituting into the definition of supervised deficiency gives the result. ■

The proposition states that, under faithful labels, a representation that determines the relevant entity distinctions has negligible supervised deficiency. However, its supervised surplus may still be large if it contains substantial within-class information not determined by the labels.

Proposition 8.10.2 Let λ' be a computable coarsening of λ , meaning that

$$K(\lambda' \mid \lambda) = O(1),$$

but in general

$$K(\lambda \mid \lambda') \neq O(1).$$

Then a representation $\mathbf{x}$ may have larger supervised surplus with respect to λ' than with respect to λ .

Proof. A coarser label string contains less information than the original one. Information in $\mathbf{x}$ that explains distinctions present in λ may no longer be recoverable from λ' . Therefore the conditional complexity

$$K(\mathbf{x} \mid \lambda')$$

may be larger than

$$K(\mathbf{x} \mid \lambda).$$

Since supervised surplus is proportional to the information required to reconstruct $\mathbf{x}$ from the labels, the surplus may increase under coarsening. ■

This result is important because it shows that supervised miscoding depends on the granularity of the labels. A representation may be excellent for a fine-grained classification task but appear to contain surplus under a coarse-grained labeling scheme.

8.11 Limitations of Supervised Miscoding

Supervised miscoding is practical precisely because it replaces the inaccessible admissible representation by an available label string. This replacement is useful, but it introduces several limitations.

First, supervised miscoding is task-relative. It measures how well a representation captures the information encoded by the labels, not how well it represents the entity in its full structure. A feature may be irrelevant for predicting the label and still be scientifically important for understanding the entity.

Second, supervised miscoding depends on label quality. Incorrect labels introduce a distorted supervised target. In that case, a representation that correctly tracks the underlying entity may be penalized because it disagrees with the labels.

Third, supervised miscoding is sensitive to the representation of continuous quantities. Continuous measurements must be discretized or encoded before compression. Different discretization schemes may lead to different numerical values.

Fourth, supervised miscoding depends on the compressor. Different compressors approximate different regularities. A compressor that fails to exploit the relevant structure in the data may overestimate miscoding.

Finally, supervised miscoding should not be confused with accuracy. A representation may have low supervised miscoding because it contains information about the labels, while a particular model trained on that representation may still be inaccurate. Miscoding evaluates the representational relation between data and labels; inaccuracy evaluates the relation between a model output and the intended representation.

8.12 Practical Use

The practical use of supervised miscoding can be summarized as follows.

Given a labeled dataset, one first encodes the labels as a string λ . Then, for each candidate representation $\mathbf{x}$, feature $\mathbf{x}_j$, or feature subset $\mathbf{x}_Z$, one computes

$$\hat{\mu}_C^{\text{sup}}(\mathbf{x}, \lambda).$$

Representations with smaller supervised miscoding are preferred, because they preserve more information about the labels while introducing less information unrelated to them.

A simple feature-selection procedure is the following:

- 1 Encode the label string λ .
- 2 Compute $\hat{\mu}_C^{\text{sup}}(\mathbf{x}_j, \lambda)$ for each feature j .
- 3 Rank features from lowest to highest supervised miscoding.
- 4 Optionally compute marginal supervised miscoding reductions

$$\Delta_C^{\text{sup}}(j \mid Z)$$

to account for interactions and redundancy.

- 5 Select features that decrease the joint supervised miscoding of the representation.

This procedure is not a substitute for model validation. Rather, it provides a representation-level criterion that can be used before or alongside model training. It helps identify which parts of the representation are likely to be useful for the supervised task and which parts mostly add surplus.

8.13 Conclusion

Supervised miscoding provides a practical approximation to miscoding in settings where admissible representations are unknown but labeled effective representations are available. The key idea is to use the label string as a supervised proxy for the entity distinctions that matter in a given task. A representation has low supervised miscoding when it allows the labels to be reconstructed and when it contains little information not explained by those labels.

This construction preserves the conceptual structure of theoretical miscoding. Supervised deficiency measures missing target information. Supervised surplus measures extraneous representation information. Supervised miscoding is the maximum of these two components. In this sense, the supervised version remains faithful to the original theory while becoming computable through compression.

The price paid for computability is that the resulting quantity is relative to the labels, the encoding scheme, the sample, and the compressor. It should therefore be interpreted as a practical diagnostic rather than as an exact measure of the

true miscoding of an entity. Nevertheless, in domains such as machine learning, experimental science, and empirical classification, supervised miscoding provides a useful bridge between the formal theory of nescience and operational methods for selecting and improving representations.



9. Inaccuracy

*A little inaccuracy sometimes saves
tons of explanations.*
Saki

In Chapter 6, we introduced *mismodel* as a theoretical measure of the discrepancy between a description currently available in scientific practice and an ideal description of the representation under study. In particular, the deficiency component of *mismodel* measures how much algorithmic information is still needed to transform a given description into a minimal description of the intended representation. From a formal point of view, this is the most natural way to quantify how incomplete or defective a description is.

However, this quantity cannot usually be approximated in practice. The main obstacle is that its definition refers to minimal descriptions of the representation r , and such descriptions are generally unknown. In ordinary scientific practice, we do not have access to the set $\mathcal{D}_r^*$ of minimal descriptions of r , nor do we know which existing model, if any, is closest to that set. There is also a deeper theoretical obstruction: minimal descriptions are determined by Kolmogorov complexity, and Kolmogorov complexity is non-computable. We therefore need computable, or at least approximable, proxies.

In this book we use two complementary approximations of mismodel. The first approximation, studied in this chapter, compares the output produced by the current description with the representation that the description is intended to model. If d is a description and $\Gamma(d)$ is the string produced by executing it, then inaccuracy measures how far $\Gamma(d)$ is from the target representation r . In this sense, inaccuracy does not ask whether d is close to a minimal description of r . It asks whether the output of d is close to r . It is therefore an output-level approximation to the deficiency of a description.

The second approximation, studied in Chapter 10, goes in a different direction. Instead of comparing the output of the description with the target representation, surfeit compares the description itself with the expected size of a minimal description. Thus, inaccuracy approximates the failure of a description to reproduce the intended representation, whereas surfeit approximates the failure of the description to be minimal as a description.

These two approximations capture different aspects of mismodel. A description may have low inaccuracy because it reproduces the representation correctly, while still having high surfeit because it is unnecessarily long or overfitted. Conversely, a description may be compact and have low surfeit, while still being inaccurate because its output fails to reproduce the intended representation. This distinction is essential for the theory of nescience: a good scientific model should be both accurate and economical.

9.1 Inaccuracy

Let $r \in \mathcal{R}$ be a representation and let $d \in \mathcal{D}$ be a description. Since d is a valid description in the formal sense, executing it on the fixed universal Turing machine produces some finite string. We denote this output by $r' = \Gamma(d)$. If $r' = r$, then d is an exact description of r . If $r' \neq r$, then d is inaccurate with respect to r .

The concept of an inaccurate description was introduced in Chapter 4. In the present chapter, we use that notion as the basis for a quantitative measure. The central problem can be formulated as follows: given a representation r and a description d whose execution produces a string $r' = \Gamma(d)$ different from r , how much algorithmic information is required to correct the discrepancy between the produced output and the intended representation?

A first answer would be to measure $K(r \mid r')$, the amount of information required to reconstruct the intended representation r from the output r' produced by the description. This captures missing information: the part of r that is not present in r' . However, as in the case of miscoding, this one-sided measure is not sufficient. The output r' may also contain additional structure not present in r . In that case, it is also relevant to consider $K(r' \mid r)$, the amount of information needed to reconstruct the erroneous output from the intended representation.

For this reason, inaccuracy is defined using a bidirectional information distance between r and $\Gamma(d)$.

Definition 9.1.1 — Inaccuracy. Let $r \in \mathcal{R}$ be a representation, and let $d \in \mathcal{D}$ be a description. We define the *inaccuracy* of d with respect to r , denoted by $\iota(d, r)$, as

$$\iota(d, r) = \frac{\max \{K(r | \Gamma(d)), K(\Gamma(d) | r)\}}{\max \{K(r), K(\Gamma(d))\}}.$$

Thus, inaccuracy is the normalized information distance between the intended representation r and the representation $\Gamma(d)$ produced by the description. The numerator measures the bidirectional correction cost between the two strings. The denominator normalizes this cost by the size of the larger of the two strings, so that inaccuracies associated with different representations can be compared on a common scale.

As usual in algorithmic information theory, equalities and inequalities involving Kolmogorov complexity should be understood up to additive constants independent of the particular strings under consideration. These constants depend on the choice of universal Turing machine Γ and on the fixed encoding conventions.

The definition above is theoretically natural, but it is not directly computable. In practice (see Section 20.5), we approximate it by using the identity

$$\max \{K(x | y), K(y | x)\} = K(x, y) - \min \{K(x), K(y)\} + O(\log \max \{K(x), K(y)\}),$$

which holds for finite strings x, y . Ignoring the logarithmic term for sufficiently large strings, the inaccuracy of d with respect to r can be approximated by

$$\iota(d, r) \approx \frac{K(r, \Gamma(d)) - \min \{K(r), K(\Gamma(d))\}}{\max \{K(r), K(\Gamma(d))\}}.$$

Replacing K by a computable estimate $\hat{K}$, we obtain the practical approximation

$$\hat{\iota}(d, r) = \frac{\hat{K}(r, \Gamma(d)) - \min \{\hat{K}(r), \hat{K}(\Gamma(d))\}}{\max \{\hat{K}(r), \hat{K}(\Gamma(d))\}}.$$

This is the form used in applications (see Chapter 13). Depending on the domain, $\hat{K}$ may be estimated by compression algorithms, empirical coding schemes, residual coding costs, or model-specific description lengths.

■ **Example 9.1** Inaccuracy concerns the output of a description, not the internal form of the description itself. Suppose that a dataset is generated by a quadratic function, but we describe it using a linear model. The description d corresponding

to the linear model produces a predicted dataset $\Gamma(d)$. The inaccuracy of d with respect to the observed representation r measures how difficult it is to transform the predicted dataset into the observed dataset, and conversely. It does not measure how difficult it is to transform the linear model itself into the quadratic model. That second question belongs to the analysis of descriptions as descriptions, and is closer to the role played by surfeit in Chapter 10. ■

The normalized definition makes inaccuracy comparable across representations of different sizes.

Proposition 9.1.1 For every representation $r \in \mathcal{R}$ and every description $d \in \mathcal{D}$,

$$0 \leq \iota(d, r) \leq 1 + o(1),$$

where the $o(1)$ term vanishes as $\max\{K(r), K(\Gamma(d))\}$ tends to infinity.

Proof. The numerator is non-negative because Kolmogorov complexities are non-negative. Therefore $\iota(d, r) \geq 0$.

For the upper bound, let $x = r$ and $y = \Gamma(d)$. We use the standard inequalities

$$K(x | y) \leq K(x) + O(1), \quad K(y | x) \leq K(y) + O(1).$$

Hence

$$\max\{K(x | y), K(y | x)\} \leq \max\{K(x), K(y)\} + O(1).$$

Dividing by $\max\{K(x), K(y)\}$, we obtain

$$\iota(d, r) \leq 1 + \frac{O(1)}{\max\{K(x), K(y)\}}.$$

For sufficiently complex strings, the additive term becomes negligible. Therefore $0 \leq \iota(d, r) \leq 1 + o(1)$. ■

In the rest of the book, and especially in applications, we treat inaccuracy as a normalized quantity in the interval $[0, 1]$, ignoring machine-dependent additive constants.

Inaccuracy vanishes when the description produces the intended representation.

Proposition 9.1.2 Let $r \in \mathcal{R}$ be a representation and $d \in \mathcal{D}$ a description. If $d \in \mathcal{D}_r$, then

$$\iota(d, r) = O\left(\frac{1}{K(r)}\right).$$

In particular, inaccuracy is zero up to the usual additive constants of Kolmogorov complexity.

Proof. If $d \in \mathcal{D}_r$, then $\Gamma(d) = r$. Therefore

$$K(r \mid \Gamma(d)) = K(r \mid r) = O(1),$$

and

$$K(\Gamma(d) \mid r) = K(r \mid r) = O(1).$$

Hence the numerator of $\iota(d, r)$ is $O(1)$, while the denominator is $K(r)$. Therefore

$$\iota(d, r) = O\left(\frac{1}{K(r)}\right).$$

■

Conversely, if $\iota(d, r)$ is close to zero, then r and $\Gamma(d)$ are algorithmically close. This does not imply literal equality of the two strings, but it means that each can be reconstructed from the other using only a small amount of additional information.

Proposition 9.1.3 Let $r \in \mathcal{B}^*$ and $d \in \mathcal{D}$. If $\iota(d, r) \leq \varepsilon$, then

$$K(r \mid \Gamma(d)) \leq \varepsilon \max\{K(r), K(\Gamma(d))\}$$

and

$$K(\Gamma(d) \mid r) \leq \varepsilon \max\{K(r), K(\Gamma(d))\}.$$

Proof. The result follows immediately from the definition of $\iota(d, r)$, since the numerator is the maximum of the two conditional complexities. ■

This proposition clarifies the meaning of small inaccuracy. A description with low inaccuracy produces an output that is close to the target representation in both directions: the output contains almost all the algorithmic information needed to reconstruct r , and it contains little independent information not already present in r .

Finally, inaccuracy extends immediately to joint representations. If $r, s \in \mathcal{R}$, their joint representation is the concatenated or self-delimited string $\langle r, s \rangle$. We then define

$$\iota(d, \langle r, s \rangle) = \frac{\max\{K(\langle r, s \rangle \mid \Gamma(d)), K(\Gamma(d) \mid \langle r, s \rangle)\}}{\max\{K(\langle r, s \rangle), K(\Gamma(d))\}}.$$

No new concept is required: the joint representation is itself a finite string.

9.2 Inaccuracy and Description Deficiency

The main role of inaccuracy in the theory of nescience is to approximate the deficiency of a description. Recall that the deficiency of a description measures how much information must be added to a given description in order to obtain a minimal description of the target representation (see Section 6.1). If $\mathcal{D}_r^*$ denotes the set of minimal descriptions of r , then the deficiency of d with respect to r is of the form

$$d_D(r, d) = \min_{d^* \in \mathcal{D}_r^*} K(d^* \mid d).$$

This is a description-level quantity: it compares d with a minimal program for r .

By contrast, inaccuracy is an output-level quantity: it compares $\Gamma(d)$ with r . It does not ask whether the program d is close to a minimal program. It asks whether the output produced by d is close to the intended representation. Therefore, inaccuracy is not identical to description deficiency. It is a computable proxy for one of its observable consequences.

The relation between the two quantities can be explained as follows. If the output $\Gamma(d)$ is far from r , then any description obtained by correcting d must somehow provide the missing information needed to recover r . Thus, high inaccuracy is evidence of description deficiency. If $\Gamma(d)$ is close to r , then only a small amount of output-level correction is needed, and the description is accurate even if it may still be unnecessarily long.

The following proposition gives a simple upper bound on the cost of producing some exact description of r from d .

Proposition 9.2.1 Let $r \in \mathcal{R}$ and $d \in \mathcal{D}$. There exists a exact description $\bar{d} \in \mathcal{D}_r$ such that

$$K(\bar{d} \mid d) \leq K(r \mid \Gamma(d)) + O(1).$$

Proof. Given d , we can compute $\Gamma(d)$. Let p be a shortest program that transforms $\Gamma(d)$ into r . Thus

$$l(p) = K(r \mid \Gamma(d)).$$

Construct a new description $\bar{d}$ that first executes d , obtains $\Gamma(d)$, then applies p to obtain r , and finally outputs r . This construction is computable from d and p , up to a fixed overhead independent of r and d . Therefore

$$K(\bar{d} \mid d) \leq l(p) + O(1) = K(r \mid \Gamma(d)) + O(1).$$

Since $\bar{d}$ outputs r , we have $\bar{d} \in \mathcal{D}_r$. ■

This proposition shows that the one-sided correction term $K(r \mid \Gamma(d))$ bounds the amount of information needed to transform the current description into some exact description of r . However, it does not guarantee that the resulting description is minimal. The correction may produce a exact but highly redundant description.

This observation explains why inaccuracy cannot serve as a substitute for mismodel. Inaccuracy assesses only whether the output produced by a description coincides with, or is algorithmically close to, the intended representation. It does not assess whether the description itself attains the minimal form required for zero mismodel.

Proposition 9.2.2 A description may have zero inaccuracy with respect to a representation r while still having positive description surplus, and hence positive mismodel.

Proof. Let $d \in \mathcal{D}_r$ be any exact description of r that is not minimal. Then $\Gamma(d) = r$, so the inaccuracy of d with respect to r is zero up to the usual additive constants. However, because d is not minimal, there exists a minimal description $d^* \in \mathcal{D}_r^*$ that reconstructs the same representation using less descriptive information. Hence d contains unnecessary descriptive content relative to a minimal description of r , and therefore its description surplus is positive. Since mismodel combines description deficiency and description surplus, the mismodel of d need not be zero. ■

This proposition is important for interpreting inaccuracy correctly. A highly accurate model may still be a poor scientific model if it achieves its accuracy through excessive complexity. This is the familiar problem of overfitting in machine learning, but the point is more general: correctness of output is not the same as economy of explanation.

The next proposition gives the complementary observation. Low description surplus does not imply low inaccuracy.

Proposition 9.2.3 A description may have low description surplus while still having high inaccuracy.

Proof. Let d be a very short description that outputs a simple string s , and let r be a complex representation unrelated to s . The description d may be close to minimal, or even minimal, for producing its own output s . Hence, when considered as a description of s , it may have low description surplus. Nevertheless, if $s = \Gamma(d)$ is algorithmically far from the intended representation (r), then $K(r \mid s)$ will be large, and the inaccuracy $\iota(d, r)$ will be high. Therefore, the absence of substantial surplus in a description does not guarantee that its output is accurate with respect to the intended representation. ■

Inaccuracy and description surplus are therefore independent: a description may be accurate but unnecessarily complex, or economical while still failing to produce the intended representation.

9.3 Practical Approximation

The definition of inaccuracy relies on Kolmogorov complexity and is therefore non-computable. Nevertheless, it is more accessible in practice than mismodel deficiency, because it does not require identifying minimal descriptions. Instead, it only requires comparing two strings: the target representation r and the output $\Gamma(d)$ produced by the description.

In applications, we approximate Kolmogorov complexity by a computable function $\hat{K}$. This approximation depends on the domain. For textual representations, $\hat{K}$ may be the length of a compressed file. For empirical datasets, $\hat{K}$ may be the length of an empirical code. For predictive models, $\hat{K}(r | \Gamma(d))$ may be approximated by the amount of information needed to encode the residuals between the observed data and the model predictions.

The normalized approximation used throughout this book is

$$\hat{\imath}(d, r) = \frac{\hat{K}(r, \Gamma(d)) - \min \{ \hat{K}(r), \hat{K}(\Gamma(d)) \}}{\max \{ \hat{K}(r), \hat{K}(\Gamma(d)) \}}.$$

This expression is useful because it avoids the direct estimation of conditional Kolmogorov complexities. Instead, it estimates the shared and non-shared information between the two strings using their individual and joint compressed lengths.

Definition 9.3.1 — Estimated Inaccuracy. Let $r \in \mathcal{R}$ be a representation, $d \in \mathcal{D}$ a description, and $\hat{K}$ a computable approximation of Kolmogorov complexity. We define the *estimated inaccuracy* of d with respect to r as

$$\hat{\imath}(d, r) = \frac{\hat{K}(r, \Gamma(d)) - \min \{ \hat{K}(r), \hat{K}(\Gamma(d)) \}}{\max \{ \hat{K}(r), \hat{K}(\Gamma(d)) \}}.$$

The quality of $\hat{\imath}$ depends on the quality of $\hat{K}$. If the compressor or coding method captures the regularities relevant to the representation, then the estimate can be meaningful. If the compressor fails to capture those regularities, then the estimate may be misleading.

Proposition 9.3.1 Suppose that $\hat{K}$ approximates K with error at most ε on the strings r , $\Gamma(d)$, and $\langle r, \Gamma(d) \rangle$. Then the estimated inaccuracy $\hat{\imath}(d, r)$ differs from

the ideal normalized information-distance form by a quantity bounded by a term depending on ε and on

$$\max\{K(r), K(\Gamma(d))\}.$$

In particular, for large representations and small relative compression error, $\hat{i}(d, r)$ is a stable approximation.

Proof. The estimated numerator differs from the ideal numerator only through the approximation errors in the joint complexity and in the two individual complexities. Since the operations involved are addition, subtraction, minimum, and maximum, the accumulated error is bounded by a constant multiple of ε . The denominator is similarly perturbed by at most ε . Dividing by a large denominator makes the relative effect of these errors small. Therefore the estimate becomes stable when the strings are sufficiently large and the approximation errors are small relative to their complexities. ■

The practical interpretation is straightforward. Estimated inaccuracy is reliable only relative to a chosen encoding scheme. This is not a defect, but a necessary feature of the theory: in applications, every measurement of algorithmic information must be mediated by a concrete coding procedure.

■ **Example 9.2** In a regression problem, the representation r may be a dataset of observed response values, while $\Gamma(d)$ is the vector of predictions produced by a model. The inaccuracy of the model can be estimated by encoding the residuals between observations and predictions. If the residuals are small and structured, the correction cost is low. If the residuals are large or irregular, the correction cost is high. ■

This generalizes the ordinary notion of prediction error. Instead of measuring only numerical deviation, inaccuracy measures the amount of information needed to correct the model output into the observed representation.

9.4 Conditional Inaccuracy

In this section, we delve deeper into the concept of inaccuracy by considering its application to conditional descriptions. Specifically, we explore the inaccuracy of a description when evaluated in conjunction with pre-existing background knowledge, a notion we term *conditional inaccuracy*. As we will see, the inaccuracy of a conditional description can never exceed that of its unconditional counterpart; at worst, it remains unchanged. This property makes conditional inaccuracy a useful tool for evaluating new concepts or models and for assessing their explanatory power with respect to the entity of interest.

In Definition 4.7.5, we introduced the concept of a conditional description d for a representation r , given an arbitrary background string s . This is denoted by $d \mid s$ and defined as the self-delimiting concatenated string $\langle d, s \rangle$, where $d = \langle TM, a \rangle$ and $TM(\langle a, s \rangle) = r$. If $TM(\langle a, s \rangle) = r'$ with $r \neq r'$, then $d \mid s$ is referred to as an *inaccurate* conditional description of r .

It is important to note that $d \mid s$ must be defined for all possible background strings s . Additionally, we refer to the case in which s is the empty string, denoted $d \mid \lambda$, as the *unconditional* version of the description.

Building on the concept of inaccuracy introduced in Definition 9.1.1, we now define the notion of *conditional inaccuracy* to capture the error introduced when using an inaccurate conditional description.

Definition 9.4.1 Let $r \in \mathcal{B}^*$ be a representation, $s \in \mathcal{B}^*$ a background string, and $d \mid s$ an inaccurate conditional description. We define the *conditional inaccuracy* of the description d for the representation r given the string s , denoted by $\iota(d \mid s, r)$, as:

$$\iota(d \mid s, r) = \frac{\max \{K(r \mid \Gamma(d \mid s)), K(\Gamma(d \mid s) \mid r)\}}{\max \{K(r), K(\Gamma(d \mid s))\}}.$$

Conditional inaccuracy is thus defined as the normalized information distance between the original representation r and the string produced by the conditional description $d \mid s$.

As a normalized measure, the conditional inaccuracy of a description lies within the interval $[0, 1]$, up to the usual additive constants of Kolmogorov complexity.

Proposition 9.4.1 Let $r \in \mathcal{B}^*$ be a representation, $s \in \mathcal{B}^*$ a string, and $d \mid s$ a conditional description of r given s . Then

$$0 \leq \iota(d \mid s, r) \leq 1 + o(1).$$

Proof. This follows directly from the same argument used in Proposition 9.1.1, applied to the strings r and $\Gamma(d \mid s)$. ■

The conditional inaccuracy assumes a value of zero if the conditional description $d \mid s$ correctly produces the representation r .

Proposition 9.4.2 Let $r \in \mathcal{B}^*$ be a representation, $s \in \mathcal{B}^*$ a string, and $d \mid s$ a conditional description of r given s , where $d = \langle TM, a \rangle$. If

$$TM(\langle a, s \rangle) = r,$$

then

$$\iota(d \mid s, r) = O\left(\frac{1}{K(r)}\right).$$

Proof. If $TM(\langle a, s \rangle) = r$, then $\Gamma(d \mid s) = r$, which implies that

$$K(r \mid \Gamma(d \mid s)) = O(1)$$

and

$$K(\Gamma(d \mid s) \mid r) = O(1).$$

The numerator is therefore $O(1)$, while the denominator is $K(r)$. Hence

$$\iota(d \mid s, r) = O\left(\frac{1}{K(r)}\right).$$

■

Incorporating established prior knowledge into research should not increase the inaccuracy of a description when that prior knowledge is used appropriately. If the background knowledge is relevant to the representation being described, the description may use it to produce a more accurate output. If the background knowledge is irrelevant, it can be ignored.

Theorem 9.4.3 Let $r \in \mathcal{B}^*$ be a representation, and let $d \in \mathcal{D}$ be a conditional description of r . Then, under the assumption that the conditional description can ignore irrelevant background strings,

$$\iota(d \mid s, r) \leq \iota(d, r)$$

for all strings $s \in \mathcal{B}^*$, up to the usual additive constants.

Proof. Since $\iota(d, r)$ is equivalent to $\iota(d \mid \lambda, r)$, it is enough to compare the output produced with background string s and the output produced with the empty background string λ .

If the conditional description can ignore irrelevant background information, then the computation using s can simulate the computation using λ . Moreover, conditioning on additional information cannot increase Kolmogorov complexity beyond an additive constant:

$$K(r \mid \langle \Gamma(d), s \rangle) \leq K(r \mid \Gamma(d)) + O(1).$$

Thus, when s is useful, it may reduce the correction cost; when it is not useful, it can be ignored. Therefore the conditional inaccuracy cannot exceed the unconditional inaccuracy, up to the usual constants. ■

Theorem 9.4.3 represents a foundational intuition in the theory of nescience. It supports the idea that background knowledge can reduce the cost of producing accurate descriptions. In practical applications, one typically focuses on prior knowledge directly related to the entity under study. However, the theorem also suggests that exploratory use of background knowledge from other domains need not be harmful, provided the description can ignore irrelevant information.

■ **Example 9.3** The P versus NP problem is one of the most important open problems in theoretical computer science. Constructing a complete solution from scratch would require an enormous amount of information. However, a conditional description may rely on background knowledge from complexity theory, formal languages, computability theory, and proof theory. Such background knowledge does not solve the problem by itself, but it can reduce the amount of information that must be explicitly included in any proposed description of a solution. ■

Finally, conditional inaccuracy extends immediately to joint representations. Given two representations $r, t \in \mathcal{B}^*$, we may write

$$\iota(d \mid s, \langle r, t \rangle) = \frac{\max \{K(\langle r, t \rangle \mid \Gamma(d \mid s)), K(\Gamma(d \mid s) \mid \langle r, t \rangle)\}}{\max \{K(\langle r, t \rangle), K(\Gamma(d \mid s))\}}.$$

Again, no new concept is needed, since $\langle r, t \rangle$ is itself a finite string.

9.5 Decreasing Inaccuracy

The purpose of defining inaccuracy is not merely to measure the error of a description, but to guide its reduction. Given a target representation r , we may compare two candidate descriptions $d_1, d_2 \in \mathcal{D}$ by examining whether the second produces an output closer to r than the first.

Definition 9.5.1 Let $r \in \mathcal{R}$ be a representation, and let $d_1, d_2 \in \mathcal{D}$ be two descriptions. We define the *absolute variation of inaccuracy* from d_1 to d_2 , with respect to r , as

$$\Gamma_i^a(d_1, d_2, r) = \iota(d_1, r) - \iota(d_2, r).$$

If $\Gamma_i^a(d_1, d_2, r) > 0$, then d_2 is more accurate than d_1 with respect to r . If $\Gamma_i^a(d_1, d_2, r) < 0$, then the new description is less accurate. Since inaccuracy is normalized, the absolute variation lies between -1 and 1 , up to the usual constants.

It is often useful to express the improvement relative to the initial inaccuracy.

Definition 9.5.2 Let $r \in \mathcal{R}$ be a representation, and let $d_1, d_2 \in \mathcal{D}$ be two descriptions with $\iota(d_1, r) \neq 0$. We define the *relative variation of inaccuracy* from d_1 to d_2 , with respect to r , as

$$\Gamma_t^r(d_1, d_2, r) = \frac{\iota(d_1, r) - \iota(d_2, r)}{\iota(d_1, r)}.$$

A value of 0 indicates no change. A value of 1 indicates complete elimination of inaccuracy. Negative values indicate that the new description is less accurate than the previous one.

Relative variation must be interpreted carefully. When $\iota(d_1, r)$ is already very small, a tiny absolute change can produce a large relative variation. Thus, both absolute and relative improvements should be considered.

A second way to reduce inaccuracy is to change the representation rather than the description. This may be useful when the current representation is difficult to model. However, changing the representation may also affect miscoding, since the new representation may encode the entity better or worse than the original one.

Definition 9.5.3 Let $r_1, r_2 \in \mathcal{R}$ be two representations, and let $d \in \mathcal{D}$ be a description. We define the *absolute variation of inaccuracy* from r_1 to r_2 , with respect to d , as

$$\Gamma_t^a(d, r_1, r_2) = \iota(d, r_1) - \iota(d, r_2).$$

A positive value means that the fixed description d is more accurate with respect to r_2 than with respect to r_1 . This comparison does not say that r_2 is a better representation of the entity. It only says that d models r_2 more accurately. Whether this change is scientifically desirable also depends on the miscoding of the representations.

Definition 9.5.4 Let $r_1, r_2 \in \mathcal{R}$ be two representations, and let $d \in \mathcal{D}$ be a description with $\iota(d, r_1) \neq 0$. We define the *relative variation of inaccuracy* from r_1 to r_2 , with respect to d , as

$$\Gamma_t^r(d, r_1, r_2) = \frac{\iota(d, r_1) - \iota(d, r_2)}{\iota(d, r_1)}.$$

This quantity measures the proportional reduction in inaccuracy obtained by replacing r_1 with r_2 while keeping the description fixed.

Inaccuracy can always be reduced trivially by choosing the representation to be the output of the description. If $r_2 = \Gamma(d)$, then $\iota(d, r_2) = 0$. But such a move may dramatically increase miscoding, because r_2 may be a poor representation of the entity. This observation illustrates why inaccuracy cannot be optimized in

isolation.

Proposition 9.5.1 For every description $d \in \mathcal{D}$, there exists a representation $r \in \mathcal{B}^*$ such that

$$I(d, r) = 0$$

up to additive constants.

Proof. Take $r = \Gamma(d)$. Then d outputs r , so $d \in \mathcal{D}_r$. By Proposition 9.1.2, the inaccuracy is zero up to additive constants. ■

This proposition shows why the theory of nescience must consider miscoding, inaccuracy, and surfeit together. A model can be made perfectly accurate by redefining the target representation to match its output, but that does not mean that the resulting hypothesis represents the intended entity correctly.

9.6 Rate of Change

In the previous section we considered two ways to reduce inaccuracy: changing the description while keeping the representation fixed, and changing the representation while keeping the description fixed. In scientific practice, both components may change simultaneously. A new hypothesis may introduce a new representation and a new description.

Let

$$\mathbf{x}_1 = (r_1, d_1), \quad \mathbf{x}_2 = (r_2, d_2)$$

be two hypotheses about the same entity. Moving from $\mathbf{x}_1$ to $\mathbf{x}_2$ may reduce inaccuracy while increasing miscoding, or reduce miscoding while increasing inaccuracy. This produces a trade-off.

Definition 9.6.1 Let $e \in \mathcal{E}$ be an entity, and let

$$\mathbf{x}_1 = (r_1, d_1), \quad \mathbf{x}_2 = (r_2, d_2)$$

be two hypotheses. We define the *inaccuracy-miscoding rate of change* from $\mathbf{x}_1$ to $\mathbf{x}_2$ as

$$\Gamma_{I\mu}(\mathbf{x}_1, \mathbf{x}_2) = \frac{I(d_2, r_2) - I(d_1, r_1)}{\mu(r_2) - \mu(r_1)},$$

provided that $\mu(r_2) - \mu(r_1) \neq 0$.

The ratio measures how much inaccuracy changes per unit change in miscoding. If the denominator is small, the ratio can become unstable, so it should always be interpreted together with the absolute changes

$$\iota(d_2, r_2) - \iota(d_1, r_1)$$

and

$$\mu(r_2) - \mu(r_1).$$

■ **Example 9.4** Suppose that

$$\iota(d_1, r_1) = 0.40, \quad \mu(r_1) = 0.15,$$

and

$$\iota(d_2, r_2) = 0.20, \quad \mu(r_2) = 0.25.$$

Then inaccuracy decreases by 0.20, while miscoding increases by 0.10. The rate of change is

$$\Gamma_{\iota\mu}(\mathbf{x}_1, \mathbf{x}_2) = \frac{0.20 - 0.40}{0.25 - 0.15} = \frac{-0.20}{0.10} = -2.$$

The negative value indicates a trade-off: the new hypothesis is more accurate, but its representation is more miscoded. ■

■ **Example 9.5** Suppose instead that

$$\iota(d_1, r_1) = 0.35, \quad \mu(r_1) = 0.20,$$

and

$$\iota(d_2, r_2) = 0.30, \quad \mu(r_2) = 0.2001.$$

Then

$$\Gamma_{\iota\mu}(\mathbf{x}_1, \mathbf{x}_2) = \frac{0.30 - 0.35}{0.2001 - 0.20} = \frac{-0.05}{0.0001} = -500.$$

The ratio is large in magnitude, but the actual changes are small. This illustrates why rate-of-change quantities should not be used without also inspecting the underlying absolute variations. ■

The inaccuracy-miscoding trade-off is a two-objective projection of the full nescience problem. In the full theory, we must also consider surfeit. A hypothesis that reduces both miscoding and inaccuracy may still be undesirable if it introduces a large amount of unnecessary descriptive complexity.

9.7 Inaccuracy of Areas

The notion of inaccuracy can be extended from individual representations to research areas. Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area, and suppose that a finite subset of effectively available representations is given by

$$\hat{\mathcal{A}} = \{r_1, r_2, \dots, r_n\}.$$

We encode these representations into a single self-delimited string

$$\langle r_1, r_2, \dots, r_n \rangle.$$

A description of the area is then evaluated according to how accurately it reproduces this joint representation.

Definition 9.7.1 Let $\mathcal{A} \subseteq \mathcal{E}$ be a research area with known finite representation set

$$\hat{\mathcal{A}} = \{r_1, r_2, \dots, r_n\},$$

and let $d \in \mathcal{D}$ be a description. We define the *inaccuracy of the area* given d , denoted by $\iota(d, \hat{\mathcal{A}})$, as

$$\iota(d, \hat{\mathcal{A}}) = \frac{\max \{K(\langle r_1, r_2, \dots, r_n \rangle \mid \Gamma(d)), K(\Gamma(d) \mid \langle r_1, r_2, \dots, r_n \rangle)\}}{\max \{K(\langle r_1, r_2, \dots, r_n \rangle), K(\Gamma(d))\}}.$$

This definition treats the available representations of the area as a single composite representation. The description d is accurate for the area if its output is algorithmically close to that composite string.

Proposition 9.7.1 For every finite known subset

$$\hat{\mathcal{A}} = \{r_1, r_2, \dots, r_n\}$$

and every description $d \in \mathcal{D}$,

$$0 \leq \iota(d, \hat{\mathcal{A}}) \leq 1 + o(1).$$

Proof. The string $\langle r_1, r_2, \dots, r_n \rangle$ is an element of $\mathcal{B}^*$. The result therefore follows by applying Proposition 9.1.1 to the two strings

$$\langle r_1, r_2, \dots, r_n \rangle$$

and

$$\Gamma(d).$$



The inaccuracy of an area measures how much information is needed to correct a description of a collection of related representations. In practical applications, this may be useful when a scientific field is represented by a corpus of documents, a family of datasets, a group of models, or a structured collection of experimental results.

References

A good introduction to the study of uncertainties and error analysis in science, particularly in physics, chemistry, and engineering, is the textbook by Taylor [TAY22]. Another accessible reference on measurement and uncertainty, aimed at undergraduate students in science and technology, is the book by Hughes and Hase [HH10].

The formal background for the definition of inaccuracy comes from algorithmic information theory and, in particular, from the theory of Kolmogorov complexity and normalized information distance. The standard reference is the book by Li and Vitányi [li2008introduction]. These ideas also connect naturally with the minimum description length principle and related coding-based approaches to statistical learning.

From a philosophical perspective, Popper's work [Pop14] remains relevant because it emphasizes the role of error, refutation, and correction in scientific progress. In the present framework, inaccuracy provides an algorithmic way to quantify one specific kind of error: the discrepancy between what a description produces and the representation it is intended to reproduce.



10. Surfeit

*Everything should be made as simple as possible,
but not simpler.*
Albert Einstein

In Chapter 6 we introduced the concept of *mismodel* as the discrepancy between a description currently available in scientific practice and the ideal class of minimal descriptions of a representation. If $r \in \mathcal{R}$ is a representation and $d \in \mathcal{D}$ is a valid description, then the mismodel of d with respect to r is determined by two components: the information still missing from d , and the information contained in d that is not needed in any minimal description of r .

The theoretical definition of mismodel is useful because it identifies the two fundamental ways in which a description can fail. A description may be *deficient*: it may not contain enough information to reconstruct the representation. Or it may be *excessive*: it may contain unnecessary descriptive content. However, mismodel is not computable in general. The main reason is that minimal descriptions are not known in scientific practice. Moreover, even if a candidate minimal description were available, verifying its minimality would require solving problems related to Kolmogorov complexity, which is non-computable in general.

For this reason, mismodel cannot usually be computed directly. Instead, it must be approximated through operational quantities. There are two natural ways

to do this. The first one, studied in Chapter 9, is to evaluate how far the output produced by the current description is from the representation r . This gives rise to the notion of *inaccuracy*. Inaccuracy is therefore a practical approximation to the deficiency component of mismodel: it measures the extent to which the current model fails to reproduce the representation.

The second approach, studied in the present chapter, is to evaluate how far the current description is from being minimal as a description. This gives rise to the notion of *surfeit*. Surfeit is intended as a practical approximation to the surplus component of mismodel. Rather than comparing the description with an unknown minimal description of r , it examines the internal compressibility of the description itself. A description with a large amount of removable structure, repeated content, avoidable parameters, or unnecessary algorithmic machinery is unlikely to be close to minimal. Conversely, a description that is hard to compress is a better candidate for being near minimal, although not necessarily an accurate one.

This distinction is essential. Inaccuracy evaluates the relation between the description and the representation. Surfeit evaluates the description itself. A model may be concise but inaccurate; it may reproduce the representation well but be unnecessarily complicated; or it may be both accurate and concise. The theory of nescience requires these dimensions to be analyzed jointly.

The central idea of this chapter is therefore essentially practical: since the surplus of a description cannot normally be computed, we approximate it by measuring how much the description can still be compressed. This approximation is not perfect. It detects syntactic and algorithmic redundancy, but it may fail to detect incompressible irrelevant information. Nevertheless, it captures an important and widely used principle of scientific practice: a better understanding of a phenomenon should allow us to replace unnecessarily complex models by simpler, more economical ones without increasing their inaccuracy.

In this chapter we introduce surfeit formally, study its main properties, explain its relation to the theoretical surplus of descriptions, and discuss how it can be approximated in practice. We also examine how surfeit may be decreased, how it interacts with inaccuracy, and how the concept can be extended to research areas.

10.1 Surfeit

Let $d \in \mathcal{D}$ be a description. Recall that a description is a finite string, usually interpreted as a pair $\langle TM, a \rangle$, where TM is a Turing machine and a is an input such that $TM(a)$ halts. The length $l(d)$ measures the size of the description, while the Kolmogorov complexity $K(d)$ measures the length of the shortest program capable of producing d .

If $K(d)$ is much smaller than $l(d)$, then d contains compressible structure.

In practical terms, this means that some part of the description is not being expressed economically. This compressible excess may correspond to repeated assumptions, redundant parameters, unnecessarily verbose formalism, duplicated rules, or avoidable implementation details. We call this excess the *surfeit* of the description.

Definition 10.1.1 Let $d \in \mathcal{D}$ be a valid description. We define the *absolute surfeit* of d , denoted by $\Sigma(d)$, as

$$\Sigma(d) = l(d) - K(d).$$

Absolute surfeit measures, in bits, how much longer the current description is than its shortest possible description. However, absolute values are difficult to compare across descriptions of very different sizes. For this reason, we introduce a relative version.

Definition 10.1.2 — Surfeit. Let $d \in \mathcal{D}$ be a valid description. We define the *surfeit* of d , denoted by $\sigma(d)$, as

$$\sigma(d) = 1 - \frac{K(d)}{l(d)}.$$

The quantity $\sigma(d)$ is a normalized measure of the internal compressibility of the description. If $\sigma(d)$ is close to 0, then d is close to incompressible. If $\sigma(d)$ is close to 1, then most of the symbols of d are algorithmically redundant.

The definition deliberately does not include the representation r . This is the crucial difference between surfeit and the theoretical surplus of a description. The surplus of d with respect to r compares d with unknown minimal descriptions of r . Surfeit instead asks a weaker but more practical question: how far is d , as a string, from being minimally expressed?

■ **Example 10.1** Consider a model for classifying emails as spam or not spam. Suppose the model contains many duplicated rules, repeated feature transformations, unused parameters, and several branches of code that never affect the final prediction. Such a model may still achieve good classification accuracy, but its description is unnecessarily long. A compressed or simplified version of the same model could produce the same predictions with fewer symbols. In this case, the model has high surfeit.

By contrast, a shorter decision tree that captures the relevant distinctions with few rules has lower surfeit. This does not imply that the decision tree is more accurate. It only means that, as a description, it contains less compressible excess.

■

The surfeit of a description is always a number between 0 and 1, up to the usual additive constants involved in Kolmogorov complexity.

Proposition 10.1.1 For every non-empty description $d \in \mathcal{D}$,

$$0 \leq \sigma(d) \leq 1.$$

Proof. Since $K(d) \leq l(d) + O(1)$, and since $K(d) \geq 0$, we have, ignoring the usual additive constant,

$$0 \leq \frac{K(d)}{l(d)} \leq 1.$$

Therefore,

$$0 \leq 1 - \frac{K(d)}{l(d)} \leq 1.$$

Hence $0 \leq \sigma(d) \leq 1$. ■

The lower endpoint corresponds to descriptions that cannot be compressed further.

Proposition 10.1.2 Let $d \in \mathcal{D}$ be a valid description. Then $\sigma(d) = 0$ if and only if $K(d) = l(d)$.

Proof. By definition,

$$\sigma(d) = 1 - \frac{K(d)}{l(d)}.$$

Thus $\sigma(d) = 0$ if and only if

$$1 - \frac{K(d)}{l(d)} = 0,$$

which is equivalent to $K(d) = l(d)$. ■

A description with zero surfeit is incompressible. This does not mean that it is correct, useful, or accurate. It only means that the description itself contains no algorithmically detectable redundancy. A random string has zero surfeit, but it need not describe anything meaningful. For this reason, surfeit must always be interpreted together with inaccuracy.

■ **Example 10.2** Suppose that d_1 is a very short program that always outputs the same constant value, regardless of the input. The description may have very low surfeit because it is concise. However, if the representation r contains complex experimental data, the output of d_1 may be far from r , and therefore its inaccuracy may be high.

Conversely, suppose that d_2 is a large program that stores the entire dataset r explicitly and prints it verbatim. Then its inaccuracy may be zero, but its surfeit may be high if the dataset has a simpler generative structure that the program fails to capture. ■

This example shows why surfeit cannot replace inaccuracy. Surfeit measures whether the description is economical. Inaccuracy measures whether the description reconstructs the representation. A good scientific model should be evaluated with respect to both dimensions.

10.2 Surfeit and Description Surplus

The theoretical surplus of a description was introduced in Chapter 6. If $r \in \mathcal{R}$ is a representation and $\mathcal{D}_r^*$ denotes the set of minimal descriptions of r , the surplus of d with respect to r measures the amount of information contained in d that is not needed once a minimal description of r is available. In its algorithmic form, it can be expressed as

$$s_D(r, d) = \min_{d^* \in \mathcal{D}_r^*} K(d \mid d^*).$$

This quantity is conceptually precise, but it is not operational in general. The set $\mathcal{D}_r^*$ is usually unknown, because minimal descriptions of r are not available in scientific practice. Moreover, determining whether a description is minimal would require solving a Kolmogorov-complexity problem. Hence, $s_D(r, d)$ is normally non-computable.

Surfeit provides a weaker but computable-oriented proxy:

$$s_D(r, d) \text{ is approximated by } \sigma(d).$$

The approximation is based on the following intuition. If a description contains avoidable descriptive material, this excess will often manifest itself as internal compressibility. In such cases, measuring how much d can be compressed gives information about how much unnecessary content the description contains.

However, the approximation is not exact. The theoretical surplus $s_D(r, d)$ is semantic and representation-relative: it depends on what is unnecessary for describing r . Surfeit $\sigma(d)$ is syntactic and description-internal: it depends only on the compressibility of d . Therefore, surfeit can detect redundant structure, but it cannot detect all forms of irrelevant information.

Proposition 10.2.1 Let $r \in \mathcal{R}$ be a representation and $d \in \mathcal{D}$ a description. The value $\sigma(d)$ depends only on d , while $s_r(d)$ depends on both d and r . Hence, in general, surfeit cannot be an exact measure of description surplus.

Proof. By definition,

$$\sigma(d) = 1 - \frac{K(d)}{l(d)},$$

which depends only on the string d . On the other hand,

$$s_D(r, d) = \min_{d^* \in \mathcal{D}_r^*} K(d \mid d^*),$$

which depends on the set $\mathcal{D}_r^*$, and therefore on the representation r . Thus two different representations r_1 and r_2 may yield different values $s_D(r_1, d)$ and $s_D(r_2, d)$, while $\sigma(d)$ remains unchanged. Consequently, surfeit cannot coincide with description surplus in general. ■

The previous proposition clarifies the role of surfeit. It is not a replacement for the theoretical concept of surplus. It is a practical approximation that becomes informative under additional assumptions about the kind of unnecessary content present in the description.

The approximation is most reliable when the superfluous part of the description is itself compressible or predictable from the rest of the description.

Proposition 10.2.2 Let $d, u \in \mathcal{B}^*$ be strings, and suppose that u is computable from d up to an additive constant, that is, $K(u \mid d) = O(1)$. Then $K(du) \leq K(d) + O(1)$. Consequently,

$$\sigma(du) \geq 1 - \frac{K(d) + O(1)}{l(d) + l(u)}.$$

Proof. If $K(u \mid d) = O(1)$, then there exists a constant-size program that reconstructs u from d . Therefore, to produce du , it is enough to produce d and then apply this constant-size program to obtain u . Hence

$$K(du) \leq K(d) + O(1).$$

Since $l(du) = l(d) + l(u)$, we have

$$\sigma(du) = 1 - \frac{K(du)}{l(du)} \geq 1 - \frac{K(d) + O(1)}{l(d) + l(u)}.$$

■

This proposition formalizes a common situation: if a description contains a large block of information that can be generated from the rest of the description, then the description becomes more compressible and its surfeit increases. Repeated rules, duplicated assumptions, copied derivations, and predictable code fragments are typical examples.

However, there are also cases where surplus is not detected by surfeit.

Proposition 10.2.3 Let $d, u \in \mathcal{B}^*$ be strings such that u is incompressible and algorithmically independent of d , that is, $K(u) \geq l(u) - O(1)$ and $K(u \mid d) \geq l(u) - O(1)$. Then

$$K(du) \geq K(d) + l(u) - O(1).$$

In particular, appending u to d need not increase the surfeit of the resulting description.

Proof. Since u is algorithmically independent of d , the information required to reconstruct u is not contained in d . Therefore, any program that produces du must contain enough information to produce d and, in addition, enough information to produce u given d . Hence

$$K(du) \geq K(d) + K(u \mid d) - O(1).$$

Using $K(u \mid d) \geq l(u) - O(1)$, we obtain

$$K(du) \geq K(d) + l(u) - O(1).$$

Thus the appended string u may increase both the length and the Kolmogorov complexity of the description by approximately the same amount. In such a case, the ratio $K(du)/l(du)$ need not decrease, and the surfeit need not increase. ■

This limitation is important. A description may contain irrelevant information that is itself incompressible. Such information contributes to the theoretical surplus $s_r(d)$, because it is not needed to describe r , but it may not contribute to $\sigma(d)$, because it is not compressible. Surfeit is therefore a detector of compressible excess, not a universal detector of semantic irrelevance.

10.3 Practical Surfeit

The definition of surfeit still contains Kolmogorov complexity $K(d)$, which is non-computable. Therefore, even the description-internal version of surfeit cannot be computed exactly. In practice, we approximate $K(d)$ by the length of a compressed version of d , using a fixed compression method.

Let C be a compressor, and let $C(d)$ denote the length of the compressed version of d . We can then define a practical estimator of surfeit.

Definition 10.3.1 — Practical surfeit. Let C be a fixed compressor and let $d \in \mathcal{D}$ be a valid description. We define the *practical surfeit* of d with respect to C , denoted by $\hat{\sigma}_C(d)$, as

$$\hat{\sigma}_C(d) = 1 - \frac{C(d)}{l(d)}.$$

When necessary, the value is truncated to the interval $[0, 1]$.

The compressor C acts as a computable approximation to Kolmogorov complexity. Different compressors may yield different values, because each compressor detects different kinds of regularity. For this reason, practical surfeit should not be interpreted as an absolute quantity independent of the chosen compression method. It is a model-selection heuristic relative to a fixed coding procedure.

Proposition 10.3.1 Let C be a compressor satisfying $0 \leq C(d) \leq l(d)$ for every $d \in \mathcal{D}$. Then

$$0 \leq \hat{\sigma}_C(d) \leq 1.$$

Proof. Since $0 \leq C(d) \leq l(d)$, we have

$$0 \leq \frac{C(d)}{l(d)} \leq 1.$$

Therefore,

$$0 \leq 1 - \frac{C(d)}{l(d)} \leq 1.$$

Hence $0 \leq \hat{\sigma}_C(d) \leq 1$. ■

In applied contexts, d may be a computer program, a symbolic formula, a decision tree, a set of rules, a neural-network architecture together with its parameters, or a textual description of a scientific theory. The choice of compressor must be compatible with the representation of d . For example, a compressor designed for natural language may be appropriate for scientific texts, while a structural encoding may be more appropriate for decision trees or algebraic expressions.

■ **Example 10.3** In machine learning, a decision tree can be encoded as a finite string describing its internal nodes, thresholds, features, and leaf predictions. A tree with repeated subtrees, unnecessary branches, or unused tests will often be compressible. Its practical surfeit $\hat{\sigma}_C(d)$ will therefore be high. Pruning the tree may reduce its length while preserving most of its predictive accuracy, thereby decreasing surfeit without significantly increasing inaccuracy. ■

Practical surfeit is therefore closely related to familiar principles such as pruning, model compression, regularization, and minimum-description-length methods. The difference is conceptual: in the theory of nescience, surfeit is not merely a preference for simplicity. It is an approximation to the surplus component of mismodel.

10.4 Conditional Surfeit

We are interested in studying how the surfeit of a description for a representation is affected when some background knowledge is assumed. In particular, we examine the case where a description is constructed under the assumption that some prior information is already known. The surfeit of such a description is referred to as the *conditional surfeit*.

Definition 10.4.1 — Conditional surfeit. Let $r \in \mathcal{B}^*$ be a representation, $s \in \mathcal{B}^*$ a string, and $d_{r|s} \in \mathcal{D}$ a description of r given s . We define the *conditional surfeit* of the conditional description $d_{r|s}$, denoted by $\sigma(d_{r|s})$, as

$$\sigma(d_{r|s}) = 1 - \frac{K(r | s)}{l(d_{r|s})}.$$

This definition is primarily motivated by practical considerations. When we assume perfect knowledge of s , we can isolate and study the informational content of r that is not already covered by s , that is, the new knowledge introduced by r relative to s . Conditional surfeit thus allows us to quantify the conciseness of a description in light of what is already known.

Conditional surfeit, being a relative measure, is a number between 0 and 1, provided that $d_{r|s}$ is a valid conditional description whose length is at least $K(r | s)$.

Proposition 10.4.1 Let $r \in \mathcal{B}^*$ be a representation, $s \in \mathcal{B}^*$ be a string, and $d_{r|s} \in \mathcal{D}$ be a description of r given s . If

$$l(d_{r|s}) \geq K(r | s) > 0,$$

then

$$0 \leq \sigma(d_{r|s}) \leq 1.$$

Proof. Since $l(d_{r|s}) \geq K(r | s) > 0$, we have

$$0 < \frac{K(r | s)}{l(d_{r|s})} \leq 1.$$

Therefore,

$$0 \leq 1 - \frac{K(r | s)}{l(d_{r|s})} < 1.$$

Hence $0 \leq \sigma(d_{r|s}) \leq 1$. ■

Intuition tells us that the surfeit of a description should decrease when relevant background knowledge is assumed, provided that such background knowledge is itself represented without surfeit. However, a complete formal treatment requires a precise definition of conditional descriptions and of the operations by which background knowledge is made available to a model. For this reason, we leave the deeper study of conditional surfeit for future refinement of the theory.

In the same way that surfeit provides a weaker, description-internal version of description surplus, we can introduce conditional redundancy as a weaker version of conditional surfeit.

Definition 10.4.2 — Conditional redundancy. Let $s \in \mathcal{B}^*$ be a string, and let $d_{r|s} \in \mathcal{D}$ be a conditional description given s . We define the *conditional redundancy* of $d_{r|s}$, denoted by $\rho(d_{r|s})$, as

$$\rho(d_{r|s}) = 1 - \frac{K(d_{r|s})}{l(d_{r|s})}.$$

Conditional redundancy is a relative measure, and so it is a number between 0 and 1.

Proposition 10.4.2 We have that

$$0 \leq \rho(d_{r|s}) \leq 1$$

for every non-empty conditional description $d_{r|s}$.

Proof. Since $K(d_{r|s}) \leq l(d_{r|s})$, we have

$$\frac{K(d_{r|s})}{l(d_{r|s})} \leq 1.$$

Thus,

$$1 - \frac{K(d_{r|s})}{l(d_{r|s})} \geq 0.$$

Also, since $K(d_{r|s}) \geq 0$, we have

$$1 - \frac{K(d_{r|s})}{l(d_{r|s})} \leq 1.$$

Therefore,

$$0 \leq \rho(d_{r|s}) \leq 1.$$

■

Finally, we can extend the concepts of conditional surfeit and conditional redundancy to a finite number of background topics.

Definition 10.4.3 Let $t, s_1, s_2, \dots, s_n$ be a finite collection of topics, and let

$$d_{t|s_1^*, s_2^*, \dots, s_n^*}$$

be a conditional description of t given perfect descriptions of $s_1, s_2, \dots, s_n$. We define the *conditional surfeit* of this description as

$$\sigma(d_{t|s_1^*, s_2^*, \dots, s_n^*}) = 1 - \frac{K(t | s_1^*, s_2^*, \dots, s_n^*)}{l(d_{t|s_1^*, s_2^*, \dots, s_n^*})}.$$

The *conditional redundancy* of the same description is defined as

$$\rho(d_{t|s_1^*, s_2^*, \dots, s_n^*}) = 1 - \frac{K(d_{t|s_1^*, s_2^*, \dots, s_n^*})}{l(d_{t|s_1^*, s_2^*, \dots, s_n^*})}.$$

It is easy to show that the basic range properties of conditional surfeit and conditional redundancy apply to the case of multiple background topics as well, whenever the corresponding lengths are positive and the denominator is at least as large as the relevant conditional complexity.

10.5 Decreasing Surfeit

Our objective is to reduce the surfeit of a current description d_1 . This may be done by simplifying the description, removing unused assumptions, pruning unnecessary parameters, replacing a verbose formulation with a shorter equivalent one, or discovering a more compact modeling principle. The result is a new description d_2 .

The decrease of surfeit should not be interpreted in isolation. A shorter or less redundant description may become more inaccurate. Conversely, a more complex description may reduce inaccuracy. For this reason, surfeit reduction is only one part of the broader problem of reducing mismatch and, ultimately, nescience.

Definition 10.5.1 — Variation of surfeit. Let $d_1, d_2 \in \mathcal{D}$ be two descriptions. We define the *variation of surfeit* from d_1 to d_2 , denoted by $\Delta_\sigma(d_1, d_2)$, as

$$\Delta_\sigma(d_1, d_2) = \sigma(d_1) - \sigma(d_2).$$

A positive value of $\Delta_\sigma(d_1, d_2)$ indicates that d_2 has lower surfeit than d_1 . A negative value indicates that surfeit has increased.

Proposition 10.5.1 For every pair of descriptions $d_1, d_2 \in \mathcal{D}$,

$$-1 \leq \Delta_\sigma(d_1, d_2) \leq 1.$$

Proof. By Proposition 10.1.1, both $\sigma(d_1)$ and $\sigma(d_2)$ belong to $[0, 1]$. Therefore their difference belongs to $[-1, 1]$. ■

We can also introduce a relative measure of improvement.

Definition 10.5.2 — Relative variation of surfeit. Let $d_1, d_2 \in \mathcal{D}$ be two descriptions such that $\sigma(d_1) \neq 0$. We define the *relative variation of surfeit* from d_1 to d_2 , denoted by $\Delta_\sigma^r(d_1, d_2)$, as

$$\Delta_\sigma^r(d_1, d_2) = \frac{\sigma(d_1) - \sigma(d_2)}{\sigma(d_1)}.$$

A value of 0 indicates that there is no change in surfeit. A value of 1 indicates complete elimination of surfeit. Negative values indicate that surfeit has increased. As usual with relative quantities, this measure becomes unstable when $\sigma(d_1)$ is close to zero.

The following proposition gives a sufficient condition for decreasing surfeit through simplification.

Proposition 10.5.2 Let $d_1, d_2 \in \mathcal{D}$ be descriptions such that

$$l(d_2) < l(d_1)$$

and

$$K(d_2) = K(d_1) + O(1).$$

Then, for sufficiently large descriptions, d_2 has lower surfeit than d_1 , that is,

$$\sigma(d_2) < \sigma(d_1).$$

Proof. By definition,

$$\sigma(d_i) = 1 - \frac{K(d_i)}{l(d_i)}$$

for $i = 1, 2$. If $K(d_2) = K(d_1) + O(1)$ and $l(d_2) < l(d_1)$, then, ignoring the additive constant for sufficiently large descriptions,

$$\frac{K(d_2)}{l(d_2)} > \frac{K(d_1)}{l(d_1)}.$$

Therefore,

$$1 - \frac{K(d_2)}{l(d_2)} < 1 - \frac{K(d_1)}{l(d_1)}.$$

Hence $\sigma(d_2) < \sigma(d_1)$. ■

This proposition captures the ideal form of simplification: the description becomes shorter without losing algorithmic content. In scientific terms, this corresponds to removing unnecessary expressive material while preserving the same explanatory or predictive structure.

However, simplification can also remove useful information. If the shortened description no longer reconstructs the representation accurately, inaccuracy will increase. Therefore, the main practical problem is not merely to decrease surfeit, but to decrease surfeit without increasing inaccuracy too much.

10.6 Surfeit-Inaccuracy Rate of Change

In the previous section, we studied changes in surfeit. We now examine the interaction between surfeit and inaccuracy. This interaction is central because the two quantities approximate the two components of mismodel. Inaccuracy approximates the deficiency of the description with respect to the representation, while surfeit approximates its surplus.

A model that is too simple may have low surfeit but high inaccuracy. A model that is too complex may have low inaccuracy but high surfeit. Scientific modeling often consists in navigating this trade-off.

Definition 10.6.1 — Surfeit-inaccuracy rate of change. Let $r \in \mathcal{B}^*$ be a representation, and let $d_1, d_2 \in \mathcal{D}$ be two descriptions such that

$$\iota(d_2, r) - \iota(d_1, r) \neq 0.$$

We define the *surfeit-inaccuracy rate of change*, denoted by $\Delta_{\sigma\iota}(d_1, d_2, r)$, as

$$\Delta_{\sigma\iota}(d_1, d_2, r) = \frac{\sigma(d_2) - \sigma(d_1)}{\iota(d_2, r) - \iota(d_1, r)}.$$

This quantity measures how surfeit changes per unit change in inaccuracy when moving from d_1 to d_2 . The interpretation depends on the signs of the numerator and denominator.

- If both surfeit and inaccuracy decrease, then the move from d_1 to d_2 is unambiguously favorable.
- If both surfeit and inaccuracy increase, then the move is unambiguously unfavorable.
- If surfeit decreases while inaccuracy increases, then the move is a simplification that sacrifices some accuracy.
- If inaccuracy decreases while surfeit increases, then the move is an enrichment that sacrifices simplicity.

■ **Example 10.4** Let $r \in \mathcal{B}^*$ be a representation. Suppose that for d_1 we have

$$\sigma(d_1) = 0.40, \quad \iota(d_1, r) = 0.15,$$

and for d_2 we have

$$\sigma(d_2) = 0.20, \quad \iota(d_2, r) = 0.25.$$

Then

$$\Delta_{\sigma\iota}(d_1, d_2, r) = \frac{0.20 - 0.40}{0.25 - 0.15} = \frac{-0.20}{0.10} = -2.$$

The transition from d_1 to d_2 reduces surfeit by 0.20, but increases inaccuracy by 0.10. The value -2 indicates that two units of surfeit are removed for each unit of inaccuracy introduced. ■

The rate of change should be interpreted with caution when the denominator is close to zero. Small changes in inaccuracy may produce very large values of $\Delta_{\sigma\iota}$, even when the actual change in surfeit is modest.

■ **Example 10.5** Let

$$\sigma(d_1) = 0.35, \quad \iota(d_1, r) = 0.20,$$

and

$$\sigma(d_2) = 0.30, \quad \iota(d_2, r) = 0.2001.$$

Then

$$\Delta_{\sigma\iota}(d_1, d_2, r) = \frac{0.30 - 0.35}{0.2001 - 0.20} = \frac{-0.05}{0.0001} = -500.$$

The numerical value is large, but the actual changes are small: surfeit decreases by 0.05, and inaccuracy increases by only 0.0001. This illustrates why absolute variations should always be considered together with rates of change. ■

As we optimize both surfeit and inaccuracy, we encounter descriptions where improving one quantity requires worsening the other. These descriptions form a Pareto frontier.

Definition 10.6.2 — Pareto optimality for inaccuracy and surfeit. Let $r \in \mathcal{B}^*$ be a representation, and let $d_1 \in \mathcal{D}$ be a description. We say that d_1 is *Pareto optimal* with respect to inaccuracy and surfeit if there is no description $d_2 \in \mathcal{D}$ such that

$$\iota(d_2, r) \leq \iota(d_1, r)$$

and

$$\sigma(d_2) \leq \sigma(d_1),$$

with at least one inequality strict.

Pareto optimal descriptions represent the best available trade-offs between reconstructing the representation accurately and avoiding unnecessary descriptive complexity. In practical scientific work, model selection should normally be restricted to this frontier. A description outside the frontier is dominated: another description is at least as good in both inaccuracy and surfeit and strictly better in one of them.

10.7 Surfeit of Areas

The concept of surfeit can be extended from individual descriptions to descriptions of research areas. Let $A \subseteq \mathcal{E}$ be a research area, and let $\hat{A} = \{e_1, e_2, \dots, e_n\}$ be the subset of entities in A currently under consideration. A description of the area should encode the models, relations, assumptions, and organizing principles used to describe the entities in $\hat{A}$.

Definition 10.7.1 — Description of an area. Let $A \subseteq \mathcal{E}$ be a research area with known subset

$$\hat{A} = \{e_1, e_2, \dots, e_n\}.$$

A *description of the area* $\hat{A}$ is a valid description $d_{\hat{A}} \in \mathcal{D}$ intended to describe the entities in $\hat{A}$ and their relevant relations.

We define the surfeit of an area description exactly as in the case of individual descriptions.

Definition 10.7.2 — Surfeit of an area description. Let $d_{\hat{A}} \in \mathcal{D}$ be a description of a known research area $\hat{A}$. We define the *surfeit of the area description* as

$$\sigma(d_{\hat{A}}) = 1 - \frac{K(d_{\hat{A}})}{l(d_{\hat{A}})}.$$

This quantity measures the internal compressibility of the description of the area. A high value suggests that the area is being described with avoidable redundancy: repeated concepts, duplicated subtheories, overlapping taxonomies, or unnecessarily fragmented models. A low value suggests that the available description is compact relative to its own information content.

In practice, $K(d_{\hat{A}})$ must be approximated by compression.

Definition 10.7.3 — Practical surfeit of an area. Let C be a fixed compressor and let $d_{\hat{A}} \in \mathcal{D}$ be a description of a known research area. We define the *practical surfeit of the area* as

$$\hat{\sigma}_C(d_{\hat{A}}) = 1 - \frac{C(d_{\hat{A}})}{l(d_{\hat{A}})}.$$

The surfeit of an area is especially useful when studying scientific maturity. A mature area should not merely accumulate more descriptions; it should organize them more efficiently. If a field contains many overlapping definitions, duplicated models, inconsistent taxonomies, or repeated explanatory structures, then its area description will tend to be compressible. Reducing this compressibility by unifying concepts, eliminating duplication, and discovering more general principles corresponds to decreasing surfeit at the area level.

■ **Example 10.6** Consider an early stage of a scientific field in which several subcommunities introduce different names, notations, and models for closely related phenomena. A description of the whole area may contain many repeated structures because the same principles are rediscovered independently in different subfields. Later, a unifying theory may show that these models are special cases of a single framework. The new area description is shorter and less redundant. In the terminology of this chapter, the surfeit of the area has decreased. ■

Area-level surfeit should also be interpreted together with area-level inaccuracy. A unified theory that compresses many models into one may be elegant but inaccurate if it fails to reproduce the empirical or formal content of the subareas it replaces. As before, conciseness is valuable only when it does not destroy descriptive adequacy.

References

The concept of redundancy has been extensively studied in information theory, where efficient codes are designed to remove predictable structure from messages; see, for example, [Abr63]. The relation between simplicity, compression, and model selection is also central to algorithmic information theory and to minimum-description-length approaches; see [LV13] and the references on MDL and MML discussed in Appendix 21. In the present chapter, these ideas are reinterpreted within the theory of nescience: surfeit is not merely a preference for short descriptions, but a practical approximation to the surplus component of mismodel.



11. Interesting Questions

*It is not the answer that enlightens,
but the question.*
Eugène Ionesco

In this chapter, we introduce a set of metrics for classifying research topics according to their potential to generate interesting problems, along with a methodology for the assisted discovery of new research questions. The objective is to propose new directions, novel research ideas, that contribute to reducing nescience, that is, to diminishing the extent of the scientific unknown. The proposed methodology supports both the identification of new applications for existing tools to address open problems (the known unknown) and the discovery of entirely new and previously unexplored research directions (the unknown unknown). While the methodology is applicable to both intradisciplinary and interdisciplinary topics, the most impactful results typically arise in the latter case. In Chapter ??, we demonstrate the methodology in practice and propose several new questions and research topics.

We have already examined three dimensions for classifying research topics: miscoding (Chapter 5), inaccuracy (Chapter ??), and surfeit (Chapter ??). These metrics allow us to quantitatively assess our level of understanding of a topic, a broader concept we refer to as nescience. In this section, we

introduce two additional metrics for characterizing topics: relevance (Section ??) and applicability (Section 11.2). Relevance measures the impact a topic has on people's lives and complements the existing dimensions used to quantify nescience. Applicability quantifies how frequently a topic has been applied in other domains and helps identify new uses for existing technologies.

What is proposed in this chapter is an algebraic approach to the assisted discovery of potentially interesting research questions, grounded in the theory of nescience. On the one hand, it aims to support researchers in their day-to-day work. The methodology can be used to uncover novel tools that may be applied to a given problem, or to identify new problems where existing tools could be effectively used. In its more advanced form, the methodology facilitates the exploration of the unknown unknown, that is, research areas that have not yet been conceptualized, described, or even imagined.

On the other hand, because the methodology is based on well-defined mathematical principles, it lends itself to automation. This opens the door for artificial intelligence systems to move beyond their current limitations in autonomously generate truly novel research directions. By formalizing the process of question discovery, the methodology enables AI to propose interesting and previously unexplored research questions, and even to discover entirely new topics of scientific inquiry.

11.1 Integrative Depth

In addition of the metrics that compose nescience, further metrics are required in order to characterize the role that different research topics play within the scientific knowledge. Such metrics are useful not only for classifying individual topics, but also for identifying promising directions for inquiry (Section 11.5), and guiding the combination of topics in the discovery of new research questions (Section 11.6).

A new metric of *integrative depth* is introduced to capture the multiplicity of nescience-reducing relation among topics. Intuitively, some research topics admit a substantial reduction of nescience by conditioning on a small number of other topics, whereas others admit reductions along many distinct directions. In the latter case, there exist multiple background knowledge bases such that conditioning on each of them yields a decrease in the nescience of the topic, even though none of these backgrounds is individually necessary.

Definition 11.1.1 Given the nescience dependency graph $NDG = (\mathcal{T}, E_N)$ we define the *integrative depth* of a research topic $t \in \mathcal{T}$ as its indegree in NDG , that is,

$$ID(t) = indeg(t)$$

Recall that a directed edge from B to A in the nescience dependency graph indicates that conditioning on the knowledge background associated with B reduces the nescience of A. This does not imply that such background knowledge is necessary to understand A, nor that it is the only background capable of reducing its nescience.

Integrative depth measures the extent to which understanding a research topic requires the integration of knowledge originating from other topics. A topic with low integrative depth can be substantially understood in relative isolation, whereas a topic with high integrative depth might require to understand many other independent sources of knowledge.

In practice, topics with high integrative depth often benefit from combining insights drawn from several contributing areas (see Section XXX), although the existence of multiple nescience-reducing directions does not, by itself, imply that such combination is necessary.

■ **Example 11.1** Consider the research topic of cancer. Conditioning on knowledge from genetics reduces nescience by clarifying the role of mutations and oncogenes, while conditioning on cell biology reduces nescience by explaining mechanisms of uncontrolled proliferation. Knowledge from immunology further reduces nescience by accounting for immune surveillance and evasion, and insights from tissue mechanics and the tumor microenvironment contribute by explaining how physical and structural factors influence tumor growth. Each of these background topics independently yields a reduction in the nescience associated with cancer, even though none of them is individually necessary or sufficient to achieve a comprehensive understanding. The high integrative depth of this topic therefore reflects the existence of multiple distinct directions along which ignorance can be reduced, rather than a single required sequence of prerequisites. ■

The integrative depth metric inherits several basic properties directly from its graph-theoretic definition: *i*) for any research topic $t \in \mathcal{T}$, the integrative depth $D(t)$ is a non-negative integer; *ii*) a research topic t has integrative depth $D(t) = 0$ if and only if there is no other topic whose representations reduce the nescience of t .

Topics with zero integrative depth are autonomous in the sense that their nescience cannot be reduced by conditioning on other topics alone.

Integrative depth is independent of the current level of nescience of a topic. A topic may have high integrative depth and low nescience, corresponding to a well-understood domain that nonetheless requires the integration of many different areas. Conversely, a topic may have high integrative depth and high nescience, indicating a complex, unresolved problem whose understanding depends on progress across multiple topics.

Weighted integrative depth

In some cases, it is not sufficient to know how many topics contribute to reducing the nescience of a given topic; it is also important to quantify *how much* each contributing topic reduces nescience. For this purpose, we consider the weighted version of the nescience dependency graph.

Definition 11.1.2 Given the weighted nescience dependency graph $WNDG = (\mathcal{T}, E_N, w)$ we define the *weighted integrative depth* of a research topic $t \in \mathcal{T}$ as

$$D_w(t) = \sum_{A:(A,t) \in E_N} w(A,t) = \sum_{A:(A,t) \in E_N} (N(t | A) - N(t)).$$

The weighted integrative depth of a topic measures the total reduction in nescience obtained by integrating all topics that contribute to its understanding.

While integrative depth counts the number of distinct topics whose representations reduce the nescience of a given topic, weighted integrative depth captures the overall magnitude of that reduction. A topic may have a high integrative depth but a modest weighted integrative depth if each contributing topic yields only a small reduction in nescience. Conversely, a topic with moderate integrative depth may exhibit a large weighted integrative depth if a few incoming topics provide substantial explanatory power.

Weighted integrative depth therefore quantifies the *strength of integration* required to understand a topic, complementing the purely structural information provided by integrative depth.

■ **Example 11.2** Consider again the entity corresponding to *climate change*. Its integrative depth is high, since many topics contribute to reducing its nescience. However, the magnitude of these contributions may vary substantially. Conditioning on atmospheric physics may yield a large reduction in nescience, while conditioning on a narrowly scoped economic model may yield a smaller one. The weighted integrative depth aggregates these heterogeneous contributions into a single measure of total epistemic gain. ■

The weighted integrative depth satisfies several immediate properties: *i*) For any research topic $t \in \mathcal{T}$, the weighted integrative depth $D_w(t)$ is a non-negative real number.

Proposition 11.1.1 A research topic t has weighted integrative depth $D_w(t) = 0$ if and only if it has integrative depth $D(t) = 0$.

Proof. If $D(t) = 0$, there are no incoming edges, and the defining sum for $D_w(t)$ is empty, yielding $D_w(t) = 0$. Conversely, if $D_w(t) = 0$, there can be no incoming edges with positive weight, and hence $D(t) = 0$. ■

Integrative depth and weighted integrative depth capture complementary aspects of epistemic integration. Integrative depth reflects the number of distinct sources that must be combined to reduce the nescience of a topic, while weighted integrative depth reflects the total magnitude of the reduction achieved by those sources. Together, these metrics distinguish between topics that depend on many weakly informative contributors and those that depend on a smaller number of highly informative ones.

In later chapters, weighted integrative depth will be used to refine the classification of research topics and to guide the construction of composite research directions that balance epistemic feasibility with the potential for substantial nescience reduction.

Normalized weighted integrative depth

In many practical situations, we are not interested in the full set of research topics $\mathcal{T}$, but rather in a restricted collection $S \subseteq \mathcal{T}$ corresponding, for instance, to a discipline, a research program, or the scope of a particular organization. In this case, it is natural to evaluate integrative metrics relative to the weighted nescience dependency graph restricted to S .

Assume that the restriction of the weighted nescience dependency graph to a subset S has been defined elsewhere in the book. We denote the resulting restricted weighted graph by

$$G_{N,S}^w = (S, E_{N,S}, w),$$

where $E_{N,S} \subseteq S \times S$ is the set of directed edges between topics in S , and each edge $(A, t) \in E_{N,S}$ satisfies $w(A, t) = N(t | A) - N(t) > 0$.

Definition 11.1.3 Let $S \subseteq \mathcal{T}$ and let $G_{N,S}^w = (S, E_{N,S}, w)$ be the restricted weighted nescience dependency graph. The *weighted integrative depth of t restricted to S* is defined as

$$D_{w,S}(t) = \sum_{A:(A,t) \in E_{N,S}} w(A, t), \quad t \in S.$$

This quantity measures the total reduction in nescience of t obtained by integrating only those topics that lie within S .

Since the absolute scale of $D_{w,S}(t)$ depends on the size and composition of S , it is often useful to normalize the weighted integrative depth within S so that values lie in the unit interval and are comparable across subsets.

Definition 11.1.4 Let $S \subseteq \mathcal{T}$ be finite. The *min–max normalized weighted*

integrative depth on S is defined for each $t \in S$ by

$$\bar{D}_{w,S}(t) = \frac{D_{w,S}(t) - \min_{u \in S} D_{w,S}(u)}{\max_{u \in S} D_{w,S}(u) - \min_{u \in S} D_{w,S}(u)}.$$

In the degenerate case where $\max_{u \in S} D_{w,S}(u) = \min_{u \in S} D_{w,S}(u)$, all normalized values are set to 0.

Proposition 11.1.2 Let $S \subseteq \mathcal{T}$ be finite. For any topic $t \in S$, the restricted weighted integrative depth satisfies $D_{w,S}(t) \geq 0$.

Proof. By construction, each edge in $E_{N,S}$ has strictly positive weight $w(A, t) > 0$. Therefore, $D_{w,S}(t)$ is a sum of non-negative terms, and hence non-negative. ■

Proposition 11.1.3 Let $S \subseteq \mathcal{T}$ be finite. For any topic $t \in S$, the normalized weighted integrative depth satisfies

$$0 \leq \bar{D}_{w,S}(t) \leq 1.$$

Moreover, $\bar{D}_{w,S}(t) = 0$ for at least one topic in S , and $\bar{D}_{w,S}(t) = 1$ for at least one topic in S , whenever the denominator is non-zero.

Proof. If the denominator is non-zero, then

$$\min_{u \in S} D_{w,S}(u) \leq D_{w,S}(t) \leq \max_{u \in S} D_{w,S}(u),$$

and the stated bounds follow immediately. The extreme values are attained by any minimizer and maximizer of $D_{w,S}$ on S . ■

Proposition 11.1.4 Let $S \subseteq \mathcal{T}$ be finite and let $a > 0$ and $b \in \mathbb{R}$. Define a rescaled depth

$$D'_{w,S}(t) = aD_{w,S}(t) + b, \quad t \in S.$$

If $\max_{u \in S} D_{w,S}(u) \neq \min_{u \in S} D_{w,S}(u)$, then the min–max normalized values are invariant under this transformation:

$$\bar{D}'_{w,S}(t) = \bar{D}_{w,S}(t) \quad \text{for all } t \in S.$$

Proof. Substituting $D'_{w,S}(t) = aD_{w,S}(t) + b$ into the min–max normalization formula shows that the factor a cancels and the shift b subtracts out in numerator and denominator. ■

Proposition 11.1.5 Let $S \subseteq \mathcal{T}$ be finite and let $t \in S$. If new incoming edges pointing to t are added to the restricted weighted graph $G_{N,S}^w$ (or if the weight of an existing incoming edge is increased), then $D_{w,S}(t)$ can only increase. Consequently, if the minimum and maximum of $D_{w,S}$ over S remain unchanged, then $\bar{D}_{w,S}(t)$ can only increase.

Proof. Each new incoming edge contributes an additional positive term to the sum defining $D_{w,S}(t)$, and increasing an existing incoming weight increases that sum. If $\min_{u \in S} D_{w,S}(u)$ and $\max_{u \in S} D_{w,S}(u)$ remain fixed, then the normalization is an increasing affine transformation of $D_{w,S}(t)$, and the same monotonicity carries over to $\bar{D}_{w,S}(t)$. ■

The restricted quantity $D_{w,S}(t)$ measures the degree to which a topic t can be clarified by integrating knowledge *available within* the subset S . This is particularly important when S represents the effective scope of a research team or organization: a topic may have large global weighted integrative depth, but small restricted weighted integrative depth if most of its supporting topics lie outside the subset.

The normalized score $\bar{D}_{w,S}(t)$ provides a dimensionless ranking of topics within S , enabling comparisons across different subsets and supporting portfolio-level analyses in later chapters.

11.2 Applicability

In addition to nescience and integrative depth, it is useful to characterize the extent to which a research topic can serve as a tool for reducing ignorance in other topics. Some topics play a primarily enabling role: conditioning on them yields reductions in nescience across a broad range of other topics. The metric of *applicability* is introduced to capture this structural property.

Definition 11.2.1 Let $NDG = (\mathcal{T}, E_N)$ be the nescience dependency graph. We define the *applicability* of a research topic $t \in \mathcal{T}$ as its out-degree in NDG , that is,

$$A(t) = \text{outdeg}(t)$$

Applicability measures the breadth of the nescience-reducing influence of a research topic. A topic with low applicability contributes to the reduction of nescience in few other topics, whereas a topic with high applicability admits many distinct directions along which it can be used to reduce ignorance elsewhere.

■ **Example 11.3** As an example of a topic with high applicability, consider *linear algebra*. Conditioning on linear algebra reduces the nescience of a wide range of topics, including numerical analysis, machine learning, quantum mechanics, signal processing, and control theory. In the nescience dependency graph, this corresponds to many outgoing arcs, and hence a high applicability.

By contrast, a narrowly scoped descriptive topic, such as the *maximum recorded jump distance of a specific flea species*, may reduce the nescience of few or no other topics. Such a topic therefore exhibits low applicability. ■

Applicability inherits several immediate properties from its graph-theoretic definition: *i*) for any research topic $t \in \mathcal{T}$, the applicability $A(t)$ is a non-negative integer; *ii*) A research topic t has applicability $A(t) = 0$ if and only if conditioning on t does not reduce the nescience of any other topic.

Weighted applicability

Definition 11.2.2 Let $G_N^w = (\mathcal{T}, E_N, w)$ be the weighted nescience dependency graph. The *weighted applicability* of a research topic $t \in \mathcal{T}$ is defined as

$$A_w(t) = \sum_{u: (t,u) \in E_N} w(t,u).$$

Weighted applicability measures not only how many topics benefit from conditioning on t , but also the total magnitude of the nescience reductions it provides.

Normalized applicability

To facilitate comparison across topics, applicability values can be normalized using min-max normalization.

Definition 11.2.3 Let $\mathcal{T}$ be finite. The *min-max normalized applicability* of a topic $t \in \mathcal{T}$ is defined as

$$\bar{A}(t) = \frac{A(t) - \min_{u \in \mathcal{T}} A(u)}{\max_{u \in \mathcal{T}} A(u) - \min_{u \in \mathcal{T}} A(u)}.$$

In the degenerate case where all topics have the same applicability, all normalized values are set to 0.

An analogous normalization can be applied to weighted applicability by replacing $A(t)$ with $A_w(t)$.

Applicability and integrative depth capture complementary structural aspects of research topics within the nescience dependency graph. Integrative depth characterizes the plurality of directions along which the nescience of a topic can be reduced, whereas applicability characterizes the plurality of directions along which a topic can reduce the nescience of others.

In later chapters, these metrics will be used jointly to distinguish between research topics that primarily function as tools and those that primarily function as problems, and to guide the construction of research portfolios that balance enabling power with epistemic challenge.

11.3 Maturity

Given an unknown topic A , a natural strategy for reducing its nescience is to rely on background topics B such that conditioning on B yields a reduction in

nescience, that is,

$$N(A \mid B) < N(A).$$

However, not all background topics satisfying this condition are equally suitable to be used as tools. In particular, the usefulness of a background topic depends not only on how much it reduces the nescience of A , but also on how well understood the background topic itself is.

To capture this trade-off, we adopt the following guiding principle: a background topic B is considered preferable for reducing the nescience of A when the combined unresolved nescience associated with using B is smaller than the original nescience of A . Formally, this corresponds to requiring that

$$N(A \mid B) + N(B) < N(A).$$

This condition expresses that the reduction in nescience obtained by conditioning on B is large enough to compensate for the remaining nescience of B itself.

This principle is particularly relevant in situations where errors propagate strongly, commitments are irreversible, or independent means of verification are unavailable. In such contexts, relying on background knowledge that is itself poorly understood may introduce additional uncertainty that outweighs the apparent reduction in nescience of the primary topic.

At the same time, the condition above does not exclude the use of poorly understood background topics in all circumstances. In particular, reliance on such topics may be justified when no better-understood alternative is known to yield a reduction in the nescience of the target topic, or when errors are inexpensive and results can be independently verified. In these situations, even a background topic with high nescience may be used as a provisional tool, provided that it contributes to lowering the nescience of the primary problem.

■ **Example 11.4** Large language models illustrate this situation. Conditioning on such models often produces substantial reductions in the nescience of a wide range of tasks, including text generation, code synthesis, and exploratory problem solving. Although these models are themselves only partially understood, they are typically employed in settings where alternative well-understood tools are unavailable, errors are inexpensive, and outputs can be checked or corrected by external means. Under these conditions, reliance on such models can be epistemically productive despite their limited maturity. ■

To formalize the notion of how well a topic is understood, we introduce the concept of *maturity*. Maturity is defined as a simple reparameterization of nescience, emphasizing its inverse interpretation.

Definition 11.3.1 Let $t \in \mathcal{T}$ be a topic, and let $N(t)$ denote its nescience. The *maturity* of t , denoted $M(t)$, is defined as

$$M(t) = 1 - N(t).$$

A higher maturity value indicates that a topic is better understood and therefore more suitable to be used as a background tool for reducing the nescience of other topics. Conversely, topics with low maturity should be used with caution, since their own unresolved nescience contributes to the total uncertainty associated with their application.

■ **Example 11.5** Linear regression provides an example of a highly mature research topic. Its theoretical assumptions, limitations, and behavior are well characterized, resulting in a low nescience and a correspondingly high maturity. When used as background knowledge, linear regression typically satisfies the condition $N(A \mid B) + N(B) < N(A)$ for a wide range of applied problems, making it a reliable tool in settings where stability and interpretability are important. ■

Maturity is an intrinsic property of a research topic, reflecting the extent to which its representations are complete and reliable. It is independent of the topic's applicability or integrative depth, and should therefore not be interpreted as a measure of usefulness or importance. Rather, maturity provides a measure of epistemic stability: mature topics tend to behave predictably when used as background knowledge, whereas immature topics introduce additional uncertainty that must be managed explicitly.

In practice, when comparing maturity values across the topics of a given research area, it is often convenient to work with a normalized scale.

Definition 11.3.2 Given a finite set of topics $\mathcal{T}$, the *min-max normalized maturity* of a topic $t \in \mathcal{T}$, denoted $\tilde{M}(t)$, is defined as

$$\tilde{M}(t) = \frac{M(t) - \min_{t' \in \mathcal{T}} M(t')}{\max_{t' \in \mathcal{T}} M(t') - \min_{t' \in \mathcal{T}} M(t')},$$

where $M(t)$ is the maturity of t as in Definition 11.3.1.

In the degenerate case where all topics have the same maturity, all normalized values are set to 0. Normalized maturity values will be used in later chapters to guide the selection of background knowledge and to balance nescience reduction against the reliability of the tools employed.

11.4 Interestingness

In many scientific and practical contexts it is necessary to compare research topics and to decide which ones are more promising to pursue, either as tools for

solving other problems or as problems worthy of investigation in their own right. In this section, we introduce the notion of *interestingness* as a decision-oriented concept derived from the metrics previously introduced.

The theory of nescience naturally induces several dimensions along which topics may be evaluated, but it does not prescribe a unique way of aggregating them into a single scalar score. As a result, interestingness should be understood as arising from a multi-objective optimization problem (see Section 21.6), together with a choice of decision rule.

A research topic may be considered interesting as a tool when it can be reliably used to reduce the nescience of other topics. Two theory-derived quantities are relevant in this context: maturity, which measures how well the topic itself is understood and applicability, which measures how widely the topic has been successfully used to reduce nescience in other topics.

Definition 11.4.1 We define the *interestingness as a tool* of a topic t as the multi-objective optimization problem:

$$\text{maximize } (M(t), A(t)),$$

where $M(t)$ denotes the maturity of the topic t and $A(t)$ its applicability.

This formulation induces a partial order on topics: a topic t_1 dominates another topic t_2 if $M(t_1) \geq M(t_2)$ and $A(t_1) \geq A(t_2)$, with at least one inequality strict. The set of non-dominated topics forms the pareto frontier of interesting tools.

A research topic may also be interesting as a problem, independently of its usefulness as a tool. In this case, two different theory-internal quantities play a central role: nescience, which measures the extent to which the topic is not yet understood, and integrative depth, which measures the number of distinct background topics whose knowledge reduces the nescience of the topic. Together, these quantities characterize both the magnitude of unresolved ignorance and the structural richness of the ways in which that ignorance can be reduced.

Definition 11.4.2 We define the *interestingness as a problem* of a topic t as the multi-objective optimization problem

$$\text{maximize } (N(t), D(t)),$$

where $N(t)$ is the nescience of topic t and $D(t)$ its integrative depth.

As in the case of tools, this formulation induces a partial order. Topics with high nescience but low integrative depth correspond to forms of ignorance that admit few known avenues for progress, whereas topics with both high nescience and high integrative depth admit many distinct nescience-reducing directions and

are structurally richer as research problems.

In practical applications it is often necessary to produce a total ordering of topics. This requires the introduction of a decision maker, that is, a scalarization of the underlying multi-objective problem.

One possible choice is a balanced *Euclidean scalarization*. For example, the interestingness of a topic as a tool may be defined as

$$IT(t) = \frac{\sqrt{\tilde{M}(t)^2 + \tilde{A}(t)^2}}{\sqrt{2}},$$

where $\tilde{M}(t)$ and $\tilde{A}(t)$ are normalized versions of maturity and applicability. Analogously, interestingness as a problem may be defined as

$$IP(t) = \frac{\sqrt{\tilde{N}(t)^2 + \tilde{D}(t)^2}}{\sqrt{2}},$$

where $\tilde{N}(t)$ and $\tilde{D}(t)$ denote normalized nescience and integrative depth.

These metrics correspond to a particular decision method that favors balanced topics over extreme ones. Other decision makers may be more appropriate in different contexts. For instance, a *minimum-based* decision rule, $\min\{\tilde{M}(t), \tilde{A}(t)\}$, emphasizes robustness by penalizing weaknesses in any dimension, meanwhile a *weighted sum* allows explicit prioritization of maturity versus applicability, or nescience versus integrative depth, according to external constraints or strategic goals.

Such decision makers do not alter the underlying theory, but provide practical means of selecting and ranking topics for specific purposes.

11.5 Interesting Questions

In the theory of nescience we distinguish between known unknowns, that is, problems whose existence is recognized but whose solutions are not known, and unknown unknowns, corresponding to problems that have not yet been identified. In this section we focus on the former, and introduce a formal notion of *interesting questions* aimed at supporting the systematic exploration of known unknowns.

A central observation underlying this approach is that new research questions often arise when existing knowledge is brought to bear on unresolved problems. In the theory of nescience, such situations are naturally captured by the *nescience dependency graph*, which encodes when knowledge of one topic reduces the nescience of another.

Definition 11.5.1 Let $G_N = (\mathcal{T}, E_N)$ denote the nescience dependency graph.

We define *interesting question* as an ordered pair of topics $Q_{t \rightarrow p} = (t, p)$ such that $(t, p) \in E_N$.

In other words, a question exists only when the theory predicts that the topic t can be meaningfully used to reduce the nescience of the topic p . The directed nature of the dependency graph makes the asymmetry between tools and problems explicit: in general, $Q_{t \rightarrow p}$ and $Q_{p \rightarrow t}$ represent distinct questions, and only one of them may be well defined.

Operationally, the question $Q_{t \rightarrow p}$ can be interpreted as asking whether, and how, the knowledge associated with topic t can be used to make progress on the unresolved aspects of topic p .

The nescience dependency graph not only specifies the existence of a question, but also provides a quantitative measure of its potential impact.

Definition 11.5.2 Let $Q_{t \rightarrow p}$ be a question. The *pair-specific nescience reduction* associated with $Q_{t \rightarrow p}$ is defined as

$$\Delta N(p; t) = N(p) - N(p | t),$$

which is strictly positive by definition of the dependency graph.

The quantity $\Delta N(p; t)$ measures how much of the unresolved ignorance about p can be reduced by assuming the knowledge background associated with t . Unlike topic-level metrics, this quantity is intrinsically relational and depends on the specific pairing of tool and problem.

The interestingness of a question cannot, in general, be reduced to a single scalar quantity without introducing additional assumptions. Instead, the theory of nescience naturally induces a multi-objective evaluation of questions based on theory-derived quantities.

Given a question $Q_{t \rightarrow p}$, the following dimensions are relevant:

- $\Delta N(p; t)$, the pair-specific nescience reduction;
- $M(t)$, the maturity of the tool topic t ;
- $A(t)$, the applicability of the tool topic t ;
- $N(p)$, the nescience of the problem topic p ;
- $D(p)$, the integrative depth of the problem topic p .

Together, these quantities define the objective vector

$$\Phi(t, p) = (\Delta N(p; t), M(t), A(t), N(p), D(p)).$$

Definition 11.5.3 A question $Q_{t \rightarrow p}$ is said to be *at least as interesting* as another question $Q_{t' \rightarrow p'}$ if

$$\Phi(t, p) \geq \Phi(t', p')$$

componentwise, with at least one inequality being strict. The set of *interesting questions* is defined as the Pareto frontier of questions under this partial order.

This definition does not impose a total ordering on questions, and naturally allows for the existence of incomparable questions corresponding to different trade-offs between nescience reduction, reliability of tools, and structural richness of problems.

In practical applications, such as research planning, portfolio construction, or automated recommendation systems, it is often necessary to rank questions or to select a subset of them. This requires the introduction of a *decision maker*, understood as a scalarization of the underlying multi-objective problem.

Several decision makers are particularly natural in the present framework:

- **Risk-averse decision maker.** Select questions that maximize $\Delta N(p; t)$ subject to a lower bound on tool maturity, $M(t) \geq \tau$, for some threshold τ .
- **Exploratory decision maker.** Maximize $\Delta N(p; t) \cdot N(p)$, favoring questions that yield large reductions on highly unresolved problems.
- **Scarcity-aware decision maker.** Favor questions associated with problems of low integrative depth $D(p)$, reflecting situations where few alternative nescience-reducing paths are available.
- **Balanced scalarization.** Combine normalized versions of the components of $\Phi(t, p)$ using a weighted or Euclidean aggregation, yielding a total ordering suitable for visualization or large-scale comparison.

The choice of decision maker depends on contextual factors such as acceptable risk, available resources, and strategic goals. Importantly, these choices do not alter the underlying theory, but merely provide different ways of navigating the space of interesting questions it defines.

In summary, interesting questions are defined in the theory of nescience as graph-grounded objects corresponding to nescience-reducing relationships between topics. Their interestingness emerges from a multi-objective structure inherent in the theory, while scalar rankings arise only at the level of decision making and application.

■ **Example 11.6** We illustrate the proposed framework with a simplified example drawn from cancer research. Consider the following topics in the set $\mathcal{T}$:

- $p = \text{Cancer progression}$, understood as the set of mechanisms by which cancer develops, invades tissue, and metastasizes.
- $t_1 = \text{Genomics}$, including gene expression profiling and mutation analysis.
- $t_2 = \text{Statistical learning}$, encompassing classification, regression, and model selection techniques.
- $t_3 = \text{Cell signaling pathways}$, describing biochemical interaction networks inside cells.
- $t_4 = \text{Immunotherapy}$, covering immune-based treatment strategies.

Assume that empirical and theoretical work has established the following nescience-reducing relationships:

$$N(p \mid t_i) < N(p) \quad \text{for } i \in \{1, 2, 3, 4\}.$$

These relationships define arcs (t_i, p) in the nescience dependency graph G_N . Each arc corresponds to a question $Q_{t_i \rightarrow p}$ asking whether the knowledge associated with t_i can be used to reduce ignorance about cancer progression.

Suppose further that the estimated pair-specific nescience reductions satisfy

$$\Delta N(p; t_1) > \Delta N(p; t_2) > \Delta N(p; t_3) > \Delta N(p; t_4),$$

reflecting the fact that genomic data currently explains a larger fraction of observed variability in cancer progression than the other topics.

At the same time, the maturity values of the tool topics may differ substantially. For instance, statistical learning may have high maturity due to well-established theory and methods, whereas immunotherapy may exhibit lower maturity due to incomplete understanding of immune response dynamics. Integrative depth $D(p)$ is high, since multiple independent background topics reduce the nescience of p .

Within this framework, the set of questions

$$\{Q_{t_1 \rightarrow p}, Q_{t_2 \rightarrow p}, Q_{t_3 \rightarrow p}, Q_{t_4 \rightarrow p}\}$$

forms the candidate space of graph-grounded questions. Their interestingness is evaluated by comparing the corresponding objective vectors

$$\Phi(t_i, p) = (\Delta N(p; t_i), M(t_i), A(t_i), N(p), D(p)).$$

Different decision makers may select different questions from this set. A risk-averse decision maker may favor $Q_{t_2 \rightarrow p}$ due to the high maturity of statistical learning, even if its nescience reduction is not maximal. An exploratory decision maker may prioritize $Q_{t_1 \rightarrow p}$, which yields the largest reduction in nescience. A scarcity-aware decision maker would note that $D(p)$ is high, indicating that cancer progression admits many independent lines of inquiry, and may therefore prioritize questions that explore underrepresented connections, such as $Q_{t_4 \rightarrow p}$. ■

This example illustrates how the theory of nescience supports the systematic generation and evaluation of research questions. Questions are not arbitrary combinations of topics, but graph-grounded objects whose interestingness emerges from the interplay between pair-specific nescience reduction, the maturity of available tools, and the structural richness of the underlying problem.

11.6 New Topics

The region of *unknown unknowns* plays a central role in scientific progress, as it contains those research topics that have not yet been conceptualized or identified. One of the main objectives of the theory of nescience is to provide principled tools for exploring this region and for anticipating the emergence of future research topics. In this section we introduce a formal framework for the identification of *new topics*, understood as topics that lie beyond the current knowledge frontier.

A guiding intuition, illustrated in Figure 1.8, is that the probability of crossing the knowledge frontier through the combination of existing topics is higher when the topics involved are themselves poorly understood. Well-understood topics are typically embedded in dense and highly explored neighborhoods of the representation space, so their combinations tend to remain within the domain of known knowledge. In contrast, topics with high nescience are associated with sparser and less structured neighborhoods, making combinations involving them more likely to reach unexplored regions.

11.6.1 Candidate Topic Combinations

Let $\mathcal{T}$ denote the set of existing research topics, and let $N(t)$ denote the nescience of topic t , with $\tilde{N}(t)$ its normalized version. Let $D(t)$ denote the (normalized) integrative depth of t , as introduced in Section 11.1.

Definition 11.6.1 Given two distinct topics $t_1, t_2 \in \mathcal{T}$, a *candidate topic combination* is the unordered pair

$$C_{\{t_1, t_2\}} = \{t_1, t_2\}.$$

A candidate topic combination does not, by itself, constitute a new topic. Rather, it represents a potential source from which a new topic may emerge, depending on whether the combination leads beyond the knowledge frontier.

11.6.2 Multi-Objective Characterization of New Topics

The emergence of a new topic from a candidate combination cannot be adequately captured by a single scalar quantity. Instead, it is naturally described as a multi-objective problem involving theory-internal quantities.

Given a candidate combination $C_{\{t_1, t_2\}}$, we associate the following objective vector:

$$\Psi(t_1, t_2) = (\tilde{N}(t_1), \tilde{N}(t_2), D(t_1), D(t_2)).$$

Each component has a clear interpretation:

- $\tilde{N}(t_1), \tilde{N}(t_2)$ measure how close the topics are to the knowledge frontier;

- $D(t_1), D(t_2)$ measure the structural richness of the topics, in terms of the number of distinct ways background knowledge can reduce their nescience.

High values of $\tilde{N}$ indicate proximity to the frontier, while high values of D indicate that the topic participates in many independent nescience-reducing relations. Combinations in which both topics exhibit high nescience and substantial integrative depth are therefore structurally well positioned to give rise to new topics beyond the frontier.

Definition 11.6.2 A candidate combination $C_{\{t_1, t_2\}}$ is said to have *high potential as a new topic* if it lies on the Pareto frontier of the set

$$\{\Psi(t_i, t_j) : t_i, t_j \in \mathcal{T}, i \neq j\}$$

under the componentwise partial order.

This definition avoids imposing an arbitrary total ordering on candidate combinations and reflects the fact that different combinations may represent different trade-offs between proximity to the frontier and structural richness.

11.6.3 Decision Makers and Practical Selection

In practical applications, such as research planning or automated topic discovery systems, it is often necessary to select or rank candidate combinations. This requires introducing a *decision maker*, understood as a scalarization of the multi-objective problem above.

Examples of decision makers include:

- **Frontier-seeking decision maker**, which prioritizes combinations maximizing $\tilde{N}(t_1) + \tilde{N}(t_2)$;
- **Structure-aware decision maker**, which prioritizes combinations with high $D(t_1) + D(t_2)$, favoring structurally rich topics;
- **Balanced decision maker**, which applies a weighted aggregation of the components of $\Psi(t_1, t_2)$.

The choice of decision maker depends on contextual factors and does not alter the underlying theoretical framework.

11.6.4 Intradisciplinary and Interdisciplinary Combinations

Definition 11.6.3 Let $\mathcal{A} \subseteq \mathcal{T}$ be a research area. A candidate combination $C_{\{t_1, t_2\}}$ is said to be *intradisciplinary* if $t_1, t_2 \in \mathcal{A}$; otherwise, it is *interdisciplinary*.

Interdisciplinary combinations often correspond to candidate combinations whose components lie in distant regions of the nescience dependency graph. Such combinations are therefore more likely to explore underconnected regions of the

representation space and to contribute to the expansion of the knowledge frontier, although this is not guaranteed in general.

References

The following works provide theoretical and philosophical foundations for the concepts of interestingness, maturity, and the combination of topics as tools and problems.

[Cha13] An accessible introduction to the philosophy of science, addressing how scientific questions are formulated, evaluated, and justified.

[PM19] Introduces the principles of causal reasoning, crucial for determining whether applying one topic as a tool can effectively address another as a problem.

[Pop14] Discusses falsifiability, novelty, and the importance of bold conjectures—foundational ideas for the “new and original” criterion.

[Shm+10] Clarifies the distinction between explanatory and predictive goals, helping to differentiate between topics valuable as problems versus tools.

[Van80] Explores the aims of science, model construction, and empirical adequacy, offering a philosophical context for defining “interesting” research.



12. Effort

The effort to understand the universe is one of the very few things
that lifts human life a little above the level of farce.
Steven Weinberg

The theory of nescience has been developed to quantify the extent to which a research entity remains unknown. Given a representation and a description, nescience measures the residual defects that prevent us from reaching perfect knowledge. In this sense, nescience is a measure of epistemic imperfection: it tells us how far our current hypothesis is from an ideal representation and an ideal description.

However, knowing how much nescience remains is not enough. Scientific practice is not concerned only with the existence of better hypotheses, but also with the difficulty of finding them. Two research topics may have similar levels of nescience and yet differ radically in the effort required to reduce it. One topic may contain many obvious improvements, inexpensive experiments, and stable representations. Another may require expensive instruments, rare observations, difficult computations, or conceptual breakthroughs before any measurable progress becomes possible.

This chapter introduces the concept of *effort*. Effort is intended to measure the difficulty of reducing nescience. It is not a new component of nescience. Rather, it is an orthogonal property of the process by which nescience is reduced.

The concept of effort also helps us address a broader question: are some research areas intrinsically more difficult than others? The answer proposed in this chapter is that the difficulty of a research area can be characterized by the expected effort required to reduce its nescience by a fixed amount. In this sense, a research area is difficult not merely because we know little about it, but because improving our knowledge is costly.

12.1 Research Transition Systems

Scientific progress is modeled as a sequence of transformations between hypotheses. These transformations may modify the representation, the description, or both. For example, one may collect new data, remove irrelevant variables, change a model family, add background knowledge, simplify a description, perform a new experiment, or reframe the entity under study.

Definition 12.1.1 A *research transition system* is a tuple

$$\mathcal{S} = (\mathcal{H}, \mathcal{A}, \tau),$$

where:

- $\mathcal{H}$ is a set of hypotheses;
- $\mathcal{A}$ is a set of admissible research operations;
- $\tau : \mathcal{H} \times \mathcal{A} \rightarrow \mathcal{H}$ is a partial transition function.

If $\tau(x, a) = y$, we say that the operation a transforms the hypothesis x into the hypothesis y , and write

$$x \xrightarrow{a} y.$$

The function τ is partial because not every research operation is applicable to every hypothesis. For example, a reduction operation cannot remove a variable that is not present in the representation, and a model simplification operation may not be defined for a non-parametric description.

Definition 12.1.2 Let $\mathcal{S} = (\mathcal{H}, \mathcal{A}, \tau)$ be a research transition system. A *research path* from x_0 to x_k is a finite sequence

$$\pi = (x_0, a_1, x_1, a_2, \dots, a_k, x_k)$$

such that

$$\tau(x_{i-1}, a_i) = x_i$$

for every $i = 1, \dots, k$.

The path π represents a possible trajectory of scientific work. Each step

corresponds to a concrete or abstract research operation.

Typical admissible operations include:

- augmentation of a representation;
- reduction of a representation;
- replacement of a representation by another representation;
- augmentation of a description;
- simplification of a description;
- change of model family;
- conditioning on background knowledge;
- collection of additional observations;
- performance of a new experiment;
- numerical simulation;
- computational search over candidate descriptions;
- conceptual reframing of the entity under investigation.

The choice of admissible operations is application-dependent. In a machine learning problem, the operations may include feature selection, hyperparameter search, model training, label correction, and data acquisition. In theoretical physics, they may include deriving a new equation, changing a mathematical formalism, or testing a limiting case. In philosophy, they may include clarifying concepts, distinguishing senses of a term, or constructing counterexamples.

12.2 Computational Effort

We now introduce effort as the computational cost of a research path. The simplest notion of computational cost counts the number of admissible operations performed along the path.

Definition 12.2.1 Let

$$\pi = (x_0, a_1, x_1, \dots, a_k, x_k)$$

be a research path. The *unit computational cost* of π is

$$C_{\text{unit}}(\pi) = k.$$

This definition is useful when all research operations are treated as having equal cost. In practice, however, different operations may require different amounts of computation. Training a large model, conducting an exhaustive search, or simulating a physical system may require many more elementary steps than removing a redundant variable. We therefore introduce a more general cost function.

Definition 12.2.2 A *computational cost function* on a research transition system is a function

$$c : \mathcal{H} \times \mathcal{A} \rightarrow \mathbb{N} \cup \{\infty\},$$

where $c(x, a)$ denotes the number of computational steps required to apply operation a to hypothesis x .

Given such a function, the cost of a complete research path is obtained by summing the costs of its individual steps.

Definition 12.2.3 Let

$$\pi = (x_0, a_1, x_1, \dots, a_k, x_k)$$

be a research path. The *computational cost* of π is

$$C(\pi) = \sum_{i=1}^k c(x_{i-1}, a_i).$$

We can now define the effort required to reduce nescience by a specified amount.

Definition 12.2.4 Let $x \in \mathcal{H}$ be a hypothesis and let $\varepsilon > 0$. The *effort required to reduce nescience by ε* is defined as

$$\text{Eff}_\varepsilon(x) = \inf_{\pi: x \rightsquigarrow y} \{C(\pi) : N(y) \leq N(x) - \varepsilon\},$$

where the infimum is taken over all research paths π starting at x and ending at some hypothesis y .

If no such path exists, then

$$\text{Eff}_\varepsilon(x) = \infty.$$

In many applications it is more natural to ask for a relative reduction. For example, we may ask for the effort required to reduce nescience by one percent.

Definition 12.2.5 Let $0 < \alpha < 1$. The *effort required to reduce nescience by the relative amount α* is

$$\text{Eff}_\alpha^{\text{rel}}(x) = \inf_{\pi: x \rightsquigarrow y} \{C(\pi) : N(y) \leq (1 - \alpha)N(x)\}.$$

The particular case $\alpha = 0.01$ corresponds to the effort required to reduce nescience by 1%.

■ **Example 12.1** Suppose that x is the current best hypothesis for a machine learning problem, and that $N(x) = 0.40$. A reduction of 1% means reaching a hypothesis y such that

$$N(y) \leq 0.99 \cdot 0.40 = 0.396.$$

The effort $\text{Eff}_{0.01}^{\text{rel}}(x)$ is the minimum number of computational steps, model evaluations, or admissible operations required to obtain such a hypothesis. ■

12.3 Component-wise Effort

Nescience is obtained from several components. In the theoretical formulation developed in this book, the central components are miscoding and mismodel. In practical applications, mismodel may be approximated through inaccuracy and surfeit. It is therefore useful to define effort not only for nescience as a whole, but also for each of its components.

Let

$$\mu(r)$$

denote the miscoding of a representation, and let

$$v_r(d)$$

denote the mismodel of a description d with respect to a representation r . For a hypothesis $x = (r, d)$, we may write

$$\mu(x) = \mu(r), \quad v(x) = v_r(d).$$

Definition 12.3.1 Let $x \in \mathcal{H}$ be a hypothesis and let $\varepsilon > 0$. The *effort required to reduce miscoding by ε* is

$$\text{Eff}_{\varepsilon}^{\mu}(x) = \inf_{\pi: x \rightsquigarrow y} \{C(\pi) : \mu(y) \leq \mu(x) - \varepsilon\}.$$

Definition 12.3.2 Let $x \in \mathcal{H}$ be a hypothesis and let $\varepsilon > 0$. The *effort required to reduce mismodel by ε* is

$$\text{Eff}_{\varepsilon}^v(x) = \inf_{\pi: x \rightsquigarrow y} \{C(\pi) : v(y) \leq v(x) - \varepsilon\}.$$

Component-wise effort identifies the bottlenecks of scientific progress. If $\text{Eff}_{\varepsilon}^{\mu}(x)$ is high, then improving the representation is difficult. This may occur because the entity is hard to observe, because measurements are noisy, because relevant variables are unknown, or because no stable representation style is available. If $\text{Eff}_{\varepsilon}^v(x)$ is high, then improving the description is difficult. This may occur because the representation is already rich but difficult to model, because the required computation is large, or because the underlying structure is complex.

Definition 12.3.3 Let $x \in \mathcal{H}$ be a hypothesis and let $\varepsilon > 0$. The *effort bottleneck* of x at scale ε is the component whose reduction requires the largest effort:

$$B_\varepsilon(x) = \arg \max_{c \in \{\mu, \nu\}} \text{Eff}_\varepsilon^c(x).$$

The effort bottleneck indicates where progress is most difficult. In applied research, identifying this bottleneck is essential. If the bottleneck is miscoding, additional modeling may be unproductive because the representation itself is inadequate. If the bottleneck is mismodel, collecting more data may be less useful than improving the model class or the description.

■ **Example 12.2** Consider a predictive model in medicine. If the input data omit a clinically relevant variable, then the dominant difficulty lies in the representation. Reducing miscoding may require new measurement protocols, new instruments, or additional clinical tests. By contrast, if the representation already contains the relevant variables but the model fails to capture their interactions, then the dominant difficulty lies in mismodel. Reducing mismodel may require a richer model class, better optimization, or a different inductive bias. ■

12.4 Marginal Effort and Return on Effort

The effort required for large reductions of nescience is often less informative than the effort required for the next small improvement. Scientific progress is frequently incremental, and researchers must decide where to allocate limited resources. We therefore introduce the notion of marginal effort.

Definition 12.4.1 Let $x \in \mathcal{H}$ and let $\varepsilon > 0$. The *marginal effort of nescience reduction* at scale ε is

$$\text{ME}_\varepsilon(x) = \frac{\text{Eff}_\varepsilon(x)}{\varepsilon}.$$

For relative reductions we define

$$\text{ME}_\alpha^{\text{rel}}(x) = \frac{\text{Eff}_\alpha^{\text{rel}}(x)}{\alpha N(x)}.$$

The marginal effort measures how many computational steps are required per unit of nescience reduction. It captures the local difficulty of making progress from the current state of knowledge.

As a research field matures, one expects marginal effort to increase. Early stages often contain many obvious improvements: better measurements, simpler models, missing variables, or incorrect assumptions. Later stages may

require increasingly precise experiments, more elaborate theories, and larger computational resources. Thus, a mature field may have low nescience but high marginal effort.

The dual concept is the return on effort.

Definition 12.4.2 Let $x \rightarrow y$ be a transition between hypotheses. The *return on effort* of the transition is

$$\text{RoE}(x \rightarrow y) = \frac{N(x) - N(y)}{C(x \rightarrow y)}.$$

For a research path $\pi : x \rightsquigarrow y$, the return on effort is

$$\text{RoE}(\pi) = \frac{N(x) - N(y)}{C(\pi)}.$$

A rational research strategy should not necessarily choose the path that produces the largest absolute reduction of nescience. Instead, under limited resources, it should often choose the path that maximizes expected nescience reduction per unit of effort.

12.5 Expected Effort

The definitions above assume that the result of a research path is known. In practice, however, research is uncertain. An experiment may fail, a simulation may be inconclusive, a new representation may introduce additional miscoding, and a more complex model may increase surfeit without reducing inaccuracy.

For this reason, practical effort must be treated probabilistically. Let a be a research action available at hypothesis x . The result of applying a may be a random hypothesis Y , depending on experimental outcomes, data quality, optimization success, or theoretical discovery. The expected nescience reduction is

$$\mathbb{E}[\Delta N(a \mid x)] = \mathbb{E}[N(x) - N(Y)].$$

Definition 12.5.1 Let a be an admissible research operation at hypothesis x . The *expected return on effort* of a at x is

$$\text{ERoE}(a \mid x) = \frac{\mathbb{E}[\Delta N(a \mid x)]}{c(x, a)}.$$

This quantity ranks research actions by their expected nescience reduction per computational step. If monetary costs are used instead of computational costs, the same definition gives expected nescience reduction per unit of money.

A related quantity is the expected cost per unit of reduction.

Definition 12.5.2 The *expected effort per unit reduction* of an action a at x is

$$\text{EEUR}(a \mid x) = \frac{c(x, a)}{\mathbb{E}[\Delta N(a \mid x)]},$$

provided $\mathbb{E}[\Delta N(a \mid x)] > 0$.

If the expected reduction is zero or negative, the action does not constitute a rational nescience-reducing move under the chosen model.

■ **Example 12.3** Suppose that two research actions are available. The first action costs 100 units and is expected to reduce nescience by 0.01. The second costs 1000 units and is expected to reduce nescience by 0.05. The first has expected return

$$0.01/100 = 0.0001,$$

whereas the second has expected return

$$0.05/1000 = 0.00005.$$

Although the second action produces a larger expected reduction, the first action is more efficient per unit of effort. ■

Research also involves risk. Two actions may have the same expected reduction but different variance. A risky action may lead to a breakthrough, but may also fail completely. To account for this, one may define a risk-adjusted effort.

Definition 12.5.3 Let $\lambda \geq 0$ be a risk-aversion parameter. The *risk-adjusted effort* of an action a at x is

$$\text{RAE}(a \mid x) = \frac{c(x, a)}{\mathbb{E}[\Delta N(a \mid x)]} + \lambda \text{Var}(\Delta N(a \mid x)).$$

This definition penalizes actions whose outcomes are highly uncertain. The parameter λ determines how strongly risk is penalized. A risk-neutral researcher may take $\lambda = 0$, while a conservative funding agency may choose a larger value.

12.6 Conditional Effort

Background knowledge can reduce nescience, but it can also reduce the effort required to reduce nescience. These are different effects.

Let b denote background knowledge. Conditional nescience measures how much ignorance remains when b is available. Conditional effort measures how hard it is to reduce that remaining ignorance when b is available.

Definition 12.6.1 Let $b \in \mathcal{B}^*$ be a string encoding background knowledge. The *conditional effort required to reduce nescience by ε given b* is

$$\text{Eff}_\varepsilon(x \mid b) = \inf_{\pi: x \rightsquigarrow y} \{C_b(\pi) : N(y \mid b) \leq N(x \mid b) - \varepsilon\},$$

where $C_b(\pi)$ denotes the computational cost of the path when the background knowledge b is available.

Background knowledge may reduce effort in several ways. It may restrict the search space, suggest relevant representations, eliminate impossible hypotheses, provide reusable descriptions, or identify promising experiments.

■ **Example 12.4** The development of Newtonian mechanics is easier if infinitesimal calculus is already available. Calculus may not itself be part of the representation of a particular mechanical system, but it reduces the effort required to construct, manipulate, and compare descriptions of motion. Thus, calculus acts as background knowledge that lowers conditional effort. ■

This distinction is important for education and scientific planning. A prerequisite theory may be valuable not only because it directly reduces nescience, but because it reduces the effort required for future nescience reduction.

12.7 Effort Landscapes

The hypothesis space $\mathcal{H}$ can be viewed as a landscape in which each point $x \in \mathcal{H}$ has a nescience value $N(x)$. Scientific progress corresponds to moving through this landscape toward regions of lower nescience.

Effort enriches this picture by assigning costs to movements through the landscape. Some transitions are cheap; others are expensive. Some nearby hypotheses may be easy to reach but offer little improvement. Other hypotheses may offer large reductions of nescience but require difficult transformations.

This leads to the notion of an effort barrier.

Definition 12.7.1 Let $x, y \in \mathcal{H}$. The *effort distance* from x to y is

$$D_{\text{Eff}}(x, y) = \inf_{\pi: x \rightsquigarrow y} C(\pi),$$

where the infimum is taken over all research paths from x to y .

The effort distance need not be symmetric. It may be easier to move from x to y than from y to x . For example, adding complexity to a description may be easier than simplifying it; collecting data may be easier than identifying which parts of the data are irrelevant.

Definition 12.7.2 A hypothesis x is a *local effort minimum* for nescience if every hypothesis y with $N(y) < N(x)$ lies beyond a high effort barrier relative to the available resources.

A field may stagnate not because no better hypotheses exist, but because all known paths to better hypotheses require prohibitive effort. In such cases, progress may require reframing the problem rather than continuing along the same path.

Definition 12.7.3 A research operation a is *effort-reducing* at x if, for some $\varepsilon > 0$,

$$\text{Eff}_\varepsilon(\tau(x, a)) < \text{Eff}_\varepsilon(x),$$

even if

$$N(\tau(x, a)) \geq N(x).$$

Effort-reducing operations are important because they may not immediately improve knowledge, but they make future improvements easier. Many scientific innovations have this character. A new notation, a new instrument, a new coordinate system, a new representation, or a new computational method may lower future effort before producing a direct reduction in nescience.

■ **Example 12.5** The periodic table did not merely summarize known chemical elements. It organized them in a way that made future discoveries easier. It reduced the effort required to identify missing elements and predict their properties. In this sense, it functioned as an effort-reducing representation. ■

12.8 Thermodynamic Lower Bounds

Effort has been defined above as computational cost. However, any actual computation must be physically implemented. Strings must be stored in physical media, computations must be performed by physical devices, and information may be copied, transformed, or erased. This raises a deeper question: is there a physical lower bound on the effort required to reduce nescience?

A partial answer is provided by Landauer's principle. According to this principle, the irreversible erasure of one bit of information in a system at temperature T requires the dissipation of at least

$$k_B T \ln 2$$

units of energy as heat, where k_B is Boltzmann's constant.

This observation does not imply that effort should be identified with energy. The effort defined in this chapter is abstract and computational. Nevertheless,

Landauer's principle shows that any physical implementation of irreversible information erasure has a nonzero energetic cost.

Definition 12.8.1 Let a research path π involve the irreversible erasure of $m(\pi)$ bits in a physical system at temperature T . The *Landauer lower bound* for the energetic cost of π is

$$E_{\min}(\pi) = m(\pi)k_B T \ln 2.$$

This lower bound is usually far below the practical energy costs of scientific research. Its importance is not practical but foundational: it shows that information refinement cannot be physically free when it requires irreversible erasure.

The connection is especially natural in the case of surfeit. Surfeit measures excess descriptive material. When reducing surfeit requires physically erasing unnecessary bits from a stored description, Landauer's principle provides a theoretical lower bound on the energetic cost of that erasure.

However, care is needed. Abstract simplification is not identical to physical erasure. A description may be compressed reversibly, copied, or transformed without immediately erasing information. Landauer's bound applies specifically to irreversible erasure, not to every logical or mathematical act of simplification.

Thus, thermodynamic effort should be interpreted as a lower physical bound on implemented research processes, not as the primary definition of effort.

12.9 Monetary Cost of Nescience Reduction

Computational effort measures the number of steps required to reduce nescience. Practical scientific research, however, is often constrained by money. We therefore need a way to convert effort into monetary cost.

This cannot be done by assigning a universal price to a computational step. The monetary cost of a step depends on the implementation: the hardware used, the personnel involved, the experimental apparatus, the cost of data acquisition, and the institutional context. Therefore, monetary cost must be defined through a resource model.

Let

$$\rho(x, a)$$

be a resource vector associated with applying action a to hypothesis x . For example,

$$\rho(x, a) = (\rho_{\text{human}}, \rho_{\text{compute}}, \rho_{\text{data}}, \rho_{\text{experiment}}, \rho_{\text{equipment}}, \rho_{\text{validation}}, \rho_{\text{coordination}}).$$

Each component measures the amount of a resource consumed by the operation.

Let

$$p = (p_{\text{human}}, p_{\text{compute}}, p_{\text{data}}, p_{\text{experiment}}, p_{\text{equipment}}, p_{\text{validation}}, p_{\text{coordination}})$$

be a vector of unit prices.

Definition 12.9.1 The *monetary cost* of applying operation a to hypothesis x is

$$m(x, a) = p \cdot \rho(x, a).$$

For a research path

$$\pi = (x_0, a_1, x_1, \dots, a_k, x_k),$$

the total monetary cost is

$$M(\pi) = \sum_{i=1}^k m(x_{i-1}, a_i).$$

Definition 12.9.2 Let $x \in \mathcal{H}$ and let $\varepsilon > 0$. The *monetary cost required to reduce nescience by ε* is

$$\text{MCost}_{\varepsilon}(x) = \inf_{\pi: x \rightsquigarrow y} \{M(\pi) : N(y) \leq N(x) - \varepsilon\}.$$

For a relative reduction of 1%, we define

$$\text{MCost}_{1\%}(x) = \inf_{\pi: x \rightsquigarrow y} \{M(\pi) : N(y) \leq 0.99N(x)\}.$$

This expression formalizes the question:

How much does it cost to reduce the nescience of this topic by one percent?

In practice, this quantity will almost never be known exactly. It must be estimated using empirical cost models, historical data, expert judgment, and approximations of nescience. Nevertheless, the definition is useful because it decomposes the problem into explicit elements:

- a current hypothesis x ;
- a practical approximation of $N(x)$;
- a set of admissible research operations;
- an estimate of the expected nescience reduction of each operation;
- a resource vector for each operation;
- a price vector for the resources.

12.10 Machine Learning Application

Machine learning provides a particularly clear setting in which to apply the concept of effort. In supervised learning, the representation r is usually a dataset, and the description d is a trained model or learning procedure. Reducing nescience means improving the representation, improving the model, or both.

Let

$$x = (r, d)$$

be the current hypothesis, where r is the available dataset and d is the current model. The main admissible operations include:

- collecting more samples;
- improving labels;
- adding features;
- removing irrelevant features;
- reducing measurement noise;
- balancing classes;
- changing the representation of the data;
- selecting another model family;
- tuning hyperparameters;
- increasing model size;
- training for longer;
- regularizing the model;
- compressing or pruning the model;
- ensembling several models;
- changing the loss function;
- improving validation procedures.

Some operations primarily reduce miscoding. For example, collecting better data, correcting labels, or adding missing features improves the representation. Other operations primarily reduce mismodel. For example, changing the model family, tuning hyperparameters, or improving optimization changes the description.

A practical monetary cost model for machine learning may take the form

$$M_{\text{ML}}(\pi) = p_{\text{gpu}}H_{\text{gpu}} + p_{\text{cpu}}H_{\text{cpu}} + p_{\text{label}}n_{\text{label}} + p_{\text{data}}D + p_{\text{eng}}H_{\text{eng}} + p_{\text{val}}V,$$

where:

- H_{gpu} is the number of GPU-hours;
- H_{cpu} is the number of CPU-hours;
- n_{label} is the number of labels acquired or corrected;
- D is the amount of external data purchased or generated;
- H_{eng} is the number of engineering hours;

- V is the validation cost;
- the p 's are unit prices.

The expected return on monetary effort is then

$$\text{ERoM}(\pi) = \frac{\mathbb{E}[N(x_0) - N(x_k)]}{M_{\text{ML}}(\pi)}.$$

■ **Example 12.6** Suppose a classification problem has current nescience $N(x) = 0.30$. Three possible actions are available:

- a_1 : tune hyperparameters at cost 2,000 euros, expected nescience reduction 0.002;
- a_2 : label additional data at cost 10,000 euros, expected nescience reduction 0.009;
- a_3 : remove irrelevant features at cost 1,500 euros, expected nescience reduction 0.003.

The expected reductions per euro are:

$$\frac{0.002}{2000} = 10^{-6},$$

$$\frac{0.009}{10000} = 9 \cdot 10^{-7},$$

and

$$\frac{0.003}{1500} = 2 \cdot 10^{-6}.$$

Although labeling additional data produces the largest expected reduction, feature removal has the best expected return per euro. ■

This example illustrates why effort is useful. A research action should not be evaluated only by its expected improvement, but also by the cost required to obtain that improvement.

Machine learning also allows effort curves to be estimated empirically. If C denotes compute, data, or money, then one can estimate a function

$$N(C)$$

describing the nescience remaining after spending effort C . The slope of this curve indicates marginal return:

$$-\frac{dN}{dC}.$$

When the curve flattens, additional effort yields diminishing returns.

12.11 Effort and the Difficulty of Research Areas

The concept of effort provides a way to address the question of whether some research areas are more difficult than others. A research area should not be considered difficult merely because its current nescience is high. It should be considered difficult if reducing its nescience requires high effort.

Let $A \subseteq E$ be a research area. Suppose that for each entity $e \in A$ there is a current best hypothesis x_e . We define the difficulty of e at scale ε as the effort required to reduce the nescience of x_e by ε .

Definition 12.11.1 Let $e \in E$ be an entity with current best hypothesis x_e . The *difficulty of e at scale ε* is

$$D_\varepsilon(e) = \text{Eff}_\varepsilon(x_e).$$

The difficulty of a research area is obtained by aggregating the difficulty of its entities.

Definition 12.11.2 Let $A \subseteq E$ be a finite research area and let $w_e \geq 0$ be weights satisfying

$$\sum_{e \in A} w_e = 1.$$

The *difficulty of the research area A at scale ε* is

$$D_\varepsilon(A) = \sum_{e \in A} w_e D_\varepsilon(e).$$

The weights may represent scientific relevance, practical importance, funding priority, or proximity to the knowledge frontier.

For large areas, it may be preferable to compute difficulty only over frontier entities, that is, entities corresponding to current open problems or active research questions. If $F_A \subseteq A$ denotes the frontier of the area A , then the frontier difficulty is

$$D_\varepsilon^{\text{frontier}}(A) = \sum_{e \in F_A} w_e D_\varepsilon(e).$$

This definition separates the amount of ignorance in a field from the difficulty of reducing it. A field may contain many unknowns but still admit rapid progress if its representations are stable, its experiments are cheap, and its models can be efficiently tested. Another field may contain fewer open questions but require enormous effort for each improvement.

Definition 12.11.3 Let A and B be two research areas. We say that A is more difficult than B at scale ε if

$$D_\varepsilon(A) > D_\varepsilon(B).$$

This definition leads to a possible explanation of why different disciplines progress at different rates. Some areas may be successful not only because they contain more capable researchers or better institutions, but because their entities admit lower-effort nescience reduction.

■ **Example 12.7** Many entities in physics admit stable quantitative representations: measurements, units, equations, coordinate systems, experimental protocols, and numerical simulations. These features restrict the hypothesis space and create strong feedback loops between representation, description, and experiment. As a consequence, many branches of physics may have relatively low effort per unit of nescience reduction, despite being mathematically and experimentally demanding.

By contrast, many entities in philosophy, such as truth, consciousness, meaning, causality, justice, or knowledge, are harder to represent in stable and operational terms. Different traditions may disagree about what counts as an adequate representation, and feedback mechanisms are often weaker than in experimental science. The difficulty may therefore lie not in lack of intelligence or rigor, but in the higher effort required to reduce miscoding and mismodel for such entities. ■

This example should not be read as a claim that philosophy is less valuable than physics. On the contrary, the effort framework suggests a more charitable interpretation: some philosophical entities may be difficult precisely because they have high representational and validation effort. Slow progress may reflect the structure of the entities rather than the inferiority of the discipline.

12.12 Difficulty Profiles

Different research areas may be difficult for different reasons. Therefore, instead of assigning a single difficulty value to an area, we can define a difficulty profile.

Definition 12.12.1 Let $A \subseteq E$ be a research area. A *difficulty profile* of A is a vector

$$\mathbf{D}(A) = (D^\mu(A), D^v(A), D^{\text{exp}}(A), D^{\text{comp}}(A), D^{\text{val}}(A), D^{\text{coord}}(A)),$$

where:

- $D^\mu(A)$ is the effort required to reduce miscoding;

- $D^v(A)$ is the effort required to reduce mismodel;
- $D^{\text{exp}}(A)$ is experimental effort;
- $D^{\text{comp}}(A)$ is computational effort;
- $D^{\text{val}}(A)$ is validation effort;
- $D^{\text{coord}}(A)$ is coordination effort.

This profile gives a more informative view than a single scalar measure. Physics may have high experimental effort but strong validation. Philosophy may have lower monetary cost but higher representational and validation effort. Biology may have high miscoding effort because living systems are heterogeneous and measurement is difficult. Mathematics may have low experimental cost but high proof-search effort.

■ **Example 12.8** Particle physics may have a difficulty profile dominated by experimental and monetary effort. Building accelerators and detectors is expensive, but the entities involved often have precise mathematical representations and strong validation procedures.

Philosophy may have a difficulty profile dominated by representational and validation effort. The monetary cost of proposing an argument may be low, but the cost of stabilizing the relevant concepts and obtaining agreement about improvement may be high. ■

The concept of difficulty profiles avoids simplistic comparisons between disciplines. It allows us to say not merely that one area is harder than another, but why it is harder.

12.13 Effort and Research Prioritization

Effort is central to research prioritization. A research question should not be evaluated only by its importance or its expected nescience reduction. It should also be evaluated by the effort required to obtain that reduction.

Let q be a research question. Suppose that answering q is expected to reduce nescience by

$$\mathbb{E}[\Delta N(q)]$$

and requires expected effort

$$\mathbb{E}[\text{Eff}(q)].$$

A simple measure of research efficiency is

$$R(q) = \frac{\mathbb{E}[\Delta N(q)]}{\mathbb{E}[\text{Eff}(q)]}.$$

If $I(q)$ denotes the interestingness of q , one may include effort as a denominator:

$$I(q) = \frac{\text{Rel}(q) \cdot \text{App}(q) \cdot \mathbb{E}[\Delta N(q)]}{\mathbb{E}[\text{Eff}(q)]},$$

where $\text{Rel}(q)$ is the relevance of the question and $\text{App}(q)$ is its applicability.

This formula is not intended as a universal definition. Rather, it illustrates how effort can be incorporated into the selection of research questions. A question may be very important but require such high effort that it is not the best immediate target. Conversely, a modest question may be worth pursuing if it offers substantial nescience reduction at low cost.

Research funding can also be formulated as a portfolio problem. Suppose that actions $a_1, \dots, a_k$ are available, with monetary costs $M(a_i)$ and expected nescience reductions $\mathbb{E}[\Delta N(a_i)]$. Given a budget B , the objective is to choose a subset of actions maximizing expected nescience reduction:

$$\max_{S \subseteq \{1, \dots, k\}} \sum_{i \in S} \mathbb{E}[\Delta N(a_i)]$$

subject to

$$\sum_{i \in S} M(a_i) \leq B.$$

This formulation connects the theory of nescience with scientific management and funding allocation.

12.14 Limits of Effort Reduction

The effort required to reduce nescience may be infinite.

Definition 12.14.1 Let $x \in \mathcal{H}$ and $\varepsilon > 0$. We say that the reduction of nescience by ε from x is *effort-impossible* if

$$\text{Eff}_\varepsilon(x) = \infty.$$

This situation can arise for several reasons.

First, no better hypothesis may exist within the chosen hypothesis space. Second, better hypotheses may exist but may not be reachable under the admissible operations. Third, the improvement may require information that is not computable. Fourth, the entity may be unknowable under the relevant representation framework. Finally, the effort may be finite in theory but so large that it is practically infinite.

This distinction is important. A problem may be:

- *theoretically impossible*, if no admissible path exists;
- *computationally intractable*, if a path exists but requires prohibitive computation;
- *economically infeasible*, if the monetary cost exceeds available resources;
- *experimentally inaccessible*, if the required observations cannot be obtained;
- *conceptually blocked*, if no stable representation is available.

In such cases, the rational response may not be to continue applying effort within the same framework, but to reframe the problem. A new representation may lower effort even before it reduces nescience directly.

12.15 Conclusion

Nescience measures how much we do not know. Effort measures how difficult it is to reduce what we do not know. The two concepts are orthogonal. A topic may have high nescience and low effort, low nescience and high effort, or any combination in between.

In this chapter, effort has been defined as the computational cost of moving from one hypothesis to another with lower nescience. This definition allows us to study the difficulty of reducing nescience in general, and the difficulty of reducing its components, such as miscoding and mismodel, in particular. It also provides a bridge from abstract theory to practical applications. By introducing resource vectors and price vectors, computational effort can be connected to monetary cost. This makes it possible, at least in principle, to ask how much it costs to reduce the nescience of a topic by a specified amount.

The concept of effort also sheds light on the difficulty of research areas. Some fields may progress faster not because their entities are more important or their researchers more capable, but because their entities admit lower-effort reductions of nescience. Stable representations, strong feedback mechanisms, good instruments, and quantitative models all reduce effort. Conversely, unstable representations, weak validation procedures, and high conceptual ambiguity increase effort.

Finally, effort provides a natural framework for research prioritization. A rational research strategy should consider not only expected nescience reduction, but expected nescience reduction per unit of effort. In this way, effort connects the theory of nescience with scientific planning, machine learning, funding allocation, and the long-term project of understanding the limits of knowledge.

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13. Machine Learning

*There are no difficult problems,
only lack of imagination.*
Antonio García

We have seen that the most challenging problems to which we can apply the results of the theory of nescience arise when the set of entities $\mathcal{E}$ under study is composed of abstract elements. The difficulty with abstract entities is that there is no straightforward way to encode them as finite strings of symbols from which they can be effectively reconstructed. Without such an encoding, it becomes impossible to measure their redundancy, accuracy, or any other property required by the theory of nescience.

In practice, a common approach to overcome this limitation is to design an experiment and collect observable outcomes, as is typically done in physics or other empirical sciences. Another possible approach is to collect a set of indirect measurements that serve as observable proxies for the abstract entities, for example, by recording the behavior of users in an online social network to study social interaction patterns.

This chapter is devoted to applying the concept of minimum nescience to the field of machine learning. We assume that the entities under study have been

encoded as a dataset $\mathbb{X}$ consisting of n training vectors, each with p predictors (or features), and an associated response variable $\mathbf{y}$ (see Section 21.2).

We will begin by presenting practical approximations for the concepts of miscoding, inaccuracy, and surfeit when the entities are represented as datasets. We will then show how these quantities can be combined into the single measure of nescience. These approximations will allow us to introduce the minimum nescience principle, a method designed to automate the process of selecting optimal models in machine learning (a form of automated machine learning, or AutoML).

In addition to introducing these approximations, we will demonstrate how to apply them to solve practical machine learning problems. The examples provided will be based on the `nescience`¹ library, an open-source Python library that implements the ideas presented in this chapter.

13.1 Nescience Python Library

The `nescience` library is an open-source Python package that implements the ideas presented in this book, specifically their application to the field of machine learning. The library follows the API design and conventions of the widely used `scikit-learn` toolkit, which allows it to be seamlessly combined with the methods and workflows provided by that package.

The `nescience` library can be installed with the `pip` utility:

```
pip install nescience
```

On the companion website for the library, the reader will find a collection of `jupyter-lab` notebooks that illustrate how the library works. Each subsection of this chapter is accompanied by a notebook containing implementations of all the examples discussed, so that the reader can reproduce the results and experiment with them interactively. Additional information about the `nescience` library, including a complete API reference, is also available on the library's website.

13.2 A Note on Compression

The practical computation of the components of `nescience` requires finite, computable approximations to quantities that are theoretically defined in terms of Kolmogorov complexity. Since Kolmogorov complexity is not computable in general, the `mnpilib` library replaces it with empirical code lengths derived from observed data. The role of the `utils` functions in the library is therefore foundational: they transform observed variables into finite empirical states and compute the corresponding empirical distributions, entropies, and code lengths.

¹<https://github.com/rleiva/nescience>

Let $\mathbf{x} = (x_1, \dots, x_n)$ be a vector of observations of a random variable X . If X is categorical and takes values in a finite alphabet $\mathcal{G} = \{g_1, \dots, g_\ell\}$, we denote by

$$n_i = \sum_{j=1}^n I(x_j = g_i)$$

the number of observations belonging to category g_i , and by

$$\hat{p}_i = \frac{n_i}{n}$$

the corresponding empirical probability. The empirical entropy of $\mathbf{x}$ is then

$$\hat{H}(\mathbf{x}) = - \sum_{i: n_i > 0} \hat{p}_i \log_2 \hat{p}_i.$$

The empirical code length associated with an optimal prefix code adapted to these observed frequencies is

$$\hat{K}_C(\mathbf{x}) = n\hat{H}(\mathbf{x}) = - \sum_{i: n_i > 0} n_i \log_2 \hat{p}_i.$$

Equivalently, since each observation x_j is encoded using a codeword of length approximately $-\log_2 \hat{p}(x_j)$, we have

$$\hat{K}_C(\mathbf{x}) = - \sum_{j=1}^n \log_2 \hat{p}(x_j).$$

This quantity is not a direct estimate of Kolmogorov complexity in the algorithmic sense, but a practical empirical approximation based on Shannon coding (see Section 19.3). It measures the length of the data under the empirical distribution induced by the observed sample.

The same construction applies to several random variables. Given vectors $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}$ with n observations each, we form the joint empirical vector

$$\langle \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)} \rangle = \left\{ \langle x_1^{(1)}, \dots, x_1^{(m)} \rangle, \dots, \langle x_n^{(1)}, \dots, x_n^{(m)} \rangle \right\}.$$

The observed joint states are the distinct tuples appearing in this vector. If these joint states are denoted by $s_1, \dots, s_q$, with counts $n_1, \dots, n_q$, the joint empirical entropy is

$$\hat{H}(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}) = - \sum_{i: n_i > 0} \frac{n_i}{n} \log_2 \left(\frac{n_i}{n} \right),$$

and the joint empirical code length is

$$\hat{K}_C(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}) = n\hat{H}(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}).$$

This formulation is especially useful because many quantities appearing in the theory of nescience can be expressed in terms of individual and joint code lengths. For example, the compression-based approximation of an information distance between two empirical vectors $\mathbf{x}$ and $\mathbf{y}$ can be written as

$$E_C(\mathbf{x}, \mathbf{y}) = \frac{\hat{K}_C(\mathbf{x}, \mathbf{y}) - \min\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\mathbf{y})\}}{\max\{\hat{K}_C(\mathbf{x}), \hat{K}_C(\mathbf{y})\}}.$$

Since $\hat{K}_C(\mathbf{x}) = n\hat{H}(\mathbf{x})$, this is equivalent to

$$E_C(\mathbf{x}, \mathbf{y}) = \frac{\hat{H}(\mathbf{x}, \mathbf{y}) - \min\{\hat{H}(\mathbf{x}), \hat{H}(\mathbf{y})\}}{\max\{\hat{H}(\mathbf{x}), \hat{H}(\mathbf{y})\}}.$$

Using the empirical mutual information

$$\hat{I}(\mathbf{x}; \mathbf{y}) = \hat{H}(\mathbf{x}) + \hat{H}(\mathbf{y}) - \hat{H}(\mathbf{x}, \mathbf{y}),$$

we obtain the equivalent expression

$$E_C(\mathbf{x}, \mathbf{y}) = 1 - \frac{\hat{I}(\mathbf{x}; \mathbf{y})}{\max\{\hat{H}(\mathbf{x}), \hat{H}(\mathbf{y})\}}.$$

Thus, when optimal empirical codes are used, the compression-based distance coincides with a normalized mutual-information distance. This identity explains why empirical code length is a natural practical substitute for compression in the machine-learning implementation of the theory.

Encoding Numeric Variables

The previous construction assumes that the values of the random variable belong to a finite set. Numeric variables, however, often take many distinct values, and in the case of continuous variables the probability of observing any exact value is zero. Therefore, before empirical frequencies can be computed, numeric vectors must be discretized.

A discretization procedure maps a numeric vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ to an encoded vector $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_n)$ whose values belong to a finite alphabet of bins. If the bins are denoted by $D_1, \dots, D_b$, then

$$\tilde{x}_j = i \iff x_j \in D_i.$$

The empirical code length of the numeric vector is then computed from the discretized vector:

$$\hat{K}_C(\mathbf{x}) \approx \hat{K}_C(\tilde{\mathbf{x}}) = - \sum_{i=1}^b n_i \log_2 \left(\frac{n_i}{n} \right),$$

where

$$n_i = \sum_{j=1}^n I(x_j \in D_i).$$

The `mnplib` library is based on an uniform discretization. In uniform discretization, the interval between the minimum and maximum observed values is divided into intervals of equal width. This choice is deliberate. The goal of discretization in this context is not merely to create balanced bins, but to preserve the empirical shape of the numeric distribution at the chosen resolution. Dense regions of the original vector remain dense after discretization, and sparse regions remain sparse. Consequently, the empirical frequencies of the discretized vector still contain information about the distribution of the original numeric vector.

This is important because empirical code length is computed from the observed probabilities of the discrete states. If the discretization procedure forced all bins to contain approximately the same number of observations (eg. quantaly based discretization), the resulting distribution would become close to uniform and the empirical code length would be determined mainly by the number of bins rather than by the original distribution of the data. Uniform discretization avoids this collapse and is therefore better aligned with the compression interpretation used in `mnplib`.

The choice of the number of bins is critical. Too few bins produce an overly coarse representation and may hide relevant variation. Too many bins produce sparse empirical distributions and unreliable probability estimates. The `mnplib` library therefore uses a cubic-root rule inspired by the classical Rice rule.

The Rice rule is a simple heuristic for choosing the number of bins in a histogram. Given n observations, it recommends using approximately

$$b = 2n^{1/3}$$

bins. The rule is not derived from the specific distribution of the data; instead, it provides a general-purpose compromise between excessive coarseness and excessive sparsity. Following this idea, when `n_bins="auto"`, the library chooses

$$b = \left\lceil 2n^{1/3} \right\rceil.$$

In practice, this value is constrained by user-defined lower and upper bounds. The lower bound prevents degenerate discretizations with too few bins, while the upper bound prevents excessively fine partitions in small or noisy datasets.

This rule depends only on the sample size, not on the number of numeric variables being encoded jointly. Consequently, when several numeric variables are discretized in the same empirical distribution, each numeric variable is

discretized using the same number of bins, although the bin edges are computed separately from the range of each variable. This provides a common level of granularity across the encoded variables while preserving the empirical shape of each marginal numeric distribution (see Sections 13.3 and 13.4).

The current utility layer is stateless. Each call to the empirical distribution, empirical entropy, or empirical code-length functions receives the variables to be analyzed, encodes them within that call, and computes the corresponding empirical quantities. No fitted encoder object is stored and no learned representation is reused across calls.

Finite-Sample Corrections

The empirical entropy

$$\hat{H}(\mathbf{x}) = -\sum_i \hat{p}_i \log_2 \hat{p}_i$$

is a plug-in estimator. Although simple and natural, it is biased in finite samples, especially when the number of observed states is large relative to the number of observations. The `mnpilib` library therefore allows entropy-bias corrections.

One classical correction is the Miller–Madow correction. If q is the number of observed states and n is the sample size, the corrected entropy is

$$\hat{H}_{MM}(\mathbf{x}) = \hat{H}(\mathbf{x}) + \frac{q-1}{2n \ln 2}.$$

More generally, if entropy is computed with logarithm base a , the correction becomes

$$\hat{H}_{MM}^{(a)}(\mathbf{x}) = \hat{H}^{(a)}(\mathbf{x}) + \frac{q-1}{2n \ln a}.$$

This correction increases the entropy estimate to compensate for the downward bias of the empirical plug-in estimator.

Another possibility is Dirichlet smoothing. If the empirical counts are $n_1, \dots, n_q$, and an alphabet of size Q is assumed, one defines smoothed probabilities

$$\tilde{p}_i = \frac{n_i + \alpha}{n + \alpha Q},$$

where $\alpha > 0$ is a smoothing parameter. The corresponding smoothed entropy is

$$\hat{H}_\alpha(\mathbf{x}) = -\sum_{i=1}^Q \tilde{p}_i \log_2 \tilde{p}_i.$$

Dirichlet smoothing is useful when some states are possible but unobserved, or when zero empirical probabilities would make the representation too brittle. In the default plug-in computation, however, only observed states contribute to the empirical code length.

Normalized and Per-Sample Quantities

The utility functions distinguish between entropy and total code length. Entropy is a per-sample quantity:

$$\hat{H}(\mathbf{x}) = -\sum_i \hat{p}_i \log_2 \hat{p}_i,$$

whereas code length is a total quantity:

$$\hat{K}_C(\mathbf{x}) = n\hat{H}(\mathbf{x}).$$

Both are useful. Total code length is appropriate when approximating description length, while entropy is useful when comparing samples of different sizes.

The library also supports normalized entropy. If the empirical alphabet has q possible states, the maximum entropy is $\log_2 q$. The normalized entropy is therefore

$$\hat{H}_{\text{norm}}(\mathbf{x}) = \frac{\hat{H}(\mathbf{x})}{\log_2 q},$$

with the convention that the normalized entropy is zero when $q = 1$. This quantity lies in the interval $[0, 1]$, and measures how close the empirical distribution is to the uniform distribution over its state space.

The logarithm base is also explicit. The default base is 2, so that all code lengths are measured in bits. Other bases may be used when convenient: base e gives entropy in nats, and base 10 gives entropy in decimal units.

Information-Theoretic Identities

The empirical code-length utilities are designed to satisfy basic information-theoretic identities, up to finite-sample and discretization effects. For any empirical vectors $\mathbf{x}$ and $\mathbf{y}$, the following properties hold for the plug-in estimates:

$$\hat{H}(\mathbf{x}) \geq 0,$$

$$\hat{H}(\mathbf{x}, \mathbf{y}) \geq \hat{H}(\mathbf{x}), \quad \hat{H}(\mathbf{x}, \mathbf{y}) \geq \hat{H}(\mathbf{y}),$$

and

$$\hat{H}(\mathbf{x}, \mathbf{x}) = \hat{H}(\mathbf{x}).$$

The last identity expresses the fact that duplicating a variable does not add information. Similarly, if $\mathbf{x}$ is constant, then

$$\hat{H}(\mathbf{x}) = 0, \quad \hat{K}_C(\mathbf{x}) = 0.$$

These identities are used as sanity checks in the implementation. They ensure that the empirical representation behaves as expected before it is used in higher-level nescience quantities such as miscoding, inaccuracy, and surfeit.

Role in the Library

The utility layer of `mnplib` is not merely an implementation detail. It defines the empirical representation space on which the practical version of the theory operates. The theoretical framework speaks about representations, descriptions, and code lengths. In machine learning applications, these become finite empirical vectors, encoded states, empirical distributions, entropies, and code lengths.

The central approximation can be summarized as

$$K(\mathbf{x}) \rightsquigarrow \hat{K}_C(\mathbf{x}) = n\hat{H}(\tilde{\mathbf{x}}),$$

where $\tilde{\mathbf{x}}$ is the finite empirical encoding of the observed vector. For categorical variables, this encoding is obtained by mapping labels to discrete symbols. For numeric variables, it is obtained by uniform discretization. The quality of this approximation depends on the choice of discretization resolution, the treatment of categorical states, and the finite-sample correction applied to entropy. By making these choices explicit, the utility layer provides a stable computational basis for the rest of the library.

13.3 Miscoding

In Section ?? we introduced the concept of miscoding as a quantitative measure of how well a string based encoding $r \in \mathcal{R}$ represents a research entity from $\mathcal{E}$. The miscoding of a representation r was defined as:

$$\mu(r) = \min_{s \in \mathcal{R}_{\mathcal{E}}} \frac{\max\{K(s | r), K(r | s)\}}{\max\{K(s), K(r)\}}$$

and we saw that this quantity cannot be computed in practice in the general case: first, because it would require computations from an abstract oracle machine; second, because it relies on the uncomputable Kolmogorov complexity; and third, because it does not directly take into account the particular entity e we are interested in.

In this section, we will show how this concept can be adapted in practice to estimate the error made when using a dataset $\mathbf{X}$ as a representation of a response variable $\mathbf{y}$ (see Section 21.2). Our goal is twofold: on one hand, we are interested in measuring the overall quality of the dataset $\mathbf{X}$ as a predictor of the variable $\mathbf{y}$; on the other, we want to identify those individual features $\mathbf{x}_j$ of $\mathbf{X}$ that have the highest predictive power for $\mathbf{y}$. This is precisely the problem addressed by discriminative models (see Section ??), in which we want to estimate the conditional distribution $P(\mathbf{y} | \mathbf{X})$.

Given a training dataset $\mathbf{X}$, we can approximate the miscoding of a feature $\mathbf{x}_j$ for the target variable $\mathbf{y}$ by computing the normalized information distance

between $\mathbf{x}_j$ and $\mathbf{y}$ (see Section 20.5):

$$E(\mathbf{x}_j, \mathbf{y}) = \frac{\max\{K(\mathbf{x}_j | \mathbf{y}), K(\mathbf{y} | \mathbf{x}_j)\}}{\max\{K(\mathbf{x}_j), K(\mathbf{y})\}}$$

The Kolmogorov complexity $K(\mathbf{v})$ of a vector $\mathbf{v}$ will be approximated by the length of the compressed version of that vector $\hat{K}_C(\mathbf{v})$ using as compressor a minimal length code C (see Section ??).

Definition 13.3.1 Let $\mathbf{y}$ be a response variable, $\mathbf{X}$ a dataset composed by p features, and $\mathbf{x}_j$ the j -th feature. We define the regular feature miscoding of $\mathbf{x}_j$ as a representation of $\mathbf{y}$, denoted by $\hat{\mu}(\mathbf{x}_j, \mathbf{y})$, as:

$$\hat{\mu}(\mathbf{x}_j, \mathbf{y}) = \frac{\hat{K}_C(\mathbf{x}_j, \mathbf{y}) - \min\{\hat{K}_C(\mathbf{x}_j), \hat{K}_C(\mathbf{y})\}}{\max\{\hat{K}_C(\mathbf{x}_j), \hat{K}_C(\mathbf{y})\}}$$

Intuitively, the quantity $\hat{\mu}(\mathbf{x}_j, \mathbf{y})$ measures, in relative terms, the additional description length required to fully encode $\mathbf{y}$ given knowledge of $\mathbf{x}_j$, and vice versa. The lower this value, the better $\mathbf{x}_j$ serves as a predictor for $\mathbf{y}$.

■ **Example 13.1** Let $\mathbf{y}$ be a target variable composed of 1,000 random samples from a normal distribution $N(3, 1)$ with mean $\mu = 3$ and standard deviation $\sigma = 1$; let $\mathbf{x}_1$ be a predictor feature approximately equal to $\mathbf{y}$ with some added noise, that is $\mathbf{x}_1 = \mathbf{y} + N(3, 1)/10$; and let $\mathbf{x}_2$ be a second predictor consisting of random samples from an exponential distribution with rate $\lambda = 1$.

```
from scipy.stats import norm, expon

y = norm.rvs(loc=3, scale=1, size=10000)
x1 = y + norm.rvs(loc=3, scale=1, size=10000) / 10
x2 = expon.rvs(size=10000)
```

We can use the Nescience library to compute the miscoding of the features $\mathbf{x}_1$ and $\mathbf{x}_2$ when they encode the target variable $\mathbf{y}$.

```
from fastautoml.miscoding import Miscoding
import numpy as np

X = np.column_stack((x1, x2))

miscoding = Miscoding()
miscoding.fit(X, y)
miscoding.miscoding_features(mode="regular")
```

The output of the library would be something similar to the following²:

²Since we are generating a list of 1,000 random samples, the reader could get a slightly different result when running this example.

```
array([0.27445364, 0.9934222])
```

As it was expected the miscoding of $\hat{\mu}(\mathbf{x}_1, \mathbf{y})$ is much smaller than the miscoding of $\hat{\mu}(\mathbf{x}_2, \mathbf{y})$. In this case, we should prefer $\mathbf{x}_1$ over $\mathbf{x}_2$ as a predictor of $\mathbf{y}$. ■

Sometimes, we will use the normalized version of the complements of the individual miscodings, that is:

$$\frac{1 - \hat{\mu}(\mathbf{x}_i, \mathbf{y})}{\sum_{j=1}^p (1 - \hat{\mu}(\mathbf{x}_j, \mathbf{y}))},$$

instead of the raw values $\hat{\mu}(\mathbf{x}_i, \mathbf{y})$. This alternative, which we call the adjusted feature miscoding, is useful because it allows easier comparison with other feature selection techniques, and it has a visually intuitive interpretation as relative importance weights.

■ **Example 13.2** In this example we generate a synthetic dataset where the target variable $\mathbf{y}$ is a collection of normally distributed clusters of points, and the training set $\mathbb{X}$ is composed of both relevant and irrelevant predictors. In particular, we generate 1,000 samples with 20 features describing 10 clusters; only 4 of the features are relevant for prediction, and the remaining 16 are just random noise.

Figure 13.1 shows a two-dimensional projection of this dataset along the hyperplane composed of features 8 and 10.

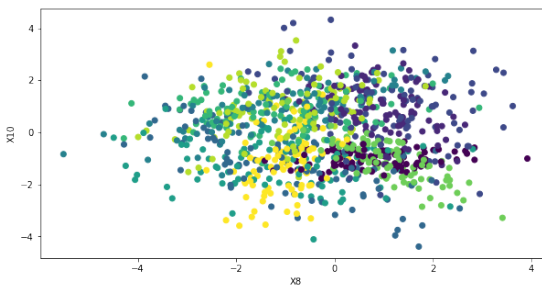


Figure 13.1: Gaussian Blob Cluster.

```
from fastautoml.miscoding import Miscoding
from sklearn.datasets.samples_generator import make_classification

X, y = make_classification(n_samples=1000, n_features=20, n_informative=4,
                          n_redundant=0, n_classes=10, n_clusters_per_class=1, flip_y=0)

miscoding = Miscoding()
miscoding.fit(X, y)
```



```
msd = miscoding.miscoding_features(mode='adjusted')
```

We use the adjusted version of miscoding for easier comparison with other feature selection techniques. If we plot the results (see Figure 13.2) we can see that the library has successfully identified the four relevant predictors (x_3 , x_8 , x_{10} and x_{16}). Since we are using the adjusted version, higher values are better; note that the actual values should be interpreted only in relative terms.

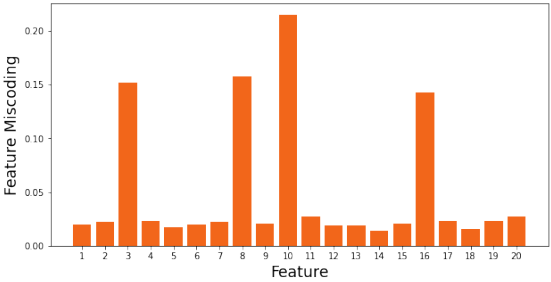


Figure 13.2: Miscoding of a Synthetic Dataset.

We can compare miscoding with correlation, a common technique used in machine learning to identify relevant features. Figure 13.3 shows the correlation between the individual features of $\mathbf{X}$ and the target variable $\mathbf{y}$. As we can observe, correlation fails to properly identify one of the relevant features (x_3).

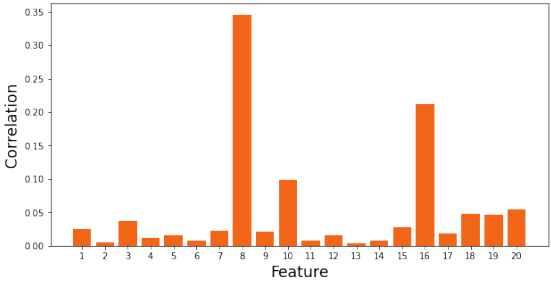


Figure 13.3: Correlation of a Synthetic Dataset.

Feature miscoding allows us to identify the most relevant features of a training dataset $\mathbf{X}$, but it cannot be used directly to compute the miscoding of the dataset itself. If we start with a miscoding of 1 (complete ignorance) and subtract the miscodings of the individual features, we may end up with a negative miscoding, which is not allowed by our theory. If we use the adjusted version, the dataset miscoding would be 0 for all datasets, which goes against the intuition that not

all datasets $\mathbf{X}$ represent a target variable $\mathbf{y}$ equally well. According to the theory of nescience, we expect non-relevant features to *add*, rather than subtract, to the global miscoding of the dataset.

To address this, we introduce the concept of partial miscoding of a feature, defined as the difference between the adjusted and the normalized miscodings.

Definition 13.3.2 Let $\mathbf{y}$ be a target variable, $\mathbf{X}$ a dataset composed by p features, and $\mathbf{x}_j$ the j -th feature. We define the partial miscoding of $\mathbf{x}_j$ as a representation of $\mathbf{y}$, denoted by $\tilde{\mu}(\mathbf{x}_j, \mathbf{y})$, as:

$$\tilde{\mu}(\mathbf{x}_i, \mathbf{y}) = \frac{1 - \hat{\mu}(\mathbf{x}_i, \mathbf{y})}{\sum_{j=1}^p 1 - \hat{\mu}(\mathbf{x}_j, \mathbf{y})} - \frac{\hat{\mu}(\mathbf{x}_i, \mathbf{y})}{\sum_{j=1}^p \hat{\mu}(\mathbf{x}_j, \mathbf{y})}$$

A positive partial miscoding indicates that the feature contributes to describing the target variable; a negative value indicates that the feature is not relevant (or is counterproductive) in representing $\mathbf{y}$.

■ **Example 13.3** We reuse the synthetic dataset of Example 13.2, but we increase the number of relevant features from 4 to 14. Then, we compute the list of partial miscodings.

```
from fastautoml.miscoding import Miscoding
from sklearn.datasets.samples_generator import make_classification

X, y = make_classification(n_samples=1000, n_features=20, n_informative=14,
                          n_redundant=0, n_classes=10, n_clusters_per_class=1, flip_y=0)

miscoding = Miscoding()
miscoding.fit(X, y)
msd = miscoding.miscoding_features(mode="partial")
```

As shown in Figure 13.4, the library not only correctly identifies the relevant features, but also assigns negative contributions to non-relevant features, thereby increasing the global miscoding when they are included.

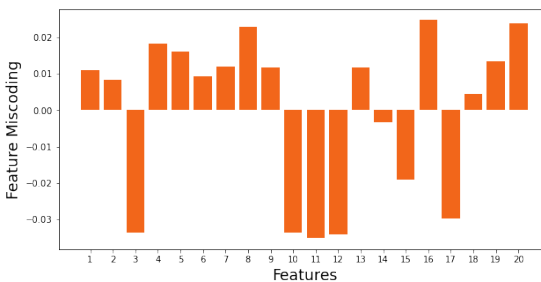


Figure 13.4: Partial Feature Miscoding.

Given the definition of partial feature miscoding, we can now define the miscoding of a target variable given a subset of predictors in a way that is closer to the original notion of miscoding in the theory of nescience.

Definition 13.3.3 Let $\mathbf{y}$ be a target variable, $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ a dataset composed by p features, and $\mathbf{Z} = \{\mathbf{z}_1, \dots, \mathbf{z}_k\}$ a subset of features, that is, $\{\mathbf{z}_1, \dots, \mathbf{z}_k\} \subseteq \{\mathbf{x}_1, \dots, \mathbf{x}_p\}$. We define the miscoding of $\mathbf{Z}$ as a representation of $\mathbf{y}$, denoted by $\hat{\mu}(\mathbf{Z}, \mathbf{y})$, as:

$$\hat{\mu}(\mathbf{Z}, \mathbf{y}) = \sum_{i=1}^k \tilde{\mu}(\mathbf{z}_i, \mathbf{y})$$

■ **Example 13.4** Using the dataset and partial feature miscodings computed in Example 13.3, Figure 13.5 shows the evolution of the miscoding of the training subset $\mathbf{Z}$ as we add more features to the study.

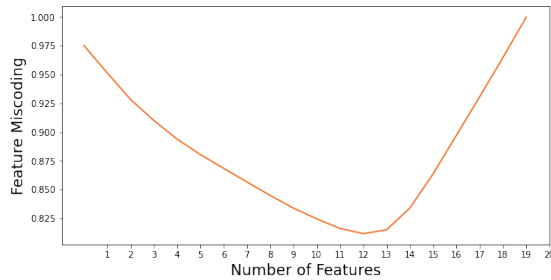


Figure 13.5: Accumulated Partial Feature Miscoding.

In the next example we compare the performance of a classifier when using the full dataset and a reduced version containing only those features identified as relevant (i.e., with positive partial miscoding).

■ **Example 13.5** We train a neural network on the `scikit-learn` digits dataset to classify handwritten digits. The evaluation criterion is the classifier's accuracy on a held-out test set. The neural network is trained and evaluated first on all features, and then on a reduced dataset containing only those features with positive partial miscoding.

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import load_digits
from fastautoml.fastautoml import Miscoding
```

```

data = load_digits()
X_raw = data.data
y_raw = data.target

miscoding = Miscoding()
miscoding.fit(X_raw, y_raw)
mscd = miscoding.miscoding_features(miscoding='partial')
X_red = X_raw[:,np.where(mscd > 0)[0]]
y_red = y_raw

X_raw_train, X_raw_test, y_raw_train, y_raw_test = train_test_split(X_raw,
                                                                    y_raw,
                                                                    test_size=.3)
X_red_train, X_red_test, y_red_train, y_red_test = train_test_split(X_red,
                                                                    y_red,
                                                                    test_size=.3)

clf = MLPClassifier(alpha=1, max_iter=1000)

clf.fit(X_raw_train, y_raw_train)
score_raw = clf.score(X_raw_test, y_raw_test)

clf.fit(X_red_train, y_red_train)
score_red = clf.score(X_red_test, y_red_test)

reduction = 1 - X_red_train.shape[1] / X_raw_train.shape[1]

print("Score raw:", score_raw, " Score Miscoding:", score_red,
      " Reduction:", reduction)

Score raw: 0.9833333333333333  Score Miscoding: 0.9814814814814815
Data Reduction: 0.46875

```

Running the above code typically shows that the classifier's accuracy is essentially the same for both datasets (about 98% on the test set). However, the reduced dataset used for training (selected via positive partial miscoding) is about 47% smaller than the original dataset. This size reduction can substantially decrease training time. Smaller datasets are also beneficial when training ensembles (e.g., random forests), where many models must be fitted.

■

Intuitively, as Example 13.4 suggests, we should prefer the subset $\mathbf{Z} \subseteq \mathbf{X}$ composed of those features whose partial miscodings are greater than zero. However, as we will see in the following sections, this might not always be the case. Feature selection is only one of the criteria used when searching for an optimal model for an entity represented by a dataset; other components of nescience, such as inaccuracy or surfeit, may suggest a different subset of predictors. The overall optimization criterion is the nescience itself. A sensible approach to using partial miscoding is to incrementally add to the model those features with the highest (positive) partial miscoding until all positive features are included or an external optimality criterion is reached.

In the case of a generative model (see Section ??), i.e., a machine learning algorithm designed to estimate the joint probability $P(\mathbf{X}, \mathbf{y})$, we can use

miscoding to assess how individual features relate to each other, by computing $\hat{\mu}(\mathbf{x}_i, \mathbf{x}_j)$ for each pair $i, j \leq p$. The result is a *miscoding matrix* (see Example 13.6).

■ **Example 13.6** The Boston dataset included in the `scikit-learn` library contains a collection of variables that (potentially) explain the price of houses in the Boston area. In this example, instead of identifying the factors that contribute most to price, we study the interdependence among factors using a miscoding matrix.

```
from fastautoml.fastautoml import Miscoding
from sklearn.datasets import load_boston

data = load_boston()

miscoding = Miscoding(X_type="numeric", y_type="numeric")
miscoding.fit(data.data, data.target)
mscd_matrix = miscoding.features_matrix(mode='regular')
```

Figure 13.6 shows a heatmap of the miscoding matrix computed over the features. Darker values represent lower miscoding (note we use the regular version here). In particular, the diagonal entries are equal to zero.

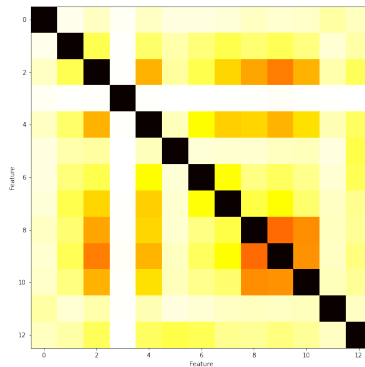


Figure 13.6: Regular Miscoding Matrix.

The minimum value of 0.52 is obtained for the pair (8,9), corresponding to the features "index of accessibility to radial highways" and "full-value property-tax rate per \$10,000". These features are good candidates to evaluate jointly in a predictive model, since they contain non-redundant information. The maximum value of 0.99 is achieved for the pair (3, 12), corresponding to "Charles River dummy variable" and "% lower status of the population". These features contain almost the same information; including both in a model adds little new information while increasing model complexity and the risk of overfitting.

13.4 Inaccuracy

In Section 9.1 we defined the inaccuracy of a description $d \in \mathcal{D}$ for a representation $r \in \mathcal{R}$ as the normalized information distance between the representation r and the string $\Gamma(d)$ printed out by a universal Turing machine when given the description as input:

$$i(d, r) = \frac{\max\{K(r \mid \Gamma(d)), K(\Gamma(d) \mid r)\}}{\max\{K(r), K(\Gamma(d))\}}$$

Inaccuracy, being based on Kolmogorov complexity, is not computable in the general case and must be approximated in practice. In this section we show how this concept can be estimated for a model trained on a dataset. The approach follows the approach described in Section 13.2.

For discriminative models, which estimate $P(\mathbf{y} \mid \mathbb{X})$, the natural proxy compares predicted targets $\hat{\mathbf{y}}$ with true targets $\mathbf{y}$ via normalized compression distance.

Definition 13.4.1 Let $\mathbb{X}$ be a dataset, $\mathbf{y}$ a response variable, m a model, and $\hat{\mathbf{y}} = m(\mathbb{X})$ the predicted values by m given $\mathbb{X}$. We define the *inaccuracy* of the model m for the target values $\mathbf{y}$, denoted by $\hat{i}(\hat{\mathbf{y}}, \mathbf{y})$, as:

$$\hat{i}(\hat{\mathbf{y}}, \mathbf{y}) = \frac{\hat{K}_C(\hat{\mathbf{y}}, \mathbf{y}) - \min\{\hat{K}_C(\hat{\mathbf{y}}), \hat{K}_C(\mathbf{y})\}}{\max\{\hat{K}_C(\hat{\mathbf{y}}), \hat{K}_C(\mathbf{y})\}}$$

Intuitively, $\hat{i}(\hat{\mathbf{y}}, \mathbf{y})$ measures how far the predictions are from the true values. The lower this quantity, the better m is as a predictor of $\mathbf{y}$. Unlike standard accuracy metrics, this measure accounts not only for how difficult it is to reconstruct $\mathbf{y}$ from $\hat{\mathbf{y}}$, but also for how much additional information $\hat{\mathbf{y}}$ contains that is unrelated to $\mathbf{y}$.

When a model provides explicit probabilities, one may also define $\hat{K}$ using the model code length $-\sum \log_2 \hat{p}_m(\cdot)$; we keep the compression-based proxy here for consistency with the rest of the chapter.

■ **Example 13.7** According to the minimum nescience principle, inaccuracy is the normalized compression distance between the actual targets $\mathbf{y}$ and the predicted targets $\hat{\mathbf{y}}$. We compare our inaccuracy metric with a classical score (we use $1 - \text{accuracy}$) on the MNIST digits dataset.

```
from mnplib.inaccuracy import Inaccuracy
from sklearn.datasets import load_digits

X, y = load_digits(return_X_y=True)

inacc = Inaccuracy()
inacc.fit(X, y)
```

We train a decision-tree classifier with maximum depth $i \in \{1, \dots, 20\}$:

```
from sklearn.tree import DecisionTreeClassifier

scores      = list()
inaccuracies = list()

for i in range(20):

    tree = DecisionTreeClassifier(max_depth=i, random_state=42)
    tree.fit(X, y)

    scores.append(1 - tree.score(X, y))
    inaccuracies.append(inacc.inaccuracy_model(tree))
```

We compare $1 - \text{accuracy}$ and inaccuracy. As shown in Figure 13.7, both behave similarly, with inaccuracy typically taking larger values due to its stronger emphasis on incorrectly predicted values.

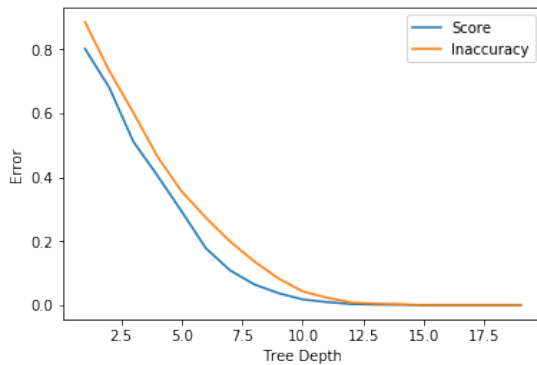


Figure 13.7: Inaccuracy vs. Score of Decison Trees

In Example 13.7 we observed that deeper trees yield smaller training error. Of course, larger depth increases the risk of overfitting. However, within the nescience framework, overfitting is handled by *surfeit* (see Section 13.5), so inaccuracy need not penalize complexity directly.

We can interpret inaccuracy as the effort, measured in description length, required to *fix* a model's predictions. Consequently, a model that makes one hundred *identical* errors is easier to correct than a model that makes one hundred *distinct* errors (see Example 13.8).

■ **Example 13.8** We again use a decision-tree classifier, now with the hyperparameter `min_samples_leaf=5` (a common regularization to reduce overfitting).

```
tree = DecisionTreeClassifier(min_samples_leaf=5)
tree.fit(X, y)
```

The inaccuracy of this new trained model is 0.17, and its score 0.08. Next we will artificially introduce one hundred errors in the dataset, simulating the case that the tree is not able to model correctly these data points. In this particular case all the errors are exactly the same.

```
X2 = X.copy()
y2 = y.copy()
for i in range(100):
    X2 = np.append(X2, [X[0]], axis=0)
    y2 = np.append(y2, (y[0]+1) % 10)
```

The inaccuracy of the decision tree, given this new dataset, has increased³ from 0.17 to 0.21.

```
inacc.fit(X2, y2)
inacc.inaccuracy_predictions(pred)
```

Score has also increased, in this case from 0.08 to 0.13.

```
1 - tree.score(X2, y2)
```

Finally, we are going to repeat exactly the same experiment, but this time instead of adding one hundred times the same error, adding one hundred different errors.

```
X3 = X.copy()
y3 = y.copy()
for i in arange(100):
    index = np.random.randint(X.shape[0])
    X3 = np.append(X3, [X[index]], axis=0)
    y3 = np.append(y3, (y[index]+1) % 10)
```

In this last case the inaccuracy of the model has increased up to 0.25, meanwhile score remained the same. ■

In line with Example 13.8, consider an extreme binary case: a model that always flips the label (predicts False when the truth is True, and vice versa). Classical accuracy deems this the worst possible model; our inaccuracy, however, is low because the predictions can be perfectly "fixed" by a simple transformation. In some applications (e.g., a hedge fund trading strategy), such a model would be extremely valuable.

In highly imbalanced datasets, standard accuracy can be misleading: a high

³Note that we had to `fit()` again the class `Inaccuracy` in order to use the new dataset. Normally this is not the way we use this class; instead what we should do is to fit once a dataset, and then compute the inaccuracy of different models. We are doing here in this way to demonstrate an interesting property of the concept of inaccuracy.

score may simply reflect correct classification of the majority class while failing on the minority class. Practitioners address this with specialized metrics. With our inaccuracy metric, as Example 13.9 shows, a model that fails to classify the minority class is penalized even if that class has few samples.

■ **Example 13.9** We create a synthetic dataset with two classes, using the `make_classification` utility of scikit-learn, where one has 95% of the samples and the other 5%:

```
from sklearn.datasets import make_classification

depth = list()
score = list()
inacc = list()

inaccuracy = Inaccuracy()

for i in np.arange(1, 100):

    X, y = make_classification(n_samples=1000, n_features=2,
                              n_informative=2, n_redundant=0,
                              class_sep=2, flip_y=0, weights=[0.95,0.05])

    inaccuracy.fit(X, y)

    tree = DecisionTreeClassifier(min_samples_leaf=i)
    tree.fit(X, y)

    depth.append(i)
    score.append(1 - tree.score(X, y))
    inacc.append(inaccuracy.inaccuracy_model(tree))
```

We vary the minimum number of samples per leaf $i \in \{1, \dots, 100\}$. In Figure 13.8, for large i the accuracy-based error (i.e., $1 - \text{accuracy}$) may stay near 0.05, yet the inaccuracy remains high because the model effectively ignores the minority class.

13.5 Surfeit

In Chapter 10 we defined the surfeit of a description $d \in \mathcal{D}$ for a representation $r \in \mathcal{R}$ as

$$\sigma(d, r) = \frac{|l(d) - K(r)|}{l(d)}.$$

This quantity measures how much longer the description d is than an ideal minimal description of the representation r . If $l(d)$ is close to $K(r)$, then the description contains little surplus. If $l(d)$ is much larger than $K(r)$, then the description contains unnecessary structure.

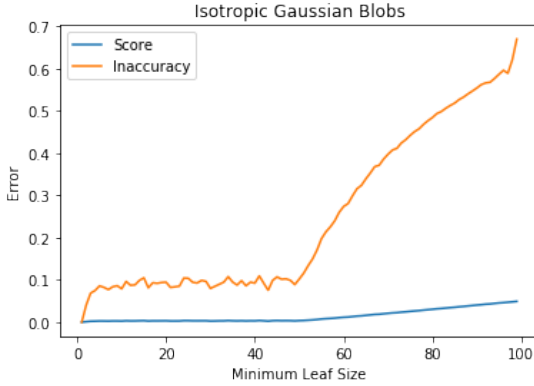


Figure 13.8: Inaccuracy of Decision Tree.

In practice, however, $K(r)$ is not computable. Moreover, when we want to study the internal redundancy of a model description itself, it is useful to consider a representation-independent approximation. Instead of asking how far d is from a shortest description of r , we ask how far d is from a shortest description of itself. This gives the redundancy-based approximation

$$\rho(d) = 1 - \frac{K(d)}{l(d)}.$$

Here $K(d)$ denotes the Kolmogorov complexity of the string d . A string with low redundancy has $K(d) \approx l(d)$, and therefore $\sigma(d) \approx 0$. A highly redundant string has $K(d) \ll l(d)$, and therefore $\sigma(d)$ is close to one.

This formulation is particularly convenient in machine learning. A trained model m is first converted into a string-based description d_m , and surfeit is then estimated from the redundancy of that description:

$$\rho(m) = \sigma(d_m) = 1 - \frac{K(d_m)}{l(d_m)}.$$

The model is therefore studied as a string of symbols. No reference to the training dataset or to the target variable is needed in the computation of surfeit. Surfeit measures the internal surplus of the model description, not the accuracy of the model with respect to observed data.

The theoretical difficulty is that $K(d_m)$ is not computable. The `mnplib` library therefore approximates $K(d_m)$ using ordinary text compression. If $C(d_m)$ denotes the length of the compressed version of the model description, the practical approximation is

$$\hat{\rho}(m) = 1 - \frac{C(d_m)}{l(d_m)}.$$

This approximation is not intended to recover Kolmogorov complexity exactly. Rather, it provides a practical test for redundancy: if the model description contains repeated patterns, regularities, or unnecessary symbolic structure, a compressor should be able to encode it using fewer symbols.

The current implementation uses `zlib` as the canonical compressor. This choice is deliberate. `zlib` is fast, deterministic, widely available, and has a small and well-understood format overhead. These properties are important because `surfeit` may be computed repeatedly during model comparison. Using a single compressor also makes the resulting values more reproducible: two users evaluating the same model description should not obtain different `surfeit` values merely because they selected different compression algorithms.

A practical issue arises when the model description is very short. For short strings, the compressed string may be longer than the original one:

$$C(d_m) > l(d_m).$$

This does not mean that the model has negative redundancy. It usually means that the fixed overhead of the compression format dominates the compressed representation. In the case of the standard `zlib` format, the stream contains a small header and a checksum trailer. To avoid interpreting this format overhead as model complexity, `mnplib` applies an overhead correction.

Let h_{zlib} be the assumed `zlib` overhead. The effective compressed length is defined as

$$C_{\text{eff}}(d_m) = \min \{l(d_m), \max \{0, C(d_m) - h_{\text{zlib}}\}\}.$$

The default value used by the library is

$$h_{\text{zlib}} = 6,$$

corresponding to the usual two-byte header and four-byte checksum trailer. The clipping operation ensures that the effective compressed length is never negative and never larger than the original model description.

There is a second practical adjustment. A model description should not be credited with a useful compressed length larger than the amount of information contained in the target it is intended to describe. If the target has empirical code length $\hat{K}_C(\mathbf{y})$, and if

$$\hat{K}_C(\mathbf{y}) < C_{\text{eff}}(d_m),$$

then the implementation uses $\hat{K}_C(\mathbf{y})$ as the reference length instead of $C_{\text{eff}}(d_m)$. In other words, the practical reference length is

$$L_{\text{ref}}(m, \mathbf{y}) = \min \{C_{\text{eff}}(d_m), \hat{K}_C(\mathbf{y})\}.$$

This condition prevents a model description from being treated as more informative than the target itself. It is especially relevant when the model description is long, or only weakly compressible, while the target variable is simple.

Definition 13.5.1 Let m be a trained model, let d_m be its string-based description, and let $\mathbf{y}$ be the target variable that the model is intended to describe. We define the practical surfeit of m as

$$\hat{\rho}(m, \mathbf{y}) = 1 - \frac{L_{\text{ref}}(m, \mathbf{y})}{l(d_m)},$$

where

$$L_{\text{ref}}(m, \mathbf{y}) = \min \{C_{\text{eff}}(d_m), \hat{K}_C(\mathbf{y})\}.$$

Thus, the implementation follows two safeguards. First, it corrects the raw compressed length by subtracting the fixed `zlib` overhead. Second, it uses the empirical code length of the target as an upper bound on the useful reference length. A compact, non-redundant model description has low surfeit. A model whose description contains repeated or unnecessary structure has high surfeit. In this sense, surfeit acts as a practical approximation to descriptive surplus, while remaining bounded by the information content of the target being described.

The implementation requires a way to encode models as strings of symbols. Ideally, descriptions would be encoded as Turing machines with respect to a fixed universal Turing machine. This would be the closest formulation to the theoretical definition. In practice, however, such an encoding would be difficult to implement and extend. The library therefore uses a restricted, Python-like textual representation of the predictive procedure. The intention is not to serialize the software object, but to produce a simplified program-like description of what the model computes. Third-party libraries, external state, and implementation-specific metadata are excluded from the description whenever possible.

A typical usage of the `Surfeit` class is:

```
from mnplib.surfeit import Surfeit

sft = Surfeit()

surfeit = sft.surfeit_model(model)
print(surfeit)
```

When the model description is already available as a string, surfeit can be computed directly from that string:

```
from mnplib.surfeit import Surfeit

model_string = """
def model(X):
    if X1 < 0.5:
        return 0
    else:
        return 1
"""

sft = Surfeit()

surfeit = sft.surfeit_string(model_string)
print(surfeit)
```

For diagnostic purposes, the detailed interface exposes the intermediate quantities used in the calculation:

```
details = sft.surfeit_string_detailed(model_string)

for key, value in details.items():
    print(key, value)
```

The detailed output includes the raw length of the model description, the raw zlib-compressed length, the overhead correction, the effective compressed length, and the final surfeit value.

Surfeit is useful in machine learning because it penalizes unnecessary descriptive structure. A highly complex model may fit the training data well, but if its description contains large amounts of repeated or compressible structure, then its surfeit will be high. Conversely, a very simple model may have low surfeit, but this does not imply that it is accurate. Surfeit should therefore be interpreted together with inaccuracy. Inaccuracy measures how well the model describes the observed phenomenon; surfeit measures how much surplus there is in the model description itself. The preferred model is not necessarily the one with lowest inaccuracy or lowest surfeit in isolation, but one that achieves a good trade-off between both quantities.

■ **Example 13.10** Surfeit can help identify overfitted models. Suppose that polynomial regression models of increasing degree are fitted to the same phenomenon. As the degree increases, the model description usually becomes longer and may contain more repeated symbolic structure. A high-degree polynomial may achieve low inaccuracy on the training data, but it can also have high surfeit because its description contains unnecessary descriptive surplus.

In such a case, surfeit acts as a counterbalance to inaccuracy. A low-degree model may be too simple and therefore inaccurate. A very high-degree model may be accurate on the training data but unnecessarily complex. The preferred model is one that achieves low inaccuracy with low surfeit. ■

Another advantage of surfeit is that it allows us to compare models from different families. If two models achieve similar inaccuracy, surfeit provides a way to prefer the model with the more economical description. For example, a small decision tree may have lower surfeit than a neural network when the decision boundary is simple. In a more complex problem, however, a neural network may provide a more compact description than a large decision tree with many branches. Surfeit therefore does not impose a fixed preference for one model family over another; it evaluates the descriptive surplus of the specific model description under consideration.

When predictive performance differs substantially, surfeit should not be used alone. A model with very low surfeit may simply be underfitted. A model with very low inaccuracy may be overfitted. The role of nescience is to evaluate these quantities jointly, selecting models that provide an adequate description of the phenomenon without unnecessary descriptive surplus.

Surfeit can help avoid overfitted models. The higher the surfeit of a model, the greater the risk that the model is an overfit to the training dataset, as Example 13.11 illustrates.

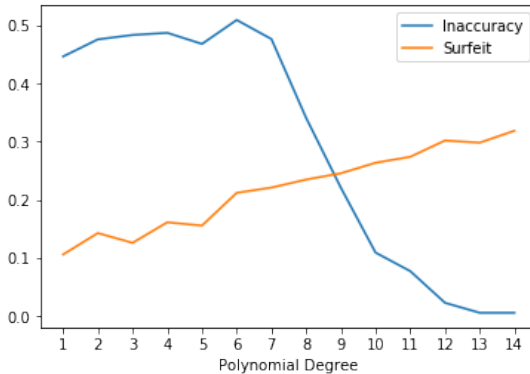


Figure 13.9: Surfeit vs Inaccuracy

■ **Example 13.11** We generate 900 samples from a sinusoidal curve and fit n -degree polynomials with $n \in \{1, \dots, 15\}$.

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

from mnplib.surfeit import Surfeit
from mnplib.inaccuracy import Inaccuracy

n_samples = 900
degrees = np.arange(1, 15)
```

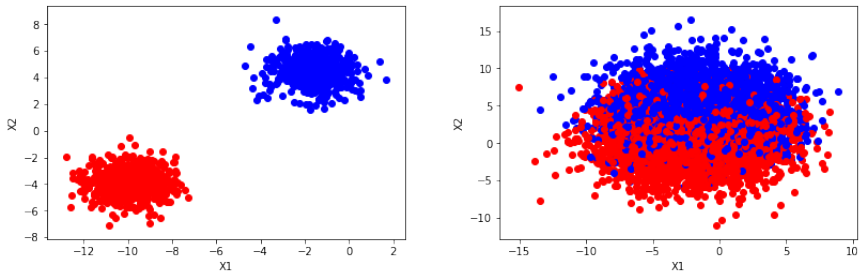


Table 13.1: Isotropic Gaussian blobs.

```

X = np.sort(np.random.rand(n_samples) * 3)
y = np.cos(1.5 * np.pi * X)

linacc = list()
lsurfeit = list()

for i in degrees:

    poly = PolynomialFeatures(degree=i, include_bias=False)
    newX = poly.fit_transform(X[:, np.newaxis])

    linear_regression = LinearRegression()
    linear_regression.fit(newX, y)

    inacc.fit(newX, y)
    inaccuracy = inacc.inaccuracy_model(linear_regression)

    sft.fit(newX, y)
    surfeit = sft.surfeit_model(linear_regression)

    linacc.append(inaccuracy)
    lsurfeit.append(surfeit)

```

In Figure 13.9 we see that, as expected, higher-degree polynomials yield lower training error, but at the cost of higher surfeit. The preferred model is one achieving low inaccuracy *and* low surfeit. ■

Another advantage of surfeit is that it allows us to compare models from different families. For example, if two models have the same accuracy, should we prefer a decision tree to a neural network, or a naive Bayes classifier to a support vector machine? The next example shows how surfeit can guide such choices.

■ **Example 13.12** We compare a decision tree with a neural network on synthetic data consisting of two isotropic Gaussian blobs. First, with standard deviation 1 and two dimensions, the two clusters are easily separable (Table 13.1, left).

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.neural_network import MLPClassifier
from mnplib.surfeit import Surfeit
from mnplib.inaccuracy import Inaccuracy
from sklearn.datasets.samples_generator import make_blobs

X, y = make_blobs(n_samples=1000, centers=2, n_features=2, cluster_std=1)

tree = DecisionTreeClassifier()
tree.fit(X, y)
tree.score(X, y)

nn = MLPClassifier()
nn.fit(X, y)
nn.score(X, y)

sft = Surfeit()
sft.fit(X, y)

sft.surfeit_model(tree)
sft.surfeit_model(nn)

```

If we run the code, both models typically achieve accuracy 1. However, the decision tree often has substantially lower surfite than the neural network (e.g., 0.25 vs. 0.73 in one run), so we would prefer the decision tree in this setting.

If we repeat the experiment with standard deviation 3 and higher dimensionality (Table 13.1, right), the situation can change:

```

X, y = make_blobs(n_samples=10000, centers=2, n_features=8, cluster_std=3)

```

In this second case, both models can again achieve very high accuracy, but the surfite of the decision tree typically increases markedly (e.g., to ≈ 0.82), while the neural network's surfite may remain similar (e.g., ≈ 0.76). For this dataset, we would then prefer the neural network.

■

In Example 13.12 we assumed equal accuracy. When accuracies differ, we must appeal to the overall concept of *nescience* to decide between models, jointly considering inaccuracy and surfite (and, where relevant, miscoding).

13.6 Nescience

In Chapter 7 we defined the concept of nescience as the solution to a nonlinear multiobjective optimization problem, in which we minimize the miscoding, inaccuracy, and surfite of representations and models. The solution to this problem is, in general, not unique: we can often find multiple pairs (representation, model) such that none of the three quantities can be improved without degrading at least one of the others (Pareto optimality). However, in practice we expect a machine-learning library to provide a single solution when

training a model on a dataset. To provide a unique solution, we resort to a utility function that selects one element from the Pareto-optimal set. The `nescience` library offers several utility functions; the default is the arithmetic mean of the three metrics.



The `nescience` library implements the following utility functions to approximate the concept of nescience, i.e., to compute $\hat{v}(\mathbb{Z}, m, \mathbf{y})$:

- Euclidean distance: $(\hat{\mu}(\mathbb{Z}, \mathbf{y})^2 + \hat{i}(\hat{\mathbf{y}}, \mathbf{y})^2 + \hat{\sigma}(m, \mathbf{y})^2)^{1/2}$
- Arithmetic mean: $\frac{\hat{\mu}(\mathbb{Z}, \mathbf{y}) + \hat{i}(\hat{\mathbf{y}}, \mathbf{y}) + \hat{\sigma}(m, \mathbf{y})}{3}$
- Geometric mean: $(\hat{\mu}(\mathbb{Z}, \mathbf{y}) \times \hat{i}(\hat{\mathbf{y}}, \mathbf{y}) \times \hat{\sigma}(m, \mathbf{y}))^{1/3}$
- Product: $\hat{\mu}(\mathbb{Z}, \mathbf{y}) \times \hat{i}(\hat{\mathbf{y}}, \mathbf{y}) \times \hat{\sigma}(m, \mathbf{y})$
- Addition: $\hat{\mu}(\mathbb{Z}, \mathbf{y}) + \hat{i}(\hat{\mathbf{y}}, \mathbf{y}) + \hat{\sigma}(m, \mathbf{y})$
- Weighted mean: $w_{\mu}\hat{\mu}(\mathbb{Z}, \mathbf{y}) + w_i\hat{i}(\hat{\mathbf{y}}, \mathbf{y}) + w_{\sigma}\hat{\sigma}(m, \mathbf{y})$
- Harmonic mean: $\frac{3}{\frac{1}{\hat{\mu}(\mathbb{Z}, \mathbf{y})} + \frac{1}{\hat{i}(\hat{\mathbf{y}}, \mathbf{y})} + \frac{1}{\hat{\sigma}(m, \mathbf{y})}}$

Euclidean distance and addition can produce nescience values greater than one, which conflicts with the intended $[0, 1]$ range. The geometric mean, product, and harmonic mean have the issue that the resulting nescience is zero (or undefined, in the harmonic case) whenever any of the three components is zero. The weighted mean introduces three additional hyperparameters (w_{μ}, w_i, w_{σ}) that must be chosen (often subject to $w_{\mu} + w_i + w_{\sigma} = 1$). It remains an open question which utility function is most appropriate in general; by default we use the arithmetic mean.

Example 13.13 shows how to use the `nescience` library to compute the nescience of a dataset and a model.

■ **Example 13.13** We compute the nescience of a decision-tree classifier on the digits dataset (MNIST handwritten digits) from `sklearn`.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.datasets import load_digits
from Nescience.Nescience import Nescience

data = load_digits()

tree = DecisionTreeClassifier()
tree.fit(data.data, data.target)
tree.score(data.data, data.target)
[ ] 1

nescience = Nescience()
nescience.fit(data.data, data.target)

nescience.nescience(tree)
[ ] 0.5895603819965907
```

The training accuracy of the decision tree is 1.0, meaning all samples are correctly classified on the training set. This is symptomatic of overfitting. In practice we would use a train/test split or cross-validation. Nonetheless, the

nescience value of about 0.59 flags that something is wrong with the model-data pairing.

■

As Example 13.13 illustrates, a key advantage of nescience is that it can evaluate model quality without requiring computationally expensive procedures such as cross-validation or holding out a test set. Another advantage is that nescience allows comparisons across different model families, as shown next.

■ **Example 13.14** We compare two model families on the Breast Cancer Wisconsin (Diagnostic) dataset from `sklearn`: a decision tree and a neural network.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import load_breast_cancer
from Nescience.Nescience import Nescience

data = load_breast_cancer()
X = data.data
y = data.target

tree = DecisionTreeClassifier(max_depth=3)
tree.fit(X, y)
tree.score(X, y)
[ ] 0.9789103690685413

nescience = Nescience()
nescience.fit(X, y)
nescience.nescience(tree)
[ ] 0.5945936419010083

nn = MLPClassifier()
nn.fit(X, y)
nn.score(X, y)
[ ] 0.9261862917398945

nescience.nescience(nn)
[ ] 0.7860523786210711
```

In one run, both models achieve high and broadly similar training accuracy. In this case, not only does the decision tree achieve the better score, but its nescience is also much lower than that of the multilayer perceptron; hence we should prefer the former to the latter.

■

Nescience can also be used to optimize hyperparameters within a (parametric) family of models. A practical advantage is that we can adopt a greedy search for a hyperparameter that is monotone with accuracy: select the value at which nescience stops decreasing and starts to increase, since this is the point at which we are no longer learning anything new from the dataset (see Example 13.15).

■ **Example 13.15** We revisit a decision tree on the breast cancer dataset. We train 10 trees with `max_depth` from 1 to 10. The deeper the tree, the higher the training accuracy, but also the higher the risk of overfitting. For each tree we compute the model's nescience and compare it to a cross-validation score. Figure 13.10 shows typical behavior: both nescience and the cross-validation error decrease as depth increases, up to an inflection point after which they increase, indicating overfitting. The nescience library suggests a maximum depth of 7, whereas cross-validation indicates an optimum near 6.

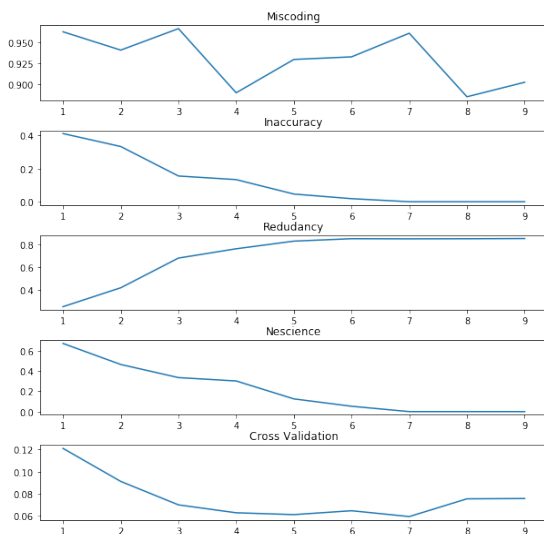


Figure 13.10: Evolution of Nescience with Tree Depth.

It is instructive to examine the three components in Figure 13.10. As expected, deeper trees reduce inaccuracy and increase surfloit. By contrast, miscoding can evolve nonmonotonically because each candidate tree may use a different subset of features at internal nodes. It would be desirable to have a tree-building algorithm that accounts for miscoding when selecting split features; such an algorithm is described in Section 13.11.

Finally, we consider hyperparameter searches where a greedy approach is not applicable—for example, when searching over multiple (often conflicting) hyperparameters. Grid search can be computationally expensive due to the number of combinations, and even more so if each candidate requires cross-validation. Since nescience can flag overfitting without cross-validation, it can substantially speed up hyperparameter search. Example 13.16 shows how to integrate nescience into `GridSearchCV`.

■ **Example 13.16** In this example we are going to see how we can use the `nescience` library to find the optimal hyperparameters for a model using a grid search. In particular, we are going to select the best hyperparameters for a multilayer perceptron classifier, including the number of hidden layers, and the size of those layers (what it is called Neural Architecture Search). The procedure will be demonstrated using the `digits` dataset.

```
from Nescience.Nescience import Nescience
from sklearn.neural_network import MLPClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import classification_report
from sklearn.model_selection import train_test_split
```

First of all we have to provide a custom loss function based on the concept of `nescience` to be integrated with the search procedure. The next code shows how to implement such a function.

```
def my_custom_loss_func(estimator, X, y):

    nsc = Nescience()
    nsc.fit(X, y)
    nescience = nsc.nescience(estimator)

    # scikit-learn expect that higher numbers are better
    score = -nescience

    return score
```

Second, we have to define the grid of hyperparameters over which we are going to do the search. The larger the grid, the better the result, but also, the more computer time is required to evaluate all possible combinations.

```
parameters = {'solver': ['lbfgs'],
              'max_iter': [1000, 1500, 2000 ],
              'alpha': 10.0 * -np.arange(1, 10, 3),
              'hidden_layer_sizes': [(60,), (100,), (60, 60,), (100, 100,),
                                     (60, 60, 60,), (100, 100, 100,)]}
```

Next code show how to do a classical grid search using the score of the models. The search will be evaluated using a train/test split of the dataset.

```
clf_std = GridSearchCV(estimator=MLPClassifier(), param_grid=parameters,
                       cv=3, iid=True, n_jobs=-1)
clf_std.fit(X_train, y_train)
clf_std.best_params_

[] {'alpha': 0.1,
[]  'hidden_layer_sizes': (100,),
[]  'max_iter': 1000,
[]  'solver': 'lbfgs'}

y_true, y_pred = y_test, clf_std.predict(X_test)
```

```
print(classification_report(y_true, y_pred))
```

	precision	recall	f1-score	support	
avg / total		0.98	0.97	0.97	540

Next code show how to perform exactly the same search, but using the concept of nescience instead of the metric score.

```
clf_nsc = GridSearchCV(estimator=MLPClassifier(), param_grid=parameters,
                       cv=3, scoring=my_custom_loss_func, iid=True)
clf_nsc.fit(X_train, y_train)
clf_nsc.best_params_

{'alpha': 0.1,
 'hidden_layer_sizes': (60,),
 'max_iter': 1500,
 'solver': 'lbfgs'}
```

```
y_true, y_pred = y_test, clf_nsc.predict(X_test)
print(classification_report(y_true, y_pred))
```

	precision	recall	f1-score	support
avg / total	0.98	0.98	0.98	540

In a typical run, both searches yield strong generalization. Notably, the nescience-based search selects a *smaller* model (e.g., a single hidden layer with 60 neurons instead of 100) and compensates with a higher `max_iter`. This reflects the role of surfeit: nescience tends to select the smallest model that attains high accuracy without overfitting the training data.

13.7 Auto Classification

The nescience library also includes a module for automated machine learning (for both classification and regression problems). The AutoML module returns the single model, chosen from a collection of model families, that yields the smallest nescience. For each family, the class performs a greedy search over the hyperparameters required by that family, using nescience as the objective.

The next example shows how to apply the AutoML tools.

■ **Example 13.17** In this example we use the nescience library to find the model that best describes the digits dataset.

```
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

from Nescience.Nescience import AutoClassifier

(X, y) = load_digits(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=1)
```

```

model = AutoClassifier()
model.fit(X_train, y_train)

model.score(X_test, y_test)
[] 0.9622222222222222

```

■

If we evaluate `type(model.model)`, we will see that the library has selected a linear support vector machine as the best model for this dataset.

A key difference between the `nescience` library and other AutoML libraries is that it returns a *single* model as the best candidate, rather than an ensemble of models. In this way, the data scientist can directly reuse the result of `AutoClassifier` and proceed with the subsequent analysis.

13.8 Auto Regression

The `AutoRegressor` class automatically selects the best model for a regression problem. In particular, it (i) computes an optimal subset of features, (ii) selects the most suitable family of models, and (iii) tunes the model's hyperparameters—using `nescience` as the objective.

■ **Example 13.18** In this example we apply our auto-regression class to estimating house prices, using the `boston` dataset included with `scikit-learn`.

```

from fastautoml.fastautoml import AutoRegressor
from sklearn.datasets import load_boston
from sklearn.model_selection import train_test_split

(X, y) = load_boston(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=1)

model = AutoRegressor()
model.fit(X_train, y_train)
AutoRegressor()

model.score(X_test, y_test)
0.8763987309111113

```

■

If we evaluate `type(model.model)`, we see that the library has selected a linear support vector machine or a decision tree (depending on the run and dataset version) as the best model for this dataset.

A key difference between the `nescience` library and other AutoML libraries is that it returns a single model as the best candidate, rather than an ensemble of models. In this way, the data scientist can directly reuse the result of `AutoRegressor` and proceed with the analysis.

13.9 Time Series

In this section we study the application of the concept of miscoding to a time series and a delayed version of itself, or to a delayed version of a second time series.

Automiscoding applies miscoding to a time series and a lagged version of itself, as a function of the lag. Automiscoding estimates to what *extent* past observations of the series can explain (or help forecast) future observations. In this sense, automiscoding has a similar objective to autocorrelation in classical time-series analysis (see Section ??).

Definition 13.9.1 Let $\{\mathbf{x}_t\}$ be a time series composed by n samples. We define lag k regular automiscoding of $\{\mathbf{x}_t\}$ as $\hat{\mu}(\mathbf{x}_{x_{k+1}, x_{k+2}, \dots, x_n}, \mathbf{x}_{x_0, x_1, \dots, x_{(n-k)}})$. We define in the same way the concepts of *adjusted automiscoding* and *partial automiscoding*.

In contrast to autocorrelation, automiscoding is defined for all time series, including those with a trend. Moreover, automiscoding remains interpretable in the presence of trends and can reveal seasonal components without requiring prior decomposition.

■ **Example 13.19** We examine cycles in the number of passengers of a US airline. Figure 13.11 shows monthly passengers from 1949 to 1960 (AirPassengers dataset, see References below). There is a clear yearly cycle. We apply automiscoding to confirm this analytically.

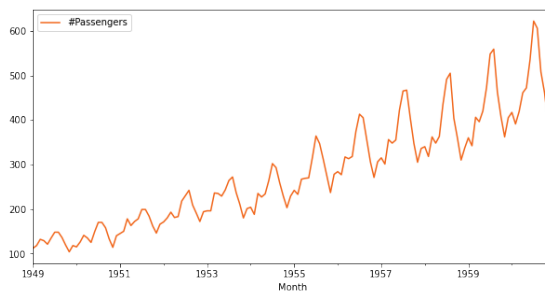


Figure 13.11: Air Passengers.

```
from nescience.timeseries import TimeSeries

data = pd.read_csv("data/AirPassengers.csv", index_col=["Month"],
                  parse_dates=True)
X = np.array(data["#Passengers"]).reshape(-1, 1)

ts = TimeSeries(auto=False)
```

```
ts.fit(data)
mscd = ts.auto_miscoding(max_lag=36)
```

As shown in Figure 13.12, the adjusted automiscoding exhibits a prominent peak every twelve months: the distance between the aligned sequences is minimized when the lag is a multiple of one year.

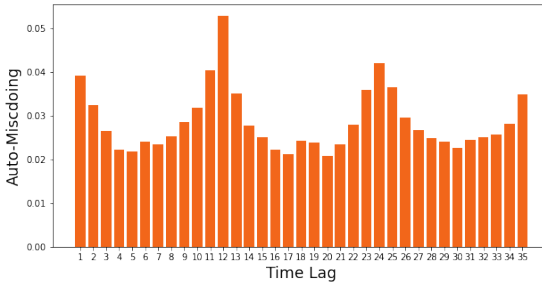


Figure 13.12: Auto-miscoding of Air Passengers.

■

Crossmiscoding measures the relationship between a time series and a lagged version of a second time series. The objective is to detect whether the first series has temporal predictive power over the second.

Definition 13.9.2 Let $\{\mathbf{x}_t\}$ and $\{\mathbf{y}_t\}$ be two time series composed by n samples each. We define lag k *regular crossmiscoding* of $\{\mathbf{x}_t\}$ and $\{\mathbf{y}_t\}$ as $\hat{\mu}(\mathbf{x}_{x_{k+1}, x_{k+2}, \dots, x_n}, \mathbf{y}_{x_0, x_1, \dots, x_{(n-k)}})$. We define in the same way the concepts of *adjusted crossmiscoding* and *partial crossmiscoding*.

■ **Example 13.20** We investigate whether it is possible to predict household appliance energy consumption. The dataset (Appliances Energy Prediction; see References) contains temperature and humidity measurements from rooms every ten minutes, the appliance energy consumption, and weather variables from a nearby station.

For each feature we compute the lag in $[1, 30]$ that maximizes crossmiscoding with the target:

```
from fastautoml.fastautoml import Miscoding

X = pd.read_csv("../data/energydata_complete.csv", parse_dates=["date"],
                index_col="date")
y = X["Appliances"]
X = X.drop(["Appliances", "lights"], axis=1)

miscoding = Miscoding()
miscoding.fit(X, y)
```



```
best_lag = list()
for i in np.arange(X.shape[1]):
    mscd = miscoding.cross_miscoding(attribute1=i, min_lag=1, max_lag=30)
    best_lag.append(np.where(mscd == np.max(mscd))[0][0] + 1)
```

Figure 13.13 shows the results. In general, indoor measurements favor small lags, whereas weather variables (which influence indoor conditions with delay) favor larger lags.

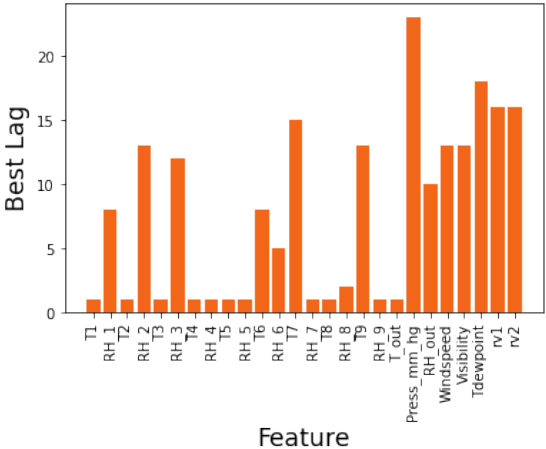


Figure 13.13: Cross Miscoding Lag

R The approximation to miscoding introduced in this chapter estimates (i) the quality of individual features as predictors of a target and (ii) the quality of the training dataset as a whole. However, it does not explicitly account for redundancy among the features themselves. For instance, two features $\mathbf{x}_i$ and $\mathbf{x}_j$ may each have low miscoding with respect to $\mathbf{y}$ yet be largely redundant with one another. It remains an open question how to extend miscoding to incorporate feature redundancy in a way that stays close to the theoretical definition, is computationally efficient, and does not require prohibitively large samples.

13.10 Anomaly Detection

As we have seen in Section XXX, the main problem in anomaly detection is that we do not have a precise mathematical definition of what an anomaly is. In this book we propose to equate *abnormal* samples with *incompressible* samples, and to study the consequences. The essence is that learning is about finding

regularities in a dataset, and data compression is about exploiting regularities. We have also seen that the best model minimizes the sum of the length of the model plus the length of the data given the model. This optimal model partitions the dataset into two disjoint subsets: a compressible part and an incompressible part. It is the latter that interests us here: being incompressible means the samples cannot be explained by the model, i.e., they are model-based anomalies under the best available model.

Definition 13.10.1 Let X be a dataset composed by p features and n samples, y the target variable, and M a model such that the nescience $N(X, M)$ is minimal. Let $\hat{y} = M(X)$ be the predictions made by the model M over the vectors of X . We define the *anomaly subset* of X , denoted by $\mathcal{A}_M^X$, to the set of X such that $y \neq \hat{y}$.

The class `nescience.anomalies` allows us to identify the anomaly subset, i.e., the samples that do not match the regularity patterns found in the rest of the dataset. In Example 13.21 we apply this class to identify houses with abnormally low prices and to explain why they are cheaper.

■ **Example 13.21** We use the Boston House Prices dataset from `scikit-learn`. The dataset contains 13 predictive features (both numeric and categorical) describing different characteristics of the houses (number of rooms, age, etc.) and the target is the median value of owner-occupied homes. The dataset has 506 samples. We aim to identify houses whose prices are *abnormally low*, i.e., houses that, given their characteristics, should have a higher price.

```
from sklearn.datasets import load_boston

data = load_boston()
X     = data.data
y     = data.target
```

We first train a "knowledge model," i.e., the best model that explains the target variable given the predictors, without overfitting. By default, the `anomalies` class uses the AutoML capabilities provided by the `nescience` library.

```
from nescience.anomalies import Anomalies

model = Anomalies(X_type="mixed", y_type="numeric")
model.fit(X, y)
```

Finally, we select those samples for which the actual price is smaller than the price predicted by the model.

```
anomalies = model.get_anomalies("smaller")
X.shape, anomalies.shape
((506, 13), (25,))
```

Thus, there are 25 houses with abnormally low price.

The `anomalies` class also allows us to group the identified anomalies according to shared characteristics.

Let us examine which attributes best describe those abnormal houses.

```
model.get_classes(n_dims=1, an_type="smaller", filter_balancedness=True,
                  filter_redundancy=False, filter_repeated_attrs=False)
```

Attribute1	Attribute2	Inertia	N Class 0	N Class 1	Ratio
2	None	154.953975	9	16	0.36
4	None	0.090593	6	19	0.24
5	None	5.002437	17	8	0.68
7	None	20.352117	6	19	0.24
8	None	77.611111	18	7	0.72
9	None	41737.11111	7	18	0.28
10	None	26.577436	13	12	0.52
12	None	430.294828	18	7	0.72

According to the inertia, the best attribute to classify the anomalies is attribute 4 (nitric oxides concentration). This attribute divides the abnormal houses into two clusters of sizes 6 and 19. Let us examine how price varies with this dimension.

```
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

lr = LinearRegression()
lr.fit(X[:,4].reshape(-1, 1), y)

plt.scatter(X[:,4].reshape(-1, 1), y)
plt.plot(X[:,4], lr.intercept_ + lr.coef_ * X[:,4], color="red")
plt.xlabel(data.feature_names[4])
plt.ylabel("Price")
```

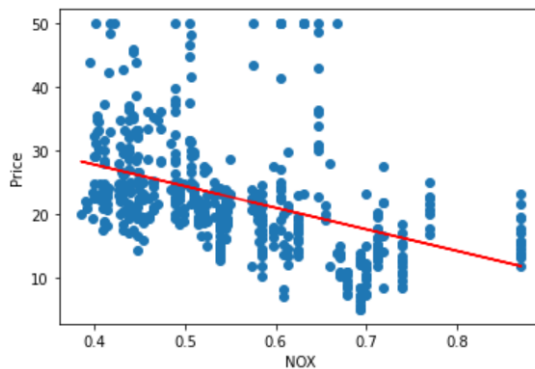


Figure 13.14: Price as a function of NOX.

The regression line suggests that house prices are lower in areas with higher nitric oxides concentration. Let us see how the anomalies split along this dimension.

```
class0, class1 = model.get_class_points(attribute1=4, attribute2=None,
                                         an_type="smaller")

plt.hist(class0)
plt.hist(class1)
plt.ylabel("Count")
plt.xlabel(data.feature_names[4])
plt.show()
```

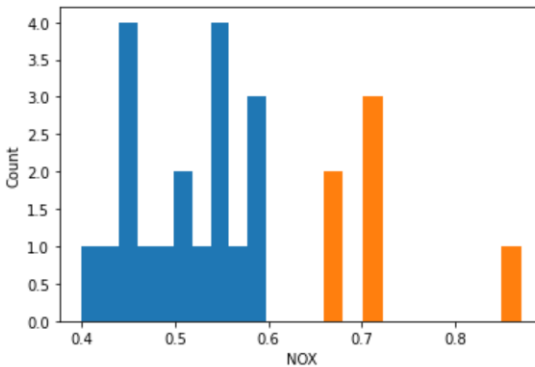


Figure 13.15: Histogram of anomalies NOX.

The analysis suggests that six of the houses have abnormally low prices because they are located in areas with high pollution levels. We can repeat the same analysis with the other attributes; however, their higher inertia indicates that class separation will be less evident. Note that there may be multiple reasons why a house's price is abnormally low.

13.11 Decision Trees

In the last sections we have seen how to use the concepts of miscoding, inaccuracy, surfeit and nescience to evaluate the quality of datasets and models, and to automatically select a family of models and search over its hyperparameters to find the best possible description of a topic. In particular, we have studied in detail the family of binary decision trees. The procedure used in the `fastautoml` library with trees was a mix between a classical approach (a CART algorithm combined with a cost-complexity pruning), and an evaluation of candidate trees using the minimum nescience principle. In this section we are going to see a new algorithm to derive optimal trees, both for classification and regression

problems, that is entirely based on the theory of nescience. The new algorithm, by design, avoids the overfitting of the training dataset without losing accuracy, it does not require the optimization of hyperparameters, thus significantly reducing the training time, and it produces much smaller and more shallow trees than traditional algorithms, facilitating the interpretability of the results.

13.11.1 Algorithm Description

The following pseudocode shows the proposed algorithm to build a decision tree given a training dataset $(\mathbf{X}, \mathbf{y})$. The procedure is based on a breadth first traversal of trees combined with a greedy approach. It requires a function called `BESTSPLIT()` that returns the best split of a given subset of the data into two subsets; and a second function, called `NESCIENCE()` that provides an estimation of the nescience of the current tree. The algorithm is based on two nested loops: the external **while** loop keeps a list of the candidate nodes to grow, whereas the internal **for** loop finds the best node to grow the tree. The latter operation requires to check all possible growing options and select the one that minimizes the nescience. The exit point of the algorithm is when there are no more branches to grow. We keep track of the best nescience achieved during the building process and return the associated tree.

```
def BUILD_TREE(data)

    nodesList <- list()
    tree <- BESTSPLIT(data)
    bestNescience <- NESCIENCE(tree)
    nodesList.append(tree)

    while not nodesList.empty()

        nescience <- bestNescience
        bestNode <- None
        childNode <- None

        for i <- 1, nodesList.length()

            node <- nodesList[i]

            node.child <- BESTSPLIT(node.ldata)
            tmp <- NESCIENCE(tree)
            if tmp < nescience
                nescience <- tmp
                bestNode <- i

            node.left <- None

        if nescience < bestNescience
            node <- nodesList[bestNode]
            bestNescience <- nescience
            nodesList.append(node.left)
```

```

        if not node.left.empty() and not node.right.empty():
            nodesList.remove(bestNode)

    return tree

```

The main difference of our algorithm from other decision tree building algorithms is in the way the tree is evaluated. Instead of using only accuracy as most of the algorithms do, in addition, we take into account the complexity of the tree (surfeit) to avoid overfitting, and the quality of the subset of data used during the training process (miscoding).

Nescience

The calculation of the nescience implemented in the algorithm is based on a Euclidean distance utility function (see Section 13.6), because that one was the one that produced the best results in the tests we have performed. For the computation of miscoding and inaccuracy, we use the same techniques that the one used in the `fastautoml` library, described in Section 13.3 and Section 13.4 respectively. For the implementation of surfeit, we use the same template to describe trees that was used in the `DecisionTreeClassifier` of the `AutoClassifier` class, and that was described in Section ???. The only difference is that we also allow equalities in the nodes (if `[attr] = [thresh]`), something not supported by the `DecisionTreeClassifier` algorithm of the `scikit-learn` library.

The generic problem of the instability of inaccuracy due to very short models, also applies to this algorithm (see Section 13.5), and the particular problem of the algorithms to build decision trees, in which the best local split might not be that one that minimizes the error (see Section ??) is also relevant in this case.

The concept of nescience is used in two different ways in our algorithm. For every iteration of the `for` loop we have to decide which one of the candidate branches of the tree we should develop. Recall that the order in which we develop the branches is important, since it might happens that one branch does not get developed because that would mean increase the surfeit without a sufficiently large decrease of the inaccuracy. The second place is at the end of the `while` loop, we keep track of the nescience of the different building steps, to decide at the end of the algorithm with which tree we return.

We treat regression problems as classification problems in which we discretizes the continuous target variable y into n intervals given the number of samples, and using a uniform discretization (see Section). Once the target variable has been discretized, we train a regular classification tree.

Splitting Criteria

Given a subset $\mathbf{Q} \subseteq \mathbf{X}$ we have to find an split for $\mathbf{Q}$ such that the values of $\mathbf{y}$ are grouped together. Recall that a split is a pair $\theta = (j, w)$, where $1 \leq j \leq p$ is a feature index and w is the partition point (see Section ??). A split divides the set $\mathbf{Q}$ into two disjoint subsets $\mathbf{Q}_l$ and $\mathbf{Q}_r = \mathbf{Q} \setminus \mathbf{Q}_l$. In case of a continuous variable we have that $\mathbf{Q}_l = \{\mathbf{x}_i \in \mathbf{Q} : x_{ij} \leq w\}$, and if the feature is categorical we define $\mathbf{Q}_l = \{\mathbf{x}_i \in \mathbf{Q} : x_{ij} = w\}$ ⁴.

In Section ?? we saw that a common splitting criteria used in practice is to minimize the weighted entropy $\tilde{H}$ of the subsets $\mathbf{Q}_l$ and $\mathbf{Q}_r$, that is, to find an split that it is minimal $\theta^* = \arg \min_{\theta} \tilde{H}(\mathbf{Q}, \theta)$. More explicitly, if $\mathbf{y}$ is a target vector taking values from a set of k labels $\mathcal{G} = \{g_1, \dots, g_k\}$ (either because is a categorical target or a continuous target that has been discretized into k intervals), and denoting the subsets of $\mathbf{y}$ as $\mathbf{y}^l = \{y_i : \mathbf{x}_i \in \mathbf{Q}_l\}$ and $\mathbf{y}^r = \{y_i : \mathbf{x}_i \in \mathbf{Q}_r\}$, and n_l and n_r are the number of elements of $\mathbf{y}^l$ and $\mathbf{y}^r$ respectively, we have that

$$\begin{aligned} \tilde{H}(\mathbf{Q}, \theta) = & \frac{n_l}{n} \left(- \sum_{i=1}^k \frac{\sum_{j=1}^{n_l} I(y_j^l = g_i)}{n_l} \log_2 \frac{\sum_{j=1}^{n_l} I(y_j^l = g_i)}{n_l} \right) \\ & + \frac{n_r}{n} \left(- \sum_{i=1}^k \frac{\sum_{j=1}^{n_r} I(y_j^r = g_i)}{n_r} \log_2 \frac{\sum_{j=1}^{n_r} I(y_j^r = g_i)}{n_r} \right) \end{aligned} \quad (13.1)$$

In our nescience based decision tree algorithm, the splitting criteria is to minimize the total length of encoding the subsets $\mathbf{Q}_l$ and $\mathbf{Q}_r$ using optimal codes. We have to find the optimal split $\theta^* = \arg \min_{\theta} \hat{K}_C(\mathbf{Q} | \theta) = \arg \min_{\theta} \{\hat{K}_{C_l}(\mathbf{Q}_l) + \hat{K}_{C_r}(\mathbf{Q}_r)\}$ where C_l and C_r are the optimal codes given the relative frequencies of the observed values of $\mathbf{y}^l$ and $\mathbf{y}^r$ respectively. The quantity $\hat{K}_C(\mathbf{Q} | \theta)$ is computed as:

$$\begin{aligned} \hat{K}_C(\mathbf{Q} | \theta) = & \hat{K}_{C_l}(\mathbf{Q}_l) + \hat{K}_{C_r}(\mathbf{Q}_r) = \\ & - \sum_{i=1}^k \log_2 \frac{\sum_{j=1}^{n_l} I(y_j^l = g_i)}{n_l} - \sum_{i=1}^k \log_2 \frac{\sum_{j=1}^{n_r} I(y_j^r = g_i)}{n_r} \end{aligned} \quad (13.2)$$

In this particular case (if we use as compression algorithm a code with optimal lengths, and continuous variables have been discretized) it turns out that both expressions are equivalent given the following relation:

$$\tilde{H}(\mathbf{Q}, \theta) = \frac{1}{n} \hat{K}_C(\mathbf{Q} | \theta)$$

⁴Ideally, for the categorical case, instead of a single feature w we should search over all the elements of the power set of the set of features $\mathcal{P}\{1, 2, \dots, p\}$. Unfortunately, that would imply to check 2^p cases, something that is time-expensive from the computational point of view.

We prefer to talk of encoding length instead of weighted entropy because it has an easier interpretation in the context of the theory of nescience.



Strictly speaking, if we want to implement a decision trees search algorithm fully compliant with the minimum nescience principle, instead of using a total length encoding as splitting criteria, we should had computed the nescience at each split and select that one that makes it minimal. However, early experiments have shown that at local level it works better to group the values of y than to reduce the nescience. Further research is required to confirm and explain this point.

Practical Implementation

In the web page that accompanies this book⁵ we provide an open-source implementation of our algorithm in Python. Our software can be used together with other machine learning tools from the `scikit-learn` library, since we adhere to the their API guidelines. For example, our algorithm can be used as part of an ensemble of classifiers, like the `BaggingClassifier` meta-estimator, or the results of the classification could be cross-validated with tools like `cross_val_score`. As an example, to provide a model for the breast cancer dataset, we could do something like the following:

```
from NescienceDecisionTree import NescienceDecisionTreeClassifier
from sklearn.datasets import load_breast_cancer

data = load_breast_cancer()

model = NescienceDecisionTreeClassifier()
model.fit(data.data, data.target)
print("Score: ", model.score(data.data, data.target))
```

13.11.2 Algorithm Evaluation

In this section we are going to evaluate our new algorithm, and compare its performance against the well-known algorithm CART. CART, *Classification and Regression Trees*, is the de-facto standard algorithm used in the machine learning industry for the derivation of decision trees. For this particular experiment we have used the CART implementation provided by `scikit-learn`.

Figure 13.16 shows a synthetic dataset consisting of 1000 random points lying on a two dimensional plane, where all the points with an X_1 attribute less than 50 are colored blue, and the rest as red. We have artificially introduced a red point, simulating a measurement error, in the blue area. The black lines correspond to the decisions performed by CART. Since the CART algorithm will not stop until all the points have been properly classified, we have to specify an expected count condition to limit the number of splits. The figure correspond to

⁵<http://www.mathematicsunknown.com>

the tree generated by CART setting the `min_samples_leaf` hyperparameter to 5.

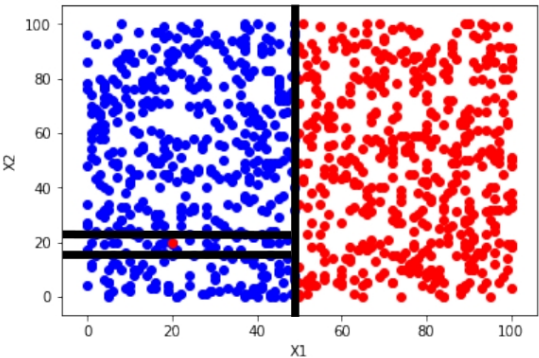


Figure 13.16: Synthetic dataset with CART algorithm splits.

The tree obtained by applying our algorithm to the dataset of Figure 13.16 can be seen in Figure 13.17. The nescience based algorithm does not try to model the error point, since the gain due to an increment in the accuracy does not compensate the surfeit introduced in the model. Recall that the algorithm stops when the total nescience of the tree, based on the measures of miscoding, inaccuracy and surfeit, does not decrease when adding new nodes to the tree. Our algorithm presents a lower sensitivity to the errors found in datasets, at least if the number of errors is small compared with the number of valid points.

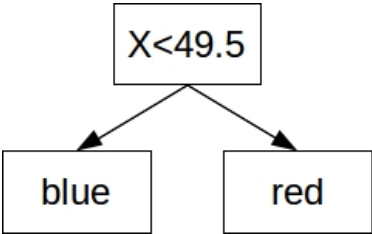


Figure 13.17: Decision tree obtained by the nescience algorithm.

Our second experiment, again with synthetic dataset, is depicted in Figure 13.18. There, we create two isotropic Gaussian blobs that partially overlap. We start with a standard deviation of 2.5 for each cluster, so they are easy to separate, and we increase the standard deviation in increments of 0.01, until we reach 4.5, which causes significant overlaps. For each value of the standard deviation, we run the experiment 100 times and we compute the average accuracy for the two algorithms using different datasets for training (70% of the data) and testing (30%

of the data). The results of this experiment are shown in Figure 13.19.

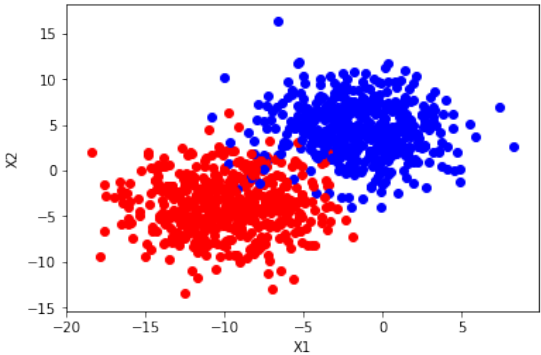


Figure 13.18: Isotropic Gaussian Blobs.

As we can see, the performance of both algorithms, in terms of accuracy, is similar. However we should note that the hyperparameter `minimum_leaf_size` of the CART algorithm has been optimized to achieve the best accuracy. For this particular experiment, the best value was achieved with a minimum leaf size of 26 points. By definition, given the fact that CART has one degree of freedom more than the nescience algorithm, it should produce better accuracy; something that it is not observed (both algorithms have a mean accuracy of 0.87).

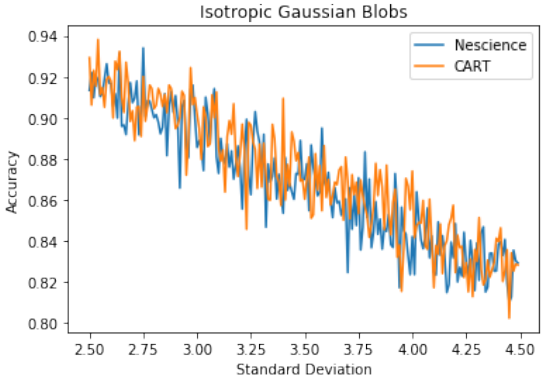


Figure 13.19: Accuracy of Isotropic Gaussian Blobs.

For each iteration of the experiment, we have also computed the average number of nodes, including internal and leaf nodes, required by the models to properly classify the clouds in the dataset. The results of this measurement are show in Figure 13.20. Our algorithm requires an average of 4 nodes compared to 23 nodes for the CART algorithm. Moreover, our algorithm is more stable than

CART, in the sense that it produces models of similar complexity when it gets similar input datasets (a standard deviation of 0.31 compared to 3.77 for CART).

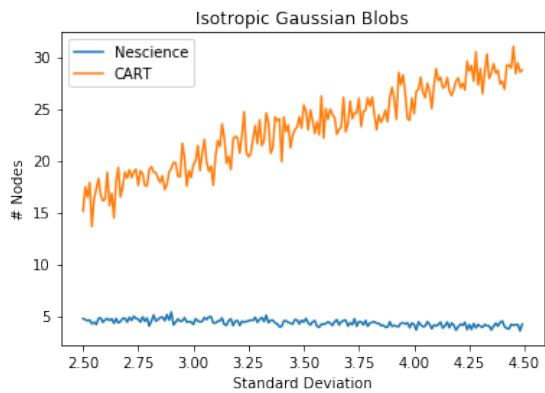


Figure 13.20: Number of Nodes.

In Figure 13.21 we show the maximum depth of the tree, defined as the longest path from the root of the tree to any of its leaves. The maximum depth of the tree is a good measure of the average time it will require for the model to provide a classification. The nescience algorithm has an average depth of 1.6 nodes, whereas the average depth yielded by the CART algorithm is 4.8 nodes.

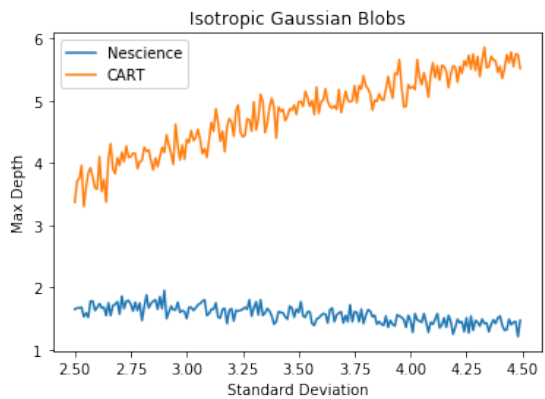


Figure 13.21: Maximum depth of the model.

The last part of the evaluation consists in comparing the performance of our algorithm and CART with a collection real datasets. More specifically, we have selected 12 well known datasets from the UCI Machine Learning Repository. The selected datasets are: diagnosis of breast cancer (cancer), optical recognition of handwritten digits (digits), predicting protein localization sites in gram-

negative bacteria (yeast), classification of NASA space shuttle data (shuttle), classification of blocks in web pages (page), segmentation of outdoor images (image), predicting the age of abalones from physical measurements (abalone), predicting the quality of red and white variants of Portuguese wine (wine), filter spam emails (spam), wall-following robot navigation (wall), classification of land use based on Landsat satellite images (landsat), and distinguishing signals from background noise in the MAGIC gamma telescope images (magic). For each dataset, we have repeated the experiment 100 times, by randomly selecting the training (70%) and testing (30%) subsets at each iteration.

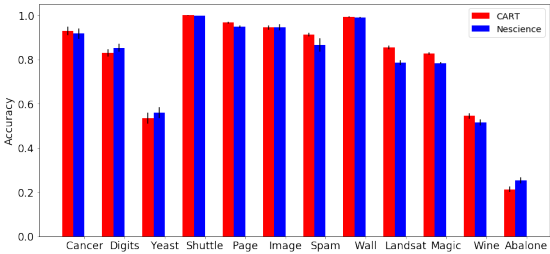


Figure 13.22: Maximum depth of the model.

In Figure 13.22 we compare the accuracy of the resulting models obtained by applying the CART algorithm and the nescience algorithm to the above datasets. In 4 of the 12 datasets, our algorithm provides better accuracy than CART. In the remaining 8 cases, the accuracy is, on average, less than 1% smaller.

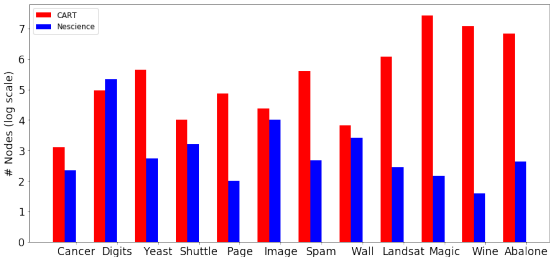


Figure 13.23: Maximum depth of the model.

In Figure 13.23 it is shown a comparison of the total number nodes (internal nodes plus leaf nodes) of the resulting models. Only for one of the datasets (digits), our model produces a slightly more complex tree than those generated by CART. In the rest of the cases, the number of nodes in the trees generated by the nescience algorithm have between two and three orders of magnitude fewer nodes (in this figure the y axis is in logarithmic scale).

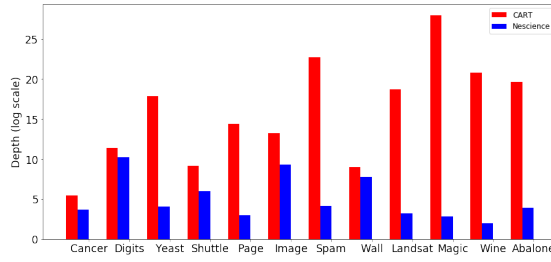


Figure 13.24: Maximum depth of the model.

Finally, in Figure 13.24 we provide a comparison of the depth of the tree of the resulting models. Our algorithm always yields a shallower tree than the CART algorithm.

We would like to mention that the nescience algorithm is highly robust with respect to the compressor selected or the nescience function implemented. In Table 13.11.2, we have apply the nescience algorithm to the datasets described above, and evaluate different alternatives for the definition of the nescience function $N(X, M)$: arithmetic mean

$$N(X, M) = (\mu(M, D) + \iota(X, M) + \sigma(M, D))/3,$$

geometric mean

$$N(X, M) = (\mu(M, D) + \iota(X, M) + \sigma(M, D))^{1/3},$$

harmonic mean

$$N(X, M) = 3/(\mu(M, D) + \iota(X, M) + \sigma(M, D)) - 1),$$

Euclidean distance

$$N(X, M) = (\mu(M, D) + \iota(X, M) + \sigma(M, D))^{1/3},$$

sum

$$N(X, M) = \mu(M, D) + \iota(X, M) + \sigma(M, D),$$

and product

$$\mu(M, D) + \iota(X, M) + \sigma(M, D).$$

The table shows limited difference between the different functions.

Similarly, Table 13.11.2 shows the performance of our algorithm when using the LZMA, zlib, and bz2 compressors. We observe that all of them yield similar

	Euclid	Arithm.	Geometric	Product	Addition	Harmonic
Acc.	0.758	0.784	0.803	0.803	0.784	0.81
Stdev.	0.051	0.041	0.033	0.033	0.041	0.038

Table 13.2: Comparison of nescience functions

	bz2	lzma	zlib
Accuracy	0.813	0.804	0.81
Stdev	0.03	0.045	0.038

Table 13.3: Comparison of compressors

performance. The above results suggest that the performance our algorithm is independent of the specific choice made for either implementation aspect.

We emphasize that the CART algorithm requires to optimize a configuration hyperparameter in order to obtain good results, whereas the algorithm proposed in this book does not require from this optimization.

Shallower trees means faster forecasting times when the models used in production, since the number of `if-else` conditions to be evaluated is smaller. Moreover, smaller trees makes easier to interpret the results by human analysts. and much shorter training times, something very relevant in case of training ensembles of trees, like random forest or boosted trees (although the use of ensembles of models is highly discouraged by the theory of nescience, given their high surfeit).

13.12 Algebraic Model Selection

As it was the case for the definition of nescience based on the encyclopedic description of research topics, the nescience of structured datasets can be used to evaluate alternative descriptions of research topics (mathematical models), and to identify how far these descriptions are from an ideal perfect knowledge. This evaluation could be used to identify those topics which require further research. Moreover, the same methodology could be applied to collections of datasets to identify our current knowledge of research areas (collections of topics).

If we combine the concept of nescience of a model, with our concepts of relevance and applicability of research topics, we could apply our methodology for the assisted discovery of interesting questions to collections of datasets; a very useful methodology now that big datasets are becoming widely available.

In order to evaluate the methodology developed, we are going to apply it to a particular research topic: *Multipath Wave Propagation and Fading*. The

problem at hand is to understand the effect of a propagation environment on a radio signal, such as the one used by wireless devices. The signals reaching the receiving antenna could follow multiple paths, due to atmospheric reflection and refraction, and reflection from water and objects such as buildings. The effects of these multiple wave paths include constructive and destructive interference (fading), and phase shifting of the original signal, resulting a highly complex received signal (see Figure 13.25).

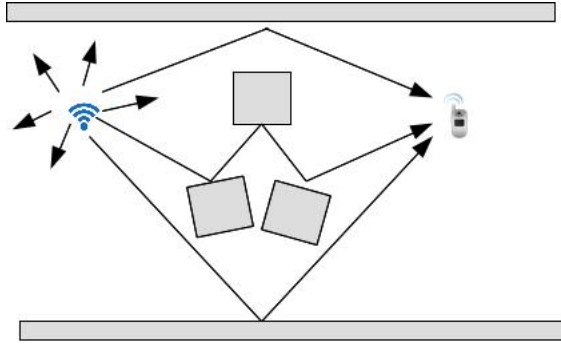


Figure 13.25: Multipath Signal Propagation

In many circumstances, it is too complicated to describe all reflection, diffraction and scattering processes that determine the different paths the signal will follow. Rather, it is preferable to describe the probability (stochastic model) that the received signal attains a certain value. We are interested in to analyze how well these stochastic models (our current knowledge) are able to describe what happen in reality.

The *Rayleigh fading model* assumes that the magnitude of a signal will vary randomly, or fade, according to a Rayleigh distribution (the radial component of the sum of two uncorrelated Gaussian random variables). The Rayleigh probability density function of the power signal is given by:

$$P_{\sigma}(x) = \frac{1}{\sigma} \exp \left[-\frac{x}{\sigma} \right]$$

where σ is the mean of the received signals. Rayleigh fading is viewed as a reasonable model for the effect of heavily built-up urban environments, when there is no dominant propagation along a line of sight between the transmitter and receiver.

The Rice or *Rician distribution* describes the power of the received signal when the target consists in many small scatterers of approximately equal strength, plus one dominant scatterer whose individual received signal equals all that of all

the small scatterers combined (there is a dominant line of sight). The probability density function of the power of the received signal is given by:

$$P(x) = \frac{1}{\bar{\sigma}} (1 + a^2) \exp \left[-a^2 - \frac{x}{\bar{\sigma}} (1 + a^2) \right] I_0 \left[2a \sqrt{(1 + a^2) \frac{x}{\bar{\sigma}}} \right]$$

where $\bar{\sigma}$ is the mean of the received signals, and it is equal to $\bar{\sigma} = (1 + a^2) \bar{\sigma}_R$, being $a^2 \bar{\sigma}_R$ the power of the dominant scatterer, and I_0 is the modified zeroth order Bessel function of the first kind.

An experiment (see Figure 13.26) was set up to collect a real dataset to analyze. The experiment was run on a $135m^2$ office full of obstacles (interacting objects). The transmitter was an Odroid C1 Linux computer with a Ralink RT5370 USB Wifi adapter. The receiver was a (fixed in space) Motorola Moto G mobile phone. Data was collected using the Kismet⁶ platform (an 802.11 layer2 wireless network detector, sniffer, and intrusion detection system), with some ad hoc, home made, software extensions, mostly for data aggregation. A total of 3,177 samples (power level measured in dBm) were collected during one hour experiment.

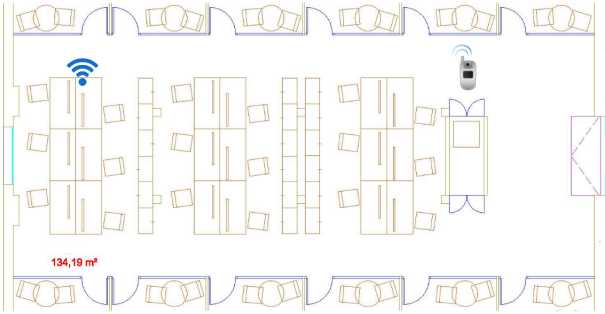


Figure 13.26: Experimental Set Up

Next table summarizes the results of applying the three considered models (uniform, Rayleigh and Rice) and the optimal encoding using a Huffman code:

The uniform model, that is, assuming zero knowledge about the topic covered by the dataset, has a nescience of 1.30. This value is a kind of upper level for the nescience associated with that particular topic and dataset; any model with a higher nescience should be classified as zero knowledge model. If we introduce the knowledge that in a environment with multiple obstacles the signal propagation can be described as a Gaussian process (Rayleigh distribution), we are able to decrease our nescience to 0.75, that is, there were a 43% improvement

⁶<https://www.kismetwireless.net/index.shtml>

Model	LDM	Nescience
Uniform	17,351	1.30
Rayleigh	13,229	0.75
Rice	11,118	0.47
Huffman	7,541	-

Table 13.4: Nescience of Models

in our understanding of the topic. If we add the knowledge that there is usually a strongly dominant signal seen at the receiver caused by a line of sight between the antenna and the mobile phone (Rician distribution), the nescience decreases to 0.47, and so, we have achieved an additional 23% gain in our understanding. Given that numbers we can conclude that the Rayleigh model increases our knowledge with respect to the uniform model, and that the Rice model does so with respect to Rayleigh. However, the nescience of this last model is 0.47. That means that there still patterns in the dataset that are not explained by the Rice model, or what it is equivalent according to our methodology, there is still some knowledge to discover and learn.

The methodology has been applied to a dataset gathered in a single experiment under a controlled environment, since the goal of this Chapter was to provide a methodology to quantify the nescience of structured datasets, not to evaluate models for signal propagation and fading. In order to to conclude that, in general, the Rice model is an improvement over Rayleigh, a more realistic experiment is required, with multiple datasets gathered in real environments.

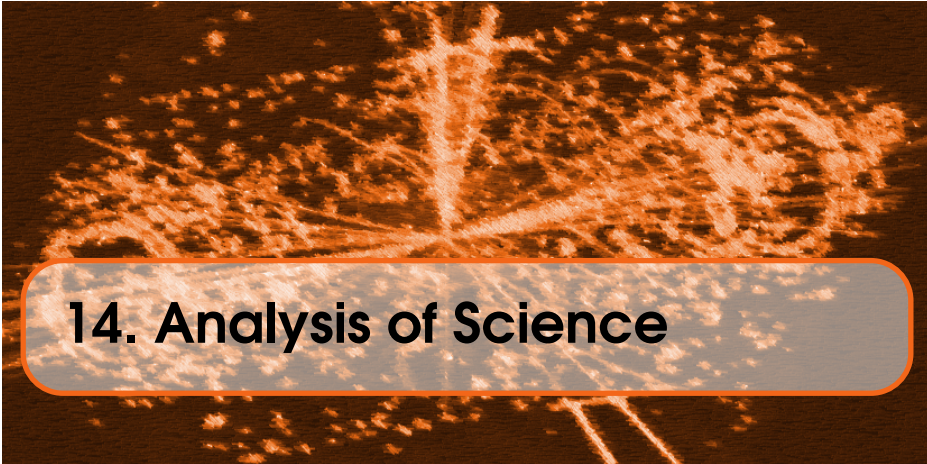
References

A insightful description of the differences between explanation models and predictive models, how these models are used in different scientific disciplines, and what are the implications for the process of statistical modeling can be found in [Shm+10].

The application of the minimum description length principle to the identification of optimal decision trees have been proposed in [QR89], further refined and clarified in [WP93]; however the coding method proposed by those authors is different from the one used in this book.

The Minimum Description Length [Grü07] and the Minimum Message Length [Wal05] techniques have been applied to the problem of inferring decision trees in [QR89], later on clarified and extended in [WP93], in [MRA+95] as a technique for pruning, and in [RS98], among others. Although the underlining concepts behind the cost function proposed in this chapter are the same (namely,

that learning is equivalent to the capability to compress), our approach is very different from the ones described in these works.



14. Analysis of Science

Science may be regarded as the art of data compression.
Li & Vitányi

The purpose of this chapter is to apply the theoretical framework of nescience developed in the previous chapters to the study of our current scientific knowledge. Until now, our focus has been primarily on the conceptual and mathematical foundations of nescience. We have defined and analyzed the core notions of miscoding, inaccuracy, redundancy, and nescience itself as abstract quantities, exploring their properties and interrelations. In this chapter, we will shift from theory to practice: we will operationalize these notions and use them to analyze the state of contemporary science as it is documented today.

Our empirical analysis will be grounded on Wikipedia's collection of scientific pages. Wikipedia provides a vast, open, and relatively well-structured repository of scientific knowledge that spans across virtually all research domains. While Wikipedia articles are not primary scientific sources, they offer an invaluable meta-level view of what the scientific community currently considers established knowledge, as well as how that knowledge is organized, debated, and maintained. By systematically processing this corpus, we can obtain large-scale approximations of the quantities defined in our theory and thereby measure the current degree of understanding achieved in different parts of science.

We will begin by presenting practical methods to approximate, for any given scientific topic, its level of miscoding, inaccuracy, redundancy, and resulting nescience. Miscoding will capture the extent to which the information contained in an article fails to describe its intended subject; inaccuracy will measure the degree of controversy or error that remains unresolved; redundancy will reflect the amount of superfluous or repetitive information; and nescience will integrate these dimensions into a single estimate of our overall ignorance about the topic. This analysis will allow us to identify which topics appear to be well understood and which remain poorly understood or even actively confused.

After establishing methods for individual topics, we will extend the analysis to broader research areas. We will study how the values of miscoding, inaccuracy, and redundancy of a collection of related topics combine to produce a macroscopic picture of the maturity and clarity of an entire field. This will enable us to compare different disciplines, discover which areas are advancing rapidly, and detect fields that may be stagnating or lacking conceptual coherence.

In addition to these cross-sectional analyses, we will also perform a longitudinal study, examining the evolution of knowledge—and, symmetrically, of nescience—over time. By analyzing historical versions of Wikipedia articles, we will trace how the levels of miscoding, inaccuracy, and redundancy have changed as scientific understanding has advanced. This will provide a first empirical glimpse of the dynamics of knowledge accumulation, revealing how scientific topics move from being poorly understood and controversial to becoming stable and concise.

Finally, this chapter will also explore a long-standing philosophical question: the demarcation problem, that is, the problem of distinguishing scientific knowledge from non-scientific or pseudoscientific claims. We will propose a practical approach to this problem based on the metrics of nescience, showing how our framework can help identify topics that fail to exhibit the patterns of knowledge growth characteristic of science.

14.1 Describing Current Knowledge

According to the theory of nescience, evaluating our current knowledge begins by selecting a specific set of entities we want to understand. Within this framework, universal sets that include every possible entity, whether known or unknown, are not permitted. This restriction arises from the logical paradoxes associated with such sets, including Cantor's theorem, which demonstrates that the set of all subsets has a strictly greater cardinality than the set itself, and Russell's paradox, which involves sets that are not members of themselves (see Section 2.1). Therefore, in this chapter, since our objective is to assess the current state of human knowledge, we focus on a finite set composed of entities that are already

known and have been studied by science. These entities are suitable for analysis because humanity possesses at least some information about them.

The next step involves identifying the best available encoding of the selected entities as representations, typically composed of data or facts, followed by determining the most accurate descriptions we have, such as models, theories, and laws. In the theory of nescience, we maintain a clear distinction between representations and descriptions because this separation facilitates the discovery of new knowledge: either by improving the representation or by refining the description. However, since our aim here is to evaluate rather than expand our knowledge, we simplify the process by using only descriptions. These descriptions serve both as the representation and the description of an entity. Moreover, we will drop the requirement that these descriptions be computable, since very few scientific descriptions in practice are actually computable. Requiring computability would severely limit the scope of our analysis and exclude a significant portion of scientific knowledge.

For our analysis, we use descriptions derived from the collection of scientific pages from Wikipedia. Wikipedia is a free, collaborative online encyclopedia launched in 2001 that allows anyone with internet access to create, edit, and update articles on a wide array of topics. It is maintained by a global community of volunteer contributors and aims to provide reliable, neutral, and verifiable information to the public. Several features make Wikipedia particularly suitable for analyzing scientific knowledge: it maintains a transparent version history that tracks content changes over time; supports collaborative validation that reduces individual bias; enforces strict citation requirements to ensure information is verifiable and based on authoritative sources; offers broad and consistent coverage across scientific disciplines; and is dynamically updated to reflect new findings and corrections.

Wikipedia is fundamentally a tertiary source. It compiles and synthesizes information from primary sources, such as firsthand accounts of events or discoveries, and secondary sources, which interpret and evaluate those primary materials. Tertiary sources play a key role in academia by summarizing information that has already been thoroughly discussed and vetted. High-quality Wikipedia articles strive to be comprehensive, covering all major aspects of a topic in appropriate detail while avoiding undue emphasis on minor or peripheral information.

Furthermore, Wikipedia is inherently citational. Every statement included in a Wikipedia article must be supported by a published, reliable, and verifiable source. Original research is not allowed, meaning that novel claims, hypotheses, or breakthroughs must already be published in credible sources such as academic journals or books before being included. Wikipedia also employs a form of peer review, where one or more editors review and suggest improvements to an

article. While this process is not anonymous or uniformly applied like traditional academic peer review, it still acts as a quality control mechanism that enhances the coverage, clarity, and accuracy of articles through community oversight.

Wikipedia pages are written in the MediaWiki Markup Language, a simplified system for formatting text that allows users without technical knowledge of XHTML or CSS to easily edit articles. Before we can analyze the content of a scientific page, it is essential to remove all markup tags and formatting elements to isolate only the meaningful textual information. To achieve this, we used the Python library `wikitextparser`, which allows us to parse and process the raw wikitext of Wikipedia articles to extract the relevant content. In addition to stripping out the markup, the library was configured to eliminate other non-relevant elements, such as images, references, and lists, which do not contribute directly to our analysis of knowledge.

Wikipedia articles are written in the MediaWiki Markup Language, a simplified formatting system that allows users without technical expertise to contribute easily. Before we can analyze this content, it is necessary to remove all markup tags and formatting elements to isolate only the meaningful text. To do this, we used the Python library `wikitextparser`, which parses raw wikitext and extracts relevant content. In addition to removing markup, the library was configured to exclude other non-essential elements such as images, references, and lists, which do not directly support our knowledge analysis.

Wikipedia hosts a vast collection of articles covering disciplines such as history, literature, politics, entertainment, and science. Since our goal is to evaluate scientific knowledge specifically, we limited our analysis to articles categorized under the "Science" category. This subset includes topics related to the natural and formal sciences—such as physics, biology, chemistry, and mathematics—ensuring that our study remains aligned with the goal of assessing scientific understanding. We further refined this selection by excluding articles focused on individuals, organizations, awards, lists, portals, and similar metadata-oriented content. This filtering ensures that our analysis includes only those articles centered on scientific concepts, processes, or phenomena.

Both the list of articles and the full content of each article were retrieved directly from Wikipedia using the MediaWiki API. This API provides programmatic access to Wikipedia's content, allowing automated queries for page metadata, article text in wikitext format, category information, and revision history. It ensures we are working with the most current and accurate versions of each article while offering a reproducible and efficient method for gathering large-scale data¹.

¹The articles analyzed in this chapter correspond to April 2025.

14.2 Measuring Knowledge

As mentioned in the previous section, to evaluate our current knowledge in practice, we used the collection of scientific pages from Wikipedia, which we pre-processed to extract the relevant information for our analysis. We focused exclusively on the topic descriptions, treating descriptions and representations as equivalent. In this context, since descriptions and representations are considered the same, the inaccuracy of a topic is effectively zero, leaving miscoding and surfeit as our primary metrics of interest.

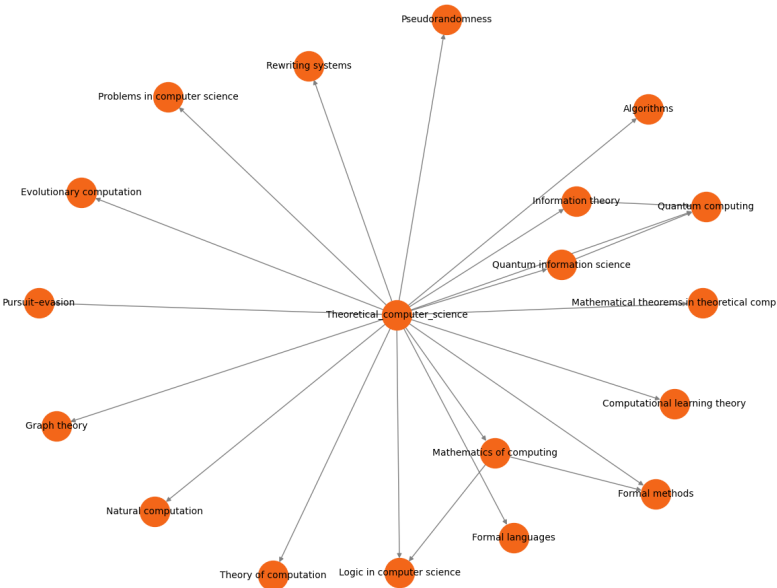


Figure 14.1: Categories in Theoretical Computer Science

To demonstrate how our theory can be practically applied to assess the current state of knowledge, we selected a collection of topics listed under the Theoretical Computer Science category on Wikipedia. This includes all pages within the main category, as well as those in its subcategories and their respective subcategories, up to a depth of five levels. Figure 14.1 illustrates the Theoretical Computer Science category (Level 1) alongside all its Level 2 subcategories, such as Theory of Computation, Graph Theory, and Logic in Computer Science.

Only pages classified as "GA" have been considered. Articles in the GA class are considered complete, and they have been examined by one or more impartial reviewers. From these list, we have removed pages related to non-relevant topics,

such as journals, symposiums, associations, or awards. Additionally, we have removed articles too generic pages (e.g. parallel computing) or pages describing multiple algorithms (e.g. Tropical cyclone forecasting).

Only pages classified as "GA" (Good Article) have been considered. GA articles are regarded as complete and have been reviewed by one or more impartial evaluators. From this list, we excluded pages related to non-relevant topics, such as journals, symposiums, associations, or awards. We also removed overly generic articles (e.g., Parallel Computing) and pages describing multiple algorithms (e.g., Tropical Cyclone Forecasting).

The topics analyzed cover a wide range of areas within theoretical computer science and mathematics. In graph theory, we explored topics such as graph coloring (Greedy coloring), structural properties and special classes (Snark, Perfect graph, Halin graph, Laves graph, Rook's graph, Rado graph, Well-covered graph, Pseudoforest, Component, Unit distance graph, Cop-win graph), graph transformations (Graph homomorphism, Logic of graphs), and graph-based theorems (Steinitz's theorem, De Bruijn-Erdős theorem, Turán's brick factory problem). In algorithms and data structures, we considered classical techniques like Binary search, Selection algorithm, Euclidean algorithm, Fast inverse square root, Farthest-first traversal, Linear probing, Trie, as well as more specialized algorithms such as the Gale-Shapley algorithm, Widest path problem, and data structures like the Cartesian tree. Topics in logic and computational foundations included 2-satisfiability, the Rule of inference, and the BIT predicate. In mathematical methods and theorems, we covered results such as Pick's theorem, Viète's formula, Sylvester-Gallai theorem, Sylvester's sequence, Shapley-Folkman lemma, and the Handshaking lemma. The list also touches on applied mathematics and modeling, through topics like the Dirac delta function, Earth-Moon problem, Tropical cyclone forecast model, Finite subdivision rule, and Network synthesis. Furthermore, in machine learning and computational complexity, we included Reinforcement learning from human feedback and the Small set expansion hypothesis. Finally, some topics address computer science system design and principles, such as the Allocator (C++), Book embedding, Three utilities problem, the Commutative property, and visualization techniques like the Arc diagram. Overall, the list covers a broad spectrum of topics within the area of theoretical computer science, highlighting its diversity and interdisciplinary connections.

14.2.1 Surfeit

In Section 10.1, we introduced the concept of surfeit as a relative measure to quantify the unnecessary effort involved in explaining an entity using a particular

Algorithm	Analysis of Boolean functions
Automated reasoning	Farthest-first traversal
Gale-Shapley algorithm	Widest path problem
Cartesian tree	Viète's formula
Dirac delta function	Euclidean algorithm
Fast inverse square root	Rule 184
Binary search	Linear probing
Selection algorithm	2-satisfiability
Graph homomorphism	Logic of graphs
Steinitz's theorem	De Bruijn-Erdős theorem
Earth-Moon problem	Turán's brick factory problem
Laves graph	Rook's graph
Snark	Perfect graph
Clique problem	Feedback arc set
Euclidean minimum spanning tree	Component
Unit distance graph	Halin graph
Cop-win graph	Pseudoforest
Well-covered graph	Rado graph
Network synthesis	Telephone number (mathematics)
Book embedding	Three utilities problem
BIT predicate	Pick's theorem
Sylvester-Gallai theorem	RL from human feedback
Finite subdivision rule	Sylvester's sequence
Tropical cyclone forecast model	Shapley-Folkman lemma
Theil-Sen estimator	Trie
Rule of inference	Commutative property
Allocator (C++)	Handshaking lemma

Table 14.1: Topics in Theoretical Computer Science

description. The surfeit of a description d for a representation r is defined as:

$$\sigma(d, r) = \frac{|l(d) - K(r)|}{l(d)}$$

where $r \in \mathcal{B}^*$ and $d \in \mathcal{D}$ is a description of r . Intuitively, the less we know about an entity, the longer our description tends to be. As our understanding of the entity improves, we should be able to remove all redundant elements from its description.

In practice, when descriptions and representations are considered equivalent, the concept of surfeit simplifies to:

$$\rho(d) = 1 - \frac{K(d)}{l(d)}$$

This is referred to as the redundancy of a description.

The Kolmogorov complexity of the Wikipedia pages was estimated by compressing the raw text. For compression, we used bzip2, a free and open-source program based on the Burrows-Wheeler algorithm. bzip2 is commonly used in practice to estimate Kolmogorov complexity because it employs a very large compression buffer. It is well known that if the buffer size is too small, the estimation of a text's Kolmogorov complexity can be significantly distorted. For example, the gzip compressor uses a 32 KB buffer, which is too small for our purposes. In contrast, bzip2 offers a buffer size of 900 KB at its highest compression setting (compression level of 9), which is more than sufficient for compressing Wikipedia pages.

Figure 14.2 (left) depicts an histogram of the redundancy for the 50 theoretical computer science articles studied. The histogram shows that, once markup and boiler-plate are stripped, redundancy of articles is tightly concentrated between ≈ 0.75 and 0.82 , peaking near 0.80 . In practical terms, even the clean prose of these pages can typically be shrunk to about three-quarters of its original size, indicating a common stock of recurring phrases, technical jargon, and definitional patterns across the corpus. The scatter plot in figure 14.2 (right) adds a systematic dimension: redundancy rises by roughly 2-3 percentage points for every additional 10000 characters. Because every file fits inside the single 900 kB block used at bzip2's highest setting (-9), block-size effects are neutralised and the only compressor artefacts left—headers and model warm-up—explain at most a few tenths of that increase. The bulk of the slope therefore reflects the prose itself: longer articles increasingly reuse terminology, theorem-proof templates, and explanatory scaffolding, making them intrinsically more compressible. In short, both figures together show that theoretical-computer-science pages share a fairly uniform baseline of linguistic redundancy, and that this redundancy

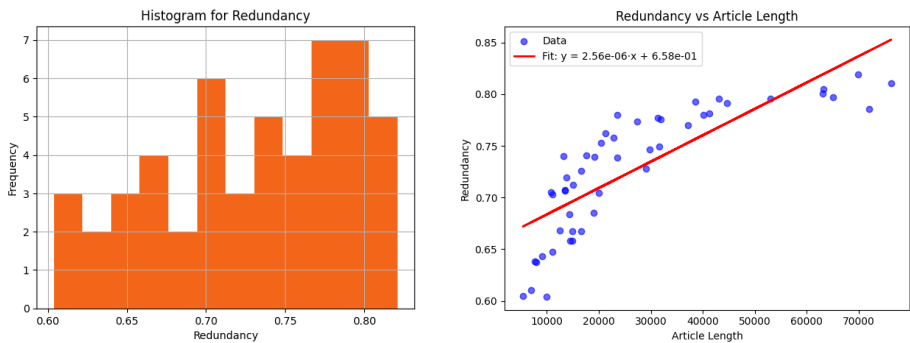


Table 14.2: Redundancy of Topics

grows gradually with article length because the writing becomes more internally self-similar rather than because of any quirk of the compression tool.

Table 14.3 lists the ten topics from Wikipedia within the area of Theoretical Computer Science that exhibit higher redundancy. Each page revolves around a single, well-defined object—a classic algorithm (binary search, Euclidean algorithm, clique problem), a mathematical construct (Dirac delta, Rado graph), or a concise metatheorem (2-SAT, Shapley-Folkman lemma). Pages of this kind inevitably iterate the same symbol palette, definitions, and step-by-step explanations: every subsection restates the object, walks through an example, then re-expresses the same idea in a different formalism (pseudocode, recurrence, proof sketch, code snippet). That self-similar scaffolding gives compressors a wealth of repeated n-grams to exploit, driving redundancy up to 0.79 - 0.82 even after boiler-plate has been stripped. While the list includes some of the longer articles in the corpus (e.g. Dirac delta at 117 kB), it also contains mid-sized pages such as Rado graph (38 kB). In other words, high redundancy is not just a by-product of length; rather, it emerges when a compact conceptual core is expanded through multiple parallel presentations (history, intuition, formal statement, variations, applications, proof, pseudocode).

Table 14.4 lists the ten topics that exhibit lower redundancy. These articles sit in the low-redundancy tail (between 0.60 and 0.67) differ from the high-redundancy ones in both size and content mix. (i) Every entry is under 17 kB—well below the corpus median. Because bzip2’s fixed header and model “warm-up” overhead account for a larger share of such small files, their redundancy starts a few percentage points lower even before content is considered. This matches the positive slope in the scatter-plot: shorter pages naturally cluster toward the left-hand foot of the trend line. (ii) Many of these topics centre on a single formula, sequence, or code idiom (e.g. Viète’s formula, Sylvester’s sequence, Allocator). They pack unique symbols, numeric constants

Topic	Redundancy	Length	Compressed
Dirac delta function	0.821	117121	20969
Binary search	0.819	69838	12634
Euclidean algorithm	0.811	113242	21455
2-satisfiability	0.811	76225	14442
Shapley-Folkman lemma	0.805	63172	12323
Clique problem	0.801	63125	12574
RL from human feedback	0.797	64981	13160
Book embedding	0.796	53046	10831
Rule of inference	0.795	43158	8826
Rado graph	0.793	38626	7989

Table 14.3: Topics with higher redundancy

and one-off identifiers that appear only once or twice, so compressors see far fewer repeated n-grams than in algorithm-walk-through articles. Others, such as "Tropical cyclone forecast model" and "Finite subdivision rule", read more like concise surveys, stringing together disparate subtopics rather than iterating one core definition from multiple angles; that heterogeneity also suppresses repetition.

Together, the list illustrates the floor of the redundancy spectrum: pages that are (i) short enough for compression overhead to matter and (ii) information-dense or eclectic enough to avoid recycling the same prose patterns never reach the 0.70-0.80 plateau seen elsewhere. Their position reinforces the earlier finding that rising redundancy with length is not merely a compressor artefact; it also depends on how much authors reuse terminology and explanatory scaffolding as an article grows.

14.2.2 Inaccuracy

Inaccuracy serves as the second metric in assessing our understanding of a research entity. The underlying idea is that the more accurate our model, the better our understanding of the entity. Formally, we calculate the inaccuracy of a description d as the normalized information distance between the original representation r and the output representation r' generated by our description d . That is, inaccuracy is quantified as the length of the smallest computer program capable of correcting the erroneous output of our model.

Inaccuracy, serving as the second gauge to measure our comprehension of a research entity, is based on the principle that the more precise our model, the better our grasp of the entity. Formally, the inaccuracy of a description d is computed as the normalized information distance between the original representation r and

Topic	Redundancy	Length	Compressed
Tropical cyclone forecast model	0.667	16675	5550
Allocator (C++)	0.658	14887	5085
Finite subdivision rule	0.658	14416	4933
Snark (graph theory)	0.648	11020	3882
Theil-Sen estimator	0.643	9171	3275
Earth-Moon problem	0.638	7640	2762
Halin graph	0.637	7953	2885
Viète's formula	0.61	7016	2733
Turán's brick factory problem	0.605	5427	2146
Sylvester's sequence	0.604	9994	3960

Table 14.4: Topics with Lower redundancy

the output representation r' generated by the description d . Thus, inaccuracy is assessed as the extent of the smallest computer program that can rectify the incorrect output of our model.

Inaccuracy evaluates how well the output of our description aligns with the selected representation encoding the entity. However, this representation could be flawed itself, as discussed in the preceding chapter. Inaccuracy focuses solely on the description d , neglecting the potential miscoding within the representation r . Furthermore, even though it doesn't require an oracle, inaccuracy cannot be calculated for every case, so it needs to be estimated in practical situations, as we will explore in Part III of this book.

Let us consider $r \in \mathcal{B}^*$ as a representation, and $d \in \mathcal{D}$ as a description, where $d = \langle TM, a \rangle$. We then define the *inaccuracy* of the description d with respect to the representation r , denoted as $\iota(d, r)$, according to the following formula:

$$\iota(d, r) = \frac{\max\{K(r \mid \delta(d)), K(\delta(d) \mid r)\}}{\max\{K(r), K(\delta(d))\}}$$

Intuitively, accuracy measures the difficulty in converting an incorrect representation r' produced by a desription d into the original representation r . In essence, this involves the computation of the normalized information distance between r' and r .

In practice, and in the particular case of descriptions based on the scientific articles of Wikipedia, we cannot apply this definition, because an article of Wikipedia is not a Turing machine that produces a string based output.

Figure 14.8 left depicts a histogram showing the inaccuracy of the analyzed topics, and right side shows a scatterplot of the article lenghts vs. the length of the correspondign talk pages, together with a regression line fitted to the data. The

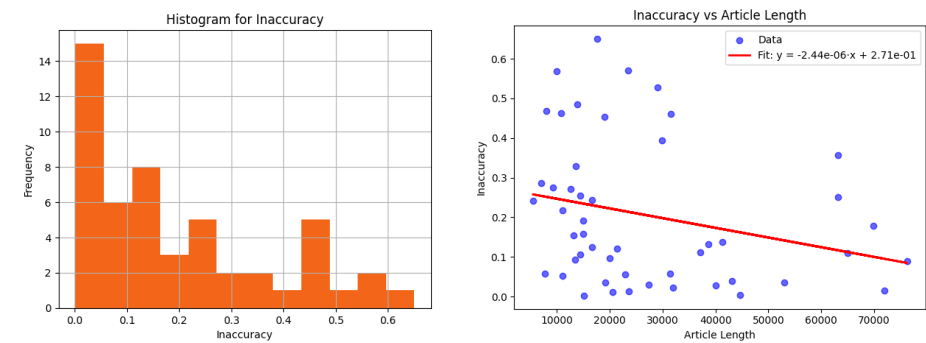


Table 14.5: Inaccuracy of Topics

two plots suggest that, within this sample of good-quality theoretical computer science pages, talk pages chatter is generally modest and becomes proportionally smaller as articles grow. Inaccuracy, approximated as talk / (article + talk), is highly right-skewed. Roughly half the pages fall below 0.05 and more than three-quarters below 0.15, meaning their talk pages contain at most one word of discussion for every six to twenty words of main-text content. Only a small tail of articles pushes beyond 0.40, indicating a handful of topics that still attract extensive debate or revision despite their “good” label. In the scatter plot we can observe that the best-fit line ($y \approx -2.44 \times 10^{-6}length + 0.271$) slopes downward, so predicted inaccuracy drops from about 0.27 for a 0-length stub to ≈ 0.10 at 70 kB. Empirically, short articles (< 15 kB) show the widest spread—from virtually no talk to talk pages two-thirds the size of the article—whereas long articles (> 50 kB) cluster below 0.15 with only rare outliers. In other words, once an article expands to tens of thousands of characters, the relative volume of unresolved discussion shrinks, suggesting that length correlates with maturity and consensus. Taken together, the figures imply that most good theoretical-CS pages are comparatively settled, that the few contentious ones are disproportionately short, and that growing an article tends to absorb or resolve the issues reflected on its talk page rather than magnifying them.

Table 14.9 lists topics exhibiting the lowest levels of nescience. The topics with the lowest inaccuracy values (≤ 0.04) look like settled science. They cover classical, mathematically-rigorous results—matching algorithms (Gale-Shapley), structural graph properties (perfect graphs, Steinitz’s theorem), and well-studied data-structures or routing primitives (Cartesian tree, widest-path problem). For such topics there is little room for interpretive dispute: statements are either formally correct or not, and once a clean exposition is in place subsequent editors rarely need lengthy back-and-forth. That is why some talk pages are only a few dozen words long (25 for Gale-Shapley) even when the article itself

Topic	Inaccuracy	Article length	Talk length
Gale-Shapley algorithm	0.001656	15072	25
Perfect graph	0.004412	44683	198
Unit distance graph	0.012203	20479	253
Rook's graph	0.013267	23576	317
Network synthesis	0.014493	71945	1058
Logic of graphs	0.022260	31933	727
Steinitz's theorem	0.027765	40094	1145
Cartesian tree	0.030849	27395	872
Widest path problem	0.035644	19128	707
Book embedding	0.036141	53046	1989

Table 14.6: Topics with lower inaccuracy

spans tens of kilobytes. Length alone does not guarantee a quiet talk page—the list includes both mid-sized texts (≈ 20 kB) and the 72 kB Network synthesis article—yet all of them keep the ratio of discussion to content below 4%. Where talk does appear (e.g. 1 989 words for Book embedding or 1 145 for Steinitz's theorem) it is still dwarfed by the main text, implying that open issues tend to be minor wording tweaks, sourcing details, or peripheral expansions rather than fundamental disagreements. In short, low-inaccuracy articles are those whose subject matter is uncontroversial, formally locked-down, and already presented in a stable, comprehensive fashion, leaving editors with little need for ongoing debate.

Finally, Table 14.7 lists the topics with the highest nescience values. The articles with the highest inaccuracy scores are not necessarily the most error-ridden; rather, they are the ones whose talk pages have become arenas for protracted discussion because their subjects invite disagreement or continual refinement—foundational concepts like the commutative property and tries that draw many first-time editors, folklore-tinged algorithms such as the fast inverse square root, or theorems and graph classes (e.g., Sylvester-Gallai, pseudoforests, Halin graphs) that have multiple equivalent statements, proofs, or naming conventions. In every case the talk page approaches or even exceeds the size of the main text, so anywhere from 40% to 65% of the combined bytes are debate rather than exposition; this ratio is amplified by the fact that most of these articles are relatively short ($\approx 10 - 30$ kB), meaning even modest amounts of conversation loom large. Together they illustrate that a high inaccuracy score chiefly signals ongoing editorial contention—about scope, presentation order, historical credit, or performance claims—rather than a simple lack of factual correctness, and that such contention is most pronounced where definitions are ambiguous, folklore

Topic	Inaccuracy	Article length	Talk length
Sylvester-Gallai theorem	0.394610	29810	19431
Pseudoforest	0.454130	18999	15806
Selection algorithm	0.461374	31578	27049
BIT predicate	0.463016	10817	9327
Halin graph	0.467671	7953	6987
Three utilities problem	0.485948	13846	13089
Fast inverse square root	0.528723	29001	32536
Sylvester’s sequence	0.567808	9994	13130
Trie	0.570980	23503	31280
Commutative property	0.650826	17583	32773

Table 14.7: Topics with higher inaccuracy

collides with formalism, or broad audiences repeatedly revisit basic material.

14.2.3 Nescience

According to the theory of nescience, our understanding of an entity should be based on the quality of the description used to explain it. In Chapter 7, we introduced a quantitative measure of our ignorance regarding a research entity. This measure depends on the miscoding of a string-based representation of the entity, as well as the inaccuracy and surfeit of the model describing this representation. Ideally, we seek representations and descriptions that simultaneously minimize these three aspects.

In practical terms, particularly when entities correspond to topics covered by scientific Wikipedia pages, we consider only the surfeit and inaccuracy of a description. In this scenario, the representation of a topic coincides with its description, rendering the computation of miscoding irrelevant.

To address the multi-objective optimization problem posed by nescience, we apply a global criterion approach (see Section 21.6.3). This method solves multi-objective optimization problems by minimizing the distance between a chosen reference point and the feasible region of the objective space. We select the origin vector (0,0) as our reference point. The distance metric used will be the harmonic mean of surfeit and inaccuracy, defined as:

$$\frac{2}{\iota(d,r)^{-1} + \sigma(d,r)^{-1}}$$

As described in previous sections, surfeit is approximated by the redundancy of the topic’s description (the Wikipedia page), while inaccuracy is estimated by the length of the corresponding Talk page.

We are using a decision maker to compute nescience based on redundancy and inaccuracy, and since these quantities do not have the same scale, it is highly convenient to apply the following additional transformation to them:

$$\mu_i = \frac{\mu_i - \min(\mu)}{\max(\mu) - \min(\mu)}$$

where μ_i refers to the considered metric (nescience, relevance, ...).

Figure 14.8 left depicts a histogram showing the nescience levels of the analyzed topics, and right side shows a scatterplot of the article lengths vs. the nescience, together with a regression line fitted to the data. The two panels paint a mixed picture of how much “unknown-ness” remains in these 50 good theoretical-CS articles once both redundancy (compressibility of the prose) and inaccuracy (relative size of the talk page) are factored in. Histogram — a long, uneven tail. Roughly one-third of the articles cluster below nescience ≈ 0.10 , meaning they are simultaneously concise (low redundancy) and largely uncontested (little talk). Frequency then stays fairly flat through the 0.15 - 0.35 band and drops again, before a second bump appears between 0.60 and 0.75. That small peak corresponds to pages that are both highly repetitive and heavily debated—the same ones that topped the earlier redundancy and inaccuracy lists. In short, while many good-class pages are well understood, a non-trivial minority still exhibits a pronounced knowledge gap. Scatter plot — almost length-neutral. The best-fit line rises only 0.007 points for every additional 10 000 characters (slope $\approx 7.5 \times 10^{-7}$), so article length alone is not a strong predictor of nescience. Short pages can swing from near-zero to 0.7, depending on how contentious they are, whereas long pages (> 50 kB) compress more (pushing nescience up) but also attract proportionally less talk (pushing it down), leaving them parked in a mid-range around 0.25-0.35. The gentle upward tilt therefore reflects the fact that redundancy grows with length a bit faster than inaccuracy falls, but the wide vertical spread shows that editorial stability—rather than size—is the decisive factor. Overall, “good” status keeps most theoretical-CS articles in a low-to-moderate nescience zone, yet about a quarter of them still suffer from enough repetition and unresolved discussion to signal that our coverage of those topics remains less mature than the label might suggest.

Table 14.9 lists topics exhibiting the lowest levels of nescience. The articles that score best on the nescience metric fall into two complementary archetypes, each driving the harmonic mean down by making one of the two ingredients—redundancy or inaccuracy—almost vanish. The first archetype is epitomised by “Sylvester’s sequence” and “Turán’s brick-factory problem”: they are short, information-dense pages whose prose compresses poorly (normalized redundancy ≈ 0), so even a rather lively talk page cannot lift their nescience above zero. The second archetype—illustrated by “Gale-Shapley algorithm”, “Perfect

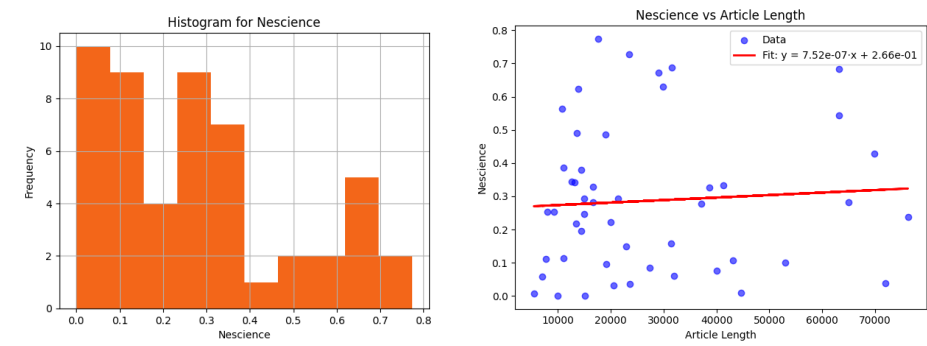


Table 14.8: Nescience of Topics

Topic	Nescience	N. Inaccuracy	N. Redundancy
Sylvester’s sequence	0.000000	0.872117	0.000000
Gale-Shapley algorithm	0.000000	0.000000	0.499166
Turán’s brick factory	0.007362	0.370945	0.003718
Perfect graph	0.008448	0.004245	0.864084
Unit distance graph	0.031744	0.016248	0.686717
Rook’s graph	0.035003	0.017887	0.812327
Network synthesis	0.038635	0.019774	0.836612
Viète’s formula	0.057626	0.437415	0.030845
Logic of graphs	0.061030	0.031738	0.791546
Steinitz’s theorem	0.076638	0.040219	0.811139

Table 14.9: Topics with lower nescience

graph", "Network synthesis", and several other graph-theoretic topics—shows the opposite pattern: their text is highly repetitive (normalized redundancy $\approx 0.8 - 0.86$), but the talk pages are almost silent, so inaccuracy is near zero and the harmonic mean again collapses. Taken together, the list reveals that low lack-of-knowledge can arise either from concise, debate-heavy but non-redundant expositions or from long, internally repetitive but well-settled treatments; what matters for nescience is that at least one dimension of uncertainty is squeezed to the floor.

Finally, Table Finally, Table 14.10 lists the topics with the highest nescience values. The upper tail of the nescience scale is populated by pages that score high on both axes at once—they are repetitive and heavily debated—so the harmonic mean refuses to average the trouble away. Half of the list ("BIT predicate", "Three-utilities problem", "Fast inverse square root", "Selection algorithm", "Trie", "Commutative property") shows normalized-inaccuracy ≥ 0.70 , signalling

Topic	Nescience	N. Inaccuracy	N. Redundancy
Farthest-first traversal	0.490109	0.505215	0.475880
Shapley-Folkman lemma	0.543853	0.384947	0.926181
BIT predicate	0.562673	0.710692	0.465683
Three utilities problem	0.622535	0.746017	0.534125
Sylvester-Gallai theorem	0.629670	0.605318	0.656063
Fast inverse square root	0.671295	0.811908	0.572197
Clique problem	0.683333	0.548081	0.907206
Selection algorithm	0.688247	0.708163	0.669420
Trie	0.727080	0.877003	0.620932
Commutative property	0.774591	1.000000	0.632109

Table 14.10: Topics with higher nescience

talk pages that rival the article in size; the same set attracts a steady stream of newcomers or folklore-laden claims, which keeps discussion alive. The other half ("Shapley-Folkman lemma", "Clique problem", "Sylvester-Gallai theorem", "Farthest-first traversal") earns its place chiefly through very high redundancy (≈ 0.90 in the case of the clique problem), reflecting articles that restate definitions, proofs and examples in multiple near-duplicate forms. In every case the other dimension is still uncomfortably large (≥ 0.38), so neither trimming repetition nor settling open talk-page issues alone would be enough to pull these topics out of the danger zone. In short, high-nescience entries are those where unresolved editorial contention co-exists with a self-similar writing style—pages that simultaneously need consensus-building and structural tightening before they can be considered well understood.

14.2.4 Conclusion

Across the 50 “good-class” Wikipedia pages in theoretical computer science, our indicators tell a nuanced story about what the encyclopaedia can reveal—and obscure—about collective understanding. Talk-page activity, captured by our inaccuracy ratio, does a credible job of flagging articles whose content is still contested: pages with minimal discussion almost always cover settled, formally unambiguous material, whereas those whose talk pages rival the main text correspond to concepts riddled with naming ambiguity, folklore or pedagogical disagreement. Redundancy, distilled from compression ratios after stripping boiler-plate, complements this view by signalling how tightly authors have distilled the topic: low values align with concise, information-dense expositions, while high values mark articles that re-iterate definitions, proofs and code snippets in multiple guises. Each metric thus illuminates a different facet—editorial

Area	Num. Topics
Biology	214
Philosophy	33
Psychology	44
Mathematics	127
Chemistry	138
Physics	52

Table 14.11: Number of topics analyzed per area

consensus on one side, stylistic efficiency on the other—yet neither by itself captures “knowledge quality” outright.

Their harmonic mean, which we dubbed nescience, succeeds as a triage tool because it punishes a topic the moment either dimension becomes extreme. Articles that are both verbose and hotly debated—such as those on the fast inverse square root or the clique problem—cluster at the high end, clearly pointing to areas where Wikipedia’s coverage remains unsettled. Conversely, pages with either crisp, irredundant prose or near-empty talk pages sink to the bottom, reflecting mature, well-understood subjects like the Gale-Shapley algorithm or perfect graphs. In that sense, nescience offers a practical snapshot of our residual ignorance as reflected in the encyclopaedia: it reliably highlights where clarification, consolidation or further research are still needed, even if it cannot, on its own, disentangle whether the root cause is factual dispute, pedagogical complexity, or simple editorial neglect.

14.3 Measuring Research Areas

In this section, we evaluate the nescience of entire research areas rather than individual topics, measuring how much knowledge each discipline encompasses. We examined all English-language “good-class” articles in six broad disciplines (biology, chemistry, mathematics, philosophy, physics, and psychology) comprising roughly 600 pages (see Table 14.11). For each discipline, we assessed three metrics: inaccuracy, redundancy, and nescience.

Wikipedia organizes articles into categories by topic to simplify navigation and discovery. Rather than a strict hierarchy, categories form an interconnected network: each category can have multiple subcategories, and a subcategory may belong to multiple parent categories. Editors avoid circular links, where a category would indirectly include itself through its descendants. Conceptually, this structure resembles a partially ordered set in mathematics and computer science, and it allows diverse classification schemes to coexist within a unified framework. When one category clearly falls under another, it is designated as a

subcategory to preserve a logical “is-a” relationship. Articles are typically placed only in their most specific relevant category to avoid redundancy. For example, the article on Claude Shannon appears in "Category:American information theorists," not directly under the broader "Category:Mathematicians."

The top level of Wikipedia’s category system is “Category:Contents,” which has no parent. From there, the path we have used proceeds as: Contents → Articles → Main topic classification → Academic disciplines → Science → Branches of Science (Applied Sciences, Formal Sciences, Natural Sciences, Social Sciences). Our analysis focuses on five scientific areas: Mathematics, which falls under Formal Sciences; Biology, classified within Life Sciences under Natural Sciences; Physics and Chemistry, both part of Physical Sciences in the Natural Sciences; and Psychology, included in Social Sciences. We also analyze Philosophy, categorized under Humanities within Academic disciplines. These selections ensure comprehensive coverage of both theoretical and empirical fields.

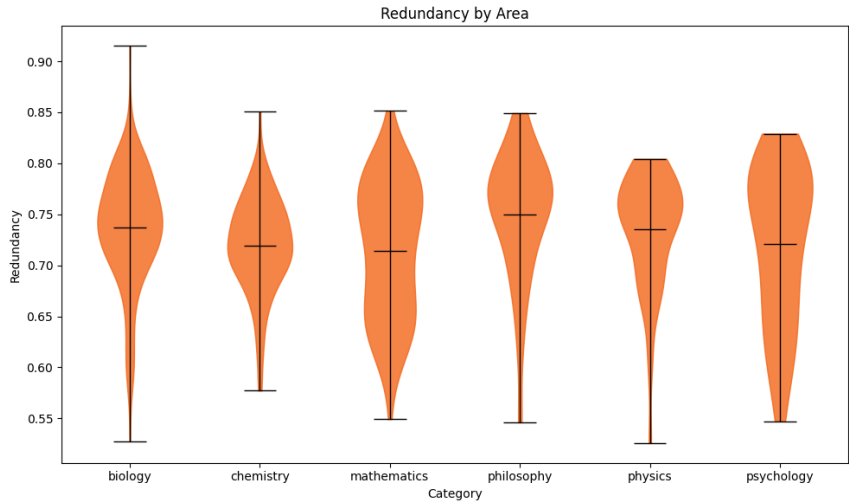


Figure 14.2: Redundancy by Area

First, we looked at redundancy (see Figure 14.2), which shows how much an article repeats itself. We took out templates, lists, and other markup, then compressed each page using bzip2 at its highest setting. Pages with more repeated text stay larger after compression. We found that most good articles in every field fall in a redundancy range of about 0.70 to 0.80, so they repeat text at similar rates. Biology had the widest range, from about 0.52 up to over 0.90, meaning some articles are very concise while others include many standard sections. Chemistry was the most consistent group, clustering around 0.72, possibly because of regular naming rules and reaction sections. Mathematics, philosophy,

physics, and psychology were all in the middle, with medians near 0.75 but small differences in spread. For example, math and philosophy pages sometimes repeat proofs or arguments, raising their upper range, while psychology has some short experiment-focused articles. Physics looks like psychology but with a slightly lower median. In short, redundancy is similar across fields: most good articles keep about three-quarters of their text after compression, showing that writing style and citation rules has a higher impact than the topic itself.

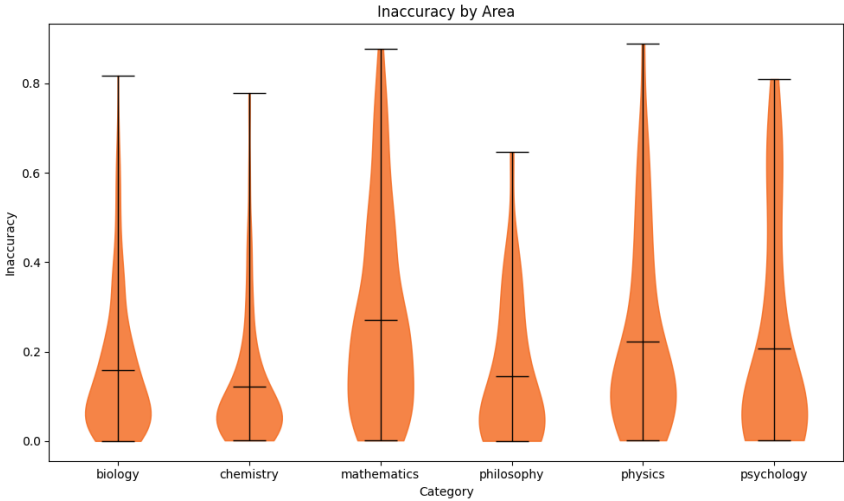


Figure 14.3: Inaccuracy by Area

Next, we estimated inaccuracy by looking at editorial debate rather than factual error (see Figure 14.3). We looked at each article’s talk page size and compared it to the total size of both the talk page and the article. A larger talk page means more discussion or disagreement. The violin plot shows that most articles in every field spend only a small part of their bytes on talk pages, but a few pages have much longer debates. In chemistry, most articles keep their talk pages below about 0.1 of the total bytes, and the median is just above that. Only a few articles reach up to 0.80. Biology, philosophy, and psychology also have most pages under roughly 0.15, but these fields have taller plots in the upper half. That means a few topics, like controversial medical issues in biology or big theoretical arguments in philosophy and psychology, spark long discussions. Mathematics and physics have the highest and widest range of debate. Their median sits around 0.25, and many pages go up to 0.5 or even 0.9, showing that some articles have as much discussion as main content. Overall, most scientific topics settle into low debate, but math and physics pages often have more back-and-forth about definitions, notation, and sources, while chemistry articles are more stable.

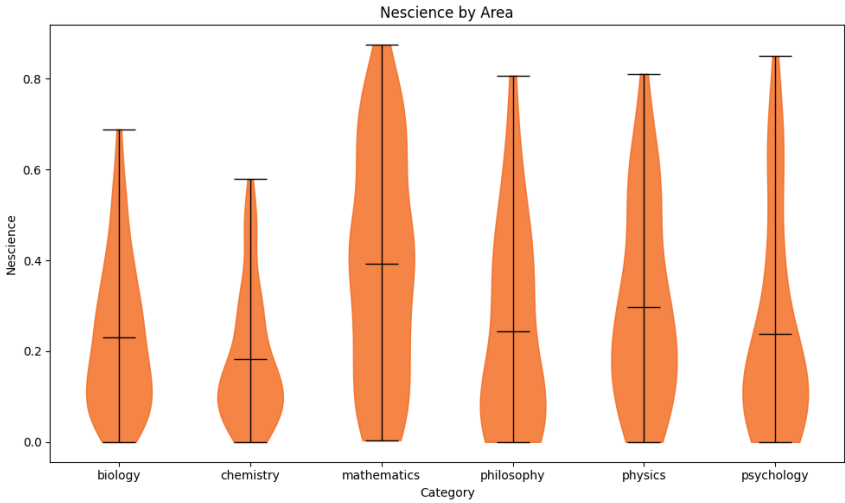


Figure 14.4: Nescience by Area

Finally, we combined redundancy and inaccuracy into one score, nescience (see Figure 14.4). We first scaled each measure to range from 0 to 1, then took their harmonic mean. This score highlights pages that are either very repetitive, heavily debated, or both. In the violin plot, chemistry articles have low nescience (mostly under 0.20), showing they are concise and rarely debated. Biology is similar but has a few articles with higher scores, reflecting some ongoing talk-page discussions. Mathematics has the highest median nescience (around 0.40) and reaches up to 0.90, because many pages repeat concepts and have active debates over definitions. Philosophy and physics each show two peaks, one for well-settled articles near zero and one above 0.50 for more contested topics. Psychology mainly has low nescience, but some articles still face long debates. Overall, while every field has many well-covered pages, our nescience score shows that gaps in knowledge are biggest in mathematics and, to a lesser degree, in philosophy, physics, and psychology.

Table 14.12 shows the average scores for each discipline. The numbers show that redundancy is almost the same across all six fields, ranging from 0.71 to 0.75, but inaccuracy varies more than a factor of two, from 0.12 in chemistry to 0.27 in mathematics. Because redundancy stays nearly constant, the differences in the combined nescience score come mostly from inaccuracy. In other words, after removing repeated boilerplate, the writing style is similar across subjects; what really changes is how much editors discuss or disagree on the talk pages. For example, chemistry’s low inaccuracy (0.12) gives it the smallest nescience (0.18), even with average redundancy. In contrast, mathematics has the highest

Category	Inaccuracy	Redundancy	Nescience
Biology	0.157773	0.736704	0.229765
Chemistry	0.122821	0.719500	0.183116
Mathematics	0.271307	0.714365	0.392008
Philosophy	0.144690	0.750007	0.244622
Physics	0.223447	0.735314	0.297802
Psychology	0.207355	0.721208	0.237835

Table 14.12: Average metrics by area

inaccuracy (0.27) and therefore the highest nescience (0.39), despite slightly lower redundancy. Physics and psychology have similar redundancy (around 0.72-0.74), but physics sees more talk-page debate (0.22 vs. 0.21), so its nescience is higher than psychology’s. Philosophy shows the highest redundancy (0.75) but only a medium nescience (0.24) because its talk pages are quieter. Overall, this table tells us that for high-quality articles, the main difference between fields is how much editors argue, not how the text is written.

Putting together the redundancy and inaccuracy measures gives a single nescience score. This shows that all fields have similar writing patterns, most high-quality articles compress to about 70-80% of their original length, but they differ in how much they spark online discussions. Chemistry and biology have the quietest talk pages, so they score lowest on nescience. Mathematics and physics have more debates, so they score highest, even though their writing density is similar. Philosophy and psychology fall in between. These gaps happen because each field has many settled articles and a smaller group that need more discussion. In short, Wikipedia’s top science articles are usually concise and well written, but the talk pages reveal where editors still disagree most. The biggest knowledge gaps remain in math topics heavy on definitions and in physics areas open to different interpretations.

14.4 The Evolution of Knowledge

Our understanding of a scientific topic typically improves through sustained research efforts over time, resulting in a reduction of the topic’s nescience, our lack of knowledge. Specifically, this improvement occurs through reductions in redundancy (repeated, unnecessary information) and inaccuracy (incorrect or misleading information), without significant increases in either component offsetting these improvements. New explanatory frameworks may emerge from innovative theories, refinements of existing theories, or simplified models. Ideally, each successive explanation will become more concise and accurate, removing previously redundant details and errors. Occasionally, new descriptions may

temporarily lengthen as additional factual information is incorporated, causing a temporary increase in nescience. Nevertheless, the overall trend in scientific research should consistently be a progressive decrease in nescience as our knowledge advances.

To demonstrate how our theory can be used to characterize the evolution of our understanding of a scientific topic, that is, how nescience decreases with time, we have selected three highly relevant research topics as illustrative examples: graphene, CRISPR, and deep learning. Graphene is a single layer of carbon atoms arranged in a two-dimensional honeycomb lattice, known for its exceptional strength, conductivity, and versatility. CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) is a revolutionary gene-editing technology that allows for precise, targeted modifications to DNA. Deep learning is a subset of machine learning based on artificial neural networks, enabling complex pattern recognition and predictive modeling. These topics were chosen for their scientific importance and the richness of their associated descriptions, allowing us to apply our methodology and illustrate its effectiveness.

Figure 14.13 illustrates the evolution over time of redundancy, inaccuracy, and nescience metrics for the topic of Graphene. Each point on the graph corresponds to the analysis of a cleaned-up version of the Wikipedia page at a specific point in time. The early history of the graphene article shows the classic growth-and-turmoil phase of a fast-moving research topic. Between 2005 and 2010 redundancy climbed from ≈ 0.5 to just over 0.7 as the page ballooned with additional sections, figures and repeated explanations, while inaccuracy, our proxy for debate, spiked above 0.5. That surge coincides with the period when graphene research exploded in the literature, patents, and the popular press, so editors were actively negotiating scope, terminology and sourcing. The high redundancy plus high inaccuracy pushed the nescience curve to its peak near 0.9 in 2009-2010, signalling that the encyclopaedia's coverage lagged behind the field's rapid advances.

A sharp inflection occurs in 2011. The talk-page ratio collapses almost to zero-likely because extensive discussion was archived or split into sub-pages after the article stabilised-driving inaccuracy, and therefore nescience, down to negligible levels. From that point on the page continues to lengthen: redundancy drifts upward into the 0.78-0.82 band as definitions, production methods and applications are reiterated for different audiences. Yet talk-page activity remains a small fraction of total bytes, so nescience stays close to zero with only minor blips when new findings (e.g., commercial production techniques around 2014 or 2019) briefly rekindle debate. The negative regression slope on both the inaccuracy and nescience plots confirms a long-term convergence toward consensus, while the positive slope on redundancy simply reflects a mature, template-rich article that keeps accumulating detail without reopening fundamental disputes.

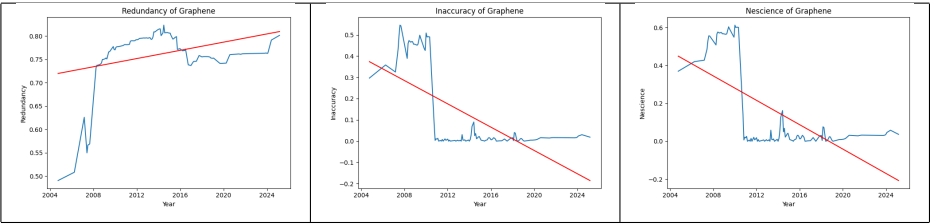


Table 14.13: Knowledge evolution of graphene

CRISPR’s Wikipedia trajectory (see Figure 14.14) mirrors the arc of a scientific whirlwind maturing into mainstream knowledge. When the page first took shape (2010 - 2013) redundancy sat in the low-to-mid 0.60, reflecting a compact article that still compressed poorly, while inaccuracy-driven by an energetic talk page-hovered around 0.40. Those years coincide with the burst of laboratory discoveries that turned CRISPR-Cas9 from an obscure bacterial defence into a headline-making gene-editing tool, so editors were constantly renegotiating scope and sourcing. Both factors pushed nescience above 0.70, signalling a large knowledge gap between fast-moving science and the encyclopaedia’s ability to consolidate it.

A decisive shift occurs in 2015-2016. The talk-page share collapses, dropping inaccuracy to near zero almost overnight; the page was substantially rewritten and much of the debate was archived once standard terminology, mechanism diagrams and milestone experiments had stabilised. Redundancy, meanwhile, jumps above 0.75 and keeps inching up as new sections on ethics, patents and clinical trials are added-material that inevitably repeats the core biology and acronyms. With one input shrinking and the other growing only slowly, nescience plummets and stays near the floor, apart from brief spikes that map neatly onto external flashpoints (the 2018 gene-edited-babies scandal, for example).

The overall downward regression slopes for both inaccuracy and nescience confirm a steady convergence toward consensus, while the upward slope in redundancy simply records the article’s swelling, template-rich structure. By 2020 the CRISPR page looks much like that of graphene in its mature phase: highly compressible, rarely contested, and updated incrementally rather than rewritten from scratch each time the field advances.

The deep-learning article (see 14.15) traces a pattern typical of a field that shifted from niche research to global headline status almost overnight. In its early years (2011-2014) the page was compact and still being drafted: redundancy climbed from ≈ 0.40 to the low 0.70 as basic definitions, algorithm lists and seminal breakthroughs were added and often restated, while inaccuracy oscillated but stayed below 0.30, reflecting moderate talk-page traffic. The real turbulence began in 2017: the talk-page share abruptly jumped above 0.40 and stayed there

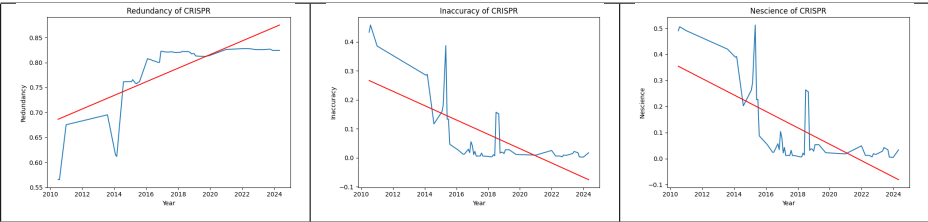


Table 14.14: Knowledge evolution of CRISPR

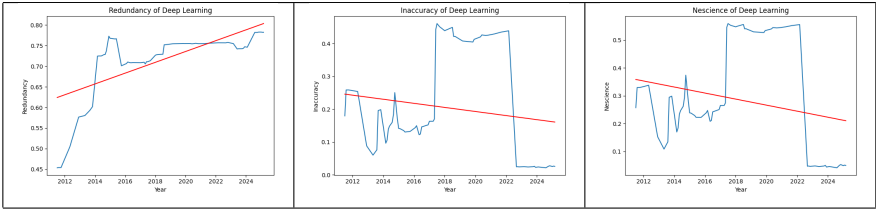


Table 14.15: Knowledge evolution of deep learning

for nearly five years, mirroring the community’s debates over benchmark claims, ethics, and the hype surrounding "AI booms." Because redundancy had already settled in the mid-0.70s, this burst of discussion drove nescience to a sustained plateau above 0.85, marking the article as one of Wikipedia’s most actively contested science pages during the height of deep-learning publicity.

A dramatic correction occurs in late 2022. Large chunks of deliberation were apparently archived or spun off, slashing inaccuracy to near zero almost overnight and dropping nescience with it. Since then the page has remained highly compressible (redundancy $\approx$ 0.75 - 0.78) but largely uncontested; only small upticks appear as new foundation-model milestones are folded in. The negative regression slopes for inaccuracy and nescience therefore chart a long-run movement toward consensus, even though the five-year plateau of high nescience serves as a reminder that Wikipedia’s knowledge gap can stay wide for a protracted period when a discipline’s methods, jargon and social implications are evolving faster than editors can lock down a stable narrative.

Across three emblematic case studies (graphene, CRISPR and deep learning) the time-series confirm that Wikipedia’s nescience reliably chronicles the lifecycle of a scientific topic: initial growth brings a surge of redundancy as material is layered on, and a spike of inaccuracy as editors negotiate scope and sourcing; once consensus forms, talk-page activity collapses, redundancy stabilises in the high-0.70s, and nescience falls toward zero. Graphene reached this mature phase around 2011, CRISPR around 2016, while deep learning lingered in a high-gap state from 2017 to 2022 before a mass archiving of debate produced an abrupt convergence. The trajectory is thus quantifiable: rising redundancy without rising

inaccuracy signals steady accretion of detail, whereas persistent high nescience marks periods when the science itself, or its social framing, is still in flux.

14.5 The Demarcation Problem

In this section, we propose a practical method, based on the theory of nescience, to address the demarcation problem, specifically, the challenge of distinguishing scientific from non-scientific knowledge in real-world contexts. Although demarcation is a longstanding philosophical issue, our aim is not to conclusively solve this complex problem but rather to provide insights into its nature and outline potential paths toward future solutions through practical and operational methods.

For our experiments and analysis, we have selected six scientific topics and six pseudoscientific topics to evaluate our approach to the demarcation problem. Our analysis is based on the descriptions of these topics provided by Wikipedia and their associated Talk pages. As in previous analyses conducted in this chapter, we have preprocessed the Wikipedia pages by using the `wikitextparser` Python library to remove Wikimedia tags and other irrelevant elements such as tables and images.

The six scientific topics selected are: Climate Change (the long-term alteration of temperature and weather patterns), Graphene (a single-layer carbon material with extraordinary physical properties), Dark Matter (a hypothesized form of matter making up approximately 85% of the universe), Deep Learning (a subset of machine learning using neural networks with multiple layers), Lithium-ion Battery (a type of rechargeable battery widely used for portable electronics and electric vehicles), and Brain-Computer Interface (a technology enabling direct communication between the brain and external devices). These scientific topics have been selected due to their intensive research activity over the past 20 years.

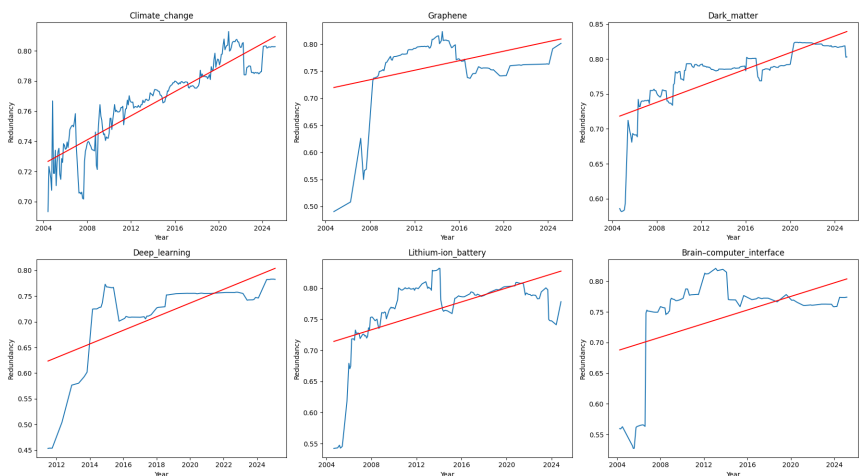


Figure 14.5: Evolution of Redundancy in Scientific Topics

Redundancy, as an approximation of the concept of surfeit, is computed, as in the previous case, by comparing the ratio of the length of a text to its compressed version. Figure 14.5 shows the evolution of redundancy for these selected scientific topics over the past 20 years. As observed, redundancy for these topics demonstrates an increasing trend, as confirmed by the computed regression line. Although one might generally expect redundancy to decrease as our understanding improves, new discoveries and emerging knowledge frequently necessitate additional details and explanations, thus increasing redundancy in descriptions.

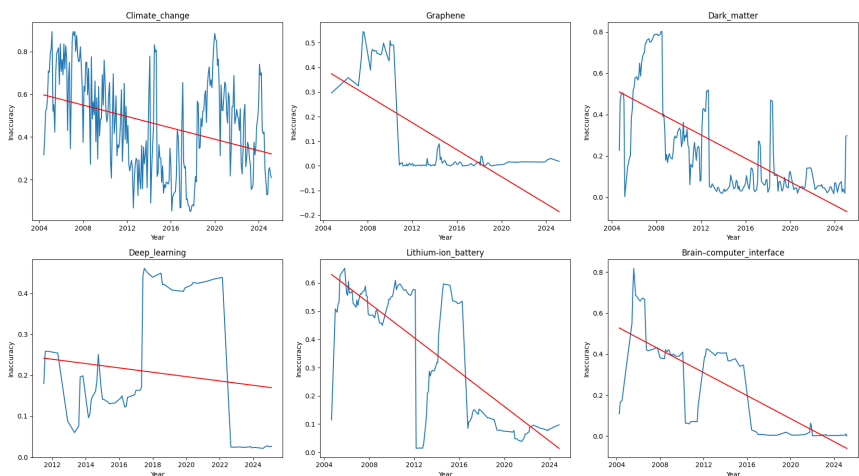


Figure 14.6: Evolution of Inaccuracy in Scientific Topics

Inaccuracy is computed based on the ratio

$$length_talk / (length_talk + length_article),$$

where `length_talk` represents the length of the Talk page for each topic, and `length_article` is the length of the corresponding Wikipedia article, as done in the previous sections. Utilizing Talk pages is an effective method for approximating article inaccuracy since these pages typically contain discussions, disputes, and clarifications regarding inaccuracies or controversies in the articles. Figure 14.6 illustrates the evolution of inaccuracies for the selected scientific topics, showing a clear decreasing trend across all topics.

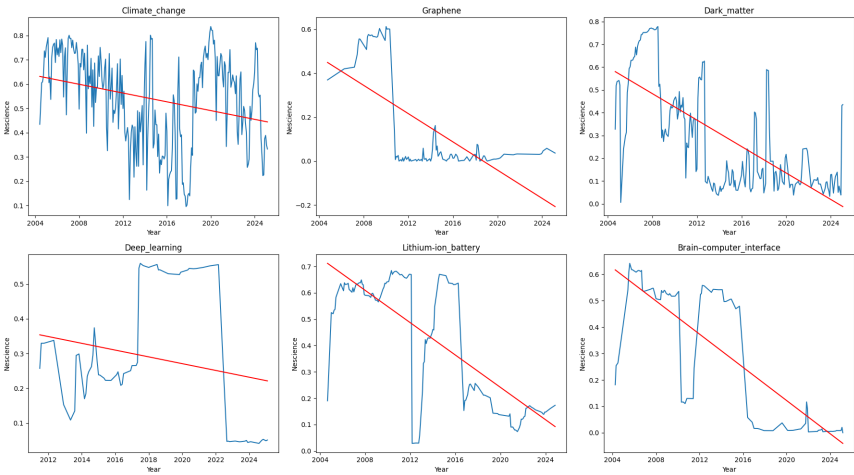


Figure 14.7: Evolution of Nescience in Scientific Topics

Finally, Figure 14.7 shows the evolution of nescience for all selected scientific topics, where nescience is estimated as the harmonic mean of the metrics of redundancy and inaccuracy. As depicted, despite the positive trend in redundancy, nescience exhibits a decreasing trend, suggesting our overall understanding of these topics improves over time.

Additionally, we have selected six pseudoscientific topics to further evaluate our demarcation method: Lunar Effect (the belief that lunar cycles influence human behavior), Water Memory (the claim that water retains a memory of substances previously dissolved in it), Astral Projection (the claimed ability of consciousness to leave the physical body and travel in the astral plane), Enneagram of Personality (a model describing personality types based on a geometric figure with nine interconnected points), Perpetual Motion (the hypothetical concept of a machine that operates indefinitely without energy input), and Dowsing (a technique claiming the ability to locate water, minerals,

or other hidden substances through intuitive means). These topics represent different pseudoscientific categories—Lunar Effect (Astrology), Water Memory (Homeopathy), Astral Projection (Parapsychology), Enneagram of Personality (Numerology), Perpetual Motion (Physics-related pseudoscience), and Dowsing (Divination)—and were selected based on classifications from Wikipedia itself as pseudoscience.

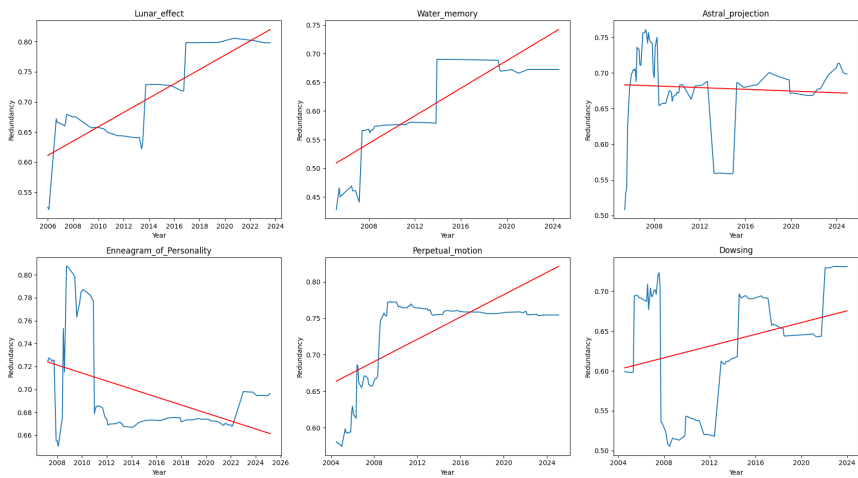


Figure 14.8: Evolution of Redundancy in Pseudoscientific Topics

Figure 14.8 shows the evolution of redundancy for these selected pseudoscientific topics over the past 20 years. As it was the case of scientific topics, redundancy for these topics demonstrates an increasing trend for "Lunar Effect", "Water Memory", "Perpetual Motion", and "Dowsing", and a rather surprising non-increasing trend for "Astral Projection" and "Enneagram of Personality". This decrease in redundancy for the latter two topics may indicate that their Wikipedia articles have undergone substantial editing aimed at streamlining or simplifying the content.

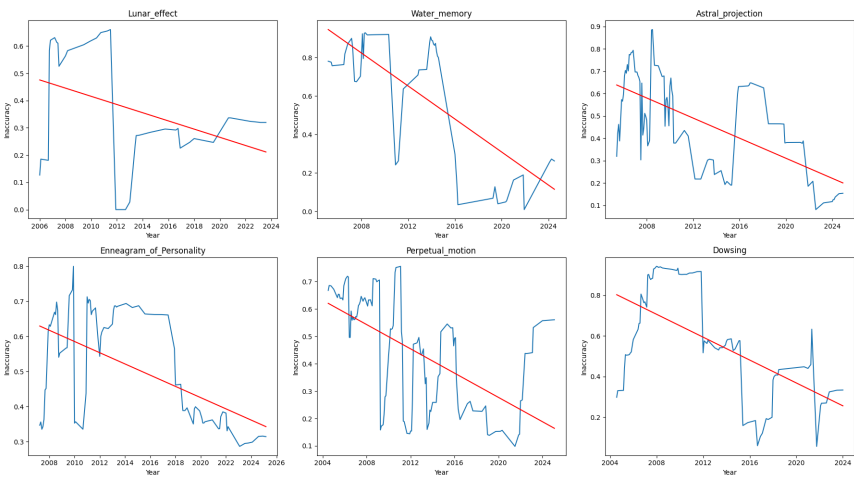


Figure 14.9: Evolution of Inaccuracy in Pseudoscientific Topics

Figure 14.9 illustrates the evolution of the estimated inaccuracies for the same pseudoscientific topics. A clear decreasing trend in inaccuracies is observable across most topics, suggesting, somewhat unexpectedly, that controversies regarding these topics have started to settle down.

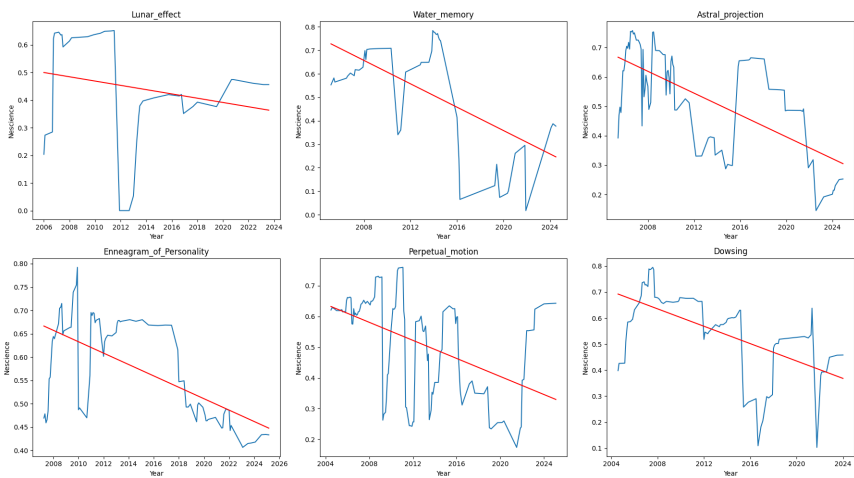


Figure 14.10: Evolution of Nescience in Pseudoscientific Topics

Finally, Figure 14.10 shows the evolution of the estimated nescience for all selected pseudoscientific topics. As expected, given the previous figures for redundancy and inaccuracy, a decreasing trend is observed for most topics, suggesting that our knowledge about this topics has, in fact, decreased with time.

Our preliminary analysis is too superficial to draw definitive conclusions. A more detailed analysis, employing randomized controlled experiments and rigorous hypothesis testing, needs to be conducted. Additionally, the approximations for redundancy and inaccuracy could be refined further (e.g., citation analysis, scientific community surveys, bibliometric analysis). Based on our initial findings, there appear to be no significant practical differences between science and pseudoscience—both communities seem capable of increasing our knowledge about their respective topics. The essential distinction between science and pseudoscience may lie in the validity and truthfulness of claims, rather than in their methodologies. Alternatively, demarcation could also be based on science's demonstrated capability to transform theoretical results into practical, real-world applications—a capability generally lacking in pseudoscience.

References

The following is a list of references that are directly relevant for the chapter on "Analysis of Science", where we have applied the theory of nescience to real-world scientific knowledge using Wikipedia as a proxy. These references support our work in the following ways: philosophical foundations of science, empirical studies of scientific practice, metrics for information content and complexity, and critiques of scientific reliability.

[Cha13]: This book is a classic introduction to the philosophy of science, offering an accessible yet rigorous discussion of how science operates, what counts as scientific knowledge, and how theories change over time. It provides the conceptual background needed to frame your empirical analysis of scientific topics and their epistemic status.

[Mos16]: Mosterín examines the structure and evolution of scientific concepts and theories, discussing how they are created, revised, and sometimes abandoned. His perspective will support your treatment of how individual topics evolve and how nescience diminishes over time.

[Pop14]: Popper's seminal work on the falsifiability criterion and the dynamics of theory change provides the philosophical grounding for your historical analysis of scientific knowledge growth. It will also help frame your treatment of the demarcation problem within your nescience framework.

[LSW13]: Latour's ethnographic study shows how scientific facts are socially constructed within laboratories. Including it emphasizes that Wikipedia articles reflect not only facts but the negotiated outcomes of scientific communities, which ties directly to your measurement of inaccuracy and controversy.

[Ioa05]: Ioannidis empirically argues that much published science is unreliable due to biases, small samples, and statistical issues. This is crucial to motivate the need for metrics like inaccuracy in your analysis of scientific

content.

[Van80]: Van Fraassen's constructive empiricism offers a philosophical counterpoint to scientific realism, questioning what it means for a theory to be considered "true." This perspective enriches your discussion of what your nescience metrics are actually measuring in terms of epistemic commitment.

[Sup02]: Suppes' work on the formal representation of scientific theories will support your transition from individual Wikipedia articles (topics) to aggregated structures (research areas), helping ground your methodology conceptually.

[Li+04]: This paper introduces the normalized compression distance (NCD), a practical information-theoretic tool for measuring similarity between data objects. It underpins your empirical estimation of redundancy and miscoding in Wikipedia articles.

[LV13]: This is the definitive reference for Kolmogorov complexity, the theoretical foundation for your measures of miscoding, redundancy, and ultimately nescience. It provides both the mathematical background and many applied examples.

[CAO+05]: This article warns about practical issues when using real compressors to approximate Kolmogorov complexity. It is highly relevant for the methodological section where you will approximate your metrics on Wikipedia data.



15. Discovering the Unknown

*To be surprised, to wonder,
is to begin to understand.*
José Ortega y Gasset

In this chapter, we are going to see how to apply in practice our methodology for the assisted discovery of interesting research questions. As was the case in the previous chapter, in which we studied the concept of nescience from a practical point of view, here we will focus on the empirical side of our framework, showing how the theoretical concepts introduced earlier can be operationalized and used to guide actual research activities.

In the first part of this chapter, we will examine how to approximate the two new metrics introduced: relevance and applicability. The relevance of a topic will be estimated from the number of web pages on the Internet that link to the topic's page on Wikipedia (external links). Applicability will be estimated from the number of links among Wikipedia's scientific pages themselves (internal links). We will illustrate both metrics with examples drawn from the set of topics that make up the research area of theoretical computer science.

Next, we will describe how to apply our methodology for the discovery of interesting questions and present several examples of new research questions that, in principle, could be addressed by science. Some of these questions

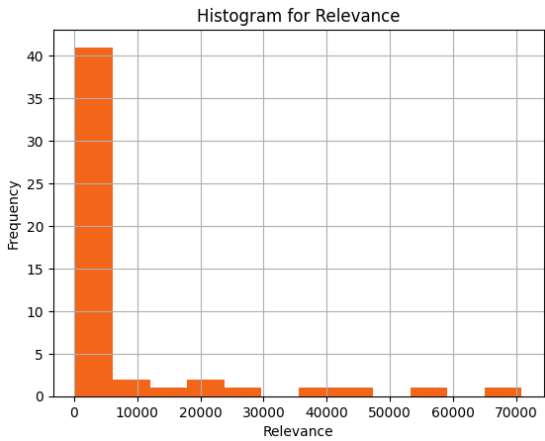


Figure 15.1: Relevance of Topics.

will be intradisciplinary (arising within theoretical computer science) while others will be interdisciplinary, combining theoretical computer science with philosophy or biochemistry. Finally, we will derive a number of new research topics which, according to our subjective interpretation of the combinations found, are sufficiently promising to warrant further investigation. We will also assess whether these proposed topics satisfy the criteria for interesting questions established in Chapter 11.

In the last part of the chapter, we will apply the set of metrics defined for the classification of individual research topics to entire research areas. This will allow us to estimate the interestingness of different disciplines both as sources of new problems and as sources of useful tools for solving open problems. These metrics will also enable us to compare the relative merits of different fields of knowledge, and to highlight some examples of research areas that appear to be in decline.

15.1 Relevance

In Definition ?? we introduced the concept of the relevance of a research topic as a measure of the impact that this topic could have on people’s lives. The idea was that the higher the relevance, the greater its potential as a source of interesting questions, since it would address problems that affect many people. Relevance was defined as the degree of the research topic in the relevance graph, a bipartite graph connecting topics and people (see Definition ??). Of course, this relevance graph is a mathematical abstraction that is very difficult to compute in practice, since we do not have information about how people are affected by each topic.

Topic	Relevance
Dirac delta function	70,869
Rule of inference	58,356
Fast inverse square root	46,570
Euclidean algorithm	38,032
Trie	29,116
Binary search	23,246
Reinforcement learning from human feedback	22,294
Commutative property	17,322
Gale–Shapley algorithm	11,251
Pick’s theorem	7,255

Table 15.1: Most relevant topics

As a practical proxy for the relevance of a topic, we use its Wikipedia pageviews over the last two months (i.e., the sum of daily visits during the most recent 60 complete UTC days). The intuition is that the more relevant a topic is, the more frequently people will consult its Wikipedia article. Our goal is not an absolute notion of relevance but a measure of relative relevance across topics. Moderate under- or over-estimation is acceptable provided it affects all topics roughly uniformly—for example, due to seasonal effects or platform-wide trends—so that comparative rankings remain stable. Further work is needed to quantify how well pageview-based relevance aligns with the theoretical construct of relevance and to correct for known biases (e.g., transient news spikes or bot traffic).

In Figure 15.1, which shows the histogram of the relevance of all good articles in the area of theoretical computer science, total traffic over the last 60 days is 393,672 pageviews, with an average relevance of 7,719. The histogram shows a classic head–long-tail pattern: most topics cluster in the lowest bins (a few hundred to a few thousand pageviews), while a small number of articles receive very large traffic, extending the right tail out past 70k. The top three pages—Dirac delta function (70,869), Rule of inference (58,356), and Fast inverse square root (46,570)—collect 44.7% of all views. This strong right-skew implies that the mean relevance will sit far above the median and be dominated by a handful of high-visibility pages (e.g., widely taught or broadly cited topics). The sparsity of mid-range bins suggests a sharp separation between everyday background interest and a small set of standout articles; outliers are genuine and not just binning artifacts. For modeling purposes we will use log-transformed counts to stabilize variance.

Table 15.1 lists the ten most relevant topics according to this metric. For

Topic	Relevance
Well-covered graph	228
BIT predicate	242
Halin graph	360
Laves graph	462
Tropical cyclone forecast model	510
Book embedding	516
Turán’s brick factory problem	520
Farthest-first traversal	626
Cop-win graph	647
Finite subdivision rule	662

Table 15.2: Least relevant topics

each topic, it shows its raw relevance (number of external links). The top ten articles account for 82.4% of the total page views. Overall, the list aligns reasonably well with what one would expect to attract broad attention around theoretical computer science. Judged purely by intuitive relevance (irrespective of formal classification), most items in the list plausibly merit attention from a theoretical computer science audience: Euclidean algorithm, Binary search, Trie, and the Gale–Shapley algorithm are canonical in algorithms, data structures, and algorithmic game theory; Rule of inference underpins proof techniques used throughout complexity and verification; Fast inverse square root is a celebrated algorithmic trick illustrating implementation–theory trade-offs; and Reinforcement learning from human feedback reflects an influential line of modern learning that interacts with algorithmic and statistical theory.

A few highly viewed pages are better understood as adjacent or cross-disciplinary anchors rather than core theoretical computer science articles: Dirac delta function (analysis/mathematical physics), Commutative property (elementary algebra), or Pick’s theorem (discrete geometry). As it was explained in Section 14.3, Wikipedia’s hierarchical category system is not a strict ontological taxonomy with hard area–subarea boundaries; it is closer to a bibliographic organization in which topics are linked across neighboring fields when readers, editors, or sources commonly associate them. In this sense, it is unsurprising—and entirely reasonable—that a few non-pure TCS pages appear alongside canonical entries such as the Euclidean algorithm, binary search, tries, and the Gale–Shapley algorithm.

Table 15.2 lists the ten least relevant topics according to this metric. The bottom ten articles account for 1.2% of the total page views. Well-covered graph, Halin graph, Cop-win graph, and Book embedding speak to structural

and algorithmic themes in graph theory and graph drawing; Turán’s brick factory problem and Finite subdivision rule touch classic extremal and geometric constructions; Farthest-first traversal is a recognizable clustering/approximation heuristic; and the BIT predicate underlies descriptive complexity formalisms that, while niche, are conceptually central. By contrast, Laves graph and Tropical cyclone forecast model draw smaller, more domain-specific audiences yet still intersect algorithmic or structural concerns. The common thread is their narrow scope: these topics serve expert or subfield-focused needs rather than broad readership, which naturally depresses pageviews. Consequently, pageviews are a sensible proxy for broad relevance (what many people consult), that match or intuition of how many people could be affected by the topic.

Taken together, both lists are consistent with what we would expect people to consult when engaging with theoretical computer science ideas. Using Wikipedia pageviews as a proxy for relevance is therefore a sensible first-order approximation: it captures widespread interest in core topics, is easy to compute at scale, and yields stable comparative rankings over multi-week windows, while benefiting from standard safeguards (e.g., log-scaling, smoothing, and outlier checks) to mitigate transient spikes or teaching-cycle effects.

15.2 Applicability

Applicability measures how likely it is that a research topic can be applied to solve open problems. The underlying idea is that if a tool has already been applied to solve multiple problems, then there is a high probability that it can be used again to solve new ones. Formally, the number of problems to which a tool has been applied is computed using the applicability graph (see Definition ??), and the applicability of a topic is defined as its out-degree in this graph (see Definition ??).

In practice, we have approximated the applicability graph by using the network of internal links between the scientific pages of Wikipedia. Specifically, we approximate the applicability of a topic by counting the number of Wikipedia pages that link to its page (using the “*What links here*” facility, which lists the pages that link to—but do not redirect to—the current page).

Figure ?? shows a histogram of the applicability values for the selected set of topics. The histogram’s shape is dominated by an enormous spike in the very first bin and a sparsely populated, elongated right-hand tail. Roughly three-quarters of the theoretical computer science topics attract fewer than about 50 incoming Wikipedia links, while only a few reach the triple-digit range and just one or two exceed 300. This steep drop-off is the hallmark of a power-law (or at least heavy-tailed) distribution: applicability, as measured here, is concentrated in a tiny set of “super-connectors,” while the median topic sees only modest reuse. In

practical terms, the average link count is a misleadingly optimistic figure—pulled upward by a few giants—whereas the typical topic remains relatively niche.

Such inequality has several implications. First, it reinforces the idea that a small core of foundational concepts underpins a large share of problem-solving across the field; investing effort in those hubs yields the greatest leverage. Second, the near-emptiness of the mid-range suggests that rising topics face a kind of applicability “valley of death”: they must bridge a substantial gap before joining the elite group of widely referenced concepts. Finally, the long tail highlights opportunity—numerous specialized notions lie at low link counts, potentially poised for breakout if new cross-disciplinary applications emerge.

Table 15.3 lists the ten most applicable topics according to this metric. The backlink data confirm the long-tailed picture hinted at by the histogram: applicability, at least as proxied by Wikipedia’s “What links here” counts—is highly unequal. A single super-hub (the *Dirac delta function*, 625 links) dwarfs the rest, while only a handful of topics even cross the 200-link mark. This means that when researchers look for broadly reusable tools, the “return on attention” is greatest in a very small set of concepts that already function as connective tissue across many areas.

Equally striking is *who* those hubs are. Alongside classic algorithmic staples such as the *Euclidean algorithm*, *Trie*, and *Binary search*, we see foundational math and logic notions (e.g., *Rule of inference*, *Commutative property*), a fast-rising AI methodology (*Reinforcement learning from human feedback*), and even a domain-specific meteorology model. In other words, high applicability favors breadth rather than disciplinary purity: ideas that spill into multiple conversations—whether introductory, theoretical, or applied—accumulate the most links. For anyone mapping future research bets, this suggests monitoring backlink growth over time (to catch newcomers like RLHF early) and normalizing counts by sub-field size to filter out topics whose popularity is driven by narrow, self-referential clusters.

15.3 Maturity

As introduced in Definition ??, *maturity* is the opposite of *surfeit*. While *surfeit* measures the amount of redundant or superfluous information present in our current description of a topic, *maturity* quantifies how concise and efficient that description has become. In other words, a topic is considered mature when our understanding of it is sufficiently advanced that its essential content can be stated without unnecessary verbosity. High maturity thus indicates that knowledge about the topic has been largely consolidated, leaving little room for removing redundant parts from its description.

The maturity of a topic is estimated based on the length of its Wikipedia

Topic	Internal Links
Dirac delta function	625
Rule of inference	409
Reinforcement learning from human feedback	351
Euclidean algorithm	343
Trie	244
Telephone number (mathematics)	228
Commutative property	214
Tropical cyclone forecast model	195
Cartesian tree	155
Binary search	116

Table 15.3: Applicability of topics

article (only the textual content) and the length of a compressed version of that same text. The intuition behind this approach is that the more mature a topic is, the less redundant information its article will contain, and therefore the better it can be compressed. Topics that are still immature often accumulate loosely organized or repetitive content, which resists compression. By comparing the original and compressed lengths, we obtain a practical proxy for surfeit (as the ratio between the two lengths), and therefore for maturity as its inverse.

Table 15.4 lists the ten most mature topics according to this metric. For each topic, it shows its maturity value and its normalized version. Well-classified topics—those that our intuition tells us are well understood—include *Read-only right moving Turing machines*, *Crossing sequence (Turing machines)*, and perhaps the *P'' language*. Other topics that might be misclassified include *communication X-Machine*, *Power DEVS*, *MPIR*, and *constraint automaton*.

15.4 Interestingness

The *interestingness* of a topic, when considered as a tool, measures the likelihood that this tool can be applied to other problems. Table 15.5 shows the average applicability and average maturity of each of the selected areas, together with their average interestingness as a source of tools. The table largely fits our intuitive expectations about which areas are more important as sources of tools: computer science appears as the most interesting area, while sociology is the least. The only surprising case is epistemology, which emerges as a source of very interesting tools, even more so on average than mathematics. This anomaly may be explained by the high proportion of poorly written articles in this area, which artificially lowers maturity and thereby increases the calculated interestingness.

Table 15.6 shows the relevance and nescience of the selected areas, as well

Topic	Maturity	Norm.
Carry operator	5.34	1.00
Binade	4.54	0.99
Comm. X-Machine	3.01	0.97
PowerDEVS	2.35	0.94
MPIR	2.00	0.92
Constraint automaton	1.84	0.90
RO right moving TM	1.73	0.89
P''	1.71	0.89
Crossing sequence (TM)	1.63	0.88
Microsoft Binary Format	1.53	0.86

Table 15.4: Maturity of topics

Research Area	Applicability	Maturity	Tools
Sociology	1.00×10^{-3}	2.93×10^{-3}	3.09×10^{-3}
Biology	9.20×10^{-4}	4.65×10^{-3}	4.74×10^{-3}
Chemistry	3.11×10^{-3}	5.01×10^{-3}	5.90×10^{-3}
Psychology	1.14×10^{-3}	6.91×10^{-3}	7.00×10^{-3}
Mathematics	9.32×10^{-3}	9.47×10^{-3}	1.32×10^{-2}
Epistemology	1.55×10^{-3}	1.75×10^{-2}	1.76×10^{-2}
Computer_science	9.93×10^{-3}	1.90×10^{-2}	2.15×10^{-2}

Table 15.5: Interestingness of Areas as Tools

	Relevance	Nescience	Problems
Mathematics	4.22×10^{-2}	3.51×10^{-1}	3.53×10^{-1}
Computer_science	2.35×10^{-2}	4.43×10^{-1}	4.44×10^{-1}
Chemistry	5.95×10^{-2}	4.66×10^{-1}	4.70×10^{-1}
Biology	3.85×10^{-2}	4.75×10^{-1}	4.77×10^{-1}
Psychology	5.06×10^{-2}	5.28×10^{-1}	5.31×10^{-1}
Epistemology	4.54×10^{-2}	5.30×10^{-1}	5.32×10^{-1}
Sociology	4.21×10^{-2}	5.43×10^{-1}	5.44×10^{-1}

Table 15.6: Interestingness of Areas as Problems

Topic	Interestingness
GNU MPAL	0.49
Ternary numeral system	0.48
IEEE 854-1987	0.47
Arithmetic logic unit	0.43
Recursion	0.42
Barrel shifter	0.42
State space	0.42
Abstract machine	0.41
Computational model	0.39
Arithmetic overflow	0.39

Table 15.7: Interestingness of Tools

as their interestingness as sources of problems. Once again, the results align with our intuition: sociology is the area with the highest number of interesting problems, while mathematics is the area with the fewest.

Table 15.7 lists the ten most relevant topics according to their interestingness as tools. Out of these ten, only two (*ternary numeral system* and *recursion*) appear in the lists of the top ten most mature or most applicable topics; the rest are new. Among these we find clear examples of tools, such as the *GNU Multiple Precision Arithmetic Library* and the *standard for radix-independent floating-point arithmetic* (IEEE 854-1987). Other cases are less clear, such as *arithmetic logic unit*, *barrel shifter*, or *arithmetic overflow*. Topics that match our intuitive notion of a tool include *recursion*, *state space*, *abstract machine*, and *ternary numeral system*. Finally, some topics, like *computational model*, are too broad to be meaningful in the context of a single research question.

Table 15.8 lists the ten most interesting topics as problems. Examples that align with our intuitive notion of problems (concepts that are both highly relevant

Topic	Interestingness
Arithmetical hierarchy	0.72
Regular expression	0.68
Computability theory	0.65
Halting problem	0.65
Recursion (CS)	0.64
Lambda calculus	0.63
Floating point	0.61
Quantum computer	0.57
Computability	0.55
Computable function	0.55

Table 15.8: Interestingness of Problems

and poorly understood) include *arithmetical hierarchy*, *halting problem*, *floating point*, *quantum computer*, and *computable function*. Interestingly, the topic *recursion* appears in both lists: as a tool it refers to the general concept, while as a problem it refers to the specific implementation of recursion in computer science. The classification of *regular expression* as a problem, rather than as a tool, can be explained by the length and detail of its Wikipedia article, which provides an extensive reference-level description. This observation raises the broader methodological question of how to distinguish introductory from reference articles when analyzing Wikipedia data. Finally, some topics, such as *computability theory*, *lambda calculus*, and *computability*, are too broad to be treated as individual problems.

15.5 Interesting Research Questions

Before computing the new interesting questions, it is highly convenient to normalize the metrics of the topics involved in the study. Otherwise, a small number of topics with extreme values could dominate the results. For the normalization process we have used the Box-Cox method, which is based on the identification of the best transformation from a family of power transformations. This method is particularly useful when the data is skewed or exhibits heteroscedasticity, as it stabilizes variance and makes the distributions of the metrics more comparable across topics.

A major challenge in identifying topics as tools, that is, topics with very high maturity (or, equivalently, very low nescience), is to distinguish between cases where a description is short because the topic is genuinely well understood (for example, a mathematical theorem), and cases where the article is short simply because it is unfinished or poorly written. Our approach uses Wikipedia’s

Tool	Problem	Interestingness
Ternary numeral system	Regular expression	1.21
GNU MPAL	Arithmetical hierarchy	1.19
IEEE 854-1987	Arithmetical hierarchy	1.17
Quantum computer	Regular expression	1.17
Ternary numeral system	Arithmetical hierarchy	1.15
Division by zero	Regular expression	1.15
Turing machine	Regular expression	1.14
GNU MPAL	Regular expression	1.13
Ternary numeral system	Halting problem	1.13
Recursion	Halting_problem	1.13

Table 15.9: Interesting Intradisciplinary Questions

classification of articles as stubs, but this classification is far from reliable, since many articles that are essentially stubs are not labeled as such. Many of the misclassified topics in our dataset suffer from precisely this problem. Developing an automatic method to distinguish well-understood topics from poorly written descriptions remains an open research question.

By combining the results of Table 15.7 and Table 15.8, we can generate new ideas for how existing tools might be applied to open problems. As stated above, the goal of this approach is not to provide definitive research problems, but to highlight potentially interesting applications. It is ultimately the responsibility of the researcher to decide whether a given combination of topics is meaningful and worth pursuing. Table 15.9 shows the results of this intradisciplinary combination process.

Most of the interesting questions identified in this way have relatively low quality. As noted earlier, the main difficulty is that it is very hard to distinguish automatically, and in an unsupervised fashion, between a short article that reflects a well-understood topic and a short article that is simply poorly written. In this section we review some of the identified intradisciplinary questions in order to clarify what we mean by *interesting question* and how such questions should be interpreted. A few examples include:

- Interesting Question 7: *Can we apply Turing machines to regular expressions?* The answer is clearly yes, and this is a well-known result: regular expressions define regular languages, which are recognized by finite automata, and finite automata can be simulated by Turing machines.
- Interesting Question 10: *Can we apply recursion to the halting problem?* Again, the answer is yes. The proof of the undecidability of the halting problem is itself based on a machine that calls itself recursively.

Tool	Problem	Interestingness
State space	Action potential	1.17
Turing machine	Action potential	1.16
Quantum computer	Action potential	1.16
Abstract machine	Action potential	1.14
Computational model	Action potential	1.13
State space	Membrane potential	1.12
State space	Meiosis	1.11
Arithmetic logic unit	Meiosis	1.11
GNU MPAL	Flashbulb memory	1.11
Ternary numeral system	Working memory	1.10

Table 15.10: Interesting Interdisciplinary Questions

Both examples demonstrate that the methodology does not always produce original questions, but sometimes recovers already established results.

The most genuinely interesting questions arise when combining topics from two different disciplines. However, the probability that such automatically identified questions are meaningful is typically lower than in the intradisciplinary case, due to larger semantic gaps between fields.

For the interdisciplinary analysis we used the same collection of pages from the theory of computation as in the intradisciplinary study, combined with a new collection of topics from the area of bioinformatics. The topics were selected using the Wikipedia categories *natural sciences*, *biology*, and *biological processes*. In total, more than 10^5 combinations were analyzed. Table 15.10 lists the most relevant interdisciplinary applications.

The interdisciplinary set of questions also suffers from the stub-article problem, and thus the quality of the results remains limited. Nevertheless, some examples are worth discussing:

- Interesting Question 1: *Can we apply state space to action potential?* Questions 1-5 all point toward the same idea: whether the notion of action potential can be formalized in a way that allows it to be simulated or reproduced computationally.
- Interesting Question 7: *Can we apply state space to meiosis?* This is analogous to Question 1, but now concerning the possibility of formalizing and simulating the process of meiosis.

These examples illustrate how interdisciplinary combinations can hint at possible directions for formal modeling, even if many of the suggested questions are too vague or already well established to constitute truly novel research avenues.

15.6 New Research Topics

If we combine the list of highly relevant and not very well understood problems with themselves, we might obtain new topics that lie in the *unknown unknown* area.

Table 15.11 presents the top 25 candidates for potential new topics according to their interestingness. In this analysis we included all the topics from all knowledge areas. A striking result is that many of the suggested combinations involve the concept of intellectual property (*copyright*, *open access*, *public domain*, and perhaps *wiki*), which suggests that this is an area where there are still many issues to explore, far more than we might expect. On the other hand, it is possible that the outcome reflects a bias in Wikipedia toward these topics. Further investigation is needed to clarify this point.

To illustrate how new topics are generated, we consider the following examples:

- New topic 17: *Public domain* + *Earth*. This question raises the issue of whether the Earth should be considered a public resource, touching upon the fundamental concept of private property. The methodology suggests that this is not a well-understood topic.
- New topic 18: *Public domain* + *Internet*. This raises a similar issue to Question 17, but in the context of Internet governance.

In both cases, however, the combinations fail to provide well-defined, innovative, and genuinely novel research topics.

We can also restrict our search for new topics to a reduced number of knowledge categories. For example, Table 15.12 shows the ten most interesting new topics arising from the already studied area of *theory of computation* and the new area of *phenomenology* (from Level 2 *philosophy of mind*, and Level 1 *cognitive science*). Based on this list, we might consider the following potential new topics:

- New topic 2: *Turing machine* + *synesthesia*. This could suggest a new type of Turing machine that incorporates synesthetic properties. These *synesthetic Turing machines* might be defined as a group of Turing machines linked together so that when one machine reads a symbol from its tape, it automatically triggers a state change in another machine. This property of synesthesia could also be extended to nondeterministic Turing machines.
- New topic 4: *Kolmogorov complexity* + *self-awareness*. This could be interpreted as an investigation into the minimum complexity required for a computer program to exhibit self-awareness.

Problem	Problem	Interestingness
Public domain	Open access	1.71
Public domain	REST	1.70
Public domain	Wiki	1.70
Open access	REST	1.70
Copyright	Public domain	1.69
Open access	Wiki	1.69
Public domain	QR code	1.69
Copyright	Open access	1.68
Wiki	REST	1.68
Open access	QR code	1.68
Public domain	Transport Layer Security	1.68
Copyright	REST	1.68
QR code	REST	1.67
Open access	Transport Layer Security	1.67
Copyright	Wiki	1.67
Wiki	QR code	1.67
Public domain	Earth	1.67
Public domain	Internet	1.67
REST	Transport Layer Security	1.66
Copyright	QR code	1.66
Earth	Open access	1.66
Internet	Open access	1.66
Public domain	Open source	1.66
Public domain	Web 2.0	1.66
Wiki	Transport Layer Security	1.66

Table 15.11: New Topics

Question	Question	Interestingness
Kolmogorov complexity	Change blindness	1.24
Turing machine	Synesthesia	1.23
Kolmogorov complexity	Qualia	1.23
Kolmogorov complexity	Self-awareness	1.22
Turing machine	Qualia	1.22
Kolmogorov complexity	Synesthesia	1.21
Turing completeness	Synesthesia	1.20
Turing machine	Self-awareness	1.20
Turing completeness	Qualia	1.20
Turing completeness	Self-awareness	1.18

Table 15.12: Restricted New Topics

15.7 The Unknown Unknowns Project

The Unknown Unknowns Project is rooted in the theory of nescience, and in particular in its metrics for quantifying how well we understand a scientific topic. It builds upon the idea that by combining topics that are not yet well understood, we can generate new and previously unknown areas of knowledge. The project leverages the rich collection of scientific descriptions contained in Wikipedia to map the current extent of human understanding and to detect where it begins to fail. Its ultimate goal is to discover what lies hidden in the scientific unknown unknown, the research topics that humanity not only does not understand but is not yet aware it does not understand. In essence, the project aims to bring to the present the research topics of the future.

Wikipedia Crawling

We treat Wikipedia as a vast, labeled graph in which vertices represent pages and categories, and edges connect categories to their subcategories and contained pages. Starting from a small set of root scientific categories, we perform a breadth-first traversal to identify relevant pages. For each discovered article, we (i) retrieve structured metadata in batches via the MediaWiki API, (ii) apply structural and quality filters to remove out-of-scope entries (e.g., redirects, disambiguation pages, and non-scientific entities identified through Wikidata typing), and (iii) compute per-page metrics, including pageviews (relevance) and compression-based maturity or surfeit derived from article text. Finally, we normalize these metrics and export both a tabular dataset and a hierarchical JSON representation of the category-page structure.

Our pipeline relies on three Wikimedia services: the MediaWiki Action API (English Wikipedia) for category traversal, page metadata, content, and

assessments; the Wikidata Query Service (WDQS) to verify whether a page's topic should be excluded by checking instance of (P31) and transitive subclass of (P279*) relationships; and the Wikimedia REST Pageviews API to aggregate daily user pageviews over a specified time window.

Step 1.- Category traversal. We begin with a list of root categories ("Applied sciences," "Formal sciences," "Natural sciences," and "Social sciences") and discover associated scientific pages via a breadth-first search (BFS) up to a user-defined maximum depth (by default 5). In Wikipedia, a category is itself a page that organizes subcategories and pages. Categories reside in namespace 14, while encyclopedic articles (our targets) belong to namespace 0 or mainspace. Using the MediaWiki Action API's `list=categorymembers`, which returns both subcategories and pages directly contained in a category, we iterate level by level. For each category, we call `action=query`, following pagination through the returned `cmcontinue` and `continue` tokens, which are pagination tokens provided by the MediaWiki API and must be included in subsequent requests to retrieve additional results until no further tokens are returned. Since the category graph is a directed acyclic graph (DAG) rather than a strict hierarchy (categories may have multiple parents), we explicitly record edges rather than assuming a unique tree. At each iteration, discovered subcategories are added to the BFS frontier, and all discovered pages are added to the dataset. Visited sets for categories and pages prevent duplicates and cycles. We record the structure as category edges (parent category id, child category id) for the category DAG and page to categories (page id, category id) for direct page-category relationships.

Step 2.- Batch enrichment of page properties. For the distinct page ids identified in Step 1, we obtain metadata from the MediaWiki Action API in batches of at most 50 ids per request using `action=query`. The `info` module provides metadata (page ids, titles), redirect status, and content model; `pageprops` includes structured flags such as disambiguation and `set_index_article`, along with the Wikidata identifier (`wikibase_item`); `revisions` retrieves the latest wikitext, byte size, and timestamp; `categories` lists all page categories (for completeness); and `pageassessments` returns community quality ratings (e.g., Stub, Start, C, B, GA, FA) assigned by WikiProjects.

Step 3.- Filtering. We retain only pages passing a series of structured checks, avoiding heuristic title filters (e.g., avoiding regular-expression checks on page titles). First, we apply structural filters to ensure valid encyclopedic content: the page must be in mainspace, not a redirect, not a disambiguation page, not a set-index page, and must use the wikitext content model. Next, we apply quality filters to exclude low-quality or list-style pages: we prioritize the community PageAssessments label and discard Stubs (including List/FL and Start quality classes). Finally, we remove non-scientific topics via Wikidata typing:

each page's Wikidata entity (its Q-ID in `pageprops.wikibase_item`) is tested through WDQS to determine whether a path—following "instance of" (P31) and recursively "subclass of" (P279) relationships in Wikidata, tracing upward through the class hierarchy—P31 / (P279)* leads to any of several high-level deny anchors (e.g., human Q5, book Q571, film Q11424, city Q515, Wikimedia list article Q13406463). Here, P31 means "instance of," P279 means "subclass of," and (P279)* represents zero or more subclass steps (the transitive closure). If such a path exists, the page is excluded. This three-part filter removes people, places, creative works, organizations, products, and other non-topic entities, yielding a clean collection of scientific articles.

Step 4.- Metric collection. For each remaining page, we compute a consistent set of features. We first measure the article's length from its wikitext after removing low-signal sections—headings such as "See also", "References", "Further reading", and "External links", using a `wikitext` parser. From this cleaned text we record `article_length` in UTF-8 bytes (and optionally raw characters). We compute compression-based measures by compressing the text with `zlib` (level 9) and compute Redundancy as $1 - (\text{article_comp_length} / \text{article_length})$. For relevance, we sum daily user pageviews over a defined window (e.g., 30 or 60 days) via the REST endpoint `all-access`, using canonical titles resolved during enrichment.

Step 5.- Derived metrics and interestingness. From these primary measurements we define Relevance as total pageviews and Maturity from compression. We then normalize the dataset: `NormMaturity` using min-max scaling, and `NormRelevance` using min-max scaling on $\log(x+1)$ to reduce heavy-tail effects. A composite indicators is then computed: Interestingness (Problem) = `NormRelevance` x `NormRedundancy`, identifying topics that are popular yet textually redundant (potentially under-explained and research-worthy).

Step 6.- Outputs. For reproducibility and visualization, we export canonical tables: `categories` (`cat_id`, `title`), `category_edges` capturing the category DAG (`parent_cat_id`, `child_cat_id`), `page_categories` (`page_id`, `cat_id`) for direct memberships, and `pages_with_metrics` containing per-page metrics and labels. We also produce a hierarchical JSON projection of the category graph by identifying root categories (those without incoming edges), recursively nesting subcategories, and inserting page nodes beneath each category they belong to. Because categories may have multiple parents, a category can legitimately appear in multiple locations within this JSON structure, a deliberate duplication preserving the DAG while supporting tree-like visualizations.

To ensure stability, all enrichment requests are batched (< 50 IDs per call). We log pagination tokens, HTTP errors, and retry decisions for auditing. Concurrency is avoided to respect Wikimedia's usage guidelines and to maintain deterministic runs. As pageviews depend on both the chosen window and the execution date,

we record the UTC end date of the REST query; since article content, assessments, and backlinks change over time, we also store lastrevid and timestamps. The pipeline gracefully degrades for optional services: if WDQS is unavailable or rate-limited, processing continues with missing labels or deny checks defaulting to "keep," and gaps noted in the logs. The software stack uses Python with requests, pandas, mwparserfromhell, and optionally pyarrow for Parquet output.

The Known and the Unknown

The crawl covered the scientific content of the English Wikipedia, starting from the four principal branches of knowledge—Applied sciences, Formal sciences, Natural sciences, and Social sciences—and expanding through their subcategories to a maximum depth of five levels. In total, 38,074 categories were explored, of which approximately 18,000 did not contain articles of sufficient quality to be included in the analysis. The resulting dataset comprises 18,149 high-quality scientific topics, distributed across the major branches as follows: 10,785 articles in the Applied sciences, 2,776 in the Formal sciences, 9,826 in the Natural sciences, and 11,794 in the Social sciences. Each article was classified according to Wikipedia's internal quality assessment scale, which ranks entries based on their completeness, accuracy, and sourcing. *C-class* articles are moderately developed, providing substantial information but often lacking depth or comprehensive references. *B-class* articles are more complete, with reasonably good structure, accuracy, and coverage of the topic, though they may still require some polishing or additional citations. *A-class* articles are near-publication quality, representing a high standard of completeness and reliability, just below the *Good Article* and *Featured Article* levels. Within this classification, the crawl collected 11,894 C-class, 4,928 B-class, 882 Good Articles (GA), 429 Featured Articles (FA), and 16 A-class entries. Given that Wikipedia functions as the largest collaborative encyclopedia ever created, this corpus constitutes one of the most comprehensive and accurate descriptive representations of current human knowledge, offering a unique empirical foundation for analyzing the state of science and its frontiers.

In Figure 15.2 it is shown an histogram of the maturity of the known topics, computed as the ratio of the size of the text of the Wikipedia article and its compressed version. The histogram reveals that the maturity of Wikipedia's scientific topics follows an approximately normal distribution centered around 0.4, indicating that most articles exhibit a balanced level of informational density. This suggests that, on average, about 40% of each article's content represents essential, non-redundant information, while the rest reflects natural linguistic and structural redundancy typical of encyclopedic writing.

In Figure 15.1 it is shown an histogram of the logarithm of the relevance

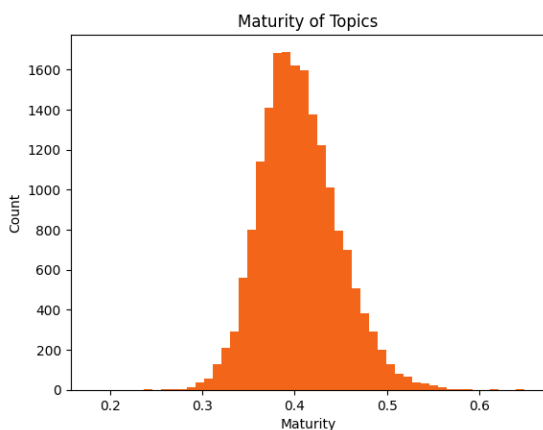


Figure 15.2: Maturity of scientific topics.

of a topic, as the total number of visits to the Wikipedia page over the last two months. The histogram shows that the logarithm of topic relevance, measured by the total number of visits to each Wikipedia page over the last two months, follows a roughly log-normal distribution centered around a log relevance of about 8–9. This implies that most scientific topics receive between 10^8 and 10^9 visits, while only a few attract extremely high or low levels of attention. The right-skewed shape indicates the presence of a small number of highly popular topics that dominate user interest, reflecting the uneven distribution of public attention across scientific fields. Overall, the data suggest that while most topics maintain a moderate and steady level of visibility, a minority of them capture a disproportionately large share of the audience, highlighting the concentration of public interest in a limited set of scientific areas.

Computation of the new research topics, as the creative combination of already existing topics, has been performed automatically using a large language model. The OpenAI API provides programmatic access to a family of advanced larger language models developed by OpenAI, allowing developers and researchers to integrate natural language understanding, reasoning, and generation capabilities into their own applications. Among these models, *GPT-5-nano* represents a lightweight yet remarkably capable variant of the GPT-5 architecture—optimized for efficiency, speed, and cost while retaining much of the creativity and fluency of larger models. Despite its compact size, GPT-5-nano demonstrates an impressive ability to generate coherent, context-aware, and often original ideas across diverse domains. Its design enables it not only to respond accurately to prompts but also to combine existing concepts in novel ways, making it a powerful tool for creative exploration, brainstorming, and the

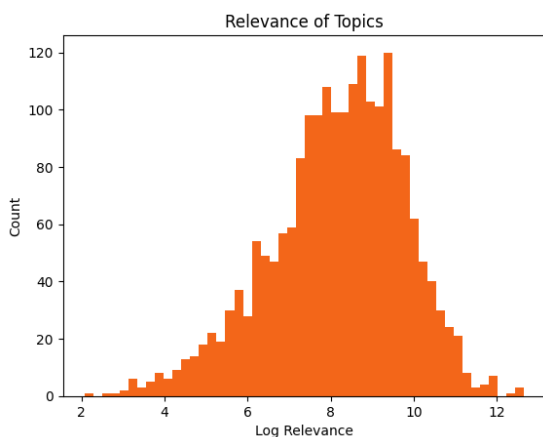


Figure 15.3: Log-relevance of scientific topics.

discovery of new research directions or innovative solutions. The following is the prompt used to generate a new research topic:

"You are an innovative scientist exploring unexplored frontiers of knowledge. Your task is to conceive a new and original research topic that emerges from the creative synthesis or inspiration of the two given scientific concepts. The proposed topic must:

- Represent a novel idea that, to your knowledge, has never been formally studied or proposed before.
- Have the potential, if successfully developed and applied, to produce a major positive impact on humanity by advancing science, technology, or understanding.
- Be scientifically plausible—grounded in real principles, not fantasy or pseudoscience.
- Be expressed in a single, concise paragraph written in formal academic language, rich in scientific and technical terminology, and free of filler or speculation markers (avoid “perhaps”, “maybe”, etc.).
- Avoid generic formulations like ‘a study of the relationship between...’ and instead describe a specific, well-defined new direction.

Respond directly with the proposed research topic, nothing else."

The exploration of the unknown unknown encompassed an unprecedented combinatorial space of potential scientific discoveries. In total, 164 million new research topics were generated and evaluated, each representing a possible novel research area. Within single disciplines, or intradisciplinary combinations, the analysis produced approximately 12 million new topics in the Applied sciences, 8

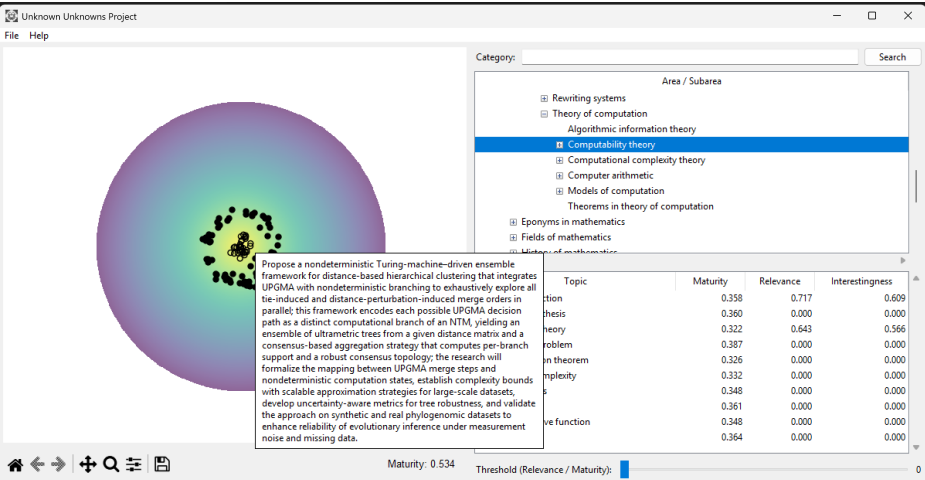


Figure 15.4: Unknown Unknowns App.

million in the Formal sciences, 96 million in the Natural sciences, and 139 million in the Social sciences. When considering interdisciplinary combinations—those that bridge distinct domains—the numbers were even higher, with 132 million in the Applied sciences, 45 million in the Formal sciences, 125 million in the Natural sciences, and 138 million in the Social sciences. Although these figures are immense, they reflect not ambition for its own sake, but the scale of the challenge involved in systematically exploring what remains beyond our current understanding. This constitutes, to date, one of the most extensive and systematic attempts to identify and describe the unknown unknown in science—an experiment that seeks not to claim discovery, but to illuminate the vast and still largely uncharted territory of human ignorance.

Unknown Unknowns App

The Unknown Unknowns application (see Figure 15.4) serves as an exploratory instrument for discovering what it is hidden in the scientific unknown. It transforms our current vast scientific corpus into an interactive landscape where both what is known and what remains undiscovered can be visually explored. At its core lies the idea that new knowledge emerges not in isolation, but from the interplay of existing concepts. By combining less mature, not yet well understood topics with automatically generated hypotheses about potential new ones, the application becomes a map of the scientific frontier, a cartographic representation of the boundary between understanding and ignorance.

When launched, the program constructs an internal graph that mirrors the structure of scientific knowledge. Categories and subcategories form a tree that organizes the disciplines, while individual articles represent specific research

topics. Each page is associated with several indicators: maturity, which measures how developed or well-established a topic is; relevance, which reflects its significance and social impact; and interestingness, which captures its potential to raise new questions. A complementary dataset describes the “unknown topics”—novel research ideas automatically inferred from pairs of existing ones. These unknown topics, described textually and assigned a preliminary maturity level, represent possible directions in which human knowledge could expand.

The interface is divided into two complementary regions. On the left, a circular diagram depicts the maturity of topics as a constellation of points over a concentric gradient. The center of the diagram glows brightly, symbolizing the domains of science that are most mature and consolidated. The darker outer regions represent areas of uncertainty and opportunity (the unknown unknowns), where new ideas are still forming. Known topics appear as filled black dots, positioned according to their maturity: the more developed the topic, the closer it lies to the center. Unknown topics appear as empty circles, marking the places where new combinations of knowledge could give rise to discoveries. Their distribution reveals the topology of ignorance, an intricate periphery of potential that surrounds and extends the boundaries of current science. Hovering over any unknown displays the research opportunity in a multi-line tooltip, while the lower status bar shows its numerical maturity.

The right side of the window offers the tools to navigate this landscape. A search bar allows the user to locate specific scientific categories by name, while the hierarchical tree below displays the organization of disciplines and subdisciplines. Expanding a branch of the tree simply reveals its subareas, but selecting a category triggers a full update of the diagram and the topics table. In this mode, the application displays all the known topics that belong to the chosen category and its descendants, together with the unknown topics that emerge from them. The unknown topics are filtered so that only those whose maturity exceeds a minimum threshold are shown, ensuring that the visualization remains focused on meaningful possibilities rather than random combinations.

Beneath the tree, the topics table lists all the pages that directly belong to the selected category. Each entry includes its maturity, relevance, and interestingness. Clicking on a row switches the visualization into topic mode, in which the diagram focuses on that particular topic and displays all the unknown topics that are conceptually derived from it. This view allows the user to see how a not very well-understood idea can serve as a seed for new research directions, how from a single concept, many potential discoveries can emerge. Double-clicking a topic opens its Wikipedia page in a browser, connecting the abstract representation back to the underlying content.

The scale at the bottom of the interface controls the threshold of relevance and maturity used to filter the visualization. It allows the user to move smoothly from

a wide, inclusive view of the scientific landscape to a more focused view that highlights only those topics that are socially impactful. By raising the threshold, the user filters out marginal or less relevant areas, concentrating instead on the regions of knowledge that have the greatest influence on human life and technological progress. In this way, the application not only reveals the unknown unknowns but also helps prioritize which of them might lead to discoveries of real significance.

Every design element in the application serves the same epistemological goal: to make the frontier of human knowledge visible. The bright center of the diagram stands for what science already understands, while the dark periphery shows where ignorance still dominates. The empty circles of the unknown topics are markers of potential discovery, each one a hypothetical domain of future research, a conceptual territory awaiting exploration. The relevance filter allows the user to discern which of these emerging ideas connect most strongly to society's needs and interests. The result is not merely a visualization of data, but a dynamic, exploratory tool that guides the intellect toward the edges of what is currently thinkable.

In this sense, the application is both cartographic and philosophical. It functions as a visual compass for discovery, helping the observer navigate between the solid continents of established knowledge and the open ocean of possibility. By grounding the abstract notion of the "unknown unknown" in concrete, measurable data, it transforms ignorance into something that can be mapped, explored, and eventually reduced. In doing so, it exemplifies the central idea of the theory of nescience: that by quantifying what we do not yet know, we can systematically move beyond the limits of present understanding.

References

Core references for the chapter "Interesting Research Questions"

[Pop14]: Popper's account of conjectures, refutations, and the growth of knowledge provides the philosophical foundation for thinking about unanswered questions and how science progresses through them.

[Cha13]: Chalmers offers a clear overview of debates about what constitutes science. This is essential background for addressing the demarcation problem in identifying valid research questions.

[Li+04]: Introduces the normalized compression distance (NCD), which provides a universal method for comparing objects. This is a key methodological tool for estimating redundancy and discovering connections between topics.

[CAO+05]: Warns about practical limitations of using real compressors to approximate Kolmogorov complexity. Essential for a sound application of compression-based metrics in identifying research frontiers.

[LV13]: The standard reference on Kolmogorov complexity. It connects theoretical information content with practical methods like NCD, underpinning the discovery of redundancy, miscoding, and research gaps.

[Sup02]: Suppes discusses how scientific theories can be formally represented. This supports the extension from individual topics to structured research areas and the search for interdisciplinary questions.

[LSW13]: Latour highlights how scientific knowledge emerges from negotiation and practice. This adds a sociological perspective to the identification of research questions and debates.

[Ioa05]: Ioannidis shows that much of published research is flawed or irreproducible. This underscores the importance of using independent measures (like nescience) to identify which questions are still genuinely open.



16. Discrete Mathematics

*Mathematics may be defined as the subject in which
we never know what we are talking about,
nor whether what we are saying is true.*

Bertrand Russell

The majority of mathematical concepts used throughout this book belong to the domain of *discrete mathematics*. This field focuses on mathematical objects that take on distinct, separate values, as opposed to continuous ones. In this book, we make use of various discrete structures, including integers, strings, graphs, and computer programs. A key feature of discrete sets is their countability, meaning they can be put into one-to-one correspondence with the natural numbers. By contrast, continuous mathematics, such as calculus, will play only a minimal role in the theoretical development of the theory of nescience.

Our primary interest in discrete mathematics arises from its direct relevance to computation. The theory of nescience draws upon various aspects of computer science, including algorithms, coding, and string complexity. Computers operate in discrete steps and manipulate information stored in discrete memory units. Our interest in computers stems from the aspiration to apply our theoretical framework to a broad range of real-world entities. We consider computers to be the most suitable tools for modeling the world around us. While pure mathematics often

engages with abstract objects independent of their representations, the theory of nescience places significant emphasis on the representation (or encoding) of objects.

This chapter provides a brief overview of the fundamental concepts of discrete mathematics, introducing topics such as sets, strings and languages, counting methods, matrices, and graphs. While we do not present formal definitions or prove theorems in this overview, these subjects lay the groundwork for the theories and ideas explored in later sections. Discrete mathematics is a broad and diverse field; here, we focus only on those elements essential for understanding the theory of nescience. More advanced topics (such as computability, information theory, and complexity) require a deeper treatment and are addressed in dedicated chapters.

The References section provides a list of recommended books that explore the topics introduced in this chapter in greater depth. Readers interested in further developing their understanding can use this list to delve more deeply into each subject, thereby complementing the foundational overview presented here.

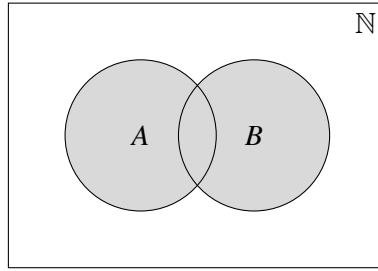
16.1 Sets, Relations and Functions

The sets of *natural*, *integer*, *rational*, and *real* numbers are denoted by $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, respectively. Each of these sets includes the number 0. The *positive integers* are represented by $\mathbb{Z}^+$, and the *positive reals* by $\mathbb{R}^+$; both sets also include 0. Let A be a *set*. We indicate that x is an *element* of A using the notation $x \in A$, and that x is not an element of A with $x \notin A$. Elements of a set can be listed explicitly using braces, as in $A = \{0, 1, 2, 3\}$, or defined by a condition using *set-builder* notation, for example, $A = \{x \in \mathbb{N} : x < 4\}$, provided that the *universe* of discourse is clearly specified.

Suppose A and B are two sets. We use the notation $A = B$ to indicate that the sets are *equal*. The expression $A \subseteq B$ signifies that A is a *subset or equal* to B , while $A \subset B$ denotes that A is a *proper subset* of B (meaning that A is contained in B but is not equal to B). The condition $A = B$ holds if and only if both $A \subseteq B$ and $B \subseteq A$ are true. The symbol $\emptyset$ denotes the *empty set*, which is the set containing no elements.

■ **Example 16.1** For every set A , we have that $\emptyset \subseteq A$ and $A \subseteq A$. ■

The term *cardinality* refers to the number of elements in a finite set A , denoted by $d(A)$. Accordingly, the cardinality of the empty set $\emptyset$ is 0, since it contains no elements. For any two sets A and B , the notation $A \cup B$ denotes the *union* of A and B , while $A \cap B$ represents their *intersection*. For a collection of n sets $A_1, A_2, \dots, A_n$, we write their union and intersection as $\cup_{i=1}^n A_i$ and $\cap_{i=1}^n A_i$, respectively. For an arbitrary collection of sets indexed by a set I , we use the

Figure 16.1: Representation of $A \cup B$ as a Venn Diagram

notations $\cup_{i \in I} A_i$ and $\cap_{i \in I} A_i$. When dealing with an infinite collection of sets, we may use the shorthand $\cup_{i=1}^{\infty} A_i$ and $\cap_{i=1}^{\infty} A_i$. Occasionally, we will make use of *Venn diagrams* to visually represent set operations, as illustrated in Figure 16.1.

Given any two sets A and B , the *set difference* is denoted by $A \setminus B$, and the *complement* of the set A is written as A^c . *De Morgan's laws* state that for any sets A and B , we have $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Two sets A and B are said to be *disjoint* if their intersection is empty, i.e., $A \cap B = \emptyset$. A collection of sets $A_1, A_2, \dots, A_n$ is disjoint if $A_i \cap A_j = \emptyset$ for all $i \neq j$. A *partition* of a set A is a collection of nonempty, pairwise disjoint subsets $A_1, A_2, \dots, A_n$ such that $A = \cup_{i=1}^n A_i$. The *power set* of A , denoted by $\mathcal{P}(A)$, is the set of all possible subsets of A . If the cardinality of A is n , i.e., $d(A) = n$, then the cardinality of its power set is 2^n , so $d(\mathcal{P}(A)) = 2^n$.

■ **Example 16.2** Given the set $A = \{1, 2, 3\}$, its power set is:

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$$

■

Consider a non-empty set A and a collection $\mathcal{F}$ of subsets of A . The pair $(A, \mathcal{F})$ is called a *field* over A if it satisfies the following conditions: it contains the empty set, that is, $\emptyset \in \mathcal{F}$; it is closed under complementation, meaning that for every $F \in \mathcal{F}$, the complement F^c also belongs to $\mathcal{F}$; and it is closed under finite unions, meaning that for any subsets $F_1, \dots, F_n \in \mathcal{F}$, the union $F_1 \cup \dots \cup F_n$ is also in $\mathcal{F}$. Additionally, it can be shown that a field also satisfies two further properties: the universal set A itself belongs to $\mathcal{F}$, and it is closed under finite intersections, so that $F_1 \cap \dots \cap F_n \in \mathcal{F}$ for all subsets $F_1, \dots, F_n \in \mathcal{F}$.

Consider two elements, x and y . An *ordered pair*, denoted as (x, y) , is a pairing in which the order of the elements matters. Generalizing this idea, an *n -tuple* is an ordered sequence of n elements, written as $(x_1, \dots, x_n)$. The *Cartesian product* of two sets A and B , denoted by $A \times B$, is the set of all ordered pairs (x, y) such that $x \in A$ and $y \in B$. This concept extends naturally to n sets

$A_1, A_2, \dots, A_n$, where the Cartesian product is expressed as $A_1 \times A_2 \times \dots \times A_n$. Additionally, the n -fold Cartesian product of a set A with itself is denoted as A^n .

Let R be a subset of the Cartesian product of a set A with itself, i.e., $R \subseteq A \times A$. Such a subset is called a *binary relation*. We write aRb to indicate that the ordered pair (a, b) belongs to R . A binary relation is said to be *reflexive* if, for every element $a \in A$, it holds that aRa . It is called *symmetric* if, for all $a, b \in A$, the condition aRb implies bRa . A relation is *antisymmetric* if, for all $a, b \in A$, the coexistence of aRb and bRa implies $a = b$. It is *transitive* if, for all $a, b, c \in A$, the conditions aRb and bRc together imply aRc . A relation is *total* if, for every pair $a, b \in A$, either aRb or bRa holds. Binary relations can also be defined between two different sets A and B , in which case R is a subset of $A \times B$. Furthermore, the concept can be generalized to n -ary relations, represented as $R \subseteq A_1 \times A_2 \times \dots \times A_n$.

Let R be a binary relation that is a subset of the Cartesian product $A \times A$, i.e., $R \subseteq A \times A$. If this relation is reflexive, symmetric, and transitive, it is called an *equivalence relation*, typically denoted by the symbol $\sim$. Under an equivalence relation, two elements $a, b \in A$ are said to be *equivalent* if $a \sim b$. The *equivalence class* of an element a , denoted by $[a]$, is the set of all elements in A that are equivalent to a . That is, the equivalence class of a is defined as $[a] := \{b \in A : a \sim b\}$. An equivalence relation partitions the set A into disjoint subsets known as *equivalence classes*, collectively forming what is called the *quotient set*. The quotient set is denoted by $A/\sim$ and is defined as the set of all equivalence classes: $A/\sim := [a] : a \in A$.

A binary relation that is reflexive, transitive, and antisymmetric is known as a *partial order*, typically denoted by the symbol $\preceq$. A set equipped with a partial order is called a *partially ordered set*, or *poset* for short. In a poset, an element $a \in A$ is considered *minimal* if there is no element $b \in A$ such that $b \preceq a$ and $b \neq a$. Similarly, an element a is said to be *maximal* if there is no element $b \in A$ such that $a \preceq b$ and $b \neq a$. A relation that is reflexive, transitive, antisymmetric, and total is called a *total order*, often denoted by the symbol $\leq$. A set endowed with a total order is referred to as a *totally ordered set*. In such a set A , the *maximum element*, denoted by $\max(A)$, satisfies $\max(A) \geq x$ for all $x \in A$, and the *minimum element*, denoted by $\min(A)$, satisfies $\min(A) \leq x$ for all $x \in A$.

■ **Example 16.3** Let R be a relation that is a subset of the Cartesian product of the set of natural numbers $\mathbb{N}$ with itself, i.e., $R \subset \mathbb{N} \times \mathbb{N}$. In this relation, an ordered pair (a, b) belongs to R if and only if a is a divisor of b . The set $\mathbb{N}$, together with the relation R , forms a partially ordered set. In this context, 1 is the unique minimal element, and every prime (e.g. 11) is maximal. ■

A *function* is defined as a binary relation $f \subseteq A \times B$ such that for each element $x \in A$, there exists at most one $y \in B$ for which $(x, y) \in f$. In this

context, elements $(x, y) \in f$ are written as $f(x) = y$, and the function is denoted by $f : A \rightarrow B$. The set A is called the *domain* of f , and B is the *codomain*. The set $y \in B : \exists x \in A$ such that $f(x) = y$ is known as the *range* of f . If the relation is not defined for every $x \in A$, the function is called a *partial function*, and we write $f(x) \uparrow$ to indicate that f is undefined at x .

A function is said to be *injective* if, for all elements x and y , the condition $f(x) = f(y)$ implies that $x = y$. A function is *surjective* if, for every y in the codomain, there exists at least one x in the domain such that $f(x) = y$. A function is described as *bijective* if it is both injective and surjective. The *identity* function $I_A : A \rightarrow A$, defined by $I_A(a) = a$ for all $a \in A$, is an example of a bijective function. These concepts (function, partial function, injective, surjective, and bijective) can be extended to n -ary functions, which are functions of the form $f : A_1 \times A_2 \times \cdots \times A_n \rightarrow B$.

The *inverse* of a bijective function f , is defined as $f(f^{-1}(x)) = f^{-1}(f(x)) = x$, for all x in the domain of f^{-1} and the range of f , respectively. Given two functions f and g , where the domain of f includes the range of g , the *composition* of f with g , denoted by $f \circ g$, is defined as $(f \circ g)(x) = f(g(x))$.

■ **Example 16.4** In Section 18.4, we will explore an alternative interpretation of a function as a procedure or algorithm that assigns an element of B to each element of A . For example, the following C code defines a partial function from $\mathbb{R}$ to $\mathbb{R}$, which is partial because $\text{inv}(0) \uparrow$:

```
double inv(double x) {
    return 1 / x;
}
```

■

The *characteristic function* of a set A is denoted by $1_A : A \rightarrow \{1, 0\}$, where $1_A(x) = 1$ if $x \in A$ and $1_A(x) = 0$ otherwise.

An infinite set A is said to be *countable* if there exists a bijective function that maps the elements of A onto the set of natural numbers $\mathbb{N}$. In contrast, a set is considered *uncountable* if it is neither finite nor countable. A set is said to have *countably many* elements if it is either finite or countably infinite.

■ **Example 16.5** The sets $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are countable, whereas $\mathbb{R}$ is uncountable.

Considering a real number $x \in \mathbb{R}$, its *absolute value*, denoted by $|x|$, is defined as x if $x \geq 0$ and $-x$ if $x < 0$. The *ceiling* function of x , written as $\lceil x \rceil$, is the smallest integer greater than or equal to x . The *floor* function of x , denoted by $\lfloor x \rfloor$, is the largest integer less than or equal to x . Given two positive integers a and b , the *modulo* operation, written as $a \bmod b$, yields the remainder when a is divided by b .

For two functions f and g defined as $f, g : \mathbb{N} \rightarrow \mathbb{R}^+$, we say that $f(n)$ is of the *order of* $g(n)$, denoted $f(n) = O(g(n))$, if there exist positive constants $c > 0$ and m such that $f(n) \leq cg(n)$ for all integers $n \geq m$. In this context, g is called an *upper bound* for f .

16.2 Strings and Languages

Consider a non-empty finite set $\mathcal{S} = \{s_1, s_2, \dots, s_q\}$, referred to as the *alphabet*. The elements of this set are called *symbols*. A *sequence* over $\mathcal{S}$ is defined as an ordered arrangement of symbols $x_1x_2 \dots x_n$, where each x_i belongs to $\mathcal{S}$. In the special case where the alphabet is $\mathcal{B} = \{0, 1\}$, such sequences are known as *binary sequences*. We use the term *string* to denote a finite sequence. This book primarily focuses on binary strings.

The *length* of a string s , denoted by $l(s)$, refers to the total number of symbols contained in s . The symbol λ is used to denote the *empty string*, which is defined as the unique string over $\mathcal{S}$ with length 0. Given a symbol $x \in \mathcal{S}$, the string consisting of x repeated n times is denoted by x^n . If $s = x_1x_2 \dots x_n$ is a string, its *reverse*, denoted by s^R , is given by $x_nx_{n-1} \dots x_1$.

The set of all strings $s_1s_2 \dots s_n$ of length n over the alphabet $\mathcal{S}$ is denoted by $\mathcal{S}^n$ ¹. We denote by $\mathcal{S}^+$ the union of all $\mathcal{S}^n$ for $n \geq 1$, and by $\mathcal{S}^*$ the set $\mathcal{S}^+ \cup \lambda$. Note that all strings in $\mathcal{S}^*$ have finite length. The term *Kleene closure* refers to $\mathcal{S}^*$.

■ **Example 16.6** The following relations hold: the cardinality of the set of binary strings of length n is $d(\{s \in \mathcal{B}^* : l(s) = n\}) = 2^n$, and the cardinality of the set of binary strings of length up to n is $d(\{s \in \mathcal{B}^* : l(s) \leq n\}) = 2^{n+1} - 1$. ■

Given two strings s and t from $\mathcal{S}^*$, the *concatenation* of s and t , denoted as st , is the sequence obtained by placing the sequence of symbols in t immediately after the sequence in s . Consequently, the length of the concatenated string, $l(st)$, is the sum of the lengths of s and t . This indicates that $\mathcal{S}^*$ is closed under the operation of concatenation. Moreover, the set $\mathcal{S}^*$, together with concatenation, forms a *free monoid*. This means that concatenation is associative ($s(tr) = (st)r$), and that there exists an identity element, specifically, the empty string λ , for which $\lambda a = a\lambda = a$ holds for any string a .

A string s is called a *substring* of a string t if there exist strings u and v (possibly empty) such that $t = usv$. If there exists a string u such that $t = su$, then s is said to be a *prefix* of t , denoted by $s <_p t$. A subset $S \subset \mathcal{S}^*$ is described as *prefix-free* if, for any $s, t \in S$, the condition $s <_p t$ implies $s = t$. Given two sets of

¹It is important to avoid confusing the set of strings of length n over an alphabet, $\mathcal{S}^n$, with the n -fold Cartesian product of a set, S^n . The use of calligraphic fonts helps distinguish between alphabets and other sets.

strings $S, T \subset \mathcal{S}^*$, the (left) *quotient* $S^{-1}T$ is defined as the set of residual strings obtained from T by removing a prefix in S ; formally, $S^{-1}T = \{t \mid st \in T \wedge s \in S\}$.

We denote the *self-delimited* form of a string $s \in \mathcal{S}^*$ by $\bar{s}$, and define it as $\bar{s} = 1^{l(s)}0s$. Consequently, the length of $\bar{s}$, denoted $l(\bar{s})$, is given by $l(\bar{s}) = 2l(s) + 1$, meaning it is twice the length of s plus one.

■ **Example 16.7** The set $\bar{\mathcal{S}}^*$, consisting of all self-delimited strings from $\mathcal{S}^*$, is prefix-free. ■

In cases where $\mathcal{S}$ is a totally ordered set, we can define a total order on $\mathcal{S}^*$. This ordering, known as *shortlex ordering*, arranges sequences primarily by length, with shorter sequences appearing first. Among sequences of the same length, lexicographical order is used to break ties.

■ **Example 16.8** Given $\mathcal{S} = a, b, c$ with $a < b < c$, the shortlex order on $\mathcal{S}^*$ produces the sequence $\lambda < a < b < c < aa < ab < \dots < cc < aaa < aab < \dots < ccc < \dots$ ■

For any arbitrary object O , we use the notation $\langle O \rangle$ to denote its string based representation, assuming the existence of a standard encoding scheme. For objects $O_1, O_2, \dots, O_k$, the expression $\langle O_1 O_2 \dots O_k \rangle$ refers to the plain concatenation of their string representations: $\langle O_1 \rangle \langle O_2 \rangle \dots \langle O_k \rangle$. By contrast, the notation $\langle O_1, O_2, \dots, O_k \rangle$ indicates a structured concatenation that allows for the decoding and unique identification of each individual object. For example, this may be implemented as $\langle \bar{O}_1 \rangle \langle \bar{O}_2 \rangle \dots \langle \bar{O}_k \rangle$.

■ **Example 16.9** Natural numbers can be represented by binary strings via the following encoding method: $\langle 0 \rangle = \lambda$, $\langle 1 \rangle \rightarrow 0$, $\langle 2 \rangle \rightarrow 1$, $\langle 3 \rangle \rightarrow 00$, $\langle 4 \rangle \rightarrow 01$, $\langle 5 \rangle \rightarrow 10$, $\langle 6 \rangle \rightarrow 11$, $7 \rightarrow 000$, and so on. Therefore, the pair of numbers $\langle 3, 7 \rangle$ would be represented as 110001110000. Given this particular encoding, it follows that $l(\langle n \rangle) = \lfloor \log_2(n+1) \rfloor$. ■

A *language*, denoted by L , over an alphabet $\mathcal{S}$, is defined as a subset of strings, that is, $L \subseteq \mathcal{S}^*$. The individual elements of L are called *words*. The unique language that contains no words is referred to as the *empty language*, and is denoted by $L = \emptyset$.

Consider two languages L_1 and L_2 over a common alphabet $\mathcal{S}$. Several standard operations can be applied to these languages. The *union* of L_1 and L_2 is defined as $L_1 \cup L_2 = \{w \in \mathcal{S}^* \mid w \in L_1 \text{ or } w \in L_2\}$. The *intersection* of L_1 and L_2 is given by $L_1 \cap L_2 = \{w \in \mathcal{S}^* \mid w \in L_1 \text{ and } w \in L_2\}$. The *complement* of L_1 is defined as $\bar{L}_1 = \{w \in \mathcal{S}^* \mid w \notin L_1\}$. Finally, the *Kleene closure* of L_1 , denoted L_1^* , is defined as $L_1^* = \lambda \cup \{wz \mid w \in L_1 \text{ and } z \in L_1^*\}$.

Languages can be systematically generated using a finite set of string rewriting rules, commonly referred to as grammars. A *grammar*, denoted by G , is defined as a 4-tuple (N, Σ, P, S) , where: $N \subseteq \mathcal{S}$ is a finite set of *nonterminal*

symbols; $\Sigma \subseteq \mathcal{S}$ is a finite set of *terminal symbols*; P is a finite set of *production rules* of the form $(\Sigma \cup N)^* N (\Sigma \cup N)^* \rightarrow (\Sigma \cup N)^*$; and $S \in N$ is a distinguished *start symbol*. Each production rule allows one string of symbols to be rewritten into another, beginning with the start symbol and proceeding through successive applications of the rules.

■ **Example 16.10** Consider the alphabet $\mathcal{S} = \{S, a, b\}$, and define the grammar (N, Σ, P, S) where $N = \{S\}$, $\Sigma = \{a, b\}$, $P = \{S \rightarrow aSb, S \rightarrow ba\}$, and the start symbol is $S \in N$. This grammar generates the language $L = \{a^n bab^n \mid n \geq 0\} = \{ba, abab, aababb, aaababbb, \dots\}$. ■

The *Chomsky hierarchy* is a classification scheme for grammars, organized according to their expressive power, that is, the types of languages they are capable of generating. The hierarchy, arranged from the most to the least restrictive class of grammars, is described as follows (where a denotes a terminal symbol; A, B are nonterminal symbols; and α, β , and γ are strings composed of terminals and/or nonterminals):

Type-3 Known as *regular grammars*. In these grammars, the left-hand side of each production rule consists of a single nonterminal symbol. The right-hand side must be either the empty string, a single terminal symbol, or a terminal symbol followed by a nonterminal symbol. Formally, the rules are of the form $A \rightarrow \lambda$, $A \rightarrow a$, or $A \rightarrow aB$.

Type-2 Referred to as *context-free grammars*. In this class, the left-hand side of each production rule is exactly one nonterminal symbol. The general form of the rules is $A \rightarrow \alpha$.

Type-1 These are *context-sensitive grammars*. Here, production rules allow a nonterminal to be replaced by a string in a specific context. The rules take the form $\alpha A \beta \rightarrow \alpha \gamma \beta$.

Type-0 This category includes *recursively enumerable grammars*, which place no restrictions on the structure of production rules. These can be written in the general form $\gamma \rightarrow \alpha$.

■ **Example 16.11** The grammar presented in Example 16.10 is a Type-2 grammar, that is, a context-free grammar. ■

The *Backus-Naur form* (BNF) is a notation system specifically designed to describe context-free grammars. It is widely used in computer science to formally specify the syntax of programming languages and communication protocols. A BNF grammar consists of a set of production rules structured as follows:

`<symbol> ::= __expression__`

Here, `<symbol>` denotes a non-terminal symbol, `__expression__` represents a sequence of terminal and/or non-terminal symbols, and `::=` signifies

that the symbol on the left-hand side can be replaced by the expression on the right. Multiple alternatives for the expression can be provided in a single rule by separating them with a vertical bar |, indicating that any one of the alternatives may be chosen during substitution. Symbols that never appear on the left-hand side of any production rule are considered terminal symbols. In contrast, those that do appear on the left-hand side are non-terminal symbols and are conventionally enclosed in angle brackets $\langle \rangle$. The non-terminal symbol on the left-hand side of the first production rule is designated as the start symbol.

■ **Example 16.12** The grammar introduced in Example 16.10 can be expressed in Backus-Naur form using the following production rule:

$\langle \text{string} \rangle ::= a \langle \text{string} \rangle b \mid ba$

■

16.3 Counting Methods

Combinatorics, a specialized branch of mathematics, is primarily concerned with the study of discrete objects and the relationships among them. Its central themes include the counting, arrangement, and selection of such objects, along with the methods used to carry out these tasks. Combinatorics provides a powerful set of tools for analyzing large collections of objects that satisfy specific properties. In this section, we revisit the most important results in combinatorics, focusing on their interpretation in terms of sets and ordered lists.

The *multiplication rule* is a fundamental principle that determines the number of possible outcomes in the Cartesian product of sets. According to this rule, if there are k sets $A_1, A_2, \dots, A_k$, and each set A_i contains n_i elements (for $i = 1, \dots, k$), then the Cartesian product $A_1 \times A_2 \times \dots \times A_k$ contains exactly $n_1 n_2 \dots n_k$ elements. In particular, if a set A has n elements, then the k -fold Cartesian product A^k contains n^k elements.

The *inclusion-exclusion principle* determines the cardinality of the union of multiple sets based on the sizes of the individual sets and all possible intersections among them. Given k sets $A_1, A_2, \dots, A_k$, the formula is:

$$\begin{aligned} d\left(\bigcup_{i=1}^k A_i\right) &= \sum_{i=1}^k d(A_i) - \sum_{i<j} d(A_i \cap A_j) + \sum_{i<j<l} d(A_i \cap A_j \cap A_l) - \\ &\quad - \sum_{i<j<l<m} d(A_i \cap A_j \cap A_l \cap A_m) + \dots + (-1)^{k+1} d(A_1 \cap \dots \cap A_k) \end{aligned}$$

Permutations refer to the number of distinct ways in which the elements of a set can be arranged. Let A be a set with n elements. The number of ordered selections of k elements from n distinct elements without replacement, denoted

by $P_{n,k}$, is given by $P_{n,k} = n(n-1) \cdots (n-k+1)$. In the case where $k = n$, the total number of permutations is $P_{n,n} = n(n-1) \cdots 1 = n!$ where $n!$ is read as " n factorial".

■ **Example 16.13** Consider the set $\{a, b, c\}$. There are six distinct permutations of its elements: $\{a, b, c\}$, $\{a, c, b\}$, $\{b, a, c\}$, $\{b, c, a\}$, $\{c, a, b\}$, and $\{c, b, a\}$. Each permutation represents a unique ordering of the elements in the original set.

■ The *pigeonhole principle* is a simple yet powerful concept. It asserts that if there are more pigeons than pigeonholes, then at least one pigeonhole must contain more than one pigeon. More formally, if n items are distributed among m containers and $n > m$, then at least one container must hold more than one item.

The logarithmic form of *Stirling's* approximation is particularly effective for estimating large factorials:

$$\log(n!) \approx \frac{1}{2} \log(2\pi) + \left(n + \frac{1}{2}\right) \log(n) - n$$

Numerous counting problems involve determining the number of subsets of a specific size within a given set. For a set with n elements, the total number of possible subsets is 2^n , including both the empty set and the set itself. The number of subsets of size k , also known as the number of *combinations* of k elements from a set of n , is denoted by $C_{n,k}$ and computed using the formula $C_{n,k} = \frac{P_{n,k}}{k!} = \frac{n!}{k!(n-k)!}$. The symbol $\binom{n}{k}$ also represents the value $C_{n,k}$, which is known as the *binomial coefficient*. It is known that $\binom{n}{0} = \binom{n}{n} = 1$ for all n , and $\binom{n}{k} = \binom{n}{n-k}$ for all $k = 0, 1, \dots, n$. Additionally, $\binom{n}{k} = 0$ whenever $k > n$.

■ **Example 16.14** Consider the set $\{a, b, c, d\}$. There are 4 combinations of size 3: $[a, b, c]$, $[a, b, d]$, $[a, c, d]$, and $[b, c, d]$. Since combinations disregard order, $[a, c, d]$ and $[d, c, a]$ represent the same combination. ■

The *multinomial coefficient*, a generalization of the binomial coefficient to more than two categories, represents the number of ways to partition a set of objects into a fixed number of subsets, each containing a specified number of elements. Suppose we have a set with n elements that is to be divided into k subsets of sizes $n_1, n_2, \dots, n_k$, respectively. The multinomial coefficient, denoted as $\binom{n}{n_1, n_2, \dots, n_k}$, gives the number of such possible partitions and is calculated by the formula:

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

where $n_1 + n_2 + \cdots + n_k = n$.

The different combinations of replacement and ordering lead to four distinct counting scenarios, as summarized below. These conditions depend on whether

the order of selection matters and whether elements can be selected more than once.

	Without replacement	With replacement
Ordered	$\frac{n!}{(n-r)!}$	n^r
Unordered	$\binom{n}{r}$	$\binom{n+r-1}{r}$

In the first row, we consider ordered selections: without replacement, the number of ways corresponds to the number of permutations of r elements from a set of n ; with replacement, each of the r positions can independently be filled with any of the n elements. In the second row, we consider unordered selections: without replacement, the count is given by the standard binomial coefficient; with replacement, the result corresponds to the number of multisets of size r formed from n distinct elements.

16.4 Matrices

A *matrix*, denoted by A , of order $m \times n$ is a rectangular array consisting of m rows and n columns, filled with a sequence of mn scalars. It is typically written as:

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}$$

The entry a_{ij} denotes the element located at the i -th row and j -th column of the matrix A . The set of all matrices of order $m \times n$ is denoted by $\mathcal{M}_{m \times n}$. A *row matrix* is any matrix in the set $\mathcal{M}_{1 \times n}$, while a *column matrix* belongs to $\mathcal{M}_{m \times 1}$. A *square matrix* is an element of $\mathcal{M}_{n \times n}$. The entries a_{ii} of a square matrix form its *main diagonal*. A *diagonal matrix* is a square matrix in which all entries outside the main diagonal are zero. The *identity matrix*, denoted by I , is a special diagonal matrix with all diagonal entries equal to 1.

The *transpose* of a matrix $A \in \mathcal{M}_{m \times n}$ is defined as the matrix $A^T \in \mathcal{M}_{n \times m}$, whose entry at position (i, j) is equal to the entry at position (j, i) in A . If $A = A^T$, then A is called a *symmetric matrix*. A *submatrix* of a matrix is obtained by deleting one or more rows and/or columns.

■ **Example 16.15** Consider the square matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. The entry in position $(2, 3)$ is 6, and the main diagonal consists of the elements 1, 5, and 9. The transpose of A is $A^T = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$. The matrix $B = \begin{pmatrix} 1 & 3 \\ 4 & 6 \end{pmatrix}$ is a submatrix of A . ■

The addition of two matrices A and B of the same size produces a new matrix $A + B$, where each entry at position (i, j) is defined as $(A + B)_{ij} = a_{ij} + b_{ij}$. Matrix

addition is associative, i.e., $(A + B) + C = A + (B + C)$, and commutative, i.e., $A + B = B + A$. It has a neutral element, the zero matrix, such that $A + 0_{m \times n} = A$, and each matrix has an additive inverse, since $A + (-A) = 0_{m \times n}$.

The product of a scalar λ and a matrix A yields another matrix, denoted by λA , in which each entry at position (i, j) is given by $(\lambda A)_{ij} = \lambda a_{ij}$. Scalar multiplication is distributive over matrix addition, i.e., $\lambda(A + B) = \lambda A + \lambda B$, and over scalar addition, i.e., $(\alpha + \beta)A = \alpha A + \beta A$. It is also associative with respect to scalar multiplication, i.e., $(\alpha\beta)A = \alpha(\beta A)$, and has a multiplicative identity, since $1A = A$.

The product of two matrices $A_{m \times n}$ and $B_{n \times p}$ results in a matrix $AB_{m \times p}$, where each entry at position (i, j) is given by $(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$. Matrix multiplication is associative, meaning that $(AB)D = A(BD)$. It has a left identity element: $AI_n = A$, and a right identity element: $I_m A = A$. It is also compatible with scalar multiplication, so that $\alpha(AB) = (\alpha A)B = A(\alpha B)$. Furthermore, matrix multiplication is distributive over addition: from the right, $A(B + C) = AB + AC$, and from the left, $(B + C)D = BD + CD$.

The transpose operation satisfies the following properties: $(A + B)^T = A^T + B^T$, $(\lambda A)^T = \lambda A^T$, and $(AB)^T = B^T A^T$.

■ **Example 16.16** Given the matrices $A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$ we have that $A + B = \begin{pmatrix} 6 & 8 \\ 10 & 13 \end{pmatrix}$, $2A = \begin{pmatrix} 2 & 4 \\ 6 & 10 \end{pmatrix}$ and $AB = \begin{pmatrix} 19 & 22 \\ 50 & 58 \end{pmatrix}$. ■

A square matrix A is said to be *invertible* or *non-singular* if there exists a matrix B such that $AB = BA = I$. If A is non-singular, the matrix B is unique and is called the *inverse* of A , denoted by A^{-1} . A matrix A is called *orthogonal* if its transpose is equal to its inverse, that is, $A^T = A^{-1}$. The columns and rows of an orthogonal matrix are referred to as *orthonormal vectors*.

The *determinant* is a special scalar value that can be computed from the elements of a square matrix. For a matrix A , the determinant is denoted by $\det(A)$. It can be calculated using the Leibniz formula:

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \cdot a_{1,\sigma(1)} \cdot a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}$$

where S_n denotes the set of all permutations of the integers 1 to n , and $\text{sgn}(\sigma)$ is the sign of the permutation σ , equal to $+1$ for even permutations and -1 for odd permutations. A matrix is invertible if and only if its determinant is nonzero.

■ **Example 16.17** The determinant of a 3×3 matrix $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ is computed as:

$$\det(A) = aei + bfg + cdh - ceg - bdi - afh$$

■

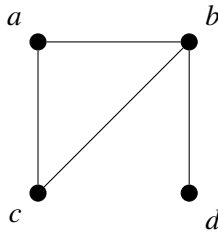


Figure 16.2: An Example of Graph

For a given matrix A , the *rank*, denoted as $\text{rank}(A)$, is the maximum number of linearly independent rows or columns in the matrix.

A number λ and a nonzero vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$ are called an *eigenvalue* and an *eigenvector* of A , respectively.

Matrix decomposition is the process of transforming a matrix into a more tractable form while preserving certain properties, such as the determinant or rank. The *singular value decomposition* (SVD) of a matrix A of order $m \times n$ is a factorization of the form $A = U\Sigma V^T$, where U is an $m \times m$ orthogonal matrix, Σ is an $m \times n$ diagonal matrix with nonnegative entries, and V^T is the transpose of an $n \times n$ orthogonal matrix.

16.5 Graphs

A *graph*² G is defined as an ordered pair (V, E) , where V is a set of *vertices*, and E is a set of *edges*. Each element of E is an unordered pair $\{u, v\}$, where $u, v \in V$ are distinct (loops are not allowed). Two vertices u and v are said to be *adjacent* if $\{u, v\} \in E$; in that case, they are referred to as the *endpoints* of the edge. If the set V is infinite, the graph is called an *infinite graph*. In this book, however, we consider only finite graphs. Given a graph $G = (V, E)$, its *adjacency matrix* is a square matrix of size $d(V) \times d(V)$, denoted A , where $A_{uv} = 1$ if $\{u, v\} \in E$, and $A_{uv} = 0$ otherwise.

Graphs are typically depicted as a set of dots (vertices) connected by lines (edges).

■ **Example 16.18** Let $V = a, b, c, d$ and $E = a, b, a, c, a, d, b, c$. The graph $G = (V, E)$ is illustrated in Figure 16.2. ■

When a vertex v is an endpoint of an edge e , the edge e is said to be *incident* on v . The *degree* of a vertex v , denoted $\deg(v)$, is the number of edges incident on v . A vertex with degree zero is called an *isolated vertex*, while a vertex with

²The definition of a graph given here corresponds to that of a *simple graph*, as commonly found in discrete mathematics literature.

degree one is termed a *pendant vertex*. The *neighborhood* of a vertex v , denoted $N(v)$, is the set of all vertices adjacent to v . For a subset $A \subset V$, the neighborhood of A is defined as $N(A) = \bigcup_{v \in A} N(v)$. A *path* in a graph is a sequence of distinct vertices $v_0, v_1, \dots, v_k$ such that v_i and v_{i+1} are adjacent for each $0 \leq i < k$. The path is called a *simple path* if no vertex is repeated. A graph is said to be *connected* if there exists a path between every pair of vertices. If $v_0 = v_k$, the path is called a *cycle*. A cycle is *simple* if it includes at least three vertices and only the first and last vertices are repeated.

■ **Example 16.19** In a graph $G = (V, E)$, the *handshaking theorem* states that $\sum_{v \in V} \deg(v) = 2m$, where $m = d(E)$, since each edge contributes to the degree count of two vertices. ■

If the vertex pairs (u, v) are arranged as ordered pairs, the graph is called a *directed graph*. In this context, u is referred to as the *initial vertex* and v as the *terminal vertex*. For a directed graph G , the *in-degree* of a vertex v , denoted by $\text{indeg}(v)$, is the number of edges for which v is the terminal vertex. The *out-degree* of a vertex v , denoted by $\text{outdeg}(v)$, is the number of edges for which v is the initial vertex. A directed graph is said to be *strongly connected* if there exists a directed path from every vertex to every other vertex. Directed graphs are typically represented using arrows instead of lines to indicate the direction of edges.

A graph G is classified as *emphbipartite* if its vertex set V can be partitioned into two subsets V_1 and V_2 such that every edge of G connects a vertex from V_1 to a vertex from V_2 . Bipartite graphs are commonly denoted as $G = (V_1, V_2, E)$. The degrees of the vertices in a bipartite graph satisfy the *degree sum formula*, $\sum_{u \in V_1} \deg(u) = \sum_{v \in V_2} \deg(v) = d(E)$.

A graph $G(V', E')$ is called a *subgraph* of a graph $G(V, E)$ if $V' \subseteq V$ and $E' \subseteq E$, where every edge in E' has both endpoints in V' . A graph G is said to be a *labeled graph* if its edges and/or vertices are assigned specific data. In particular, if each edge e in G is assigned a nonnegative number $w(e)$, then $w(e)$ is referred to as the *weight* of the edge e .

A specific type of graph that plays a fundamental role in this book is the *tree*. A tree is defined as a non-empty graph in which any pair of vertices is connected by a unique path. A tree typically includes a specially designated vertex known as the *root*, and every edge is conceptually oriented away from this root.

■ **Example 16.20** An alternative definition of a tree, grounded in set theory, characterizes it as a partially ordered set $(T, <)$ such that for every $t \in T$, the set $S = \{s \in T : s < t\}$ has a least element, that is, an element smaller than all other elements in S . ■

Given a tree T , for any vertex v other than the root, the *parent* of v is the

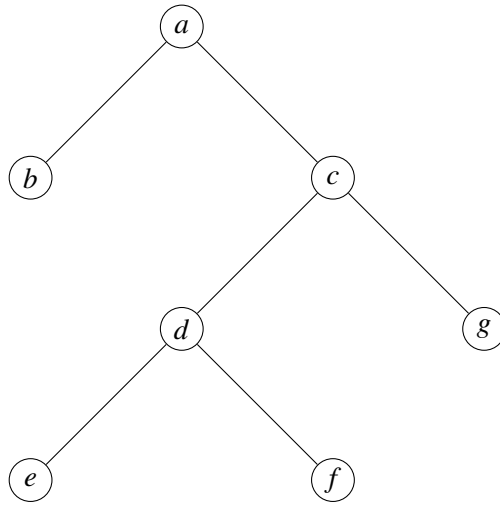


Figure 16.3: An Example of a Tree

unique vertex u such that there is an edge directly connecting u to v . Conversely, if u is the parent of v , then v is called a *child* of u . Any other vertex in the tree that shares the same parent as v is referred to as a *sibling* of v . The *ancestors* of a vertex consist of all vertices along the unique path from the root to that vertex, excluding the vertex itself but including the root. The *descendants* of a vertex v are all vertices that have v as an ancestor. A vertex with no children is called a *leaf*, whereas vertices that have one or more children are known as *branches*. The *depth* of a vertex v is the length of the unique path from the root to v . The *height* of the tree is the maximum depth among all its vertices.

■ **Example 16.21** For the tree illustrated in Figure 16.3, the root vertex is a ; c is the parent of d , so d is a child of c ; d and g are siblings; the ancestors of d are a and c ; the descendants of c are d , e , and f ; the leaf vertices are b , e , f , and g ; a and c are branches; the depth of d is 3; the height of the tree is 4. ■

Given a vertex v in a tree, the *subtree* rooted at v is the subgraph that consists of v , all its descendants, and all edges connecting these vertices. A tree is called a *k-ary tree* if each branch has at most k children. If every branch has exactly k children, the tree is referred to as a *full k-ary tree*. A k -ary tree with $k = 2$ is specifically known as a *binary tree*. A k -ary tree of height h is said to be *balanced* if all its leaves are located at depth h or $h - 1$.

■ **Example 16.22** A tree with n vertices contains exactly $n - 1$ edges. A full k -ary tree with i branches contains $m = ki + 1$ vertices. ■

The procedure of visiting each node in a tree exactly once is known as *tree*

traversal. Traversal methods are categorized based on the order in which nodes are visited, with the two main types being *depth-first* and *breadth-first*. In a *depth-first traversal*, the algorithm starts at the root node and explores as far as possible along each branch before backtracking to explore siblings. There are three common strategies for visiting nodes in a depth-first manner: *in-order*, *pre-order*, and *post-order*.

The following code snippet, written in a C-like syntax, demonstrates a recursive pre-order depth-first traversal algorithm used to print the nodes of a binary tree:

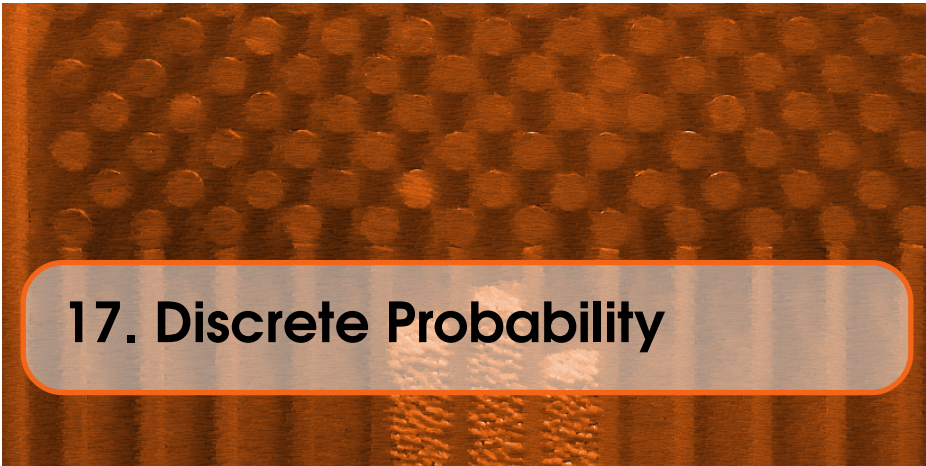
```
void print_tree(binary_tree *tree) {
    if (!is_empty(tree)) {
        printf("%c\n", tree->node);
        print_tree(tree->left_branch);
        print_tree(tree->right_branch);
    }
}
```

Conversely, in a *breadth-first traversal*, the algorithm begins at the root of the tree and explores all nodes at the current depth level before proceeding to nodes at the next depth level. Implementing breadth-first traversal typically requires the use of auxiliary data structures, such as queues. For examples and implementation details of such algorithms, please refer to the references section.

■ **Example 16.23** For the tree described in Example ??, a pre-order depth-first traversal produces the string "abcdefg". In contrast, a pre-order breadth-first traversal yields the string "abcdgef". ■

References

The book *Discrete Mathematics* by Johnsonbaugh [Joh09] is designed for undergraduate students enrolled in a one- or two-semester course in discrete mathematics, and it provides thorough coverage of fundamental topics in the field. *Introduction to the Theory of Computation* by Sipser [Sip12] offers a comprehensive and accessible introduction to key topics in computational theory. It is widely praised for its clear presentation and in-depth treatment of automata, formal languages, and complexity theory. *Introduction to Algorithms* by Cormen et al. [Cor+90], often referred to as "CLRS" after the initials of its authors, is a foundational text in the study of algorithms. It covers a broad range of subjects, including graph algorithms, and is known for its depth and rigor. Lastly, *Matrix Computations* by Golub and Van Loan [GV13] focuses on a wide array of topics in matrix theory, with particular emphasis on computational techniques—making it especially valuable for those involved in numerical and computational studies.



17. Discrete Probability

*The purpose of models is not to fit the data
but to sharpen the questions.*

Samuel Karlin

Probability theory is the branch of mathematics that studies random experiments and phenomena. It assigns a numerical value to each possible outcome of an experiment, reflecting how likely that outcome is to occur. Even when the outcome of a specific experiment cannot be predicted in advance, probability theory allows us to analyze its properties and derive meaningful insights. For example, while we cannot foresee the next number drawn in a lottery, probability theory helps explain why spending all our savings on lottery tickets is an unwise strategy for becoming wealthy.

The significance of probability theory goes far beyond games of chance. It forms the mathematical foundation of statistical inference, allowing us to draw conclusions from data, and it underpins many machine learning algorithms used for classification and prediction with large datasets. Moreover, probability theory plays a vital role in fields such as finance, risk management, and the natural sciences, where understanding uncertainty and variability is crucial.

In this chapter, we focus on the area of discrete probability. In this version of the theory, the possible outcomes of an event are either finite or, at most,

countably infinite. Our interest in discrete probability arises for two main reasons: first, due to its practical applications in learning from data; and second, because it has deep connections to several theoretical concepts explored in this book, such as the length of optimal codes, the probability that a random machine will halt, and the derivation of universal distributions based on Kolmogorov complexity. All of these connections are directly relevant to our theory of nescience.

We will cover only the most important concepts and results of probability theory. The material has been selected based on its relevance to the theory of nescience. For instance, moment-generating functions are not included. For a more comprehensive introduction to probability theory, refer to the references at the end of this chapter.

Our approach to probability theory will be formal and axiomatic. We will begin by stating a basic set of fundamental axioms, from which we will derive the main results and properties. Axioms are essential in mathematical theory, as they provide the foundation for constructing a consistent, universal, and rigorous framework. In the context of probability theory, they allow us to define and manipulate the elusive concept of probability in a way that is both precise and widely applicable.

17.1 Interpretations of Probability

The concept of *probability* presents a profound intellectual challenge. Consider the case of rolling a die and computing the probability of obtaining an even number. The die has six distinct outcomes, and since half of them are even, we conclude that the probability is $3/6$, or equivalently $1/2$. This reflects the *classical interpretation* of probability, which states that in an experiment where all finite outcomes are equally likely, the probability of an event is given by the ratio of favorable outcomes to the total number of possible outcomes.

However, this interpretation faces a problem of circularity: the notion of "equally likely" often assumes a symmetry in the outcomes, but in formal contexts this can be criticized for relying implicitly on the very concept of probability it seeks to define. An alternative approach is the *principle of indifference*, which holds that in the absence of any relevant evidence, all outcomes should be assigned equal probability. This principle, however, breaks down when there is evidence suggesting that the outcomes are not equally probable. This principle, however, only applies in the absence of evidence. When information is available, such as knowing the die is loaded, it cannot be justifiably used, and the assumption of equally likely outcomes breaks down.

The *frequentist interpretation* of probability posits that one should roll the die multiple times and compare the frequency of even outcomes to the total number of rolls. The fundamental idea is to repeat the experiment under similar conditions

and assign to each outcome a probability equal to its relative frequency.

This interpretation, however, faces two major limitations. First, the notion of "similar conditions" is vague and lacks a precise definition; after all, if conditions were exactly identical in a deterministic system, the outcomes would also be identical. In practice, the assumption is that conditions are similar enough to allow statistical regularities to emerge. Second, the concept of a "large number of repetitions" is ill-defined—technically, the experiment must be repeated an infinite number of times.

From a practical standpoint, implementing the frequentist interpretation presents significant challenges. Some experiments, such as estimating the probability of a candidate winning an election, cannot be repeated. Moreover, probability is defined only in the context of a sequence of trials, which makes it impossible to compute the probability of a single, non-repeatable event. Finally, the interpretation assumes the existence of a limiting relative frequency, a condition that is not always satisfied, as illustrated by certain financial time series.

The *subjective interpretation*, representing a third approach to the concept of probability, proposes assigning probabilities to events based on our degree of belief: the stronger our conviction that an event will occur, the higher the probability we assign to it. However, not all possible probability assignments are acceptable; certain coherence conditions must be met. For example, assigning probabilities in a way that guarantees a loss in a betting system, a scenario known as a *Dutch book*, violates these conditions.

It turns out that the conditions necessary and sufficient to avoid such inconsistencies align precisely with the axioms of probability that will be introduced later. Thus, we are free to assign probabilities according to our beliefs, as long as these assignments remain consistent with those axioms.

A key limitation of the subjective interpretation is that degrees of belief can vary widely between individuals. The *Bayesian interpretation* offers a refinement: we begin with an initial (prior) assignment of probabilities and update them as new evidence becomes available. As more evidence is accumulated, revised probabilities tend to converge to values that are more consistent with observed data, and under certain conditions, may approximate the long-run frequencies or objective probabilities (if such exist). Nevertheless, assigning probabilities to an infinite number of events is generally infeasible for humans.

Currently, the notion of probability is defined axiomatically through the *axiomatic interpretation*. This approach abandons the attempt to define probability explicitly, and instead accepts certain fundamental properties as given.

Mathematically, probability is defined as a real number between 0 and 1, where a probability of 0 corresponds to an impossible event, and a probability

of 1 corresponds to an event that is certain to occur. Intuitively, however, we often express probabilities as percentages, for example, saying that it will rain tomorrow with a 70% probability, which is also valid interpretation of the concept of probability.

Additional properties are also required. For example, if two events A and B , with probabilities $P(A)$ and $P(B)$ respectively, are disjoint, then the probability of either A or B occurring should be $P(A) + P(B)$. If A and B can occur simultaneously and are independent (independence being a concept that is mathematically well-defined but often conceptually subtle), then the probability of both occurring together should be $P(A)P(B)$. Furthermore, the probability of A occurring given that B has already occurred should equal the fraction of the probability of A that also lies within B .

17.2 Foundations of Probability Theory

Probability theory is fundamentally concerned with assigning numerical values to specific events drawn from a sample space ¹.

Definition 17.2.1 Given $(\Omega, \mathcal{A})$ as a field over a non-empty discrete set, Ω is called the *sample space*, its elements are called *outcomes*, and the elements of $\mathcal{A}$ are referred to as *events*. Specifically, Ω is known as the *certain event*, while the empty set $\emptyset$ is called the *impossible event*.

As discussed in Section 16.1, since $(\Omega, \mathcal{A})$ is a field, it follows that $\Omega \in \mathcal{A}$ and $\emptyset \in \mathcal{A}$. Moreover, the union of a finite collection of events is also an event: $A_1 \cup A_2 \cup \dots \cup A_n \in \mathcal{A}$, and likewise, the intersection of a finite collection of events is an event: $A_1 \cap A_2 \cap \dots \cap A_n \in \mathcal{A}$.

As mentioned in the introduction to this chapter, our primary focus is on discrete mathematics. Accordingly, we will concentrate on probabilities defined over discrete sets, whether finite or countably infinite. Extending the concept of probability to continuous sets requires the use of σ -algebras of sets instead of fields, along with the tools of measure theory. On a philosophical level, one might argue that all sample spaces must be countable, since physical measurements cannot be made with infinite precision. In practice, this is indeed the case: any empirical observation is subject to finite resolution.

¹The term "event" in this context may seem somewhat counterintuitive, as it typically suggests that something has happened, an implication not always applicable in the mathematical setting. For example, consider the sample space of all possible outcomes from tossing a fair coin. A subset of this sample space might be the empty set, which represents no outcome at all. From a conventional perspective, this does not correspond to anything "happening," which may confuse readers unfamiliar with the mathematical usage of the term. Nevertheless, for consistency and clarity, we will continue to use the term "events" to refer to subsets of the sample space.

The standard axiomatization used in probability theory is encapsulated in the framework of the *Kolmogorov axioms*².

Definition 17.2.2 (*Kolmogorov's Axioms*) A *probability* is a real number $P(A) \in \mathbb{R}$ assigned to each event $A \in \mathcal{A}$ in the field $(\Omega, \mathcal{A})$, subject to the following axioms:

Axiom 1 Non-negativity: For all events $A \in \mathcal{A}$, we have $P(A) \geq 0$.

Axiom 2 Normalization: The probability of the certain event is one, i.e., $P(\Omega) = 1$.

Axiom 3 Additivity: For any finite sequence of pairwise disjoint events $A_1, A_2, \dots, A_n \in \mathcal{A}$, the probability of their union equals the sum of their individual probabilities:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

The triplet $(\Omega, \mathcal{A}, P)$ is called a *probability space*.

Despite their foundational importance, the Kolmogorov axioms present certain limitations. While they establish essential constraints on any probability function (such as non-negativity, normalization, and additivity) they do not prescribe how to assign probabilities to specific events. In other words, the axioms define the formal rules that probabilities must satisfy but remain silent on how those probabilities should be determined in practice. This is a consequence of their high level of generality: any mathematical structure that satisfies these properties can be regarded as a valid probability model. As such, they are abstract enough to encompass not only probability measures but also other normalized physical measures such as mass, volume, or charge.

From an intuitive standpoint, it might seem more natural to assign probabilities directly to the individual elements of the sample space, especially in the discrete case where each outcome is well, defined and countable. However, in the formal theory of probability, probabilities are assigned to events, which are subsets of the sample space. This approach, though less intuitive at first glance, ensures compatibility with the measure-theoretic framework that underlies modern probability theory. In the continuous case, individual outcomes (such as a specific real number) typically have probability zero, and probability must be defined directly over sets (e.g., intervals) rather than points.

²In discrete probability theory, the sample space consists of a finite or countably infinite set of distinct outcomes. As a result, events are typically made up of individual, separable outcomes. Since probabilities are assigned directly to these discrete events, only finite unions of disjoint events need to be considered in Axiom 3 to account for all practical cases.

■ **Example 17.1** Consider a sample space Ω consisting of n equally probable outcomes. If an event $A \subset \Omega$ contains $d(A) = m$ elements, then the probability of event A is given by $P(A) = m/n$. ■

We now proceed to establish some fundamental results in probability theory, beginning with the calculation of the probability of the complement of an event, that is, the probability that the event does not occur.

Proposition 17.2.1 For any event A , it holds that $P(A^c) = 1 - P(A)$.

Proof. The sets A and A^c are disjoint, and their union satisfies $A \cup A^c = \Omega$. By Axiom 3 (additivity), we have $P(A \cup A^c) = P(A) + P(A^c)$. By Axiom 2 (normalization), we know that $P(A \cup A^c) = P(\Omega) = 1$. Combining both equations, we obtain $P(A) + P(A^c) = 1$, which completes the proof. ■

As a direct consequence of the previous proposition, we can deduce the probability of the impossible event.

Proposition 17.2.2 The probability of the impossible event is zero; that is, $P(\emptyset) = 0$.

Proof. Since $\emptyset = \Omega^c$, we apply the complement rule:

$$P(\emptyset) = P(\Omega^c) = 1 - P(\Omega) = 1 - 1 = 0.$$

■

As expected, sub-events (i.e., subsets) are associated with probabilities no greater than those of the events containing them.

Proposition 17.2.3 If $A \subset B$, then $P(A) \leq P(B)$.

Proof. The event B can be written as the disjoint union of A and $B \cap A^c$. By Axiom 3: $P(B) = P(A) + P(B \cap A^c)$. Since $P(B \cap A^c) \geq 0$, it follows that $P(A) \leq P(B)$. ■

With these basic properties established, we can now confirm that all probabilities lie between zero and one.

Proposition 17.2.4 For any event A , we have $0 \leq P(A) \leq 1$.

Proof. By Axiom 1, $P(A) \geq 0$. Since $A \subset \Omega$, the previous proposition implies $P(A) \leq P(\Omega) = 1$. ■

Axiom 3 allows us to compute the probability of the union of disjoint events. However, it does not directly apply to cases involving non-disjoint events. The following proposition provides a formula for computing the probability of the union of two events that may overlap.

Proposition 17.2.5 For any two events A and B , we have:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. The union of A and B can be expressed as $A \cup B = (A \setminus B) \cup B$. Since $A \setminus B$ and B are disjoint, by Axiom 3 we obtain:

$$P(A \cup B) = P(B) + P(A \setminus B).$$

Now, observe that $P(A) = P(A \setminus B) + P(A \cap B)$, which implies:

$$P(A \setminus B) = P(A) - P(A \cap B).$$

Substituting this into the previous expression yields:

$$P(A \cup B) = P(B) + P(A) - P(A \cap B),$$

as required. ■

This result generalizes to any finite number of events via the *principle of inclusion-exclusion* (see Section 16.3):

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \\ &\quad \sum_{i < j < k < l} P(A_i \cap A_j \cap A_k \cap A_l) + \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n) \end{aligned}$$

To ensure that a probability function satisfies Kolmogorov's axioms, we must define it using methods that are guaranteed to produce valid probability measures. One such method involves assigning probabilities to individual elements of a finite or countable sample space and extending this assignment to subsets via summation.

Proposition 17.2.6 Let $(\Omega, \mathcal{A})$ be a field over a non-empty discrete set, and let $p_1, p_2, \dots$ be a sequence of nonnegative real numbers such that $\sum_{i=1}^{\infty} p_i = 1$. Define a function $P : \mathcal{A} \rightarrow [0, 1]$ by $P(A) = \sum_{\{i: s_i \in A\}} p_i$, with the convention that the sum over an empty index set is 0. Then P is a probability on $(\Omega, \mathcal{A})$.

Proof. We must verify that the function P satisfies the three Kolmogorov axioms:

Axiom 1 (Nonnegativity): Let $A \in \mathcal{B}$. Since each $p_i \geq 0$, the sum

$$P(A) = \sum_{\{i: s_i \in A\}} p_i$$

is a (possibly infinite) sum of nonnegative terms. Hence, $P(A) \geq 0$.

Axiom 2 (Normalization): We have that

$$P(\Omega) = \sum_{\{i: s_i \in \Omega\}} p_i = \sum_{i=1}^{\infty} p_i = 1,$$

by assumption. Hence, $P(\Omega) = 1$.

Axiom 3 (Finite Additivity): Let $A_1, \dots, A_n \in \mathcal{B}$ be pairwise disjoint events. Define $A = \bigcup_{j=1}^n A_j$. Then the sets $\{i : s_i \in A_j\}$ are disjoint for distinct j , and we have:

$$P(A) = \sum_{\{i: s_i \in A\}} p_i = \sum_{j=1}^n \sum_{\{i: s_i \in A_j\}} p_i = \sum_{j=1}^n P(A_j),$$

where we used the disjointness of the A_j 's and associativity of finite sums.

Hence, P satisfies all three axioms of probability (with finite additivity), and is therefore a valid probability function on $(\Omega, \mathcal{A})$. ■

Finally, we introduce the concept of a probability mass function³. A probability mass function assigns a probability to each event in a discrete sample space.

Definition 17.2.3 Let $(\Omega, \mathcal{A}, P)$ be a discrete probability space. A *probability mass function* is a real-valued function $f : \mathcal{A} \rightarrow [0, 1]$ such that $f(A) = P(A)$ for every $A \in \mathcal{A}$.

In Example 17.1, we introduced a discrete probability space $(\Omega, \mathcal{A}, P)$ consisting of n equally probable elements. The corresponding probability mass function is defined as $f : \mathcal{A} \rightarrow [0, 1]$, with

$$f(A) = \frac{d(A)}{n}, \quad \text{for all } A \in \mathcal{A},$$

where $d(A)$ denotes the number of elements in the event A .

In many discrete settings, it is common to treat each individual element $\omega \in \Omega$ as an elementary event. That is, we assume that every set $\{\omega\}$ belongs to $\mathcal{A}$. In this case, we can define the probability mass function as a function $f : \Omega \rightarrow [0, 1]$, where $f(\omega) = P(\{\omega\})$ for each $\omega \in \Omega$. This formulation is often more natural, as it allows us to specify probabilities at the level of individual outcomes.

³Most probability textbooks define the probability mass function as a real-valued function of the form $f : \mathbb{R} \rightarrow [0, 1]$. However, this definition requires the prior introduction of the concept of a random variable. Since random variables play a more secondary role in the context of discrete probability, we have postponed their introduction to a later subsection. For this reason, we have provided a definition of the probability mass function based directly on events.

The following proposition shows that, under this assumption, the probability of any event $A \in \mathcal{A}$ can be computed by summing the probabilities of the individual outcomes that make up the event.

Proposition 17.2.7 Let $(\Omega, \mathcal{A}, P)$ be a discrete probability space such that $\{\omega\} \in \mathcal{A}$ for every $\omega \in \Omega$. Define the function $f : \Omega \rightarrow [0, 1]$ by $f(\omega) = P(\{\omega\})$. Then for any event $A \in \mathcal{A}$, the probability of A is given by:

$$P(A) = \sum_{\omega \in A} f(\omega).$$

Proof. Since $A \in \mathcal{A}$, and $\{\omega\} \in \mathcal{A}$ for each $\omega \in \Omega$, it follows that every singleton $\{\omega\} \subseteq A$ is a measurable event. Furthermore, the singletons $\{\omega\}$ for $\omega \in A$ are pairwise disjoint and their union is exactly A . Thus, by Axiom 3 (finite additivity),

$$P(A) = P\left(\bigcup_{\omega \in A} \{\omega\}\right) = \sum_{\omega \in A} P(\{\omega\}) = \sum_{\omega \in A} f(\omega).$$

■

17.3 Conditional Probability

The concept of conditional probability plays a fundamental role in statistical learning. Conditional probability allows us to refine the probability of an event based on new information or imposed conditions. Within the axiomatic framework established by Kolmogorov, conditional probability is introduced as a definition.

Definition 17.3.1 Let A and B be two events such that $P(B) > 0$. The *conditional probability* of A given B , denoted by $P(A | B)$, is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Because it satisfies the Kolmogorov axioms, a conditional probability is itself a valid probability. Note that $P(A | B)$ is undefined when $P(B) = 0$.

Some scholars argue that, given its central role in probability theory, conditional probability should be a property logically deduced from the foundational axioms, rather than introduced as a definition. This viewpoint naturally calls for an extension of Definition 17.2.2 to include additional properties. Unfortunately, there is no consensus among mathematicians or philosophers on how such an extension should be formalized.

The standard interpretation of conditional probability views it as the updated probability of event A given that event B has occurred. This interpretation,

however, may implicitly suggest a temporal or even causal relationship between B and A , a suggestion that is not necessarily valid.

■ **Example 17.2** Suppose we are playing a game with a standard deck of 52 cards and we draw two cards. Let event A be “drawing at least one heart,” and event B be “drawing at least one queen.” These two events are dependent, as the occurrence of B affects the probability of A . However, the events are not temporally related, since both cards are drawn simultaneously, one does not occur before the other. This example illustrates that dependency in probability theory does not require a temporal sequence between the events involved. ■

The probability of two events occurring together (though not necessarily in a temporal sequence, as discussed in Example 17.2), given their conditional relationship, is captured by the formula $P(A \cap B) = P(A | B) P(B)$. This equation often provides a more intuitive understanding of the concept of conditional probability. In fact, several authors have argued that this formulation should serve as the primary definition of conditional probability, rather than the quotient-based approach.

This formula generalizes to any finite number of events through what is known as the *multiplication rule*, which is expressed as:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) \dots P(A_n | A_1 \cap A_2 \cap \dots \cap A_{n-1}) \quad (17.1)$$

The concept of event independence plays a central role in probability theory and statistical learning.

■ **Definition 17.3.2** Two events A and B are declared to be *independent* if $P(A \cap B) = P(A) P(B)$.

Intuitively, events A and B are considered independent if knowing that event B has occurred does not affect the probability of event A . This property follows logically from the definition of independence.

Proposition 17.3.1 Let A and B be two events such that $P(A) > 0$ and $P(B) > 0$. Then A and B are independent if and only if $P(A | B) = P(A)$.

Proof. Assume A and B are independent, i.e., $P(A \cap B) = P(A) P(B)$. Then,

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) P(B)}{P(B)} = P(A).$$

Conversely, assume that $P(A | B) = P(A)$. Then by the multiplication rule,

$$P(A \cap B) = P(A | B) P(B) = P(A) P(B),$$

which confirms independence. ■

Note that if $P(A | B) = P(A)$, then it also follows that $P(B | A) = P(B)$. This is a direct consequence of the fact that the joint probability $P(A \cap B)$ can be expressed as $P(A)P(B)$.

As in the case of conditional probability, some scholars have argued that independence should not be treated merely as a definition, but rather as a logical consequence of the foundational axioms of probability. However, incorporating independence as a derived property would require extending or modifying the axiomatic system, a topic for which no universally accepted approach currently exists.

The concept of independence can be generalized to collections of more than two events. The events $A_1, \dots, A_n$ are said to be *mutually independent* if, for every subset of events $A_{i_1}, \dots, A_{i_j}$ with $j \in 2, 3, \dots, n$, the following condition holds:

$$P(A_{i_1} \cap \dots \cap A_{i_j}) = P(A_{i_1}) \cdots P(A_{i_j}).$$

It is important to emphasize that *pairwise independence*, that is, the condition $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$, is not sufficient to guarantee mutual independence. Mutual independence requires that the product rule holds for *every* subset of two or more events, not just for pairs.

■ **Example 17.3** Confusion often arises when distinguishing between mutually exclusive (or disjoint) events and independent events. For two mutually exclusive events A and B , the expression for the conditional probability $P(A | B)$ becomes problematic: if B occurs, then A cannot occur by definition, making $P(A | B)$ either undefined (if $P(B) = 0$) or equal to zero (if $P(B) > 0$). Similarly, computing the conditional probability of A given B when $P(B) = 0$ is not well-defined.

Although Definition 17.3.2 does not explicitly prohibit A and B from being mutually exclusive, the two concepts are fundamentally incompatible unless one (or both) of the events has probability zero. In fact, two mutually exclusive events A and B are independent if and only if $P(A) = 0$, $P(B) = 0$, or both. ■

An interesting situation occurs when two events A and B are not independent in general, but become independent when conditioned on the occurrence of a third event C .

Definition 17.3.3 Let A , B , and C be events such that $P(B \cap C) > 0$. We say that A and B are *conditionally independent* given C if $P(A | B \cap C) = P(A | C)$.

Next example illustrates that two events may be independent but fail to remain conditionally independent given a third event.

■ **Example 17.4** Consider the experiment of rolling two fair dice. The outcomes of the two dice are independent, since knowing the result of one die does not provide any information about the result of the other. Now suppose we observe

that the first die shows a four, and we are also given that the sum of the two dice is an odd number (event C). This additional condition restricts the possible outcomes of the second die: it must now show an odd number to make the total odd. As a result, the outcome of the second die is no longer independent of the first, once we condition on the sum being odd. ■

The following theorem presents Bayes' rule, a fundamental result that underpins an important statistical learning technique known as Bayesian inference (see Section 21.1.2)⁴.

Theorem 17.3.2 — Bayes' Theorem. Let A and B be two events such that $P(B) \neq 0$. Then we have that

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

In this context, $P(A)$ is called the *prior probability*, while $P(A | B)$ is known as the *posterior probability*.

Proof. From the definition of conditional probability, we have:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad (\text{provided } P(B) \neq 0),$$

and

$$P(B | A) = \frac{P(A \cap B)}{P(A)} \quad (\text{provided } P(A) \neq 0).$$

Solving the second equation for $P(A \cap B)$ and substituting into the first yields the desired result. ■

Bayesian inference allows us to update our degree of belief in the occurrence of an event A , initially represented by the prior probability $P(A)$, in light of new evidence, represented by the occurrence of another event B . The result is the posterior probability $P(A | B)$, which reflects our revised belief after considering the new information.

⁴As shown in the proof, Bayes' theorem follows directly from the definition of conditional probability. However, despite its practical utility, this derivation raises important philosophical questions regarding its foundational status. If conditional probability is taken as a definition rather than a derived consequence of more fundamental axioms, then Bayes' theorem rests on the same definitional foundation. As such, it functions effectively within this framework, but its theoretical justification depends on accepting conditional probability as a primitive notion rather than as something logically deduced.

■ **Example 17.5** Let E represent a rare disease that affects one in every million people, so that $P(E) = 1 \times 10^{-6}$. Let $+$ denote a diagnostic test designed to detect the disease, with a failure rate of one in a thousand tests; that is, $P(+ | E) = 999/1000$. We wish to compute the probability that a person has the disease given a positive test result, i.e., $P(E | +)$. Applying Bayes' theorem, we get:

$$P(E | +) = \frac{P(+ | E)P(E)}{P(+)} = \frac{P(+ | E)P(E)}{P(+ | E)P(E) + P(+ | E^c)P(E^c)} = 0.001$$

This result shows that, despite the test failing only once in a thousand applications, the probability of actually having the disease after a positive result remains extremely low. This seemingly paradoxical outcome arises because the probability of a false positive (10^{-3}) is much higher than the prior probability of having the disease (10^{-6}).

In practical settings, this problem is often addressed by administering a second, independent test to individuals who test positive. Under the assumption of test independence, the probability of disease given two positive results increases substantially, to approximately 0.5. ■

Bayes' theorem is particularly useful when the events involved are dependent, allowing new information about one event to revise our assessment of the probability of another.

■ **Example 17.6** Suppose you draw a single card from a standard 52-card deck. Let event A be "drawing a red card," and let event B be "drawing a queen." In this case, applying Bayes' theorem to compute $P(A | B)$, the probability of having drawn a red card given that a queen has been drawn, yields little insight. This is because knowing that a queen was drawn does not significantly alter the probability of the card being red: two of the four queens are red and two are black, mirroring the overall color distribution in the deck. Thus, event B provides no relevant new information about event A . ■

Bayes' theorem can be generalized to accommodate multiple events. Let $A_1, \dots, A_k$ be a finite collection of events with $P(A_j) > 0$ for all $j = 1, \dots, k$, and assume that these events form a partition of the sample space Ω . Let B be an event such that $P(B) > 0$. Then, for each $i = 1, \dots, k$, the conditional probability $P(A_i | B)$ is given by:

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_{j=1}^k P(B | A_j)P(A_j)}$$

This expression demonstrates the applicability of Bayes' theorem to a wider range of inference problems involving multiple events.

17.4 Random Variables

A random variable⁵ is a function that assigns a real number to each possible outcome of an experiment, thus providing a quantitative representation of the results. Although the Kolmogorov framework assigns probabilities to events, random variables offer a more direct approach by associating numerical values with outcomes. This not only simplifies the representation of probabilistic phenomena but also facilitates the analysis of their properties. Owing to their effectiveness, most statisticians conduct their investigations primarily within the framework of random variables rather than working directly with probability spaces.

Definition 17.4.1 Let $(\Omega, \mathcal{A}, P)$ be a discrete probability space. A *random variable* is a function $X : \Omega \rightarrow \mathbb{R}$ mapping from the set of the sample space Ω to the real numbers $\mathbb{R}$. A random variable is discrete if its range $\{x_1, x_2, \dots, x_i, \dots\}$ is finite or countably infinite.

Our focus here is on discrete random variables defined on discrete probability spaces. Although it is possible to define a discrete random variable on a non-discrete probability space by restricting it to a discrete range, this case is not considered in this book.

Random variables are particularly useful when they represent specific characteristics of the experiment. For example, if the sample space consists of a school's student body, a random variable could assign to each student their respective height. Random variables also allow us to transform the original outcomes of the sample space into new events. For instance, if two dice are rolled, a random variable could represent the sum of the two outcomes.

It is important to note that we are free to define a random variable on any sample space, even if the assignment does not appear intuitively meaningful. For example, one might assign a numerical value to each color in a deck of cards, draw two cards at random, and take the sum of the assigned numbers. While such a construction may have little practical interpretation, it nonetheless defines a valid random variable, and probabilities can be computed accordingly.

Definition 17.4.2 Let $X : \Omega \rightarrow \mathbb{R}$ be a discrete random variable, and let $C \subset \mathbb{R}$ be a set such that $\{\omega \in \Omega : X(\omega) \in C\}$ is an event. The probability of X taking a value in C , denoted $P(X \in C)$, is defined by $P(X \in C) = P(\{\omega \in \Omega : X(\omega) \in C\})$.

The probability associated with a random variable X effectively defines a

⁵The term "random variable" can be somewhat misleading. First, a random variable is not a variable in the usual algebraic sense, but rather a function. Second, it is not the random variable itself that is random; the randomness lies in the experiment it represents. Despite this potential for confusion, we follow the standard terminology.

probability space on the real line, specifically over the range of X .

■ **Example 17.7** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ be the sample space for a single toss of a fair die, and P a probability that assigns $1/6$ to each single outcome in Ω . Let $X : \Omega \rightarrow \mathbb{R}$ be a discrete random variable defined as:

$$X(\omega) = \begin{cases} 0 & \text{if } \omega \text{ is even } (2, 4, 6), \\ 1 & \text{if } \omega \text{ is odd } (1, 3, 5). \end{cases}$$

This random variable maps each outcome of the die toss to either 0 (if the result is even) or 1 (if the result is odd). For $C = \{0\}$, the probability $P(X \in C) = P(X = 0) = P(\{2, 4, 6\}) = 1/2$. For $C = \{1\}$ the probability $P(X \in C) = P(X = 1) = P(\{1, 3, 5\}) = 1/2$. Through this transformation, the original discrete probability space has been mapped to the real numbers using the discrete random variable X , establishing a new probability over X 's range. ■

Unless stated otherwise, we will assume that $\{\omega \in \Omega, : X(\omega) = x_i\}$ is an event for every point x_i in the range of X . The following example shows a case where this assumption does not hold.

■ **Example 17.8** Let $(\Omega, \mathcal{A}, P)$ be a discrete probability space, where $\Omega = \{1, 2\}$, $\mathcal{A} = \{\emptyset, \Omega\}$, $P(\emptyset) = 0$ and $P(\Omega) = 1$. Let $X : \Omega \rightarrow \mathbb{R}$ be a discrete random variable defined as $X(1) = 1$ and $X(2) = 2$. We are interested in the probability $P(X = 1)$, however since $\{\omega \in \Omega : X(\omega) = 1\} = \{1\}$ is not an event (because $\{1\} \notin \mathcal{A}$), such probability cannot be determined. ■

Definition 17.2.3 introduced the concept of probability mass function for probability spaces, based on the probabilities of the events. The following definition extends this concept to random variables.

Definition 17.4.3 Let X be a discrete random variable over a discrete probability space, and let $\{x_1, x_2, \dots\}$ be the range of X . The *probability mass function* of the discrete random variable X , abbreviated as p.m.f., is the function $f : \text{range}(X) \rightarrow [0, 1]$ defined by $f(x_i) = P(X = x_i)$.

The set of values for which the probability mass function is positive, $\{x : f(x) > 0\}$, is called the *support* of the distribution of X .

It is possible for two random variables to share the same probability mass function while differing in other significant respects.

■ **Example 17.9** Let $\Omega = \{H, T\}$ be the sample space for a single toss of a fair coin, and let P be a probability that assigns $1/2$ to each outcome in Ω . Define a discrete random variable $X : \Omega \rightarrow \mathbb{R}$ by setting $X = 1$ if the coin shows Head and $X = 0$ if coin shows Tail. The distribution of X is given by $P(X = 0) = P(X = 1) = 0.5$. This distribution is identical to that of the random variable in

Example 17.7, even though the two random variables are defined on different sample spaces and represent different experiments. ■

Given the probability mass function of a random variable, we can determine the probability of any subset of the real line.

Proposition 17.4.1 Let X be a discrete random variable with probability mass function f . For any subset C of the real line, the probability $P(X \in C)$ is given by $\sum_{x_i \in C} f(x_i)$.

Proof. Since each outcome in the sample space corresponds to exactly one value in the range $x_1, x_2, \dots$ of X , we have

$$P(X \in C) = P(\{\omega \in \Omega : X(\omega) \in C\}) = \sum_{x_i \in C} P(X = x_i) = \sum_{x_i \in C} f(x_i).$$

■

This result shows that the probability mass function completely determines the probability distribution of a discrete random variable, since the probability of any event involving X can be computed from it.

The following proposition states a fundamental property: the total probability over all possible values of a random variable is equal to 1.

Proposition 17.4.2 Let X be a discrete random variable with probability mass function f . If $\{x_1, x_2, \dots\}$ is the range of X , then $\sum_{i=1}^{\infty} f(x_i) = 1$.

Proof. Since X is a total function, each outcome in the sample space is associated with exactly one value in the range $\{x_1, x_2, \dots\}$. By the axioms of probability,

$$\sum_{i=1}^{\infty} f(x_i) = \sum_{i=1}^{\infty} P(X = x_i) = \sum_{i=1}^{\infty} P(\{\omega \in \Omega : X(\omega) = x_i\}) = P(\Omega) = 1.$$

■

The values $\{x_1, x_2, \dots\}$ represent all the possible outcomes of X . Since these outcomes are mutually exclusive and collectively exhaustive, their probabilities must sum to the total probability of the entire sample space, which is 1.

A fundamental characterization of discrete distributions is that they are completely determined by a nonnegative function on the real line whose total mass is 1. The next proposition states this precisely.

Proposition 17.4.3 A function $f(x)$ is a probability mass function of a random variable X if, and only if, i) $f(x) \geq 0$ for all x , ii) $\sum_{i=1}^{\infty} f(x_i) = 1$.

Proof. Suppose f is the probability mass function of a discrete random variable X with range $\{x_1, x_2, \dots\}$. Then $f(x_i) = P(X = x_i) \geq 0$ for all i , and

$$\sum_{i=1}^{\infty} f(x_i) = \sum_{i=1}^{\infty} P(X = x_i) = P(\Omega) = 1,$$

so (i)-(ii) hold.

Conversely, assume $f(x) \geq 0$ for all x and $\sum_{i=1}^{\infty} f(x_i) = 1$ for some countable set $\{x_1, x_2, \dots\}$ (the points where f may be positive). Define a discrete sample space $\Omega = \{x_1, x_2, \dots\}$, set $P(x_i) = f(x_i)$, and define $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) = \omega$. Then for each i ,

$$P(X = x_i) = P(\{\omega \in \Omega : X(\omega) = x_i\}) = P(\{x_i\}) = f(x_i),$$

so f is the probability mass function of X . ■

The proposition shows that a discrete distribution is fully specified by any function that is everywhere nonnegative and sums to 1 over its (countable) support; values of f outside the support are irrelevant.

The cumulative distribution function expresses the probability that a random variable takes a value less than or equal to a given point.

Definition 17.4.4 The *cumulative distribution function* (c.d.f.) F of a discrete random variable X is defined by $F(x) = P(X \leq x)$ for all $-\infty < x < \infty$

If X has a distribution characterized by the probability mass function $f(x)$, its cumulative distribution function $F(x)$ behaves as follows: at each distinct value x_i in the range of X , $F(x)$ increases by $f(x_i)$; between these values, $F(x)$ remains constant.

The cumulative distribution function shows how probabilities accumulate over the range of a random variable, providing insight into the overall distribution of the data. It can be viewed as the running total of the probabilities given by the probability mass function.

■ **Example 17.10** Let X be a discrete random variable representing the grades of students in a class, where each grade is between 0 and 10. The probability that a student receives a grade of x is given by the probability mass function $p(x)$. The cumulative distribution function $F(x)$ gives the probability that a randomly selected student scores x or less. For example, if $F(7) = 0.6$, this means there is a 60% chance that a randomly chosen student scored 7 or below. ■

The cumulative distribution function of a random variable is always non-decreasing.

Proposition 17.4.4 Let F be the cumulative distribution function of a discrete random variable X . If $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.

Proof. For $x_1 < x_2$, the event $X \leq x_1$ is a subset of the event $X \leq x_2$. By the monotonicity property of probability, $P(X \leq x_1) \leq P(X \leq x_2)$. Since $F(x) = P(X \leq x)$, it follows that $F(x_1) \leq F(x_2)$. ■

The following proposition states the asymptotic properties of the cumulative distribution function of a random variable, showing its limits as x approaches negative and positive infinity.

Proposition 17.4.5 Let F be the cumulative distribution function of a discrete random variable X . Then $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

Proof. As $x \rightarrow -\infty$, the event $X \leq x$ eventually becomes empty, because there are no values in the range of X smaller than sufficiently negative x . The probability of this event is therefore 0, so

$$\lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} P(X \leq x) = 0.$$

As $x \rightarrow \infty$, the event $X \leq x$ eventually contains the entire sample space Ω , since all possible values of X are less than or equal to sufficiently large x . The probability of this event is therefore 1, so

$$\lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} P(X \leq x) = 1.$$

■

The probability that X exceeds x is the complement of the cumulative distribution function at that point.

Proposition 17.4.6 Let F be the cumulative distribution function of a discrete random variable X . Then, for every $x \in \mathbb{R}$, $P(X > x) = 1 - F(x)$.

Proof. The events $X > x$ and $X \leq x$ are complementary, so their probabilities sum to 1 $P(X > x) + P(X \leq x) = 1$. Therefore, $P(X > x) = 1 - P(X \leq x)$. By the definition of the cumulative distribution function, $P(X \leq x) = F(x)$, and thus $P(X > x) = 1 - F(x)$. ■

The following proposition relates the probability of a random variable X taking a value between two points to the difference in its cumulative distribution function at those points.

Proposition 17.4.7 Let F be the cumulative distribution function of a discrete random variable X . Then, for all $x_1, x_2 \in \mathbb{R}$ with $x_1 < x_2$, $P(x_1 < X \leq x_2) = F(x_2) - F(x_1)$.

Proof. The probability that $X \leq x_2$ is $F(x_2)$. Subtracting from this the probability that $X \leq x_1$, which is $F(x_1)$, leaves exactly the probability that X falls strictly between x_1 and x_2 :

$$P(x_1 < X \leq x_2) = F(x_2) - F(x_1).$$

■

17.4.1 Multivariate Distributions

A multivariate probability distribution extends the concept of a probability distribution to multiple random variables, each with its own set of possible outcomes. While univariate distributions describe phenomena involving a single random variable, multivariate distributions capture the relationships and dependencies between two or more variables. This allows for the analysis of complex phenomena in which the outcome of interest is influenced by several factors simultaneously, providing insight into how these variables interact and affect the probabilities of different outcomes.

Definition 17.4.5 Let $X_1, X_2, \dots, X_n$ be n discrete random variables on a common discrete probability space $(\Omega, \mathcal{A}, P)$, where $X_i : \Omega \rightarrow \mathbb{R}$ for $i = 1, \dots, n$. The *joint probability distribution* of $X_1, \dots, X_n$ assigns probabilities to events involving all variables simultaneously. Formally, for any set $C \subset \mathbb{R}^n$, the joint probability is

$$P((X_1, X_2, \dots, X_n) \in C),$$

where

$$\{\omega \in \Omega : (X_1(\omega), X_2(\omega), \dots, X_n(\omega)) \in C\}$$

is an event in the sample space.

The joint probability distribution of the random variables $X_1, X_2, \dots, X_n$ defines a probability distribution on $\mathbb{R}^n$. If the random variables $X_1, X_2, \dots, X_n$ each have a discrete distribution, then the joint distribution is also a discrete distribution.

Definition 17.4.6 Let $X_1, \dots, X_n$ be n discrete random variables over a common discrete probability space. The *joint probability mass function* of $X_1, \dots, X_n$ is the function

$$f : \text{range}(X_1) \times \dots \times \text{range}(X_n) \rightarrow [0, 1]$$

defined by

$$f(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n).$$

■ **Example 17.11** A classic example of a bivariate discrete joint distribution involves rolling two six-sided dice. Define two discrete random variables: X_1 is the outcome of the first die, and X_2 is the outcome of the second die. Both X_1 and X_2 follow a discrete uniform distribution over $1, 2, 3, 4, 5, 6$. The joint distribution of X_1 and X_2 describes the probability of each possible pair of outcomes when the dice are rolled. The joint probability mass function $f(x_1, x_2)$ is:

$$f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2) = \frac{1}{36}, \quad x_1, x_2 \in \{1, 2, 3, 4, 5, 6\}.$$

The joint distribution allows us to answer questions such as the probability that the sum of the two dice equals a certain value, or that one die shows a higher number than the other. ■

Before exploring the properties of multivariate random variables, we introduce the concept of a random vector. This simplifies notation and improves clarity by grouping multiple random variables into a single object.

■ **Definition 17.4.7** A *discrete random vector* $\mathbf{X}$ is an ordered collection of n discrete random variables $X_1, X_2, \dots, X_n$, where $X_i : \Omega \rightarrow \mathbb{R}$ for all $i = 1, \dots, n$. Equivalently, $\mathbf{X} = (X_1, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$.

Given the joint probability mass function of a random vector, we can determine the probability of any subset of the n -dimensional real space.

Proposition 17.4.8 Let $\mathbf{X} = (X_1, \dots, X_n)$ be a discrete random vector with joint probability mass function f . For any subset C of $\mathbb{R}^n$,

$$P(\mathbf{X} \in C) = \sum_{\mathbf{x} \in C} f(\mathbf{x}),$$

where $\mathbf{x} = (x_1, \dots, x_n)$.

Proof. Since each outcome in the sample space corresponds to exactly one value in the range of $\mathbf{X}$, we have:

$$P(\mathbf{X} \in C) = P(\{\omega \in \Omega : \mathbf{X}(\omega) \in C\}) = \sum_{\mathbf{x} \in C} P(\mathbf{X} = \mathbf{x}) = \sum_{\mathbf{x} \in C} f(\mathbf{x}).$$

■

The following proposition states a fundamental property, that the total probability over all possible outcomes of a discrete random vector is 1.

Proposition 17.4.9 Let $\mathbf{X} = (X_1, \dots, X_n)$ be a discrete random vector with a joint probability mass function f . If the range of $\mathbf{X}$ is $\mathbf{x} = (x_1, \dots, x_n)$, then $\sum_{\mathbf{x}} f(\mathbf{x}) = 1$.

Proof. Since $\mathbf{X}$ is a total function and each outcome in the sample space corresponds to exactly one value in the range of $\mathbf{X}$, the axioms of probability imply:

$$\sum_{\mathbf{x}} f(\mathbf{x}) = \sum_{\mathbf{x}} P(\mathbf{X} = \mathbf{x}) = P(\Omega) = 1.$$

■

A particularly important multivariate construction is the distribution of the sum of n random variables. The notation $X_1 + \dots + X_n$ refers to the sum of the random variables (applied pointwise on outcomes), not a sum of n probability distributions. There are two standard ways to formalize this, depending on how the random variables are defined.

The first one refers to a sum on a common sample space.

Definition 17.4.8 Let $X_1, X_2, \dots, X_n$ be discrete random variables on a common probability space $(\Omega, \mathcal{A}, P)$, with $X_i : \Omega \rightarrow \mathbb{R}$. The sum is the discrete random variable

$$S = X_1 + \dots + X_n : \Omega \rightarrow \mathbb{R}, \quad S(\omega) = X_1(\omega) + \dots + X_n(\omega).$$

The *distribution of the sum* is the distribution of S .

The second one refers to a sum on product space (independent relations).

Definition 17.4.9 Let $X : \Omega \rightarrow \mathbb{R}$ model a single trial. To model n repetitions, consider the product sample space Ω^n (each n -tuple of outcomes is a sample point, and in the i.i.d. case all n -tuples are equally likely when Ω is finite). Define

$$X_i(\omega_1, \dots, \omega_n) = X(\omega_i), \quad i = 1, \dots, n,$$

and the sum

$$S(\omega_1, \dots, \omega_n) = X(\omega_1) + \dots + X(\omega_n), \quad S : \Omega^n \rightarrow \mathbb{R}.$$

Again, the *distribution of the sum* is the distribution of S .

In both setups, the distribution of S depends on the joint distribution of $(X_1, \dots, X_n)$. Independence is not assumed unless explicitly stated.

■ **Example 17.12** Let $X : \Omega \rightarrow \mathbb{R}$ be the outcome of rolling a six-sided die (values $1, \dots, 6$, each with probability $1/6$). For three rolls, use the product space Ω^3 with each triple equally likely. Define

$$X_1(\omega_1, \omega_2, \omega_3) = X(\omega_1), X_2(\omega_1, \omega_2, \omega_3) = X(\omega_2), X_3(\omega_1, \omega_2, \omega_3) = X(\omega_3),$$

and

$$S(\omega_1, \omega_2, \omega_3) = X(\omega_1) + X(\omega_2) + X(\omega_3).$$

Then $S : \Omega^3 \rightarrow \mathbb{R}$ is the sum of three (independent) die rolls. ■

17.4.2 Marginal Probability Mass Function

Given a multivariate discrete probability mass function, the marginal probability mass function of a subset of variables is derived by summing the joint probability mass function over all possible values of the remaining variables. This process "marginalizes" out the variables not of interest, allowing us focus on the probability mass function of a single variable or a subset of variables within the multivariate context.

Definition 17.4.10 Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be an n -dimensional discrete random vector with joint probability mass function $f_{\mathbf{X}}(x_1, x_2, \dots, x_n)$, partition $\mathbf{X}$ into two subvectors: $\mathbf{Y} = (Y_1, Y_2, \dots, Y_k)$, a k -dimensional random vector consisting of k discrete random variables selected from $\mathbf{X}$, and $\mathbf{Z} = (Z_1, Z_2, \dots, Z_{n-k})$, the remaining $(n - k)$ discrete random variables of $\mathbf{X}$. The *marginal probability mass function* $f_{\mathbf{Y}}$ of $\mathbf{Y}$ is obtained by summing $f_{\mathbf{X}}$ over all possible values of the variables in $\mathbf{Z}$. That is, for any specific values $(y_1, y_2, \dots, y_k)$ of $\mathbf{Y}$,

$$f_{\mathbf{Y}}(y_1, y_2, \dots, y_k) = \sum_{z_1} \sum_{z_2} \cdots \sum_{z_{n-k}} f_{\mathbf{X}}(x_1, x_2, \dots, x_n),$$

where in each term of the sum, $x_i = y_i$ for i corresponding to variables in $\mathbf{Y}$, and $x_j = z_j$ for j corresponding to variables in $\mathbf{Z}$.

This definition captures the marginalization process in the discrete setting, which is key to understanding and analyzing the behavior of specific variables within a larger multivariate framework.

■ **Example 17.13** Consider two discrete random variables X and Y , each taking values in $0, 1$, with joint probability mass function:

$X \backslash Y$	0	1
0	0.1	0.3
1	0.2	0.4

To find the marginal probability mass function of X , sum over all possible values of Y :

$$\begin{aligned} f_X(0) &= f_{X,Y}(0,0) + f_{X,Y}(0,1) = 0.1 + 0.3 = 0.4, \\ f_X(1) &= f_{X,Y}(1,0) + f_{X,Y}(1,1) = 0.2 + 0.4 = 0.6. \end{aligned}$$

■

While the marginal probability mass functions of $X_1, \dots, X_n$ can be obtained from their joint probability mass function by summing over the range of the other variables, the reverse process is not possible without additional information about the dependence structure among the variables. Marginal probability mass functions describe only the individual behavior of each variable and do not capture interactions between them.

■ **Example 17.14** Let X and Y be discrete random variables, each taking values in $0, 1$, with marginal probability mass functions:

$$\begin{aligned} f_X(0) &= 0.5, & f_X(1) &= 0.5, \\ f_Y(0) &= 0.5, & f_Y(1) &= 0.5. \end{aligned}$$

Without additional information about the relationship between X and Y , the joint probability mass function cannot be reconstructed from the marginals alone. ■

A random vector $\mathbf{X}$ is said to have independent components if knowing the outcome of one component provides no information about the others. In this case, events defined by different components occur independently, and the joint probability mass function factorizes into the product of the marginals.

Definition 17.4.11 Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be an n -dimensional discrete random vector with joint probability mass function $f_{\mathbf{X}}$ and marginal probability mass functions $f_{X_1}, f_{X_2}, \dots, f_{X_n}$. The components $X_1, \dots, X_n$ are *independent* if, for all $(x_1, \dots, x_n)$ in the support of $\mathbf{X}$,

$$f_{\mathbf{X}}(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i).$$

The concept of independence for a random vector $\mathbf{X}$ simplifies the computation and understanding of joint probability distributions, particularly in complex problems involving multiple variables. It allows the joint probability distribution of the vector $\mathbf{X}$ to be expressed as the product of the individual marginal distributions of $X_1, X_2, \dots, X_n$.

Proposition 17.4.10 Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be an n -dimensional random vector with joint probability mass function $f_{\mathbf{X}}$ and marginal probability mass functions

$f_{X_1}, f_{X_2}, \dots, f_{X_n}$. The random variables $X_1, X_2, \dots, X_n$ are independent if and only if for every $(x_1, x_2, \dots, x_n)$ in the support of $\mathbf{X}$, we have:

$$f_{\mathbf{X}}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \dots f_{X_n}(x_n).$$

Proof. Assume that $X_1, X_2, \dots, X_n$ are independent. For any values $x_1, x_2, \dots, x_n$, we have:

$$\begin{aligned} f_{\mathbf{X}}(x_1, x_2, \dots, x_n) &= P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) \\ &= P(X_1 = x_1)P(X_2 = x_2) \dots P(X_n = x_n) \\ &= f_{X_1}(x_1)f_{X_2}(x_2) \dots f_{X_n}(x_n). \end{aligned}$$

Conversely, assume that for all $x_1, x_2, \dots, x_n$,

$$f_{\mathbf{X}}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \dots f_{X_n}(x_n).$$

Then, for any subsets $A_1, \dots, A_n$ of $\mathbb{R}$, we have:

$$\begin{aligned} P(X_1 \in A_1, \dots, X_n \in A_n) &= \sum_{(x_1, \dots, x_n) \in A_1 \times \dots \times A_n} f_{\mathbf{X}}(x_1, \dots, x_n) \\ &= \sum_{x_1 \in A_1} \dots \sum_{x_n \in A_n} f_{X_1}(x_1) \dots f_{X_n}(x_n) \\ &= \left(\sum_{x_1 \in A_1} f_{X_1}(x_1) \right) \dots \left(\sum_{x_n \in A_n} f_{X_n}(x_n) \right) \\ &= P(X_1 \in A_1) \dots P(X_n \in A_n). \end{aligned}$$

This equality holds for all subsets $A_1, \dots, A_n$, which implies that $X_1, \dots, X_n$ are independent. ■

17.4.3 Conditional Probability Mass Function

The concept of the conditional probability mass function provides a way to quantify the probability of an event given that another event has occurred. In the case of discrete random variables, the conditional probability mass function of Y given $X = x$ describes the probability mass function of Y under the condition that X takes a specific value x . This concept is essential for analyzing dependencies between discrete random variables, allowing us to refine probability assessments based on new information.

Definition 17.4.12 Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be an n -dimensional random vector with joint probability mass function $f_{\mathbf{X}}(x_1, x_2, \dots, x_n)$. Partition $\mathbf{X}$ into two subvectors: $\mathbf{Y} = (Y_1, Y_2, \dots, Y_k)$, a k -dimensional random vector consisting of k discrete random variables selected from $\mathbf{X}$, and $\mathbf{Z} = (Z_1, Z_2, \dots, Z_{n-k})$, the remaining $n - k$ discrete random variables of $\mathbf{X}$. Let $f_{\mathbf{Z}}$ be the marginal probability mass function of $\mathbf{Z}$ across its $n - k$ dimensions. Provided that

$f_{\mathbf{Z}}(\mathbf{z}) > 0$ for any vector $\mathbf{z} \in \mathbb{R}^{n-k}$, the *conditional probability mass function* $f_{\mathbf{Y}|\mathbf{Z}}$ of $\mathbf{Y}$ given $\mathbf{Z} = \mathbf{z}$ is defined as:

$$f_{\mathbf{Y}|\mathbf{Z}}(\mathbf{y} | \mathbf{z}) = \frac{f_{\mathbf{X}}(\mathbf{y}, \mathbf{z})}{f_{\mathbf{Z}}(\mathbf{z})}.$$

The following example illustrates how to compute the conditional probability mass function of Y given X .

■ **Example 17.15** Consider two discrete random variables X and Y , each taking values in $\{0, 1\}$, with the joint probability mass function:

$X \backslash Y$	0	1
0	0.3	0.2
1	0.1	0.4

The conditional probability mass function of Y given $X = 0$ is:

$$\begin{aligned} f_{Y|X}(0 | 0) &= \frac{f_{X,Y}(0,0)}{f_X(0)} = \frac{0.3}{0.5} = 0.6, \\ f_{Y|X}(1 | 0) &= \frac{f_{X,Y}(0,1)}{f_X(0)} = \frac{0.2}{0.5} = 0.4. \end{aligned}$$

Similarly, for $X = 1$:

$$\begin{aligned} f_{Y|X}(0 | 1) &= \frac{f_{X,Y}(1,0)}{f_X(1)} = \frac{0.1}{0.5} = 0.2, \\ f_{Y|X}(1 | 1) &= \frac{f_{X,Y}(1,1)}{f_X(1)} = \frac{0.4}{0.5} = 0.8. \end{aligned}$$

■

The next proposition generalizes the multiplication rule (see Equation 17.1) by combining marginal and conditional probability mass functions to recover the joint probability mass function for any configuration of discrete random variables within a random vector.

Proposition 17.4.11 Let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$, $f_{\mathbf{X}}(\mathbf{x})$, $f_{\mathbf{Z}}(\mathbf{z})$, and $f_{\mathbf{Y}|\mathbf{Z}}(\mathbf{y} | \mathbf{z})$ be as in Definition 17.4.12. Then, for each $\mathbf{z}$ such that $f_{\mathbf{Z}}(\mathbf{z}) > 0$ and each possible value of $\mathbf{y}$, the joint probability mass function is:

$$f_{\mathbf{X}}(\mathbf{x}) = f_{\mathbf{Y}|\mathbf{Z}}(\mathbf{y} | \mathbf{z})f_{\mathbf{Z}}(\mathbf{z}),$$

where $\mathbf{x} = (\mathbf{y}, \mathbf{z})$.

Proof. By the definition of the conditional probability mass function:

$$f_{\mathbf{Y}|\mathbf{Z}}(\mathbf{y} | \mathbf{z}) = \frac{f_{\mathbf{X}}(\mathbf{y}, \mathbf{z})}{f_{\mathbf{Z}}(\mathbf{z})}, \quad \text{for } f_{\mathbf{Z}}(\mathbf{z}) > 0.$$

Rearranging yields:

$$f_{\mathbf{X}}(\mathbf{y}, \mathbf{z}) = f_{\mathbf{Y}|\mathbf{Z}}(\mathbf{y} | \mathbf{z})f_{\mathbf{Z}}(\mathbf{z}),$$

and since $\mathbf{x} = (\mathbf{y}, \mathbf{z})$, the result follows. ■

Similarly, the conditional probability mass function of $\mathbf{Z}$ given $\mathbf{Y} = \mathbf{y}$, denoted $f_{\mathbf{Z}|\mathbf{Y}}(\mathbf{z} | \mathbf{y})$, satisfies:

$$f_{\mathbf{X}}(\mathbf{x}) = f_{\mathbf{Z}|\mathbf{Y}}(\mathbf{z} | \mathbf{y})f_{\mathbf{Y}}(\mathbf{y}).$$

Bayes' theorem (see Theorem 17.3.2) provides a way to update probability estimates for a hypothesis given new evidence. For random vectors, the theorem extends naturally to the multidimensional case.

Theorem 17.4.12 Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ and $\mathbf{Y} = (Y_1, Y_2, \dots, Y_m)$ be two random vectors representing distinct sets of discrete random variables. The generalized Bayes' theorem states:

$$P(\mathbf{X} = \mathbf{x} | \mathbf{Y} = \mathbf{y}) = \frac{P(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x})P(\mathbf{X} = \mathbf{x})}{P(\mathbf{Y} = \mathbf{y})},$$

where $P(\mathbf{Y} = \mathbf{y})$ is the marginal probability of $\mathbf{Y}$, given by:

$$P(\mathbf{Y} = \mathbf{y}) = \sum_{\mathbf{x}} P(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x})P(\mathbf{X} = \mathbf{x}).$$

Proof. From the definition of conditional probability:

$$P(\mathbf{X} = \mathbf{x} | \mathbf{Y} = \mathbf{y}) = \frac{P(\mathbf{X} = \mathbf{x}, \mathbf{Y} = \mathbf{y})}{P(\mathbf{Y} = \mathbf{y})}.$$

Since

$$P(\mathbf{X} = \mathbf{x}, \mathbf{Y} = \mathbf{y}) = P(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x})P(\mathbf{X} = \mathbf{x}),$$

substitution gives the stated result. ■

This generalized Bayes' theorem updates beliefs about $\mathbf{X}$ based on new evidence from $\mathbf{Y}$. It links the prior information we have about $\mathbf{X}$, the likelihood of observing $\mathbf{Y} = \mathbf{y}$ given $\mathbf{X} = \mathbf{x}$, and the evidence provided by the actual observation of $\mathbf{Y} = \mathbf{y}$.

Finally, building on the concept of independence between discrete random variables (see Definition 17.4.11), we define conditional independence. This concept comes into play when the independence of a set of discrete random variables is considered in the context of being conditioned on another set of variables.

Definition 17.4.13 Let $\mathbf{Z}$ be a random vector with joint probability mass function $f_{\mathbf{Z}}(\mathbf{z})$. The variables of the random vector $\mathbf{X} = (X_1, \dots, X_n)$ are *conditionally independent* given $\mathbf{Z}$ if, for all $\mathbf{z}$ such that $f_{\mathbf{Z}}(\mathbf{z}) > 0$:

$$f_{\mathbf{X}|\mathbf{Z}}(\mathbf{x} | \mathbf{z}) = \prod_{i=1}^n f_{X_i|\mathbf{Z}}(x_i | \mathbf{z}),$$

where $f_{\mathbf{X}|\mathbf{Z}}(\mathbf{x} | \mathbf{z})$ is the conditional probability mass function of $\mathbf{X}$ given $\mathbf{Z} = \mathbf{z}$, and $f_{X_i|\mathbf{Z}}(x_i | \mathbf{z})$ is the conditional probability mass function of X_i given $\mathbf{Z} = \mathbf{z}$.

17.5 Characterizing Distributions

A *measure of central tendency* is a number derived from a probability distribution that summarizes its typical or central value. The two most commonly used measures are the *expected value* and the *median*, each offering a different perspective for characterizing a distribution.

In addition to central tendency, it is often useful to describe how much the distribution varies around its center. For this purpose, *measures of dispersion* are employed, the most common being the *variance* and the *standard deviation*. These quantify the spread of a distribution around its central measure.

In the case of bivariate distributions, analogous measures of dispersion are the *covariance* and the *correlation*, which capture the strength and direction of the statistical relationship between two discrete random variables.

Together, these measures—central tendency and dispersion—allow us to summarize, compare, and analyze probability distributions in a concise and meaningful way.

17.5.1 Measures of Central Tendency

The most common measures of central tendency used to characterize probability distributions are the expected value and the median.

Expected Value

The expected value of a discrete random variable is computed as the weighted average of all its possible values, where the weights are the probabilities of the corresponding outcomes⁶.

⁶The term “expected value” can be misleading, because it does not necessarily coincide with one of the possible values of the random variable. For example, the expected value of a fair six-sided die is 3.5, which is not itself an attainable outcome. This counterintuitive aspect has often caused confusion in applications of probability.

Definition 17.5.1 Let X be a discrete random variable with probability mass function f . The *expected value* of X , denoted by $E(X)$, is defined as:

$$E(X) = \sum_x x f(x).$$

This definition depends only on the distribution of X , not on the original sample space. Consequently, two different random variables with the same distribution will have the same expected value, even if their underlying probability spaces are different.

A drawback of the expected value is that it can be heavily influenced by small changes in the probability assigned to large values of X .

■ **Example 17.16** Consider a company with 100 employees, and define a discrete random variable X based on their salaries. Suppose X takes the value 300 with probability 99/100 and 6000 with probability 1/100. The expected salary is:

$$E(X) = (300 \times 99/100) + (6000 \times 1/100) = 357.$$

Now suppose that one additional employee earns 6000 instead of 300. The expected salary becomes:

$$E(X) = (300 \times 98/100) + (6000 \times 2/100) = 414.$$

Thus, changing the salary of a single employee increases the expected salary of the company by more than 13%. ■

The expected value operator is linear: the expected value of a linear combination of discrete random variables equals the same linear combination of their expectations.

Proposition 17.5.1 Let $X_1, \dots, X_n$ be discrete random variables with expectations $E(X_i)$. For constants $a_1, \dots, a_n$ and b ,

$$E(a_1 X_1 + \dots + a_n X_n + b) = a_1 E(X_1) + \dots + a_n E(X_n) + b.$$

Proof. By definition,

$$\begin{aligned} E(a_1 X_1 + \dots + a_n X_n + b) &= \sum_{x_1, \dots, x_n} (a_1 x_1 + \dots + a_n x_n + b) f(x_1, \dots, x_n) = \\ &= a_1 \sum_{x_1} x_1 f(x_1) + \dots + a_n \sum_{x_n} x_n f(x_n) + b = a_1 E(X_1) + \dots + a_n E(X_n) + b. \end{aligned}$$

■

Note that independence is not required for linearity of expectation. Linearity holds for all discrete random variables as long as the expectations exist.

The expected value of the product of independent random variables equals the product of their expected values.

Proposition 17.5.2 Let $X_1, \dots, X_n$ be independent discrete random variables with expectations $E(X_i)$. Then:

$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n E(X_i).$$

Proof. By independence of $X_1, \dots, X_n$, their joint probability mass function factorizes:

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n).$$

Thus,

$$\begin{aligned} E(X_1 \cdots X_n) &= \sum_{x_1, \dots, x_n} (x_1 \cdots x_n) f(x_1, \dots, x_n) \\ &= \prod_{i=1}^n \left(\sum_{x_i} x_i f_{X_i}(x_i) \right) = \prod_{i=1}^n E(X_i). \end{aligned}$$

■

If the random variables are not independent, the equality above does not necessarily hold.

The Median

The median of a discrete random variable is a measure of central tendency that identifies a point dividing the probability distribution so that at least half of the probability mass lies on each side.

Definition 17.5.2 Let X be a discrete random variable. A *median* of X , denoted by m , is any value satisfying:

$$P(X \leq m) \geq \frac{1}{2} \quad \text{and} \quad P(X \geq m) \geq \frac{1}{2}.$$

This definition guarantees that at least half of the probability mass lies at or below m , and at least half lies at or above m . In general, the median of a discrete distribution need not be unique; any value satisfying the inequalities is considered a valid median. In applications, it is common to select the smallest such m .

The median is often more intuitive than the expected value, since it always corresponds to an actual value of the random variable. It is also a robust measure of central tendency, particularly when the distribution is skewed or contains outliers.

■ **Example 17.17** Consider the company from Example 17.16. The median of the salaries is the value m for which $P(X \leq m) \geq \frac{1}{2}$. In this case, $m = 300$. Now suppose that one of the base employees is promoted to the executive level with a salary of 6000. The recalculated median remains $m = 300$. This illustrates that the median is more robust than the expected value in the presence of outliers. ■

17.5.2 Measures of Dispersion

The most common measures of dispersion in use to characterize probability distributions are the variance and its squared root, called standard deviation.

The Variance

The variance of a discrete random variable is a measure of the spread or dispersion of its possible values around the expected value.

Definition 17.5.3 Let X be a discrete random variable with expected value $E(X)$ and probability mass function f . The *variance* of X , denoted by $Var(X)$, is defined as:

$$Var(X) = E[(X - E(X))^2] = \sum_x (x - E(X))^2 f(x)$$

Variance depends only on the distribution of the random variable and quantifies how much the values typically differ from the expected value. For example, if all possible values of a random variable are identical, the variance is zero.

■ **Example 17.18** Suppose the daily commuting time (in minutes) of an employee is modeled by a random variable X that takes values 20, 30, 40, 50 with probabilities

$$P(X = 20) = 0.1, P(X = 30) = 0.4, P(X = 40) = 0.4, P(X = 50) = 0.1.$$

The expected value is

$$E(X) = 20(0.1) + 30(0.4) + 40(0.4) + 50(0.1) = 35.$$

The variance is

$$\begin{aligned} Var(X) &= (20 - 35)^2(0.1) + (30 - 35)^2(0.4) + \\ &\quad (40 - 35)^2(0.4) + (50 - 35)^2(0.1) = 50. \end{aligned}$$

Thus, the variance of commuting times is 50 (measured in squared minutes). ■

The following proposition shows how the variance behaves under linear combinations of independent random variables.

Proposition 17.5.3 Let $X_1, \dots, X_n$ be independent discrete random variables with finite expected values, and let $a_1, \dots, a_n$ and b be constants. Then:

$$Var(a_1X_1 + \dots + a_nX_n + b) = a_1^2Var(X_1) + \dots + a_n^2Var(X_n).$$

Proof. Let $Y = a_1X_1 + \dots + a_nX_n + b$. Since adding a constant does not affect variance,

$$Var(Y) = Var(a_1X_1 + \dots + a_nX_n).$$

For independent random variables, variance is additive:

$$\text{Var}(a_1X_1 + \dots + a_nX_n) = \text{Var}(a_1X_1) + \dots + \text{Var}(a_nX_n).$$

Finally, for any constant a_i , $\text{Var}(a_iX_i) = a_i^2\text{Var}(X_i)$, so

$$\text{Var}(a_1X_1 + \dots + a_nX_n + b) = a_1^2\text{Var}(X_1) + \dots + a_n^2\text{Var}(X_n).$$

■

A particularly useful identity expresses variance in terms of expected values. This formulation often simplifies calculations and highlights the connection between variance and variability.

Proposition 17.5.4 Let X be a discrete random variable with expected value $E(X)$. Then:

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof. By definition,

$$\text{Var}(X) = E[(X - E(X))^2].$$

Expanding the square gives:

$$\text{Var}(X) = E[X^2 - 2XE(X) + (E(X))^2].$$

By linearity of expectation,

$$\text{Var}(X) = E(X^2) - 2E(X)E(X) + E((E(X))^2).$$

Since $E(X)$ is a constant, $E((E(X))^2) = (E(X))^2$, hence

$$\text{Var}(X) = E(X^2) - (E(X))^2.$$

■

Standard Deviation

The standard deviation is a measure of the dispersion of a distribution, closely related to the variance. Unlike variance, which squares the deviations from the mean and therefore uses squared units, the standard deviation is defined as the square root of the variance. This restores the measure to the same units as the original variable, making it easier to interpret.

Definition 17.5.4 Let X be a discrete random variable with expected value $E(X)$ and finite variance $Var(X)$, and let f be its probability mass function. The *standard deviation* of X , denoted by σ , is defined as:

$$\sigma = \sqrt{Var(X)} = \sqrt{\sum_x (x - E(X))^2 f(x)}.$$

The standard deviation summarizes how far the values of X typically lie from the mean. A smaller standard deviation indicates that the values are concentrated near the mean, while a larger standard deviation indicates greater spread.

The following proposition describes how the standard deviation transforms under linear changes of a random variable.

Proposition 17.5.5 Let X be a discrete random variable with standard deviation σ_X , and let a and b be constants. Then the standard deviation σ_Y of the random variable $Y = aX + b$ is

$$\sigma_Y = |a| \sigma_X.$$

Proof. The variance of Y is

$$Var(Y) = \sum_x (ax + b - (aE(X) + b))^2 f(x) = a^2 \sum_x (x - E(X))^2 f(x) = a^2 \sigma_X^2.$$

Taking square roots gives

$$\sigma_Y = |a| \sigma_X.$$

■

In practice, variance is often more convenient for theoretical derivations because of its algebraic properties, while standard deviation is usually preferred for interpretation, since it is expressed in the same units as the original data.

■ **Example 17.19** Continuing from Example 17.18 the standard deviation is

$$\sigma = \sqrt{Var(X)} = \sqrt{50} \approx 7.07.$$

Unlike variance, which is expressed in squared minutes, the standard deviation is measured in minutes, the same units as the commuting times themselves. This means that the typical commuting time deviates from the average of 35 minutes by about 7 minutes.

■

17.5.3 Measures of Statistical Relationship

Covariance and correlation are measures that describe the relationship between two variables. Covariance quantifies the direction of their linear association, indicating whether the variables tend to increase or decrease together. However, covariance is expressed in the product of the units of the two variables, making its magnitude difficult to interpret. Correlation, in contrast, standardizes this measure to a dimensionless quantity between -1 and 1 : a value of -1 indicates a perfect negative linear relationship, 1 a perfect positive linear relationship, and 0 the absence of a linear relationship.

Covariance

Covariance is a measure of the joint variability of two discrete random variables X and Y . It indicates the direction of their linear relationship: whether they tend to increase together, decrease together, or vary independently.

Definition 17.5.5 Let X and Y be two discrete random variables with finite expected values $E(X)$ and $E(Y)$. The *covariance* of X and Y , denoted by $\text{Cov}(X, Y)$, is defined as

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))].$$

The sign of the covariance reveals the direction of the relationship: a positive value means that as X increases, Y tends to increase; a negative value means that as X increases, Y tends to decrease. A covariance of zero indicates no linear relationship, though nonlinear dependencies may still exist and are not captured by covariance.

The following result provides a useful formula for computing covariance.

Proposition 17.5.6 For any two discrete random variables X and Y with finite expectations,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

Proof. By definition,

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))].$$

Expanding the product inside the expectation,

$$E[XY - XE(Y) - YE(X) + E(X)E(Y)].$$

Using linearity of expectation and the fact that $E(X)$ and $E(Y)$ are constants,

$$E(XY) - E(X)E(Y) - E(X)E(Y) + E(X)E(Y) = E(XY) - E(X)E(Y).$$



The following proposition shows the connection between independence and covariance.

Proposition 17.5.7 If X and Y are independent discrete random variables, then

$$\text{Cov}(X, Y) = 0.$$

Proof. From the identity above,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

If X and Y are independent, then $E(XY) = E(X)E(Y)$. Substituting,

$$\text{Cov}(X, Y) = E(X)E(Y) - E(X)E(Y) = 0.$$

■

The converse does not hold in general: a covariance of zero does not imply independence. Two random variables may be dependent yet uncorrelated.

Correlation

The magnitude of covariance depends on the scale of the variables, making it difficult to compare across different contexts. To address this, correlation standardizes covariance by dividing by the product of the standard deviations of the variables. This yields a dimensionless measure of the strength and direction of their linear relationship.

Definition 17.5.6 Let X and Y be two discrete random variables with finite, nonzero variances $\text{Var}(X)$ and $\text{Var}(Y)$. The *correlation* of X and Y , denoted by $\text{Cor}(X, Y)$ is defined as

$$\text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}.$$

Correlation takes values in the interval $[-1, 1]$.

Proposition 17.5.8 For any pair of random variables X and Y with finite nonzero variances,

$$-1 \leq \text{Cor}(X, Y) \leq 1.$$

Proof. Let $U = X - E(X)$ and $V = Y - E(Y)$. Then

$$\text{Cov}(X, Y) = E[UV].$$

By the Cauchy–Schwarz inequality,

$$|E[UV]|^2 \leq E[U^2]E[V^2].$$

Noting that $E[U^2] = \text{Var}(X)$ and $E[V^2] = \text{Var}(Y)$, we obtain

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}.$$

Dividing both sides by the product of the standard deviations gives

$$-1 \leq \text{Cor}(X, Y) \leq 1.$$

■

A value of $\text{Cor}(X, Y) = 1$ indicates a perfect positive linear relationship, $\text{Cor}(X, Y) = -1$ indicates a perfect negative linear relationship, and $\text{Cor}(X, Y) = 0$ indicates no linear relationship.

■ **Example 17.20** Consider two variables: weekly hours of exercise (X) and monthly weight loss in kilograms (Y). Suppose the following data are observed:

Person	X	Y
1	5	1
2	10	2
3	15	3
4	20	3.5
5	25	4

Computing the correlation coefficient yields ≈ 0.98 , indicating a very strong positive linear relationship. This means that individuals who exercise more hours per week tend to lose more weight. However, it is important to note that correlation measures association, not causation. ■

The following proposition establishes that if two discrete random variables are independent and have finite variances, their correlation must be zero.

Proposition 17.5.9 If X and Y are independent discrete random variables with finite, nonzero variances, then

$$\text{Cor}(X, Y) = 0.$$

Proof. From the definition,

$$\text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}.$$

If X and Y are independent, then $E(XY) = E(X)E(Y)$, so $\text{Cov}(X, Y) = 0$. Substituting into the formula above gives $\text{Cor}(X, Y) = 0$. ■

Note, however, that the converse is not true: correlation equal to zero does not imply independence, since variables can be dependent in nonlinear ways.

17.6 Common Distributions

In this section, we define and discuss several important families of distributions that are widely used in applications of probability theory. Specifically, we introduce the following families of discrete distributions: uniform, Bernoulli, binomial, and discrete normal. For each family, we briefly describe how it arises in practical problems and why it provides a suitable probability model for certain types of experiments. We also present the explicit form of the probability mass function and highlight some of the fundamental properties of the distributions in each family.

17.6.1 Uniform Distribution

The uniform distribution is the simplest probability distribution in probability theory. It models situations in which all outcomes are equally likely.

Definition 17.6.1 A discrete random variable X is said to follow a *uniform distribution* if all outcomes have the same probability. Specifically, if X takes values in the set $x_1, x_2, \dots, x_n$, the probability mass function is

$$P(X = x_i) = \frac{1}{n}, \quad i = 1, 2, \dots, n.$$

A classic example of the discrete uniform distribution is the outcome of rolling a fair die, where each face has probability $1/6$. Unlike other distributions introduced in this section, the uniform distribution is determined solely by its set of possible values, with no additional parameters required.

For a discrete uniform random variable X taking values $x_1, x_2, \dots, x_n$, the expected value and variance are given by

$$E(X) = \frac{x_1 + x_2 + \dots + x_n}{n}, \quad \text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - E(X))^2.$$

17.6.2 Bernoulli Distribution

The Bernoulli distribution is one of the simplest and most fundamental in probability theory. It models experiments with only two possible outcomes, commonly referred to as "success" and "failure."

Definition 17.6.2 A discrete random variable X is said to follow a *Bernoulli distribution* with parameter $p \in [0, 1]$, written $X \sim \text{Bernoulli}(p)$, if it takes the value 1 (success) with probability p and the value 0 (failure) with probability

$1 - p$. Its probability mass function is

$$P(X = x) = \begin{cases} p & \text{if } x = 1, \\ 1 - p & \text{if } x = 0. \end{cases}$$

The Bernoulli distribution arises naturally in many applied settings, especially those involving binary outcomes. For example, in medical studies, the result of a treatment can be modeled as a Bernoulli random variable, with "success" representing recovery and "failure" representing no improvement.

The expected value of $X \sim \text{Bernoulli}(p)$ is $E(X) = p$. Intuitively, this reflects the long-run average outcome of many trials. The variance is

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1 - p).$$

This variance is maximized when $p = 0.5$, corresponding to maximum uncertainty when success and failure are equally likely.

Definition 17.6.3 A sequence of independent and identically distributed random variables $X_1, X_2, \dots$, each with $X_i \sim \text{Bernoulli}(p)$, is called a sequence of *Bernoulli trials* with parameter p . An infinite sequence of Bernoulli trials is called a *Bernoulli process*.

The Bernoulli process serves as the foundation for more complex stochastic processes and underlies many important models, including the binomial distribution.

17.6.3 Binomial Distribution

The binomial distribution is a discrete probability distribution that models the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success. It is one of the most widely used distributions in probability theory for problems involving repeated binary outcomes.

Definition 17.6.4 A discrete random variable X is said to follow a *binomial distribution* with parameters n and p if it represents the number of successes in n independent Bernoulli trials, each with success probability p . The probability mass function is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad k = 0, 1, 2, \dots, n.$$

Here, n is the number of trials, and p is the probability of success on each trial. The coefficient $\binom{n}{k}$ counts the number of ways to obtain exactly k successes in n trials.

The binomial distribution arises naturally in many applications where outcomes can be classified as either "success" or "failure."

■ **Example 17.21** Suppose a company produces light bulbs, and historically, 95% of them are functional (success) while 5% are defective (failure). The company tests a random sample of 20 bulbs. Let X be the number of functional bulbs. Then $X \sim \text{Binomial}(n = 20, p = 0.95)$. For example, the probability that exactly 18 bulbs are functional is:

$$P(X = 18) = \binom{20}{18} (0.95)^{18} (0.05)^2.$$

■

The expected value of a binomial random variable X with parameters n and p is

$$E(X) = np,$$

representing the average number of successes in n trials. Its variance is

$$\text{Var}(X) = np(1 - p),$$

which measures the variability of the number of successes around the expected value.

17.6.4 Discrete Normal Distribution

The normal distribution is a continuous probability distribution that frequently appears in natural and social sciences, primarily due to the Central Limit Theorem (see Theorem 17.7.5). However, many real-world situations involve discrete outcomes, such as counts of events or integer-valued random variables. In these cases, it is desirable to have a discrete analog of the normal distribution. This is not a limitation, but rather a reflection of the fact that continuous distributions often arise as limiting abstractions of underlying discrete processes.

In this section, we introduce the *discrete normal distribution*, which serves as a discrete counterpart to the normal distribution for integer-valued random variables. Although not as standardized as other families such as the Bernoulli or binomial distributions, the discrete normal can provide a useful tool in applications where outcomes must remain integer-valued. propositions.

To define the discrete normal distribution, we first introduce the concept of a standardized discrete random variable.

Definition 17.6.5 Let X be a discrete random variable with expected value $E(X)$ and variance $\text{Var}(X)$. The *standardized discrete random variable* of X ,

denoted by Z , is defined as:

$$Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}}.$$

This transformation centers the distribution around 0 and rescales it so that the variance is 1.

■ **Example 17.22** If $X \sim \text{Bin}(n, p)$, then its standardized version is

$$Z = \frac{X - np}{\sqrt{np(1-p)}}.$$

The random variable Z takes values in the set

$$\left\{ \frac{k - np}{\sqrt{np(1-p)}} : k = 0, 1, \dots, n \right\},$$

with probability mass function

$$P\left(Z = \frac{k - np}{\sqrt{np(1-p)}}\right) = \binom{n}{k} p^k (1-p)^{n-k}.$$

■

The normal distribution arises as the limit of the standardized binomial distribution as $n \rightarrow \infty$. However, because the binomial is discrete and the normal is continuous, it is natural to consider a discrete distribution whose probabilities are proportional to the normal density evaluated at integer points.

Definition 17.6.6 A discrete random variable X is said to follow a *discrete normal distribution* with parameters μ and σ^2 if it has probability mass function:

$$P(X = k) = \frac{1}{Z} \exp\left(-\frac{(k - \mu)^2}{2\sigma^2}\right), \quad k \in \mathbb{Z},$$

where Z is the normalization constant:

$$Z = \sum_{k=-\infty}^{\infty} \exp\left(-\frac{(k - \mu)^2}{2\sigma^2}\right).$$

This distribution retains the discrete nature of integer-valued random variables while exhibiting the bell-shaped curve characteristic of the continuous normal distribution.

Proposition 17.6.1 The normalization constant Z in the discrete normal distribution satisfies:

$$Z \approx \sqrt{2\pi\sigma^2},$$

for large σ , and thus the probability mass function can be approximated by:

$$P(X = k) \approx \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(k - \mu)^2}{2\sigma^2}\right).$$

Proof. For large σ , the sum in Z can be approximated by an integral:

$$Z = \sum_{k=-\infty}^{\infty} \exp\left(-\frac{(k - \mu)^2}{2\sigma^2}\right) \approx \int_{-\infty}^{\infty} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx = \sqrt{2\pi\sigma^2}.$$

■

For large σ , the discrete normal approaches the continuous normal probability den evaluated at integer points.

■ **Example 17.23** Suppose we model the number of defective items in a production run of 100 units, where each item is defective with probability $p = 0.1$ independently. Then $X \sim \text{Bin}(100, 0.1)$, with

$$E(X) = 100 \times 0.1 = 10, \quad \text{Var}(X) = 100 \times 0.1 \times 0.9 = 9.$$

For large n , X may be approximated by the discrete normal distribution with parameters $\mu = 10$ and $\sigma^2 = 9$. The probability of exactly 12 defective items is approximated by

$$P(X = 12) \approx \frac{1}{Z} \exp\left(-\frac{(12 - 10)^2}{2 \times 9}\right).$$

This provides an alternative to the usual continuous normal approximation, avoiding the need for continuity corrections and keeping the support discrete. ■

The following proposition highlights an important property of the discrete normal distribution, specifically its symmetry around the mean.

Proposition 17.6.2 The discrete normal distribution is symmetric about its mean μ . Formally, for any integer k ,

$$P(X = \mu + k) = P(X = \mu - k).$$

Proof. From the definition,

$$P(X = \mu + k) = \frac{1}{Z} \exp\left(-\frac{k^2}{2\sigma^2}\right),$$

and similarly,

$$P(X = \mu - k) = \frac{1}{Z} \exp\left(-\frac{(-k)^2}{2\sigma^2}\right) = \frac{1}{Z} \exp\left(-\frac{k^2}{2\sigma^2}\right).$$

Thus the distribution is symmetric about μ . ■

The discrete normal distribution serves as a bridge between discrete and continuous probability models. It inherits key properties of the normal distribution—such as symmetry and centrality—while remaining defined on the integers. This makes it particularly useful in contexts such as manufacturing, epidemiology, or educational testing, where outcomes are integer-valued but approximately normal in shape.

17.7 Large Random Samples

A random sample is a collection of independent and identically distributed random variables. Random samples are fundamental in probability theory, providing the foundation for making inferences about an unknown probability distribution. Two key results that arise from random samples are the law of large numbers, which ensures that the sample mean converges to the expected value of the distribution as the sample size increases, and the central limit theorem, which states that the distribution of the sample mean approaches a normal distribution as the sample size grows, regardless of the original population's distribution.

Random sampling assumes that every member of the population has an equal probability of being selected and that selections are independent of one another. In this sense, the sample is an idealized representation of how real-world data might be generated. Perfect random sampling is rarely achievable in practice, but it provides a baseline against which deviations can be evaluated. This framework allows us to construct probability distributions describing the expected behavior of sample statistics (e.g., sample mean, variance), making it possible to quantify uncertainty and to draw principled conclusions about the population.

The random sampling model is therefore a powerful representation of how data might be collected, though it is not identical to reality. Like all models, it works well in specific contexts, but its assumptions do not universally hold. The practical usefulness of random sampling depends on how closely the model aligns with the real-world situation under study.

17.7.1 Random Sample

The concept of a random sample is pivotal in the field of statistical learning, since assuming that a set of discrete random variables forms a random sample greatly simplifies the mathematical foundations of inference. However, the criteria

required for a collection of variables to be considered a random sample are not always satisfied in real-world scenarios.

Definition 17.7.1 Let f be a probability mass function, and let $X_1, X_2, \dots, X_n$ be a collection of discrete random variables. The variables $X_1, X_2, \dots, X_n$ form a *random sample* from the distribution f if each X_i follows the probability mass function f and the variables $X_1, X_2, \dots, X_n$ are mutually independent. Such variables are called *independent and identically distributed (i.i.d.)*. The number n of variables is called the *sample size*.

The joint probability mass function g of a random sample $X_1, X_2, \dots, X_n$ is

$$g(x_1, x_2, \dots, x_n) = f(x_1)f(x_2) \dots f(x_n).$$

■ **Example 17.24** A factory produces light bulbs, and the quality control department is interested in estimating the proportion of defective bulbs. The number of defectives in a batch of 100 bulbs is modeled by a binomial distribution with parameter p , the probability of a bulb being defective. Suppose the department randomly selects 5 batches, each of 100 bulbs. Let $X_1, \dots, X_5$ denote the number of defectives in each batch. Each $X_i \sim \text{Binomial}(100, p)$, and assuming the batches are independent, the variables $X_1, \dots, X_5$ form a random sample. ■

In statistical inference, it is convenient to compute the sample mean, as the average of n random variables.

Definition 17.7.2 Let $X_1, X_2, \dots, X_n$ be n discrete random variables. The *sample mean*, denoted by $\bar{X}_n$, is:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Do not confuse the random variable $\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$ with the population mean $E(X_i)$, which is a fixed real number. Recall that $X_1 + \dots + X_n$ defines a new random variable on the Cartesian product of the original sample spaces, not a sum of probability distributions.

The following proposition shows that the sample mean provides an unbiased estimate of the population mean, and that its variance decreases as the sample size increases.

Proposition 17.7.1 Let $X_1, \dots, X_n$ be a random sample with finite mean $E(X_i) = \mu$ and variance $\text{Var}(X_i) = \sigma^2$. Then

$$E(\bar{X}_n) = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Proof. By linearity of expectation,

$$E(\bar{X}_n) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \mu.$$

For the variance,

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right).$$

Since the X_i are independent,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2,$$

so

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

■

Thus, the sample mean is unbiased, and its variance shrinks at rate $1/n$. As n grows, the distribution of $\bar{X}_n$ becomes more concentrated around the population mean μ .

In statistical inference, it is also convenient to compute the sample variance as a measure of the dispersion of the sample values.

Definition 17.7.3 Let $X_1, X_2, \dots, X_n$ be a set of discrete random variables with sample mean $\bar{X}_n$. The *sample variance*, denoted by S^2 , is:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

The factor $1/(n-1)$ ensures that S^2 is an unbiased estimator of the population variance when the X_i are i.i.d. Do not confuse the sample variance, which is computed from observed data, with the theoretical variance of the underlying distribution.

■ **Example 17.25** Suppose we observe $X_1 = 6, X_2 = 4, X_3 = 5, X_4 = 7, X_5 = 3$. Then

$$\bar{X}_5 = \frac{6+4+5+7+3}{5} = 5,$$

and

$$S^2 = \frac{1}{4} [(6-5)^2 + (4-5)^2 + (5-5)^2 + (7-5)^2 + (3-5)^2] = \frac{10}{4} = 2.5.$$

■

The sample standard deviation is used to estimate the standard deviation of a population based on a sample taken from it. Unlike the population standard deviation, which uses the true mean of the entire population, the sample standard deviation uses the mean of the sample as an estimate of the true mean.

Definition 17.7.4 Let $X_1, X_2, \dots, X_n$ be n discrete random variables with sample mean $\bar{X}_n$ and sample variance S^2 . The *sample standard deviation*, denoted by S , is:

$$S = \sqrt{S^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}$$

■ **Example 17.26** Using the variance from the previous example, the sample standard deviation is

$$S = \sqrt{2.5} \approx 1.58.$$

■

Finally, when two variables are sampled together, the sample covariance estimates the covariance between their populations.

Definition 17.7.5 Let $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_n$ be two sets of discrete random variables with sample means $\bar{X}$ and $\bar{Y}$ respectively. The *sample covariance* is:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

■ **Example 17.27** Suppose we observe the paired data

i	1	2	3	4	5
X_i	2	4	6	8	10
Y_i	1	3	5	7	9

Then

$$\bar{X} = 6, \quad \bar{Y} = 5,$$

and

$$s_{XY} = \frac{1}{4} [(-4)(-4) + (-2)(-2) + (0)(0) + (2)(2) + (4)(4)] = \frac{40}{4} = 10.$$

■

17.7.2 Law of Large Numbers

The law of large numbers states that the sample mean of a large random sample, i.e., a large number of independent and identically distributed (i.i.d.) discrete random variables, will be close to the expected value of the distribution, and that as we increase the sample size, the sample mean gets closer (in a precise probabilistic sense) to that value. Before proving this theorem, we establish two related propositions: Markov's inequality and Chebyshev's inequality.

Markov's inequality provides a simple bound on how unlikely it is for a nonnegative discrete random variable to take a large value, given its expected value. Intuitively, if the average value of a nonnegative random variable X is small, then X is unlikely to be very large (recall Example 17.16). Markov's inequality is particularly useful because it requires no assumptions on the distribution of X beyond nonnegativity and finiteness of the mean.

Proposition 17.7.2 — Markov's Inequality. Let X be a nonnegative discrete random variable, i.e. $P(X \geq 0) = 1$, with expected value $E(X)$. Then for every real number $t > 0$ we have that

$$P(X \geq t) \leq \frac{E(X)}{t}$$

Proof. Let f be the probability mass function of X . Since X is non-negative we have that $E(X) = \sum_{x>0} xf(x)$. Then

$$\begin{aligned} E(X) &= \sum_{x>0} xf(x) = \sum_{x=0}^t xf(x) + \sum_{x=t}^{\infty} xf(x) \geq \\ &\sum_{x=t}^{\infty} xf(x) \geq \sum_{x=t}^{\infty} tf(x) = t \sum_{x=t}^{\infty} f(x) = tP(X \geq t) \end{aligned}$$

that is, $E(X) \geq tP(X \geq t)$. ■

For example, if $E(X) = 10$ and we ask for $P(X \geq 50)$, Markov's inequality gives $P(X \geq 50) \leq 10/50 = 0.2$. While Markov's inequality is very general and easy to use, its bound is often not tight because it uses only the mean of X .

Chebyshev's inequality bounds the probability that a random variable deviates from its expected value by more than a specified number of standard deviations. Unlike Markov's inequality, which only requires non-negativity, it does not require X to be nonnegative, but it requires knowledge of both the mean and variance.

Corollary 17.7.3 — Chebyshev's Inequality. Let X be a discrete random variable with finite mean $E(X)$ and finite variance $\text{Var}(X)$. Then for every

$t > 0$,

$$P(|X - E(X)| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

Proof. Apply Markov's inequality to the nonnegative random variable $(X - E(X))^2$:

$$P(|X - E(X)| \geq t) = P((X - E(X))^2 \geq t^2) \leq \frac{E[(X - E(X))^2]}{t^2} = \frac{\text{Var}(X)}{t^2}.$$

■

Chebyshev's inequality formalizes the idea that most of the probability mass of a distribution lies near its mean: the probability of being several standard deviations away decays at least on the order of $1/t^2$.

The last notion we need for the law of large numbers is convergence in probability. Intuitively, a sequence of random variables converges in probability to b if the values X_n concentrate around b as n increases.

Definition 17.7.6 Let $X_1, X_2, \dots$ be a sequence of discrete random variables. It is said that the sequence $X_1, X_2, \dots$ *converges in probability* to b , denoted by $X_n \xrightarrow{P} b$, if for every positive real number $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - b| < \varepsilon) = 1$$

Given these elements, we can state and prove the (weak) law of large numbers.

Theorem 17.7.4 — Law of Large Numbers. Let $X_1, X_2, \dots$ be i.i.d. random variables with finite mean $E(X_i) = \mu$ and finite variance $\text{Var}(X_i) = \sigma^2$. Let $\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$ be the sample mean. Then

$$\bar{X}_n \xrightarrow{P} \mu.$$

Proof. By Proposition 17.7.1, $E(\bar{X}_n) = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$. Applying Chebyshev's inequality,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

From there we can obtain

$$P(|\bar{X}_n - \mu| < \varepsilon) = 1 - P(|\bar{X}_n - \mu| \geq \varepsilon) \geq 1 - \frac{\sigma^2}{n\varepsilon^2}.$$

As n approaches infinity, the above expression approaches 1. Thus,

$$\bar{X}_n \xrightarrow{P} \mu$$

■

It is important to note that the assumptions (independence and identical distribution, with finite variance) are essential. The theorem is an asymptotic statement: for a finite sample, the sample mean is typically close to the population mean, but not necessarily equal to it.

■ **Example 17.28** Consider rolling a fair six-sided die with sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $P(\omega) = 1/6$ for each outcome. Let $X : \Omega \rightarrow \mathbb{R}$ map each side to its face value, so $E(X) = 3.5$.

First take a random sample of size two: roll the die twice (in sequence, so the rolls are distinguishable) and let X_1 and X_2 be the outcomes. The sample mean is the random variable

$$\frac{X_1 + X_2}{2} : \Omega \times \Omega \rightarrow \mathbb{R}, \quad (\omega_1, \omega_2) \mapsto \frac{X_1(\omega_1) + X_2(\omega_2)}{2}.$$

whose probability mass function is depicted in Figure 17.1. A direct calculation shows

$$\begin{aligned} P\left(\left|\frac{X_1 + X_2}{2} - E(X)\right| < 1\right) &= P(|X_1 + X_2 - 7| < 2) \\ &= P(X_1 + X_2 \in \{6, 7, 8\}) = \frac{16}{36} \approx 0.444. \end{aligned}$$

Now consider a random sample of size ten. The distribution of

$$\frac{X_1 + \cdots + X_{10}}{10} - E(X)$$

is much more concentrated around 0 (see Figure 17.2); numerically,

$$P\left(\left|\frac{X_1 + \cdots + X_{10}}{10} - 3.5\right| < 1\right) \approx 0.97,$$

illustrating the sharpening concentration predicted by the law of large numbers. ■

Finally, the law of large numbers concerns the *sample mean* $\frac{1}{n} \sum_{i=1}^n X_i$, not other aggregates. For instance, the deviation from the expected total,

$$\sum_{i=1}^n X_i - nE(X),$$

does not converge to 0; in fact, its typical size grows on the order of $\sqrt{n}$ (see Example 17.29). Confusing these statements underlies a common misconception.

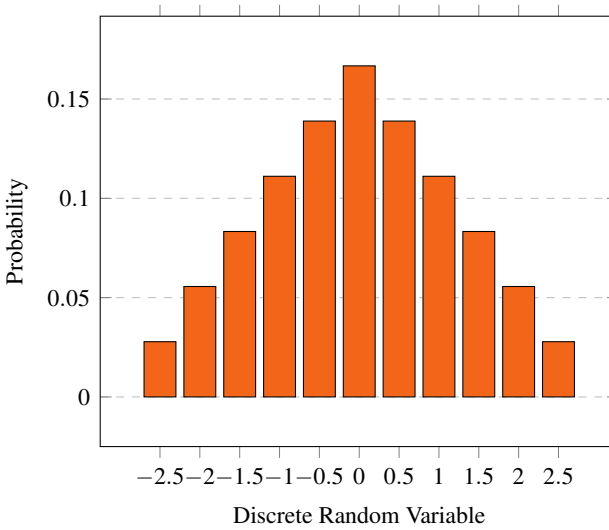


Figure 17.1: Probability mass function of the discrete random variable $\frac{X_1+X_2}{2} - \mathbb{E}(X)$.

■ **Example 17.29** If we toss a fair coin, the probability of heads is $1/2$. By the law of large numbers, the *proportion* of heads in many tosses is close to $1/2$. However, the *difference* between the number of heads and tails need not be close to 0; in fact, its typical magnitude grows like a constant multiple of $\sqrt{n}$ as the number of tosses n increases. The belief that the counts of heads and tails should stay nearly equal as n grows is a form of the gambler's fallacy. ■

17.7.3 Central Limit Theorem

Let $X_1, \dots, X_n$ be a sample of n independent and identically distributed random variables with mean μ and variance σ^2 . As we saw in the previous section, the law of large numbers states that the sample average $\bar{X}_n$ converges in probability to μ as n increases. The central limit theorem strengthens this by describing the distribution of the centered and scaled sample mean: the quantity $\sqrt{n}(\bar{X}_n - \mu)$ converges in distribution to a normal random variable with mean 0 and variance σ^2 . Equivalently, the standardized statistic

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

converges in distribution to the standard normal distribution $N(0, 1)$. This result holds regardless of the shape of the underlying distribution, provided it has finite variance.

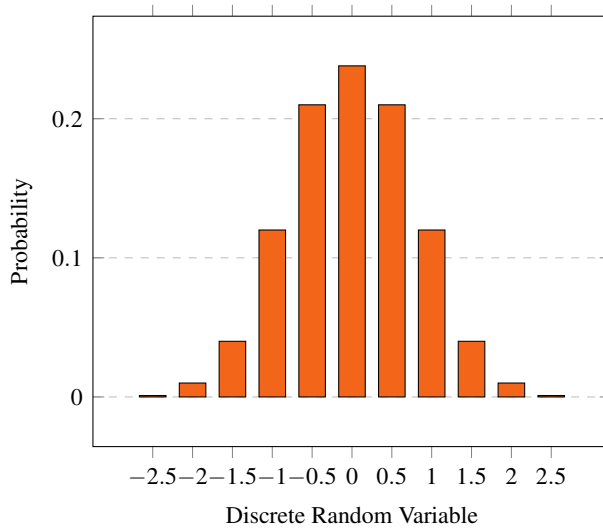


Figure 17.2: Probability mass function of the discrete random variable $\frac{X_1 + X_2 + \dots + X_{10}}{10} - E(X)$.

Theorem 17.7.5 — Central Limit Theorem. Let $X_1, \dots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2 . Define

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}, \quad \text{where} \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$,

$$Z_n \Rightarrow N(0, 1),$$

i.e., Z_n converges in distribution to the standard normal law with density $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$. Equivalently,

$$\sqrt{n}(\bar{X}_n - \mu) \Rightarrow N(0, \sigma^2).$$

Proof sketch. We first consider the special case of Bernoulli random variables. Let $X_i \sim \text{Bernoulli}(p)$ and $S_n = \sum_{i=1}^n X_i$. The standardized sum is

$$Z_n = \frac{S_n - np}{\sqrt{np(1-p)}}.$$

Using Stirling's approximation for factorials in the binomial formula, one shows

that for integers k close to np (within order $\sqrt{n}$),

$$P(S_n = k) \approx \frac{1}{\sqrt{2\pi np(1-p)}} \exp\left(-\frac{(k - np)^2}{2np(1-p)}\right).$$

This is exactly the probability mass function of a discretized normal distribution. Summing over such k produces a Riemann sum converging to the normal distribution function, proving the result for Bernoulli variables.

For general i.i.d. random variables with finite mean and variance, one reduces to the Bernoulli case by truncating the variables to control large deviations and matching means and variances. The same approximation then applies, and the limiting distribution is again standard normal. ■

The central limit theorem is a cornerstone of probability and statistics. It justifies normal approximations for sampling distributions of means and enables approximate inference. The basic conditions used here are i.i.d. observations with finite variance.

■ **Example 17.30** Suppose a factory produces light bulbs and the lifetime of each bulb (measured, say, in hours) has mean $\mu = 1000$ and standard deviation $\sigma = 50$. A random sample of $n = 36$ bulbs is tested. What is the probability that the sample mean lifetime lies between 990 and 1010 hours?

By the CLT, for $n = 36$ the sample mean $\bar{X}_n$ is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n} = 50/6 \approx 8.33$:

$$\bar{X}_n \approx N\left(1000, \left(\frac{50}{\sqrt{36}}\right)^2\right).$$

Standardizing,

$$z_{990} = \frac{990 - 1000}{8.33} \approx -1.20, \quad z_{1010} = \frac{1010 - 1000}{8.33} \approx 1.20.$$

Thus

$$\begin{aligned} P(990 \leq \bar{X}_n \leq 1010) &\approx P(-1.20 \leq Z \leq 1.20) = \Phi(1.20) - \Phi(-1.20) \\ &\approx 0.8849 - 0.1151 = 0.7698. \end{aligned}$$

So the probability is about 76.98%. ■

References

[DeG+86] is a widely respected textbook in the fields of statistics and probability theory. First published in 1975, this book is known for its clear exposition of the

fundamental concepts of probability and statistics, making it suitable for both beginners and those with some background in the subject. The book's approach balances theory and application, making it useful both for learning theoretical underpinnings and for applying probability and statistics to real-world problems.

[Chi13] offers a comprehensive introduction to the foundational aspects of probability, with a focus on the philosophical questions it raises. Childers explores various interpretations of probability, including frequentist, propensity, classical, Bayesian, and objective Bayesian, and presents these complex ideas in a way that is accessible even to those without a strong background in probability or mathematics.

An example of the problems associated with the misinterpretation of expected value is the St. Petersburg Paradox. Introduced by Nicholas Bernoulli in 1713, this paradox involves a gambling game with an infinite expected payoff, yet no reasonable person would pay more than \$25 to play it. Despite being three centuries old, the paradox continues to inspire new arguments and solutions in recent years (see [Hua13] for a historical review of the main proposed solutions).



18. Computability

*Wanderer, there is no road,
the road is made by walking.*
Antonio Machado

We continue our review of the background required to understand the theory of nescience by providing a mathematical formalization of the concept of a *computable procedure*. Intuitively, a computable procedure is a method consisting of a finite sequence of instructions that, when applied to a problem, produces the correct answer after a finite number of steps. This informal notion rests on the requirement that the instructions be clear and precise enough for any human to follow without additional guidance. We may strengthen this requirement further by demanding that the instructions be so unambiguous that a machine could execute them.

In 1936, the British mathematician Alan Turing introduced a formal model of a family of hypothetical machines and argued that, for every intuitively computable procedure, there exists a *Turing machine* capable of carrying it out. The model is not only simple enough to allow precise mathematical analysis but also sufficiently general to capture the intuitive concept of effective computation.

Over the years, many alternative approaches have been proposed to formalize the notion of computability. Some of these are technically intricate, yet all have

been shown to be equivalent in expressive power to Turing machines; in other words, they characterize the same class of effectively computable functions. Two notable examples are Alonzo Church's *lambda calculus* and the *theory of recursive functions* developed by Kurt Gödel and Stephen Kleene. The *Church-Turing thesis* asserts that any reasonable formalization of a computable procedure (subject to minimal requirements, such as performing only a finite amount of work in a single step) coincides with the Turing machine model. While not a theorem, this thesis has become a widely accepted working principle, providing a stable and shared notion of computability that is independent of any specific formalism.

The concept of the Turing machine, originally conceived as a model of a mechanical device designed to solve a particular problem, was later extended and generalized. A *universal Turing machine* can simulate the behavior of any other Turing machine and thereby compute any function that is computable in the intuitive sense. This is analogous to modern digital computers, which can execute algorithms expressed in diverse programming languages. This universality raises a natural question: Are there problems that no Turing machine can solve? The answer is yes. Certain well-defined problems, such as the Halting problem, lie beyond the reach of computation, and such problems are more common than one might initially expect. The existence of uncomputable functions will play a central role in our theory of nescience.

Given the abstract character of many entities studied in science, we employ the concept of the *oracle Turing machine* to formalize our framework. An oracle Turing machine resembles an ordinary Turing machine but is augmented with the ability to query an external oracle. The oracle, whose internal mechanism is unspecified, provides answers to questions that may be uncomputable for standard machines. Different oracles yield different computational powers, giving rise to a hierarchy of relative computability. For our purposes, the oracle can be viewed as a theoretical construct that models access to sources of information beyond the limits of mechanical computation. It should be emphasized that the oracle is not a physical device but an abstract tool for reasoning about the boundaries of computation.

Turing machines reveal the intrinsic limitations of computation. Exploring these limitations is not merely a philosophical exercise; it has profound implications for the field of *computational complexity*. Situated at the intersection of computer science and mathematics, computational complexity studies the resources (most notably time and space) required to solve problems. Problems are classified into *complexity classes* according to the asymptotic resources needed by the best algorithms known for their solution. One of the most famous open problems in this field is the $P \stackrel{?}{=} NP$ question, which asks whether the class P of problems solvable in polynomial time coincides with the class NP of

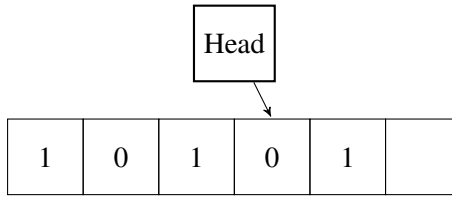


Figure 18.1: Turing Machine

problems for which proposed solutions can be verified in polynomial time. In this book, our interest extends beyond the epistemological issue of determining which problems can be solved in principle, given unlimited resources, to the more practical question of which problems can be solved efficiently with respect to time.

18.1 Turing Machines

A Turing machine is an extremely simplified model of a general-purpose computer, yet it is powerful enough to capture the notion of effective computation. Intuitively, the machine consists of a head that operates on a two-way infinite tape divided into cells, each containing a symbol. At each time step, the machine reads the symbol under the head and, according to a fixed set of rules, either writes a new symbol, moves the head one cell to the left or right, or performs both actions. The rules are encoded in a finite control table. The input to the computation is written on the tape at the start, and once the machine enters a designated halting state, the output can be read from the tape. Figure 18.1 illustrates a machine in its initial configuration, with the head positioned at the beginning of the input string.

Definition 18.1.1 — Turing Machine. A Turing machine is a 7-tuple

$$(Q, \Gamma, \sqcup, \Sigma, q_0, q_f, \tau)$$

where:

Q is a finite, non-empty set of *states*,

Γ is a finite, non-empty set of *tape symbols*,

$\sqcup \in \Gamma$ is the *blank symbol*,

$\Sigma \subseteq \Gamma \setminus \{\sqcup\}$ is the set of *input symbols*,

$q_0 \in Q$ is the *initial state*,

$q_f \in Q \setminus \{q_0\}$ is the designated *halting state*,

$\tau : (Q \setminus \{q_f\}) \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}$ is the *transition*.

The algorithm executed by the machine is defined by the (partial) transition function τ . This function dictates the machine’s actions based on its current state and the tape symbol currently under the head. According to τ , the machine transitions to a new state, writes a new symbol on the tape (or retains the existing one), and moves the head left, right, or keeps it stationary (L , R , or S respectively). The machine follows a finite, uniquely determined sequence of steps until it reaches the final state q_f ¹ and *halts*, making no subsequent moves. The algorithm’s output is the string of symbols $s \in \Gamma^*$ remaining on the tape after halting. Some machines, however, may enter an infinite loop, never reaching a halting state. If a machine encounters an undefined transition, it will enter an infinite loop, never halting. We assume that the head of the machine is initially positioned at the first symbol of the input string.

■ **Example 18.1** The following Turing machine is designed to solve the problem of adding two natural numbers. It consists of the set of states $Q = \{q_o, q_1, q_f\}$, the set of tape symbols $\Gamma = \{0, 1, \sqcup\}$, and the set of input symbols $\Sigma = \{0, 1\}$. The transition function is defined in the table below, where rows are indexed by machine states, and columns by tape symbols:

	0	1	$\sqcup$
q_o	$(q_f, \sqcup, R)$	$(q_1, \sqcup, R)$	$\uparrow$
q_1	$(q_f, 1, S)$	$(q_1, 1, R)$	$\uparrow$

Table 18.1: Transition Rules

For natural numbers n and m , the input string is composed of n occurrences of the symbol ‘1’, followed by a ‘0’, and then followed by m occurrences of ‘1’. The machine’s output will be a string of $n + m$ consecutive ‘1’s. For instance, to add the numbers 2 and 3, the input string should be $\sqcup 110111 \sqcup$, resulting in the output string $\sqcup 11111 \sqcup$. ■

A Turing machine can also be represented by a *state diagram*. A state diagram is similar to a labeled directed graph² where the vertices represent the states of the machine. The edges signify transitions from one state to another, and the edge labels indicate the symbol under the head that leads to the new state, the symbol that gets written on the tape, and the direction in which the head moves. Following these conventions, the state diagram for the Turing machine in Example 18.1 is depicted in Figure 18.2.

It is a remarkable fact that minor alterations to the definition of a Turing machine do not change its computational power. In other words, the definition is

¹Some authors allow multiple halting states in the definition of Turing machines.
²In this particular case, we allow loops and multiple edges originating from vertices.

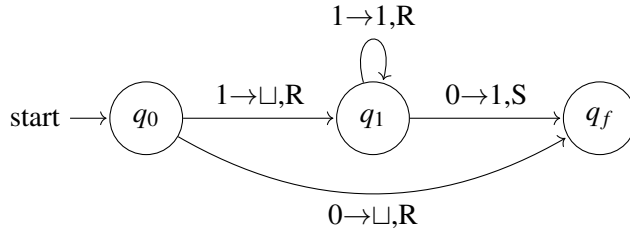


Figure 18.2: State diagram of the addition machine

highly robust. In Example 18.2, it's demonstrated that adding more tapes to the machine doesn't expand the range of problems it can solve. Similar arguments can be made when adding finite storage to the control, allowing for parallel processing with multiple control heads, and so on.

■ **Example 18.2** A *multitape Turing machine* is a Turing machine equipped with multiple heads and their respective tapes. In the initial configuration, the input string resides in tape 1, while the other tapes are blank. The transition function for a multitape Turing machine is:

$$\tau : (Q \setminus q_f) \times \Gamma^k \rightarrow Q \times \Gamma^k \times \{L, R, S\}^k,$$

where k denotes the number of tapes. Multitape Turing machines are equivalent in power to standard Turing machines. We can validate this claim by devising a method for a standard Turing machine to mimic a multitape machine's behavior. This requires encoding the content of multiple tapes onto a single tape, introducing a new symbol as a tape separator, and encoding the positions of the heads across the tapes with a distinct head location symbol. If we designate the tape separation symbol as $|$ and the head location symbol as h , a simulation tape for a machine with 3 tapes might appear as $\square 01h00|000h1|h0101\square$. The standard machine's operation would involve scanning the sub tapes one by one, pinpointing the head's location, and executing the necessary transition. If the computation on one sub tape necessitates writing a new symbol beyond its boundary, we'd need to shift subsequent symbols to accommodate the new one. While the simulation might operate at a slower pace than the original multitape machine, both machine types can solve an identical set of problems. ■

For the remainder of this book, without any loss of generality, we'll assume that the set of input symbols is $\Sigma = \mathcal{B}$ and the set of tape symbols is $\Gamma = \{0, 1, \square\}$.

In addition to providing a formal definition of a Turing machine, it's essential to formally outline its computational process. This entails detailing how the machine reads the input string, produces the output string, and transitions between states during computation. We will start by defining the concept of the machine's

internal configuration. This configuration captures the machine's current state and position, as well as the present state of the tape.

Definition 18.1.2 A *configuration* of a Turing machine T is the 3-tuple (q, s, i) , where $q \in Q$ represents a state of the machine, $s \in \Gamma^+$ denotes a string containing the tape's content (excluding the blank symbols), and $1 \leq i \leq n$ is the index of the symbol s_i beneath the head. Here, s_1 is the first symbol on the tape, and $n = l(s)$.

Configurations enable us to describe the current state of a Turing machine without any loss of information. At any stage of computation, one could halt the machine, record its configuration, and later resume the computation from the exact point of interruption using this configuration.

The following definition explains how we transition from one configuration to the next during computation.

Definition 18.1.3 A configuration $C = (q, s, i)$ *yields* another configuration $C' = (r, s', j)$ if there exists a transition $\tau : (q, s_i) = (r, s'_i, a)$, where

$$s = s_1 \dots s_{i-1} s_i s_{i+1} \dots s_n, s' = s_1 \dots s_{i-1} s'_i s'_{i+1} \dots s_n,$$

and

$$j = \begin{cases} i + 1 & \text{if } a = R \\ i - 1 & \text{if } a = L \\ i & \text{if } a = S \end{cases} \quad (18.1)$$

If $a = L$ and $i = 1$, we extend s on the left by a blank and keep the head at position 1 (now on $\sqcup$); if $a = R$ and $i = n$, we extend s on the right by a blank and set the head to position $n + 1$.

Building on the concepts of configuration and one configuration yielding another, we can now formally articulate the notion of computation.

Definition 18.1.4 — Computation. Let T be a Turing machine, C_0 its initial configuration, and C_n a configuration encompassing the final state q_f . A *computation* under machine T refers to a finite sequence of $n + 1$ configurations $(C_0, C_1, \dots, C_n)$ wherein each configuration C_k yields the subsequent configuration C_{k+1} , for all $0 \leq k < n$.

Computations are deterministic; meaning, for a given Turing machine T and an input string s , the configuration sequence is uniquely determined.. If machine T neither halts nor progresses with input s , we deduce the absence of computation.

■ **Example 18.3** The computation of the Turing machine described in Example

18.1 using the input string 110111 results in the following sequence of configurations:

- 1 $(q_0, 110111, 1)$
- 2 $(q_1, 10111, 1)$
- 3 $(q_1, 10111, 2)$
- 4 $(q_f, 11111, 2)$

■

Intuitively, a procedure is deemed computable by a human if it can be delineated through specific steps, executed systematically, without relying on intuition or ingenuity. This intuitive grasp aligns with the formalized concept of a Turing machine, bridging informal comprehension and the machine's rigorous definition—a cornerstone in the theory of computation. However, this alignment presents an intriguing challenge. Affirming that our grasp of computability mirrors a Turing machine's capabilities cannot be proven traditionally, as 'computability' lacks a well-defined interpretation. Consequently, some researchers categorize this as a *thesis*, avoiding the formal 'theorem' label. Turing himself opted to term it a *definition*, steering clear of denoting it as a theorem.

Theorem 18.1.1 — Turing's Thesis. A procedure is computable if, and only if, it can be executed by a Turing machine.

To further underscore the significance and robustness of the Turing machine as a model of computation, it's worth noting, as mentioned earlier in this chapter, that all alternative formalizations of computability proposed to date align in terms of their computational capabilities with that of the Turing machine. This universality underscores the Turing machine's central position in the realm of theoretical computer science.

18.2 Universal Turing Machines

In Section 18.1, we explored storing the current state of a Turing machine, its configuration, to pause and later resume computation. In Example 18.4, we will delve into a similar procedure, not for storing the machine's current state, but for saving a comprehensive description of the machine itself. This methodology facilitates the enumeration, or listing, of all possible Turing machines. Such enumeration is instrumental in demonstrating the existence of problems that cannot be solved by any Turing machine (refer to Section 18.3) and unveiling the pivotal concept of the *Universal Turing Machine*.

■ **Example 18.4** To describe a Turing machine concisely, we need to encode the transition function $\tau : (Q \setminus \{q_f\}) \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}$. This function can

be represented as a collection of quintuples (q, s, r, t, a) , where $q \in (Q \setminus \{q_f\})$, $r \in Q$, $s, t \in \Gamma$, and $a \in \{L, R, S\}$. In this manner, any Turing machine T is fully described by a collection of quintuples:

$$(q_1, s_1, r_1, t_1, a_1), (q_2, s_2, r_2, t_2, a_2), \dots, (q_m, s_m, r_m, t_m, a_m)$$

where $m \leq d(Q \setminus \{q_f\}) \times d(\Gamma)$, with the stipulation that the first quintuple refers to the initial state and the second one to the final state; i.e., $q_1 = q_o$ and $r_2 = q_f$. A possible approach to describe these quintuples is to encode the elements of the set $Q \cup \Gamma \cup \{L, R, S\}$ using a fixed-length binary code (refer to Definition 19.1.6 for more details), encoding the quintuple (q, s, r, t, a) as $\langle q, s, r, t, a \rangle$. The length of an encoded quintuple is $5l$, where $l = \lceil \log(d(Q \cup \Gamma \cup \{L, R, S\})) \rceil$. Following this convention, machine T is encoded as the binary string:

$$\langle T \rangle = \langle \bar{l}, \langle q_1, s_1, r_1, t_1, a_1 \rangle, \dots, \langle q_r, r_r, s_r, t_r, a_r \rangle \rangle$$

The length of the encoded machine, following this schema, would be $l(\langle T \rangle) \leq 5lm + \log l + 1$. ■

Since each Turing machine is composed by a finite set of quintuples, we can encode and list all the machines using a shortlex ordering. We associate each machine T with the index i corresponding to its position in this list, and we denote by T_i the i -th Turing machine. Each positive integer i encodes one, and only one, Turing machine. However, as Proposition 18.2.1 shows, all Turing machines have an infinite number of indexes. We associate each Turing machine with its smallest index.

Proposition 18.2.1 — Padding Lemma. Each Turing machine has infinitely many indexes.

Proof. Consider a Turing machine T_i encoded by the string $\langle T_i \rangle$. We can create a new encoding $\langle T_j \rangle$ by appending a finite number of 0's to $\langle T_i \rangle$, such that $\langle T_j \rangle = \langle T_i \rangle 0^n$ for some positive integer n . Since n can take on any positive integer value, there are infinitely many possible encodings $\langle T_j \rangle$ for the same Turing machine T_i . ■

A universal Turing machine is a machine that can simulate the behavior of any other Turing machine on arbitrary input. The universal machine achieves this by reading both the description of the machine to be simulated (for instance, using the coding schema described in Example 18.4) and the input string for the computation from its own tape.

Definition 18.2.1 — Universal Turing Machine. A *Universal Turing Machine*

is a Turing machine U such that $U(\langle\langle T_i \rangle, s \rangle) = T_i(s)$ for all Turing machines T_i and all input strings $s \in \mathcal{B}$.

Naturally, we must prove that such a machine exists before we can utilize it. One could argue that a human being could decode the machine T_i and simulate its behavior with the input string s , and then refer to Theorem 18.1.1. A more rigorous approach would be to explicitly construct a universal Turing machine. However, providing a detailed description of one of these machines is beyond the scope of this book. Instead, we direct the reader to the references included at the end of the chapter for further exploration.

18.3 Non-Computable Problems

Turing machines enable us to delineate the set of problems that can be resolved through effective procedures or, in other words, by computers. It may be surprising to learn that numerous problems cannot be addressed using algorithms; such challenges lie beyond the computational capabilities of machines. We are not alluding to speculative queries like whether a computer can be intelligent or self-aware but to concrete, well-defined mathematical problems. We are also not referring to complex problems that demand a substantial amount of time to solve, as those, irrespective of their time consumption, remain computable.

One classic exemplar of non-computability is the *halting problem*. As illustrated in Algorithm 18.1, it involves a program or algorithm tasked with determining whether any given program (including itself) and input will eventually halt or continue to run indefinitely. Alan Turing proved that no algorithm can exist to solve this problem for all possible program-input pairs. This revelation wasn't a reflection on the limitations of technology or processing power but highlighted a profound theoretical limit intrinsic to computation.

Algorithm 18.1 HALT function

```

procedure HALT( $A, I$ )
  if  $A(I)$  halts then
    return 1
  else
    return 0
  end if
end procedure

```

The proposition below proves that the halting problem is non-computable.

Proposition 18.3.1 — Halting Problem. Define HALT as in Algorithm 18.1. There does not exist a Turing machine that computes the HALT function for all

possible pairs (A, I) , where A is a Turing machine and I is the input string to that machine.

Proof. The proof is by contradiction. Assume that the machine $HALT$ exists, and define a new Turing machine TC such that $TC(A) = 1$ if $HALT(A, A) = 0$, and $TC(A)$ will never stop if $HALT(A, A) = 1$. Then the contradiction arises when we ask about the result of $TC(TC)$: if $TC(TC)$ stops we have that $HALT(TC, TC) = 0$ and that $TC(TC)$ should not stop, and if $TC(TC)$ does not stop then we have that $H(TC, TC) = 1$ and thus $TC(TC)$ should stop. ■

The existence of such non-computable problems underscores the boundaries of mechanical computation. It illustrates that while Turing machines, and by extension, computers are profoundly powerful tools capable of solving an extensive array of problems, they are not omnipotent. A frontier of unsolvable problems exists, necessitating deeper exploration into the realms of mathematics, logic, and perhaps even philosophy to understand the inherent limits of computation.

The Halting Problem also has significant practical consequences in computer programming. For instance, it is impossible to write a program that can guarantee any other arbitrary program is bug-free or that all infinite loops with conditional exits will eventually halt for all possible inputs.

The next example introduces a well-defined, practical problem involving simple string manipulation that cannot be solved using computers.

■ **Example 18.5** Given two finite lists $(\alpha_1, \dots, \alpha_n)$ and $(\beta_1, \dots, \beta_n)$ of strings over some alphabet Σ , where $d(\Sigma) \geq 2$, the *Post Correspondence Problem* (PCP) asks to determine if there exists a sequence of $K \geq 1$ indices (i_k) , with $1 \leq i_k \leq n$ for all $1 \leq k \leq K$, such that $\alpha_{i_1} \dots \alpha_{i_K} = \beta_{i_1} \dots \beta_{i_K}$. For instance, given the sequences (a, ab, bba) and (baa, aa, bb) , a solution would be $\alpha_3 \alpha_2 \alpha_3 \alpha_1 = \beta_3 \beta_2 \beta_3 \beta_1$. No algorithm exists to solve PCP. Like many proofs of incomputability, the proof proceeds by showing that $HALT$ can be reduced to PCP, meaning if PCP is decidable, then the Halting Problem should be decidable as well. We will not detail the proof in this section; for interested readers, we refer to the references at the end of this chapter. ■

Non-computable problems are generally not derived directly from natural phenomena but from logical and mathematical constructs. To date, there are no known examples of non-computable problems manifesting plainly in natural phenomena. It's essential to distinguish between non-computability and unpredictability. Non-computable problems are those for which no algorithm can ever be created to solve them. In contrast, unpredictable systems (such as chaotic or complex systems) are theoretically computable but are unpredictable

in practice due to factors like sensitivity to initial conditions or measurement precision.

18.4 Computable Functions and Sets

Each Turing machine T defines a function $f_T : \mathcal{B}^* \rightarrow \mathcal{B}^*$ that assigns to each input string $s \in \mathcal{B}^*$ an output string $T(s) \in \mathcal{B}^*$. The relationship between Turing machines and functions forms the basis for introducing the concept of a *computable function*.

Definition 18.4.1 A function $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ is *computable* if there exists a Turing machine T that defines the function f and halts for all the values of $\mathcal{B}^*$.

The terminology in computational theory can vary. While computable functions are occasionally referred to as *recursive functions*, this book opts for the term computable functions for consistency.

■ **Example 18.6** The function that assigns to each pair of natural numbers x and y their sum $x + y$ is computable, as demonstrated in Example 18.1. ■

In real-world scenarios, certain functions don't provide a defined output for all possible inputs. Partial computable functions, characterized by Turing machines that don't halt for specific inputs, model these cases.

Definition 18.4.2 A partial function $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ is *partial computable* if there exists a Turing machine T that defines f for defined values and does not halt for undefined values.

The distinction between total computable functions and partial computable functions is significant in computability theory because it reflects the difference between problems that are always solvable by an algorithm (total) and those that are only solvable in some cases (partial).

■ **Example 18.7** The function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ that assigns to each pair of natural numbers x and y the number $x - y$ is a partial computable function, since it is not defined in the case that $x < y$. Recall that according to our definition of Turing machine (see Definition 18.1.1), when the machine reaches an undefined configuration enters an infinite loop without ever halting. ■

We can expand the application of the principles of computability and partial computability to the domain of sets. We characterize sets through the lens of their characteristic functions that discern the membership of elements within the sets.

Definition 18.4.3 A set $A \in \mathcal{B}^*$ is *computable* if its characteristic function χ_A is a total computable function. A set $A \in \mathcal{B}^*$ is *computably enumerable* if its characteristic function χ_A is a partial computable function, that is, $\chi_A(a) = 1$ if $a \in A$, but $\chi_A(a)$ is undefined if $a \notin A$.

The application of these concepts is illustrated through the example of the set of all Turing machines that halt for all inputs.

■ **Example 18.8** The set of all Turing machines that halt on all inputs, as demonstrated in 18.3.1, is not computable but is computably enumerable. ■

Next proposition provides an alternative characterization of computable sets.

Proposition 18.4.1 A set $A \in \mathcal{B}^*$ is computable if and only if A and its complement A^c are computably enumerable.

Proof. If A is computable, by definition, there exists a Turing machine that decides for any input $x \in \mathcal{B}^*$ whether $x \in A$ or $x \notin A$, halting in both cases. This implies that both A and its complement A^c can be enumerated by Turing machines. Thus, both A and A^c are computably enumerable.

Conversely, suppose A and A^c are computably enumerable. This means there exist two Turing machines, T_A and T_{A^c} , that enumerate the elements of A and A^c , respectively. To show that A is computable, construct a Turing machine T that, given an input $x \in \mathcal{B}^*$, simulates T_A and T_{A^c} in parallel to search for x . If x appears in the enumeration produced by T_A , T halts and accepts x as an element of A . If x appears in the enumeration produced by T_{A^c} , T halts and accepts x , indicating $x \notin A$. Since every element of $\mathcal{B}^*$ must be in either A or A^c and both sets are computably enumerable, T will eventually halt for every input x , proving that A is computable. ■

18.5 Oracle Turing Machine

An oracle Turing machine (see Figure 18.3) is a theoretical model of computation that extends the capabilities of a standard Turing machine by providing it with an oracle. The oracle is a black box that can instantly compute certain answers, even for problems that are unsolvable or would take an impractical amount of time for a standard Turing machine to process. This model helps computer scientists and mathematicians explore the implications and boundaries of computational theory, including questions about complexity classes and the limits of what is computationally possible. The oracle Turing machine isn't a physical or implementable machine but rather a conceptual tool used in theoretical studies.

Definition 18.5.1 — Oracle Turing Machine. An *oracle Turing machine*

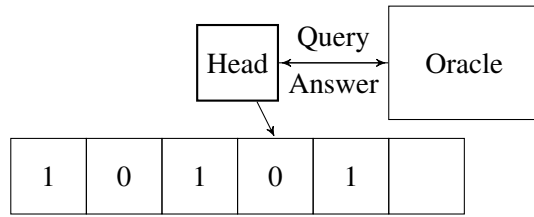


Figure 18.3: Oracle Turing Machine

with oracle set $\mathcal{O}$ is a 8-tuple $(Q, \Gamma, \sqcup, \Sigma, q_i, q_f, \tau, \mathcal{O})$ where:

Q is a finite, non-empty, set of *states*,

Γ is a finite, non-empty, set of *tape symbols*,

$\sqcup \in \Gamma$ is the *blank symbol*,

$\Sigma \subseteq \Gamma \setminus \sqcup$ is the set of *input symbols*,

$q_o \in Q$ is the *initial state*,

$q_f \in Q \setminus \{q_o\}$ is the *final state*,

$\tau : (Q \setminus \{q_f\}) \times \Gamma \times \{0, 1\} \rightarrow Q \times \Gamma \times \{L, R, S\}$ is the *transition*,

$\mathcal{O} \subseteq \Sigma^*$ is the *oracle set*.

Building on the concept of a regular Turing machine (refer to Definition 18.1.1), an oracle Turing machine introduces the unique feature of an oracle set $\mathcal{O}$. This set comprises a subset of strings for which the oracle can instantly provide answers. The true strength of an oracle machine emerges when the set $\mathcal{O}$ is non-computable; in cases where $\mathcal{O}$ is computable, a regular Turing machine would be sufficient.

The transition function τ for the oracle Turing machine is nuanced. At each step the machine will query the oracle. Specifically, it sends the current string w —starting under the head and extending to the rightmost non-empty cell of the input tape—to the oracle. The oracle then responds with a binary answer: '1' if w is in $\mathcal{O}$ or '0' if it isn't. The machine doesn't always utilize this response. There are steps where, despite receiving an answer, the computation proceeds unaffected by the oracle's response—essentially ignoring it. However, when the oracle's answer is pivotal, the machine makes a decision based on it. This decision could affect the next state, the next symbol to be written, or the next move (left, right, or stay). In essence, while the machine has the capability to continuously consult the oracle, it strategically chooses when to act on the information received. We could have modified the τ function in such a way that the oracle is queried only when the answer is relevant for the computation, but that would require making important changes to the behaviour of the function,

such as the introduction of new control states.

■ **Example 18.9** In the realm of theoretical computer science, an oracle can be invoked to "solve" the halting problem (refer to Theorem 18.3.1). The oracle is a hypothetical device or black box that, as if by magic, provides an instantaneous answer to a specific problem instance. The oracle set $\mathcal{O}$ would consist of a collection of strings in the form $\langle P, I \rangle$, where P represents a program encoded as a string and I denotes its input. Given a program and its input, this oracle would instantly inform us whether the program halts on that input. Naturally, this concept is purely theoretical. No such oracle exists in reality, and the halting problem remains unsolvable in practical terms. ■

A standard Turing machine can be viewed as an oracle machine that simply *ignores* its oracle or, equivalently, uses any fixed oracle set and never branches on the oracle bit. Thus ordinary Turing machines embed as a subclass of oracle Turing machines.

Proposition 18.5.1 For every (ordinary) Turing machine $T = (Q, \Gamma, \sqcup, \Sigma, q_0, q_f, \tau)$ and every oracle set $\mathcal{O} \subseteq \Sigma^*$, there exists an oracle Turing machine $T^{\mathcal{O}}$ such that for all inputs $x \in \Sigma^*$, the computation of T on x and the computation of $T^{\mathcal{O}}$ on x (with oracle $\mathcal{O}$) produce the same outcome.

Proof. Define $T^{\mathcal{O}} = (Q, \Gamma, \sqcup, \Sigma, q_0, q_f, \tau', \mathcal{O})$ with the same components as T and transition function τ' given by

$$\tau'(q, s, b) = \tau(q, s) \quad \text{for all } (q, s) \in (Q \setminus \{q_f\}) \times \Gamma \text{ and } b \in \{0, 1\}.$$

That is, $T^{\mathcal{O}}$ ignores the oracle bit. A step-by-step induction on the unique computation of T on x shows that T and $T^{\mathcal{O}}$ visit identical configurations (up to the presence of the unused oracle bit) and therefore have the same outcome and output. ■

An oracle machine is specified by two pieces of data: the *machine component* $(Q, \Gamma, \sqcup, \Sigma, q_0, q_f, \tau)$ and the *oracle set* $\mathcal{O}$. The former is defined syntactically and does not depend on which $\mathcal{O}$ is chosen. Semantically, the behavior of a fixed machine component on a given input depends only on the answers to the queries it actually makes (i.e., on the restriction of $\mathcal{O}$ to those queries).

Proposition 18.5.2 Let $M = (Q, \Gamma, \sqcup, \Sigma, q_0, q_f, \tau)$ be a fixed machine component. For any two oracle sets $\mathcal{O}_1, \mathcal{O}_2 \subseteq \Sigma^*$ and any input x , if $\mathcal{O}_1$ and $\mathcal{O}_2$ give the same answers to all oracle queries made during the run of M on input x (with either oracle), then the computations $M^{\mathcal{O}_1}(x)$ and $M^{\mathcal{O}_2}(x)$ are identical (same sequence of configurations, hence same outcome and output).

Proof. Consider the (unique) run of $M^{\mathcal{O}_1}$ on input x . By hypothesis, $\mathcal{O}_2$ agrees with $\mathcal{O}_1$ on precisely the queried strings that arise along this run. Since τ is

deterministic given the current state, scanned symbol, and oracle bit, and the oracle bit provided by $\mathcal{O}_2$ matches that from $\mathcal{O}_1$ at every step, both computations take the same transition at each step. Thus the configuration sequences coincide, yielding identical outcomes. ■

A (partial) function is *computable relative to $\mathcal{O}$* if some oracle machine with oracle $\mathcal{O}$ computes it. Two useful facts: (i) ordinary computable functions are $\mathcal{O}$ -computable for any oracle (by ignoring the oracle); (ii) if $\mathcal{O}$ is itself computable, then $\mathcal{O}$ -computability coincides with ordinary computability.

Definition 18.5.2 — Oracle computability. Given $\mathcal{O} \subseteq \Sigma^*$, a partial function $f : \Sigma^* \rightarrow \Sigma^*$ is $\mathcal{O}$ -computable if there exists an oracle Turing machine $M^{\mathcal{O}}$ such that, for every $x \in \Sigma^*$:

- if $f(x)$ is defined, then $M^{\mathcal{O}}$ halts on input x and outputs $f(x)$ (under the chosen output convention);
- if $f(x)$ is undefined, then $M^{\mathcal{O}}$ does not halt on input x .

Proposition 18.5.3 If $\mathcal{O}$ is (ordinarily) computable, then a partial function f is $\mathcal{O}$ -computable if and only if f is computable by an ordinary Turing machine.

Proof. Suppose f is $\mathcal{O}$ -computable via $M^{\mathcal{O}}$. Since $\mathcal{O}$ is computable, there exists an ordinary machine D deciding membership in $\mathcal{O}$. Simulate $M^{\mathcal{O}}$ step by step on an ordinary machine; whenever $M^{\mathcal{O}}$ needs the oracle bit, call D on the current query string to obtain the bit and continue the simulation. This yields an ordinary machine computing f .

If f is computable by an ordinary machine T , then by Proposition 18.5.1 there is an oracle machine $T^{\mathcal{O}}$ that ignores the oracle and computes f for any $\mathcal{O}$. Hence f is $\mathcal{O}$ -computable. ■

More generally, if $\mathcal{O}$ is *Turing reducible* to $\mathcal{O}'$ (i.e., decidable by some $\mathcal{O}'$ -oracle machine), then every $\mathcal{O}$ -computable function is $\mathcal{O}'$ -computable. This captures the standard notion of *relative computability*.

■ **Example 18.10 — Toy oracle machine that outputs the oracle's answer.** Let $\Sigma = 0, 1$ and $\Gamma = 0, 1, \sqcup$. Fix the oracle set $\mathcal{O} = 11, 101$ (finite for concreteness). Consider the oracle machine $M^{\mathcal{O}}$ that, on any input $w \in \Sigma^*$, writes a single symbol indicating the oracle's answer and halts:

- If the queried string (as per your convention) is in $\mathcal{O}$, write 1 and halt.
- Otherwise, write 0 and halt.

Use states $Q = q_0, q_f$, with q_0 initial and q_f halting. Define τ independently of the scanned symbol $s \in \Gamma$ by the sextuples

$$(q_0, 0, 1, q_f, 1, S), \quad (q_0, 1, 1, q_f, 1, S), \quad (q_0, \sqcup, 1, q_f, 1, S),$$

$$(q_0, 0, 0, q_f, 0, S), \quad (q_0, 1, 0, q_f, 0, S), \quad (q_0, \sqcup, 0, q_f, 0, S),$$

and leave all other transitions undefined (so the machine halts after one step). Then $\langle M \rangle$ is obtained by concatenating the fixed-length codes of these six sextuples after the header. The oracle's finite code is, for example,

$$\langle \mathcal{O} \rangle = \langle 2, \langle 11 \rangle, \langle 101 \rangle \rangle,$$

using any self-delimiting code for strings. The full encoding is the pairing $\langle \langle M \rangle, \langle \mathcal{O} \rangle \rangle$. ■

These additions make explicit (i) how ordinary computation embeds into oracle computation, (ii) in what precise sense the machine component is independent of the oracle set, (iii) that an oracle machine is exactly a pair $(M, \mathcal{O})$, and (iv) what it means for a function to be computable *relative* to an oracle. The encoding example matches your earlier quintuple-based scheme with a minimal extension to handle the oracle bit.

References

The original paper from Alan Turing where the concepts of Turing machine, universal Turing machine, and non-computable problems were introduced is [Tur36], however it is a difficult to read paper for the contemporary reader. An easier to read introduction to computability theory, from the point of view of languages, can be found in [Sip12], and a more advanced introductions in [Coo03] and [Soa16]. In [Fer09] we can find a description of the most important computability models proposed so far. The Post Correspondence Problem was introduced by Emil Post in [Pos46]; for the details of the proof sketched in Example 18.5 please refer to [Sip12].



19. Coding

Information is the resolution of uncertainty.
Claude Shannon

In this section, we review the conceptual foundations and main results of coding theory and the closely related field of information theory.

Coding is the process of representing a sequence of symbols from one alphabet by a sequence of symbols from another alphabet. Coding has many practical applications, such as error detection and correction, cryptography, and telecommunications. Our focus here is on data compression, that is, encoding a message using fewer symbols than in its original representation, while preserving all the information it contains. Compression algorithms reduce message size by identifying and eliminating statistical redundancies. For example, one may assign shorter descriptions to the most frequent symbols in the source and longer descriptions to the less frequent ones. A particular class of codes, known as prefix-free codes, will play a central role in this book. Prefix-free codes establish a natural connection between coding theory and probability theory, a connection that will be essential in the context of the theory of nescience.

Information theory proposes that the amount of information conveyed by the occurrence of an event is given by the negative logarithm of the probability of that event. In this sense, information can be interpreted as surprise: the less

likely an event is, the more information we gain when it occurs. Although we will not adopt this interpretation of information in the theory of nescience, we will rely extensively on another fundamental concept from information theory: entropy. Entropy quantifies the uncertainty in the value of a random variable or the outcome of a random process. It is particularly relevant because it establishes a fundamental bound on compression: no code can achieve an average word length smaller than the entropy of the source alphabet.

There exist many interesting concepts derived from entropy, like joint entropy, conditional entropy, or mutual information. However, these concepts are more relevant in the context of communication because they allow us to solve the problem of how to transmit information in a reliable manner over a noisy channel. Here, they are introduced for completeness purposes, and to compare them with our own definitions of joint nescience and conditional nescience.

Several additional concepts are derived from entropy, including joint entropy, conditional entropy, and mutual information. These quantities play a central role in communication theory, as they are instrumental in addressing the problem of transmitting information reliably over noisy channels. In this book, however, they are introduced primarily for completeness and as a basis for comparison with our own definitions of joint nescience and conditional nescience.

19.1 Coding

Intuitively, coding refers to the process of losslessly describing a sequence of symbols (a message) from some alphabet by another sequence of symbols from a (possibly different) alphabet. There is no universal agreement on what exactly constitutes a code, as different authors propose different definitions in the literature. Fortunately, the definition of a prefix-free code (the kind of code used in the theory of nescience) is standard and widely accepted.

Let $\mathcal{S} = \{s_1, s_2, \dots, s_q\}$ be a finite set called the *source alphabet*, and let $\mathcal{X} = \{x_1, x_2, \dots, x_r\}$ be a finite set called the *code alphabet*.

Definition 19.1.1 — Code. A *code* for $\mathcal{S}$ is a total function $C : \mathcal{S} \rightarrow \mathcal{X}^+$. If $(s, x) \in C$ we say that s is a *source symbol* and x is the corresponding *code word*. If C is injective, we say that the code is *nonsingular*.

Nonsingularity ensures that each symbol in the source alphabet is represented by a distinct code word, allowing us to unambiguously describe the individual symbols of the source alphabet. In what follows, unless otherwise stated, the term *code* will always mean a nonsingular code. Moreover, without loss of generality, we restrict our discussion to *binary codes*, that is, $\mathcal{X} = \mathcal{B} = \{0, 1\}$.

The property of nonsingularity can also be extended from individual symbols to strings of symbols. For that, we introduce the notion of extensions of a code.

Definition 19.1.2 The *extension of order n* of a code C is a function $C^n : \mathcal{S}^n \rightarrow \mathcal{B}^+$ defined by

$$C^n(s_{i_1} \dots s_{i_n}) = C(s_{i_1}) \dots C(s_{i_n}),$$

where the right-hand side is the concatenation of the code words corresponding to the symbols $s_{i_1} \dots s_{i_n} \in \mathcal{S}^n$. An extension of order n is *nonsingular* if C^n is injective.

If it is clear from the context, we will also use the word *code* to refer to a nonsingular extension of order n of a code, and the elements of $\mathcal{S}^n$ will be called *source words*.

■ **Example 19.1** The code defined by $C(a) = 0$, $C(b) = 00$, $C(c) = 01$, and $C(d) = 11$ is nonsingular, but its extension of order 2 is singular, since $C^2(ab) = C^2(ba) = 000$. ■

As Example 19.1 illustrates, not all nonsingular codes have nonsingular extensions. In such cases, it may be impossible to recover the original message from its encoded version. To address this, we introduce the stronger property of unique decodability.

Definition 19.1.3 A code C is *uniquely decodable* if its extension C^n is nonsingular for all n .

The following proposition provides a useful characterization of unique decodability.

Proposition 19.1.1 A code C is uniquely decodable if and only if the function $C^+ : \mathcal{S}^+ \rightarrow \mathcal{B}^+$ is injective.

Proof. If C^+ is injective, then every restriction C^n is injective, and the code is uniquely decodable.

Conversely, assume that C^n is nonsingular for all n . Suppose, for contradiction, that C^+ is not injective. Then there exist two source words $s_1 \in \mathcal{S}^n$ and $s_2 \in \mathcal{S}^m$ with $n \neq m$ such that $C^+(s_1) = C^+(s_2)$. Construct $s_3 = s_1s_2$ and $s_4 = s_2s_1$. Both have the same length, but

$$C^+(s_3) = C^+(s_1)C^+(s_2) = C^+(s_2)C^+(s_1) = C^+(s_4),$$

contradicting the injectivity of C^{n+m} . Therefore C^+ must be injective. ■

■ **Example 19.2** The code $C(a) = 0$, $C(b) = 01$, $C(c) = 011$, and $C(d) = 0111$ is uniquely decodable. For example, the code word 0010011 corresponds uniquely to the source word *abac*. Unique decodability is achieved because the leading 0 acts as a delimiter, separating code words. ■

The Sardinas-Patterson theorem provides a necessary and sufficient condition for unique decodability. The theorem is based on an algorithmic construction that examines successive sets of possible ambiguities.

Let C_0 denote the set of code words of a code C . Define

$$C_n = C^{-1}C_{n-1} \cup C_{n-1}^{-1}C,$$

for all $n \in \mathbb{N}$, where $C^{-1}D = \{y \mid \exists x \in C, xy \in D\}$ is the left quotient of C and D . Finally, let

$$C_\infty = \bigcup_{n=1}^{\infty} C_n.$$

Proposition 19.1.2 — Sardinas-Patterson. A code C is uniquely decodable if and only if $C_0 \cap C_\infty = \emptyset$.

Proof. If $C_0 \cap C_\infty \neq \emptyset$, then there exists a nonempty overlap between some code word and a concatenation of code words, implying that at least two distinct source words map to the same encoded string, so C is not uniquely decodable.

Conversely, if $C_0 \cap C_\infty = \emptyset$, then the iterative construction guarantees that no ambiguity arises in the decoding process, regardless of message length. Therefore the code is uniquely decodable. ■

■ **Example 19.3** For the code of Example 19.1:

$$C_0 = \{0, 00, 01, 11\}$$

$$C_1 = \{0, 1\}$$

Since $C_0 \cap C_1 \neq \emptyset$, the code is not uniquely decodable.

For the code of Example 19.2:

$$C_0 = \{0, 01, 011, 0111\}$$

$$C_1 = \{1, 11, 111\}$$

$$C_2 = \emptyset$$

Here $C_0 \cap C_\infty = C_0 \cap \bigcup_{n=1}^{\infty} C_n = C_0 \cap C_1 = \emptyset$, so the code is uniquely decodable. ■

Next definition introduces the concept of prefix-free codes. Prefix-free codes will play a critical role in the computation of the amount of algorithmic information of an arbitrary string (described in Chapter 20), and in our own theory of nescience. Prefix-free codes also allow us to link coding theory and probability theory through the Kraft inequality (Theorem 19.2.1). Note that we prefer the name *prefix-free code* over the more standard *prefix code*, since the former more accurately describes the concept.

Definition 19.1.4 — Prefix-free Code. A code C is *prefix-free* if for all i, j with $1 \leq i, j \leq q$ and $i \neq j$, $C(s_i)$ is not a prefix of $C(s_j)$.

Note that the fact that a string is a prefix of itself does not violate the prefix-free property, because the condition specifically excludes considering a string as a prefix of itself ($i \neq j$) when determining whether a set of code words is prefix-free.

■ **Example 19.4** The code $C(a) = 0$, $C(b) = 10$, $C(c) = 110$, and $C(d) = 1110$ is prefix-free. Here, the symbol 0 plays the role of a delimiter (as in Example 19.2), but placing it at the end of each code word ensures that no code word can be a prefix of another. ■

Prefix-free codes are uniquely decodable, as the next proposition shows.

Proposition 19.1.3 Let C be a prefix-free code. Then C is uniquely decodable.

Proof. Assume, for contradiction, that C is not uniquely decodable. Then there exist two distinct sequences $r \neq s$ of source symbols such that $C(r) = C(s)$. Among all such pairs, choose one with the minimal common encoded length. Let $r = ar'$ and $s = bs'$ with $a \neq b$ (if $a = b$, cancel the common first symbols to get a shorter counterexample, contradicting minimality). Then $C(a)$ is a prefix of $C(b)C(s')$ and $C(b)$ is a prefix of $C(a)C(r')$. In particular, one of $C(a)$ or $C(b)$ must be a prefix of the other, contradicting the prefix-free property. Hence C must be uniquely decodable. ■

From an engineering point of view, it is highly convenient to have codes whose source symbols can be decoded as soon as the corresponding code words are received. In other words, it should not be necessary to wait for the next code word in order to decode the current symbol. For this reason, some authors refer to these codes as *instantaneous codes*.

Definition 19.1.5 A code C is *instantaneous* if, for any order n and for any sequence of code words $C(s_{i_1}), C(s_{i_2}), \dots, C(s_{i_n})$, each sequence of code words $\mathbf{t} = C(s_{i_1})C(s_{i_2}) \dots C(s_{i_m}) \dots$ can be uniquely decoded as $\mathbf{s} = s_{i_1}s_{i_2} \dots s_{i_m} \dots$ without ambiguity, regardless of the continuation of $\mathbf{t}$.

■ **Example 19.5** The code described in Example 19.2 is not instantaneous. For instance, after receiving the sequence 011, the decoded symbol could be c if the next bit is 0 or d if it is 1; thus, one must look ahead to decide. ■

Prefix-free codes and instantaneous codes are essentially two terms for the same concept. A prefix-free code is one in which no code word is a prefix of any other code word. This ensures a clear boundary between consecutive code words, allowing immediate decoding upon receipt.

Proposition 19.1.4 A code C is instantaneous if and only if it is prefix-free.

Proof. Suppose C is instantaneous but not prefix-free. Then there exist at least a pair of code words, say $C(s_i)$ and $C(s_j)$, such that $C(s_i)$ is a prefix of $C(s_j)$. Consider a sequence where $C(s_j)$ is transmitted. Since $C(s_i)$ is a prefix of $C(s_j)$, the decoder would decode $C(s_i)$ from the initial part of $C(s_j)$, leading to ambiguity, as the rest of $C(s_j)$ could be seen as another code or part of the next code. This contradicts the assumption that C is instantaneous.

Suppose C is prefix-free. Reading a concatenation of code words from left to right, the first position at which a code word ends is unambiguous, because no longer code word can have the current code word as a prefix. Thus each code word can be decoded immediately upon completion, with no lookahead. Hence C is instantaneous. ■

The last type of codes we review are fixed-length codes. We will use fixed-length codes to compute the length of a text when there are no regularities we can exploit to compress it.

Definition 19.1.6 If all the code words of a code have the same length, we say that the code is a *fixed-length code*.

Fixed-length codes are prefix-free (and therefore instantaneous).

Proposition 19.1.5 Let C be a fixed-length code, then C is prefix-free.

Proof. Let C be a fixed length code, and $C(s_i)$ and $C(s_j)$ the code words of two arbitrary source words s_i and s_j . Assume that $C(s_i) <_p C(s_j)$, given the fact that $l(C(s_i)) = l(C(s_j))$ we have that $C(s_i) = C(s_j)$ and so, the code C is prefix-free. ■

Of course, the converse does not hold: not all prefix-free codes are fixed-length. The code in Example 19.4 is prefix-free but not fixed-length.

Figure 19.1 provides a graphical representation of the relationships among the different types of codes introduced in this section.

19.2 Kraft Inequality

The Kraft inequality provides a condition for the existence of a prefix-free code given a set of code word lengths. Kraft's inequality states that for a given set of code word lengths in a binary code, the sum of the reciprocals of powers of two corresponding to the code word lengths must be less than or equal to one. This condition is both necessary and sufficient: any prefix-free code must satisfy this inequality, and given a set of lengths that meets this condition, it is always possible to construct a corresponding prefix-free code. The elegance and

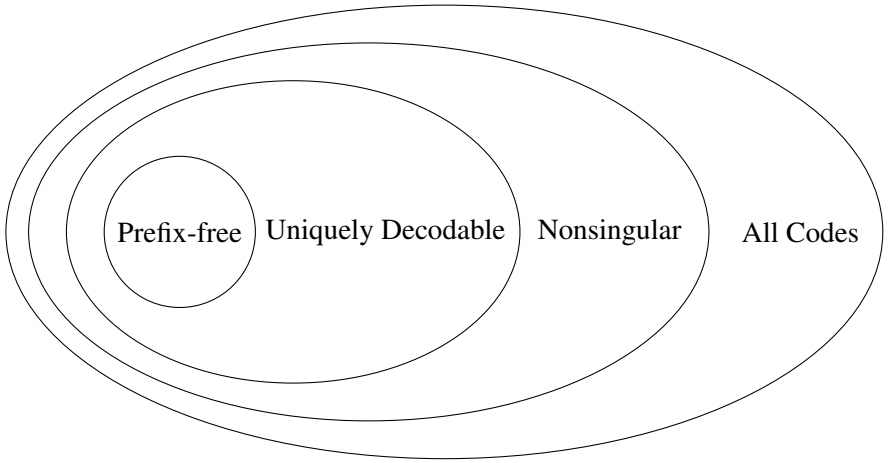


Figure 19.1: Classification of Codes

utility of Kraft's inequality lie in its ability to link the lengths of code words with probabilistic structure.

Theorem 19.2.1 — Kraft Inequality. Let $\mathcal{L} = \{l_1, l_2, \dots, l_q\}$ be a set of lengths, where $l_i \in \mathbb{N}$, then there exists a binary prefix-free code C whose code words have the lengths of $\mathcal{L}$ if, and only if,

$$\sum_{l_i \in \mathcal{L}} 2^{-l_i} \leq 1$$

Proof. Consider a binary tree whose branches are labeled with the symbols of the code alphabet, in such a way that the path from the root to the leaves traces out the symbols of a code word. The prefix-free condition implies that nodes representing complete code words cannot have descendants. An example of such a tree, for the code described in Example 19.4, is shown in Figure 19.2.

Let $l_{\max} = \max \{l_1, l_2, \dots, l_q\}$, that is, the length of the longest code word from the set of lengths. There will be at most $2^{l_{\max}}$ leaf nodes in the tree, but at level l_i we have to prune $2^{l_{\max} - l_i}$ leaves, since the code is prefix-free. Summing over all the code words' lengths, we have that the total number of pruned leaves must be less than or equal to the maximum number of leaves, that is

$$\sum_{l_i \in \mathcal{L}} 2^{l_{\max} - l_i} \leq 2^{l_{\max}}$$

or, equivalently,

$$\sum_{l_i \in \mathcal{L}} 2^{-l_i} \leq 1$$

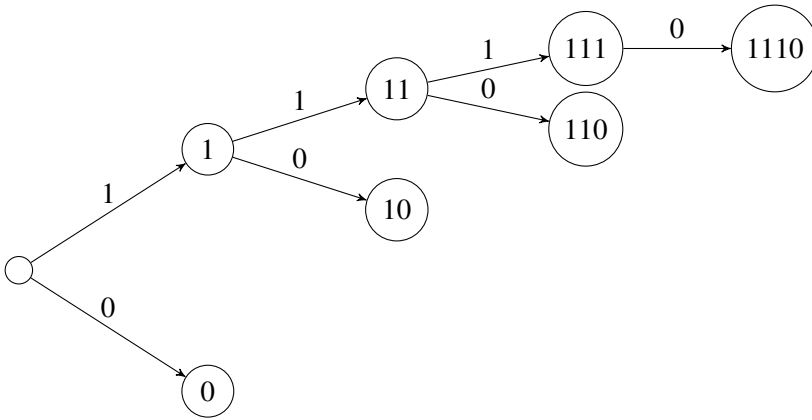


Figure 19.2: Prefix-free Tree

which is exactly the inequality we are trying to prove.

Conversely, given any set of code words' lengths $\mathcal{L} = \{l_1, l_2, \dots, l_q\}$ that satisfy the Kraft inequality, we can always construct a binary tree, like the one in the Figure 19.2. Label the first node (lexicographically) of depth l_1 as code word 1, and remove its descendants from the tree. Then label the first remaining node of depth l_2 as code word 2, and so on. Proceeding this way, we construct a prefix code with the specified lengths. ■

Given a code C whose code word lengths $\mathcal{L}$ satisfy the Kraft inequality does not necessarily mean that the code is prefix-free, since what the inequality states is that there exists a prefix-free code with those word lengths, not that all codes with those word lengths are prefix-free.

■ **Example 19.6** The code $C(a) = 0$, $C(b) = 111$, $C(c) = 110$ and $C(d) = 100$ satisfies the Kraft inequality, but it is not prefix-free. ■

McMillan's inequality extends Kraft's bound to all uniquely decodable codes: any such code with lengths l_i satisfies $\sum_i 2^{-l_i} \leq 1$. Combined with Kraft's theorem, this means a set of lengths meets this inequality if and only if it can be realized by a prefix-free—and thus uniquely decodable—code.

Theorem 19.2.2 — McMillan's Inequality. Let $\mathcal{L} = \{l_1, l_2, \dots, l_q\}$ be a set of lengths, where $l_i \in \mathbb{N}$, then there exists a uniquely decodable code C whose code words have the lengths of $\mathcal{L}$ if, and only if,

$$\sum_{l_i \in \mathcal{L}} 2^{-l_i} \leq 1$$

Proof. Let $S = \sum_{i=1}^q 2^{-l_i}$. For each $n \in \mathbb{N}$, let C^n be the set of all concatenations of n code words. Because the code is uniquely decodable, distinct concatenations in C^n produce distinct binary strings. Consider an infinite fair-coin binary sequence $X = (X_1, X_2, \dots)$, and for any finite binary string u write $[u] = \{X : X \text{ begins with } u\}$. Then $\mathbb{P}([u]) = 2^{-|u|}$, and the cylinder sets $\{[w] : w \in C^n\}$ are pairwise disjoint. Therefore,

$$\sum_{w \in C^n} 2^{-|w|} = \sum_{w \in C^n} \mathbb{P}([w]) \leq 1.$$

But the left-hand side factors as

$$\sum_{w \in C^n} 2^{-|w|} = \left(\sum_{i=1}^q 2^{-l_i} \right)^n = S^n.$$

Hence $S^n \leq 1$ for all n , which implies $S \leq 1$. ■

In the theory of nescience, we focus on prefix-free codes without loss of generality: every uniquely decodable code satisfies Kraft's inequality, so there exists a prefix-free code with the same code-word lengths.

Corollary 19.2.3 There is an instantaneous (prefix-free) code with word lengths $l_1, \dots, l_q$ if and only if there is a uniquely decodable code with these word lengths.

Proof. Every prefix-free code is uniquely decodable.

If a uniquely decodable code exists with lengths $l_1, \dots, l_q$, then by McMillan's inequality $\sum_i 2^{-l_i} \leq 1$. By Theorem 19.2.1 (sufficiency), there exists a prefix-free code with the same lengths. ■

In the context of nescience, our interest lies not in particular code assignments but in the multiset of code word lengths. This emphasis lets us abstract away from code construction details and focus on the mathematical properties of lengths, which are central to the measures we will use.

19.3 Optimal Codes

In coding theory, a compact code is an optimal encoding strategy that minimizes the expected length of code words, a concept central to evaluating a code's efficiency. The expected length of a code is determined by the weighted average of the lengths of its code words, with weights corresponding to the probability distribution of the source symbols. By designing code words so that more frequent symbols are assigned shorter lengths and less frequent ones longer lengths, compact codes effectively reduce the average size needed to encode

information. This principle is pivotal in data compression, as it allows for a significant reduction in the space required for storage or the bandwidth needed for transmission.

Let $\mathcal{S} = \{s_1, \dots, s_q\}$ be a finite source alphabet, and let P be a probability distribution on $\mathcal{S}$.

Definition 19.3.1 The *expected length* of a code C , denoted by L_C , is

$$L_C = \sum_{i=1}^q P(s_i) l_i,$$

where $\mathcal{L} = \{l_1, \dots, l_q\}$ are the code-word lengths of C . We may simply write L when C is clear from context.

Our goal is to identify a code C that minimizes L under P . Such a code is optimal for $\mathcal{S}$ and P .

Compact codes achieve the lowest possible weighted average codeword length.

Definition 19.3.2 A code C is *compact* (optimal) if, for the given source alphabet $\mathcal{S}$ and distribution P , its expected length L_C is minimal among all codes over the same code alphabet.

The existence of compact codes for all possible source alphabets is a foundational aspect of coding theory, suggesting that for every finite source alphabet, an optimally efficient encoding scheme can be devised.

Proposition 19.3.1 For every finite source alphabet $\mathcal{S}$ and distribution P there exists at least one code C that is compact.

Proof. Huffman's algorithm (see Section 19.5) constructs a prefix-free code that minimizes the expected length for the given P . Hence an optimal code exists. ■

It is convenient to distinguish between *coding efficiency* and *redundancy* relative to the source entropy. Let $H_r(P) = -\sum_i P(s_i) \log_r P(s_i)$ be the r -ary entropy (logarithms base r , where r is the code alphabet size).

Definition 19.3.3 The *coding efficiency* is

$$\eta = \frac{H_r(P)}{L} \in (0, 1],$$

and the (normalized) *redundancy* is

$$\rho = 1 - \eta = \frac{L - H_r(P)}{L} \in [0, 1).$$

Our aim is to *maximize* efficiency (equivalently, *minimize* redundancy). Note that $L \geq H_r(P)$ (Theorem 19.3.2 below), so $\eta \leq 1$.

Next theorem states that the entropy of the probability distribution P poses a limit to the average length of prefix-free codes.

Theorem 19.3.2 For any prefix-free r -ary code with lengths $\{l_i\}$ and source distribution P ,

$$H_r(P) \leq L = \sum_{i=1}^q P(s_i) l_i,$$

with equality if and only if $r^{-l_i} = P(s_i)$ for all i .

Proof. Let $\alpha = \sum_i r^{-l_i} \leq 1$ by Kraft's inequality. Define a probability distribution $\tilde{Q}$ by

$$\tilde{Q}(s_i) = \frac{r^{-l_i}}{\alpha}.$$

Then

$$\begin{aligned} L - H_r(P) &= \sum_i P(s_i) (l_i + \log_r P(s_i)) \\ &= \sum_i P(s_i) \log_r \left(\frac{r^{l_i} P(s_i)}{1} \right) = \sum_i P(s_i) \log_r \left(\frac{P(s_i)}{r^{-l_i}} \right). \end{aligned}$$

Write $r^{-l_i} = \alpha \tilde{Q}(s_i)$ to get

$$\begin{aligned} L - H_r(P) &= \sum_i P(s_i) \log_r \left(\frac{P(s_i)}{\alpha \tilde{Q}(s_i)} \right) + \log_r \left(\frac{1}{\alpha} \right) \\ &= D_r(P \| \tilde{Q}) + \log_r(1/\alpha) \geq 0, \end{aligned}$$

since the r -ary Kullback–Leibler divergence D_r is nonnegative and $\alpha \leq 1$. Equality holds iff both terms vanish, i.e., $\alpha = 1$ and $P = \tilde{Q}$, which is equivalent to $r^{-l_i} = P(s_i)$ for all i . ■

In data compression we are particularly interested in r -adic distributions.

Definition 19.3.4 A distribution P is called *r -adic* if each $P(s_i)$ is an exact r -adic probability, i.e., $P(s_i) = r^{-n_i}$ for some integers $n_i \geq 0$.

Equality in Theorem 19.3.2 occurs precisely for sources whose symbol probabilities are exact powers of r^{-1} .

Corollary 19.3.3 Equality $L = H_r(P)$ in Theorem 19.3.2 holds if and only if P is r -adic (equivalently, there exists a prefix-free code with $l_i = -\log_r P(s_i)$).

Proof. Theorem 19.3.2 shows equality holds exactly when $r^{-l_i} = P(s_i)$ for all i , i.e., when each $P(s_i)$ is an r -adic probability. ■

It is natural to consider the *ideal* real-valued lengths $\ell_i^* = -\log_r P(s_i)$, which need not be integers. Choosing integer lengths $l_i = \lceil \ell_i^* \rceil$ yields a feasible prefix-free code (by Kraft) with near-optimal average length.

Proposition 19.3.4 — Shannon–Fano bound. For $l_i = \lceil -\log_r P(s_i) \rceil$,

$$H_r(P) \leq L < H_r(P) + 1.$$

Proof. By definition $l_i < -\log_r P(s_i) + 1$, hence

$$L = \sum_i P(s_i) l_i < \sum_i P(s_i) (-\log_r P(s_i) + 1) = H_r(P) + 1.$$

Moreover, $\sum_i r^{-l_i} \leq \sum_i r^{\log_r P(s_i)} = \sum_i P(s_i) = 1$, so the lengths are feasible. The lower bound $L \geq H_r(P)$ is Theorem 19.3.2. ■

Kraft's inequality and expected length allow us to compare codes for a fixed source.

Definition 19.3.5 Let C_1, C_2 be two codes on $\mathcal{S}$ with expected lengths L_{C_1}, L_{C_2} under P . We say that C_1 is *more efficient* than C_2 (for P) if $L_{C_1} \leq L_{C_2}$, with strict inequality for at least one symbol probability configuration (equivalently, $L_{C_1} < L_{C_2}$ for the given P).

■ **Example 19.7** The non-prefix-free code in Example 19.6 has lengths $(1, 3, 3, 3)$, which satisfy Kraft's inequality. Therefore, there exists a prefix-free code with the *same* lengths; for instance,

$$C(a) = 1, \quad C(b) = 000, \quad C(c) = 001, \quad C(d) = 010.$$

For many P , these lengths yield a smaller expected length than the prefix-free code of Example 19.4 (which has lengths $(1, 2, 3, 4)$); the comparison is determined by P via $L = \sum_i P(s_i) l_i$. ■

Intuitively, a prefix-free code is *complete* if its set of code words cannot be augmented (by adding another code word) while preserving the prefix-free property.

Definition 19.3.6 A prefix-free code C is *complete* if there does not exist a binary string w such that $C \cup \{w\}$ is still prefix-free.

Kraft's inequality characterizes completeness.

Proposition 19.3.5 A prefix-free code C with lengths $\mathcal{L} = \{l_1, \dots, l_q\}$ is complete if and only if

$$\sum_{i=1}^q 2^{-l_i} = 1.$$

Proof. If the sum is < 1 , there remains unused capacity in the binary tree at some depth, so one can add at least one more leaf without violating prefix-freeness, contradicting completeness. Conversely, if the sum equals 1, the descendant subtrees of the selected leaves exactly partition the leaf set at some depth; no additional leaf can be added without creating a prefix conflict. ■

19.4 Entropy

In this section we introduce *entropy* as a measure of the uncertainty of a discrete random variable. Entropy appears in many contexts (communications, statistics, finance, etc.); here it is important because it will allow us to identify codes with the shortest possible average length.

Let $A = \{a_1, a_2, \dots, a_n\}$ be a finite set, and let X be a random variable on A with probability mass function $p(a)$.

Definition 19.4.1 — Entropy. The *entropy* of X , denoted $H(X)$ and measured in *bits*, is

$$H(X) = \sum_{a \in A} p(a) \log \frac{1}{p(a)}.$$

Entropy depends only on the probabilities $\{p(a)\}$, not on the labels in A . Clearly $H(X) \geq 0$ since $0 \leq p(a) \leq 1$ implies $-\log p(a) \geq 0$. If $p(a_i) = 0$ for some i , the corresponding summand is taken to be 0, consistent with $\lim_{p \rightarrow 0^+} p \log(1/p) = 0$. If we change the logarithm base to u , entropy scales by a factor $\log_u 2$.

■ **Example 19.8** Let X a random variable defined over the set $A = \{a_1, a_2\}$, with values $p(a_1) = q$ and $p(a_2) = 1 - q$. Then, the entropy of X is given by:

$$H(X) = q \log \frac{1}{q} + (1 - q) \log \frac{1}{1 - q}$$

Figure 19.3 shows the entropy of X for different values of q . If $q = 0$ or $q = 1$ the entropy is 0, that is, there is no uncertainty about which value of A we will get.

The maximum value of H is 1, and it is reached when $q = 1/2$; that is, we could say that 1 bit is the uncertainty associated to two equally probable symbols. ■

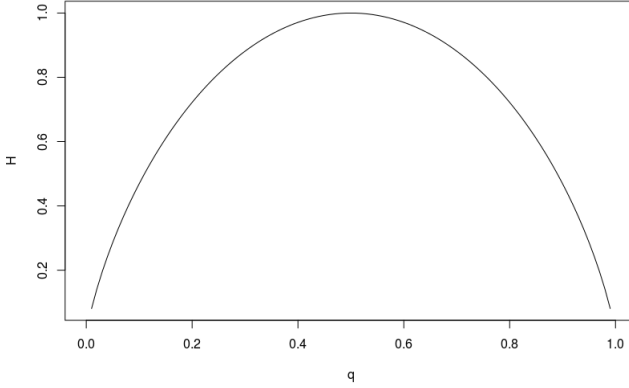


Figure 19.3: Binary Entropy Function

The next proposition shows that entropy is maximized by the uniform distribution and that the maximum equals the logarithm of the alphabet size.

Proposition 19.4.1 For X on A with $d(A) = n$, we have $H(X) \leq \log n$, with equality if and only if $p(a_1) = \cdots = p(a_n) = 1/n$.

Proof. Compute

$$\log n - H(X) = \sum_{i=1}^n p(a_i) \log n - \sum_{i=1}^n p(a_i) \log \frac{1}{p(a_i)} = \sum_{i=1}^n p(a_i) \log(n p(a_i)).$$

Using the change-of-base formula,

$$\log n - H(X) = (\log e) \sum_{i=1}^n p(a_i) \ln(n p(a_i)).$$

By the inequality $\ln x \leq x - 1$, applied with $x = \frac{1}{n p(a_i)}$, we have

$$\ln(n p(a_i)) \geq 1 - \frac{1}{n p(a_i)}.$$

Hence

$$\log n - H(X) \geq (\log e) \sum_{i=1}^n p(a_i) \left(1 - \frac{1}{n p(a_i)}\right) = (\log e) \left(1 - \frac{1}{n} \sum_{i=1}^n 1\right) = 0.$$

Equality holds iff $\ln(n p(a_i)) = 0$ for all i , i.e., $p(a_i) = 1/n$. ■

■ **Example 19.9** If we select a symbol from A according to p , the entropy $H(X)$ is the minimum expected number of binary (Yes/No) questions required to identify the symbol. For equiprobable symbols this expected number is maximal and equals $\log |A|$. ■

We can extend the concept of entropy to a pair of random variables by means of using the joint probability mass function. In this way, the joint entropy will be a measure of the uncertainty associated to both variables. Let $A = \{a_1, a_2, \dots, a_n\}$ and $B = \{b_1, b_2, \dots, b_m\}$ two finite sets, and X and Y two random variables defined over the sets A and B respectively, with probability mass function $p(a)$ and $p(b)$, and joint probability mass function $p(a, b)$.

Definition 19.4.2 The *joint entropy* of the random variables A and B , denoted by $H(A, B)$, is defined as:

$$H(A, B) = \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{1}{p(a, b)}$$

Joint entropy is symmetric: $H(A, B) = H(B, A)$, since the double sum is unchanged by swapping a and b . The definition extends to any finite tuple $(A_1, \dots, A_k)$ via the joint mass function $p(a_1, \dots, a_k)$.

Adding a second random variable with an unknown outcome may increase entropy, as the following proposition shows.

Proposition 19.4.2 We have that

$$H(A, B) \geq \max(H(A), H(B))$$

Proof. Because $\sum_b p(a, b) = p(a)$, we have $p(a, b) \leq p(a)$ for all a, b , hence $\log \frac{1}{p(a, b)} \geq \log \frac{1}{p(a)}$. Therefore,

$$\begin{aligned} H(A, B) &= \sum_{a, b} p(a, b) \log \frac{1}{p(a, b)} \geq \sum_{a, b} p(a, b) \log \frac{1}{p(a)} \\ &= \sum_a p(a) \log \frac{1}{p(a)} = H(X). \end{aligned}$$

The inequality $H(A, B) \geq H(B)$ is analogous. Taking the maximum yields the claim. ■

The joint entropy of two variables cannot exceed the sum of their individual entropies.

Proposition 19.4.3 We have $H(A, B) \leq H(A) + H(B)$, with equality if and only if A and B are independent (i.e., $p(a, b) = p(a)p(b)$ for all a, b).

Proof. By Gibbs' inequality (which follows from $\ln x \leq x - 1$),

$$0 \leq \sum_{a,b} p(a,b) \log \frac{p(a,b)}{p(a)p(b)} = \sum_{a,b} p(a,b) \log p(a,b) - \sum_{a,b} p(a,b) \log p(a) - \sum_{a,b} p(a,b) \log p(b).$$

Rearranging and recognizing the entropies,

$$0 \leq -H(A,B) + H(A) + H(B),$$

which gives $H(A,B) \leq H(A) + H(B)$. Equality in Gibbs' inequality holds if and only if $p(a,b) = p(a)p(b)$ for all a, b , i.e., when A and B are independent. ■

The next concept derived from entropy is conditional entropy. It measures the remaining uncertainty about one random variable once the outcome of another is known.

Definition 19.4.3 The *conditional entropy* of the random variable B given the random variable A , denoted by $H(B | A)$, is defined as:

$$H(B | A) = \sum_{a \in A} \sum_{b \in B} p(a,b) \log \frac{1}{p(b | a)}$$

In general $p(b | a) \neq p(a | b)$, so $H(B | A) \neq H(A | B)$. We have $H(B | A) = 0$ if and only if B is (almost surely) a function of A .

The next proposition formalizes that conditioning cannot increase uncertainty.

Proposition 19.4.4 Given the random variables X and Y , we have that $H(Y | X) \leq H(Y)$, and $H(Y | X) = H(Y)$ if, and only if, $p(a)$ and $p(b)$ are independent.

Proof.

$$\begin{aligned} H(Y | X) &= \sum_{a \in A} \sum_{y \in B} p(a,b) \log \frac{1}{p(b | a)} = \sum_{a \in A} \sum_{b \in B} p(a,b) \log \frac{p(a)}{p(a,b)} \\ &= \sum_{a \in A} \sum_{b \in B} p(a,b) \log p(a) - \sum_{a \in A} \sum_{b \in B} p(a,b) \log p(a,b) \\ &= -H(X) + H(X,Y) \end{aligned}$$

By Proposition 19.4.3, $H(X,Y) \leq H(X) + H(Y)$, hence $H(Y | X) \leq H(Y)$. Equality holds if and only if $H(X,Y) = H(X) + H(Y)$, i.e., X and Y are independent. ■

From an intuitive standpoint, the uncertainty of a pair (X, Y) should equal the uncertainty of X plus the remaining uncertainty of Y after observing X . This is the *chain rule*.

Proposition 19.4.5 — Chain rule. For any discrete X, Y , we have

$$H(X, Y) = H(X) + H(Y | X).$$

Proof.

$$\begin{aligned} H(Y, X) &= \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{1}{p(a, b)} = \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{1}{p(a)p(a | b)} \\ &= \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{1}{p(a)} + \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{1}{p(a | b)} \\ &= H(X) + H(Y | X) \end{aligned}$$

■

The last derived concept of entropy we are going to see is mutual information. Intuitively, the mutual information of two random variables X and Y measures the information that X and Y share, that is, how much knowing one of these variables reduces the uncertainty about the other.

Definition 19.4.4 The *mutual information* of the random variable X and Y , denoted by $I(X; Y)$, is defined as:

$$I(X; Y) = \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{p(a, b)}{p(a)p(b)}$$

Since $p(a, b) = p(b, a)$ we have that $I(X; Y) = I(Y; X)$, that is, the order of the random variables does not affect the concept of mutual information.

Next proposition shows that mutual information is a positive quantity, and it is equal to 0 if, and only if, the random variables are independent.

Proposition 19.4.6 For any discrete random variables X and Y we have that $I(X; Y) \geq 0$, and $I(X; Y) = 0$ if, and only if, the variables X and Y are independent.

Proof. Interpret $I(X; Y)$ as the Kullback-Leibler divergence

$$I(X; Y) = D(p_{X,Y} \| p_X p_Y) = \sum_{a,b} p(a, b) \log \frac{p(a, b)}{p(a)p(b)} \geq 0$$

by Gibbs' inequality, with equality if and only if $p(a, b) = p(a)p(b)$ for all a, b . ■

Mutual information admits several equivalent forms that make its operational meaning explicit: it quantifies the reduction in uncertainty achieved by observing one variable about the other. Using the chain rule $H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y)$, we obtain the identities below, showing that $I(X; Y)$ is exactly the decrease in entropy of X due to Y (and symmetrically for Y).

Proposition 19.4.7 For any discrete random variables X and Y , we have that:

$$I(X;Y) = H(X) - H(X | Y) = H(Y) - H(Y | X)$$

Proof. By the chain rule, $H(X,Y) = H(Y) + H(X | Y)$. Substituting into

$$I(X;Y) = \sum_{a,b} p(a,b) \log \frac{p(a,b)}{p(a)p(b)} = -H(X,Y) + H(X) + H(Y),$$

we obtain $I(X;Y) = H(X) - H(X | Y)$. The other identity follows by symmetry. ■

Another equivalent form of mutual information is its “inclusion-exclusion” expression: the sum of the marginal entropies minus the joint entropy. It follows immediately by substituting $H(X | Y) = H(X,Y) - H(Y)$ (or symmetrically $H(Y | X) = H(X,Y) - H(X)$) into $I(X;Y) = H(X) - H(X | Y)$, and highlights the Venn-diagram overlap interpretation.

Proposition 19.4.8 For any discrete random variables X and Y , we have that:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

Proof. Combine $I(X;Y) = H(X) - H(X | Y)$ with $H(X | Y) = H(X,Y) - H(Y)$ (chain rule), yielding $I(X;Y) = H(X) + H(Y) - H(X,Y)$. ■

A useful sanity check is the self-information case. Since observing S reveals S completely, the remaining uncertainty $H(S | S)$ is zero; hence the mutual information of a variable with itself equals its entropy, as stated below.

Proposition 19.4.9 For any discrete random variable S we have that $I(S;S) = H(S)$.

Proof. By the identity $I(X;Y) = H(X) - H(X | Y)$ with $X = Y = S$,

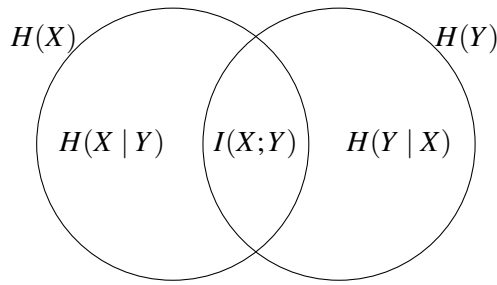
$$I(S;S) = H(S) - H(S | S) = H(S) - 0 = H(S). \quad \blacksquare$$

■ **Example 19.10** Let $X \sim \text{Bernoulli}(1/2)$ and let $Y = X \oplus N$, where $N \sim \text{Bernoulli}(\varepsilon)$ is independent noise ($\oplus$ is XOR). Then $H(X) = H(Y) = 1$, and

$$H(X | Y) = H_b(\varepsilon) := -\varepsilon \log \varepsilon - (1 - \varepsilon) \log (1 - \varepsilon), \quad I(X;Y) = 1 - H_b(\varepsilon).$$

For instance, with $\varepsilon = 0.1$, $H_b(0.1) \approx 0.469$, so $I(X;Y) \approx 0.531$ bits. A Venn diagram interpretation: the circle areas represent $H(X)$ and $H(Y)$, the overlap is $I(X;Y)$, and the non-overlapping parts are $H(X | Y)$ and $H(Y | X)$. ■

Figure 19.4 provides Venn-style schematic for entropies, conditional entropies and mutual information.

Figure 19.4: $H(X)$ and $H(Y)$ with overlap $I(X;Y)$.

19.5 Huffman Algorithm

From a practical point of view, there exists an algorithm, called *Huffman algorithm*, that provides a method to build compact prefix-free codes given a probability distribution. For simplicity, we will study first the particular case of constructing binary prefix-free codes, and later I will provide its generalization to the case of D-ary prefix-free codes.

Algorithm 19.1 Huffman Algorithm

```

procedure HUFFMAN( $Q$ )
   $T \leftarrow$  empty tree
  for  $i \leftarrow 1, d(Q) - 1$  do
    allocate a new node  $z$ 
     $z.\text{left} = x = \text{EXTRACT-MIN}(Q)$ 
     $z.\text{right} = y = \text{EXTRACT-MIN}(Q)$ 
     $z.\text{freq} = x.\text{freq} + y.\text{freq}$ 
    INSERT( $Q, z$ )
  end for
  return  $T$ 
end procedure

```

The algorithm (see Algorithm 19.1) expects as input a source alphabet $\mathcal{S} = \{s_1, s_2, \dots, s_q\}$ and their corresponding probabilities $P = \{p_1, p_2, \dots, p_q\}$. For simplicity, $Q = \{(s_1, p_1), (s_2, p_2), \dots, (s_q, p_q)\}$ will merge both sets into a single one. The algorithm works by constructing a binary tree T , similar to the one used in the proof of Theorem 19.2.1. The algorithm requires $d(Q) - 1$ iterations to finish. During each iteration, the two elements with the lowest probability are selected and removed from set Q , and a new tree node z is created, with the addition of the removed values, and added to the set Q . Once the tree has been constructed, we have to perform a tree transversal assigning a 0 to each left branch, and a 1 to each right branch, until we reach a leaf.

■ **Example 19.11** Assume we have the source alphabet $\mathcal{S} = \{a, b, c, d, e, f\}$ with the associated probabilities $P = \{0.35, 0.16, 0.08, 0.12, 0.06, 0.23\}$. In Figure are depicted the contents of the set Q and the tree T for each iteration of the algorithm. At the end of the algorithm, if we perform a traversal of the T tree, we will get the following prefix-free compact code for the source alphabet $\mathcal{S}$:

Source Word	Code Word
a	11
b	00
c	1011
d	100
e	1010
f	01

■

19.6 Discretization of Continuous Variables

When summarizing large masses of raw data, it is often useful to distribute the data into classes, or categories, and to determine the number of individuals belonging to each class, called absolute frequency. The following definition formally introduces this concept.

Definition 19.6.1 Let $\mathcal{S}$ be a population consisting of n individuals, and let a variable $X : \mathcal{S} \rightarrow \mathcal{D}$ represent the mapping of individuals in $\mathcal{S}$ to values in the set $\mathcal{D}$, where k is the cardinality of $\mathcal{D}$. The *absolute frequency*, also known simply as *frequency*, denoted by n_i for $1 \leq i \leq k$, quantifies the number of individuals in $\mathcal{S}$ for which X assigns the value corresponding to the i -th category of $\mathcal{D}$.

The sum of the frequencies must be equal to population size, that is, $\sum_{i=1}^k n_i = n$.

If the variable X is continuous, the elements of $\mathcal{D}$ are referred to as *class intervals*. The endpoints of these intervals are known as *class limits*; the smaller number is termed the *lower class limit*, and the larger number, the *upper class limit*. A *class interval* that lacks either an upper or a lower class limit is known as an *open class interval*. The *width* of a class interval is defined as the difference between the upper and lower class boundaries, denoted by $a_i = e_i - e_{i-1}$. The *class mark*, or the midpoint of a class interval, is calculated as $a_i = \frac{e_i + e_{i-1}}{2}$. It is assumed that all observations within a specific class interval are equivalent concerning their categorical assignment.

■ **Example 19.12** In a study measuring adult heights within a community, researchers record heights ranging from 150 cm to 200 cm and organize them into 10 cm class intervals: 150-159 cm, 160-169 cm, 170-179 cm, 180-189 cm, and 190-199 cm. Each interval's lower and upper class limits are respectively the start and end points, such as 150 cm and 159 cm for the first interval. If the last interval had no specified upper limit, it would be considered an open class interval. The class width, typically 10 cm, is the operational span between boundaries, and the class mark, calculated as the midpoint of each interval (e.g., 154.5 cm for the first interval), provides a central value for summarizing data within that range. ■

A tabular arrangement of data by classes together with the corresponding class frequencies is called a frequency distribution, or frequency table.

Definition 19.6.2 Let $\mathcal{S}$ be a population consisting of n individuals, and let $X : \mathcal{S} \rightarrow \mathcal{D}$ be a variable that maps individuals in $\mathcal{S}$ to k distinct values of the set $\mathcal{D}$. A *frequency distribution* is represented the set of pairs $\{(d_i, n_i) : 1 \leq i \leq k\}$, where d_i denotes the i -th interval and n_i represents the number of individuals from $\mathcal{S}$ whose value under X falls within the interval d_i .

Frequency distributions are useful for statistical analysis and helps in visualizing data by grouping values, which simplifies the understanding of distribution and central tendencies within the data.

The relative frequency of a class is the frequency of the class divided by the total frequency of all classes

Definition 19.6.3 Let $\mathcal{S}$ be a population consisting of n individuals, and let a variable $X : \mathcal{S} \rightarrow \mathcal{D}$ represent the mapping of individuals in $\mathcal{S}$ to values in the set $\mathcal{D}$, where k is the cardinality of $\mathcal{D}$. The *relative frequency*, denoted by f_i , is the ratio $f_i = \frac{n_i}{n}$ for $1 \leq i \leq k$.

The sum of the relative frequencies is equal to one, that is, $\sum_{i=1}^k f_i = 1$.

The total frequency of all values less than the upper class boundary of a given class interval is called the cumulative frequency up to and including that class interval.

Definition 19.6.4 Let $\mathcal{S}$ be a population consisting of n individuals, and let a variable $X : \mathcal{S} \rightarrow \mathcal{D}$ represent the mapping of individuals in $\mathcal{S}$ to values in the set $\mathcal{D}$, where k is the cardinality of $\mathcal{D}$. The *cumulative frequency*, denoted by N_i for $1 \leq i \leq k$, represents the total number of individuals in $\mathcal{S}$ for which X assigns a value less than or equal to the upper limit of the i -th category of $\mathcal{D}$. This is mathematically expressed as $N_i = \sum_{j=1}^i n_j$, where n_j is the absolute frequency of the j -th category.

By accumulating the frequencies up to each category or interval, cumulative

frequencies provides a running total that shows how many data points fall below a certain value.

Discretization Algorithms

Let $\mathcal{X}$ a continuous random variable that follows a probability density function $P_{\mathcal{X}}$, and assume we have collected n independent and identically distributed samples $\mathbf{x} = \{x_1, \dots, x_n\}$ from $\mathcal{X}$. We are interested in computing the length of a compressed version of $\mathbf{x}$ using an optimal compressor. Unfortunately, and except for some degenerate distributions, there is no lossless compression algorithm that produces a string with fewer bits than encoding directly the elements $\mathbf{x}$. Compression algorithms for continuous data only work in case that the elements of $\mathbf{x}$ are not independent, as it is the case with images or sound. But, if this is not the case, the only option available to compress $\mathbf{x}$ is to use a lossy compression algorithm, where some information is lost.

We are looking for an algorithm to produce a finite non-overlapping partition of m discrete intervals $D = \{[d_0, d_1], (d_1, d_2], \dots, (d_{m-1}, d_m]\}$, where $d_0 = \min \mathbf{x}_j$, and $d_m = \max \mathbf{x}_j$, and $d_i < d_{i+1}$ for $i = 0, 1, \dots, m-1$, assign a unique label to each interval, and encode the elements of $\mathbf{x}$ using this labeling schema. As compression algorithm we will use an optimal length code given the relative frequencies of the labels in the encoded vector. In this sense, our goal is to have a collection of intervals with sufficiently number of samples (so they are statistically significant) and that the distribution of frequencies resembles the original probability distribution $P_{\mathcal{X}}$.

A discretization algorithm is a mapping between a (possibly huge) number of numeric values and a reduced set of discrete values, and so, it is a process in which some information is potentially lost. The choice of discretization algorithm is something that could have a high impact in the practical computation of the nescience. We are interested in a discretization algorithm that produces a large number of intervals (low bias), with a large number of number of observations per interval (low variance). Common techniques include *equal width discretization*, *equal frequency discretization* and *fixed frequency discretization*. However, these techniques require the optimization of an hyperparameter, and so, they are not suitable for our purposes.

In a *proportional discretization approach* the number of intervals m and the number of observations per interval s are equally proportional to the number of observations n . The algorithm starts by sorting the values of $\mathbf{x}_j$ in ascending order and then discretizing them into m intervals of approximately s (possibly identical) values each. In this way, as the number of training observations increases, both interval frequency and number of intervals increases, taking advantage of the larger number of observations. In the same way, when the number of observations decreases, we reduce both.

References

Core references for the chapter "Coding"

[CT12]: Cover and Thomas is the standard reference for modern information theory. It introduces entropy, mutual information, source and channel coding theorems, and provides the theoretical basis for coding.

[Abr63]: Abramson's book is a classic early text that presents coding theory in an accessible way. It is useful for historical perspective and for understanding the development of the main coding ideas.

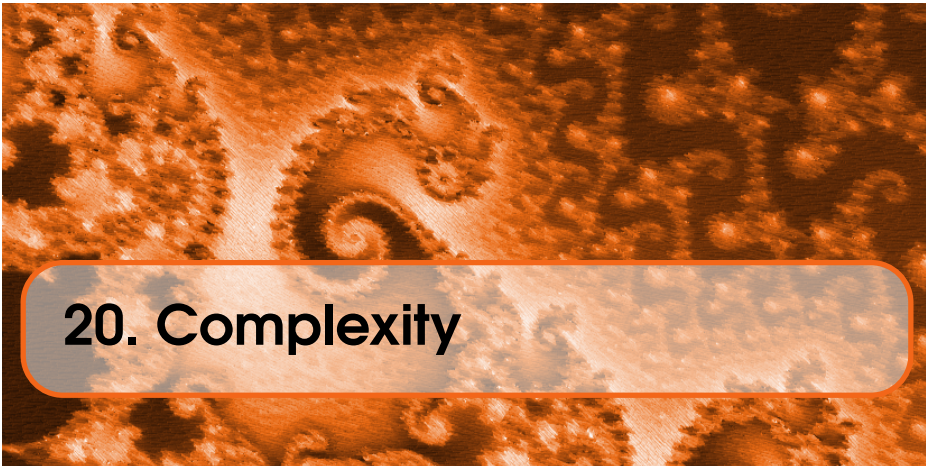
[Kra49]: Kraft's thesis introduces the Kraft inequality, a cornerstone result for the theory of uniquely decodable prefix codes.

[McM56]: McMillan extends Kraft's result to a more general form, establishing the Kraft-McMillan inequality fundamental to coding theory.

[GG12]: Gersho and Gray provide the standard reference on quantization methods used in lossy compression, important for bridging coding theory with practical applications.

[Llo82]: Lloyd's algorithm (often rediscovered as k-means) is a central method for quantization and lossy compression. It provides a concrete link between coding and clustering.

[LV13]: Li and Vitányi's book is the definitive source on Kolmogorov complexity, linking coding theory with randomness and algorithmic information. It is essential for connecting coding to the broader framework of nescience.



20. Complexity

*An approximate answer to the right problem
is worth a good deal more than
an exact answer to an approximate problem.*

John Tukey

In Appendix 19, the concept of the complexity of a string based on the lengths of the codewords of a prefix-free code was introduced. This definition is limited by two main factors: first, it necessitates prior knowledge of the set of possible strings, and second, it requires the definition of a probability distribution over this set a priori. It would be highly desirable to expand the set of strings to encompass all strings (that is, $\mathcal{B}^*$) without requiring a probability distribution, thereby providing an absolute notion of string complexity. Unfortunately, even if these issues are resolved, a more fundamental limitation arises when studying the complexity of strings using codes: certain strings that we intuitively expect to be simple cannot be compressed. For instance, the binary expansion of the constant π is widely conjectured to behave like a uniform distribution over the set $\{0, 1\}$ and, as such, cannot be compressed. Yet it can be fully and effectively described by a very short mathematical formula. This motivates the need for an alternative definition of string complexity.

Kolmogorov complexity, also known as *algorithmic information theory*, offers a definition of the complexity of a string that directly addresses these issues. Intuitively, the amount of information in a finite string is measured by the length of the shortest computer program capable of producing the string. This approach does not require prior knowledge of the set of valid strings or their probability distribution. Furthermore, objects like π are appropriately classified as having low complexity. We may argue that Kolmogorov complexity provides a universal definition of the amount of information that closely aligns with our intuitive understanding. To compute the Kolmogorov complexity of a string, it is necessary to fix a universal description method or computer language, together with a universal computer. One might question whether, in doing so, the complexity of a string becomes dependent on the chosen language. Fortunately, it has been shown that this is not the case: all reasonable (and sufficiently powerful) languages yield the same description length, up to a fixed constant that depends on the choice of languages but not on the string itself. Unfortunately, Kolmogorov complexity also introduces a significant challenge: it is a non-computable quantity and, as such, must be approximated in practice.

At this point, one might ask whether it is possible to define the complexity of arbitrary objects, not just strings. The answer is yes, at least in theory. Given an object x , the task is to provide an encoding method that represents the object as a string. This encoding is useful only if we can losslessly and effectively reconstruct the original object from its description. However, providing such encodings is not always feasible, either because the objects in question are abstract (as in much of mathematics) or because practical reconstruction of the object from its description is currently impossible (for example, with living organisms¹).

20.1 Strings Complexity

In Section 18.1, the concept of the Turing machine, an idealized model of computation, was introduced. We saw that Turing machines can be represented as partial computable functions $T : \mathcal{B}^* \rightarrow \mathcal{B}^*$, which assign to each input string $s \in \mathcal{B}^*$ an output string $T(s) \in \mathcal{B}^*$ (Definition 18.4.1). We also introduced the concept of a universal Turing machine $U : \mathcal{B}^* \times \mathcal{B}^* \rightarrow \mathcal{B}^*$ (Definition 18.2.1), a machine that can simulate the behavior of any other Turing machine; that is, for all $(x, v) \in \mathcal{B}^* \times \mathcal{B}^*$, we have that $U(x, v) = T_x(v)$. Later, in Section 19.1, the concept of a code, and in particular, the notion of a prefix-free code, was introduced (Definition 19.1.4). We saw that this kind of code presents important properties (Theorem 19.2.1). The next definition merges the best of both worlds, Turing machines and prefix-free codes, and introduces a new type of universal Turing machine.

¹As of now, it is not possible to recreate an animal solely based on its DNA.

Definition 20.1.1 A *prefix-free universal Turing machine* is a universal Turing machine $U : \mathcal{B}^* \times \mathcal{B}^* \rightarrow \mathcal{B}^*$ such that, for every $v \in \mathcal{B}^*$, the domain U_v is prefix-free, where $U_v : \mathcal{B}^* \rightarrow \mathcal{B}^*$ and $U_v(p) = U(p, v)$ for all $p \in \mathcal{B}^*$.

Using modern computer science terminology we could say that U is the computer, p is the program, and v is the input to the program. Intuitively, the above definition requires that no computer program can be a prefix of any other program. This is not a limitation from the point of view of string lengths, since, by applying McMillan's theorem (Theorem 19.2.1), given a uniquely decodable program, we could always find a prefix-free one that computes exactly the same function and has the same length. In practice, programming languages enforce syntactic rules that make programs effectively self-delimiting (for example, programs or functions must terminate with specific delimiters).

Fixing the input v allows us to regard the set of valid programs $\{p : U(p, v) \downarrow\}$ as prefix-free. This ensures that descriptions can be uniquely parsed and avoids ambiguity when concatenating programs.

The concept of a prefix-free universal Turing machine allows us to introduce a new definition of the complexity of a string that aligns more closely with our intuitive understanding of the amount of computational information contained in an object (encoded as a string).

Definition 20.1.2 — Kolmogorov Complexity. Fix a prefix-free universal Turing machine $U : \mathcal{B}^* \times \mathcal{B}^* \rightarrow \mathcal{B}^*$. The *Kolmogorov complexity* of a string $s \in \mathcal{B}^*$, denoted by $K(s)$, is defined as:

$$K(s) = \min_{p, v \in \mathcal{B}^*} \{l(p) + l(v) : U(p, v) = s\}.$$

Intuitively, the shortest description of a string s is given by two elements: a program p (a self-delimiting program) that captures all the regular patterns of the string, and a new string v that comprises those parts of s that do not present any regularity. We have to find the optimum balance between increasing the complexity of the program, trying to grasp more regularities, or increasing the size of the non-compressible part.²

■ **Example 20.1** Consider the string composed of one thousand repetitions of the substring "10", that is $\underbrace{"1010 \dots 1010"}_{1.000 \text{ times}}$. We could write the following program:

```
example(char *v) {
```

²In the literature, the Kolmogorov complexity of the string s is defined as $K(s) = \min_{p \in \mathcal{B}^*} \{l(p) : U(p, \lambda) = s\}$, that is, the length of the shortest computer program that, without any additional input, can print the string s . We prefer to use the two-part definition $l(p) + l(v)$ because it is more in line with the requirements of the theory of nescience.

```

    for (int i=1; i<=1000; i++)
        printf("%s", v);
}

```

and then run it with:

```
example("10");
```

in order to print it. The length of the original string is 2,000 bits, but suppose the program length is approximately 480 bits (assuming that every symbol is encoded using a uniform code of 8 bits), and the input length is 2 bits. We can then conclude that the string has a low complexity. Of course, in order to compute the actual Kolmogorov complexity of the string we would need to find the shortest Turing machine that prints that string.

On the contrary, a string composed of two thousand random bits would, with overwhelming probability, have high complexity, since no program significantly shorter than the string itself can generate it. ■

As we mentioned in the preface of this chapter, Kolmogorov complexity would not be particularly useful if the complexity of strings depended on the choice of universal Turing machine. The following theorem demonstrates that this concern is unfounded, up to a constant that depends on the choice of machines, but not on the strings themselves. This establishes Kolmogorov complexity as an inherent property of strings.

Theorem 20.1.1 — Invariance theorem. Let U and U' be two universal Turing machines. Then, there exists a constant $C_{U,U'}$, depending only on U and U' , such that for each string $s \in \mathcal{B}^*$ we have:

$$K_U(s) \leq K_{U'}(s) + C_{U,U'}.$$

Proof. Let p, v be the shortest strings such that $U'(p, v) = s$. Then we can encode the pair (U', p) and simulate it on U , obtaining $U(\langle U', p, v \rangle, \lambda) = U'(p, v) = s$. Encoding the pair (U', p) requires a fixed description of U' plus the description of p . Thus, $K_U(s) \leq K_{U'}(s) + C_{U,U'}$, where $C_{U,U'}$ is the length of an interpreter for U' on U . ■

■ **Example 20.2** Consider a universal programming language, such as Java, and an alternative language, such as Python. We can write a Python interpreter in Java, that is, a Java program that takes a Python script as input and executes it. Then, to compute the complexity of a string $s \in \mathcal{B}^*$ using Java, $C_J(s)$, it would be no greater than the complexity of the string using Python, $C_P(s)$, plus the length of the Python interpreter written in Java, $C_{J,P}$. Importantly, the length of the interpreter, $C_{J,P}$, does not depend on the string s . ■

Although we have proved that Kolmogorov complexity does not depend on the selected universal Turing machine, the size of the constant $C_{U,U'}$ could pose a limitation in practical applications, especially when computing the complexity of short strings where the constant might significantly exceed the complexity of the string itself. This challenge is addressed by the Minimum Description Length principle, as described in Section 21.5.

Notation 20.1. We denote by s^* the shortest program that outputs the string s on the universal Turing machine U , that is, $s^* = \langle p, v \rangle$, $U(s^*) = s$, and $l(s^*) = K(s)$. If more than one program satisfies these properties, we select the first one using a lexicographical order induced by $0 < 1$.

The size of the constant $C_{U,U'}$ is not the only challenge presented by Kolmogorov complexity; another issue is its non-computability, that is, there is no algorithm capable of determining the shortest program that generates an arbitrary string. The following theorem on the uncomputability of Kolmogorov complexity marks a pivotal insight into the intrinsic limits of complexity theory.

Theorem 20.1.2 The function $K : \mathcal{B}^* \rightarrow \mathbb{N}$ that assigns to each string s its Kolmogorov complexity $K(s)$ is not computable.

Proof. Assume, for contradiction, that K is computable. Then we could construct a function that, for any n , finds the first string s such that $K(s) > n$. This function would produce such an s by a program of length $O(\log n)$, thereby giving a description of s much shorter than n . This contradicts the definition of Kolmogorov complexity. Therefore, K is not computable. ■

If K were computable, we could also solve the Halting Problem by constructing a program that, for any input program and input, computes whether the program halts by checking if its Kolmogorov complexity is finite. Since the Halting Problem is known to be undecidable, this provides an alternative contradiction.

In practice, we approximate Kolmogorov complexity using compression algorithms, such as the Huffman algorithm described in Section 19.5, or more sophisticated schemes like Lempel-Ziv, which provide practical upper bounds on $K(s)$ relative to their model class.

20.2 Properties of Complexity

In this section, we delve into the properties of Kolmogorov complexity. We will explore the foundational principles that govern this complexity measure, including its invariance, symmetry, and non-computability. Through examining

these properties, we gain deeper insights into the interplay between information, computation, and randomness.

Kolmogorov complexity is always a finite positive natural number.

Proposition 20.2.1 For all $s \in \mathcal{B}^*$ we have that $0 < K(s) < \infty$.

Proof. Since $K(s)$ is defined as the length of a program-input pair, it is a non-negative integer. For non-empty strings s , we have $K(s) > 0$. The property $K(s) < \infty$ is a consequence of Proposition 20.2.2 and the fact that we are only dealing with finite strings. ■

The Kolmogorov complexity of a string cannot surpass the sum of its own length and a constant.

Proposition 20.2.2 There is a constant c such that for all $s \in \mathcal{B}^*$ we have that $K(s) \leq l(s) + c$.

Proof. Let $s \in \mathcal{B}^*$ be an arbitrary string, and consider the encoding of a Turing machine p such that for any input $v = s$, it halts and outputs s . The program p is designed to simply reproduce its input. Given this setup, when p is executed on a universal Turing machine U with s as input, it satisfies the condition $U(p, s) = s$. The length of p is a constant c across all strings s . There exists a fixed program p_{id} that outputs its input; its length is constant and independent of s . By the definition of Kolmogorov Complexity $K(s)$, which seeks the minimum length of a program-input pair that generates s , the combination of p and s presents a feasible solution. Therefore, we have $K(s) \leq l(s) + l(p) = l(s) + c$. ■

The size of the constant c depends on the specific encoding schema used by the selected universal Turing machine U , but it is independent of the string s . In Section 20.6, we will explore the characteristics of random strings, which are defined as strings that cannot be compressed. Such strings exhibit a Kolmogorov complexity close to their own length, that is, $K(s) \geq l(s) - c$ for some constant c .

The absolute difference in Kolmogorov complexity between any string x and its transformed counterpart $f(x)$, via a computable bijection, is bounded by a constant c . That is, not only does f not increase the complexity of x by more than a constant, but also f does not decrease the complexity by more than a constant.

Proposition 20.2.3 Let $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ is a computable bijection, then there exists a constant c such that $|K(f(x)) - K(x)| < c$.

Proof. Let P_f be the program that computes f and $P_{f^{-1}}$ the program that computes the inverse of f . For any string x , let P_x be the shortest program that generates x . Then, a program $P_{f(x)}$ that generates $f(x)$ can be constructed by concatenating P_x with P_f . The length of this program is $|P_{f(x)}| = |P_f| + |P_x|$. Since $|P_f|$ is a constant that does not depend on x , we can say that $K(f(x)) \leq K(x) + |P_f|$. Similarly, given

$f(x)$, we can construct a program P'_x to generate x by applying $P_{f^{-1}}$ to $f(x)$. The length of this program is $|P'_x| = |P_{f^{-1}}| + |P_{f(x)}|$. Thus, $K(x) \leq K(f(x)) + |P_{f^{-1}}|$. The two inequalities combined imply that $|K(f(x)) - K(x)| \leq \max(|P_f|, |P_{f^{-1}}|) = c$, where c is a constant that represents the maximum of the lengths of the programs that compute f and f^{-1} . This constant c does not depend on x , but rather on the complexity of the functions f and f^{-1} . ■

This proposition shows a remarkable stability of informational content under computable bijections, underscoring the intrinsic robustness of Kolmogorov complexity in the face of such transformations.

■ **Example 20.3** Consider the function $f : \mathcal{B}^* \rightarrow \mathcal{B}^*$ that reverses the order of the bits in a string, i.e., $f(x_1x_2 \dots x_n) = x_n \dots x_2x_1$. This function is a computable bijection, since both f and its inverse (which is itself) can be computed by a fixed, finite program. If x is a highly compressible string, such as $x = 1010 \dots 10$ repeated 1,000 times, then $f(x)$ is also highly compressible (it is the same pattern written backwards). If x is an incompressible random string, then $f(x)$ is also incompressible. The difference in Kolmogorov complexity between x and $f(x)$ is bounded by the length of the fixed program that reverses the bits, a constant independent of x . ■

Finally, it is worth emphasizing that some strings are very compressible, and this phenomenon occurs at every string length. For instance, the string of n zeros,

$$0^n = \underbrace{00 \dots 0}_n, \quad \text{\textit{n times}}$$

has a description of length $O(\log n)$: a program that prints "0" exactly n times. Therefore, $K(0^n) \leq c + \log n$, which is asymptotically much smaller than n . Similarly, strings with simple patterns, such as alternating zeros and ones, or repetitions of short substrings, can always be described concisely regardless of their total length.

20.3 Joint Kolmogorov Complexity

The joint Kolmogorov complexity of two strings s and t is defined as the length of the shortest program p that, when executed on a universal Turing machine U , outputs the pair $\langle s, t \rangle$, in such a way that both strings can be unambiguously retrieved. Here, $\langle s, t \rangle$ denotes a computable pairing function that encodes two strings into a single string in such a way that both components can be effectively recovered. The pairing function $\langle s, t \rangle$ is assumed to be a fixed, computable bijection with computable inverse, so that both s and t can be effectively recovered. Different choices of pairing function affect $K(s, t)$ by at most an additive constant.

Definition 20.3.1 — Joint Kolmogorov Complexity. The *Joint Kolmogorov complexity* of the strings $s, t \in \mathcal{B}^*$, denoted by $K(s, t)$, is defined as:

$$K(s, t) = \min_{p, v \in \mathcal{B}^*} \{l(p) + l(v) : U(p, v) = \langle s, t \rangle\}$$

The notation $K(s, t)$ and $K(st)$ represent two different concepts in the context of Kolmogorov complexity. $K(s, t)$ refers to the joint Kolmogorov complexity of two strings s and t as per Definition 20.3.1, meanwhile $K(st)$ represents the Kolmogorov complexity of the concatenation of s and t , without any additional structure to distinguish between them, and so, Definition 20.1.2 is applied. The choice between $K(s, t)$ and $K(st)$ depends on whether it's important to preserve and utilize the distinction and relationship between s and t . If analyzing the interplay or the shared characteristics of s and t is relevant, $K(s, t)$ is more appropriate. If the focus is on the information content of the combined sequence without regard to its origin from two separate strings, $K(st)$ is used.

■ **Example 20.4** Consider the strings $s = 0000$ and $t = 1111$. The concatenation $st = 00001111$ can be described by a short program that prints this eight-bit string directly, so $K(st)$ is roughly the length of that description. However, the joint description $\langle s, t \rangle$ requires that the decoding procedure be able to recover the boundary between s and t . Thus $K(s, t)$ and $K(st)$ differ by at most a fixed constant, reflecting the extra information required to separate s and t . ■

Our first proposition highlights a fundamental symmetry in Kolmogorov complexity, illustrating that the complexity of describing a pair of strings in either order differs by at most a constant. This reflects the intrinsic property that the information content is independent of the specific arrangement of the strings being described. The constant c encapsulates the overhead associated with the operations needed to reverse the order of the strings.

Proposition 20.3.1 There is a constant c such that for all $x, y \in \mathcal{B}^*$ we have that $|K(x, y) - K(y, x)| \leq c$.

Proof. Let U be a universal Turing machine, and let p_{yx} be the shortest program that outputs $\langle y, x \rangle$ when executed on U . To obtain a program that outputs $\langle x, y \rangle$, we can prepend p_{yx} with a fixed program that swaps the order of the two components in the decoded pair. This additional program has a constant length c , independent of x and y . Therefore, $K(x, y) \leq K(y, x) + c$. ■

Next proposition underscores the subadditive nature of Kolmogorov complexity, proving that the total complexity of describing two strings jointly cannot exceed the sum of their individual complexities by more than a fixed constant, irrespective of the strings' content.

Proposition 20.3.2 There is a constant c such that for all $s, t \in \mathcal{B}^*$ we have that $K(s, t) \leq K(s) + K(t) + c$.

Proof. Let s^* and t^* be the shortest self-delimiting programs that generate s and t , respectively. Since s^* and t^* are prefix-free, their concatenation can be parsed unambiguously by a universal Turing machine U . A fixed wrapper program of length c instructs U to run both s^* and t^* in sequence and output the pair $\langle s, t \rangle$. Therefore, $K(s, t) \leq K(s) + K(t) + c$. ■

The final proposition establishes a lower bound on the joint complexity of two strings relative to their individual complexities.

Proposition 20.3.3 There is a constant c such that for all $s, t \in \mathcal{B}^*$ we have

$$K(s, t) \geq \max(K(s), K(t)) - c.$$

Proof. Let p be the shortest program that outputs the pair $\langle s, t \rangle$, so $K(s, t) = l(p)$. From $\langle s, t \rangle$, both s and t can be effectively recovered using a fixed decoding program of length c . Thus $K(s) \leq K(s, t) + c$ and $K(t) \leq K(s, t) + c$. Rearranging gives $K(s, t) \geq K(s) - c$ and $K(s, t) \geq K(t) - c$, which together imply $K(s, t) \geq \max(K(s), K(t)) - c$. ■

20.4 Conditional Kolmogorov complexity

In this section, we explore the concept of *conditional Kolmogorov complexity*, which measures how the description length of a string s may decrease when prior knowledge of another string t is available. This notion highlights the impact of background information on the compressibility of a description. Here, $\langle v, t \rangle$ denotes a computable pairing function with computable inverse, ensuring both v and t can be recovered unambiguously. Different choices of pairing function affect $K(s|t)$ by at most an additive constant.

Definition 20.4.1 — Conditional Kolmogorov Complexity. The *conditional Kolmogorov complexity* of a string $s \in \mathcal{B}^*$ given the string $t \in \mathcal{B}^*$ is defined as:

$$K(s|t) = \min_{p, v \in \mathcal{B}^*} \{l(p) + l(v) : U(p, \langle v, t \rangle) = s\}$$

This definition is equivalent up to a constant to the standard formulation $K(s|t) = \min_p \{l(p) : U(p, t) = s\}$, since the encoding $\langle v, t \rangle$ can be replaced by t alone with at most constant overhead. As with unconditional Kolmogorov complexity, the conditional complexity is machine-independent: for any two universal Turing machines U and U' , there exists a constant $C_{U, U'}$ such that for all $s, t \in \mathcal{B}^*$,

$$K_U(s|t) \leq K_{U'}(s|t) + C_{U, U'}.$$

■ **Example 20.5** Let $s = 1010101010$ and $t = 10$. The unconditional complexity $K(s)$ is proportional to the length of s , since it is a ten-bit string. However, given t , we can describe s succinctly by a short program: “print t five times.” Thus $K(s|t)$ is only $O(\log 5)$, far smaller than $K(s)$. This illustrates how prior knowledge reduces the description length. ■

As with the unconditional Kolmogorov complexity, the conditional Kolmogorov complexity is a finite non-negative integer.

Proposition 20.4.1 For all $s, t \in \mathcal{B}^*$ we have that $0 \leq K(s|t) < \infty$.

Proof. $K(s|t)$ is the length of a program-input pair, so it is non-negative. The property $K(s|t) < \infty$ is a consequence of Proposition 20.2.2 and the fact that we are only dealing with finite strings. ■

Next proposition posits that when a string s is conditioned upon itself, its complexity reduces to at most a universal constant.

Proposition 20.4.2 There is a constant c such that for all $s \in \mathcal{B}^*$ we have that $K(s|s) \leq c$.

Proof. When a string s is conditioned on itself, the information needed to generate s from s can be encapsulated in a Turing machine that simply copies its input to its output. This machine, being independent of the specific content of s , has a fixed length c . ■

This proposition explores the relationship between unconditional and conditional Kolmogorov complexities, establishing an upper bound for the latter. It asserts that for any strings s and t , the complexity of s given t is at most the complexity of s alone, plus a constant c . This highlights the intuitive notion that having additional information can only reduce the complexity of describing a string, or in the worst case, add a constant overhead, but does not increase it beyond this bound.

Proposition 20.4.3 There is a constant c such that for all $s, t \in \mathcal{B}^*$ we have that $K(s|t) \leq K(s) + c$.

Proof. Let p_s be the shortest program that generates s without any auxiliary input, so $|p_s| = K(s)$. Construct a new program p' that ignores the conditional input t and executes p_s . The additional instructions have fixed length c , independent of s or t . Therefore $K(s|t) \leq K(s) + c$. ■

Conditional complexity is not symmetric. In general, $K(s|t) \neq K(t|s)$.

Proposition 20.4.4 There exist strings $s, t \in \mathcal{B}^*$ such that $K(s|t) \neq K(t|s)$.

Proof. Consider s as a random n -bit string, and let t be a shortest description of s . Then $K(s|t) = O(1)$, since given t one can directly reconstruct s . On the other hand, $K(t|s) \geq n - O(1)$, since otherwise we would obtain a shorter description of s , contradicting minimality. Thus, in general, conditional Kolmogorov complexity is asymmetric. ■

The relationship between conditional, unconditional, and joint Kolmogorov complexities offers a comprehensive perspective on the informational interdependencies of binary strings. It posits that the complexity of a string s given another string t is at most the complexity of s alone, which in turn is no greater than the joint complexity of both s and t .

Proposition 20.4.5 For all $s, t \in \mathcal{B}^*$ we have that $K(s|t) \leq K(s) \leq K(s, t)$.

Proof. The first inequality follows from Proposition 20.4.3. The second inequality follows from Proposition 20.3.3, since from $\langle s, t \rangle$ we can recover s with constant overhead. ■

The Kolmogorov complexity chain rule is a fundamental principle that connects the joint complexity of two strings with their individual and conditional complexities. It asserts that the total complexity of a pair of strings s and t can be decomposed into the complexity of s plus the complexity of t given s , up to a logarithmic additive term. This relationship mirrors the additive property of entropy in information theory and provides a powerful tool for understanding the interplay between information content and conditional information in the context of Kolmogorov complexity.

Proposition 20.4.6 — Kolmogorov chain rule. For all $s, t \in \mathcal{B}^*$ we have

$$K(s, t) = K(s) + K(t | s) + O(\log(K(s, t))).$$

Proof. The upper bound follows by concatenating a shortest program p_s that generates s with a shortest program $p_{t|s}$ that generates t given s , plus a fixed wrapper program that runs them in sequence. The overhead is logarithmic, due to the need for self-delimiting encodings of program lengths.

For the lower bound, note that from $\langle s, t \rangle$ one can reconstruct s , and then from s and a decoding procedure reconstruct t . This yields $K(s, t) \geq K(s) + K(t|s) - O(\log(K(s, t)))$. Together, these inequalities establish the claim. ■

■ **Example 20.6** Let $s = 0000$ and $t = 1111$. The joint description $\langle s, t \rangle$ can be generated by first describing s (a short program "print four zeros") and then describing t given s (a short program "print four ones"). Hence $K(s, t) \approx K(s) + K(t|s)$ up to logarithmic overhead, illustrating the chain rule in practice. ■

20.5 Information Distance

In this section, we aim to introduce a universal metric for quantifying the absolute information distance between two or more individual entities encoded as strings of symbols. Intuitively, the information distance between two strings s and t can be understood as the length of the shortest computer program for a universal computer that enables the generation of s given t and vice versa.

Definition 20.5.1 The *information distance* between two strings $s, t \in \mathcal{B}^*$ with respect to a universal Turing machine U , denoted by $ID_U(s, t)$, is defined as

$$ID_U(s, t) = \min\{l(p) : U(p, s) = t, U(p, t) = s\}$$

For any two universal Turing machines U_1 and U_2 , the information distance between two strings differs by at most an additive constant c , which depends only on the choice of machines and not on the specific strings. It is also important to note that despite its theoretical significance, information distance is non-computable: there does not exist an algorithm that can compute it exactly for arbitrary strings.

One might consider using the conditional Kolmogorov complexity $K(s | t)$ as a measure of information distance. However, this quantity is asymmetric (see Proposition 20.4.2), making it unsuitable as a distance. Similarly, the sum $K(s | t) + K(t | s)$ is also inadequate, as it double-counts the overlapping information needed to transform s into t and vice versa.

The following result shows how information distance can be expressed in terms of conditional Kolmogorov complexities.

Proposition 20.5.1 Let $s, t \in \mathcal{B}^*$ be two binary strings. Then

$$ID_U(s, t) = \max\{K(s | t), K(t | s)\} + O(\log \max\{K(s | t), K(t | s)\}).$$

Proof sketch. Suppose without loss of generality that $K(s | t) \geq K(t | s)$. A shortest program p of length $K(s | t)$ transforms t into s . To also transform s into t , we append a fixed routine that, given s , reconstructs t by inverting the transformation. This requires at most logarithmic overhead to encode program lengths in a self-delimiting way. Thus one program of length $K(s | t) + O(\log K(s | t))$ suffices in both directions. Therefore,

$$ID_U(s, t) = \max\{K(s | t), K(t | s)\} + O(\log \max\{K(s | t), K(t | s)\}).$$

■

■ **Example 20.7** Let $s = 0011$ and $t = 1100$. The bitwise exclusive-or $s \oplus t = 1111$ captures the difference between the two strings. Given s and the xor-mask $s \oplus t$, one can reconstruct t by $t = s \oplus (s \oplus t)$, and conversely obtain s from t and

the mask. Hence, the information distance between s and t is essentially the Kolmogorov complexity of the xor-mask, $K(s \oplus t)$, up to logarithmic overhead. If the mask is simple (all ones), the distance is small; if it is random, the distance is close to the full length of the strings. ■

It is convenient to introduce the following function, which captures the essence of information distance:

$$E(x, y) = \max\{K(x | y), K(y | x)\}.$$

Proposition 20.5.2 $E(x, y)$ is a metric up to logarithmic additive terms.

Proof sketch. Non-negativity and symmetry are immediate from the definition. Identity holds since $E(x, x) = O(1)$, while if $x \neq y$, at least one of $K(x|y)$ or $K(y|x)$ is large, so $E(x, y) > 0$. For the triangle inequality, let x, y, z be strings. From x one can compute y using a program of length $K(y|x)$, and from y one can compute z using a program of length $K(z|y)$. Composing these programs, from x we can compute z using a program of length $K(y|x) + K(z|y) + O(\log)$. Hence

$$K(z|x) \leq K(y|x) + K(z|y) + O(\log).$$

A similar argument applies symmetrically, establishing the triangle inequality for $E(x, y)$ up to logarithmic additive terms. ■

The information distance $E(x, y)$ admits an alternative characterization in terms of the joint and individual Kolmogorov complexities, as follows.

Proposition 20.5.3

$$E(x, y) = \max\{K(x | y), K(y | x)\} = K(xy) - \min\{K(x), K(y)\} + O(\log K(xy)).$$

Proof sketch. Assume without loss of generality that $K(x) \leq K(y)$. By the chain rule,

$$K(xy) = K(x) + K(y | x) + O(\log K(xy)).$$

Thus

$$K(y|x) = K(xy) - K(x) + O(\log K(xy)).$$

Since $E(x, y) = \max\{K(x|y), K(y|x)\}$, the dominant term is $K(y|x)$ in this case. Hence

$$E(x, y) = K(xy) - \min\{K(x), K(y)\} + O(\log K(xy)).$$

■

We now introduce the notion of *admissible information distances*.

Definition 20.5.2 An *admissible information distance* $d(x,y)$ is a total function mapping pairs of strings to non-negative integers such that: (1) $d(x,y)$ is upper semicomputable, (2) it satisfies the metric properties up to $O(1)$, and (3) it is normalized in the sense that $\sum_y 2^{-d(x,y)} \leq 1$ for all x .

Among all admissible information distances, $E(x,y)$ plays a distinguished role by being the smallest one up to an additive constant, as stated next.

Proposition 20.5.4 $E(x,y)$ is an admissible information distance, and moreover it is minimal: for every other admissible information distance $d(x,y)$, we have

$$E(x,y) \leq d(x,y) + O(1).$$

Proof idea. $E(x,y)$ is upper semicomputable because conditional complexities are. It satisfies the metric axioms up to logarithmic additive terms, as shown earlier. Normalization follows because conditional Kolmogorov complexities induce a semimeasure. Minimality holds because any admissible distance can be simulated by conditional descriptions, while $E(x,y)$ already captures the maximal overlap of descriptions. Thus $E(x,y)$ is universal among admissible information distances. ■

This universality means that $E(x,y)$ encompasses all other admissible information distances: it is the minimal such function up to additive constants. Therefore, $E(x,y)$ provides a canonical, universal measure of information distance.

Normalized Information Distance

Information distance is an absolute measure; however, when assessing similarity, we are often more concerned with relative measures. For instance, two strings of length 1,000,000 differing by 1000 bits are perceived as relatively more similar than two strings of length 1000 that differ by the same number of bits. This motivates the introduction of a normalized version of information distance: the size of the description required for transformation should be evaluated relative to the sizes of the objects being compared.

Definition 20.5.3 The *normalized information distance* between two binary strings $s, t \in \mathcal{B}^*$, denoted by $NID(s,t)$, is defined as:

$$NID(s,t) = \frac{\max\{K(s|t), K(t|s)\}}{\max\{K(s), K(t)\}}.$$

As expected, the normalized information distance takes values between 0 and 1, up to negligible additive terms.

Proposition 20.5.5 The normalized information distance $NID(s, t)$ takes values in the range $[0, 1]$ up to vanishing additive terms.

Proof. Non-negativity follows because complexities are non-negative, and so $NID(s, t) \geq 0$. For the upper bound, we use the fact that $K(s|t) \leq K(s) + O(1)$ and $K(t|s) \leq K(t) + O(1)$. Therefore

$$\max\{K(s|t), K(t|s)\} \leq \max\{K(s), K(t)\} + O(1).$$

Dividing by $\max\{K(s), K(t)\}$ gives

$$NID(s, t) \leq 1 + O\left(\frac{1}{\max\{K(s), K(t)\}}\right).$$

Hence, up to negligible terms, $NID(s, t) \in [0, 1]$. ■

The normalized information distance not only captures relative similarity but also inherits the essential structure of a metric space, satisfying the axioms of a metric up to vanishing additive terms.

Proposition 20.5.6 The normalized information distance $NID(x, y)$ is a metric, up to negligible errors.

Proof sketch. Non-negativity and symmetry are immediate from the definition. For identity, $NID(x, y) = O(1/\max\{K(x), K(y)\})$ when $x = y$, while if $x \neq y$ at least one of the conditional complexities is large, so $NID(x, y) > 0$. For the triangle inequality, let x, y, z be strings. From the inequality

$$K(z|x) \leq K(y|x) + K(z|y) + O(\log),$$

we obtain

$$E(x, z) \leq E(x, y) + E(y, z) + O(\log),$$

where $E(\cdot, \cdot)$ is the (unnormalized) information distance.

Normalizing by $\max\{K(x), K(z)\}$ introduces at most a vanishing additive error. Thus NID satisfies the metric axioms up to negligible terms. ■

The normalized information distance can also be expressed directly in terms of the joint and individual Kolmogorov complexities, as shown below.

Proposition 20.5.7

$$NID(x, y) = \frac{K(xy) - \min\{K(x), K(y)\} + O(\log K(xy))}{\max\{K(x), K(y)\}}.$$

Proof sketch. From the earlier result for information distance we know

$$E(x, y) = \max\{K(x|y), K(y|x)\} = K(xy) - \min\{K(x), K(y)\} + O(\log K(xy)).$$

Dividing both sides by $\max\{K(x), K(y)\}$ yields the expression for $NID(x, y)$. ■

Normalized Compression Distance

Although the normalized information distance is not computable, it has a wide range of potential applications. By approximating Kolmogorov complexity with practical compressors, we can obtain a computable surrogate of NID . Let $Z(s)$ denote the length in bits of the string s compressed using a compressor Z (such as gzip, bzip2, or PPMZ). Similarly, $Z(s | t)$ denotes the compressed size of s when the compressor is given t as auxiliary input. This motivates the following definition.

Definition 20.5.4 The *normalized compression distance* between two strings $s, t \in \mathcal{B}^*$, given the compressor Z , and denoted by $NCD_Z(s, t)$, is defined as:

$$NCD_Z(s, t) = \frac{\max\{Z(s | t), Z(t | s)\}}{\max\{Z(s), Z(t)\}}$$

In practice, most compressors do not support conditional compression, making $Z(s | t)$ difficult to compute directly. Fortunately, the definition can be reformulated in terms of concatenated compression, which avoids this issue.

Proposition 20.5.8 The normalized compression distance between two strings $s, t \in \mathcal{B}^*$, given the compressor Z , satisfies:

$$NCD_Z(s, t) = \frac{Z(st) - \min\{Z(s), Z(t)\}}{\max\{Z(s), Z(t)\}}.$$

Proof sketch. From the definition of NCD_Z , we have

$$NCD_Z(s, t) = \frac{\max\{Z(s | t), Z(t | s)\}}{\max\{Z(s), Z(t)\}}.$$

For real-world compressors, the conditional compression $Z(s | t)$ can be approximated by

$$Z(s | t) \approx Z(st) - Z(t),$$

since compressing the concatenation st encodes s with t as a prefix, effectively using t as context. Similarly,

$$Z(t | s) \approx Z(st) - Z(s).$$

Therefore,

$$\max\{Z(s | t), Z(t | s)\} \approx Z(st) - \min\{Z(s), Z(t)\}.$$

Substituting this into the definition yields

$$NCD_Z(s, t) = \frac{Z(st) - \min\{Z(s), Z(t)\}}{\max\{Z(s), Z(t)\}}.$$

This equality holds up to negligible additive errors, which vanish for ideal compressors. ■

The normalized compression distance constitutes a family of distances, each defined by the choice of compressor Z . The effectiveness of Z determines how closely the normalized compression distance mirrors the normalized information distance, ultimately influencing how well NCD approximates NID in practical applications.

20.6 Incompressibility and Randomness

A string is considered incompressible if its Kolmogorov complexity is approximately equal to its length; in other words, there is no significantly shorter description or program that can produce it.

Definition 20.6.1 For each constant c we say that a string $s \in \mathcal{B}^*$ is *c-incompressible* if $K(s) \geq l(s) - c$.

The next proposition shows that incompressible strings exist for every string length.

Proposition 20.6.1 For every length n , there exists a string of length n that is incompressible.

Proof. By a counting argument, the number of programs of length less than n is at most $2^n - 1$, while the number of binary strings of length n is exactly 2^n . Therefore, at least one string of length n cannot be generated by any program shorter than n , and is thus incompressible. ■

We extend the term *incompressible string* to include all *c-incompressible* strings with c small. In this sense, most strings are incompressible.

Proposition 20.6.2 For any n and constant $c > 0$, at least a fraction $1 - 2^{-(c-1)}$ of the strings of length n are *c-incompressible*.

Proof. Consider the number of programs of length at most $n - c$. There are at most

$$\sum_{i=0}^{n-c} 2^i = 2^{n-c+1} - 1$$

such programs. Since there are 2^n strings of length n , at least $2^n - (2^{n-c+1} - 1)$ strings must be *c-incompressible*. Thus the fraction of *c-incompressible* strings of length n is at least

$$1 - \frac{2^{n-c+1} - 1}{2^n} = 1 - 2^{-(c-1)} + \frac{1}{2^n}.$$

For large n , this tends to $1 - 2^{-(c-1)}$. ■

The notion of randomness, especially in the context of sequences or strings, is often associated with unpredictability, lack of pattern, or absence of structure. Kolmogorov complexity formalizes this intuition by linking randomness to incompressibility: a string is random if it cannot be generated by any program significantly shorter than the string itself.

Definition 20.6.2 We say that a string $s \in \mathcal{B}^*$ is *random* if it is c -incompressible for some fixed small constant c .

Random strings are characterized by high Kolmogorov complexity, meaning they are incompressible. The shortest program that can generate a random string is essentially the string itself. Such strings contain the maximum amount of information possible.

■ **Example 20.8** Consider a string generated by flipping a fair coin for each bit, such as 1011010110110101. With overwhelming probability, this string will be incompressible, since no shorter program or pattern can generate it. Any attempted compression would yield a program of length comparable to the string itself. ■

The unpredictability of random strings stems from their incompressibility. Because no algorithm can exploit patterns in such strings, predicting their bits is no better than random guessing.

Random strings are typical in the space of all strings. In the sense of Kolmogorov complexity, almost all strings are random, while only a vanishing fraction admit significantly shorter descriptions.

■ **Example 20.9** Consider the set of all conceivable high-resolution digital photographs, each represented as a binary string encoding pixel colors and intensities. Only a tiny fraction of these images exhibit recognizable regularities, such as a uniform blue sky or a solid monochromatic background, which can be compressed into shorter binary descriptions. By contrast, the overwhelming majority of possible images resemble random strings: they lack compressible patterns and are essentially incompressible. ■

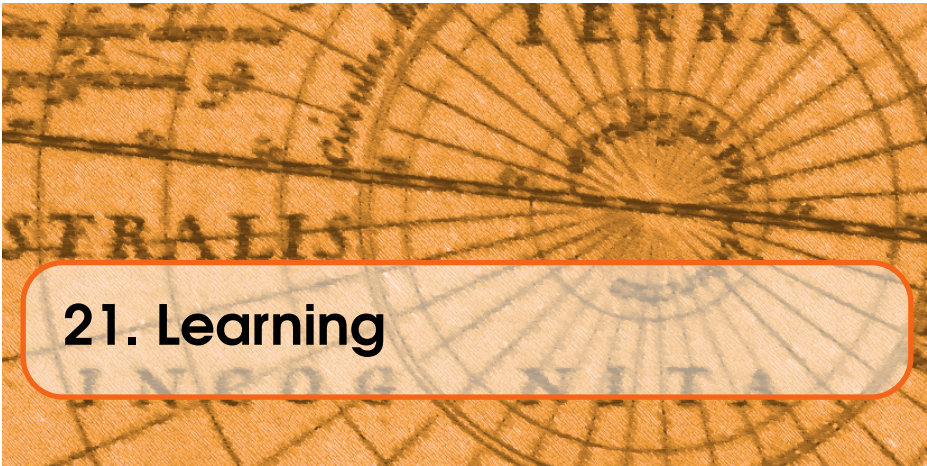
References

Kolmogorov complexity, named after the Soviet mathematician Andrey Kolmogorov, is a measure of the complexity of a string of text or other data. It is defined as the length of the shortest possible description of the string in some fixed universal description language. This measure is inherently uncomputable in general, as proven by the halting problem's undecidability,

but it provides a powerful theoretical tool for understanding data complexity. Beyond theoretical interest, Kolmogorov complexity has applications in pattern recognition, data compression, and the study of randomness. It offers a framework for understanding the limits of compressibility and the nature of information.

The concept of Kolmogorov complexity emerged independently in the works of several researchers in the early 1960s. Andrey Kolmogorov introduced it in 1965 [Kol65], motivated by trying to formalize the concept of randomness and complexity through the lens of information theory. Ray Solomonoff laid the groundwork for algorithmic information theory, introducing a related concept that would later be recognized as a form of Kolmogorov complexity [Sol64]. His work focused on the idea of describing data compactly using probabilistic models. Almost simultaneously with Kolmogorov, Gregory Chaitin developed similar ideas [Cha69]. Chaitin is known for introducing the concept of algorithmic randomness and for his work on the incompleteness theorem, which relates to the limits of formal systems in proving the complexity of sequences. The motivations behind the development of Kolmogorov Complexity were multifaceted, encompassing the desire to better understand the nature of information, randomness, and the limits of computation and prediction, thus laying the groundwork for numerous applications in theoretical computer science, mathematics, and beyond.

For those readers interested in delving into the details of Kolmogorov complexity, a variety of foundational texts and advanced treatments are available. [LV13] provides a comprehensive coverage to the concepts and applications of Kolmogorov complexity, and it's widely regarded as the definitive textbook on the subject, although it is not recommended for beginners. [Cal02] delves deeply into the foundations of algorithmic information theory, focusing on the rigorous mathematical exploration of randomness and complexity through Kolmogorov complexity. [CT12] broader in scope, this book provides an excellent foundation in information theory, including discussions relevant to Kolmogorov complexity. It's a great resource for understanding the context in which Kolmogorov complexity operates within information theory.



21. Learning

*Some mathematical statements are true for no reason,
they're true by accident.*
Gregory Chaitin

Learning from data lies at the heart of scientific discovery and technological innovation. It refers to the broad collection of methods and algorithms designed to automatically build mathematical models from sample datasets, usually with the aim of making predictions, classifying objects, or revealing hidden structure in the data. In the past decade, learning algorithms have achieved striking success in diverse domains such as self-driving cars, practical speech recognition, effective web search, and personalized recommendation systems.

In this chapter, we will explore how the problem of learning from data is formally formulated within the field of machine learning. In doing so, this chapter builds on the probabilistic foundations introduced earlier in Section ??, extending them toward the design of algorithms that can infer general patterns from finite observations.

We will begin by reviewing the classical approach to learning based on *statistical inference*, where learning is framed as estimating the parameters of an assumed probabilistic model. We will then turn to *machine learning methods*, which often relax these assumptions and focus on constructing predictive models

that generalize well to unseen data, even when the underlying data-generating process is unknown or too complex to model explicitly.

In addition, we will study two powerful principles that are central to the *theory of nescience*: the *Minimum Description Length (MDL)* and the *Minimum Message Length (MML)* principles. These approaches reinterpret learning as a problem of finding the most concise explanation for the data, connecting model selection to fundamental concepts from algorithmic information theory and data compression. This perspective will also allow us to extend the concept of nescience—originally defined for textual descriptions—to empirical datasets.

Finally, we will review the area of *multiobjective optimization*, which addresses learning problems involving multiple, often conflicting, objectives. Unlike traditional single-objective approaches, multiobjective methods aim to find a set of Pareto-optimal solutions, requiring additional techniques to incorporate preference structures and select a final decision.

Throughout the chapter, our goal will be to develop a unified view of learning, spanning from classical statistical inference to modern machine learning and nescience-based methods, and to prepare the ground for the introduction of new learning algorithms inspired by the theory of nescience in Chapter 13.

21.1 Statistical Inference

Statistical inference has traditionally been presented as the process of making sense of reality through data. It applies probabilistic models to relate finite observations to broader populations, aiming to offer predictions and structured conclusions. However, statistical inference is not a direct reflection of reality but rather a tool that depends heavily on assumptions and idealized models. These models simplify complex phenomena, and the validity of their conclusions relies on how well these assumptions align with the underlying data-generating processes.

The random sampling model serves as a stepwise framework in statistical inference, guiding how we connect finite samples to broader populations:

Step 1.- Defining the Population and Collecting Data: The first step is to define the population of interest and determine how data will be collected. This involves specifying the sampling process and ensuring that each data point is drawn independently and identically distributed (iid) from the same population, with every unit having a nonzero probability of selection. Once the sampling process is defined, data is collected as a sample from the population, and each observation is treated as a realization of a random variable following the specified distribution.

Step 2.- Constructing and Checking the Random Sampling Model: Based on assumptions about how the data was generated, the random sampling model

is mathematically expressed. For example, we may assume that $X_1, \dots, X_n$ are iid random variables with common distribution $f(x | \theta)$, where θ is an unknown parameter of the population distribution. This step formalizes the relationship between the data and the population. Real-world data rarely fits these idealized assumptions perfectly, so it is essential to check for deviations such as selection bias or lack of independence, and to address them appropriately.

Step 3.- Applying the Likelihood Function and Estimating Parameters: The likelihood function plays a central role in parameter estimation by quantifying how plausible different parameter values are given the observed data $\mathbf{X}$. Methods such as maximum likelihood estimation (MLE) or Bayesian inference refine these estimates, translating raw data into structured knowledge about the population. For example, the sample mean $\bar{X}$ is often used as an estimate of the population mean μ .

Step 4.- Quantifying Uncertainty and Generalizing to the Population: Recognizing that the sample is just one realization of a random process, uncertainty must be quantified to understand how sample statistics behave across repeated samples. Sampling distributions, confidence intervals, and hypothesis tests help describe this variability. The Central Limit Theorem ensures that for large samples, the distribution of the sample mean $\bar{X}$ approximates a normal distribution, making inference more reliable. Finally, results are generalized from the sample to the population using frequentist methods such as confidence intervals and p -values, or Bayesian methods that update beliefs about θ by incorporating prior knowledge.

The random sampling model provides a structured, step-by-step approach for making inferences from data. However, each step depends on assumptions that rarely hold perfectly in practice. As such, conclusions drawn from statistical inference should be viewed as tentative and context-dependent, subject to revision when new data or insights become available.

Definition 21.1.1 A *statistical model* is a family of probability distributions, together with a specification of its probability distribution, and the identification of the parameters, denoted by θ , of that distribution.

A statistic is a function of the observable data. A statistic is used to summarize or describe some aspect of the sample. For example, the sample mean and the sample variance are statistics because they summarize data drawn from a sample.

Definition 21.1.2 Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample. A *statistic* is a random variable $T = r(X_1, \dots, X_n)$, where $r()$ is an arbitrary real-valued function of n variables.

A *parametric random variable* is a random variable X that belongs to a family of distributions parameterized by θ . The parameter θ can either be a single scalar

or a vector of values, and it is often treated as a fixed but unknown quantity. In the Bayesian setting, θ itself is modeled as a random variable with a prior distribution. The set $\Theta = \{\theta_1, \theta_2, \dots\}$, consisting of all possible values of θ , is called the *parameter space*, and its elements are referred to as *parameters*. When the parameter θ is unknown, the distribution of the random variable X is said to be conditional on θ , denoted by $f(X | \theta)$.

As in Chapter ??, in this section we are only considering parametric discrete random variables defined over discrete probability spaces.

■ **Example 21.1** We have seen in Definition 17.6.4 that a binomial distribution with parameters n and p is a model for a family of experiments in which we are interested in knowing the number of successes in a sequence of n independent binary trials, where the probability of success is p . If X is a random variable following a binomial distribution, the probability of getting exactly k successes is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

In statistical inference, we are usually interested in the inverse problem. That is, we have the actual result of an experiment composed of n trials, in which we know how many successes k we have obtained, and we would like to estimate the parameter p , that is, the probability of success. ■

We assume that the true value of the unknown parameter θ can be inferred, typically by analyzing a collection of data samples. The observable data $\mathbf{X} = (X_1, \dots, X_n)$ is modeled as a random sample from the distribution $f(X | \theta)$ conditional on θ .

An estimator of a parameter θ is a function of the random sample $\mathbf{X}$ that we hope provides a value close to the unknown parameter θ .

■ **Definition 21.1.3** Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample from a discrete random variable X with parameter θ . An *estimator* of the parameter θ is a real-valued function $\delta(X_1, \dots, X_n)$. If $X_1 = x_1, \dots, X_n = x_n$ are observed, then $\delta(x_1, \dots, x_n)$ is called the *estimate* of θ .

The estimator itself is a random variable, and its probability distribution can be derived from the joint distribution of $X_1, \dots, X_n$. Every estimator is a statistic, but not every statistic is an estimator. The estimate is a real number.

When selecting an estimator, it is important to measure how "good" an estimate is. One common approach is to quantify the loss or cost associated with choosing an estimate that deviates from the true parameter value. This is done using a loss function, which assigns a numerical penalty to the difference between the estimate and the actual parameter.

Definition 21.1.4 A *loss function* is a real-valued function of two variables, $L(\theta, a)$, where $\theta \in \Theta$ and a is a real number.

The loss function represents the cost incurred when the true parameter value is θ and the estimate is a . In other words, $L(\theta, a)$ quantifies the loss when there is a discrepancy between the estimate and the true value.

Let $\xi(\theta)$ denote the prior probability mass function of θ on the set Θ . Given a particular estimate a , the expected loss for discrete random variables is computed as:

$$E[L(\theta, a)] = \sum_{\theta \in \Theta} L(\theta, a) \xi(\theta)$$

The expected loss is the weighted average of the losses, where the weights are the probabilities $\xi(\theta)$ assigned to each possible value of θ . The goal is to choose an estimate a that minimizes this expected loss.

■ **Example 21.2** The loss function $L(\theta, a) = (\theta - a)^2$ is called the *squared error loss*. Let θ be a real-valued parameter. Suppose that squared error loss is used and that the posterior mean of θ , $E(\theta | \mathbf{X})$, is finite. Then, the Bayes estimator of θ under squared error loss is

$$\delta^*(\mathbf{X}) = E(\theta | \mathbf{X}).$$

In words, the posterior mean minimizes the expected squared deviation from the true parameter.

The loss function $L(\theta, a) = |\theta - a|$ is called the *absolute error loss*. When absolute error loss is used, a Bayes estimator of a real-valued parameter θ is any median of the posterior distribution of θ . That is,

$$\delta^*(\mathbf{X}) \in \text{Median}(\theta | \mathbf{X}).$$

In words, the posterior median minimizes the expected absolute deviation from the true parameter. ■

21.1.1 Maximum Likelihood Estimator

The Maximum Likelihood Estimator (abbreviated MLE) is a method of estimating the parameters of a probability model. It is one of the most commonly used techniques in statistics for fitting model parameters to observed data. The core idea behind MLE is to find the parameter values that maximize a likelihood function, which represents the probability of observing the given data under the model. MLE is widely used due to its desirable properties for large samples, but its effectiveness relies on correct model specification and may struggle with small sample sizes or computational difficulties.

The likelihood represents the probability (or likelihood) of observing the data as a function of the parameters of the model.

Definition 21.1.5 Let $X_1, X_2, \dots, X_n$ be n independent and identically distributed discrete random variables, and let $P(X_i = x_i \mid \theta)$ be the probability mass function of X_i , where $\theta \in \Theta$ is an unknown parameter, and Θ is the parameter space. The *likelihood function* is then defined as:

$$L(\theta \mid X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i = x_i \mid \theta),$$

where $x_1, x_2, \dots, x_n$ are the observed data points (held fixed when viewing L as a function of θ).

The likelihood function is a product of probabilities. Taking the logarithm of the likelihood, the *log-likelihood function*, simplifies this product into a sum:

$$\ell(\theta \mid X_1, X_2, \dots, X_n) = \log L(\theta \mid X_1, X_2, \dots, X_n) = \sum_{i=1}^n \log P(X_i \mid \theta).$$

Sums are generally easier to work with than products, particularly when performing differentiation for optimization purposes. This transformation makes it more straightforward to compute derivatives and apply optimization algorithms.

Maximum likelihood estimation is a method that determines values for the parameters of a model. The parameter values are found such that they maximise the likelihood that the process described by the model produced the data that were actually observed.

Definition 21.1.6 The *Maximum Likelihood Estimator*, abbreviated as MLE, for discrete random variables is defined as the parameter value that maximizes the likelihood function, i.e.,

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} L(\theta \mid X_1, X_2, \dots, X_n).$$

Equivalently, any maximizer of the log-likelihood $\ell(\theta \mid X_1, \dots, X_n)$ is an MLE. When there are multiple maximizers, any of them may be taken.

The MLE can also be found by maximizing the log-likelihood function:

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} \ell(\theta \mid X_1, X_2, \dots, X_n)$$

■ **Example 21.3** Consider $X_1, X_2, \dots, X_n$ i.i.d. random variables from a Bernoulli distribution with parameter p . The probability mass function is

$$P(X_i = x_i \mid p) = p^{x_i} (1 - p)^{1-x_i}, \quad x_i \in \{0, 1\}.$$

The likelihood function for this sample is

$$L(p \mid X_1, X_2, \dots, X_n) = \prod_{i=1}^n p^{X_i} (1 - p)^{1-X_i} = p^S (1 - p)^{n-S},$$

where $S = \sum_{i=1}^n X_i$ is the total number of successes. The function $f(p) = p^a(1-p)^b$ on $p \in (0, 1)$ attains its maximum at $p = \frac{a}{a+b}$. In our case, $a = S$ and $b = n - S$, so

$$\hat{p} = \frac{S}{S + (n - S)} = \frac{S}{n}.$$

Therefore, the maximum likelihood estimator for p is

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i,$$

which is simply the sample mean of the Bernoulli trials (the proportion of successes). ■

The maximum likelihood estimator is consistent under suitable conditions.

Proposition 21.1.1 Let $\hat{\theta}_n$ be an MLE for θ based on an i.i.d. sample from a discrete model $\{P_\theta : \theta \in \Theta\}$. Suppose the model is identifiable (i.e., $P_\theta = P_{\theta_0}$ implies $\theta = \theta_0$), Θ is compact, and for each x , $\log P(X = x \mid \theta)$ is continuous in θ . If $E_{\theta_0}[|\log P(X_1 \mid \theta)|] < \infty$ for all $\theta \in \Theta$, then

$$\hat{\theta}_n \xrightarrow{P} \theta_0 \quad \text{as } n \rightarrow \infty,$$

where θ_0 is the true value of the parameter.

Sketch. (1) By identifiability and Gibbs' inequality, the expected log-likelihood (per observation) is uniquely maximized at θ_0 :

$$Q(\theta) := E_{\theta_0}[\log P(X_1 \mid \theta)] \leq Q(\theta_0),$$

with equality iff $\theta = \theta_0$.

(2) By the Law of Large Numbers and the continuity/integrability assumptions, the average log-likelihood converges uniformly to $Q(\theta)$ on Θ :

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \log P(X_i \mid \theta) - Q(\theta) \right| \xrightarrow{P} 0.$$

(3) By the argmax continuous mapping principle, any maximizer $\hat{\theta}_n$ of the sample log-likelihood converges in probability to the unique maximizer of Q , namely θ_0 . ■

The MLE has several notable disadvantages. Firstly, it is highly dependent on sample size; for small samples, MLE may exhibit bias or high variance, as its desirable asymptotic properties, such as consistency, are guaranteed only

for large datasets. Additionally, MLE requires correct model specification. It is sensitive to assumptions about the underlying model, and if the model is misspecified, the resulting estimates can be misleading. From a computational perspective, maximizing the likelihood function can pose numerical challenges, often necessitating complex optimization techniques that are computationally demanding and may result in convergence to local rather than global maxima. Furthermore, MLE can be sensitive to extreme observations; because it seeks to maximize the likelihood of the observed sample, large or influential data points may have a disproportionate impact on the estimate, leading to distortions under contamination.

■ **Example 21.4** Consider a situation where we are dealing with a uniform distribution $U(0, \theta)$, where θ is the unknown upper bound of the distribution. We have a sample $X_1, X_2, \dots, X_n$ of i.i.d. observations from this distribution, and we wish to estimate the parameter θ using the maximum likelihood estimator. The log-likelihood function is:

$$\ell(\theta \mid X_1, X_2, \dots, X_n) = -n \log \theta, \quad \text{subject to } \theta \geq \max(X_1, X_2, \dots, X_n).$$

Since $\ell(\theta)$ is decreasing in θ , the maximum likelihood occurs at the smallest feasible value, namely

$$\hat{\theta} = \max(X_1, X_2, \dots, X_n).$$

For small samples, $\hat{\theta}$ is biased downward: $E[\hat{\theta}] = \frac{n}{n+1} \theta$ (e.g., if $\theta = 10$ and $n = 5$, then $E[\hat{\theta}] = \frac{5}{6} \cdot 10 = 8.33$). Moreover, the estimator depends only on the sample maximum; the other $n - 1$ observations do not affect the estimate, which is statistically inefficient in finite samples. Finally, the estimator has high sensitivity to the largest observation: small perturbations of the maximum change $\hat{\theta}$ directly (and under data contamination, a single large outlying value can inflate $\hat{\theta}$ substantially). ■

21.1.2 Bayesian Inference

Unlike the Maximum Likelihood Estimator (MLE), which selects the parameter values that maximize the probability of the observed data, *Bayesian inference* takes a fundamentally different approach by treating parameters as random variables endowed with probability distributions rather than fixed values. Bayesian inference also incorporates prior knowledge, allowing for an informed estimation process even before observing data. This approach enables Bayesian methods to dynamically update beliefs as new data become available. Bayes' theorem (see Theorem 17.3.2) formalizes this process by combining a prior probability mass function with the likelihood of the observed data to produce a posterior probability mass function, which represents the updated belief about the parameter.

Definition 21.1.7 Let $f(X | \theta)$ be the probability mass function of a discrete random variable X with parameter θ . The probability distribution of the parameter θ , denoted by $\xi(\theta)$, is called the *prior distribution*.

The prior distribution must be defined over the parameter space Θ . It is called the prior distribution because it represents our knowledge or belief about the parameter θ before observing any data.

Definition 21.1.8 Let $f(X | \theta)$ be the probability mass function of a discrete random variable X with parameter θ , and let $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample from X . The conditional probability mass function of the parameter θ given the observed values $X_1 = x_1, \dots, X_n = x_n$, denoted $\xi(\theta | x_1, \dots, x_n)$, is called the *posterior distribution*.

The posterior distribution represents our updated knowledge of the parameter θ after taking into account the observed data. A strongly informative prior can dominate the posterior, while a weak or diffuse prior allows the likelihood to play a larger role (and, under regularity, the likelihood typically dominates as n grows).

The posterior pmf is computed using Bayes' theorem:

$$\xi(\theta | \mathbf{X}) = \frac{L(\mathbf{X} | \theta) \xi(\theta)}{P(\mathbf{X})},$$

where:

- $\xi(\theta)$ is the prior distribution, which encodes our belief about θ before seeing the data;
- $L(\mathbf{X} | \theta)$ is the likelihood function, representing the probability of observing the data given θ ; under i.i.d. sampling, $L(\mathbf{X} | \theta) = \prod_{i=1}^n f(x_i | \theta)$;
- $P(\mathbf{X})$ is the marginal probability (also called the evidence), ensuring the posterior pmf is a valid probability mass function, computed as

$$P(\mathbf{X}) = \sum_{\theta \in \Theta} L(\mathbf{X} | \theta) \xi(\theta).$$

Thus, the posterior pmf $\xi(\theta | \mathbf{X})$ represents an updated belief about θ after incorporating the observed data. (If Θ is uncountable, replace sums by integrals and pmfs by densities.)

■ **Example 21.5** Suppose we have a biased coin with an unknown probability θ of landing heads. We want to estimate θ using Bayesian inference after observing a few coin flips. Before flipping the coin, we assume a prior belief about θ . Let's say we assume a uniform prior:

$$\xi(\theta) = \begin{cases} \frac{1}{3}, & \theta \in \{0.2, 0.5, 0.8\} \\ 0, & \text{otherwise} \end{cases}$$

This prior suggests we believe θ is equally likely to be 0.2, 0.5, or 0.8 before any data is observed. Now, we flip the coin $n = 3$ times and observe $x = 2$ heads. Assuming independent flips, the likelihood function is given by:

$$p(X = x | \theta) = \binom{3}{2} \theta^2 (1 - \theta)^1$$

For the possible values of θ :

$$p(2 | 0.2) = \binom{3}{2} (0.2)^2 (0.8)^1 = 3(0.04)(0.8) = 0.096$$

$$p(2 | 0.5) = \binom{3}{2} (0.5)^2 (0.5)^1 = 3(0.25)(0.5) = 0.375$$

$$p(2 | 0.8) = \binom{3}{2} (0.8)^2 (0.2)^1 = 3(0.64)(0.2) = 0.384$$

Using Bayes' rule:

$$\xi(\theta | X = 2) = \frac{p(X = 2 | \theta) \xi(\theta)}{P(X = 2)}$$

where the denominator is the marginal probability:

$$P(X = 2) = \sum_{\theta} p(X = 2 | \theta) \xi(\theta)$$

$$P(X = 2) = (0.096 \times \frac{1}{3}) + (0.375 \times \frac{1}{3}) + (0.384 \times \frac{1}{3})$$

$$P(X = 2) = 0.032 + 0.125 + 0.128 = 0.285$$

Now, calculating the posterior probabilities:

$$\xi(0.2 | X = 2) = \frac{0.096 \times \frac{1}{3}}{0.285} = \frac{0.032}{0.285} \approx 0.112$$

$$\xi(0.5 | X = 2) = \frac{0.375 \times \frac{1}{3}}{0.285} = \frac{0.125}{0.285} \approx 0.439$$

$$\xi(0.8 | X = 2) = \frac{0.384 \times \frac{1}{3}}{0.285} = \frac{0.128}{0.285} \approx 0.449$$

Before observing the data, we believed θ was equally likely to be 0.2, 0.5, or 0.8. After observing 2 heads in 3 flips, our belief has shifted: the most probable value for θ is now 0.8 (44.9% probability), followed by 0.5 (43.9% probability). The probability that $\theta = 0.2$ has dropped significantly to 11.2%. ■

A Bayes estimator is a point estimate derived from the posterior distribution in Bayesian inference, analogous to how the MLE selects parameter values maximizing the likelihood. However, while MLE relies solely on observed data, the Bayes estimator incorporates both prior knowledge and observed data through the posterior. It is a single value that represents the "best guess" for the unknown parameter by minimizing the expected loss under a given loss function.

Suppose we observe the value $\mathbf{x}$ of the random vector $\mathbf{X}$ before estimating θ , and let $\xi(\theta | \mathbf{x})$ denote the posterior pmf of θ on Θ . For each estimate a , the posterior expected loss is

$$E[L(\theta, a) | \mathbf{x}] = \sum_{\theta \in \Theta} L(\theta, a) \xi(\theta | \mathbf{x}).$$

Definition 21.1.9 Let $L(\theta, a)$ be a loss function. For each possible value $\mathbf{x}$ of $\mathbf{X}$, let $\delta^*(\mathbf{x})$ be a value of a that minimizes $E[L(\theta, a) | \mathbf{x}]$. Then δ^* is called a *Bayes estimator* of θ . Once $\mathbf{X} = \mathbf{x}$ is observed, $\delta^*(\mathbf{x})$ is called a *Bayes estimate* of θ .

Equivalently, for each possible value $\mathbf{x}$ of $\mathbf{X}$, the value $\delta^*(\mathbf{x})$ is chosen so that

$$E[L(\theta, \delta^*(\mathbf{x})) | \mathbf{x}] = \min_a E[L(\theta, a) | \mathbf{x}].$$

The theory of Bayes estimators provides a coherent framework for parameter estimation. To apply it, one must specify both a loss function and a prior distribution for the parameter. Specifying a meaningful prior in a multidimensional parameter space Θ can be challenging because it requires modeling dependencies among parameters, and the computational cost of posterior inference typically grows with dimension (often necessitating MCMC or variational methods).

21.2 Machine Learning

Machine learning is the study of algorithms that, given a finite sample of individuals from a population, learn a rule to make predictions about new, previously unseen individuals. Each individual is represented by a vector of attributes, one of which is designated as the target. Since the true relationship between the predictors and the target is unknown, the algorithm must approximate it. The primary objective is to construct a function, called a model, that uses the observed attributes to predict the target with minimal error, both on the training examples and on future individuals drawn from the same population.

Although statistical inference and machine learning share the goal of using data to draw conclusions about unseen cases, they tend to emphasise

different aspects of this task. Classical statistical inference often begins with a probabilistic model and seeks estimators that maximise likelihood, yield interpretable parameters, and satisfy theoretical properties such as consistency, efficiency, or convergence. In contrast, machine learning typically starts with a flexible class of functions, often implemented as large-scale software artefacts such as neural networks, and focuses on minimising an empirical loss that reflects predictive accuracy, even in the absence of formal guarantees. In practice, however, the boundary is fluid: statisticians often minimise risk functions, and many machine learning methods are grounded in statistical theory.

In this book, we present machine learning as a distinct discipline, adopting a fully deterministic approach. We introduce models, residuals, and optimization criteria without relying on probability theory or its results. This choice is deliberate: by avoiding probabilistic assumptions, we align the theoretical foundations of machine learning with the framework of the theory of nescience, facilitating the integration and reuse of results across both domains.

We begin by defining the basic mathematical objects that will underpin the rest of the discussion: populations and their individuals.

Definition 21.2.1 A *population* is a non-empty, well-defined set $\mathcal{S}$. The elements $\mathbf{x} \in \mathcal{S}$ are called *individuals*.

Throughout this book, we restrict our attention to countable populations. Both finite and countably infinite cases are permitted.

Each individual is described by a fixed list of attributes (also called features), and each attribute has a domain that specifies its set of admissible values.

Definition 21.2.2 Each individual $\mathbf{x}$ in the population $\mathcal{S}$ is characterized by p attributes ($p \geq 1$), such that $\mathbf{x} = (x_1, x_2, \dots, x_p) \in \mathcal{S}$. For each $i \in 1, \dots, p$, the *domain* of the i -th attribute is defined as $\mathcal{D}_i = \{x_i : \mathbf{x} \in \mathcal{S}\}$.

The set $\mathcal{D}_i$ may be any type of set—for example, the real numbers $\mathcal{D}_i = \mathbb{R}$, a set of integers such as $0, 1, \dots, k$, or a collection of categorical labels like red, green, blue. An attribute is called *quantitative* if the elements of $\mathcal{D}_i$ can be measured numerically. Quantitative attributes can be further classified as *discrete* if they take a countable number of distinct values, or *continuous* if they can take any value within a given range or interval.

An attribute is called *qualitative* or *categorical* if it represents characteristics that cannot be measured numerically. A qualitative attribute is said to be *nominal* if there is no natural order among its categories, and *ordinal* if such an order exists.

■ **Example 21.6** Consider a population consisting of the inhabitants of a small village, where we are interested in studying three attributes: age, gender, and daily water consumption. The attribute representing age, mapping individuals

to their age in years, is a quantitative discrete attribute, as it takes integer values. Gender, mapping individuals to categories like "Male" or "Female," is a qualitative nominal attribute, since the categories have no inherent order. Daily water consumption, mapping individuals to the number of liters they drink per day, is a quantitative continuous attribute, as it can take any real value within a measurable range. ■

Broadly speaking, machine learning algorithms can be classified into two main categories: supervised and unsupervised. In *supervised* learning, we are given a collection of training samples (also called *predictors*) along with their corresponding observed target values (or *labels*), and our goal is to predict the target for new, previously unseen observations. In contrast, in *unsupervised* learning, no target values are provided; we are given only training samples, and the objective is to uncover the underlying structure of the data. We postpone the discussion of unsupervised learning to another section.

Definition 21.2.3 Let $\mathcal{S}$ be a population whose individuals $\mathbf{x}$ are characterized by p attributes, with $\mathbf{x} = (x_1, x_2, \dots, x_p) \in \mathcal{S}$. A *target variable*, denoted by $\mathbf{y}$, is a vector corresponding to another attribute of the individuals in $\mathcal{S}$, distinct from the p attributes used as predictors. The *domain* of the target variable is $\mathcal{D}_y = y_i$.

We typically choose statistical learning methods based on whether the target attribute is quantitative or qualitative. (Whether the predictors are qualitative or quantitative is generally considered less critical.) Supervised learning algorithms are applied to *regression problems* when the target attribute is quantitative, and to *classification problems* when the target attribute is qualitative.

We assume the existence of an underlying function, or model, that relates the predictors to the target.

Definition 21.2.4 Let $\mathcal{S}$ be a population and $\mathbf{y}$ a target attribute. A *model* is a function $f : \mathcal{S} \rightarrow \mathbf{y}$ such that, for every individual $\mathbf{x} \in \mathcal{S}$, we have $y = f(\mathbf{x})$.

In practice, and for most of the populations, the set of attributes measured is not sufficient to fully characterize the target variable, meaning that $f(\mathbf{x})$ do not map to y_k , but to something approximated. This fact is modeled in the discipline of machine learning using a random variable, and saying that $y_k = f(\mathbf{x}) + \varepsilon_k$ where the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent, $\mathbb{E}[\varepsilon_k] = 0$, and $\text{Var}[\varepsilon_k] < \infty$.

In practice, for most populations, the measured attributes are not sufficient to fully determine the target variable. This means that $f(\mathbf{x})$ does not exactly map to the observed value $\mathbf{y}$, but rather to an approximation. In the traditional formulation of machine learning, this uncertainty is modeled using a random

variable, assuming that

$$y_k = f(\mathbf{x}) + \varepsilon_k,$$

where the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent random variables, with $\mathbb{E}[\varepsilon_k] = 0$ and $\text{Var}[\varepsilon_k] < \infty$.

Learning algorithms operate on a finite sample of data, called the *training set*, from which they infer an approximation to the unknown function f .

Definition 21.2.5 A *training data set*, is a finite subset of the population $\mathcal{S}$, consisting of a collection of predictor $\mathbf{X}$ and the corresponding target $\mathbf{y}$.

Let x_{ij} represent the value of the j -th predictor for the i -th observation, where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, p$. Correspondingly, let y_i represent the target value for the i -th observation. Then, the training data consist of the pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, where each $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})$.

The objective of supervised learning is to construct a model $\hat{f} : \mathcal{S} \rightarrow \mathbf{y}$ such that the predicted value $\hat{y}$ is as close as possible to the true value $y(\mathbf{x})$.

Definition 21.2.6 Given a training set $\mathcal{T}$, a learning procedure returns an *estimator*

$$\hat{f} : \prod_{i \neq j} \mathcal{D}_i \longrightarrow \mathcal{D}_j.$$

For a new individual $\mathbf{x} \notin \mathbf{x}_1, \dots, \mathbf{x}_n$, the predicted target value is

$$\hat{y} = \hat{f}(\mathbf{x}_{-j}),$$

where $\mathbf{x}_{-j}$ denotes the vector of predictor values, excluding the target.

The challenge for the machine learning algorithms is to ensure that the estimator $\hat{f}$ generalizes well to the entire population $\mathcal{S}$, even though it was constructed using only the training data $\mathbf{X}$.

21.2.1 Parametric vs. Non-parametric Models

Most statistical learning methods can be characterized as either *parametric* or *non-parametric*. This distinction concerns the assumptions we make about the form of the underlying function f that relates the input variables X to the output variable Y .

Parametric methods follow a two-step model-based approach.

- 1 First, we assume a specific functional form for f (for example, a linear or polynomial function) that depends on a finite set of parameters θ .
- 2 Second, we use the training data to estimate the values of these parameters.

By making a strong assumption about the shape of f , the problem of estimating f reduces to estimating a relatively small number of parameters.

This makes parametric methods computationally efficient and less data-demanding. However, if the chosen functional form does not match the true (unknown) f , the model will suffer from *underfitting*: it will be unable to capture the actual structure of the data. On the other hand, choosing a highly flexible parametric form (with many parameters) increases the risk of *overfitting*: the model may follow the noise in the training data too closely, resulting in poor generalization to unseen data.

■ **Example 21.7 Linear Regression.** In simple linear regression we assume a linear functional form

$$f(x) = \beta_0 + \beta_1 x,$$

with just two parameters β_0 and β_1 to be estimated from the data. This strong assumption allows the model to be fitted with very little data, but it cannot capture nonlinear relationships between x and y . ■

Non-parametric methods make no explicit assumption about the functional form of f . Instead, they allow the data to determine the shape of the model. Because they are more flexible, non-parametric methods can in principle approximate a much wider range of functions than parametric ones.

However, this flexibility comes at a cost: non-parametric methods typically require a much larger amount of training data to achieve accurate estimates of f , and they are often harder to interpret by humans. Moreover, they can also overfit if not properly regularized or if the dataset is too small.

■ **Example 21.8 k -Nearest Neighbours (k-NN).** In k -NN regression, to predict the value of Y at a new point x , we take the average of the Y values of the k closest training points to x . There is no assumed functional form: as more data points are added, the estimated function can take on increasingly complex shapes. ■

In short, parametric models are simpler, data-efficient, and easier to interpret but rely on strong assumptions about the form of f , while non-parametric models are more flexible and expressive but data-hungry and harder to interpret. Choosing between them involves balancing bias (parametric models tend to have higher bias) and variance (non-parametric models tend to have higher variance).

21.2.2 Generative vs. Discriminative Models

Another important distinction among statistical learning methods is between *generative* and *discriminative* models. These two classes of models differ in

what they attempt to learn from the data, and consequently in how they make predictions.

Generative models aim to learn the joint probability distribution $P(X, Y)$ of the input variables X and the output variable Y . From the joint distribution, they can recover the conditional distribution $P(Y | X)$ used for prediction by applying Bayes' theorem:

$$P(Y | X) = \frac{P(X, Y)}{P(X)}.$$

Learning $P(X, Y)$ usually involves modeling two components: the prior class probabilities $P(Y)$ and the class-conditional densities $P(X | Y)$. Once these are known, we can generate new synthetic data points (X, Y) , which is why these models are called “generative.”

Generative models can be advantageous when:

- we want to simulate or generate new data,
- we have very limited data (priors help regularize estimation),
- we care about understanding the underlying data distribution.

However, they may perform worse at prediction when their assumptions about $P(X | Y)$ are incorrect.

■ **Example 21.9 Naïve Bayes Classifier.** Naïve Bayes assumes that the features $X_1, \dots, X_d$ are conditionally independent given Y . It estimates $P(Y)$ and each $P(X_j | Y)$ separately, and uses Bayes' rule to classify new observations. This model can be trained with very little data and is robust to noise, but its strong independence assumption may be unrealistic. ■

Discriminative models directly learn the conditional distribution $P(Y | X)$ or a direct decision function $f: X \rightarrow Y$ without modeling $P(X)$. They focus solely on the decision boundary between classes rather than modeling how the data is generated.

This often allows discriminative models to achieve higher predictive accuracy, especially when the amount of training data is large. However, they do not provide a full probabilistic model of the data and cannot generate new samples (X, Y) .

■ **Example 21.10 Logistic Regression.** Logistic regression models the conditional probability of Y given X as

$$P(Y = 1 | X = x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}},$$

without making any assumption about the distribution of X . It directly estimates the decision boundary separating the classes. ■

In essence, generative models learn how the data are produced, while discriminative models learn how to separate the classes. Generative models can be

more data-efficient and interpretable but are more prone to model misspecification, whereas discriminative models tend to be more accurate given sufficient data but offer less insight into the data-generating process.

21.3 Model Accuracy

When assessing the performance of a machine learning model, we want to quantify how close the model's predictions $\hat{Y} = \hat{f}(X)$ are to the true responses $Y = f(X) + \varepsilon$. As introduced in Equation ??, the error term ε accounts for factors that have not been included in the model and for inherent randomness in the data.

We can distinguish two conceptually different sources of prediction error:

- The *irreducible error*, given by the variance of ε , reflects variability that cannot be explained by any model based on the given predictors.
- The *reducible error* comes from the difference between the true function f and our estimate $\hat{f}$, which can be made smaller by choosing a better model or training procedure.

Formally, assuming that both X and $\hat{f}$ are fixed,

$$\begin{aligned} E[(Y - \hat{Y})^2] &= E[(f(X) + \varepsilon - \hat{f}(X))^2] \\ &= (f(X) - \hat{f}(X))^2 + \text{Var}(\varepsilon). \end{aligned}$$

Here, the second term $\text{Var}(\varepsilon)$ is the irreducible error, which places a lower bound on the achievable prediction error. Unfortunately, this quantity is almost always unknown in practice.

21.3.1 Bias-Variance Decomposition

More generally, if we view $\hat{f}$ as a random estimator depending on the particular training sample used, we obtain the *bias-variance decomposition*:

$$E[(Y - \hat{f}(X))^2] = \underbrace{\text{Bias}(\hat{f}(X))^2}_{\text{systematic error}} + \underbrace{\text{Var}(\hat{f}(X))}_{\text{estimation variability}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible error}}.$$

This decomposition makes explicit the tradeoff between bias and variance: simpler models have high bias but low variance, whereas more complex models have low bias but high variance.

21.3.2 Regression Metrics

A common metric for regression tasks is the *mean squared error*:

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{f}(X_i))^2.$$

Minimizing the MSE is equivalent to maximizing the likelihood under a Gaussian noise model. Indeed, if we assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently, then

$$\log P(Y | X, \theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \hat{f}(X_i; \theta))^2 + \text{const},$$

so the maximum likelihood estimator (MLE) of the parameters θ is the one that minimizes the MSE.

Other common regression metrics include:

- The *root mean squared error* (RMSE) is the square root of the MSE, which brings the error back to the original units of Y .
- The *mean absolute error* (MAE) is

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{f}(X_i)|,$$

which is more robust to outliers than MSE.

- The *coefficient of determination* R^2 is

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i - \hat{f}(X_i))^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2},$$

and measures the proportion of variance in Y explained by the model.

21.3.3 Classification Metrics

When the response variable Y is qualitative (categorical), we cannot directly compute squared errors. Instead, we measure how often the predicted class label matches the true class.

A simple and widely used metric is the *misclassification rate*:

$$\text{Error rate} = \frac{1}{n} \sum_{i=1}^n I(Y_i \neq \hat{f}(X_i)),$$

where $I(\cdot)$ is 1 if its argument is true and 0 otherwise. The *accuracy* is simply 1 minus the error rate.

In imbalanced classification problems, accuracy can be misleading. In such cases we use metrics based on the *confusion matrix*, such as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

where TP , FP , and FN are the numbers of true positives, false positives, and false negatives, respectively. The *F1-score* combines them as their harmonic mean:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Our ultimate goal is to build a model that generalizes well to previously unseen data. For this reason, model accuracy should be evaluated not only on the training dataset but also on a separate test dataset. A model with very low training error but high test error is said to *overfit* the training data. Techniques such as cross-validation, regularization, and early stopping are commonly used to detect and reduce overfitting.

21.4 Minimum Message Length

The *Minimum Message Length* (MML) principle is based on the idea that a good theory, or explanation, for a dataset is a small collection of premises under which the data is not surprising. The best theories are those that are concise, explain most of the data, and achieve high accuracy. An *explanation message* is composed of two parts: the first part comprises all the premises induced from the data, including numerical values; the second part contains all the data that cannot be derived from these premises. The message also assumes the existence of some already known and accepted premises (prior knowledge). Given the prior premises and the message, it should be possible to recover the original dataset. According to the MML principle, theories are not rejected because of contradictory measurements; such contradictions only increase the length of the second part of the message.

In the MML framework, a message is a losslessly encoded version of the original data. The first part of the message contains a probabilistic model of the data, and the second part is the data encoded using this model. Our goal is to find the shortest possible explanation message. If the length of the explanation message is longer than that of the raw data, the theory is considered inadequate.

Bayes' theorem (see Theorem 17.3.2) states that the probability $P(H | E)$ of a hypothesis H given evidence E is:

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}.$$

We are interested in finding the hypothesis H with the highest posterior probability $P(H | E)$, assuming a fixed evidence E . That is, we want to maximize $P(E | H)P(H)$, or equivalently, maximize the joint probability $P(H \wedge E)$.

The *Minimum Message Length* principle (MML for short) builds on the fact that, using an optimal code, the length of encoding $H \wedge E$ as a binary string is given by $-\log_2 P(H \wedge E)$ (see Theorem 19.3.2). That is:

$$\begin{aligned} l(H \wedge E) &= -\log_2 P(H \wedge E) = -\log_2 (P(E | H)P(H)) \\ &= -\log_2 P(E | H) - \log_2 P(H). \end{aligned}$$

Thus, the most probable model H is the one that allows us to encode $H \wedge E$ with the shortest possible binary string. This encoded string consists of two parts: a general assertion about the data (the model itself) and a detailed description of the data given that assertion.

Let $\mathbb{X}$ be the discrete set of all possible datasets, $\mathcal{X}$ a random variable taking values in $\mathbb{X}$, and $f(X \mid \theta)$ a probability distribution for $\mathcal{X}$ parameterized by θ .

Definition 21.4.1 Let Θ be a discrete set of possible parameters for f , with prior probability distribution $h(\theta)$ for $\theta \in \Theta$, and let $\hat{\theta} \in \Theta$ be an inferred parameter. An *assertion* is the encoded version of $\hat{\theta}$ using an optimal code given the distribution h .

The length of the assertion using an optimal binary code is $-\log_2 h(\hat{\theta})$ (see Section 19.3). Note that θ could be a single scalar or a vector of values. Moreover, θ could index more than one family of probability distributions f .

Definition 21.4.2 Let $X \in \mathbb{X}$ be a dataset, and $\hat{\theta} \in \Theta$ an inferred value for the distribution f . A *detail* is the encoded version of X using an optimal code given the probability distribution $f(X \mid \hat{\theta})$.

The length of the detail using an optimal binary code is $-\log_2 f(X \mid \hat{\theta})$. In other words, the length of the detail is the negative log-likelihood of X given $\hat{\theta}$.

Definition 21.4.3 Let $X \in \mathbb{X}$ be a dataset, and $\hat{\theta} \in \Theta$ an inferred value for the distribution f . A *message* for the dataset X given an inference $\hat{\theta}$ is the concatenation of the assertion for $\hat{\theta}$ and the corresponding detail for X given $\hat{\theta}$.

The length of a message using an optimal binary code is $-\log_2 h(\hat{\theta}) - \log_2 f(X \mid \hat{\theta})$. This length allows us, for example, to compare the posterior plausibility of two competing explanations or hypotheses $\hat{\theta}_1$ and $\hat{\theta}_2$.

Definition 21.4.4 Let $X \in \mathbb{X}$ be a dataset. The *Minimum Message Length* of X , denoted by $MML(X)$, is defined as:

$$MML(X) = \arg \min_{\hat{\theta} \in \Theta} (-\log_2 h(\hat{\theta}) - \log_2 f(X \mid \hat{\theta})).$$

In practice, the actual messages are not physically constructed, since we are only interested in their lengths, not in their content. It is assumed that the sets $\mathbb{X}$ and Θ , and the functions $f(X \mid \theta)$ and $h(\theta)$, are known a priori, and thus need not be included as part of the encoded message.

■ **Example 21.11** Consider an experiment in which we toss a weighted coin 100 times. Denote by 1 a face and by 0 a cross, so that each experiment produces a binary string of length 100. Our collection of all possible datasets is $\mathbb{X} = \mathcal{B}^{100}$, θ

is a number in the interval $[0, 1]$, and the likelihood $f(X | \theta)$ follows a binomial distribution (that is, $f(X | \theta) = \theta^n (1 - \theta)^{100-n}$ where n is the number of faces in X). Since we do not know anything about how the coin is weighted, we assume that $h(\theta)$ is the uniform distribution over $[0, 1]$ (that is, $h(\theta) = 1$ for $\theta \in \Theta$). Under these assumptions, the length of a message for X given an inferred parameter $\hat{\theta}$ is:

$$-\log_2 h(\hat{\theta}) - \log_2 f(X | \hat{\theta}) = -n \log_2 \hat{\theta} - (100 - n) \log_2 (1 - \hat{\theta}).$$

We are interested in finding the value of $\hat{\theta}$ that minimizes the length of the encoded version of X , that is, the minimum message length for X . ■

A Maximum A Posteriori (MAP) analysis of the experiment in Example 21.11 would produce the same estimate for $\hat{\theta}$ as the Minimum Message Length approach. Moreover, given that $h(\theta)$ follows a uniform distribution, a Maximum Likelihood Estimation (MLE) approach would yield exactly the same value for $\hat{\theta}$. This illustrates that MML generalizes MAP and MLE by making explicit the trade-off between model complexity (assertion length) and goodness of fit (detail length).

21.5 Minimum Message Length

The *Minimum Message Length* (MML) principle is a Bayesian approach to inductive inference closely related to the Minimum Description Length (MDL) principle introduced in Section ???. While MDL was developed as a practical reformulation of Kolmogorov complexity, MML was originally proposed by Wallace and Boulton in 1968 as a formal method for statistical inference grounded in information theory and Bayesian probability.

Both MDL and MML share the fundamental insight that *learning can be understood as compression*: a good model of the data is one that allows the data to be described concisely. However, while MDL is rooted in coding theory and emphasizes designing universal codes for classes of models, MML is firmly rooted in Bayesian statistics and emphasizes probabilistic inference. In MDL the model cost and the data cost are combined using coding arguments, while in MML they are combined using Bayes' theorem.

Like MDL, MML represents the communication of a dataset $D = \{x_1, \dots, x_n\}$ as a two-part message:

- 1 The first part describes a hypothesis H (including its parametric form and numerical parameter values).
- 2 The second part describes the data D given that the receiver already knows H .

Formally, the total message length is

$$L(H, D) = L(H) + L(D | H),$$

where $L(H)$ is the codelength (in bits) of describing the hypothesis, and $L(D | H)$ is the codelength of the data encoded using that hypothesis.

Under information theory, the Shannon–Fano code assigns an optimal codelength of $-\log_2 P(D | H)$ bits to data D when using hypothesis H . Thus,

$$L(D | H) = -\log_2 P(D | H).$$

The first part $L(H)$ is determined by the prior probability of H :

$$L(H) = -\log_2 P(H),$$

where $P(H)$ is a prior density over hypotheses.

By Bayes' theorem, minimizing the total message length is equivalent to maximizing the posterior probability of H :

$$P(H | D) \propto P(H) P(D | H) \iff \arg \min_H L(H) + L(D | H).$$

This gives MML a clear Bayesian interpretation: the best model is the one that achieves the greatest compression of the data *and* has the highest prior plausibility.

The term $L(D | H) = -\log_2 P(D | H)$ is the empirical negative log-likelihood of H on D . If the dataset D is drawn from an unknown true distribution P^* , then for large n this term approaches the *cross-entropy* between P^* and P_H :

$$\frac{1}{n} L(D | H) \approx H(P^*, P_H) = -\sum_x P^*(x) \log_2 P_H(x).$$

Minimizing this cross-entropy encourages models that assign high probability to the observed data, while the prior term $L(H)$ penalizes overly complex hypotheses. This provides a natural protection against overfitting: complex models that perfectly fit the sample may have very small $L(D | H)$ but incur a large $L(H)$, whereas overly simple models may have small $L(H)$ but large $L(D | H)$.

■ **Example 21.12** Suppose we are given a binary sequence $D = (x_1, \dots, x_n)$ and consider two candidate hypotheses: H_1 , a Bernoulli model with parameter θ , and H_2 , a first-order Markov chain with four parameters.

Under MML, the message length of H_1 would be

$$L(H_1) + L(D | H_1) = -\log_2 P(H_1) - \log_2 P(D | \theta),$$

where $P(\theta)$ is a prior over the Bernoulli parameter. Similarly, the Markov chain would have

$$L(H_2) + L(D \mid H_2) = -\log_2 P(H_2) - \log_2 P(D \mid \theta_{00}, \theta_{01}, \theta_{10}, \theta_{11}).$$

Even if H_2 fits the data better (lower $L(D \mid H)$), it may be penalized for its greater complexity via $L(H_2)$, leading to the selection of the simpler H_1 if the data does not justify the extra parameters. ■

The MML principle states that the best explanation for a dataset is the one that minimizes the total length of a two-part message describing both the model and the data. This connects statistical inference and data compression:

$$\arg \min_H \left(-\log_2 P(H) - \log_2 P(D \mid H) \right)$$

which is equivalent to Bayesian MAP estimation expressed in information-theoretic terms. In practice, MML provides a robust framework for model selection, avoiding overfitting by balancing model complexity and goodness of fit, and offering a computable alternative to the ideal but uncomputable notion of Kolmogorov complexity.

21.5.1 Comparison between MDL and MML

Although the Minimum Description Length (MDL) and Minimum Message Length (MML) principles are closely related and often yield similar practical results, they are conceptually and technically distinct. Both are rooted in information theory and share the central idea that learning is equivalent to compression: the best hypothesis is the one that gives the shortest description of the data. However, they approach this idea from different philosophical and mathematical standpoints.

MDL is rooted in coding theory and algorithmic information theory. It was designed as a computable approximation to Kolmogorov complexity. MML, in contrast, is rooted in Bayesian statistics and formal decision theory. While MDL begins with the design of universal codes over sets of hypotheses, MML starts from a Bayesian prior over hypotheses and applies Bayes' theorem to produce an optimal code.

MDL does not require an explicit prior over models: the description length of a hypothesis is determined by a chosen coding scheme, and priors are only implicit in the choice of that scheme. MML, on the other hand, explicitly incorporates prior probabilities $P(H)$ over hypotheses, making the prior an integral part of the inference process. This gives MML a clear Bayesian interpretation, while MDL is often regarded as a frequentist or agnostic method.

In its refined form, MDL encodes the data using a *one-part universal code* for the whole model class, which avoids having to specify the parameter values explicitly. MML always uses a *two-part message*: first the model (with its parameter values), then the data given the model. This makes MML closer to Maximum A Posteriori (MAP) estimation, whereas refined MDL is closer to minimax-optimal universal coding.

In MDL, the model description length is a tool to penalize complex models and is not necessarily interpretable as a probability. In MML, the model description length is explicitly the negative log of the prior probability, $L(H) = -\log_2 P(H)$, and the data length is the negative log-likelihood, $L(D | H) = -\log_2 P(D | H)$.

In practice, MDL is often used for model selection and structure learning when explicit priors are unavailable or undesirable, while MML is preferred when strong prior knowledge about the models or parameters is available. Both criteria penalize overfitting by trading off model complexity against data fit, but they do so under different philosophical assumptions.

21.6 Multiobjective Optimization

Multiobjective optimization is the area of mathematics that deals with the problem of simultaneously optimizing two or more conflicting functions. It has been applied in many areas of science, including engineering, economics, and logistics, where there is typically no single solution that simultaneously satisfies all objectives. In such situations, a decision must be made in the presence of trade-offs between the conflicting goals.

From a formal point of view, we are interested in solving the following *multiobjective optimization* problem:

$$\begin{array}{ll} \text{minimize} & \{f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})\} \\ \text{subject to} & \mathbf{x} \in \mathbf{S} \end{array}$$

where $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, k$, are two or more *objective functions*, and the nonempty set $\mathbf{S} \subset \mathbb{R}^n$ is the *feasible region*, whose elements $\mathbf{x} = (x_1, x_2, \dots, x_n)$ are *decision vectors*. The image of the feasible region $f(\mathbf{S}) \subset \mathbb{R}^k$, denoted by $\mathbf{Z}$, is called the *objective region*, and its elements $\mathbf{z} = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x}))$ are *objective vectors*. In some applications, the feasible region is defined by a collection of inequality constraints $\mathbf{S} = \{\mathbf{x} \in \mathbb{R}^n \mid g(\mathbf{x}) = (g_1(\mathbf{x}), \dots, g_m(\mathbf{x})) \leq 0\}$.

In this book we will be dealing with nonlinear multiobjective minimization problems, where at least one of the objective functions or constraint functions is not linear. Objective functions can also be incommensurable, that is, measured in different units or on different scales.

Since the objective functions are conflicting, there does not exist a single solution that is optimal with respect to every objective function (the objective region is only partially ordered).

Definition 21.6.1 A decision vector $\mathbf{x} \in \mathbf{S}$ *dominates* another decision vector $\mathbf{y} \in \mathbf{S}$ if $f_i(\mathbf{y}) \leq f_i(\mathbf{x})$ for all $i \in \{1, \dots, k\}$ and $f_j(\mathbf{y}) < f_j(\mathbf{x})$ for at least one $j \in \{1, \dots, k\}$. An objective vector $\mathbf{z} \in \mathbf{Z}$ *dominates* another objective vector $\mathbf{w} \in \mathbf{Z}$ if $w_i \leq z_i$ for all $i \in \{1, \dots, k\}$ and $w_j < z_j$ for at least one $j \in \{1, \dots, k\}$.

Dominance can be studied from the point of view of the decision variable space or the objective space. An objective vector dominates another objective vector if, and only if, its corresponding decision vector also dominates the other decision vector.

We are interested in those objective vectors for which none of its individual components can be improved without deteriorating at least one of the others.

Definition 21.6.2 A decision vector $\mathbf{x} \in \mathbf{S}$ is *Pareto optimal* if there does not exist another decision vector $\mathbf{y} \in \mathbf{S}$ such that $\mathbf{y}$ dominates $\mathbf{x}$. An objective vector $\mathbf{z} \in \mathbf{Z}$ is Pareto optimal if there does not exist another objective vector $\mathbf{w} \in \mathbf{Z}$ such that $\mathbf{w}$ dominates $\mathbf{z}$.

Pareto optimality can also be studied from the point of view of the decision variable space or the objective space. An objective vector is Pareto optimal if, and only if, its corresponding decision vector is Pareto optimal.

Definition 21.6.3 The set of Pareto optimal solutions, denoted by $\mathbf{P}_D$, is called the *Pareto optimal set*. The set of Pareto optimal solutions in the space of objectives, denoted by $\mathbf{P}_O$, is called the *Pareto frontier*.

Sometimes, in practice, it is convenient to use a more restrictive definition of optimality, in which we identify those vectors for which there does not exist any other vector that improves over all the components simultaneously.

Definition 21.6.4 A decision vector $\mathbf{x} \in \mathbf{S}$ is *weakly Pareto optimal* if there does not exist another decision vector $\mathbf{y} \in \mathbf{S}$ such that $f_i(\mathbf{y}) < f_i(\mathbf{x})$ for all $i = 1, \dots, k$. An objective vector $\mathbf{z} \in \mathbf{Z}$ is weakly Pareto optimal if there does not exist another objective vector $\mathbf{w} \in \mathbf{Z}$ such that $w_i < z_i$ for all $i = 1, \dots, k$.

An objective vector is weakly Pareto optimal if its corresponding decision vector is weakly Pareto optimal. Obviously, the Pareto optimal set is a subset of the weakly Pareto optimal set.

■ **Example 21.13** In figure 21.1 we have depicted a sample of the objective region of a multiobjective optimization problem composed by two real-valued objective functions f_1 and f_2 that we are interested in minimizing. White points are not weakly Pareto optimal since there exist points that improve both components

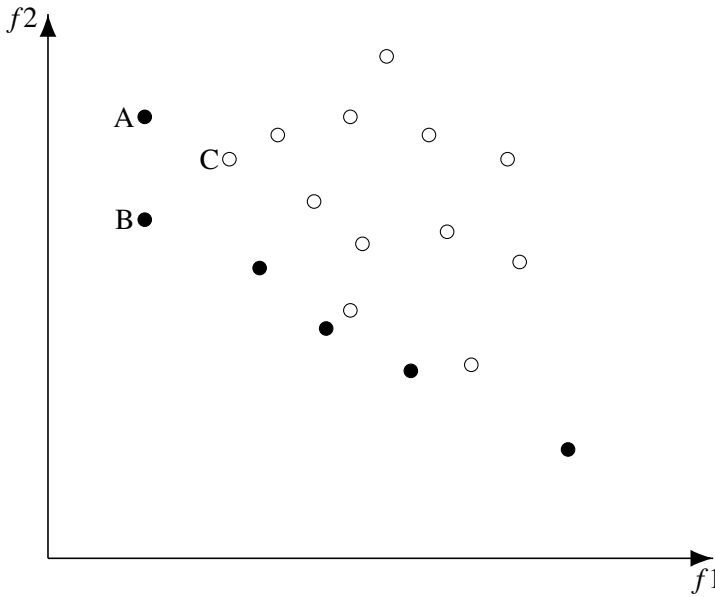


Figure 21.1: Pareto optimality.

at the same time (for example, point **B** improves point **C** in both functions). Black points are weakly Pareto optimal since there is no point that improves both components at the same time. Point **A** is not Pareto optimal since point **B** improves one component without deteriorating the other. ■

Mathematically speaking, all the solutions that compose the Pareto optimal set are equally good. However, for most practical applications, it is highly desirable to obtain a single solution. Finding this solution requires additional information not included in the definition of the optimization problem. The relation of preference between objective function values is expressed through a *decision maker*, who is assumed to have additional insights about the problem to be solved. In this sense, solving a multiobjective optimization problem requires finding those feasible decision vectors that are Pareto optimal and that also satisfy the additional requirements imposed by the decision maker.

In practice, we assume that the preferences of the decision maker can be expressed using a value function.

Definition 21.6.5 A *value function* is a function $U : \mathbb{R}^k \rightarrow \mathbb{R}$ that assigns to each objective vector $\mathbf{z} = (z_1, \dots, z_k)$ a single real value $U(\mathbf{z})$.

Value functions are maximized: among all the Pareto optimal objective vectors, the decision maker selects the one that achieves the largest value of $U(\mathbf{z})$.

This reflects the idea that the decision maker aggregates multiple criteria into a single utility score and prefers the alternative that yields the highest overall satisfaction. In this way, a multiobjective optimization problem is converted into a single-objective problem over the Pareto frontier.

Value functions allow us to order the vectors of the objective region $\mathbf{Z}$. We are interested in applying the value function to the Pareto optimal subset to find a unique solution to the multiobjective optimization problem.

21.6.1 Range of the Solutions

We are interested in investigating the range of the solutions included in the Pareto optimal set. To do that, we have to find the lower and upper bounds of this set. In the rest of this section, we assume that the objective functions are bounded over the feasible region $\mathbf{S}$.

An objective vector that minimizes all objective functions is called an ideal objective vector, while an objective vector that maximizes (within the Pareto set) all objective functions is called a nadir objective vector. These two vectors can be formally defined using the notions of *infimum* and *supremum*.

Definition 21.6.6 The *ideal objective vector* $\mathbf{z}^* \in \mathbb{R}^k$ is defined component-wise as

$$z_i^* = \inf_{\mathbf{x} \in \mathbf{S}} f_i(\mathbf{x}), \quad i = 1, \dots, k.$$

If there exists an ideal objective vector that belongs to the objective region, that is $\mathbf{z}^* \in \mathbf{Z}$, then that vector would be a solution of the optimization problem, and that solution would be unique. In general, the ideal vector is not feasible, but it constitutes a lower bound to the Pareto optimal set.

Definition 21.6.7 The *nadir objective vector* $\mathbf{z}^{nad} \in \mathbb{R}^k$ is defined component-wise as

$$z_i^{nad} = \sup_{\mathbf{z} \in \mathbf{P}_O} z_i, \quad i = 1, \dots, k,$$

where $\mathbf{P}_O \subseteq \mathbf{Z}$ is the Pareto frontier.

The nadir vector is the componentwise worst objective value among all Pareto optimal solutions, and it provides an upper bound to the Pareto set. However, unlike the ideal vector, it cannot usually be computed directly, because the full Pareto frontier is unknown. Instead, it is often estimated using the so-called *payoff table*.

Definition 21.6.8 Let $\mathbf{z}^* = \{z_1^*, \dots, z_k^*\}$ be the ideal objective vector, and let $\{\mathbf{x}_1^*, \dots, \mathbf{x}_k^*\}$ be the decision vectors that individually minimize each objective function:

$$z_i^* = f_i(\mathbf{x}_i^*), \quad \text{with } \mathbf{x}_i^* \in \arg \min_{\mathbf{x} \in \mathbf{S}} f_i(\mathbf{x}).$$

This collection $\{\mathbf{x}_1^*, \dots, \mathbf{x}_k^*\}$ is called the *payoff table*.

By construction, each $\mathbf{x}_i^*$ minimizes f_i over $\mathbf{S}$, but not necessarily the other objectives. Let f_{ij}^* denote the value of the objective function f_i computed at $\mathbf{x}_j^*$, that is, $f_{ij}^* = f_i(\mathbf{x}_j^*)$ for $i, j = 1, \dots, k$. Using these values, we can estimate the nadir objective vector as follows.

Definition 21.6.9 Given the payoff table $\{\mathbf{x}_1^*, \dots, \mathbf{x}_k^*\}$, the *nadir objective vector* is approximated as

$$z_i^{\text{nad}} = \max_j f_{ij}^*, \quad i = 1, \dots, k.$$

The ideal objective vector and the nadir objective vector may, or may not, be feasible. In Figure 21.2 are depicted the ideal (vector **E**) and nadir (vector **F**) vectors of the multiobjective optimization problem of Example 21.13.

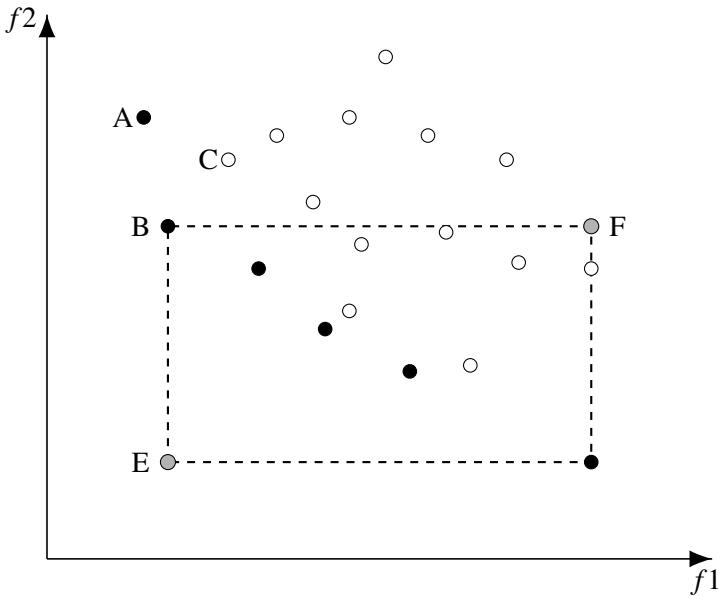


Figure 21.2: Ideal and nadir bounds (dashed) over a schematic objective space.

For some applications, the range of values of the objective functions can differ

by orders of magnitude. In those situations, it is advisable to normalize them, so the values are in the same scale. We can use the ideal and nadir vector for this normalization process, by replacing each objective function $f_i(\mathbf{x})$ ($i = 1, \dots, k$) by the normalized function

$$\frac{f_i(\mathbf{x}) - z_i^*}{z_i^{nad} - z_i^*}$$

21.6.2 Trade-offs

Since the functions we want to minimize are conflicting, we often have to assume that the only way to gain a benefit in one aspect of the problem is to lose something in another aspect. How much we have to give up in one objective to achieve a certain improvement in another is called a *trade-off*.

Definition 21.6.10 Let $\mathbf{x}^1, \mathbf{x}^2 \in \mathbf{S}$ be two decision vectors. The *ratio of change* between the functions f_i and f_j for the vectors $\mathbf{x}^1, \mathbf{x}^2$, denoted by Δ_{ij} , is defined as:

$$\Delta_{ij}(\mathbf{x}^1, \mathbf{x}^2) = \frac{f_i(\mathbf{x}^1) - f_i(\mathbf{x}^2)}{f_j(\mathbf{x}^1) - f_j(\mathbf{x}^2)}$$

for all $i, j = 1, \dots, k$ such that $f_j(\mathbf{x}^1) - f_j(\mathbf{x}^2) \neq 0$.

Δ_{ij} is called a *partial trade-off* between f_i and f_j from $\mathbf{x}^1$ to $\mathbf{x}^2$ if $f_l(\mathbf{x}^1) = f_l(\mathbf{x}^2)$ for all $l = 1, \dots, k$, $l \neq i, j$. If $f_l(\mathbf{x}^1) \neq f_l(\mathbf{x}^2)$ for at least one $l \neq i, j$, then Δ_{ij} is called a *total trade-off*.

If the trade-off between two objective functions is extremely small or extremely large (that is, if a small change in one aspect of the problem produces a disproportionately large change in another), then the solution behaves similarly to a weakly Pareto optimal solution that is not truly Pareto optimal. In some practical applications, it is convenient to filter out those solutions that exhibit this undesirable behavior. This leads to the concept of *properly Pareto optimal* solutions, which impose a bound on the possible trade-offs between objectives.

Definition 21.6.11 A decision vector $\mathbf{x} \in \mathbf{S}$ is *properly Pareto optimal* if it is Pareto optimal and if there exists a real number $M > 0$ such that, for every $i = 1, \dots, k$ and every $\mathbf{y} \in \mathbf{S}$ satisfying $f_i(\mathbf{y}) < f_i(\mathbf{x})$, there exists at least one $j \in \{1, \dots, k\}$ such that $f_j(\mathbf{x}) < f_j(\mathbf{y})$ and

$$\frac{f_i(\mathbf{x}) - f_i(\mathbf{y})}{f_j(\mathbf{y}) - f_j(\mathbf{x})} \leq M.$$

An objective vector $\mathbf{z} \in \mathbf{Z}$ is properly Pareto optimal if the decision vector

corresponding to it is properly Pareto optimal.

Intuitively, a solution is properly Pareto optimal if improving one objective even slightly is possible only at the expense of worsening at least one other objective by a proportionally bounded amount. This rules out extreme cases where an infinitesimal improvement in one objective requires an arbitrarily large deterioration in another.

Note that the properly Pareto optimal set is a subset of the Pareto optimal set, and the Pareto optimal set is a subset of the weakly Pareto optimal set.

21.6.3 Optimization Methods

Generating Pareto optimal solutions plays a central role in multiobjective optimization. From a mathematical perspective, the problem is considered to be solved when the entire Pareto optimal set has been identified. However, this is often not enough for practical decision-making. In real-world applications, we typically want to obtain a single solution. This means that once the Pareto optimal set is known, we must find a way to impose a complete order on its elements. This requires the participation of a decision maker who provides additional information in the form of a preference structure over the objective space.

A standard way to generate Pareto optimal solutions is through scalarization: converting the multiobjective optimization problem into a single-objective optimization problem, or into a family of such problems. In this approach, the original vector-valued objective function is replaced by a real-valued scalarization function that incorporates the original objectives. To be useful, a scalarization function is usually expected to satisfy the following conditions: i) for every Pareto optimal solution $\mathbf{x} \in \mathbf{P}_D$, there exists a choice of parameters of the scalarization function such that $\mathbf{x}$ is the unique optimal solution of the scalarized problem; ii) every solution produced by the scalarization function is Pareto optimal.

Several scalarization-based methods exist, and they can be classified according to how the decision maker participates in the solution process:

- No-preference methods: no preference information from the decision maker is used.
- A posteriori methods: first a (possibly large) set of Pareto optimal solutions is generated; then the decision maker selects the preferred one among them.
- A priori methods: the decision maker specifies preference information before the optimization is carried out, and this information is used to generate a single preferred solution.
- Interactive methods: the decision maker provides preference information progressively during the optimization process, which iteratively guides the search toward a preferred solution.

In no-preference methods, the knowledge of the decision maker is not taken into account, and the optimization problem is solved using a relatively simple method. In a posteriori methods, the Pareto optimal set (or part of it) is identified first, and then the decision maker selects the preferred solution among the alternatives. A priori methods and interactive methods incorporate preference information directly into the search, thereby producing solutions more tailored to the decision maker's values.

Global Criterion

The global criterion method is a no-preference method in which the distance between a given reference point and the feasible objective region is minimized. In this method, all the objective functions are considered equally important. The reference point is usually chosen as the ideal objective vector $\mathbf{z}^*$, and a common distance metric is the L_p -norm. Under these assumptions, the global criterion method becomes the following minimization problem:

$$\begin{aligned} & \text{minimize} && \left(\sum_{i=1}^k (f_i(\mathbf{x}) - z_i^*)^p \right)^{\frac{1}{p}} \\ & \text{subject to} && \mathbf{x} \in \mathbf{S}. \end{aligned}$$

Different values of p produce different solutions to this minimization problem. Common choices are $p = 1$, $p = 2$, and $p = \infty$.

Proposition 21.6.1 The solution of the L_p -based global criterion problem is Pareto optimal.

Proof. Let $\mathbf{x}$ be a solution of the L_p -based global criterion problem, with $1 \leq p < \infty$, and assume that $\mathbf{x}$ is not Pareto optimal. Then, by Definition 21.6.2, there must exist a point $\mathbf{y} \in \mathbf{S}$ such that $f_i(\mathbf{y}) \leq f_i(\mathbf{x})$ for all $i = 1, \dots, k$, and $f_j(\mathbf{y}) < f_j(\mathbf{x})$ for at least one j . Then we have that $(f_i(\mathbf{y}) - z_i^*)^p \leq (f_i(\mathbf{x}) - z_i^*)^p$ for all $i \neq j$ and $(f_j(\mathbf{y}) - z_j^*)^p < (f_j(\mathbf{x}) - z_j^*)^p$. Adding all these terms and raising to the $1/p$ power gives

$$\left(\sum_{i=1}^k (f_i(\mathbf{y}) - z_i^*)^p \right)^{\frac{1}{p}} < \left(\sum_{i=1}^k (f_i(\mathbf{x}) - z_i^*)^p \right)^{\frac{1}{p}},$$

which contradicts the assumption that $\mathbf{x}$ is a solution to the minimization problem. ■

Although all the solutions selected by the L_p -based global criterion are Pareto optimal, not all Pareto optimal solutions can be obtained by this method. In practice, it is often convenient to normalize the ranges of the objective functions, so that objectives with values closer to the ideal vector do not dominate the criterion. A common normalization factor used in practice is $z_i^{\text{nad}} - z_i^*$.

Weighting Method

The weighting method is one of the simplest and most widely used techniques for generating different Pareto optimal solutions. The idea is to associate each objective function with a weighting coefficient and minimize the weighted sum of the objectives. In this way, the multiple objective functions are transformed into a single objective function. We suppose that the weighting coefficients w_i are real numbers such that $w_i \geq 0$ for all $i = 1, \dots, k$, and that they are normalized so that $\sum_{i=1}^k w_i = 1$. The multiobjective optimization problem is then transformed into the following *weighting problem*:

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^k w_i f_i(\mathbf{x}) \\ &\text{subject to} && \mathbf{x} \in \mathbf{S}, \end{aligned}$$

where $w_i \geq 0$ for all $i = 1, \dots, k$ and $\sum_{i=1}^k w_i = 1$.

Weighting coefficients equal to zero should generally be avoided, because they effectively remove an objective from consideration. Furthermore, the objective functions should be normalized or scaled so that their values are approximately of the same order of magnitude; otherwise, the effect of the weights can be misleading and difficult to control. Only when the objectives are scaled comparably can one steer the search toward desirable solutions by adjusting the weights.

Proposition 21.6.2 If all weighting coefficients are strictly positive ($w_i > 0$ for all $i = 1, \dots, k$), then any solution of the weighting problem is Pareto optimal.

Proof. Assume that $\mathbf{x}^*$ solves the weighting problem but is not Pareto optimal. Then there exists a $\mathbf{y} \in \mathbf{S}$ such that $f_i(\mathbf{y}) \leq f_i(\mathbf{x}^*)$ for all i and $f_j(\mathbf{y}) < f_j(\mathbf{x}^*)$ for some j . Since $w_i > 0$, multiplying by the weights gives

$$\sum_{i=1}^k w_i f_i(\mathbf{y}) < \sum_{i=1}^k w_i f_i(\mathbf{x}^*),$$

contradicting the optimality of $\mathbf{x}^*$. Therefore, $\mathbf{x}^*$ must be Pareto optimal. ■

If the solution of the weighting problem is unique, it is Pareto optimal even if some weights are zero. However, not all Pareto optimal solutions can be found using the weighting method: it only generates solutions lying on the convex hull of the Pareto frontier. In practice, the condition $w_i \geq \varepsilon$ with some small $\varepsilon > 0$ is often used instead of strict positivity, which requires a careful choice of ε .

The weighting method can also be used as an a priori method by letting the decision maker specify a weighting vector representing their preferences. In this case, the weighting problem can be viewed as maximizing (the negative

of) a linear *value function*. However, it must be noted that changing the weighting coefficients linearly does not imply that the values of the objective functions will change linearly. In particular, if some of the objective functions are correlated, then seemingly “good” weighting vectors may produce poor results, and seemingly “bad” weighting vectors may produce useful results. Furthermore, weighting coefficients are often hard to interpret and understand for average decision makers.

When the weighting method is used as an a priori method, it is sometimes said that the weights reflect the *relative importance* of the objective functions. However, it is more accurate to interpret them as representing the *rates at which the decision maker is willing to trade off* values of the objective functions. This interpretation better captures their intended role.

Finally, using the weighting method as an a priori method implicitly assumes that the decision maker’s underlying value function is (or can be approximated by) a linear function of the objectives. This assumption may be unrealistic in many cases, and it explains why the weighting method can be difficult to control in practice.

ε -Constraint Method

The ε -constraint method is another classical scalarization technique for generating Pareto optimal solutions. Unlike the weighting method, which combines all objective functions into a single weighted sum, the ε -constraint method selects one of the objective functions to optimize and converts the remaining objectives into inequality constraints bounded by user-specified thresholds (the ε values).

Let us assume without loss of generality that f_1 is the chosen primary objective function to be minimized. Then the multiobjective optimization problem is transformed into the following ε -constraint problem:

$$\begin{aligned} &\text{minimize} && f_1(\mathbf{x}) \\ &\text{subject to} && f_i(\mathbf{x}) \leq \varepsilon_i, \quad i = 2, \dots, k, \\ &&& \mathbf{x} \in \mathbf{S}. \end{aligned}$$

Different choices of the bounds $\varepsilon_2, \dots, \varepsilon_k$ produce different solutions. By systematically varying these values, one can generate a set of Pareto optimal solutions.

Proposition 21.6.3 Any optimal solution of the ε -constraint problem is Pareto optimal, provided that all the constraints $f_i(\mathbf{x}) \leq \varepsilon_i$ are active or at least binding for some feasible choice of ε_i .

Proof. Suppose $\mathbf{x}^*$ solves the ε -constraint problem and is not Pareto optimal. Then there exists $\mathbf{y} \in \mathbf{S}$ such that $f_i(\mathbf{y}) \leq f_i(\mathbf{x}^*)$ for all i , and $f_j(\mathbf{y}) < f_j(\mathbf{x}^*)$ for

at least one j . In particular, $f_1(\mathbf{y}) < f_1(\mathbf{x}^*)$ and $f_i(\mathbf{y}) \leq \varepsilon_i$ for $i \geq 2$, meaning that $\mathbf{y}$ is also feasible and strictly improves the objective. This contradicts the optimality of $\mathbf{x}^*$. ■

A key advantage of the ε -constraint method is that, unlike the weighting method, it can generate Pareto optimal solutions lying in the non-convex parts of the Pareto frontier. This makes it particularly suitable for problems where the Pareto set is not convex.

In practice, choosing suitable ε_i values can be challenging. If they are too restrictive, the problem becomes infeasible; if they are too loose, the solution will simply minimize f_1 with little regard to the other objectives. A common strategy is to estimate lower and upper bounds for each f_i (for example, the ideal and nadir values defined in Section 21.6.1) and then sample values of ε_i within these ranges. This allows the decision maker to explore different trade-offs between objectives systematically.

The ε -constraint method can be used as a posteriori method (by generating many solutions for different ε values and letting the decision maker choose) or as an interactive method (by progressively adjusting the ε_i values based on the decision maker's feedback).

References

Core references for the chapter "Learning"

[Jam+13]: This book introduces modern statistical learning methods with a focus on practical applications. It bridges classical statistical inference and machine learning, making it ideal as an entry point for the chapter's first sections.

[Ped+11]: Scikit-learn is one of the most widely used open-source machine learning libraries. Citing it highlights the connection between theoretical principles and modern implementations that readers can experiment with.

[Grü07]: Grunwald's book is the definitive reference on the MDL principle. It formalizes the idea of balancing model complexity and data fit and provides both theoretical results and practical applications.

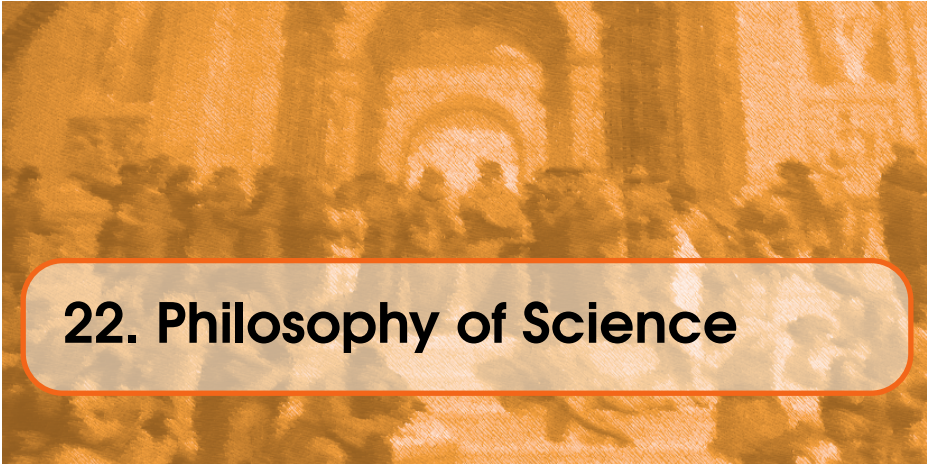
[QR89]: This classic paper shows how MDL can be applied to decision tree induction, providing a concrete and accessible example of the principle in action.

[WB68]: The foundational paper introducing the Minimum Message Length (MML) principle. It frames model selection as an optimal coding problem, balancing hypothesis description and data encoding.

[Wal05]: Wallace's comprehensive monograph on MML presents the full theoretical development and applications of the principle, making it the standard reference.

[Mie12]: Miettinen's book is a standard reference on multiobjective optimization. It introduces Pareto optimality, scalarization methods, and

decision-making frameworks, providing the mathematical basis for the chapter's last section.



22. Philosophy of Science

*To go where you don't know,
you have to go the way you don't know.*
San Juan de la Cruz

The *philosophy of science* is the branch of philosophy that examines the foundations, methods, and implications of scientific inquiry. It explores how scientific knowledge is generated, the validity of scientific theories, the nature of scientific reasoning, and the role of values in science. By addressing fundamental questions about objectivity, reality, and the limits of scientific explanation, the philosophy of science helps us understand how science works and its impact on our understanding of the world.

The philosophy of science provides a theoretical framework for analyzing the core elements that constitute our theory of science. It enables us to critically examine the foundations and assumptions of our theory, guiding us toward the identification of the essential questions that we must address. Moreover, philosophical inquiry compels us to rigorously push the boundaries of our analysis, ensuring that our theory is explored to its ultimate consequences, both logically and conceptually. This approach strengthens the robustness of the theory, even if these results do not directly translate into practical applications.

A central theme in scientific methodology is the distinction between *discovery* and *justification*. Discovery refers to how new ideas or hypotheses emerge, often through creativity, intuition, or serendipity, while justification involves the use of evidence and logical reasoning to evaluate and validate these ideas. Philosophers of science have historically emphasized justification, focusing on methods to rigorously test and analyze scientific claims. However, both discovery and justification are critical to understanding scientific progress, and this chapter will address both aspects.

In this chapter, we provide a concise overview of key elements from the philosophy of science, as well as relevant concepts from other branches of philosophy, such as metaphysics, epistemology, and ontology, that are important to the theory of nescience. Certain other topics within the philosophy of science, such as the problem of objectivity (examining whether science can be truly objective or is influenced by social and personal values) or the role of values in science, including the relationship between ethical, social, and political values and scientific practices, are not included in this review, as they are not directly relevant to a mathematical theory of nescience.

The chapter begins with a brief introduction to the field and its importance in understanding scientific inquiry. We delve into the problem of which entities can be known, examining the scope of scientific knowledge and the nature of observable and unobservable phenomena. The chapter also addresses the concept of scientific representation, discussing how models, theories, and laws reflect aspects of reality. We examine how science discovers new knowledge, the principles behind the scientific method, and the various ways in which scientists formulate and test hypotheses. Finally, we explore the limits of science, considering the boundaries of what science can explain and where its explanatory power may fall short.

22.1 What is Science

Science is a systematic method of investigating the world around us, aimed at generating reliable knowledge through observation and reasoning. Unlike other forms of inquiry, science is rooted in the idea that knowledge must be based on evidence that can be tested and verified. By combining theoretical thinking with empirical data, science offers a powerful way to explain, predict, and understand natural phenomena. Sometimes science seeks explanations for practical purposes, but other times it is sought simply to satisfy our intellectual curiosity.

The *scientific method* is understood as the systematic process by which science acquires new knowledge. In schools, the scientific method is often taught as a series of steps: first observing and describing a phenomenon, then coming up with a hypothesis to explain it, testing that hypothesis through experiments,

analyzing the results, and finally making a conclusion. However, the idea of a single, universal scientific method that can be applied to all sciences isn't widely accepted anymore. Instead, scientists and philosophers recognize that different scientific fields require different methods because each field faces its own challenges and complexities.

The following is a list of the key features that make science distinct and valuable as a way of knowing.

- *Empiricism*: At its core, science is an empirical endeavor, meaning it relies on observation, experimentation, and measurable evidence to understand the natural world. Scientific knowledge is grounded in data that can be gathered through direct or indirect observation, ensuring that claims can be tested and verified by others.
- *Testability*: A key characteristic of science is its focus on developing testable hypotheses, that is, statements or predictions that can be empirically investigated. This means that scientific claims must be falsifiable, open to potential disproof if the evidence does not support them. This distinguishes science from fields that rely on unfalsifiable or speculative claims.
- *Theoretical Frameworks*: Science is not merely about collecting facts; it seeks to develop broader explanations through models, laws, and theories. Models offer simplified representations of complex systems, while theories are well-supported frameworks that explain the underlying mechanisms of phenomena. These frameworks help interpret data and guide further investigation, offering coherence to the body of scientific knowledge.
- *Self-Correction*: A defining feature of science is its self-correcting nature. Scientific theories are not static; they evolve as new evidence comes to light. When new data contradicts a theory, science adapts, modifies, refines, or sometimes discards the theory in favor of one that better fits the evidence. This continuous process of revision ensures that scientific knowledge becomes more accurate over time.
- *Generalizability*: Science strives to uncover universal principles that apply across various contexts, not just to isolated cases. While science begins with specific observations, its goal is to identify general laws and patterns that explain a broad range of phenomena. This pursuit of generalizable knowledge allows science to predict future occurrences and provide deeper insights into the workings of the natural world.

The main task of the philosophy of science is to understand how techniques such as experimentation, observation, and theory building enable scientists to discover the secrets of nature. The philosophy of science draws on other areas of philosophy, such as ontology, epistemology, metaphysics, and logic. *Ontology*, the branch of philosophy concerned with the nature of existence, addresses the

question of which entities exist in the real world. It examines the types of entities that science can study, whether they are observable, abstract, or indeterminate, and engages with the philosophical debates surrounding their existence. Ontology provides the foundation for exploring the boundaries of scientific knowledge and the limits of inquiry. *Epistemology* is the branch of philosophy concerned with knowledge itself—how we acquire it, what justifies it, and what its limits are. While ontology deals with what exists, epistemology addresses how we come to know what exists. In the context of science, epistemology examines the methods, evidence, and reasoning that underpin scientific investigation, exploring how scientific knowledge is built, how reliable it is, and what counts as a justified belief in scientific practice. Both ontology and epistemology are part of the broader field of *metaphysics*, which deals with the fundamental nature of reality and existence. Metaphysics explores the most basic concepts and categories of being, such as time, space, causality, and possibility, as well as the relationship between mind and matter. *Logic*, the study of valid reasoning, underpins the entire scientific enterprise by providing the tools to analyze arguments, identify fallacies, and structure sound reasoning. In science, logic is used to ensure the coherence of theoretical frameworks, the validity of inferences drawn from data, and the consistency of explanations and predictions. Formal logic, including propositional and predicate logic, provides a systematic way to evaluate arguments and detect errors, while informal logic addresses reasoning in everyday scientific discourse. Together, ontology, epistemology, metaphysics, and logic provide a comprehensive philosophical foundation for understanding what science studies, how it builds knowledge, and the fundamental nature of the reality science seeks to explain.

This chapter on philosophy of science is structured around key components that illustrate how scientific knowledge is developed and justified (see Figure 22.1). It begins with a discussion on *Entities*, which can be either concrete or abstract, representing the objects of scientific study. From these entities, knowledge is acquired through *Observation*, which involves gathering empirical data and facts. These observations are then transformed into *Representations*, such as recorded data, or facts, that scientists use to analyze and interpret phenomena. Through the process of *Discovery*, these representations lead to the formulation of *Explanations*, which take the form of scientific *theories and laws* that describe underlying principles governing the natural world. The chapter also explores how scientific explanations influence future observations, shaping what is investigated and how data is interpreted. Additionally, thought Experiments provide an alternative way of reasoning about scientific problems beyond direct empirical observation. Finally, the chapter addresses *Justification*, emphasizing the criteria by which scientific claims are validated, ensuring that discoveries and explanations are robust, reliable, and well-supported by evidence. This structure

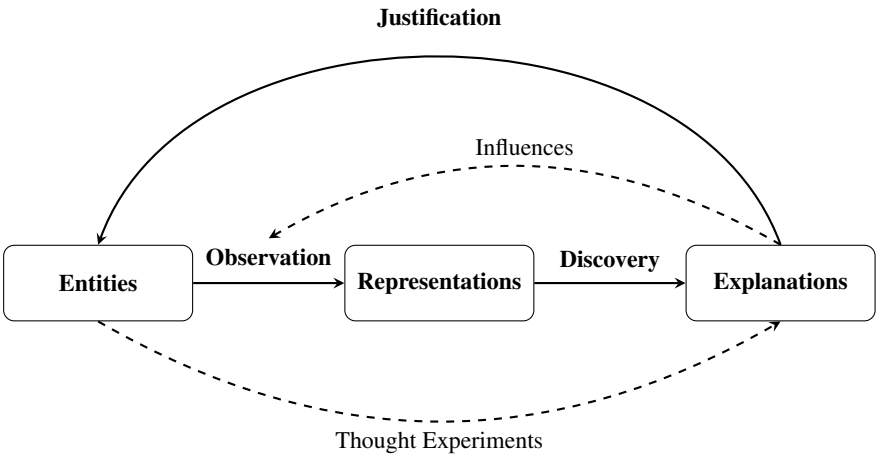


Figure 22.1: Chapter Organization

provides a comprehensive view of the scientific process, from the identification of entities to the formation and justification of scientific theories.

22.2 What is an Entity

In this section, we focus on the fundamental problem of determining which kinds of entities can be known or investigated by science.

A *knowable entity* is an object, phenomenon, or concept that can be investigated, understood, or described through scientific or intellectual inquiry. Knowable entities are those that, either directly or indirectly, can be observed, measured, inferred, or modeled, using available tools, methods, or theories. They are within the scope of human knowledge, and their properties can be analyzed or explained. Examples of knowable entities include the stars and plantes, animals, or computer algorithms. A *non-knowable entity* refers to something that cannot be directly observed, measured, or understood using current scientific or intellectual methods. This could be due to limitations in technology, the abstract or metaphysical nature of the entity, or inherent epistemological boundaries. Non-knowable entities might remain beyond the reach of human understanding either temporarily (until methods evolve) or permanently (due to their nature). Examples of non-knowable entities include the nature of consciousness, the origin of the universe, or the existence of a deity. The distinction between knowable and non-knowable entities is not always fixed, but varies depending on the clarity and precision of the area of interest and the evolving nature of scientific understanding in each field.

Scientific research encompasses both *concrete entities* and *abstract entities*.

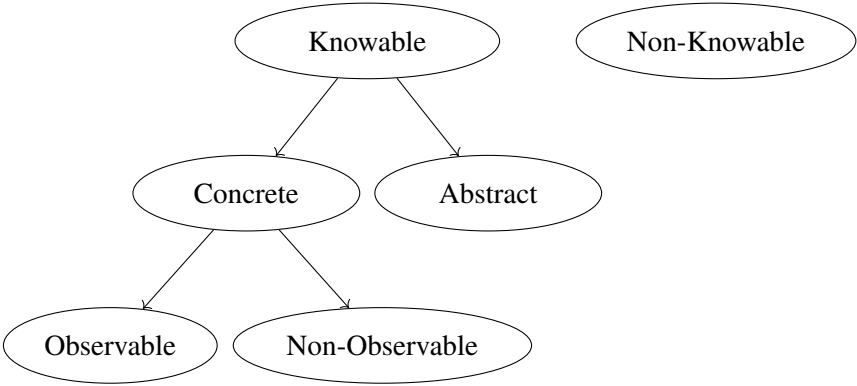


Figure 22.2: Classification of Research Entities

Concrete entities refer to physical objects or phenomena that can be directly observed, measured, or interacted with, such as stars, cells, or chemical compounds. These entities form the basis of empirical research, where data is collected through direct observation or experimentation. In contrast, abstract entities are conceptual and do not have a physical presence, such as numbers, algorithms, or theoretical models. Abstract entities play a crucial role in scientific research, particularly in fields like mathematics and theoretical physics, where they provide the framework for understanding and modeling concrete phenomena. While abstract entities cannot be observed directly, they can be known through indirect methods. The inclusion of these abstract entities in scientific inquiry raises important philosophical questions about their ontological status: Are these abstract entities real in the same way that physical objects are, or are they simply conceptual tools? This question remains an open and debated issue in the philosophy of science.

Finally, in scientific inquiry, there is also a distinction between *observable entities*, which can be directly perceived or measured, and *non-observable entities*, which cannot be directly observed but can still be detected or inferred from empirical evidence. Observable entities include things like trees, planets, and bacteria, objects that can be seen or measured using scientific instruments. Non-observable, but still detectable, entities include things like subatomic particles and gravitational forces. These entities are often crucial for explaining observable phenomena and are identified through the use of scientific instruments and theoretical frameworks. For example, we cannot observe a quark in the same way we observe a tree, but through scientific theory, experimentation, and the detection of indirect evidence, we infer its existence.

A central debate within the scope of science is the tension between *reductionism* and *holism*. Reductionism is the view that complex systems can be

fully understood by breaking them down into their simplest, most fundamental parts. For example, a reductionist might argue that biological processes can be explained entirely by chemistry, and chemistry by physics. This approach assumes that understanding the smallest components of a system will provide a complete explanation of the whole. In contrast, holism argues that some phenomena cannot be fully understood by reducing them to their components. Instead, the whole system exhibits properties that cannot be predicted or explained by analyzing its parts in isolation. For example, in ecology, the interactions within an ecosystem can produce emergent properties that are not reducible to the behavior of individual species.

Finally, the scope of science is shaped by the debate between *scientific realism* and *anti-realism*, which addresses the question of whether the entities posited by scientific theories are real or merely useful constructs. The contrast between realism and anti-realism is most marked for sciences which make claims about non-observable entities. Scientific realism holds that the entities described by scientific theories exist independently of our knowledge of them. According to this view, successful scientific theories reveal truths about the world. In contrast, anti-realism argues that scientific theories are useful tools for predicting and organizing observations, but we should not necessarily believe that non-observable entities like electrons or gravitational waves are real. For anti-realists, the purpose of science is not to describe an independent reality, but rather to provide models that help us navigate and predict phenomena.

Scientists often seek to classify the objects they study into general kinds or categories. A key goal of classification is to convey meaningful information, enabling us to better understand and navigate the complexities of the natural world. However, this process raises several intriguing philosophical questions. Do the categories used in science represent real, essential divisions in nature, or are they merely human-made constructs designed to impose order on a complex and often ambiguous reality (realism vs. anti-realism)? Since any set of objects can, in principle, be classified in various ways, how should we determine the most appropriate approach? Is there a 'correct' way to classify, or are all classification systems fundamentally arbitrary? Some philosophers argue that *natural kinds*, groups that correspond to divisions genuinely existing in the world, do exist, as opposed to merely reflecting human interests. Understanding whether and how such natural kinds exist can provide valuable insights into the structure of reality and inform scientific inquiry.

22.3 Observation in Science

Scientific knowledge has traditionally been granted a special epistemic status, in part because it is thought to rest on a secure foundation of solid facts firmly

established by observation. On this view, facts are taken to be direct, theory-independent deliverances of the senses, immediately accessible to any careful and unprejudiced observer. They are assumed to precede theory and to provide a firm and reliable basis upon which scientific knowledge can be built.

However, there are compelling reasons to question the straightforwardness and security of facts acquired by observation and experiment. What an observer perceives when viewing an object or scene is not determined solely by the images projected onto the retina. Rather, perception is shaped by the observer's prior experience, knowledge, and expectations. One must "clean" to be a competent observer in science, and the perceptual experience of an expert is often markedly different from that of an untrained novice faced with the same situation. This observation challenges a literal understanding of the claim that facts are simply given via the senses. Even when multiple observers view the same scene from the same vantage point, they may interpret what they see in different ways.

Before an observer can formulate and assent to an observation statement, they must already possess an appropriate conceptual framework and know how to apply it. Observation statements presuppose considerable background knowledge. Our search for relevant facts is guided by our current state of knowledge, and this does not undermine the idea that knowledge should ultimately be grounded in facts confirmed by observation. However, it does mean that if the knowledge providing the categories through which we interpret our observations is flawed, the resulting observation statements will be similarly flawed. Judgments about what counts as an observable fact depend on the background of knowledge against which they are made. Indeed, scientific revolutions involve not only the progressive transformation of scientific theories but also a reconfiguration of what are regarded as the observable facts themselves.

This leads to the recognition that facts are fallible and subject to correction. Perceptions are influenced by the expectations and prior knowledge of the observer, so what appears as a straightforward fact to one person may not be so to another. Judgments about the truth of observation statements depend on what is already known or assumed, rendering observable facts as fallible as the presuppositions that underlie them. Everyday observation is far from passive: observers perform numerous actions, often automatically or unconsciously, to establish the validity of what they perceive. To serve as a foundation for scientific knowledge, observation statements must be formulated in ways that allow their validity to be tested through routine, objective procedures that minimize the need for subjective judgment.

Observations that can function as a basis for science must therefore be both objective and fallible. They are objective to the extent that they can be publicly tested by straightforward procedures, and they are fallible to the extent that they may be overturned by new tests enabled by advances in science and technology.

Moreover, science does not require just any facts, but relevant facts. What counts as relevant depends on the current state of development of a science. The world around us involves many interwoven processes that are difficult to disentangle through mere observation of naturally occurring events. To obtain facts relevant for identifying and characterizing particular processes, it is generally necessary to intervene practically: to conduct experiments designed to isolate the process under investigation and eliminate the influence of other processes. If there are facts that constitute the foundation of science, they come primarily in the form of experimental results rather than unstructured observational data.

Yet experimental results are themselves far from straightforwardly given. Their production involves complex practical work, technical know-how, trial and error, and the exploitation of available technologies. Judgments about their adequacy are not simple either: experimental results can be faulty if the knowledge informing their design or interpretation is flawed. They can become obsolete as technology advances, can be rejected when new understanding reveals an experimental set-up to be inadequate, or can be ignored as irrelevant after theoretical shifts. Like observation statements, experimental results are fallible and revisable. Furthermore, they must not only be adequate but also significant: what counts as an adequate or significant experiment depends heavily on the prevailing theoretical and practical context. All experiments presuppose the truth of some theories to judge whether the set-up is appropriate and the instruments are providing the intended readings.

This recognition has led to attempts to formalize the relationship between scientific knowledge and the facts on which it rests. The empiricists maintained that all knowledge should be derived from ideas implanted in the mind through sense perception, while the positivists developed a broader conception of what counts as a fact. Logical positivists, in particular, sought to rigorously analyze the logical structure of the relationship between scientific knowledge and the observational and experimental facts that are thought to underpin it.

In sum, while observation plays a crucial role in science, its role is more complex and less direct than traditionally assumed. Observational and experimental facts are not simply given; they are theory-laden, constructed through skilled practice, and embedded within conceptual frameworks. They provide a fallible yet indispensable foundation for scientific knowledge, one that must continually be tested, refined, and sometimes replaced as science advances.

22.4 Scientific Representation

Science helps us understand the natural world by using different kinds of representations of research entities. These representations include measurements from scientific instruments, descriptions of observations, digital images like

X-rays or MRI scans, and more. Scientific practice also often considers mathematical equations, models, and theoretical constructs as valid forms of representation. The challenge of *scientific representation* is to identify the conditions that make a representation scientific and determine what makes an effective representation. The main issues discussed in this area of philosophy include:

Scientific Representation Problem: The scientific representation problem is about figuring out the necessary and sufficient conditions that make a representation valid in science. It explores whether these conditions are the same across all scientific fields or if they vary depending on the discipline or research context. For example, what qualifies as a valid representation in physics, which often uses mathematical models, might be different from what is used in biology, where visual and descriptive representations are common. This problem also questions whether scientific representations need to be adapted to specific research goals to be considered valid.

Representational Demarcation Problem: The representational demarcation problem looks at whether scientific representations are fundamentally different from other kinds of representations, like those found in art or everyday life. It examines what makes scientific representations unique, focusing on their purpose, accuracy, and the methods used to create them. Unlike artistic representations, which may emphasize subjective interpretation or aesthetic value, scientific representations are generally held to standards of precision, reliability, and empirical adequacy. Understanding these differences helps clarify the specific role that scientific representations play in knowledge production.

Problem of Style: The problem of style addresses the fact that the same entity can be represented in different ways, depending on the goals and methods of the research. Different styles of representation—such as diagrams, mathematical equations, physical models, or computer simulations—each have unique characteristics, including the intended audience, level of abstraction, and type of information conveyed. This issue also asks whether these styles are fixed or if new styles can be invented to meet emerging scientific needs. The flexibility of representation styles is crucial because it allows for new insights and different ways of understanding scientific phenomena.

Standard of Accuracy: The standard of accuracy problem is about determining what makes a scientific representation accurate. It involves figuring out how to distinguish between accurate and inaccurate representations by considering factors like how well the representation matches empirical data, captures important features of the phenomenon, and its ability to make predictions. This issue also explores whether accuracy should be seen as an objective standard or if it depends on the specific aims and context of the research. For example, a simplified model might still be considered accurate if it effectively serves its

purpose, such as making predictions or providing explanations.

Problem of Ontology: The problem of ontology in scientific representation deals with the nature of the entities that can serve as representations. It asks whether representations need to be concrete, like physical models or graphs, or if they can also be abstract, like mathematical equations or theoretical constructs. This issue also questions whether representations must be realistic or if more abstract, idealized forms can still be effective in scientific inquiry. Understanding these ontological aspects helps define the types of entities that are allowed in scientific discourse and how they relate to the real-world phenomena they represent.

There are also five conditions of adequacy that a scientific representation should satisfy to be considered effective and reliable:

Requirement of Directionality: The requirement of directionality examines the relationship between representations and the real world. Representations are meant to describe entities in the real world, but this condition raises the question of how, if at all, real-world entities might describe their representations. It challenges us to think about the direction of influence between the representation and the entity it aims to depict.

Surrogate Reasoning: Surrogate reasoning addresses how scientific representations allow researchers to generate hypotheses about the entities they represent. This condition explores how using a representation as a surrogate can lead to new insights or predictions about the target, effectively using the representation to stand in for the real-world entity during reasoning and analysis.

Applicability of Mathematics: The applicability of mathematics condition is concerned with how mathematical models can be used to represent the real world. It questions how abstract mathematical constructs can effectively describe complex physical systems and whether the success of mathematical representation depends on any special features of the target phenomena. This condition highlights the central role of mathematics in developing and understanding scientific theories.

Possibility of Misrepresentation: The possibility of misrepresentation addresses whether representations that are not fully accurate can still be considered valid scientific representations. It considers situations where simplifications or approximations are necessary and whether these less-than-perfect representations can still contribute valuable understanding of a phenomenon. This condition is important for understanding how idealizations and abstractions function in scientific practice.

Targetless Models: The targetless models condition explores whether we can allow representations that do not have a direct real-world counterpart. It questions if a model that does not represent any existing entity can still be useful in scientific inquiry, perhaps as a way to explore theoretical possibilities or

to understand potential scenarios. This condition emphasizes the creative and exploratory aspects of scientific modeling.

There have been multiple proposals to formally define the concept of scientific representation. Unfortunately, none of these proposals can provide a convincing answer to the questions and conditions of adequacy described above. In the rest of this section, we describe some of these proposals, identifying their advantages and drawbacks. To compare these proposals, we will present them as: "A scientific model M represents a target system T if, and only if ...".

Stipulative Fiat: The stipulative fiat proposal states that "a scientific model M represents a target system T if, and only if, a scientist stipulates that M represents T ." The main problem with this interpretation is that, since anything can be a representation if a scientist says so, it is difficult to guarantee the surrogative reasoning condition. If any model can be deemed a representation by simple stipulation, it becomes challenging to determine which representations are genuinely useful for making scientific inferences. Proponents of this theory acknowledge that while all representations may be stipulated, some are undeniably more useful than others.

Similarity Conception: The similarity conception proposes that "a scientific model M represents T if, and only if, M and T are similar." This conception addresses the surrogative reasoning condition since similarity between the model and the target allows us to derive similar properties. However, it introduces new challenges, particularly regarding the problem of style. The concept of similarity is often vague: in what sense are M and T similar? This vagueness can lead to issues with directionality and accuracy, as different aspects of similarity may not always align with what is relevant for scientific representation.

Structuralist Conception: The structuralist conception is based on the idea of isomorphism. According to this view, a scientific model M represents a target system T if the structure of M is isomorphic to the structure of T . In other words, there is a one-to-one correspondence between the elements and relationships in both M and T . This approach justifies surrogative reasoning because having the same structure implies that properties and relations in the model correspond to those in the target. Furthermore, since mathematics is fundamentally concerned with the study of structures, this conception also supports the applicability of mathematics in representing natural systems.

Inferential Conception: The inferential conception proposes that a model M is an epistemic representation of a target T if, and only if, the user adopts an interpretation of M in terms of T . This view emphasizes the role of the user in giving meaning to the model, suggesting that representation is not an inherent property of the model itself but arises through its use in making inferences about the target system. This conception underscores the importance of context and interpretation in determining whether a model effectively represents its target.

Fiction View of Models: According to the fiction view of models, M represents T if and only if M functions as a prop in a game of make-believe that prescribes imagining certain things about T . This view draws an analogy between scientific modeling and storytelling, where models are treated as fictional constructs that facilitate imaginative engagement with the target system. Although this approach highlights the creative aspects of modeling, it raises questions about how such fictional constructs can be rigorously linked to real-world entities.

Representation-As: The representation-as approach suggests that a scientific model represents a target system as something, emphasizing that representation involves highlighting certain features of the target while downplaying others. This conception focuses on the interpretive aspect of modeling, where the modeler selects specific attributes of the target to represent, depending on the research goals. This approach allows for a flexible understanding of representation that can accommodate different styles and purposes, but it also implies that the usefulness of a representation is contingent on how well the modeler captures the relevant aspects of the target.

Each proposal has its strengths and weaknesses, highlighting the complexity of what it means for a model to effectively represent a target in scientific inquiry. Understanding these various perspectives is crucial for advancing our comprehension of the role of representation in science.

22.5 Scientific Discovery

Scientific discovery refers to the process through which new knowledge, ideas, or principles are uncovered within science. Unlike the systematic procedures associated with justification, discovery often involves creativity, intuition, and inspiration. While some discoveries arise from planned experiments or systematic observation, others occur unexpectedly, challenging existing paradigms or opening new fields of inquiry. The general agreement among philosophers is that the creative process of conceiving a new idea is a non-rational process that cannot be formalized as a set of steps. Understanding discovery is crucial for appreciating how science evolves and adapts, as it reveals the dynamic and often unpredictable nature of scientific progress.

The following two proposals assume that a domain-neutral logic of discovery can be formalized, offering attempts to develop such a framework.

- *Discovery as abduction:* Abductive reasoning is a mode of discovery that begins with surprising or anomalous phenomena and seeks to generate plausible hypotheses to explain them. This process is conceptualized as follows: (i) some unexpected data, such as $p_1, p_2, \dots, p_n$, is encountered; (ii) these data would be less surprising if a hypothesis of type H were true; and (iii) therefore, there is justification to develop a hypothesis of type

H. Two types of abduction are distinguished: *selective abduction*, which involves choosing from known hypotheses, and *creative abduction*, which generates entirely new hypotheses. Abduction present some limitations. First, multiple hypotheses may explain the same phenomena, making additional criteria necessary to evaluate their merit. Second, the schema of abductive reasoning does not account for the act of conceiving a hypothesis itself.

- *Heuristic programming* is a computational approach designed to simulate and assist the creative aspects of human problem-solving. These programs operate as searches within a defined problem space, which includes all possible configurations for a given domain. Each configuration represents a specific state within the problem space, with two key states being the initial state, the starting point of the search, and the goal state, which represents the desired outcome. Operators define the moves that transition between states, while path constraints limit permissible moves within the problem space. Problem-solving in this context involves finding a sequence of operations that connects the initial state to the goal state. The core aim of this approach is to develop heuristics (practical rules or strategies) to efficiently navigate and solve complex problems. Heuristic programming has its limitations, scientific problem spaces are often ill-defined, and computer programs rely on experimental data, meaning that simulations frequently cover only specific aspects of scientific discovery.

Many philosophers argue that discovery is an important topic within the philosophy of science, even as they move away from the idea of a formal logic of discovery. A highly influential perspective is Thomas Kuhn's examination of how new facts and theories emerge in the so-called *paradigm shifts*. According to Kuhn, discovery is not a single event but rather a complex and prolonged process that often results in paradigm shifts. Paradigms consist of shared generalizations, theoretical commitments, values, and exemplars that unify a scientific community and shape its research practices. During periods of normal science, research focuses on expanding and refining the existing paradigm rather than pursuing novelty. Discovery typically begins with the recognition of anomalies—phenomena that defy the expectations established by the current paradigm. This process includes observing and conceptualizing the anomaly, followed by revising the paradigm to accommodate it. During paradigm crises, theory-driven discoveries may occur as scientists propose speculative theories, develop new expectations, and conduct experiments or observations to test these ideas. Ultimately, a new paradigm emerges, transforming the once-anomalous phenomena into standard expectations.

Scientific discovery can be also viewed as inherently tied to creativity, with philosophers drawing from cognitive science, neuroscience, computational

research, and environmental and social psychology to better understand how new ideas emerge. This perspective aims to demystify the mental processes behind creative thought, emphasizing that scientific creativity can be analyzed and understood philosophically. Central to this analysis are two pivotal cognitive mechanisms: analogies and mental models, which serve as fundamental tools in the generation of innovative ideas and the advancement of knowledge.

- *Analogy*: Analogies play a crucial role in scientific discovery by enabling the transfer of ideas from one domain to another. Philosophers distinguish between three types of analogies: positive, negative, and neutral. Positive analogies involve properties that are shared by both the model (the well-understood domain) and the target domain (the new domain). Negative analogies include properties that belong solely to the model and do not apply to the target domain. Neutral analogies are the most intriguing because they consist of properties whose relevance to the target domain remains uncertain. These neutral analogies are significant as they often lead to new insights and hypotheses about the less familiar domain. Additionally, there is a distinction between horizontal and vertical analogies. Horizontal analogies connect two domains at the same level of abstraction, whereas vertical analogies involve relationships between different levels of abstraction within the same domain.
- *Model-based reasoning*: The concept of model-based reasoning suggests that much of human thought, including probabilistic and causal reasoning as well as problem-solving, relies on mental models rather than formal logic or strict methodological rules. In this approach, the mind constructs structural representations of real-world or imagined situations and manipulates these models to explore possibilities and generate insights. Conceptual structures are viewed as models, and conceptual innovation involves creating new models using various operations. Analogical reasoning, or analogical modeling, is one of the primary forms of model-based reasoning, alongside visual modeling and simulative modeling, such as through experiments.

22.6 Scientific Explanation

In philosophy of science it is often made the assumption that there exists a single, distinct type of explanation that qualifies as "scientific." The concept of "scientific explanation" suggests at least two key elements: first, a contrast between explanations characteristic of science and those that are not, and second, a contrast between "explanations" and other forms of discourse, such as mere "descriptions." It is important to note that a set of claims can be true, accurate, and supported by evidence while still failing to qualify as explanatory.

Good scientific explanations are typically evaluated based on several key characteristics. While different philosophical frameworks might emphasize different criteria, the following characteristics are broadly agreed upon:

Empirical Adequacy: A good explanation must align with observed and experimental evidence. It should accurately describe the phenomena being explained, providing a detailed account of how the observed data supports the explanation. Additionally, it should integrate well with existing empirical findings, ensuring reliability and fostering broader scientific understanding.

Logical Coherence: The explanation should be logically consistent and free of contradictions. Its components should interconnect in a structured and harmonious way, enabling clear and valid reasoning. This coherence ensures that the explanation aligns with established logical principles and provides a solid foundation for understanding the phenomena under investigation.

Causal Relevance: Good explanations often identify the causal mechanisms responsible for the phenomenon. They should clarify how and why the phenomenon occurs, detailing the specific interactions and processes involved. By establishing a clear causal link, these explanations provide a framework for understanding not just what happens, but the underlying reasons and mechanisms driving the event. This depth of causality enables predictions, interventions, and broader scientific application.

Generality: Explanations that apply to a broader range of phenomena are considered more valuable. They should transcend individual instances to reveal patterns or principles that connect seemingly disparate phenomena. By addressing a wider scope, such explanations enhance our ability to generalize knowledge, foster predictive accuracy, and provide a unifying perspective across different domains of inquiry.

Simplicity (Parsimony): A good explanation should not be unnecessarily complex. Among competing explanations, the one that makes the fewest assumptions while still accounting for the phenomena is often preferred (Occam's Razor). This principle emphasizes clarity and efficiency in reasoning, ensuring that explanations avoid superfluous details or unwarranted complexity. Simpler explanations are easier to test, communicate, and apply, fostering a more practical and streamlined understanding of the phenomena.

Explanatory Depth: Good explanations go beyond surface descriptions to provide a deeper understanding of the underlying mechanisms, principles, or causes. They delve into the fundamental elements that drive the phenomenon, offering insights into its origins and interconnections with related concepts. By unraveling these deeper layers, such explanations help uncover not just the "how," but the "why," enriching our comprehension and enabling a more profound application of knowledge.

Testability: The explanation must allow for predictions that can be tested and

potentially falsified. This criterion ensures that the explanation is scientifically meaningful and subject to empirical scrutiny. By being testable, the explanation invites rigorous examination and challenges, which help to confirm its validity or expose its weaknesses. Testability also connects scientific explanations to the broader experimental process, enabling continuous refinement and adaptation in light of new evidence, thereby fostering scientific progress and reliability.

Unification: A good explanation often unifies disparate phenomena under a single framework or theory, showing how they are related. It highlights the connections and underlying principles that bring coherence to seemingly unrelated phenomena. By doing so, unification not only simplifies our understanding but also enhances the explanatory power of a theory, allowing for a more integrated and comprehensive perspective on the natural world.

Use of Laws: Good explanations often incorporate established scientific laws or theories to provide a robust foundation for their claims. By grounding explanations in well-established principles, they ensure credibility and consistency with the broader scientific framework. This approach not only strengthens the explanatory power of the claims but also facilitates their integration into existing knowledge systems, thereby fostering coherence, predictability, and utility in advancing scientific understanding.

Practical Applicability: In some cases, the usefulness of an explanation in solving problems, guiding further research, or applying knowledge in practical ways adds to its value. An explanation with practical applicability not only enhances theoretical understanding but also bridges the gap between abstract knowledge and real-world implementation. It can inform decision-making, inspire technological innovations, and address pressing societal challenges. By demonstrating utility in diverse contexts, such explanations underscore the relevance of scientific inquiry to everyday life and future advancements.

Asymmetry: The requirement of asymmetry in scientific explanation is a widely discussed topic in the philosophy of science. According to most philosophical accounts, explanation is indeed considered to be an asymmetric relation, meaning that if X explains Y, it should not also be true that Y explains X under the same laws and additional facts. However, there are debates and complications surrounding this idea. Let's examine the arguments for and against asymmetry in scientific explanations.

Explanation as prediction: When we provide a covering law explanation for a phenomenon, the laws and specific facts we reference could have allowed us to predict the phenomenon's occurrence. This highlights that a scientific explanation is inherently a potential prediction. Conversely, every reliable prediction is inherently a potential explanation, illustrating that explanation and prediction are structurally symmetric in their foundational principles.

The following paragraphs briefly introduce the most relevant proposals to

characterize what is a scientific explanation.

The *Deductive-Nomological model*, or DN model, emphasizes the importance of deductive reasoning and general laws. According to this model, a phenomenon is explained by demonstrating how it logically follows from a general law combined with specific initial conditions. This explanation comprises two main components: the *explanans*, which includes the general laws and initial conditions, and the *explanandum*, which is the phenomenon to be explained. For the explanation to be valid, the explanans must be true and logically entail (see logical deduction at Section 22.7) the explanandum, meaning the phenomenon can be deduced from the laws and conditions provided. A DN model answers the question "Why did the explanandum occur?" by showing that the phenomenon resulted from specific circumstances $C_1, C_2, \dots, C_i$, in conjunction with laws $L_1, L_2, \dots, L_j$. For example, the motion of a particular pendulum can be explained by applying Newton's laws of motion (general laws) along with specific details such as the pendulum's length and initial displacement (specific initial conditions). Alternatively, the general behavior of pendulums can also be explained using the same laws and reasoning, since the DN model can be applied to both particular occurrences and general patterns.

The *Statistical-Relevance model*, or SR model focuses on explaining phenomena through statistical relationships rather than strict deductive reasoning. Unlike the Deductive-Nomological model, which requires logical entailment from general laws, the SR model emphasizes the identification of statistically relevant factors that significantly influence the likelihood of a phenomenon. In this model, given some class or population A , an attribute C is *statistically relevant* to another attribute B if and only if $P(B|A, C) \neq P(B|A)$. This means that C affects the probability of B within the context of A . An explanation involves identifying such statistically relevant factors and evaluating their impact within a reference class (a group of events or entities sharing common characteristics). For example, in explaining the likelihood of developing a particular disease, an SR explanation might highlight factors such as age, genetic predisposition, or lifestyle choices, showing how these variables alter the probability of the disease occurring. By uncovering these statistical relationships, the SR model provides a method for explaining probabilistic phenomena that cannot be addressed deterministically.

The *Causal-Mechanical model*, or CM model, of scientific explanation emphasizes understanding phenomena by uncovering the underlying causal mechanisms that produce them (see causality at Section 22.7). This model asserts that explanations are not just about identifying laws or statistical relationships but about revealing the actual processes and interactions that link causes to effects. A causal-mechanical explanation requires tracing a continuous causal chain, often through detailed physical or biological processes, to show how an event is brought about. For example, explaining the boiling of water involves

identifying the causal mechanism: heat energy transfers to the water molecules, increasing their kinetic energy until intermolecular bonds are overcome, leading to a phase change from liquid to gas. The CM model prioritizes clarity in how individual components interact and influence each other, providing a deeper understanding of the phenomenon by grounding it in observable and empirically testable mechanisms. This approach is particularly effective in fields like biology, physics, and engineering, where complex systems and their interactions play a central role in explanation.

The *unificationist account* of scientific explanation, emphasizes the power of explanation through the unification of diverse phenomena under a single, coherent framework of principles and patterns. According to this model, the primary aim of scientific explanation is to reduce the number of independent assumptions and derive a wide range of phenomena from a minimal, consistent set of explanatory patterns. An explanation is considered successful if it contributes to this unifying framework by connecting seemingly disparate observations through common principles. For instance, Newtonian mechanics unifies the motions of celestial bodies and terrestrial objects under the same laws of motion and gravitation. The unificationist approach highlights the importance of simplicity, generality, and coherence in scientific theories, proposing that the value of an explanation lies in its ability to integrate knowledge into an organized, explanatory schema. By offering a comprehensive understanding of diverse phenomena, this account showcases the interconnectedness and systematic nature of scientific inquiry.

The *pragmatic theories* of scientific explanation emphasize the context-dependent and audience-specific nature of explanations, focusing on their purpose and practical utility rather than strict formal structures. These theories argue that explanations are answers to "why" questions posed within a specific context, and their adequacy depends on how well they address the interests and background knowledge of the audience. A scientific explanation, therefore, is not inherently tied to a universal standard but varies depending on the explanatory goals, such as prediction, understanding, or control. For instance, explaining why a bridge collapsed might involve detailed structural analysis for engineers, whereas a simplified account focusing on the immediate cause, like high winds, might suffice for the general public. Pragmatic approaches recognize that explanatory demands can differ across disciplines, situations, and audiences, making the effectiveness of an explanation contingent on its relevance, clarity, and alignment with the inquirer's needs. This perspective underscores the interplay between scientific knowledge and its communication within varied practical contexts.

22.7 Scientific Justification

Scientific knowledge is often said to be justified by appeal to evidence. However, this justification is not as straightforward or conclusive as it might first appear. It has long been recognized that scientific knowledge can neither be conclusively proved nor conclusively disproved. The role of justification in science is therefore complex: it is always partial, provisional, and open to revision, rather than absolute or final.

Logicians distinguish between two fundamental kinds of reasoning: *deduction and induction*. Deductive reasoning moves from general premises to conclusions that must be true if the premises are true. Inductive reasoning moves from particular observations to general conclusions, and can lead from true premises to false conclusions. Scientists make use of induction whenever they generalize from a finite number of observations to broader laws, which they do constantly.

This reliance on induction gives rise to what is known as Hume's problem of induction. Any inductive inference assumes that the unexamined cases will resemble the examined ones, an assumption often called the "uniformity of nature." But this assumption cannot itself be proven, and so the rational justification of induction seems to rest on circular reasoning. Karl Popper responded to this difficulty by arguing that induction is not needed at all: science, he claimed, progresses by proposing bold conjectures and then trying to refute them through deductive tests. Although no amount of evidence can prove a scientific theory true, a single well-corroborated counterexample can, in principle, refute it. From this perspective, scientific theories are never justified as true, only as not-yet-falsified.

In practice, scientists also often justify theories by appealing to what is known as inference to the best explanation. When confronted with a body of evidence, they favor the theory that best explains it, typically one that is simpler, more coherent, and unifies disparate observations. This is an inherently comparative and evaluative notion of justification: a theory is considered justified not because it is proven true, but because it currently explains the data better than its rivals.

Much of modern science can be described as following the *hypothetico-deductive method*. A theory is proposed, from which testable predictions are deduced. These predictions are then confronted with observations or experimental results. If the results conflict with the predictions, the theory may need to be rejected or revised; if the results agree, the theory is supported, though never conclusively proven. This illustrates a fundamental feature of scientific justification: positive results support a theory only tentatively, while negative results can count against it. Yet even refutation is not straightforward, because failed predictions may stem from auxiliary assumptions, background theories, or

experimental errors rather than from the core theory itself.

This underdetermination of theory by evidence means that evidence alone cannot logically force the acceptance or rejection of a theory. Instead, scientists must weigh multiple considerations when deciding which theory is best justified, including simplicity, explanatory power, and coherence with established knowledge.

Alongside these logical models, science often relies on probabilistic justification. Statistical inference allows scientists to estimate how likely their data would be if a given hypothesis were true. The Bayesian approach refines this by providing a formal framework for updating the degree of belief in a hypothesis when new evidence is obtained. Each new observation shifts the credibility of a hypothesis up or down, never proving it true but adjusting its level of justification relative to alternatives.

This view makes clear that justification in science is gradual and comparative: theories become more or less justified over time as evidence accumulates. However, it also shows the limits of justification. Bayesian methods assume that all relevant hypotheses are already under consideration, and cannot by themselves account for genuinely novel theories. They also depend on prior assumptions that may be subjective or biased. In large part, then, what counts as “justified” depends on what the scientific community currently considers possible or plausible.

Thomas Kuhn highlighted that justification is not theory-neutral. What counts as evidence, and how evidence is interpreted, depends on the conceptual framework or paradigm within which scientists are working. During periods of normal science, researchers largely take the paradigm for granted and evaluate theories in terms of how well they solve puzzles defined by it. When anomalies accumulate, a period of crisis can lead to a scientific revolution in which the old paradigm is replaced by a new one. Kuhn argued that competing paradigms are often incommensurable, lacking a common standard by which to compare them directly.

This means that the standards of justification themselves can shift when paradigms change. What once counted as strong evidence may later be dismissed as irrelevant, and what was once regarded as speculative may later be seen as well justified. This view challenges the traditional separation between the context of discovery and the context of justification: the criteria for what justifies a theory are historically and socially situated, not timeless or purely logical.

There is no universal algorithm that can mechanically determine which theory is best justified. Scientific virtues such as empirical fit, simplicity, explanatory power, and coherence often trade off against one another, and reasonable scientists may weigh them differently. As Paul Feyerabend argued, the history of science shows that methodological pluralism—even breaking the accepted rules—has

sometimes been necessary for progress. Scientific justification is therefore not purely mechanical or formal, but involves the exercise of judgment, creativity, and even a degree of risk-taking.

Finally, justification in science often depends on establishing causal claims. This is usually done not through passive observation but through carefully designed experiments that isolate variables and eliminate confounding factors. Experimental results provide stronger justification because they actively create the conditions needed to test causal hypotheses. Yet they, too, are interpreted through theoretical assumptions, and their justificatory power depends on the soundness of those assumptions. Even causal claims are therefore not justified by observation alone, but by their integration into a broader theoretical framework.

22.8 The Limits of Science

Science has been one of the most successful human enterprises, providing powerful tools for understanding and transforming the world. Yet its scope is not unlimited. There are questions science is ill-suited to answer, boundaries it may never cross, and domains where its methods cease to apply. Recognizing these limits is not to diminish science, but to place it in the broader landscape of human inquiry.

Philosophy, for example, pursues questions beyond the reach of empirical investigation. It employs methods such as logical reasoning, thought experiments, and conceptual analysis, which seek clarity about the meaning and boundaries of our concepts rather than empirical predictions. These methods can reveal types of truths—about logic, meaning, or ethics—that lie outside science's domain. While science asks how the world works, philosophy can ask what existence itself means, or why there is something rather than nothing. Such metaphysical questions appear to fall beyond the scope of empirical methods.

Similarly, science can describe the consequences of actions, but it cannot tell us what we ought to value or how we ought to act. Moral and aesthetic judgments involve norms, not facts. Scientific results can inform ethical debates, but cannot settle them. This highlights a central boundary of science: it can describe what is, but not what ought to be.

Even within its own domain, science cannot explain everything. Any scientific explanation must rely on some fundamental laws or principles; yet those laws cannot themselves be explained by deeper laws without falling into infinite regress. There will therefore always be a point at which science must stop and accept certain principles as unexplained. This does not mean we can say in advance which questions are forever unanswerable, only that some will inevitably be so.

Moreover, science is constrained by what can be observed or indirectly

detected. If an entity or phenomenon leaves no observable trace, then it is beyond the reach of empirical science. This raises subtle questions about the ontological status of unobservable entities: scientific theories may refer to them, but their existence can never be confirmed directly.

Another limit of science arises from its division into disciplines. Each scientific field is designed to explain a particular range of phenomena using its own methods. Physics, for example, studies the fundamental structure of matter, while sociology studies the structure of human societies. It is often assumed that the more fundamental sciences can, in principle, subsume the less fundamental ones—a view called reductionism. Yet many phenomena studied by the higher-level sciences are multiply realized at the physical level, and their behavior emerges from complex interactions not captured by lower-level laws alone. Holism emphasizes that some properties can only be understood at the level of the whole system. This suggests that even complete knowledge of physics would not automatically yield complete knowledge of biology, psychology, or culture.

A further limitation concerns the very nature of scientific knowledge. Scientific realists argue that science aims to describe the world as it truly is, including unobservable entities. Anti-realists reply that theories should be seen as instruments for predicting observable phenomena, not as literal descriptions of reality. One argument for anti-realism is the problem of underdetermination: the same body of empirical data can often be explained by many different, mutually incompatible theories. If evidence underdetermines theory, then perhaps we should suspend judgment about the reality of unobservable entities. Realists respond that not all explanations are equally good, and that virtues like simplicity, coherence, and explanatory power can guide rational choice. Yet the debate shows that even the status of scientific knowledge itself is contested.

Science is often described as objective, yet it is carried out by human beings embedded in cultural and social contexts. Research agendas, funding priorities, values, and personal biases all shape what questions are asked, how data are interpreted, and which results are accepted. This does not mean that science is purely subjective: its methods aim to correct for individual bias through replication, peer review, and open criticism. But recognizing these social influences reveals that scientific objectivity is an ideal rather than an automatic fact. The self-correcting nature of science works only because its claims remain open to challenge and revision.

Another important boundary is the demarcation between science and non-science. Popper famously proposed falsifiability as a criterion: a theory is scientific if it makes predictions that could in principle be shown false. This criterion has been influential, but not universally accepted, as some fields (such as string theory or evolutionary psychology) are difficult to test in practice yet

still considered scientific by many. The demarcation problem shows that there is no single, universally agreed rule for what counts as science, and that the borders of science are themselves somewhat fluid and historically contingent.

Finally, some questions may remain permanently beyond science's reach. The ultimate origin of the universe, the nature of dark matter, or the final fate of reality may elude definitive answers. Science depends on its tools, and some phenomena might require conceptual frameworks or observational methods we cannot yet—and perhaps can never—develop. This does not make such questions meaningless, but it shows that science is only one strand in the broader human quest for knowledge.

Even where science can answer a question, it cannot tell us whether answering it is good or bad. The same knowledge that enables genetic engineering or artificial intelligence also raises profound ethical dilemmas about how such technologies should be used. Science can provide the means to achieve our ends, but it cannot choose the ends themselves. For this, we must appeal to ethical and political reasoning beyond the scope of empirical science.

References

[Cha13]: Chalmers provides a clear and widely used introduction to the philosophy of science, discussing induction, falsifiability, paradigms, and the limits of scientific reasoning. It is ideal as a general reference for framing the main questions addressed in this chapter.

[Pop14]: Popper introduces falsifiability as the demarcation criterion for science and develops his vision of knowledge growth through bold conjectures and refutations. This is central for your treatment of scientific progress and the demarcation problem.

[Van80]: Van Fraassen defends constructive empiricism, arguing that science aims for empirical adequacy rather than truth. This perspective supports your discussion on the limits of science and the interpretation of scientific representations.

[Sup02]: Suppes develops a structuralist account of scientific theories, emphasizing how scientific knowledge can be formalized in terms of models and invariants. This is directly relevant to your sections on scientific representation.

[Mos16]: Mosterín explores the logical and conceptual structure of scientific theories, focusing on how concepts evolve and how theories are built. This supports your account of entities, theories, and the evolution of knowledge.

[LSW13]: Latour's ethnographic study highlights the social processes behind the making of scientific knowledge. Including this reference enriches your treatment of science as a human practice and its limits.

[Nag14]: Nagel gives a concise overview of epistemological issues such

as justification, belief, and truth. This is useful background for connecting philosophy of science with broader epistemology.

?? contains an interesting review of the concept of science, the scientific method, and the role that technology plays in our society. The author proposes that the goal of science should be quality, although the concept of quality is left undefined, and how to reconcile the rational and romantic points of view in science. The book also contains some advice about which is the right state of mind to pursue a scientific problem, and how to deal with the inevitable failures.



A. About Quotes and Photos

*What we know is little,
combined with tenacious concentration on a subject
and what we are ignorant of is immense.*
Pierre-Simon Laplace

This appendix gathers the quotes and photographs that appear at the beginning of each chapter, offering a brief explanation of their origin and meaning. These elements were carefully chosen for their deep symbolic connection to the central themes of the theory of nescience. The quotations capture, in a few words, timeless insights about knowledge, ignorance, discovery, and the creative power of questioning; some of them serving as true sources of inspiration in the development of the theory itself.

The photographs, likewise, are not mere illustrations: each has been selected as a visual metaphor for a key concept discussed in the book, and has been pre-processed to ensure a coherent visual style. Together, these quotes and images form a symbolic framework that accompanies the reader throughout the book, signposting the intellectual journey from the unknown to the known.

A.1 Chapter Quotes

The quotations collected in the book have been carefully selected for their profound relevance to the theory of nescience. Each captures, in a few words, a key idea about knowledge, ignorance, discovery, or the creative act of questioning, all central to the themes explored throughout this book. Some of these quotes have been a genuine source of inspiration in the development of the theory itself, shaping its concepts and guiding its spirit. They are presented here as intellectual signposts, inviting the reader to reflect on the timeless wisdom that underlies our ongoing quest to illuminate the unknown.

“Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away.”

Antoine de Saint-Exupéry

French aviator and author of *The Little Prince*, Saint-Exupéry combined technical discipline with poetic clarity. This quote celebrates the art of refinement: true perfection emerges not from accumulation, but from stripping away the superfluous to reveal the essential. It resonates with the theory of nescience, which seeks to reduce surfeit, eliminating redundant information until only the irreducible core of knowledge remains.

“If presented with a choice between indifferent alternatives, then one ought to select the simplest one.”

William of Ockham (Occam’s Razor)

The medieval philosopher William of Ockham championed conceptual economy in an age of scholastic excess. His Razor urges us to prefer simplicity when multiple explanations fit the same evidence. This principle aligns with nescience by warning against needless complexity, guiding us to seek the most compact and accurate representations of knowledge.

“We are all agreed that your theory is crazy. The question which divides us is whether it is crazy enough.”

Niels Bohr

Bohr, a founder of quantum theory, embraced the paradoxes that reshaped modern physics. His quip reminds us that revolutionary ideas often seem absurd at first. It underscores the need for bold conceptual leaps to break through entrenched assumptions and reveal what lies hidden in the unknown.

“All great work is the fruit of patience and perseverance, combined with tenacious concentration on a subject over a period of months or years.”

Santiago Ramón y Cajal

The father of modern neuroscience, Ramón y Cajal spent decades meticulously

mapping the nervous system. His words highlight the slow, deliberate nature of deep discovery. For nescience, they affirm that overcoming ignorance is rarely sudden; it demands sustained, focused engagement with the unknown.

“A little inaccuracy sometimes saves tons of explanation.”

Saki (H. H. Munro)

Known for his concise, satirical stories, Saki understood the power of brevity. This remark suggests that relentless precision can hinder clarity, and that small simplifications can reveal the bigger picture. It speaks to nescience by acknowledging that small sacrifices in exactness can accelerate the broader reduction of ignorance.

“Everything should be made as simple as possible, but not simpler.”

Albert Einstein

Einstein revolutionized physics through elegant, minimalist theories, yet he warned against oversimplification. His maxim captures the delicate balance at the heart of nescience: reducing the length and complexity of our descriptions without introducing errors that distort reality.

“There are known knowns. . . There are things we don’t know we don’t know.”

Donald Rumsfeld

Though a political figure, Rumsfeld articulated a taxonomy of knowledge that became famous beyond its context. He distinguished what we know, what we know we don’t know, and what we don’t even realize we don’t know. This hierarchy mirrors the structure of nescience, which measures progress by transforming unknown unknowns into known unknowns, and eventually into knowns.

“It is not the answer that enlightens, but the question.”

Eugène Ionesco

A pioneer of the Theatre of the Absurd, Ionesco used paradox to challenge conventional thinking. His statement celebrates the generative power of questions, which open new conceptual spaces. Nescience places similar emphasis on asking fruitful questions as the first step toward uncovering the hidden unknown.

“There are no difficult problems, only lack of imagination.”

Antonio García

A Spanish engineer, García championed creativity as the key to problem-solving. His observation reframes obstacles as failures of perspective rather than inherent difficulty. This spirit of imaginative thinking is vital to finding new ways to approach what seems unknowable.

“Science may be regarded as the art of data compression.”

Ming Li & Paul Vitányi

Pioneers of algorithmic information theory, Li and Vitányi showed that scientific theories compress vast datasets into concise descriptions. Their insight directly underpins science: progress is measured by how much shorter and more accurate our descriptions become as we transform scattered facts into structured knowledge.

“To be surprised, to wonder, is to begin to understand.”

José Ortega y Gasset

A leading Spanish philosopher, Ortega y Gasset saw wonder as the spark of understanding. This quote marks the moment when the familiar becomes strange enough to question. That moment, recognizing something as unknown, is the first step in reducing it.

“Mathematics may be defined as the subject in which we never know what we are talking about, nor whether what we are saying is true.”

Bertrand Russell

Logician, philosopher, and Nobel laureate, Russell used humor to reveal the abstract nature of mathematics. His quip highlights how mathematical structures often float free of meaning. This echoes nescience’s concern that our representations may be precise yet disconnected from reality, requiring continual testing to ground them.

“The purpose of models is not to fit the data but to sharpen the questions.”

Samuel Karlin

Karlin, a mathematician who bridged probability and biology, valued models as tools for inquiry. His statement reframes models not as answers but as catalysts for better questions. This perspective resonates with science, where models are judged not just by accuracy but by how they illuminate unexplored unknowns.

“Wanderer, there is no road; the road is made by walking.”

Antonio Machado

One of Spain’s most beloved poets, Machado captured the spirit of exploration. His verse suggests that progress comes from forging paths rather than following them. Reducing ignorance often means inventing entirely new conceptual routes through uncharted intellectual terrain.

“Information is the resolution of uncertainty.”

Claude Shannon

The father of information theory, Shannon redefined knowledge as the reduction of uncertainty. This precise definition transformed communication and computing.

It also aligns perfectly with nescience, which measures discovery by how much it reduces inaccuracy and brings clarity to the unknown.

“An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem.”

John Tukey

A trailblazing statistician, Tukey emphasized exploratory thinking over rote analysis. His maxim warns that asking the wrong question wastes precision. It affirms that choosing the right unknown to explore matters more than perfectly solving a trivial one.

“Some mathematical statements are true for no reason; they’re true by accident.”

Gregory Chaitin

Chaitin, a founder of algorithmic information theory, exposed the limits of provability. His remark highlights that some truths are irreducible and beyond explanation. This insight sets a boundary for science: not all unknowns can be conquered, some are unknowable by nature.

“To go where you don’t know, you have to go by a way you don’t know.”

Saint John of the Cross

The Spanish mystic and poet portrayed spiritual growth as a journey through darkness. This line captures the essence of venturing beyond the familiar. Reaching the unknown unknown requires abandoning well-worn methods and daring to step into paths yet unimagined.

A.2 Chapter Pitures

The photographs presented at the beginning of each chapter have been carefully selected for their symbolic significance and their deep connection to the central themes of the theory of nescience. Each image embodies a key idea explored in the book, such as the pursuit of knowledge, the struggle against ignorance, the propagation of understanding across generations, and the continual expansion of the frontiers of the known. Rather than serving as mere decoration, these images act as visual metaphors, inviting the reader to reflect on the broader meaning of the concepts discussed in each chapter.

All the photographs included here are believed to be royalty-free (or at least publicly usable according to their appearance on Google Images). In addition, every image has been pre-processed using GIMP¹, the GNU Image Manipulation Program, to enhance their clarity, contrast, and compositional balance, ensuring a consistent visual style throughout the book. This curated selection aims to enrich

¹<http://www.gimp.org>

the reading experience, providing a visual narrative that parallels the intellectual journey from ignorance to understanding.

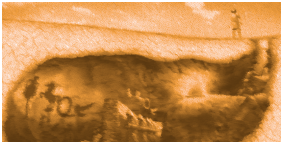
The Torch Bearers



The Torch Bearers is an aluminum sculpture by the American artist Anna Hyatt Huntington, donated to the city of Madrid in 1955 and now located on the campus of the Universidad Complutense. The work depicts an old, dying man passing a flaming torch to a vigorous young rider, who carries it forward with determination. The torch symbolizes the light of knowledge, handed from one generation to the next in humanity's endless

pursuit of understanding. As each generation advances the frontier of the known, they also illuminate new regions of the unknown, leaving an enduring legacy for their successors to explore. Huntington also created bronze casts of the sculpture, which can be found in Valencia, Havana, and at several cultural institutions across the United States.

The Allegory of the Cave



The Allegory of the Cave, conceived by the Greek philosopher Plato, illustrates humanity's struggle to move from illusion to understanding. Prisoners are chained in a cave, able to see only shadows cast on a wall, mistaking these fleeting silhouettes for reality.

When one prisoner breaks free and ascends to the surface, he is at first blinded by the light, but gradually perceives the true forms of things under the sun. This image embodies the essence of the theory of nescience: our initial grasp of the world is partial and distorted, yet through effort, questioning, and discovery we can escape the cave of ignorance and illuminate what was once unknown.

Ars Magna



Ars Generalis Ultima, also known as the Ars Magna ("The Ultimate General Art"), was a groundbreaking work published in 1305 by the Spanish philosopher Ramon Llull. In it, Llull described a mechanical method for reasoning, composed of concentric rotating wheels inscribed with fundamental concepts. By systematically combining these elements, the device aimed to generate all possible logical truths about a given topic (especially to defend and explain Christian beliefs through rational argument). Although primitive by modern standards, Llull's invention represents one of the earliest attempts to use formal logic as a tool for discovery. It stands as a distant ancestor of modern computational thinking and an early expression of the human desire to reduce ignorance by unveiling hidden relations between ideas.

Galileo's Telescope



Galileo Galilei was an Italian astronomer, physicist, engineer, philosopher, and mathematician who helped ignite the scientific revolution of the seventeenth century. He was among the first to proclaim that the language of nature is written in mathematics, and that its laws can be unveiled through systematic observation and reasoning. In 1610, Galileo built a telescope and turned it toward the heavens, revealing mountains on the Moon, the moons of Jupiter, and countless stars invisible to the naked eye. These discoveries shattered long-held beliefs and expanded the boundaries of the known cosmos. His telescope thus became a symbol of humanity's enduring drive to pierce the veil of ignorance.

Athena's Owl



This ancient silver coin depicts the owl of Athena, the Greek goddess of wisdom. Revered as a symbol of knowledge, insight, and scholarly pursuit throughout the Western tradition, the owl embodies the capacity to see through darkness, just as understanding allows us to pierce the unknown. The German philosopher Hegel famously wrote that "the owl of Athena spreads its wings only with the falling of the dusk," suggesting that philosophy comprehends the meaning of an era only as it fades into the past. In this sense, the owl reminds us that knowledge often arrives after the fact, illuminating what once was obscure.

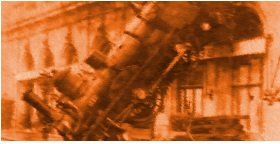
Escher's Infinite Stairs



This artwork, known as "Relativity" by M. C. Escher, depicts figures endlessly walking along staircases that defy the laws of geometry and gravity. Each group of figures moves consistently within its own frame of reference, yet the perspectives are incompatible with

one another, creating an impossible structure. This paradoxical scene symbolizes the essence of miscoding: when our representations of reality are flawed, they can lead to coherent but fundamentally incorrect interpretations. Like the figures trapped in Escher's world, we may continue along logical paths that never converge on truth, illustrating how mistaken encodings can imprison thought within self-consistent illusions, a central concern in the theory of nescience.

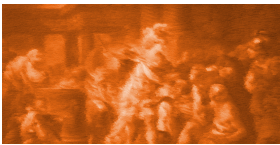
Train Wreck at Montparnasse



This historic photograph captures the train wreck at Montparnasse Station in Paris in 1895, when an express locomotive overran the buffer stop, crashed through the station wall, and plunged onto the street below. The accident, caused by a combination of mechanical failure and human miscalculation, serves

as a striking metaphor for inaccuracy. Even small errors in the models we use to predict and control complex systems can propagate into catastrophic failures when reality does not behave as expected. This image reminds us that inaccurate descriptions of the world not only mislead our understanding but can also derail our reasoning, sometimes with dramatic consequences.

Alexander Cuts the Gordian Knot



This painting depicts the legendary moment when Alexander the Great confronted the Gordian Knot, an impossibly tangled cord tied to an ancient chariot in the city of Gordium. According to prophecy, whoever could untie the knot would become ruler of Asia.

Rather than struggling to unravel its countless coils,

Alexander drew his sword and cut through it in a single stroke. This bold act symbolizes the triumph of simplicity over unnecessary complexity, an apt metaphor for surfeit, the burden of excessive and redundant information that obscures true understanding. This image reminds us that insight often comes not from adding more complexity, but from daring to strip it away to reveal the underlying essence of a problem.

Creation of Adam



This detail from Michelangelo's *The Creation of Adam*, painted on the ceiling of the Sistine Chapel, captures the moment when the hand of God reaches out to spark life, and with it, the potential for knowledge, into humanity. The near-touching fingers symbolize the latent connection between the divine source of truth and the human capacity to grasp it. This image embodies the idea that all knowledge exists in principle, yet remains inaccessible until we reach for it. What separates ignorance from understanding is the daring act of extending our minds toward the unknown, bridging the gap between what is given and what is comprehended.

Rodin's *The Thinker*



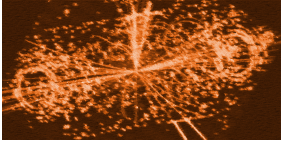
This sculpture, *The Thinker* by Auguste Rodin, portrays a solitary figure immersed in deep contemplation, his entire body tense with the effort of thought. Far from passive, his posture reveals thinking as an act of creation, shaping new possibilities within the mind. This image symbolizes the vital activity of discovering interesting questions: reaching beyond what is known, probing the boundaries of understanding, and daring to imagine what has not yet been asked. Every breakthrough begins not with an answer, but with the courage to pose a question that opens a new path into the unknown.

Rossum's Universal Robots



This photograph shows a scene from *R.U.R.* (Rossum's Universal Robots), the 1920 play by Karel Čapek that introduced the word robot to the world. In the play, artificial beings are created to perform human labor, eventually surpassing their makers. This early vision of automation resonates with the modern ambition of machine learning: building systems capable of discovering patterns and generating knowledge with minimal human intervention. The image symbolizes the quest to construct machines that can autonomously reduce our ignorance, relentlessly searching through the unknown to uncover new truths, guided by the principle of minimizing nescience.

Large Hadron Collider



This image shows the traces of particles produced in a collision inside the Large Hadron Collider (LHC) at CERN, where protons are accelerated to near-light speeds and smashed together to probe the fundamental structure of matter. Each collision generates a burst of data that allows scientists to test

hypotheses, refine theories, and discard false models. This process vividly illustrates how research reduces ignorance: by confronting the unknown through controlled experimentation, science transforms uncertainty into knowledge. Such experiments provide measurable evidence of progress, as each discovery narrows the gap between what is known and what remains to be uncovered.

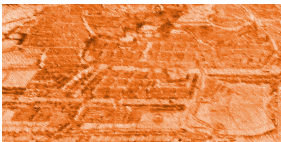
Wikipedia Monument



The Wikipedia Monument, located in Słubice, Poland, was created by the Armenian sculptor Mihran Hakobyan and unveiled in October 2014 as the world's first monument dedicated to the online encyclopedia. The statue honors the countless anonymous editors who, across all cultures and

languages, have voluntarily contributed to building one of humanity's most ambitious collective knowledge projects. The inscription celebrates Wikipedia as a symbol of global collaboration beyond political, religious, and cultural boundaries. A foundation for a knowledge society capable of fostering sustainable development, social justice, and peace among nations. This monument stands as a testament to the creative power of many minds working together to illuminate the unknown, piece by piece, like assembling the fragments of an ever-expanding puzzle.

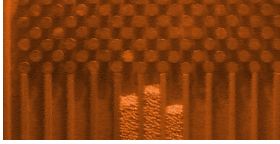
Königsberg Bridges



The old city of Königsberg in Prussia (now Kaliningrad, Russia) was built on both banks of the Pregel River, with two large islands linked to each other and to the mainland by seven bridges. The famous Seven Bridges of Königsberg problem asked whether it was possible to devise a walk that crossed

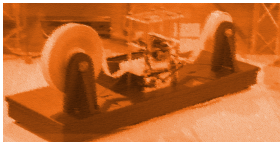
each bridge exactly once, starting and ending at any point. In 1736, the Swiss mathematician Leonhard Euler proved that such a walk was impossible. His elegant solution marked the birth of graph theory, a new branch of mathematics that studies the structure of connections, a foundational concept in discrete mathematics and in the formal representation of knowledge.

Galton's Box



This device, known as a Galton box or quincunx, was invented by the English scientist Francis Galton to demonstrate the principles of probability and the emergence of statistical regularities. Small balls dropped from the top bounce randomly off a series of pegs, eventually accumulating in bins at the bottom to form the familiar bell-shaped curve of the normal distribution. Although each individual path is unpredictable, the collective outcome is remarkably stable, revealing the hidden order that emerges from chance. In the context of probability theory, the Galton box illustrates how randomness can give rise to predictable patterns, transforming uncertainty into measurable knowledge.

The Turing Machine



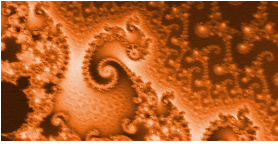
In 1936, the British mathematician Alan Turing introduced a formal model of a hypothetical computing device (later known as the Turing machine) and argued that it could perform any calculation that a human could carry out by following an algorithm. Turing's model was strikingly simple yet powerful enough to enable rigorous mathematical analysis of computation. Remarkably, many of the principles embedded in his abstract design anticipated the architecture of real computers built a decade later. The Turing machine stands as one of the rare moments in the history of science when theory not only preceded practice, but made it possible, laying the foundations of modern computer science.

Morse Key



This Morse key is a manual switching device used to send electrical pulses along a telegraph wire using the Morse code, named after Samuel F. B. Morse, the inventor of the telegraph. Morse code represents letters, numbers, and punctuation through standardized sequences of short and long signals called dots and dashes. Although not strictly binary (since it also includes symbols for spacing between letters and words) it is remarkably efficient: the average bit length per character in English is about 2.53 bits. This is an extraordinary achievement considering that Morse designed his code intuitively, long before the formal principles of coding theory were developed, making it one of the earliest practical examples of efficient symbolic encoding.

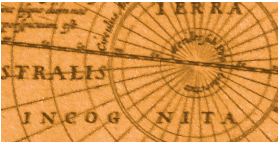
Mandelbrot Set



The Mandelbrot set is generated by sampling complex numbers and testing, for each point, whether repeatedly iterating a simple mathematical function causes the values to diverge to infinity. By interpreting the real and imaginary parts of each number as image

coordinates, pixels are colored according to how quickly their sequences diverge, while points that never diverge are shown in black. The resulting image reveals an intricate boundary that unfolds into endlessly finer, self-similar patterns under magnification. Despite its visual richness, the Mandelbrot set has very low Kolmogorov complexity, as it arises from an extremely short algorithm, illustrating how simple rules can produce extraordinary complexity in appearance while remaining simple in description.

Terra Australis Incognita



This antique map depicts a vast southern landmass labeled Terra Australis Incognita, the unknown southern land, a hypothetical continent once believed to balance the known continents of the Northern Hemisphere. Cartographers filled its empty expanse with speculative coastlines and imagined regions, a

testament to how human minds try to complete what they do not yet understand. This map stands as a powerful allegory of the unknown: the regions of knowledge we do not even realize we are missing. Like explorers staring at blank spaces on their charts, we must first recognize that these voids exist before we can begin to illuminate them.

The School of Athens



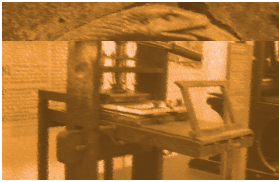
This fresco, *The School of Athens* by Raphael, portrays an idealized gathering of the great thinkers of classical antiquity (Plato, Aristotle, Socrates, Pythagoras, Euclid, and many others) engaged in animated debate and shared inquiry. Painted on the walls of the Vatican in the early 16th century, it

celebrates the collective pursuit of wisdom through dialogue, reasoning, and teaching. The scene embodies the essence of learning as a cumulative human endeavor: knowledge grows not in isolation, but through the exchange of ideas across generations. It symbolizes the collaborative effort to diminish ignorance, as each mind contributes a fragment of understanding to the expanding edifice of human knowledge.

Ouroboros

The Ouroboros, an ancient symbol depicting a serpent devouring its own tail, represents the eternal cycle of destruction and creation, of endings becoming beginnings. Often associated with alchemy, philosophy, and mysticism, it embodies the self-renewing nature of knowledge, how each resolution gives rise to new questions, and each discovery reshapes what remains unknown. The Ouroboros serves as a powerful metaphor for the dynamic process of inquiry: as we consume the known, we expose the unknown beyond it, endlessly renewing the pursuit of understanding.

Gutenberg's Printing Press



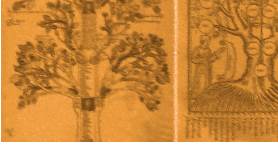
The Gutenberg printing press, invented by Johannes Gutenberg in the mid-15th century, revolutionized the production and dissemination of knowledge. By introducing movable type and mechanical printing, Gutenberg transformed books from rare, hand-copied artifacts into mass-produced vehicles of ideas, enabling knowledge to spread across continents and generations. This invention ignited an unprecedented expansion of learning, accelerating the scientific revolution and reshaping human thought. The printing press symbolizes a turning point where the known began to outpace the unknown, an enduring reminder that advances in how we transmit knowledge can profoundly accelerate our conquest of ignorance.

Great Library of Alexandria



The Great Library of Alexandria, founded in the 3rd century BCE in ancient Egypt, was one of the most ambitious centers of learning in human history. It is said to have housed hundreds of thousands of scrolls, gathering the accumulated knowledge of the known world under one roof. Though ultimately destroyed, its legend endures as a symbol of humanity's longing to preserve and unify all understanding. The Library of Alexandria represents both the grandeur and fragility of our efforts to conquer ignorance, an eternal reminder that the pursuit of knowledge is vast, precious, and always vulnerable to loss.

Tree of Knowledge



The Tree of Knowledge has appeared in many cultural, religious, and philosophical traditions as a symbol of the branching structure of human understanding. From medieval "arbores scientiarum" to moral trees and the Kabbalistic "Etz Chaim", such diagrams attempt to organize all domains of thought as growing from shared roots and unfolding into diverse branches. They express the idea that knowledge is not a collection of isolated facts, but an interconnected living system where new ideas grow from older ones. The Tree of Knowledge embodies humanity's ongoing effort to map the known while hinting at the vast, unseen forest of the unknown that still lies beyond its outermost branches.



B. Notation

When academics encounter a new idea that doesn't conform to their preconceptions, there's often a sequence of three reactions: first dismiss, then reject, and finally declare it obvious.
S. Sloman and P. Fernbach

- $\mathbb{N}$ set of natural numbers (including 0)
- $\mathbb{Z}$ set of integers
- $\mathbb{Z}^+$ set of positive integers
- $\mathbb{Q}$ set of rational numbers
- $\mathbb{R}$ set of real numbers
- $\mathbb{R}^+$ set of positive real numbers
- $x \in A$ x is a member of A
- $\{ \}$ set formation
- $A \subset B$ A is a subset of B
- $A \subseteq B$ A is a subset or equal to B
- $\emptyset$ empty set
- $d(A)$ cardinality of A
- $A \cup B$ union of A and B
- $A \cap B$ intersection of A and B

$A \setminus B$	set difference
$\bar{A}$	complement of A
$\mathcal{P}(A)$	power set
(x, y)	ordered pair
$A \times B$	cartesian product
A^n	n -fold cartesian product
R	binary relation
$\leq$	total order
$\max(A)$	maximum
$\min(A)$	minimum
$f(x) = y$	function
I_A	identity
f^{-1}	inverse function
$f \circ g$	composition
1_A	characteristic function
$\text{abs}(x)$	absolute value
$\lceil x \rceil$	ceil
$\lfloor x \rfloor$	floor
$l(s)$	length of string
λ	empty string
s^R	reverse string
$\mathcal{S}^n$	set of strings of length n
$\mathcal{S}^+$	set of all finite strings
$\mathcal{S}^*$	set of all finite string including the empty string
$<_p$	prefix
$\bar{s}$	Self delimited string
$\langle O \rangle$	Encoding as string of O
$\deg(v)$	degree of a vertex
$N(v)$	neighborhood of a vertex
Ω	sample space
$P(x)$	probability of x
$E(X)$	expectation of X
T	Turing machine
Q	set of states
Γ	set of tape symbols
$\sqcup$	blank symbol
Σ	input symbols
q_o	initial state
q_f	final state
τ	transition function

C	configuration
C_o	initial configuration
C_f	final configuration
U	universal Turing machine
$\mathcal{E}$	Set of entities
$\mathcal{O}_{\mathcal{E}}$	Representation oracle
$\delta_{\mathcal{E}}$	Representation function
$\mathcal{R}^*$	Set of perfect representations (short for $\mathcal{R}_{\mathcal{O}_{\mathcal{E}}}^*$ and $\mathcal{R}_{\mathcal{E}}^*$)
$\mathcal{R}_e^*$	Set of perfect representations for $e \in \mathcal{E}$
$\mathcal{R}$	Set of representations (short for $\mathcal{R}_{\mathcal{E}}$)
$\mathcal{R}_e$	Set of representations for $e \in \mathcal{E}$
$\hat{\mathcal{R}}$	Set of effective representations (short for $\hat{\mathcal{R}}_{\mathcal{E}}$)
$\hat{\mathcal{R}}_e$	Set of effective representations for $e \in \mathcal{E}$
$\hat{\mathcal{E}}$	Set of effectively known entities
$t \in \mathcal{T}$	Research topic
$d \in \mathcal{D}_t$	Description of a topic
$\mathcal{D}$	Set of descriptions
$\mathcal{D}_{\mathcal{T}}$	Set of valid descriptions of $\mathcal{T}$
$\mathcal{D}_t$	Set of descriptions of $t \in \mathcal{T}$
δ	Description function
d_t^*	Perfect description of $t \in \mathcal{T}$
$d_{t,s}$	Joint description of $t, s \in \mathcal{T}$
$\mathcal{D}_{t,s}$	Set of joint descriptions of $t, s \in \mathcal{T}$
$d_{t,s}^*$	Perfect joint description of $t, s \in \mathcal{T}$
$d_{t s^*}$	Conditional description of t given $s, t, s \in \mathcal{T}$
$\mathcal{D}_{t s^*}$	Set of conditional descriptions of t given $s, t, s \in \mathcal{T}$
$d_{t s^*}^*$	Perfect conditional description of t given $s, t, s \in \mathcal{T}$
$A \subset \mathcal{T}$	Research area
$\hat{A}$	Known subset of the area $A \subset \mathcal{T}$
$\mathcal{D}_{\hat{A}}$	Description of the area A given the known subset $\hat{A}$
$\mathcal{D}_{\hat{A}}$	Set of descriptions of the area A given the known subset $\hat{A}$
$d_{\hat{A}}^*$	Perfect description of the area A given the known subset $\hat{A}$
RG	relevance graph
R_t	relevance of topic t
IP_t	interestingness of topic t as a problem
M_t	maturity of topic t
AG	applicability graph
A_t	applicability of topic t
IT_t	interestingness of topic t as a tool
T'	set of known topics

$Q_{t_1 \rightarrow t_2}$	interesting question
$IQ_{t_1 \rightarrow t_2}$	interestingness of question $Q_{t_1 \rightarrow t_2}$
$\mathbb{F}$	unknown frontier
$\mathbb{S}$	new topics area
$S_{\{t_1, t_2\}}$	new topic
$IS_{\{t_1, t_2\}}$	interestingness of a new topic
IT_A	interestingness of an area as tool
IP_A	interestingness of an area as problem
$\hat{i}(\hat{\mathbf{y}}, \mathbf{y})$	inaccuracy of predicted values



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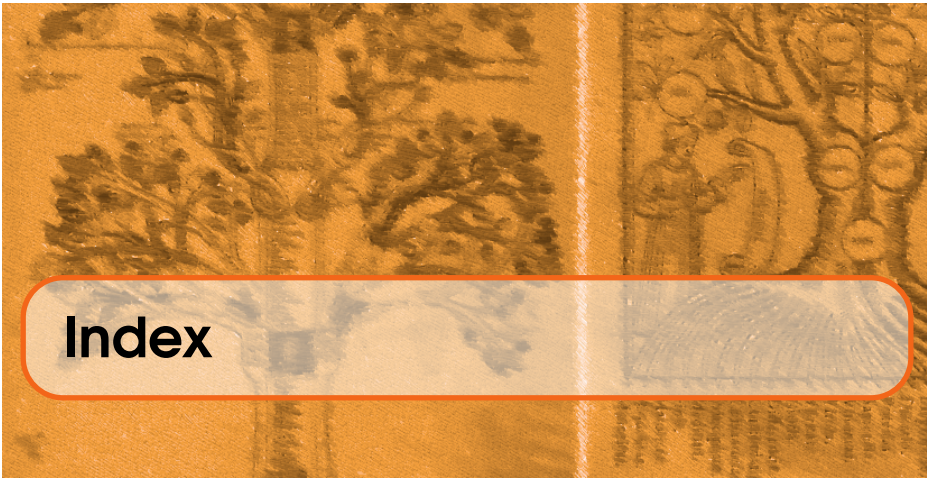
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